

# EXPLICIT BASE POINTS AND OBSTRUCTIONS TO PROPAGATION FOR WEIL CLASSES ON ABELIAN VARIETIES

DEEP BHATTACHARJEE<sup>†</sup>, PRIYABRATA MANDAL<sup>‡</sup>, AND USHASHI BHATTACHARYA

**ABSTRACT.** Every family of abelian varieties of Weil type, of any field, dimension and discriminant, contains a member whose Weil classes are explicit polynomials in divisor classes, and its Hodge conjecture is the algebraicity of one class on one connected period domain. Propagation from that member would follow from a semiregular perfect complex carrying the Weil class. We show that no complex whose Chern character is exactly a Weil class is semiregular, in any dimension: it would deform to non-algebraic Weil tori, where, extending a theorem of Voisin, coherent sheaves have no Chern character strictly between degree zero and the top degree. The complex must also carry classes, such as powers of the polarisation, that are Hodge only on the polarised family; for such a complex we compute in closed form the dimension that  $\text{Ext}^2$  must reach, and one complex reaching it suffices. By a descent lemma, the split families of the CM fields of degree at least four, in infinitely many dimensions, give every Weil family; for a quartic field at  $n = 2$ , the smallest of them, divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, of codimension two when the discriminant is trivial, and one complex must reach  $\dim \text{Ext}^2 = 112$ . Fourteen further routes are closed. This route needs three further statements. The last, the conjecture for varieties that are not abelian, is the whole conjecture through  $A \times \mathbb{P}^1$ ; its form modulo abelian varieties holds for varieties reached from abelian ones by correspondences and is open beyond, and with it the route is one of five minimal routes found by machine-checked computation. The Lefschetz standard conjecture holds for the total space of every Mumford family. The first classes beyond the Weil lines, on the square of a Mumford fourfold, are algebraic at every CM point, rigid along the base curve, and represented by cycles at every closed point of every reduction modulo a prime; they propagate to every member exactly when it holds for one ninefold, or when those cycles have bounded degree on a dense set, and once one complex has  $\text{Ext}^2$  of dimension 119; if they are algebraic, so is every Hodge class on every power of the fourfold.

---

*Date:* September 27, 2026.

*2020 Mathematics Subject Classification.* Primary 14C30, 14K05; Secondary 14C25, 14D07, 14F08, 14J28, 11G10, 13D07, 32G20.

*Key words and phrases.* Hodge conjecture, Weil classes, semiregularity.

<sup>†‡</sup>Corresponding authors: Deep Bhattacharjee<sup>†</sup>, at [d.bhattacharjee@erl-forschung.de](mailto:d.bhattacharjee@erl-forschung.de) and [itsdeep@live.com](mailto:itsdeep@live.com), and Priyabrata Mandal<sup>‡</sup>, at [priyabrata@manit.ac.in](mailto:priyabrata@manit.ac.in).

Every numerical assertion in this paper is reproduced by the scripts of [Appendix D](#), which use exact arithmetic, over  $\mathbb{Q}$ , over its finite extensions or modulo a prime, and no floating point at any step, and the finite arithmetic on which many of the paper’s theorems turn is separately checked by the Lean 4 kernel as described in [Appendix E](#). Both are distributed with the paper.

An earlier and substantially different version of part of this material was posted on Preprints.org, the preprint server of MDPI, on 5 August 2026, as *Relative Secant Cycles and Hodge Classes*, [doi:10.20944/preprints202602.0462.v5](https://doi.org/10.20944/preprints202602.0462.v5). That version proposed a construction of algebraic representatives for Weil classes. [Theorems 1.1](#) and [1.3](#) below show that no construction of that shape can produce a Weil class, and the present paper supersedes it. We thank the referees of that version for pressing on the identification step, which is where the obstruction lives. The authorship has changed since that MDPI preprint: Ushashi Bhattacharya, its second author, is the third author of this paper, and Priyabrata Mandal has been added as the second and corresponding author.

## CONTENTS

|                                                                  |     |
|------------------------------------------------------------------|-----|
| 1. Introduction                                                  | 2   |
| 2. Conventions                                                   | 16  |
| 3. Abelian varieties of Weil type                                | 18  |
| 4. Weil classes in coordinates                                   | 20  |
| 5. The character grading and the position of the Weil line       | 28  |
| 6. The cohomological Fourier–Mukai transform                     | 31  |
| 7. The divisor obstruction                                       | 34  |
| 8. The numerical count gives no obstruction                      | 36  |
| 9. The semiregularity target                                     | 38  |
| 10. The hyperplane obstruction                                   | 50  |
| 11. The Kuga–Satake route and where it stops                     | 52  |
| 12. What is left                                                 | 57  |
| 13. What a construction must break, and how Markman breaks it    | 59  |
| 14. An explicit secant plane in every dimension                  | 63  |
| 15. Base points in every Weil family                             | 71  |
| 16. The algebraic locus and the propagation criteria             | 86  |
| 17. The semiregularity criterion in its pure and corrected forms | 113 |
| 18. Secant objects on the split member                           | 131 |
| 19. Weil classes of every CM field                               | 168 |
| 20. Scope                                                        | 182 |
| 21. Hodge classes beyond the Weil lines                          | 219 |
| 22. Closure                                                      | 253 |
| Appendix A. The exterior algebra model in coordinates            | 259 |
| Appendix B. The locus of Hodge classes                           | 260 |
| Appendix C. Endomorphism algebras and where Weil type sits       | 264 |
| Appendix D. Verification                                         | 268 |
| Appendix E. Machine verification in Lean 4                       | 285 |
| Declarations                                                     | 290 |
| References                                                       | 290 |

## 1. INTRODUCTION

Let  $X$  be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}_X^p : \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) := H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$$

is surjective for every  $p$ . For  $p = 1$  this is the theorem of Lefschetz on  $(1, 1)$  classes.<sup>1</sup> For  $p \geq 2$  the conjecture is open, and the two attacks that are attempted most often are the following.

The first attack is algebraic and runs through abelian varieties. By the structure theory of Hodge classes on abelian varieties, the Hodge ring of an abelian variety of Weil type is generated by divisor classes together with finitely many exceptional classes in middle degree,

---

<sup>1</sup>The rational coefficients are not a convenience. Lefschetz proved the integral statement in degree two, and the integral analogue in higher degree is false: Atiyah and Hirzebruch produced torsion classes of type  $(p, p)$  that are not classes of cycles, and Kollár later produced non-torsion ones. So the conjecture has to be stated after tensoring with  $\mathbb{Q}$ , and every statement in this paper is a statement about  $\mathbb{Q}$ -coefficients.

the Weil classes of [Wei77].<sup>2</sup> Divisor classes are algebraic, so for these varieties the whole question is whether the Weil classes are algebraic. Markman [Mar25a, Mar25b] settled this for abelian sixfolds of Weil type of trivial discriminant,<sup>3</sup> which are the sixfolds of split Weil type, for every imaginary quadratic field, and deduced it for all abelian fourfolds of Weil type, for every discriminant and every field, using a degeneration argument of Schoen that he quotes; the Hodge conjecture for abelian fourfolds follows, and with the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [MZ99] it is quoted for abelian fivefolds as well, a deduction we have not checked ourselves. The sheaf he produces has positive rank, is built from the geometry of the abelian variety itself through a pair of secant sheaves on an abelian threefold, and satisfies an equivariant semiregularity condition.<sup>4</sup> For  $n \geq 4$ , and at  $n = 3$  for the discriminants not covered, the question is open. It is natural to hope that the higher cases can be reached by importing an object from an ambient projective space, restricting it to the dual abelian variety, and transporting the result to  $A$  by the Fourier–Mukai transform. Secant varieties of the dual are the usual choice of ambient object, because a dimension count makes the codimension come out right.

The second attack is geometric and runs through induction on dimension. One places the variety in a family of hyperplane sections, uses the algebraicity of the Hodge locus [CDK95] to control where the class stays of type  $(p, p)$ , appeals to the conjecture in one dimension lower on the sections, and attempts to transport the resulting cycles back up. Each ingredient here is a theorem. The question is whether they compose.

This paper proves that neither attack can work, and it is precise about the reasons, because the reasons are different in kind. The first fails for an algebraic reason that no refinement of the construction can repair: the classes the method produces live in the wrong subring, and they live there for reasons of  $K^\times$ -equivariance that are visible before any geometry is chosen. The second fails for a cohomological reason that is elementary once stated: the restriction map annihilates exactly the classes the induction would need to carry.

**1.1. The algebraic obstruction.** Throughout,  $A$  is a complex abelian variety of dimension  $g$ ,  $A^\vee$  is its dual,  $\mathcal{P}$  is the Poincaré bundle on  $A \times A^\vee$ , and  $\Phi_{\mathcal{P}}: D^b(A^\vee) \rightarrow D^b(A)$  is the

<sup>2</sup>Weil introduced them to show that the Hodge ring of an abelian variety is not always generated in degree one, which at the time was an open possibility. He did not conjecture that they are algebraic and did not claim they are not; the examples were offered as a warning, and they have been the standard test case ever since.

<sup>3</sup>In the normalisation of Definition 15.22 the discriminant is the class of  $(-1)^n \det H$  in  $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$ . Markman uses the class of  $\det H$  itself, in which the same sixfolds have discriminant  $(-1)^3 = -1$ .

<sup>4</sup>The hypothesis he verifies is stronger than injectivity of the semiregularity map alone, and he asks in [Mar25b, Question 11.4] whether it can be relaxed to injectivity on the image of an evaluation map, recording in his Section 12 that an affirmative answer would carry the method to abelian varieties of dimension at least eight and to fields of larger degree. Corollary 9.18 below is a statement about the unrelaxed hypothesis; Theorem 9.19 shows that for objects with Chern character concentrated on the Weil line the relaxed hypothesis fails as well, by the same  $n^2$  dimensions, and Proposition 18.18 that for the explicit split object of Section 14 it fails by the whole kernel; Theorem 9.20 that it fails for every object with a line bundle summand off the polarisation line, at every member and in every dimension, so that no sum of line bundles can run either argument (Corollary 18.22); and Remark 9.21 says what remains, which is the positive rank, indecomposable case the question was asked for. For that case Theorem 18.29 computes the classes the relaxed hypothesis is about, for every secant object on an abelian  $n$ -fold with  $n \geq 3$ , and reduces the hypothesis for Markman’s own candidate in dimension eight to one identity for a rank one complex on the fourfold; Theorem 18.34 shows that the identity fails, so that candidate does not satisfy the relaxed hypothesis either.

Fourier–Mukai transform with kernel  $\mathcal{P}$ . We write  $\theta$  for a principal polarisation class on  $A$  and  $\theta'$  for the dual class on  $A^\vee$ .

**Theorem 1.1** (Divisor obstruction). *Let  $A$  be a principally polarised complex abelian variety of dimension  $g$ , let  $i: A^\vee \rightarrow \mathbb{P}^{N-1}$  be any morphism defined by a linear system, and let  $\mathcal{F}$  be any object of  $D^b(\mathbb{P}^{N-1})$ . Put  $\mathcal{E} = \Phi_{\mathcal{P}}(Li^*\mathcal{F})$ . Then*

$$\mathrm{ch}_k(\mathcal{E}) \in \mathbb{Q} \cdot \theta^k \quad \text{for every } k \geq 0.$$

The proof has two ingredients and is short. The Chern character commutes with derived pullback, which confines  $\mathrm{ch}(Li^*\mathcal{F})$  to the subring of  $H^*(A^\vee, \mathbb{Q})$  generated by the hyperplane class, and the hyperplane class of a linear system on an abelian variety is a multiple of  $\theta'$ . The cohomological Fourier–Mukai transform then carries each power of  $\theta'$  to a rational multiple of a power of  $\theta$ ; we prove this in closed form in [Theorem 6.5](#). Neither ingredient uses anything about  $\mathcal{F}$ .

Since a Weil class is by construction not in the subring generated by divisor classes, the consequence is immediate.

**Corollary 1.2.** *With notation as in [Theorem 1.1](#), suppose in addition that  $g = 2n$  and that  $A$  is of  $(K, -1, n)$ -Weil type. Then  $\mathrm{ch}_n(\mathcal{E})$  is a Weil class only if  $\mathrm{ch}_n(\mathcal{E}) = 0$ .*

We then give a second proof of [Corollary 1.2](#) that does not pass through [Theorem 1.1](#) and that explains, in representation-theoretic terms, why the first proof had to succeed. The endomorphism  $\iota(\tau)$  attached to  $\tau \in K^\times$  acts on  $H^{2n}(A, \mathbb{Q})$ , and the  $K^\times$ -isotypic decomposition of that space separates the two objects in question.

**Theorem 1.3** (Character obstruction). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and write  $H^1(A, \mathbb{Q}) \otimes \mathbb{C} = V_+ \oplus V_-$  for the decomposition into eigenspaces of  $\iota(\sqrt{-d})$ . Then*

$$H^{2n}(A, \mathbb{Q}) \otimes \mathbb{C} = \bigoplus_{r+s=2n} \wedge^r V_+ \otimes \wedge^s V_-,$$

with  $\iota(\tau)$  acting on the  $(r, s)$  summand by  $\tau^r \bar{\tau}^s$ . The Weil line  $W(A)$  spans the two extreme summands  $(r, s) = (2n, 0)$  and  $(0, 2n)$ , on which  $K^\times$  acts by  $\tau^{2n}$  and  $\bar{\tau}^{2n}$ , while  $\eta^n$  lies in the middle summand  $(r, s) = (n, n)$ , on which  $K^\times$  acts by the norm character  $N_{K/\mathbb{Q}}(\tau)^n$ . In particular, for every  $n \geq 1$ ,

$$W(A) \cap H^{2n}(A, \mathbb{Q})^{N^n} = 0,$$

so a class that is a  $K^\times$ -eigenclass for  $N^n$  has zero component along the Weil line.

[Theorem 1.3](#) is what makes the obstruction feel inevitable rather than accidental. Any construction that is equivariant for the  $K$ -action, and whose output is therefore an eigenclass, must choose a character in advance, and the norm character is the one that the polarisation already occupies. The Weil line sits at the two ends of the  $K^\times$ -grading, as far from the middle as the grading allows. [Figure 3](#) draws the grading and marks the two positions.

It is tempting to look for a third obstruction of a numerical kind, and we record in [Section 8](#) why there is none. A secant numerology producing a class of codimension exactly  $n$  on an abelian variety of dimension  $2n$  requires a polarisation on  $A^\vee$  of Euler characteristic  $2n^2 + 2n$ , which for  $n \geq 2$  is smaller than  $2^{2n}$  and very much smaller than  $3^{2n}$ . Those two quantities are the Euler characteristics at which the theorem of Lefschetz guarantees that a linear system is free of base points and that it is very ample, and it is easy to read them as thresholds below which no such system exists. They are not. Both are sufficient conditions and neither is necessary: an abelian surface with a polarisation of type  $(1, 3)$

has  $\chi = 3 < 2^2$  and a base-point-free system, and one with a polarisation of type  $(1, 5)$  has  $\chi = 5 < 3^2$  and a very ample system, the latter being the Horrocks–Mumford embedding in  $\mathbb{P}^4$  [HM73, GP98]. We give the details in Section 8, because the erroneous reading appears in an earlier version of part of this material and we wish to retract it explicitly. Nothing else in the paper depends on it: Theorems 1.1 and 1.3 are independent of any numerical count.

**1.2. The geometric obstruction.** The second half of the paper concerns induction on dimension. Let  $j: Y \hookrightarrow X$  be a smooth ample hypersurface in a smooth projective variety of dimension  $d$ , and let  $L = c_1(\mathcal{O}_X(Y))$ .

**Theorem 1.4** (Hyperplane obstruction). *Let  $d = 2p$  and let  $\alpha \in H^{2p}(X, \mathbb{Q})$  be primitive. Then  $j^*\alpha = 0$  for every smooth ample hypersurface  $j: Y \hookrightarrow X$ .*

The proof is three lines from the projection formula and the Lefschetz hyperplane theorem, and we give it in Section 10. Its force is that the primitive middle degree is exactly where the Hodge conjecture is hard, and it is exactly where the restriction map returns nothing. Below the middle degree the restriction map is injective, but then the return trip is the problem.

**Theorem 1.5** (The return trip). *Let  $2p \leq d - 2$ , so that  $j^*: H^{2p}(X, \mathbb{Q}) \rightarrow H^{2p}(Y, \mathbb{Q})$  is an isomorphism. If  $j^*\alpha$  is algebraic on  $Y$ , then  $L\alpha$  is algebraic on  $X$ . The implication “ $L\alpha$  algebraic  $\Rightarrow \alpha$  algebraic”, asked of all smooth projective  $X$  and all  $\alpha$ , is equivalent to the Lefschetz standard conjecture.*

Putting the two together gives the statement we want.

**Corollary 1.6** (No dimensional induction through hyperplane sections). *Let  $X$  be smooth projective of dimension  $d$  and let  $\alpha \in \text{Hdg}^p(X)$ . An argument that deduces the algebraicity of  $\alpha$  from the Hodge conjecture for smooth ample hypersurfaces of  $X$ , using only  $j^*$  and  $j_*$ , returns no information when  $2p = d$  and  $\alpha$  is primitive, and otherwise establishes the algebraicity of  $L\alpha$  rather than of  $\alpha$ .*

Section 12 then states what is left. By hard Lefschetz every Hodge class is determined by primitive classes in degrees at most  $d$ , and the Lefschetz standard conjecture is the exact obstruction to moving between degrees algebraically. The conclusion is a clean equivalence, Theorem 12.2: the Hodge conjecture holds for all smooth complex projective varieties if and only if it holds for primitive classes of middle degree and the Lefschetz standard conjecture holds. This is a restatement and not a reduction, and we say so; its value is that it isolates the primitive middle degree as the only place where new ideas can enter, and Theorem 1.4 says that the classical inductive handle is blind precisely there.

**1.3. What is not obstructed.** We are careful in Section 20 about the limits of these results, because an obstruction theorem is only useful if its edges are sharp.

Theorem 1.1 does not apply to Markman’s construction [Mar25a, Mar25b], and Section 13 is devoted to saying exactly why, because that section is the useful one for anybody who wants to build something new. In outline: a rational class in the Weil line has a  $K^\times$ -orbit spanning a plane rather than a line (Proposition 13.1), so any construction reaching it must be bilinear in a pair of inputs interchanged by  $\text{Gal}(K/\mathbb{Q})$  (Corollary 13.2). Markman’s construction is a secant construction, and it satisfies this because it sites the secant geometry on an abelian  $n$ -fold  $X$ , attaches two conjugate secant sheaves to the two points where a plane of Hodge classes meets the spinor variety, and pairs them on  $X \times \widehat{X}$ . The

secant idea is therefore not what fails; siting it on  $A^\vee$  inside a projective space is what fails, and it fails three times over (Proposition 13.5). Question 13.7 states what an answer above the known range would have to supply, and Section 14 supplies three of its four ingredients in closed form for every  $n$  and every  $K$ : the abelian  $2n$ -fold of Weil type is  $X \times \widehat{X}$  with a  $K$ -action written down by one element of  $\text{NS}(X)$  (Theorem 14.2), and the plane is  $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t$  with  $u_t$  and  $v_t$  the even and odd parts of  $e^{\sqrt{-d}t}$  (Theorem 14.5). The third ingredient becomes an explicit equation: a secant sheaf has rank  $a$ ,  $c_1 = bt$ ,  $c_2 = \frac{ad+b^2}{2}t^2$ , and Bogomolov discriminant equal to the norm form  $N_{K/\mathbb{Q}}(b + a\sqrt{-d})$  times  $t^2$  (Theorem 14.7), so the numerical obstruction vanishes identically. At  $n = 3$  the equation forces the sheaf to be  $\mathcal{I}_Z(\Theta)$  with  $Z$  a union of  $d + 1$  translates of the Abel–Jacobi curve of a genus three Jacobian (Proposition 14.10), which is the shape of the object used in [Mar25b]. That equation is then solved in every dimension. Because the secant plane has no preferred scale, one is free to pass to a positive multiple of a rational point of it, and after clearing denominators the Chern characters of the powers of a single line bundle already span the subring  $\mathbb{Q}[t]$ , the matrix that expresses them in the basis  $t^k/k!$  being a Vandermonde matrix. A class of positive rank on a smooth projective variety is the class of a coherent sheaf, so the sheaf exists (Lemma 14.12 and Theorem 14.13), with an explicit integer witness in line bundles: at  $n = 4$ ,  $d = 3$  and  $a = b = 1$  the multiple is 6 and the witness is  $-23[\mathcal{O}_X] + 66[\mathcal{O}_X(t)] - 54[\mathcal{O}_X(2t)] + 20[\mathcal{O}_X(3t)] - 3[\mathcal{O}_X(4t)]$ . The fourth ingredient, the deformation argument, is untouched, and Remark 14.17 says so plainly; nothing here is a proof of the Hodge conjecture at  $n \geq 4$ , and Remark 14.15 separates what has been closed from what has not.

Theorem 1.3 applies to any construction that is  $K$ -equivariant with the norm character. A construction that is equivariant for the character  $\tau \mapsto \tau^{2n}$  would land in the right eigenspace, and we record in Remark 5.7 what such a construction would have to look like. We do not know of one, and the difficulty is that  $\tau \mapsto \tau^{2n}$  is not  $\mathbb{Q}$ -valued, so it cannot be realised by a correspondence defined over  $\mathbb{Q}$  without additional structure.

Theorem 1.4 concerns hyperplane sections. It does not obstruct induction through other families, for instance through degenerations, through blow-ups, or through the Kuga–Satake correspondence, which changes the variety rather than cutting it. Section 11 sets that construction out in full and marks its boundary. For a weight-two Hodge structure of K3 type it transports the question to an abelian variety and turns the class into an endomorphism, and Proposition 11.9 shows why nothing here bears on it; but on an abelian variety of Weil type the hypothesis  $h^{2,0} = 1$  holds only for  $n = 1$ , by Proposition 11.11, and  $n = 1$  is exactly the case in which the Weil line already consists of divisor classes.

**1.4. What the Weil classes look like, and how many there are.** Two further results are proved along the way, and they are the reason the paper is longer than the first two obstructions require.

Theorem 4.9 writes the Weil classes down. On the standard model of an abelian variety of  $(K, -1, n)$ -Weil type, built in Section 4 from a  $K$ -vector space of dimension  $2n$  and a hermitian form of signature  $(n, n)$ , the Weil line has the integral basis

$$\omega_1 = \sum_{|T| \text{ even}} (-1)^{|T|/2} d^{n-|T|/2} w^{(T)}, \quad \omega_2 = \sum_{|T| \text{ odd}} (-1)^{(|T|-1)/2} d^{n-(|T|+1)/2} w^{(T)},$$

where  $T$  runs over the subsets of  $\{1, \dots, 2n\}$  and  $w^{(T)}$  is the monomial taking  $y_j$  for  $j \in T$  and  $x_j$  otherwise. Each has  $2^{2n-1}$  terms. Written this way the classes are objects one can compute with, and two things become visible at once: the monomials occurring in them are

disjoint from those occurring in the powers of the polarisation, which gives an elementary second proof of [Proposition 3.10](#); and the balanced signature in the definition of Weil type is forced, since for any other signature the same two classes are of type  $(p, 2n - p)$  and are not Hodge classes at all.

[Theorem 4.14](#) counts what is there. For  $A$  of  $(K, -1, n)$ -Weil type with  $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ ,

$$\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = \begin{cases} 3, & k = n, \\ 1, & k \neq n, \end{cases} \quad \mathrm{Hdg}^n(A) = \mathbb{Q}\eta^n \oplus W(A),$$

so the Hodge conjecture for such an  $A$ , in every degree at once, reduces to the algebraicity of the single class  $\omega_1$ . That is [Corollary 4.15](#), and it is what makes the obstruction theorems worth proving: they are about one explicitly written class and not about a space of unknown size.

[Theorem 15.16](#) then produces that class from divisors whenever a certain condition holds. Write  $R$  for the rational structure on  $\Lambda^2 V_+ \oplus \Lambda^2 V_-$ , the part of  $H^2(A, \mathbb{Q})$  on which  $K^\times$  acts through  $\tau^2$  and  $\bar{\tau}^2$ . If  $R$  contains a Hodge class whose component in  $\Lambda^2 V_-$  is a nondegenerate form  $\Theta$ , then  $W(A)$  is spanned by the real and imaginary parts of  $\Theta^n$ , and these are polynomials in two divisor classes, hence algebraic. On  $A = X \times \hat{X}$  with the  $K$ -action of [Section 14](#) the condition holds with  $\Theta = d\beta - \hat{\beta} - \sqrt{-d}\ell$ , which is [Theorem 15.2](#). [Proposition 15.18](#) shows the condition fails whenever  $\mathrm{Hg}(A)$  is the full special unitary group, so the criterion is exactly a Noether–Lefschetz condition on  $R$ ; and [Proposition 15.19](#) determines what the split members supply: they have trivial discriminant, whatever polarisation is chosen, they form a locus of codimension  $n(n-1)/2$ , and at such a point the three classes  $\beta, \hat{\beta}, \ell$  already exhaust both the Néron–Severi group and the Hodge classes in the middle degree.

[Lemma 15.23](#) identifies the criterion: a Hodge class in  $R$  is the same thing as an endomorphism  $\mu$  with  $\mu(\tau) = \iota(\bar{\tau})\mu$ , so the criterion asks for a quaternion algebra  $(-d, \mu^2)_{\mathbb{Q}}$  inside  $\mathrm{End}^0(A)$  extending  $K$ . Those abelian varieties can be written down. For  $B = (-d, b)_{\mathbb{Q}}$  with  $b > 0$  acting on  $B^n$ , [Theorem 15.25](#) shows the balanced signature is forced rather than imposed, computes the hermitian form as  $\mathrm{diag}(2a_k d, -2a_k b d)$ , identifies the discriminant as the class of  $b^n$ , and verifies the criterion with  $\mu$  left multiplication by  $j$ . Since  $b$  may be any positive rational, and since discriminants multiply over products, [Theorem 15.31](#) follows: every Weil family, for every  $K$ , every  $n$  and every discriminant, contains a member whose Weil classes are algebraic, with a representative that is a polynomial in two divisor classes. What is then left is propagation and nothing else, which is [Corollary 15.32](#); and [Corollary 15.35](#) puts that in its sharpest form. Deligne’s theorem makes  $\omega_s$  absolutely Hodge at every point, so the Hodge conjecture for abelian varieties of Weil type, for every field, every dimension and every discriminant, is equivalent to the single implication that this one absolutely Hodge class is algebraic.

**1.5. What the last implication reduces to.** The set  $\Sigma \subseteq \mathcal{D}_{n,n}$  of points at which  $\omega$  is algebraic is then examined directly, in [Section 16.1](#). Because the relative Hilbert scheme is proper and the cycle class is locally constant in a flat family, the part of  $\Sigma$  cut out by one tuple of Hilbert polynomials is Zariski closed;  $\Sigma$  is the countable union of those pieces ([Theorem 16.4](#)). Because algebraicity survives an isogeny and  $\mathrm{SU}(V, H)$  acts trivially on the Weil line,  $\Sigma$  is stable under  $\mathrm{SU}(V, H)(\mathbb{Q})$ , and that group has dense orbits, so  $\Sigma$  is dense ([Proposition 16.7](#)). The two facts together give a dichotomy with nothing between its halves: either  $\Sigma$  is everything, and the Hodge conjecture holds for all abelian varieties of

Weil type in the family, or  $\Sigma$  has empty interior and the conjecture fails outside a meagre set (Corollary 16.5).

One further operation preserves  $\Sigma$  and moves up between dimensions; the descent of Proposition 20.13, described below, moves down. If  $A_1$  and  $A_2$  are of Weil type for the same  $K$ , with parameters  $n_1$  and  $n_2$ , then  $A_1 \times A_2$  is of Weil type with parameter  $n_1 + n_2$ , and the Weil class of the product is a fixed rational combination of external products of the Weil classes of the factors (Theorem 15.7). Written in terms of the  $K$ -valued invariant  $\omega = \omega_1 + \sqrt{-d}\omega_2$  the combination is simply  $\omega(A_1)\omega(A_2) = 2\omega(A_1 \times A_2)$ , so the Weil class is multiplicative and algebraicity passes from the factors to the product (Corollary 15.8). At  $n = 1$  the Weil classes lie in  $H^2$  and are therefore algebraic by Lefschetz with no hypothesis at all, so taking  $n$  abelian surfaces of Weil type gives, in every dimension and for every  $d$ , an abelian  $2n$ -fold of Weil type whose Weil classes are written as an explicit integral polynomial in  $2n$  divisor classes (Theorem 15.10). Like Corollary 15.4 and Theorem 15.31, it quotes nothing beyond the Lefschetz theorem on  $(1, 1)$  classes, and its reach is small: the locus of such products has dimension  $n$  inside a family of dimension  $n^2$ , smaller even than the quaternionic locus, and a general member of the family is simple and lies on none of them (Remark 15.11).

That is what turns the last implication into two statements of finite character. Theorem 16.9: if at every point of the rational orbit of a base point of Theorem 15.31 the Weil class is represented by a cycle whose Hilbert data comes from one finite set, the conjecture follows for the whole family; at  $n = 2$ , where the quaternionic locus is a divisor, infinitely many repetitions suffice rather than all but finitely many (Remark 16.25). Theorem 16.28: the base point carries a perfect complex whose Chern character is  $N\omega_1$  in degree  $n$  and zero elsewhere (Proposition 16.26), and if its semiregularity map is injective, the conjecture again follows for the whole family. Of the two, the second cannot be run on the object that supplies it: by Proposition 9.10 the complex of Proposition 16.26 has semiregularity kernel of dimension at least  $n^2$  at every base point constructed here, so its hypothesis fails for the object that supplies it (Remark 9.13), and by Corollary 9.18 the kernel is a canonical copy of the tangent space to the family the deformation was to cross, so no choice of base point inside the family can improve it. The degree bound is untouched by this, and Proposition 16.11 sharpens it: the leading coefficient of the Hilbert polynomial is constant along the orbit, because  $G(\mathbb{Q})$  preserves both the Weil class and the polarisation, so what has to be bounded is only the degree of the effective part of a representing cycle (Remark 16.12). The divisorial constructions cannot supply that bound: products of effective divisors of bounded degree represent  $\omega$  only on a nowhere dense set (Proposition 16.22), and Proposition 16.24 shows the degrees growing along an explicit pencil of quaternionic loci in one fixed polarised lattice. Neither reduction mentions a general point and neither assumes the variational Hodge conjecture; the passage from a formal deformation at one point to cycles everywhere, which is the part that is usually missing, is supplied by Theorem 16.4. Neither hypothesis is available as stated: the first is equivalent to its conclusion (Theorem 16.17), and the second is false (Theorem 17.5) and survives only in the corrected form of Remark 17.6, both described below; Remark 17.7 says what each of them asks for.

Section 18.1 then says what the object of the second criterion has to be. For a sum of line bundles the semiregularity map is computed in closed form (Lemma 18.1): it annihilates every off-diagonal extension class and the traceless part of every diagonal one, so injectivity forces multiplicity one, a vanishing of cohomology that the shifts can arrange, and the injectivity of  $(\zeta_i) \mapsto \sum_i e^{\lambda_i} \zeta_i$ . The Chern character condition then fails

outright. [Theorem 18.5](#): at any member whose only divisor classes of norm character are the multiples of  $\eta$ , if the Chern character of an object is an effective combination  $\sum_i m_i e^{\lambda_i}$  of exponentials of divisor classes and  $\text{ch}_2$  is a flat Hodge class, then the coefficient of  $\theta^+ \theta^-$  is  $\sum_i m_i N_{K/\mathbb{Q}}(u_i)$  and must vanish; the norm form is positive definite and the  $m_i$  are positive, so every  $u_i$  is zero and the Weil component of the Chern character is zero as well. The separation of  $\theta^+ \theta^-$  from  $\eta^2$  that this needs is [Lemma 18.4](#), and it is a statement about tensor rank: in the isotypic component for the square of the norm character the first class has rank one and the second has rank  $\binom{2n}{2}$ .

The hypothesis on the Chern character holds for every sheaf filtered by line bundles, for every semi-homogeneous bundle in the sense of Mukai [[Muk78](#)], and for every iterated extension of those; the hypothesis on the divisor classes holds at every split member and at every quaternionic base point. So the candidates one would reach for first do not exist, and the reason is positivity rather than a computation. What is left must be indecomposable in the derived category with a class in  $K_0$  that is not effective, which is a sharper description of the remaining construction than we started with.

The same subsection then asks where such an object could live at all, and the answer is arithmetic. Writing the candidate as a sum of line bundles indexed by elements  $u_i$  of the ring of integers of  $K$ , the Chern character conditions become  $T_{q,r} = \sum_i m_i u_i^q \bar{u}_i^r = 0$  for all  $q, r \leq n$  but three, and grouping the  $u_i$  by norm turns the diagonal part of that system into  $\sum_j M_j N_j^r = 0$  for  $r = 1, \dots, n$ , where  $N_1 < \dots < N_\nu$  are the norms that occur and  $M_j$  is the sum of the multiplicities at level  $N_j$ . Fewer than  $n + 1$  norms forces every  $M_j$  to vanish and the rank with it ([Corollary 18.13](#)); at exactly  $n + 1$  the level sums are forced by Lagrange interpolation, and since multiplicity one allows each element of the ring only once, they are capped by the number of elements of each norm. [Theorem 18.20](#) runs that competition to the end. For  $4 \leq n \leq 8$ , six imaginary quadratic fields and norms up to the bounds in [Appendix D](#), no set of norms meets the two demands, so in that range the split route closes before any geometry is used. At  $n = 3$  it does not: exactly eleven sets of four norms survive, all at  $d = 1$  or  $d = 2$ , and the smallest of them is  $(9, 18, 27, 36)$  over  $\mathbb{Q}(\sqrt{-2})$ , where the level sums are  $(4, -6, 4, -1)$  and the object would have rank one. For all eleven the rest of the system is then shown to have no solution, by an exhaustive search over sign patterns that covers  $4.29 \times 10^{14}$  cases in the largest instance.

**1.6. Verification.** Every numerical claim in this paper is checked by exact arithmetic, and the output is reproduced in [Appendix D](#). The scripts verify the closed form of the cohomological Fourier–Mukai transform on powers of  $\theta$ , the stability of  $\mathbb{Q}[\theta]$  under the transform, the numerical question of [Corollary 8.4](#) for  $2 \leq n \leq 8$ , the identity  $\sum_q h^{q,q+2} = \binom{4n}{2n-2}$  for the semiregularity target of [Section 9](#), the character separation of [Theorem 1.3](#), the coordinate model and the two generators (11), the invariant count behind [Theorem 4.14](#), the Clifford arithmetic of [Section 11](#), and the family and secant plane of [Section 14](#). All arithmetic is over  $\mathbb{Q}$  or in the exterior algebra of a finite-dimensional vector space with integer structure constants; no step is numerical in the floating-point sense. [Appendix E](#) describes a second, independent check of the finite arithmetic by the Lean 4 kernel.

**1.7. Organisation.** [Section 2](#) fixes conventions. [Section 3](#) develops the structure theory of abelian varieties of Weil type from the definitions, [Section 4](#) writes the Weil classes out in coordinates and counts the Hodge classes, and [Section 5](#) proves [Theorem 1.3](#). [Section 6](#) establishes the closed form of the cohomological Fourier–Mukai transform, and [Section 7](#) proves [Theorem 1.1](#) and [Corollary 1.2](#). [Section 8](#) settles the numerical question in the

negative, and [Section 9](#) computes the semiregularity target, explains why it is not the binding constraint either, and then proves [Theorem 9.7](#): no object whose Chern character is concentrated on the Weil line is semiregular, because the annihilator of the Weil line in  $H^{0,2}$  has dimension  $n^2$  ([Lemma 9.6](#)). It then identifies that annihilator: by [Theorem 9.17](#) the polarisation carries the tangent space to the Weil family isomorphically onto it, and by [Proposition 9.14](#) a polarisation-preserving deformation kills the whole Weil line exactly when it commutes with  $K$ , so the family is also the first-order Hodge locus of the class. The kernel and the family are therefore the same space, which is [Corollary 9.18](#); and [Theorem 9.19](#) places that space inside the image of the evaluation map of Hochschild cohomology, so that the weakened semiregularity criterion of [[Mar25b](#), Question 11.4] is unavailable for such objects too; and [Theorem 9.20](#) shows, from the infinitesimal rigidity of divisor classes, that any object with a line bundle summand off the polarisation line is obstructed along the family in a direction the semiregularity map cannot see, which removes every sum of line bundles in every dimension. This is the fourth obstruction, and it is the one that bears directly on the fourth ingredient of [Remark 14.17](#). [Section 10](#) proves [Theorem 1.4](#) and [Theorem 1.5](#), and [Section 11](#) marks the boundary of the one classical route that gets past them. [Section 12](#) proves the equivalence, [Section 13](#) identifies which hypotheses a construction must break, and [Section 14](#) gives a family of examples in which three of the four ingredients are supplied in closed form, settles the base case in every family, and closes with [Section 16.1](#) on the structure of the algebraic locus and [Section 18.1](#) on the object the second criterion requires, where [Proposition 18.16](#) computes the first-order obstruction of the explicit split object against the family: it is nonzero exactly in the directions normal to the split locus, it lies in the kernel of the semiregularity map, and it is transverse to the copy of the tangent space that [Corollary 9.18](#) places there; and [Proposition 18.18](#) writes out the evaluation map of Hochschild cohomology on that object and finds that the whole kernel of the semiregularity map is in its image; and [Proposition 18.25](#) computes the Chern character of Markman’s candidate object in dimension eight, places it at the point  $(1, 3)$  of the secant plane, and reduces its verification to one condition on one sheaf; and [Theorem 18.29](#) moves that condition from the eightfold to the fourfold, by computing the Hochschild classes that preserve any secant Chern character,  $n^2$  of them, so that the weakened criterion for the transform of  $\mathcal{F} \boxtimes \mathcal{F}$  becomes the single identity  $\text{At}(\mathcal{F})^2 = -d\Theta^2 \cdot \text{id}$  in  $\text{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega^2)$  for  $\mathcal{F}$  itself ([Corollary 18.30](#)); and [Theorem 18.34](#) decides that identity in the negative, by a local computation at an isolated double point of the support, so that the candidate of Markman’s [Section 12](#) does not satisfy the weakened criterion, and [Corollary 18.36](#) extends the obstruction to every dimension. What that argument cannot reach is a support with no such point, and [Section 18.3](#) takes up that case. At a smooth point the sheaf  $\mathcal{E}xt^2$  of the ideal vanishes, so both sides of the identity are locally zero and the germ comparison has nothing to compare ([Proposition 18.38](#)); the question moves from a point to the normal bundle. In exchange the surface itself becomes rigid. Since the Todd class of an abelian variety is trivial, Grothendieck–Riemann–Roch turns the Chern character into the whole numerical package of the support, and for  $\text{ch}(\mathcal{F}) = u_\Theta + b v_\Theta$  on a fourfold it gives

$$\chi(\mathcal{O}_S) = (b^2 + d)(3b^2 - d), \quad K_S^2 = 3(b^2 + d)(7b^2 - d), \quad e(S) = 3(b^2 + d)(5b^2 - 3d),$$

with nothing left free. Because  $\Omega_X^1$  is a quotient of the trivial bundle  $\Omega_X^1|_S$ , it is globally generated, so  $K_S$  is nef and  $e(S) \geq 0$ ; then  $K_S$  is also big,  $S$  is minimal of general type, and the Bogomolov–Miyaoaka–Yau inequality applies. Its defect is  $3e(S) - K_S^2 = 24(b^4 - d^2)$ , so the inequality says exactly  $d \leq b^2$  ([Theorem 18.40](#)). At the value  $b = 3$  that the

fourfold forces, this leaves  $d \in \{1, 3, 5, 7\}$  (Corollary 18.41): for every imaginary quadratic field beyond those four there is no secant object of rank one with smooth support at all, and for those four the invariants are the six numbers of Table 3 and nothing else. The same subsection then identifies what the local argument is really testing. It is not smoothness and it is not the complete intersection property: at any point where the support is Cohen–Macaulay the sheaf  $\mathcal{E}xt^2$  of its ideal vanishes, because Auslander–Buchsbaum makes the ideal of projective dimension one there, so both sides of the identity are locally zero (Theorem 18.43). A Macaulay2 computation over ten germs confirms the line, and shows it is sharp: the fat plane and the cone over the twisted cubic are not complete intersections and still carry no obstruction, while two planes meeting at a point, three planes meeting pairwise at a point, and a plane with an embedded point all do. So the criterion splits cleanly (Corollary 18.45): a support that is not Cohen–Macaulay is removed outright, and a support that is Cohen–Macaulay has no local obstruction at all, leaving one global vanishing in  $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ . That also explains why Markman’s candidate cannot be repaired by normalising its double points: separating the branches is what restores the depth, and Remark 18.46 shows it moves the Chern character off the secant plane. A Cohen–Macaulay support of codimension two has an ideal sheaf of projective dimension one, by the same Auslander–Buchsbaum argument, so it is resolved by two vector bundles of ranks  $r + 1$  and  $r$ . That is a shape one can search. Section 18.5 does so. When the two bundles are sums of line bundles the Chern character condition becomes four equations on the twists, and at  $r = 1$ , the complete intersections, they close outright for every  $b$  and every  $d$ : weighting by  $w(x) = x^2 + d$ , which is positive because  $d > 0$ , turns three of the equations into the statement that two positive measures share their mass, mean and variance, and a one atom measure has variance zero, so the two divisor degrees would have to coincide and the resolution would cancel (Theorem 18.49). At  $r = 2$  and  $r = 3$  an exact search over the twists finds nothing either. At  $r = 4, 5, 6$  solutions appear, but only at  $d = 15$  and  $d = 23$  (Theorem 18.50). None of them survives: a morphism from a secant object that satisfies the criterion to a power of the polarisation is killed by  $H^2(\mathcal{O})$  (Lemma 18.51), and for the cone of a map between sums of such powers the summand inclusions are not, because the relevant  $H^2$  of a nontrivial power vanishes; so no split resolution runs the criterion at any rank, any twist or any discriminant (Theorem 18.52), and the same holds when the line bundles are replaced by simple semi-homogeneous bundles, the bundles with scalar Atiyah class (Theorem 18.55). What remains is a support that is either smooth, with  $d$  one of four values, or singular and Cohen–Macaulay, in both cases with a Hilbert–Burch resolution over a bundle that is not projectively flat and a restriction  $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_Z)$  that is injective (Corollary 18.57). Remark 18.58 writes the two remaining questions in the form a construction would have to answer. After that, Section 19 removes the restriction to quadratic fields. The four theorems that turn the conjecture for a Weil family into a propagation statement, namely the reduction to one class, the closedness of the algebraic locus, the density of the rational orbits and the base point, use nothing about the degree of the field. Carrying them over needs one new input, a base point, and we supply it for every CM field  $F$  of degree  $2m$ , every  $n$  and every discriminant: the abelian variety is isogenous to a product of CM abelian  $m$ -folds, the Weil classes there are polynomials with rational coefficients in  $F$ -balanced divisor classes, written out in Theorem 19.4, and they are algebraic by the Lefschetz theorem on  $(1, 1)$  classes. The density of the rational orbits needs the hermitian form to be isotropic over the totally real subfield, which the Hasse principle gives, and then the unipotent radicals of the parabolic subgroups generate (Theorem 19.6). So the Weil classes of the higher

CM fields are not a separate problem standing beside the quadratic ones: they are the same propagation problem, and the semiregularity criterion, met at one base point of each family, closes all of them at once. Nothing in that section is conditional on Question 11.4 or on any secant construction. [Section 20.5](#) then sets down, once and in full, what separates all of this from the conjecture, and [Section 20.9](#) decides which parts of it matter. The implications proved here, together with those the literature supplies, form a finite rule set, and taking consequences in that rule set is a computation rather than a judgement. The conjecture is not among the consequences, which is the machine statement that nothing here proves it. The route that passes through this paper needs three open statements: a propagation statement, the semiregularity criterion of [Theorem 16.28](#) in the corrected form of [Remark 17.6](#), carried to every CM field in [Theorem 19.6](#) and by [Proposition 20.13](#) needed only for the split families of the CM fields of degree at least four, together with the Hodge classes on abelian varieties that divisor and Weil classes do not generate, and the conjecture for the varieties that are not abelian ([Corollary 20.41](#)). The last of the three is not a remainder: the varieties that are not abelian include  $A \times \mathbb{P}^1$  for every abelian variety  $A$ , and the conjecture for  $A \times \mathbb{P}^1$  gives it for  $A$ , so that statement is equivalent to the whole conjecture ([Proposition 20.35](#)). The statement that belongs in its place is weaker: every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences ([Definition 20.36](#)). It holds on every variety whose cohomology is reached from abelian varieties in that way, a class containing curves and abelian varieties and closed under products and surjective images, in degrees 0, 2,  $2n - 2$  and  $2n$ , and in dimension at most three ([Proposition 20.37](#)); beyond that it is open, and its standard open case, the square of a K3 surface, turns on the same Kuga–Satake correspondence as the Mumford classes below ([Remark 20.38](#)). With both statements in the rule set there are exactly five minimal sets of open statements that yield the conjecture ([Theorem 20.40](#)). Two are the conjecture restated: the conjecture for the varieties that are not abelian alone, and the Lefschetz standard conjecture for every variety together with the motivatedness of every Hodge class ([Proposition 20.17](#)). The other three use the weaker statement, together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes that it implies ([Proposition 20.18](#), [Theorem 20.19](#)), or, along the route of this paper, with the propagation statement and the classes that divisor and Weil classes do not generate. Only that last route uses a theorem of this paper. Every statement in the five except the propagation statement is a consequence of the conjecture, and the Lefschetz standard conjecture, which lies in two of them and gives the whole abelian half, is the target at the centre ([Remark 20.43](#)). None of these statements is proved here in general. One case of the Lefschetz standard conjecture, by a mechanism going back to Tankeev [[Tan03](#)], is: for an abelian scheme over a curve whose invariant cycles are algebraic, the operator  $\Lambda$  is the sum of a relative Pontryagin product with  $\ell^{g-1}/(g-1)!$  and an operator along the base built from the invariant cycles, so it is algebraic; this covers the total space of every Mumford family ([Proposition 20.21](#), [Theorem 20.22](#), [Corollary 20.23](#)). With Markman’s theorem it gives the same for every abelian scheme over a curve of relative dimension at most four whose invariant classes are Hodge, and for every fibre power of a family of Weil fourfolds with connected monodromy  $SU(V, H)$  ([Corollary 20.26](#), which uses the Hodge conjecture for all powers of such fourfolds, [Proposition 20.25](#)); and over its Shimura curve the total space of a Mumford family satisfies the Hodge conjecture ([Proposition 20.27](#)). The same decomposition shows that, for such a total space, the Lefschetz standard conjecture is exactly the statement that algebraicity propagates along the curve ([Theorem 20.29](#)). With the base points of this paper that turns

each of the two remaining targets into the Lefschetz standard conjecture for one variety: one Weil family is equivalent to it for the total space of the family over a curve through a base point (Corollary 20.32), and the first classes beyond the Weil lines are equivalent to it for the ninefold  $W \times_C W$  of a Mumford family (Corollary 20.31). Two statements are proved outright on the way: the Hodge conjecture for  $X_c \times X_c$  at every CM point  $c$  of a Mumford family, from the Hodge conjecture for abelian fourfolds (Theorem 20.30), and the fact that no finite direct sum of shifted line bundles, or of the bundles  $\pi_* L$  pushed forward along isogenies, among them the simple semi-homogeneous bundles, meets the semiregularity criterion in the corrected form described below (Proposition 17.14); the corresponding statement for the first form of the criterion, Theorem 20.33, is vacuous by Theorem 17.5. The bounded criterion, which asks that the cycles representing the class along the orbit have bounded complexity, might look like a second propagation statement, but by Theorem 16.17 it is equivalent to the conjecture for the family, a restatement and not a route; read on the transported cycle it is simply false, because the transport multiplies the degree of a component by  $c^{2n}$  and divides it by the order of the stabiliser the kernel meets, which is bounded for any component with finite stabiliser (Lemma 16.14, Theorem 16.15). Nor are the last two statements reductions, to Weil classes and to abelian varieties, in any sense that would help: the self-product of an abelian fourfold of Mumford type has eight Hodge classes in  $H^4$  against six products of divisor classes, and the two classes left over are Weil classes of nothing (Proposition 20.44); and the  $H^3$  of a very general quintic threefold is not of abelian type, because the Hodge cocharacter acts on the Lie algebra of its Mumford–Tate group with a weight that no tensor construction on abelian varieties can produce (Proposition 20.45). Each of the two is the conjecture for a class of Hodge classes that no Weil line reaches. The first of them can be pressed one step further. The two classes on the Mumford fourfold are the real multiplication by a totally real cubic field on the primitive part of  $H^2$ , and they are algebraic as soon as the Kuga–Satake class of one K3 surface of Picard number thirteen attached to the fourfold is; the Clifford relations of the Kuga–Satake construction are what bring the whole field into play (Proposition 21.2, Theorem 21.4, Remark 21.5). That Kuga–Satake class is not known to be algebraic. If the two classes are algebraic, so is every Hodge class on every power of the fourfold (Theorem 21.7), the one class that pull-backs and products do not reach being obtained by composing correspondences. The Kuga–Satake class asks for more than the two classes: it is algebraic exactly when they are and some algebraic cycle links the surface to a power of the fourfold (Proposition 21.12), and no such link passes through an abelian variety of dimension at most five (Proposition 21.13). The same classes are rigid: in each of the 46 weight blocks of  $HT^2$  of the square the Gram determinant of contraction into a rational exceptional class is a product of sums of squares and of linear forms that do not vanish at the conjugates of an irrational element of the cubic field, so the only first order deformation keeping the class of Hodge type is the direction of the compact Mumford curve, and no degeneration or larger family reaches it (Theorem 21.21, Corollary 21.23). Propagation along the curve follows from one perfect complex at one CM point with  $\text{Ext}^2$  of dimension 119, the least its Chern character allows, or from a degree bound at infinitely many CM points (Theorem 21.27, Corollary 21.28). Such an object is heavily constrained: for a dense set of classes its self-extensions have dimension at least 1, 16, 119, 328, 560, 328, 119, 16, 1 in degrees 0 to 8, the Euler characteristic forces at least 800 in odd degrees when there are none in negative degrees, its Chern character lies outside the subring generated by divisor classes, and it extends along no Hochschild direction but that of the curve (Theorem 21.31, Proposition 21.32, Corollary 21.36). No sequence of

cones, tensor products, pull-backs, push-forwards and Fourier–Mukai functors applied to line bundles reaches the classes, at any point of the curve, because every such operation preserves invariance under the Lefschetz group ([Theorem 21.33](#)). Among all complex tori, algebraic or not, the component of the Hodge locus through the square is the Mumford curve itself, so no twistor line through a member keeps the classes of Hodge type, and Markman’s twistor mechanism for Weil classes cannot move them; the twistor lines on which they are of Hodge type carry non-algebraic tori ([Proposition 21.24](#)). The square is a holomorphic symplectic variety whose symplectic class generates a copy of the structure of K3 type in  $H^1 \otimes H^1$ , and the classes are exactly the dual forms of that copy twisted by the real multiplication; so the mechanism of van Geemen and Rapagnetta [[vGR26](#)], pull-back from a hyperkähler manifold, would need a rational map onto a symplectic subvariety of a hyperkähler manifold of dimension at least ten, such as  $S^{[n]}$  with  $n \geq 5$  for a K3 surface of Picard number thirteen, and no hyperkähler eightfold serves ([Proposition 21.25](#)). After Li’s proof of the Tate conjecture for powers of abelian fourfolds over finite fields [[Li26](#)], the classes are represented by algebraic cycles at every closed point of every reduction of the curve modulo a prime, and the target is equivalent to a bound on the Hilbert data of those cycles on a Zariski dense set of closed points of the arithmetic model, with no lifting step ([Proposition 21.37](#), [Theorem 21.38](#)). [Remark 21.29](#) audits the ten routes that avoid the Lefschetz standard conjecture, and none is carried out. The five that are not closed off are not five obligations: three are the target in other forms and two imply it, so any one would suffice ([Corollary 21.40](#)). And the target is only the first case of the Hodge classes that divisor and Weil classes do not generate, so a proof of it would leave open the propagation statement, the other such classes, and the conjecture for the varieties that are not abelian, which is the whole conjecture ([Remark 21.41](#), [Proposition 20.35](#)).

The propagation statement itself is constrained from several sides, and the constraints describe the object it asks for closely. Its semiregularity form forces the evaluation map of Hochschild cohomology to vanish on the  $4n^2$ -dimensional annihilator of the Weil class, and so forbids the representative to be a sheaf on any smooth germ of codimension two of its support ([Theorem 16.30](#), [Corollary 16.31](#)). That annihilator has a shape:  $HH^1(A)$  splits as  $P \oplus Q$  into the annihilators of the two conjugate pieces  $\alpha_{\pm}$  of the Weil class, each of dimension  $2n$ , and the annihilator in degree two is exactly  $P \wedge Q$  ([Theorem 16.33](#)). So a complex meeting the criterion has its whole Hochschild action factoring through  $\bigwedge^* P \oplus \bigwedge^* Q$ , of dimension  $2^{2n+1} - 1$  in a ring of dimension  $2^{4n}$ : it sees the two halves of the class separately and their interaction not at all. With no hypothesis on semiregularity, every perfect complex with Chern character on the Weil line has  $\dim \operatorname{Ext}^2(E, E) \geq 2n(2n - 1)$  ([Corollary 16.34](#)), and the minimum is itself enough: a single complex at a single base point with  $\dim \operatorname{Ext}^2(E, E) = 2n(2n - 1)$  has injective semiregularity map and closes the family ([Theorem 16.36](#)). This turns the criterion from a rank into a number, which is the form in which a candidate can be tested. It fixes more than itself: by the Hodge–Riemann relations  $\chi(E, E) > 0$ , an indecomposable summand carries the class, and at  $n = 2$ , and at  $n = 3$  granting a compatibility the paper already uses, it has at least three independent endomorphisms, so no simple object meets the criterion ([Theorem 16.40](#)). Finally the support is placed. Two independent jet classes on a complex of finite length multiply to a nonzero class whenever its Euler characteristic is nonzero, because on a torus the semiregularity map sees that product through the Euler characteristic ([Lemma 16.37](#)); applied on a slice transverse to a component of the support, this shows that every component of codimension at least two with nonzero multiplicity is a union of cosets of an abelian subvariety tangent to a  $K$ -eigenspace, and since the Weil line misses every class pulled

back from such a quotient, the support must have a component of codimension less than  $n$  (Theorem 16.38). No complex carried by a cycle that represents the Weil class can serve, at any member of any family. The hypothesis on the Euler characteristic is sharp: the cone of the product itself, on the Koszul complex of a point, has Euler characteristic zero and kills it.

All of this concerns a complex whose Chern character is exactly a multiple of the Weil class, and no such complex is semiregular (Theorem 17.5). Its semiregularity would make it extend along every  $K$ -linear deformation of the abelian variety, and those reach complex tori that are not algebraic. On a very general such Weil torus the only Hodge classes strictly between degree zero and the top degree are the Weil classes, there are no subvarieties but points, and every coherent sheaf has vanishing Chern character in degrees 1 to  $2n - 1$ : this is Voisin's theorem in dimension four [Voi02b], and the same argument through the Hermite–Einstein metrics of Bando and Siu gives it in every dimension (Lemma 17.2, Lemma 17.3, Proposition 17.4). So the numerical criterion above is never met, and the results that describe its object are vacuous. What survives is the corrected statement in which the complex may also carry powers of the polarisation, which are Hodge on the polarised family and not beyond it; the propagation theorem holds for it unchanged and, except for the shapes below, it escapes the obstruction; the semiregular objects in Markman's proofs are not of the pure form either (Remark 17.6). Its numerical form is computed in Theorem 17.9: for a Chern character  $N\omega + \sum_k c_k \theta^k$  the annihilator in  $HT^2$  has dimension  $n^2(4 - \rho)$  for  $n \geq 3$ , with  $\rho \leq 3$  the rank of the Hankel matrix of the numbers  $k!c_k$ ,<sup>5</sup> and for a general shape it is exactly the tangent space of the polarised Weil family; so one complex with  $\dim \operatorname{Ext}^2(E, E) = 7n^2 - 2n$  at one base point would close the family, 24 at  $n = 2$  and 57 at  $n = 3$ . The shapes  $Ae^{t\theta} + B\theta^{2n}$  with  $t\theta$  the class of a line bundle, of Hankel rank at most two, are excluded, by a twist and Theorem 17.5; for every other shape the Hodge locus of the character is, for  $n \geq 3$ , the polarised family itself, so the argument through non-algebraic tori has nothing to act on (Proposition 17.11);<sup>6</sup> and a weaker flatness form of the criterion suffices (Proposition 17.15). The Hochschild profile of a corrected character is Proposition 17.10. The Mumford classes escape the obstruction for a different reason: they are rigid, and their Hodge locus is the Mumford curve.

The secant route, the one this paper spends most of its length narrowing, lies outside every minimal set: granted every demand it makes, it reaches the families of trivial discriminant, because the member it is built on is a product and a product has trivial discriminant, and through descent every imaginary quadratic family, but no CM field of higher degree (Remark 20.47).<sup>7</sup> Descent is the one link between the Weil families (Proposition 20.13): a product with a Weil surface of any discriminant and a push-forward against its Weil class, which is a divisor class, carry the Weil classes of one family onto those of every family of lower dimension and every discriminant, and a scalar extension carries those of a CM field of degree at least four onto those of the imaginary quadratic

<sup>5</sup>Precisely,  $\rho$  is the rank of the  $(2n - 1) \times 3$  matrix with entries  $\mu_{i+j}$ ,  $0 \leq j \leq 2n - 2$ ,  $0 \leq i \leq 2$ , where  $\mu_k = k!c_k$  (Setup 17.8). A vector in its kernel is a linear recurrence of order at most two among the  $\mu_k$ , and  $\rho$  does not change when the character is multiplied by  $e^{t\theta}$ .

<sup>6</sup>At  $n = 2$  the same holds with one exception: when  $c_1 = c_3 = 0$  and  $c_2^2 \int_A \theta^4 = \frac{3}{4} \int_A (N\omega)^2$  the locus has dimension five, its stabiliser is  $\mathfrak{so}(4, 3)$ , and it reaches tori that are not algebraic; the argument of Theorem 17.5 can be started there but does not finish (Proposition 17.11(iv), Remark 17.12).

<sup>7</sup>Markman has since extended the secant construction to CM fields of higher degree, on  $X \times \widehat{X}$  for an abelian variety  $X$  with real multiplication, with the semiregularity of the resulting sheaves not yet addressed [Mar25c]; see (C3) in Section 20.5. The secant route of this paper, and of its rule set, is the imaginary quadratic one.

fields it contains. So along the route of this paper the propagation statement is needed only for the split families of the CM fields of degree at least four, and the base points of nontrivial discriminant, though they exist, are not needed on any route. The computation behind the frontier is item (XXXIII) of [Appendix D](#), repeated in the Lean kernel.

The smallest case of that propagation statement, a quartic CM field at  $n = 2$ , is analysed in [Section 19.4](#). Divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, of codimension two when the discriminant is trivial, which is governed by a quaternion algebra over the real quadratic subfield; for a biquadratic field Markman’s theorem reaches them on a further locus of dimension four; the Casimir class of the natural quadratic form in degree two is algebraic exactly when the Weil classes are; correspondences with abelian varieties of dimension at most seven cannot help at a member with full Hodge group; and the corrected criterion asks  $\dim \operatorname{Ext}^2(E, E) = 112$  of one complex, every complex of that shape having  $\dim \operatorname{Ext}^2(E, E) \geq 68$  ([Proposition 19.10](#), [Proposition 19.12](#), [Proposition 19.13](#), [Proposition 19.15](#), [Theorem 19.16](#)). For the imaginary quadratic families the criterion is a statement about one indecomposable object, and no sum of objects with exterior Ext algebras meets it ([Lemma 17.16](#), [Corollary 17.17](#)); the Orlov products of explicit secant objects meet its twisted form with equality at  $n = 2$  and  $n = 3$ , which gives the injectivity of the semiregularity map from a dimension count, and cannot at  $n = 4$  ([Theorem 17.19](#), [Proposition 17.20](#)); and at  $n = 4$  a combination of natural objects with a Weil part needs at least eight of them, on one conic of a quadric of Lagrangians ([Theorem 17.23](#)). None of this is a new case of the conjecture.

[Section 22](#) then states the closure itself, as a single theorem in seven clauses, every one of them unconditional: the reduction to one class on one connected domain and its converse, a base point with an explicit cycle in every family for every field and every discriminant, one criterion for propagation, which is a number, and one that restates the conjecture, the fourteen routes that are closed and the structural reason each of them fails, the same criterion for every CM field, and the three statements that remain along this route, the last of them the conjecture itself and its weaker form open, the second not a reduction and its smallest case implied by the algebraicity of a single Kuga–Satake class, beside the other minimal routes. [Section 20](#) delimits all of it, [Appendix D](#) describes the verification and [Appendix E](#) the Lean certificate. [Appendix A](#) carries the coordinate computation behind the transform, [Appendix B](#) the theorem of Cattani, Deligne and Kaplan on the locus of Hodge classes, [Appendix C](#) the place of Weil type inside Albert’s classification, and [Appendix D](#) the output of the scripts.

## 2. CONVENTIONS

All varieties are over  $\mathbb{C}$  and all cohomology is singular cohomology with the indicated coefficients. For a smooth projective variety  $X$  we write  $H^{p,q}(X)$  for the Hodge components of  $H^{p+q}(X, \mathbb{C})$ , and

$$\operatorname{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$$

for the space of rational Hodge classes of type  $(p, p)$ . A class is *algebraic* if it lies in the image of the cycle class map  $\operatorname{cl}_X^p: \operatorname{CH}^p(X)_{\mathbb{Q}} \rightarrow \operatorname{Hdg}^p(X)$ ; the subspace of algebraic classes is a subring of  $H^*(X, \mathbb{Q})$ , since the cycle class map is a ring homomorphism on the level of intersection products.<sup>8</sup>

<sup>8</sup>The intersection product on  $\operatorname{CH}^*(X)_{\mathbb{Q}}$  exists because  $X$  is smooth, and the compatibility of the cycle class map with it is [\[Ful98, Chapter 19\]](#). The integral form of the question, with  $\mathbb{Z}$  in place of  $\mathbb{Q}$ , is a different problem and is not considered anywhere in this paper.

An abelian variety  $A$  of dimension  $g$  has  $H^1(A, \mathbb{Q})$  of dimension  $2g$  and a canonical isomorphism

$$H^k(A, \mathbb{Q}) \cong \bigwedge^k H^1(A, \mathbb{Q}) \quad (1)$$

of Hodge structures, with the product on the left corresponding to the wedge product on the right. We use (1) without further comment; it is the single structural fact that makes every computation below finite.<sup>9</sup> The dual abelian variety is  $A^\vee = \text{Pic}^0(A)$ , and  $H^1(A^\vee, \mathbb{Q}) \cong H^1(A, \mathbb{Q})^\vee$ .

A polarisation of  $A$  is a class  $\theta \in \text{NS}(A) \otimes \mathbb{Q}$  that is ample; equivalently an alternating form  $E$  on  $H_1(A, \mathbb{Q})$  satisfying the Riemann relations. The polarisation is principal when  $E$  is unimodular on  $H_1(A, \mathbb{Z})$ , and then  $\theta^g/g!$  is the class of a point. We write  $\theta'$  for the polarisation on  $A^\vee$  induced by  $\theta$  through the isogeny  $\phi_\theta: A \rightarrow A^\vee$ ; when  $\theta$  is principal,  $\phi_\theta$  is an isomorphism and  $\phi_\theta^* \theta' = \theta$ .

For an endomorphism  $f$  of  $A$  we write  $f^*$  for the induced map on cohomology. When  $K$  is a field acting on  $A$  through  $\iota: K \hookrightarrow \text{End}^0(A) = \text{End}(A) \otimes \mathbb{Q}$ , the element  $\iota(\tau)$  for  $\tau \in K$  is an element of  $\text{End}^0(A)$ ; it acts on  $H^*(A, \mathbb{Q})$ , and we allow ourselves to speak of the action of  $\tau$  when  $\iota$  is understood. For  $\tau$  in the order  $\iota^{-1}(\text{End}(A))$  the element  $\iota(\tau)$  is an honest isogeny, and the general case follows by  $\mathbb{Q}$ -linearity; no statement below depends on the distinction.

The Chern character of a bounded complex of coherent sheaves is  $\text{ch}(\mathcal{E}^\bullet) = \sum_i (-1)^i \text{ch}(\mathcal{E}^i)$  for any bounded locally free resolution, and this is well defined on  $D^b$ ; on a smooth projective variety every coherent sheaf admits such a resolution. We write  $\text{ch}_k$  for the component in  $H^{2k}$ .<sup>10</sup>

For the Fourier–Mukai transform we follow [Muk81]. The Poincaré bundle  $\mathcal{P}$  on  $A \times A^\vee$  is normalised so that  $\mathcal{P}|_{A \times \{0\}}$  and  $\mathcal{P}|_{\{0\} \times A^\vee}$  are trivial. With  $p: A \times A^\vee \rightarrow A$  and  $q: A \times A^\vee \rightarrow A^\vee$  the projections, the transform is

$$\Phi_{\mathcal{P}}: D^b(A^\vee) \rightarrow D^b(A), \quad \Phi_{\mathcal{P}}(\mathcal{G}) = Rp_*(Lq^*\mathcal{G} \otimes^L \mathcal{P}).$$

On cohomology we write the induced map with the same symbol,

$$\Phi_{\mathcal{P}}: H^*(A^\vee, \mathbb{Q}) \longrightarrow H^*(A, \mathbb{Q}), \quad \Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot \text{ch}(\mathcal{P})),$$

and the Grothendieck–Riemann–Roch theorem gives  $\text{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\text{ch}(\mathcal{G}))$  because the Todd class of an abelian variety is trivial; see Proposition 6.1.

We write  $N = N_{K/\mathbb{Q}}$  for the norm of an imaginary quadratic field and  $\bar{\tau}$  for the conjugate of  $\tau \in K$ . Characters of  $K^\times$  are written multiplicatively;  $N^n$  denotes  $\tau \mapsto N(\tau)^n$ .

Finally, a word on what we mean by a construction being *of ambient type*. We mean precisely the hypothesis of Theorem 1.1: a morphism  $i: A^\vee \rightarrow \mathbb{P}^{N-1}$  defined by a linear system, an object  $\mathcal{F}$  of  $D^b(\mathbb{P}^{N-1})$ , and the class  $\text{ch}_k(\Phi_{\mathcal{P}}(Li^*\mathcal{F}))$ . Nothing is assumed about how  $\mathcal{F}$  is chosen, and in particular  $\mathcal{F}$  may be the structure sheaf of a secant variety, of a join, of a determinantal locus, or of any other subscheme, and it may be twisted by any line bundle from the ambient space.

<sup>9</sup>Topologically  $A = V/\Lambda$  is a real torus of dimension  $2g$ , so  $H^*(A, \mathbb{Z}) = \bigwedge^* \text{Hom}(\Lambda, \mathbb{Z})$  by the Künneth formula for  $(S^1)^{2g}$ , with the cup product corresponding to the wedge product. The Hodge decomposition of  $H^k$  is the  $k$ -th exterior power of  $H^1 = H^{1,0} \oplus H^{0,1}$ , because the translation invariant forms  $dz_I \wedge d\bar{z}_J$  are harmonic for the flat metric and represent every class.

<sup>10</sup>On an abelian variety the tangent bundle is trivial and so is the Todd class. Grothendieck–Riemann–Roch therefore reads  $\text{ch}(f_*x) = f_* \text{ch}(x)$  for a morphism  $f$  between abelian varieties, and  $\text{ch}: K_0(A)_{\mathbb{Q}} \rightarrow \text{CH}^*(A)_{\mathbb{Q}}$  is an isomorphism of rings [Ful98, Chapter 15]. Both facts are used below without further mention; for the inclusion of a smooth subvariety, whose normal bundle need not be trivial, the Todd class of the normal bundle does enter, and it is written out where it matters.

### 3. ABELIAN VARIETIES OF WEIL TYPE

This section is self-contained. We fix the notion of Weil type, construct the Weil line, prove that it consists of Hodge classes, and record the two facts about it that the obstruction theorems use: its dimension, and its position relative to the subring generated by the polarisation. Standard references for the background are [BL04] for abelian varieties and [Voi02] for Hodge theory; nothing below is quoted without proof except the Riemann relations and the Lefschetz theorem on  $(1, 1)$  classes.

**3.1. Weil type.** Let  $K = \mathbb{Q}(\sqrt{-d})$  be an imaginary quadratic field,  $d$  a squarefree positive integer, and let  $-$  denote the nontrivial element of  $\text{Gal}(K/\mathbb{Q})$ .

**Definition 3.1** (Weil type). Let  $A$  be a complex abelian variety of dimension  $2n$ , let  $\iota: K \hookrightarrow \text{End}^0(A)$  be an embedding, and let  $\eta$  be a polarisation class on  $A$ . The triple  $(A, \iota, \eta)$  is of  $(K, -1, n)$ -Weil type if<sup>11</sup>

- (i)  $\dim A = 2n$ ;
- (ii) the endomorphism  $\iota(\sqrt{-d})$  acts on the tangent space  $H^0(A, \Omega_A^1)^\vee \cong H^{1,0}(A)^\vee$  with the two eigenvalues  $\pm\sqrt{-d}$  each occurring with multiplicity  $n$ ;
- (iii) the Rosati involution attached to  $\eta$  restricts on  $\iota(K) \subset \text{End}^0(A)$  to complex conjugation, that is  $\iota(\tau)^\dagger = \iota(\bar{\tau})$  for all  $\tau \in K$ .

Condition (ii) is the balanced signature condition; it is what distinguishes Weil type from the generic situation, and it is what makes the Weil line consist of Hodge classes. Condition (iii) says that the polarisation is compatible with the  $K$ -action.

**Lemma 3.2** ( $H^1$  is a  $K$ -space). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and put  $V = H^1(A, \mathbb{Q})$ . Then  $\iota$  makes  $V$  a  $K$ -vector space of dimension  $2n$ , and  $\dim_{\mathbb{Q}} V = 4n$ .*

*Proof.* The map  $\iota$  embeds  $K$  into  $\text{End}^0(A)$ , and  $\text{End}^0(A)$  acts  $\mathbb{Q}$ -linearly on  $V$  by pullback. Since  $K$  is a field,  $V$  becomes a  $K$ -vector space, and  $\dim_{\mathbb{Q}} V = 2 \dim A = 4n$ , so  $\dim_K V = 4n/[K : \mathbb{Q}] = 2n$ .  $\blacksquare$

**Notation 3.3.** Write  $V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C}$ . Since  $\sqrt{-d}$  acts on  $V$  with  $(\sqrt{-d})^2 = -d$ , the operator  $\iota(\sqrt{-d}) \otimes 1$  on  $V_{\mathbb{C}}$  has the two eigenvalues  $\pm\sqrt{-d}$ , and

$$V_{\mathbb{C}} = V_+ \oplus V_-, \quad V_{\pm} = \ker(\iota(\sqrt{-d}) \mp \sqrt{-d}), \quad (2)$$

with  $\dim_{\mathbb{C}} V_{\pm} = 2n$  each, because  $V$  is free of rank  $2n$  over  $K$  and  $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ . Complex conjugation on  $V_{\mathbb{C}}$ , coming from the real structure  $V \otimes \mathbb{R}$ , interchanges  $V_+$  and  $V_-$ . For general  $\tau = a + b\sqrt{-d} \in K$  the operator  $\iota(\tau)$  acts on  $V_+$  by the scalar  $\tau$  and on  $V_-$  by  $\bar{\tau}$ .

**Remark 3.4.** The two scalars in [Notation 3.3](#) are different functions of  $\tau$ : the operator  $\iota(\tau)$  is  $\mathbb{Q}$ -linear on  $V$  but not a scalar, and only after extending to  $\mathbb{C}$  and splitting does it become scalar on each piece. This is the whole source of the character phenomenon in [Section 5](#). A construction defined over  $\mathbb{Q}$  sees only the  $\mathbb{Q}$ -linear operator, and hence only those of its eigenvalues that are  $\mathbb{Q}$ -rational functions of  $\tau$ ;  $N(\tau) = \tau\bar{\tau}$  is one such, and  $\tau^{2n}$  is not.

<sup>11</sup>The middle entry records the sign in the compatibility between the polarisation and the  $K$ -action, and is  $-1$  here because the Rosati involution restricts to conjugation rather than to the identity. Some authors suppress it and write simply  $K$ -Weil type of dimension  $2n$ ; nothing below depends on the convention, and by [Lemma C.4](#) the sign is in any case forced.

**3.2. The Hodge structure.** The Hodge decomposition of  $V_{\mathbb{C}}$  is  $V_{\mathbb{C}} = H^{1,0} \oplus H^{0,1}$ , and the  $K$ -action preserves it because  $\iota(\tau)$  is induced by an endomorphism of  $A$  and hence is a morphism of Hodge structures. Therefore each of  $V_+$  and  $V_-$  splits,

$$V_{\pm} = V_{\pm}^{1,0} \oplus V_{\pm}^{0,1}. \quad (3)$$

**Lemma 3.5** (Balanced signature). *For  $(A, \iota, \eta)$  of  $(K, -1, n)$ -Weil type,*

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_-^{1,0} = n, \quad \dim_{\mathbb{C}} V_+^{0,1} = \dim_{\mathbb{C}} V_-^{0,1} = n.$$

*Proof.* Condition (ii) of Definition 3.1 says that  $\iota(\sqrt{-d})$  acts on the tangent space with each eigenvalue of multiplicity  $n$ . The tangent space is dual to  $H^{1,0}(A) = V_+^{1,0}$ , so  $\iota(\sqrt{-d})$  acts on  $V_+^{1,0}$  with each eigenvalue of multiplicity  $n$ ; that is,  $\dim V_+^{1,0} = \dim V_-^{1,0} = n$ . Since  $\dim_{\mathbb{C}} V_{\pm} = 2n$  by Notation 3.3 and  $V_{\pm} = V_{\pm}^{1,0} \oplus V_{\pm}^{0,1}$ , the remaining two dimensions are  $n$  each. ■

### 3.3. The Weil line.

**Definition 3.6** (Weil line). Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and  $V = H^1(A, \mathbb{Q})$ . The *Weil line* is

$$W(A) = \left( \bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_- \right) \cap H^{2n}(A, \mathbb{Q}),$$

the intersection being taken inside  $H^{2n}(A, \mathbb{C}) = \bigwedge_{\mathbb{C}}^{2n} V_{\mathbb{C}}$ .

**Proposition 3.7** (Structure of the Weil line). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type. Then*

- (i)  $W(A)$  is a  $\mathbb{Q}$ -subspace of  $H^{2n}(A, \mathbb{Q})$  of dimension 2, stable under  $\iota(K)$ , and free of rank one as a  $K$ -module;
- (ii)  $W(A) \subset H^{n,n}(A)$ , so every element of  $W(A)$  is a Hodge class;
- (iii)  $W(A)$  is canonically isomorphic to  $\bigwedge_K^{2n} V$  as a  $K$ -module.

*Proof.* (i) Write  $W_{\mathbb{C}} = \bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_-$ . Each summand is one-dimensional over  $\mathbb{C}$  because  $\dim_{\mathbb{C}} V_{\pm} = 2n$ , so  $\dim_{\mathbb{C}} W_{\mathbb{C}} = 2$ . Complex conjugation on  $H^{2n}(A, \mathbb{C})$  interchanges  $V_+$  and  $V_-$  and hence interchanges the two summands, so  $W_{\mathbb{C}}$  is defined over  $\mathbb{R}$ ; it is in fact defined over  $\mathbb{Q}$ , because the splitting (2) is the eigenspace decomposition of an operator  $\iota(\sqrt{-d})$  defined over  $\mathbb{Q}$ , and the pair of extreme exterior powers is permuted by  $\text{Gal}(\mathbb{C}/\mathbb{Q})$  acting through  $\text{Gal}(K/\mathbb{Q})$ . Therefore  $W(A) = W_{\mathbb{C}} \cap H^{2n}(A, \mathbb{Q})$  has  $\dim_{\mathbb{Q}} W(A) = 2$  and  $W(A) \otimes \mathbb{C} = W_{\mathbb{C}}$ . Stability under  $\iota(K)$  is clear since  $\iota(\tau)$  preserves each of  $V_{\pm}$ , hence each summand. A two-dimensional  $\mathbb{Q}$ -space with a faithful  $K$ -action is free of rank one over  $K$ .

(ii) The Hodge decomposition of  $\bigwedge_{\mathbb{C}}^{2n} V_+$  is computed from (3): the single line  $\bigwedge_{\mathbb{C}}^{2n} V_+$  equals  $\bigwedge^n V_+^{1,0} \otimes \bigwedge^n V_+^{0,1}$ , which by Lemma 3.5 is of Hodge bidegree  $(n, n)$ . The same computation applies to  $\bigwedge_{\mathbb{C}}^{2n} V_-$ . Hence  $W_{\mathbb{C}} \subset H^{n,n}(A)$  and so  $W(A) \subset H^{2n}(A, \mathbb{Q}) \cap H^{n,n}(A) = \text{Hdg}^n(A)$ .

(iii) The natural map  $\bigwedge_K^{2n} V \rightarrow \bigwedge_{\mathbb{Q}}^{2n} V$  obtained by antisymmetrising over a  $K$ -basis identifies  $\bigwedge_K^{2n} V$  with a  $K$ -line in  $H^{2n}(A, \mathbb{Q})$ , and after  $\otimes \mathbb{C}$  the  $K$ -structure splits this line into the  $\tau$ -eigenline and the  $\bar{\tau}$ -eigenline, which are  $\bigwedge^{2n} V_+$  and  $\bigwedge^{2n} V_-$  respectively. Both sides are two-dimensional over  $\mathbb{Q}$  and the map is injective, hence an isomorphism. ■

**Remark 3.8.** Some authors call a generator of  $W(A)$  *the Weil class* and speak of a one-dimensional space; the discrepancy is the usual one between a  $K$ -line and its underlying  $\mathbb{Q}$ -plane. Nothing below depends on the convention, because both obstruction theorems are statements about the whole of  $W(A)$ .

**3.4. Weil classes are not polynomials in the polarisation.** The point of the Weil line is that it is not accounted for by divisors. We record the statement in the form we use and prove it from the character computation of the next section, which is the cleanest route.

**Definition 3.9.** The *divisor subring*  $D^*(A) \subset H^*(A, \mathbb{Q})$  is the  $\mathbb{Q}$ -subalgebra generated by  $\text{NS}(A) \otimes \mathbb{Q} \subset H^2(A, \mathbb{Q})$ .

Every element of  $D^*(A)$  is algebraic, by the Lefschetz theorem on  $(1, 1)$  classes together with the fact that the algebraic classes form a subring. The Hodge conjecture for  $A$  in degree  $n$  is therefore equivalent to the algebraicity of a complement of  $D^{2n}(A)$  in  $\text{Hdg}^n(A)$ , and by the structure theory of Hodge classes on abelian varieties of Weil type the Weil line is that complement.

**Proposition 3.10.** *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $\text{End}^0(A) = \iota(K)$ , and suppose  $n \geq 1$ . Then*

$$W(A) \cap D^{2n}(A) = 0.$$

*Proof.* By Proposition 3.7(i) and Theorem 1.3, proved independently in Section 5, the space  $W(A) \otimes \mathbb{C}$  is the sum of the  $K^\times$ -eigenspaces for the characters  $\tau \mapsto \tau^{2n}$  and  $\tau \mapsto \bar{\tau}^{2n}$ . By Lemma 5.2 below, every class in  $\text{NS}(A) \otimes \mathbb{Q}$  is a  $K^\times$ -eigenclass for the norm character  $N$ , so every homogeneous element of  $D^{2n}(A)$  is an eigenclass for  $N^n$ .<sup>12</sup> By Lemma 5.4 the characters  $\tau \mapsto \tau^{2n}$  and  $\tau \mapsto N(\tau)^n$  are distinct, and eigenspaces for distinct characters intersect in zero. ■

We prove Lemma 5.2 and Lemma 5.4 in the next section; the forward reference is harmless because nothing in Section 5 uses Proposition 3.10.

#### 4. WEIL CLASSES IN COORDINATES

Section 3 defines the Weil line as an intersection inside  $H^{2n}(A, \mathbb{C})$ . That description is the right one for proofs, and it is the wrong one for getting a feel for the object, because it never exhibits a single class. This section writes the Weil classes out. We build an abelian variety of  $(K, -1, n)$ -Weil type from a  $2n$ -dimensional  $K$ -vector space and a hermitian form, produce two explicit integral generators  $\omega_1, \omega_2$  of the Weil line as sums of  $2^{2n-1}$  monomials each, display them in full for  $n = 2$  and tabulate them for  $n = 3$ , and then use the coordinates for two things the abstract description does not give: an elementary second proof that the Weil line misses the divisor subring, and a complete count of the Hodge classes in every degree.

Everything in this section is verified symbolically in `explicit_weil.py` and `hodge_invariants.py`; the checks are listed in Appendix D, items (VI) and (VII).

##### 4.1. The standard model.

**Setup 4.1** (The model). Fix a squarefree positive integer  $d$ , let  $K = \mathbb{Q}(\sqrt{-d})$ , and fix the embedding  $K \hookrightarrow \mathbb{C}$  carrying  $\sqrt{-d}$  to  $\delta := i\sqrt{d}$ . Fix  $n \geq 1$  and let

$$V = K^{2n}, \quad u_1, \dots, u_{2n} \text{ the standard } K\text{-basis,}$$

regarded as a  $\mathbb{Q}$ -vector space of dimension  $4n$  with the basis

$$x_j := u_j, \quad y_j := \sqrt{-d} u_j, \quad 1 \leq j \leq 2n.$$

<sup>12</sup>This is where the whole argument sits, so we put it in words. A divisor class is a rank one object, so the field acts on it through a quantity a rank one object can see, and over  $\mathbb{Q}$  that quantity is the norm. The Weil line is not of rank one over  $\mathbb{Q}$ ; it is of rank one over  $K$ , and a  $K$ -line of exterior degree  $2n$  carries the  $2n$ -th power of  $\tau$  itself.

Let  $S: V \rightarrow V$  be multiplication by  $\sqrt{-d}$ , so that in this basis

$$Sx_j = y_j, \quad Sy_j = -dx_j. \quad (4)$$

Let  $\epsilon_j = -1$  for  $1 \leq j \leq n$  and  $\epsilon_j = +1$  for  $n < j \leq 2n$ ,<sup>13</sup> and let  $Q$  be the alternating  $\mathbb{Q}$ -bilinear form on  $V$  with

$$Q(x_j, y_j) = \epsilon_j, \quad Q(x_j, x_k) = Q(y_j, y_k) = 0, \quad Q(x_j, y_k) = 0 \quad (j \neq k). \quad (5)$$

Finally let  $\eta \in \bigwedge^2 V$  be the class with the same coefficients,

$$\eta = \sum_{j=1}^{2n} \epsilon_j x_j \wedge y_j. \quad (6)$$

Over  $\mathbb{C}$  the operator  $S$  has the two eigenvalues  $\pm\delta$ , and the eigenvectors are written down at once.

**Lemma 4.2** (Eigenbasis). *For  $1 \leq j \leq 2n$  and  $s \in \{+, -\}$  put*

$$u_j^s = \frac{1}{2} \left( x_j - s \frac{\delta}{d} y_j \right) \in V_{\mathbb{C}}. \quad (7)$$

*Then  $Su_j^s = s\delta u_j^s$ , the vectors  $u_1^+, \dots, u_{2n}^+$  form a basis of  $V_+$  and  $u_1^-, \dots, u_{2n}^-$  a basis of  $V_-$ , and complex conjugation on  $V_{\mathbb{C}}$  interchanges  $u_j^+$  and  $u_j^-$ .*

*Proof.* Using (4) and  $\delta^2 = -d$ ,

$$Su_j^s = \frac{1}{2} (y_j + s \frac{\delta}{d} dx_j) = \frac{s\delta}{2} (x_j + \frac{1}{s\delta} y_j) = \frac{s\delta}{2} (x_j - s \frac{\delta}{d} y_j) = s\delta u_j^s,$$

where the third equality is  $1/\delta = -\delta/d$ . The  $2n$  vectors  $u_j^+$  are linearly independent because the change of basis  $(x_j, y_j) \mapsto (u_j^+, u_j^-)$  is block diagonal with invertible  $2 \times 2$  blocks of determinant  $\delta/(2d) \neq 0$ , and the eigenspaces have dimension  $2n$  each by [Notation 3.3](#). Conjugation fixes  $x_j$  and  $y_j$  and sends  $\delta$  to  $-\delta$ , which interchanges the two formulas. ■

**Definition 4.3** (The complex structure). Define a Hodge structure of weight one on  $V$  by

$$V^{1,0} = \langle u_1^+, \dots, u_n^+, u_{n+1}^-, \dots, u_{2n}^- \rangle_{\mathbb{C}}, \quad V^{0,1} = \overline{V^{1,0}}. \quad (8)$$

The choice is the whole point: the first  $n$  basis directions contribute to  $V^{1,0}$  through their  $+$  eigenvector and the last  $n$  through their  $-$  eigenvector, so each of  $V_{\pm}$  meets  $V^{1,0}$  in exactly  $n$  dimensions.

**Theorem 4.4** (The model is of Weil type). *With the data of [Setup 4.1](#) and [Definition 4.3](#), the pair  $(V, V^{1,0})$  is a polarised rational Hodge structure of weight one with polarising form  $Q$ , so  $V = H^1(A, \mathbb{Q})$  for an abelian variety  $A$  of dimension  $2n$ ; the multiplication  $S$  makes  $K$  act on  $A$  up to isogeny;  $\eta$  is the class of the polarisation; and  $(A, \iota, \eta)$  is of  $(K, -1, n)$ -Weil type in the sense of [Definition 3.1](#).*

*Proof.* We verify the two Riemann bilinear relations, then the three conditions of [Definition 3.1](#).

<sup>13</sup>The signs are not free. The proof of [Theorem 4.4](#) shows that the second Riemann relation holds for this choice and fails as soon as one of them is reversed, so the signature  $(n, n)$  of the sign vector is the balanced condition in disguise. Readers who prefer the opposite global sign should replace  $Q$  by  $-Q$  throughout and interchange  $V^{1,0}$  with  $V^{0,1}$ .

The form vanishes on  $V^{1,0}$ . By (5) the form  $Q$  is block diagonal for the decomposition of  $V$  into the planes  $\langle x_j, y_j \rangle$ , so  $Q(u_j^s, u_k^t) = 0$  whenever  $j \neq k$ . Within one block,  $Q(u_j^s, u_j^s) = 0$  because  $Q$  is alternating, and

$$Q(u_j^+, u_j^-) = \frac{1}{4} Q(x_j - \frac{\delta}{d} y_j, x_j + \frac{\delta}{d} y_j) = \frac{1}{4} \cdot \frac{2\delta}{d} Q(x_j, y_j) = \frac{\delta \epsilon_j}{2d}. \quad (9)$$

The basis (8) of  $V^{1,0}$  contains, for each  $j$ , exactly one of  $u_j^+, u_j^-$ . Hence any two of its members either sit in different blocks or are equal, and in both cases  $Q$  vanishes on them. So  $Q(V^{1,0}, V^{1,0}) = 0$ .

*Positivity.* Let  $0 \neq v = \sum_j a_j v_j \in V^{1,0}$ , where  $v_j$  is  $u_j^+$  for  $j \leq n$  and  $u_j^-$  for  $j > n$ . Then  $\bar{v} = \sum_j \bar{a}_j \bar{v}_j$  with  $\bar{v}_j$  the other eigenvector in the same block, so again only the diagonal terms survive:

$$i Q(v, \bar{v}) = \sum_{j \leq n} i |a_j|^2 Q(u_j^+, u_j^-) + \sum_{j > n} i |a_j|^2 Q(u_j^-, u_j^+).$$

By (9) and  $\delta = i\sqrt{d}$  the first kind of term is  $i \cdot \delta \epsilon_j / (2d) = -\epsilon_j / (2\sqrt{d})$  and the second is  $+\epsilon_j / (2\sqrt{d})$ . With  $\epsilon_j = -1$  for  $j \leq n$  and  $+1$  for  $j > n$  both are  $+1/(2\sqrt{d})$ , so  $i Q(v, \bar{v}) = \|a\|^2 / (2\sqrt{d}) > 0$ . This is the second Riemann relation, and it is what forces the signature of  $\epsilon$  to be  $(n, n)$ : reversing any single  $\epsilon_j$  makes the corresponding term negative.

*Weil type.* Condition (i) is  $\dim A = \frac{1}{2} \dim_{\mathbb{Q}} V = 2n$ . For (ii),  $V^{1,0}$  contains  $u_j^+$  for the  $n$  indices  $j \leq n$  and  $u_j^-$  for the  $n$  indices  $j > n$ , so  $\dim V_+^{1,0} = \dim V_-^{1,0} = n$ , which is Lemma 3.5, equivalent to the balanced eigenvalue condition on the tangent space by duality. For (iii), write  $\tau = a + b\sqrt{-d}$ , so that  $\iota(\tau)$  acts by

$$x_j \mapsto a x_j + b y_j, \quad y_j \mapsto -b d x_j + a y_j,$$

and  $\iota(\bar{\tau})$  by the same formulas with  $b$  replaced by  $-b$ . Since  $Q$  is block diagonal it suffices to compare on one block, where the three pairs give

$$Q(\iota(\tau)x_j, y_j) = a\epsilon_j = Q(x_j, \iota(\bar{\tau})y_j),$$

$$Q(\iota(\tau)x_j, x_j) = -b\epsilon_j = Q(x_j, \iota(\bar{\tau})x_j),$$

$$Q(\iota(\tau)y_j, y_j) = -bd\epsilon_j = Q(y_j, \iota(\bar{\tau})y_j),$$

so  $\iota(\tau)^\dagger = \iota(\bar{\tau})$ . Equivalently, and this is the form used below,  $\iota(\tau)^* \eta = N(\tau) \eta$ , because  $x_j \wedge y_j = \frac{2d}{\delta} u_j^+ \wedge u_j^-$  by (7) and  $\iota(\tau)$  multiplies  $u_j^+ \wedge u_j^-$  by  $\tau \bar{\tau}$ .

Finally  $\eta$  is of type  $(1, 1)$ , since in each block one of  $u_j^+, u_j^-$  lies in  $V^{1,0}$  and the other in  $V^{0,1}$ , and  $\eta^{2n} \neq 0$ , so  $\eta$  is a polarisation class.  $\blacksquare$

**Lemma 4.5** (The hermitian form). *On the model of Setup 4.1 there is a unique  $K$ -hermitian form  $H$  on  $V$ , linear in the first variable, with*

$$Q(v, w) = \frac{1}{2} \operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1} H(v, w)), \quad \text{namely} \quad H(v, w) = Q(\sqrt{-d} v, w) + \sqrt{-d} Q(v, w), \quad (10)$$

and in the basis  $u_1, \dots, u_{2n}$  it is  $H = \operatorname{diag}(-\epsilon_1, \dots, -\epsilon_{2n})$ , of signature  $(n, n)$ .

*Proof.*  $K$ -linearity in the first variable is the computation

$$H(\sqrt{-d} v, w) = Q(-dv, w) + \sqrt{-d} Q(\sqrt{-d} v, w) = \sqrt{-d} H(v, w),$$

and the hermitian symmetry  $\overline{H(w, v)} = H(v, w)$  follows from  $Q(v, \sqrt{-d} w) = -Q(\sqrt{-d} v, w)$ , which is condition (iii) of Definition 3.1 read backwards. Applying  $\delta^{-1}$  and taking the trace kills the first summand of (10), since  $Q$  is  $\mathbb{Q}$ -valued and  $\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}) = 0$ , and doubles the second; this gives  $Q$  and also the uniqueness, since  $H$  is determined by  $Q$

and the  $K$ -structure. Finally  $H(u_j, u_j) = Q(y_j, x_j) = -\epsilon_j$  and  $H(u_j, u_k) = 0$  for  $j \neq k$ , so the signature is the number of indices with  $\epsilon_j = -1$  against the number with  $\epsilon_j = +1$ , that is  $(n, n)$ .  $\blacksquare$

*Remark 4.6.* Two signatures have now appeared and they are the same signature seen twice. The hermitian form  $H$  has signature  $(n, n)$  by [Lemma 4.5](#), and the  $K$ -action on the tangent space has balanced eigenvalue multiplicities  $(n, n)$  by [Definition 3.1\(ii\)](#). The Riemann positivity computation in the proof of [Theorem 4.4](#) is exactly the passage between them: the sign  $\epsilon_j$  decides both which eigenvector of the  $j$ -th block is holomorphic and the sign of  $H$  on that block.

*Remark 4.7.* Taking the lattice  $\mathcal{O}^{2n}$  for an order  $\mathcal{O} \subset K$  and scaling  $Q$  to be integral on it produces an actual abelian variety rather than a Hodge structure; nothing below depends on the lattice, so we work with the rational Hodge structure throughout. The  $n^2$ -dimensional family of complex structures obtained by varying (8) over the period domain of the hermitian form is the Weil family, of dimension  $n^2$  inside the  $n(2n+1)$ -dimensional moduli of principally polarised abelian  $2n$ -folds; see [Appendix B](#).

## 4.2. The two generators.

*Notation 4.8.* For  $T \subseteq \{1, \dots, 2n\}$  write

$$w^{(T)} = w_1^{(T)} \wedge w_2^{(T)} \wedge \dots \wedge w_{2n}^{(T)}, \quad w_j^{(T)} = \begin{cases} y_j, & j \in T, \\ x_j, & j \notin T. \end{cases}$$

The  $2^{2n}$  elements  $w^{(T)}$  are part of the monomial basis of  $\bigwedge^{2n} V$ , which has dimension  $\binom{4n}{2n}$ ; they are exactly the monomials containing precisely one of  $x_j, y_j$  for every  $j$ .

**Theorem 4.9** (The Weil line in coordinates). *Let  $(A, \iota, \eta)$  be the model of [Theorem 4.4](#) and let*

$$\omega_1 = \sum_{\substack{T \subseteq \{1, \dots, 2n\} \\ |T| \text{ even}}} (-1)^{|T|/2} d^{n-|T|/2} w^{(T)}, \quad \omega_2 = \sum_{\substack{T \subseteq \{1, \dots, 2n\} \\ |T| \text{ odd}}} (-1)^{(|T|-1)/2} d^{n-(|T|+1)/2} w^{(T)}. \quad (11)$$

Then

- (i)  $\omega_1$  and  $\omega_2$  have integer coefficients and each has exactly  $2^{2n-1}$  nonzero terms;
- (ii)  $\omega_1, \omega_2$  is a  $\mathbb{Q}$ -basis of the Weil line  $W(A)$  of [Definition 3.6](#);
- (iii) writing  $\omega_{\pm} = u_1^{\pm} \wedge \dots \wedge u_{2n}^{\pm}$  for the generators of  $\bigwedge^{2n} V_{\pm}$ ,

$$\omega_+ + \omega_- = \frac{\omega_1}{2^{2n-1}d^n}, \quad \frac{\omega_+ - \omega_-}{\delta} = -\frac{\omega_2}{2^{2n-1}d^n};$$

- (iv)  $\iota(\tau)^*\omega_+ = \tau^{2n}\omega_+$  and  $\iota(\tau)^*\omega_- = \bar{\tau}^{2n}\omega_-$  for all  $\tau \in K^{\times}$ , so on the plane  $\langle \omega_1, \omega_2 \rangle$  the element  $\iota(\tau)$  acts by the  $\mathbb{Q}$ -rational matrix of multiplication by  $\tau^{2n}$  on  $K$ ;
- (v)  $\omega_1$  and  $\omega_2$  are of Hodge type  $(n, n)$ , hence are Hodge classes.

*Proof.* (iii) and (i). Expanding (7) in the product defining  $\omega_s$  and collecting the monomials  $w^{(T)}$ ,

$$\omega_s = \frac{1}{2^{2n}} \sum_T \left( -s \frac{\delta}{d} \right)^{|T|} w^{(T)}, \quad (12)$$

because the factor from index  $j$  contributes  $x_j$  with coefficient  $1/2$  and  $y_j$  with coefficient  $-s\delta/(2d)$ , and the monomials  $w^{(T)}$  appear in the order  $j = 1, \dots, 2n$  with no reordering

sign. Therefore

$$\omega_+ + \omega_- = \frac{1}{2^{2n-1}} \sum_{|T| \text{ even}} \left(\frac{\delta}{d}\right)^{|T|} w^{(T)}, \quad \frac{\omega_+ - \omega_-}{\delta} = -\frac{1}{2^{2n-1}} \sum_{|T| \text{ odd}} \frac{\delta^{|T|-1}}{d^{|T|}} w^{(T)},$$

the odd and even parts cancelling in the respective sums. Now  $\delta^{2m} = (-d)^m$ , so for  $|T| = 2m$  the coefficient in the first sum is  $(-1)^m d^{-m}$ , and for  $|T| = 2m + 1$  the coefficient in the second is  $-(-1)^m d^{-m-1}$ . Multiplying by  $2^{2n-1} d^n$  clears denominators, since  $m \leq n$  in the first case and  $m + 1 \leq n$  in the second, and yields exactly (11). The term counts are  $\sum_m \binom{2n}{2m} = \sum_m \binom{2n}{2m+1} = 2^{2n-1}$ .

(ii). The two displayed combinations are fixed by complex conjugation, because conjugation interchanges  $\omega_+$  and  $\omega_-$  by Lemma 4.2 and sends  $\delta$  to  $-\delta$ ; and they visibly have rational coefficients. They are linearly independent over  $\mathbb{C}$  since  $\omega_+, \omega_-$  are, so they span  $(\wedge^{2n} V_+ \oplus \wedge^{2n} V_-) \cap H^{2n}(A, \mathbb{Q})$ , which is  $W(A)$  by Definition 3.6 and Proposition 3.7(i).

(iv). The first two identities are immediate, since  $\iota(\tau)$  multiplies each of the  $2n$  factors  $u_j^+$  by  $\tau$ . For the last clause, by (iii) the complex line  $\mathbb{C}\omega_+$  is the  $\tau^{2n}$ -eigenline inside  $W(A) \otimes \mathbb{C}$ , so under the isomorphism  $W(A) \cong K$  of Proposition 3.7(iii) the action of  $\iota(\tau)$  is multiplication by  $\tau^{2n}$ , whose matrix in the basis  $\omega_1, \omega_2$  has rational entries.

(v). By (8) the monomial  $\omega_+ = u_1^+ \wedge \cdots \wedge u_{2n}^+$  has exactly  $n$  factors in  $V^{1,0}$ , namely those with  $j \leq n$ , and  $n$  factors in  $V^{0,1}$ ; so  $\omega_+$  is of type  $(n, n)$ , and likewise  $\omega_-$ . Hence so are the two rational combinations. ■

*Remark 4.10* (The balance is what makes them Hodge classes). Step (v) is the only place the balanced condition enters, and we isolate it. Suppose we keep everything else and replace (8) by a complex structure for which  $\dim V_+^{1,0} = p$  and  $\dim V_-^{1,0} = 2n - p$ . Then  $\omega_+$  has  $p$  holomorphic factors and  $\omega_+$  is of type  $(p, 2n - p)$ , so  $W(A)$  is a rational sub-Hodge structure of  $H^{2n}(A, \mathbb{Q})$  of type  $\{(p, 2n - p), (2n - p, p)\}$ , and it contains a nonzero Hodge class if and only if  $p = n$ . The balanced signature in Definition 3.1(ii) is thus not an extra hypothesis imposed for convenience; it is exactly the condition under which the Weil line has anything to do with the Hodge conjecture. This is verified for all  $p$  and all  $n \leq 3$  in `explicit_weil.py`, check (W4), and is proved again in Appendix C, Proposition C.7.

**4.3. The case  $n = 2$ .** Here  $\dim A = 4$ ,  $\dim_{\mathbb{Q}} V = 8$ ,  $H^4(A, \mathbb{Q})$  has dimension  $\binom{8}{4} = 70$ , and each generator has  $2^3 = 8$  terms. Writing the monomials in the order  $j = 1, 2, 3, 4$  as in Notation 4.8, (11) reads

$$\omega_1 = d^2 x_1 x_2 x_3 x_4 - d \sum_{|T|=2} w^{(T)} + y_1 y_2 y_3 y_4, \quad (13)$$

$$\omega_2 = d \sum_{|T|=1} w^{(T)} - \sum_{|T|=3} w^{(T)}, \quad (14)$$

with the wedge signs omitted from monomials. For  $d = 1$ , written out in full,

$$\begin{aligned} \omega_1 &= x_1 x_2 x_3 x_4 - y_1 y_2 x_3 x_4 - y_1 x_2 y_3 x_4 - y_1 x_2 x_3 y_4 - x_1 y_2 y_3 x_4 - x_1 y_2 x_3 y_4 - x_1 x_2 y_3 y_4 \\ &\quad + y_1 y_2 y_3 y_4, \\ \omega_2 &= y_1 x_2 x_3 x_4 + x_1 y_2 x_3 x_4 + x_1 x_2 y_3 x_4 + x_1 x_2 x_3 y_4 \\ &\quad - y_1 y_2 y_3 x_4 - y_1 y_2 x_3 y_4 - y_1 x_2 y_3 y_4 - x_1 y_2 y_3 y_4. \end{aligned}$$

The polarisation power is  $\eta^2 = 2 \sum_{j < k} \epsilon_j \epsilon_k x_j y_j x_k y_k$ , a sum of six monomials, and these are visibly different monomials from the sixteen above. That observation is the content of the next subsection.

4.4. **The case  $n = 3$ .** Here  $\dim A = 6$ ,  $\dim_{\mathbb{Q}} V = 12$ ,  $H^6(A, \mathbb{Q})$  has dimension  $\binom{12}{6} = 924$ , and each generator has  $2^5 = 32$  terms. The coefficients depend only on  $|T|$ , so the two classes are recorded completely by [Table 1](#).

| $\omega_1$ |       |             |            | $\omega_2$ |       |             |            |
|------------|-------|-------------|------------|------------|-------|-------------|------------|
| $ T $      | terms | coefficient | at $d = 2$ | $ T $      | terms | coefficient | at $d = 2$ |
| 0          | 1     | $+d^3$      | +8         | 1          | 6     | $+d^2$      | +4         |
| 2          | 15    | $-d^2$      | -4         | 3          | 20    | $-d$        | -2         |
| 4          | 15    | $+d$        | +2         | 5          | 6     | $+1$        | +1         |
| 6          | 1     | $-1$        | -1         |            |       |             |            |
| 32         |       |             |            | 32         |       |             |            |

TABLE 1. The Weil classes at  $n = 3$ . Each row gives the number of index sets  $T$  of that size and the common coefficient of the corresponding monomials  $w^{(T)}$  in (11). The two classes together involve all 64 monomials containing exactly one of  $x_j, y_j$  for each  $j$ , out of the 924 monomials of  $H^6$ .

For  $d = 1$  the coefficients are the alternating pattern  $+, -, +, -$  and  $+, -, +$ , and the two classes are the even and odd halves of  $\prod_{j=1}^6 (x_j - y_j)$  read as a formal product; for general  $d$  the powers of  $d$  record the failure of  $\sqrt{-d}$  to be a root of unity, which is the arithmetic behind [Lemma 5.4](#).

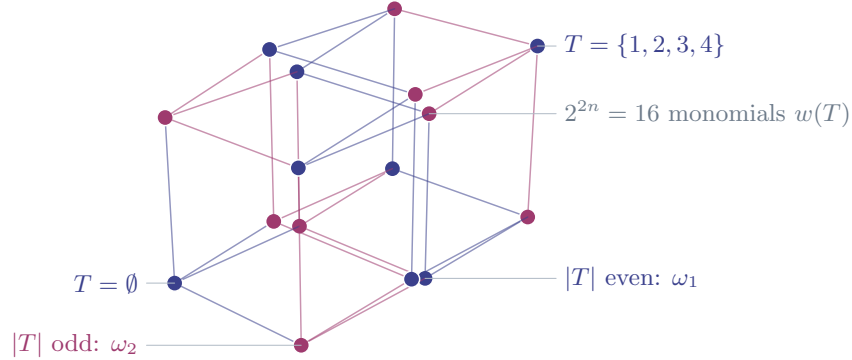


FIGURE 1. The  $2^{2n}$  monomials  $w(T)$  of [Notation 4.8](#), at  $n = 2$ , drawn as the vertices of the 4-cube in a projection that keeps all sixteen distinct. An edge joins two monomials whose index sets differ in one element. The colouring is the bipartition by the parity of  $|T|$ : the eight even vertices carry  $\omega_1$  and the eight odd ones carry  $\omega_2$ , so the two generators of the Weil line are the two halves of the cube and the coefficient of a monomial depends only on the distance from the empty set.

4.5. **An elementary proof that the Weil line misses the divisors.** [Proposition 3.10](#) was deduced from the character computation. In the model it has a proof with no

representation theory in it at all, and the proof explains the phenomenon in one line: the Weil classes and the powers of the polarisation are supported on disjoint sets of monomials.

**Proposition 4.11** (Disjoint supports). *In the model of Theorem 4.4, every monomial occurring in  $\omega_1$  or in  $\omega_2$  contains exactly one of  $x_j, y_j$  for every  $j \in \{1, \dots, 2n\}$ , and every monomial occurring in  $\eta^k$  contains either both or neither, for every  $j$  and every  $k \geq 0$ . Consequently*

$$W(A) \cap \mathbb{Q} \eta^n = 0,$$

and more generally  $W(A)$  meets the span of the monomials occurring in powers of  $\eta$  in zero.

*Proof.* The first assertion is Notation 4.8. For the second,  $\eta$  is the sum (6) of the  $2n$  two-forms  $x_j \wedge y_j$ , which pairwise commute and square to zero, so

$$\eta^k = k! \sum_{\substack{S \subseteq \{1, \dots, 2n\} \\ |S|=k}} \left( \prod_{j \in S} \epsilon_j \right) \bigwedge_{j \in S} (x_j \wedge y_j),$$

and the monomial indexed by  $S$  contains both  $x_j$  and  $y_j$  for  $j \in S$  and neither for  $j \notin S$ . A monomial cannot satisfy both patterns for a given  $j$ , so the two families of monomials are disjoint, and  $2n \geq 1$  guarantees that there is at least one index at which to test. Since  $\omega_1, \omega_2$  span  $W(A)$  by Theorem 4.9(ii), a nonzero element of  $W(A)$  has a nonzero coefficient on some  $w^{(T)}$ , which does not occur in  $\eta^n$ . ■

*Remark 4.12.* The two proofs of Proposition 3.10 measure different things. Proposition 4.11 is a statement about one model and a choice of basis, and it is short because the basis was chosen to make the  $K$ -action diagonal. The character proof is basis free and applies to every abelian variety of Weil type, including those not of the form above, and it identifies what the obstruction is rather than only that there is one. We record both because the first is checkable by hand and the second is what Theorem 1.3 generalises.

**4.6. All the Hodge classes.** The coordinates also settle how much of the Hodge conjecture for  $A$  is at stake. Recall that the Hodge classes of an abelian variety are the invariants of the Hodge group,  $\text{Hdg}^k(A) = H^{2k}(A, \mathbb{Q})^{\text{Hg}(A)}$ , where  $\text{Hg}(A) \subseteq \text{SL}(V)$  is the special Mumford–Tate group.

**Lemma 4.13** (The Hodge group lands in the special unitary group). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and let  $H$  be the  $K$ -hermitian form of Lemma 4.5. Then*

$$\text{Hg}(A) \subseteq \text{SU}(V, H).$$

*Proof.* By definition  $\text{Hg}(A)$  is a  $\mathbb{Q}$ -algebraic subgroup of  $\text{SL}(V)$  which fixes every Hodge class in every tensor construction on  $V$ . It fixes  $\eta \in \bigwedge^2 V$ , hence preserves  $Q$ ; and it commutes with  $\iota(K)$ , because each  $\iota(\tau)$  is a Hodge class in  $V \otimes V^\vee$ . A  $\mathbb{Q}$ -linear automorphism of  $V$  commuting with the  $K$ -action and preserving  $Q$  preserves  $H$ , by the formula (10) expressing  $H$  through  $Q$  and the  $K$ -action. Hence  $\text{Hg}(A) \subseteq \text{U}(V, H)$ . Now  $W(A)$  consists of Hodge classes by Proposition 3.7(ii), so  $\text{Hg}(A)$  fixes  $\omega_+$ ; and  $g \in \text{U}(V, H)$  acts on  $\bigwedge^{2n} V_+$  by  $\det_K(g)$ , so  $\det_K(g) = 1$ , which is the definition of  $\text{SU}(V, H)$ . ■

For a very general member of the Weil family the containment of [Lemma 4.13](#) is an equality;<sup>14</sup> this is Weil's computation [[Wei77](#)], in the form given by Moonen and Zarhin [[MZ99](#)]. Granting it, the Hodge classes can be counted exactly.

**Theorem 4.14** (The full Hodge ring of a general Weil variety). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ . Then for every  $k$  with  $0 \leq k \leq 2n$ ,*

$$\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = \begin{cases} 3, & k = n, \\ 1, & k \neq n, \end{cases} \quad \text{and} \quad \mathrm{Hdg}^n(A) = \mathbb{Q} \eta^n \oplus W(A).$$

*Proof.* Complexify. The group  $\mathrm{SU}(V, H) \otimes \mathbb{C}$  is  $\mathrm{SL}(V_+) \cong \mathrm{SL}_{2n}(\mathbb{C})$ , acting on  $V_+$  tautologically and on  $V_-$  through the isomorphism  $V_- \cong V_+^\vee$  induced by  $H$ . By [Proposition 5.1](#),

$$H^{2k}(A, \mathbb{C}) = \bigoplus_{r+s=2k} \wedge^r V_+ \otimes \wedge^s V_+^\vee,$$

so it suffices to find the invariants in each summand. An invariant is a weight-zero vector annihilated by every raising operator. In the weight lattice of  $\mathrm{SL}_{2n}$  the weight of  $v_T \otimes v_U^\vee$  is  $\mathbf{1}_T - \mathbf{1}_U$  modulo  $(1, 1, \dots, 1)$ ; since the entries of  $\mathbf{1}_T - \mathbf{1}_U$  lie in  $\{-1, 0, 1\}$ , the only multiples of  $(1, \dots, 1)$  it can equal are 0 and  $\pm(1, \dots, 1)$ , giving

$$T = U \quad (\text{so } r = s), \quad (T, U) = (\{1, \dots, 2n\}, \emptyset), \quad (T, U) = (\emptyset, \{1, \dots, 2n\}).$$

The last two occur only for  $(r, s) = (2n, 0)$  and  $(0, 2n)$ , each with a one-dimensional weight-zero space consisting of the invariant lines  $\wedge^{2n} V_+$  and  $\wedge^{2n} V_-$ , which are  $\mathrm{SL}$ -invariant because  $\mathrm{SL}$  has trivial determinant. For  $r = s$  the weight-zero space has dimension  $\binom{2n}{r}$  and its invariant subspace is the line spanned by the image of the identity under  $\mathrm{End}(\wedge^r V_+) \cong \wedge^r V_+ \otimes \wedge^r V_+^\vee$ , which is one dimensional because  $\wedge^r V_+$  is an irreducible  $\mathrm{SL}_{2n}$ -module and Schur's lemma applies. The kernel computation confirming that the invariant space is exactly this line, for all  $r, s$  and all  $n \leq 4$ , is item (VII) of [Appendix D](#).

Collecting: for  $k \neq n$  the only contribution is  $(r, s) = (k, k)$ , giving dimension one, which is  $\mathbb{Q} \eta^k$  since  $\eta^k$  is a nonzero invariant lying in that summand by [Lemma 5.2](#). For  $k = n$  there are three contributions,  $(n, n)$ ,  $(2n, 0)$  and  $(0, 2n)$ , of total dimension three; the first is  $\mathbb{Q} \eta^n$  and the last two span  $W(A) \otimes \mathbb{C}$  by [Lemma 5.3](#). Descending to  $\mathbb{Q}$  preserves dimensions.  $\blacksquare$

**Corollary 4.15** (What the Hodge conjecture asks for here). *Let  $(A, \iota, \eta)$  be as in [Theorem 4.14](#). The Hodge conjecture holds for  $A$  in every degree if and only if the single class  $\omega_1$  of [\(11\)](#) is algebraic.*

*Proof.* For  $k \neq n$  the Hodge classes are the multiples of  $\eta^k$ , which are algebraic, being powers of a divisor class. For  $k = n$  they are spanned by  $\eta^n$ ,  $\omega_1$  and  $\omega_2$  by [Theorem 4.14](#). Finally  $\omega_2$  is algebraic as soon as  $\omega_1$  is: by [Theorem 4.9\(iv\)](#) the plane  $W(A)$  is a free  $K$ -module of rank one generated by  $\omega_1$ , and the action of  $\iota(\tau)$  is induced by an endomorphism of  $A$ , hence carries algebraic classes to algebraic classes; some  $\tau$  moves  $\omega_1$  to a class with a nonzero  $\omega_2$  component, and the algebraic classes form a  $\mathbb{Q}$ -subspace.  $\blacksquare$

*Remark 4.16.* [Corollary 4.15](#) is the reason the Weil line is the standard test case. On a general abelian variety of Weil type the Hodge conjecture is not a statement about a large space of classes with some unknown structure; it is a statement about one explicitly

<sup>14</sup>Very general means outside a countable union of proper closed analytic subsets of  $\mathcal{D}_{n,n}$ , which is the usual meaning in this subject and is the best that can be asked: the locus where the endomorphism algebra jumps is a countable union of such subsets, by [Theorem B.6](#), and it is dense in the classical topology.

written class with  $2^{2n-1}$  integer coefficients. All three obstruction theorems of this paper are about the difficulty of producing that one class, and the construction of [Section 14](#) is an attempt to produce it.

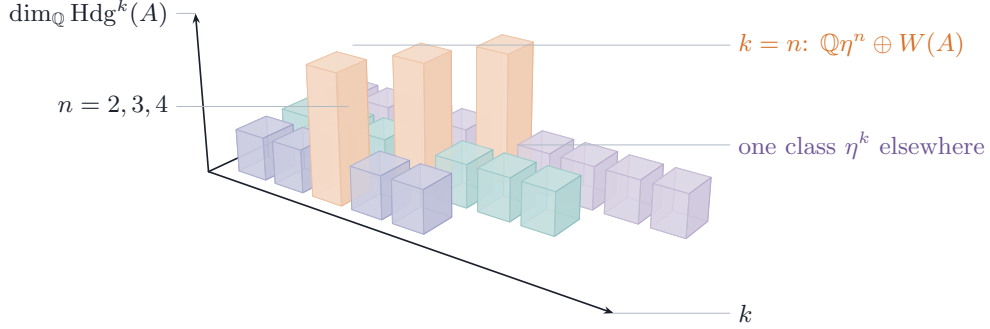


FIGURE 2. The dimensions of  $\text{Hdg}^k(A)$  for a general abelian variety of  $(K, -1, n)$ -Weil type, drawn as a bar field over the two axes  $k$  and  $n$ , at  $n = 2, 3, 4$ , as computed in [Theorem 4.14](#). Every degree carries the one class  $\eta^k$ ; only the middle degree  $k = n$  carries anything else, and the excess there is exactly the two dimensions of the Weil line. The profile is the same for every  $n$ , which is why the problem does not get structurally harder as the dimension grows, only larger.

## 5. THE CHARACTER GRADING AND THE POSITION OF THE WEIL LINE

This section proves [Theorem 1.3](#). The content is a single observation made precise: the group  $K^\times$  acts on  $H^{2n}(A, \mathbb{Q})$  through the endomorphisms  $\iota(\tau)$ , this action grades the cohomology, and the two objects whose relation is at issue, the Weil line and the power  $\eta^n$  of the polarisation, sit at different places in the grading. Once the grading is written down the obstruction is a two-line argument. What takes work is writing the grading down correctly, because the natural first guess, that the Weil line is where  $K^\times$  acts through the norm, is wrong, and it is wrong in exactly the way that makes a construction look as if it should succeed.

**5.1. The grading.** Throughout the section  $(A, \iota, \eta)$  is of  $(K, -1, n)$ -Weil type,  $V = H^1(A, \mathbb{Q})$ , and  $V_{\mathbb{C}} = V_+ \oplus V_-$  is the splitting of [Notation 3.3](#).

**Proposition 5.1** (Character grading). *For every  $k \geq 0$ ,*

$$H^k(A, \mathbb{C}) = \bigoplus_{r+s=k} \left( \wedge^r V_+ \otimes \wedge^s V_- \right), \quad (15)$$

*and  $\iota(\tau)^*$  acts on the summand indexed by  $(r, s)$  by the scalar  $\tau^r \bar{\tau}^s$ . The summands are the isotypic components for the action of  $K^\times$ , and the characters  $\tau \mapsto \tau^r \bar{\tau}^s$  for distinct pairs  $(r, s)$  with  $r + s = k$  are pairwise distinct.*<sup>15</sup>

<sup>15</sup>Distinctness is a statement about characters of the torus  $\text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$ , not about their values at one element. At a fixed  $\tau$  two of the values agree exactly when  $\tau/\bar{\tau}$  is a root of unity of order dividing  $r - r'$ , and for  $K = \mathbb{Q}(\sqrt{-1})$  or  $K = \mathbb{Q}(\sqrt{-3})$  this happens for the obvious choice  $\tau = 1 + \sqrt{-d}$ ; [Appendix E](#) records both failures.

*Proof.* By (1) we have  $H^k(A, \mathbb{C}) = \bigwedge^k V_{\mathbb{C}}$ , and the exterior power of a direct sum decomposes as  $\bigwedge^k (V_+ \oplus V_-) = \bigoplus_{r+s=k} \bigwedge^r V_+ \otimes \bigwedge^s V_-$ , which is (15). By Notation 3.3 the operator  $\iota(\tau)^*$  acts on  $V_+$  by  $\tau$  and on  $V_-$  by  $\bar{\tau}$ , so on  $\bigwedge^r V_+ \otimes \bigwedge^s V_-$  it acts by  $\tau^r \bar{\tau}^s$ , as claimed, since it is an algebra map for the wedge product.

For the last statement, suppose  $\tau^r \bar{\tau}^s = \tau^{r'} \bar{\tau}^{s'}$  for all  $\tau \in K^\times$ , with  $r + s = r' + s' = k$ . Dividing,  $(\tau/\bar{\tau})^{r-r'} = 1$  for every  $\tau \in K^\times$ . If  $r \neq r'$  this says that the image of the map  $c(\tau) = \tau/\bar{\tau}$  lies in the group of  $(r - r')$ -th roots of unity of  $K$ , which is finite because  $K$  is an imaginary quadratic field and its group of roots of unity has order 2, 4 or 6. But  $c$  has infinite image: two elements have the same image exactly when their ratio lies in  $\mathbb{Q}^\times$ , and the elements  $m + \sqrt{-d}$  for  $m \in \mathbb{Z}$  are pairwise non-proportional over  $\mathbb{Q}$ . Hence  $r = r'$  and so  $s = s'$ . ■

**Lemma 5.2** (The polarisation sits in the middle). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type. Then*

$$\iota(\tau)^* \eta = N_{K/\mathbb{Q}}(\tau) \eta \quad \text{for all } \tau \in K^\times,$$

*so  $\eta$  lies in the summand  $(r, s) = (1, 1)$  of (15) and  $\eta^n$  lies in the summand  $(r, s) = (n, n)$ , on which  $K^\times$  acts by  $N^n$ .*

*Proof.* Let  $E$  be the alternating form on  $H_1(A, \mathbb{Q})$  attached to  $\eta$ . Condition (iii) of Definition 3.1 says that the Rosati involution  $\dagger$  of  $\eta$  satisfies  $\iota(\tau)^\dagger = \iota(\bar{\tau})$ , and by the defining property of the Rosati involution,

$$E(\iota(\tau)x, y) = E(x, \iota(\tau)^\dagger y) = E(x, \iota(\bar{\tau})y) \quad \text{for all } x, y.$$

Therefore

$$\begin{aligned} (\iota(\tau)^* E)(x, y) &= E(\iota(\tau)x, \iota(\tau)y) = E(x, \iota(\bar{\tau})\iota(\tau)y) \\ &= E(x, \iota(N(\tau))y) = N(\tau) E(x, y), \end{aligned}$$

using  $\bar{\tau}\tau = N(\tau) \in \mathbb{Q}$  and the  $\mathbb{Q}$ -linearity of  $\iota$ . This is the displayed identity. By Proposition 5.1 the character  $N$  on  $H^2$  is  $\tau\bar{\tau}$ , which is the summand  $(r, s) = (1, 1)$ ; and since  $\iota(\tau)^*$  is a ring map,  $\iota(\tau)^* \eta^n = N(\tau)^n \eta^n$ , placing  $\eta^n$  in the summand  $(n, n)$ . ■

**Lemma 5.3** (The Weil line sits at the ends). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type. Then*

$$W(A) \otimes_{\mathbb{Q}} \mathbb{C} = \left( \bigwedge^{2n} V_+ \right) \oplus \left( \bigwedge^{2n} V_- \right),$$

*the summands  $(r, s) = (2n, 0)$  and  $(0, 2n)$  of (15) in degree  $2n$ , on which  $K^\times$  acts by  $\tau \mapsto \tau^{2n}$  and  $\tau \mapsto \bar{\tau}^{2n}$ .*

*Proof.* This is Definition 3.6 together with Proposition 3.7(i), which says that  $W(A)$  is the  $\mathbb{Q}$ -structure on that pair of lines; the characters are read off from Proposition 5.1 with  $(r, s) = (2n, 0)$  and  $(0, 2n)$ . ■

**Lemma 5.4** (The two characters differ). *For every  $n \geq 1$  the characters  $\tau \mapsto \tau^{2n}$  and  $\tau \mapsto N_{K/\mathbb{Q}}(\tau)^n = (\tau\bar{\tau})^n$  of  $K^\times$  are distinct, as are  $\tau \mapsto \bar{\tau}^{2n}$  and  $N^n$ .*

*Proof.* The two characters agree if and only if  $(\tau/\bar{\tau})^n = 1$  for every  $\tau \in K^\times$ , since  $\tau^{2n} = (\tau\bar{\tau})^n$  is equivalent to  $(\tau/\bar{\tau})^n = 1$  for  $\tau \neq 0$ .

Consider the map  $c: K^\times \rightarrow K^\times$ ,  $c(\tau) = \tau/\bar{\tau}$ . Two elements have the same image exactly when their ratio is fixed by conjugation, that is when it lies in  $\mathbb{Q}^\times$ . The elements  $m + \sqrt{-d}$  for  $m \in \mathbb{Z}$  are pairwise non-proportional over  $\mathbb{Q}$ , so  $c$  takes infinitely many values.

On the other hand  $\{\zeta \in K : \zeta^n = 1\}$  is contained in the group  $\mu(K)$  of roots of unity of  $K$ , and for an imaginary quadratic field  $\mu(K)$  is finite: it has order 4 for  $d = 1$ , order 6 for

$d = 3$ , and order 2 otherwise. An infinite set cannot be contained in a finite one, so some  $\tau \in K^\times$  has  $(\tau/\bar{\tau})^n \neq 1$ , and the characters differ. The second pair is the image of the first under conjugation. ■

*Remark 5.5* (The witness has to be chosen). The proof just given is non-constructive on purpose, for the following reason. One is tempted to fix a convenient  $\tau$  once and for all, and the convenient choice  $\tau = 1 + \sqrt{-d}$  fails: for  $d = 1$  the ratio  $\tau/\bar{\tau}$  equals  $\sqrt{-1}$ , a primitive fourth root of unity, so the two characters do agree at that single  $\tau$  whenever  $4 \mid n$ ; and for  $d = 3$  the ratio is  $(-1 + \sqrt{-3})/2$ , a primitive cube root of unity, so they agree whenever  $3 \mid n$ . The choice  $\tau = 3 + \sqrt{-d}$  fails as well, at  $d = 3$  with  $6 \mid n$ . These are not counterexamples to the lemma, which quantifies over all  $\tau$ ; they are counterexamples to the shortcut. We found them by formalising the statement, and both are recorded as machine-checked facts in [Appendix E](#), theorems `bad_witness` and `bad_witness_exceptions`. The choices  $\tau = 2 + \sqrt{-d}$ ,  $\tau = 4 + \sqrt{-d}$  and  $\tau = 3 + 2\sqrt{-d}$  do work for every  $d$  and every  $n \leq 12$  tested, but the finiteness argument above is what proves the lemma.

## 5.2. The obstruction.

*Proof of Theorem 1.3.* The grading (15) and the action on its summands are [Proposition 5.1](#). The placement of  $W(A)$  is [Lemma 5.3](#) and the placement of  $\eta^n$  is [Lemma 5.2](#). For the last assertion, write  $H^{2n}(A, \mathbb{Q})^{N^n}$  for the space of  $x$  with  $\iota(\tau)^*x = N(\tau)^n x$  for all  $\tau$ . After  $\otimes \mathbb{C}$  this is the isotypic component of the character  $N^n$ , which by [Proposition 5.1](#) is the single summand  $(r, s) = (n, n)$ . By [Lemma 5.4](#) the characters  $\tau^{2n}$  and  $\bar{\tau}^{2n}$  are different from  $N^n$ , so by [Lemma 5.3](#) the space  $W(A) \otimes \mathbb{C}$  meets the  $(n, n)$  summand in zero. Intersecting with the rational structure gives  $W(A) \cap H^{2n}(A, \mathbb{Q})^{N^n} = 0$ . ■

**Corollary 5.6** (What an equivariant construction can produce). *Let  $x \in H^{2n}(A, \mathbb{Q})$  satisfy  $\iota(\tau)^*x = N(\tau)^n x$  for all  $\tau \in K^\times$ . Then the component of  $x$  along  $W(A)$  in any  $K^\times$ -stable complement is zero. In particular  $x$  is a Weil class only if  $x = 0$ .*

*Proof.* Immediate from [Theorem 1.3](#), since the isotypic decomposition is a direct sum decomposition into  $K^\times$ -stable subspaces and  $x$  lies in a single summand that meets  $W(A)$  in zero. ■

[Corollary 5.6](#) is the statement that applies to constructions. A construction that is natural in the  $K$ -action produces an eigenspace, and the computation of which character it produces is usually easy, because it is determined by how the input transforms. The point of [Lemmas 5.2](#) and [5.4](#) together is that the norm character, which is the one a  $\mathbb{Q}$ -rational construction naturally produces, is the character already occupied by the polarisation.

*Remark 5.7* (Which character a construction would need). To land on the Weil line a construction must produce an eigenspace for  $\tau \mapsto \tau^{2n}$  or for  $\tau \mapsto \bar{\tau}^{2n}$ . Neither character is  $\mathbb{Q}$ -valued:  $\tau^{2n} \in K$  and is not fixed by  $\text{Gal}(K/\mathbb{Q})$  unless it is real. A correspondence defined over  $\mathbb{Q}$  acts on  $H^*(A, \mathbb{Q})$  by a  $\mathbb{Q}$ -linear map that commutes with the  $\mathbb{Q}$ -structure, so any eigenvalue attached to it in this setting is  $\text{Gal}(K/\mathbb{Q})$ -stable as a function of  $\tau$ , which forces the norm character or a power of it. The way out of this is not to make the construction more clever but to break the naturality: one must produce a class whose  $K^\times$ -orbit is two-dimensional, spanning the pair  $\tau^{2n}$  and  $\bar{\tau}^{2n}$  together, rather than an eigenspace. Both  $W(A)$  and its  $\mathbb{Q}$ -structure are of that shape, and this is precisely what makes the Weil line hard to reach: it is the one place in the grading that no single  $\mathbb{Q}$ -rational eigenvalue condition can isolate.

*Remark 5.8* (A remark on the failure mode). It is easy to arrive at the opposite conclusion by a plausible sequence of words. The Weil line is  $\bigwedge_K^{2n} V$ , a rank-one free  $K$ -module; a rank-one free  $K$ -module is two-dimensional over  $\mathbb{Q}$ ; and  $K^\times$  acts on a two-dimensional  $\mathbb{Q}$ -space by matrices of determinant  $N(\tau)$ . The last clause is true and irrelevant: the determinant of the action is the norm, but the action itself is multiplication by  $\tau^{2n}$ , and a space on which  $K^\times$  acts by multiplication by  $\tau^{2n}$  is not the same as a space on which it acts by the scalar  $N(\tau)^n$ . The first is a  $K$ -line, the second is a  $\mathbb{Q}$ -eigenspace; they agree only when  $\tau^n = \bar{\tau}^n$  for all  $\tau$ , which never happens. Figure 3 is drawn to make the distinction visible.

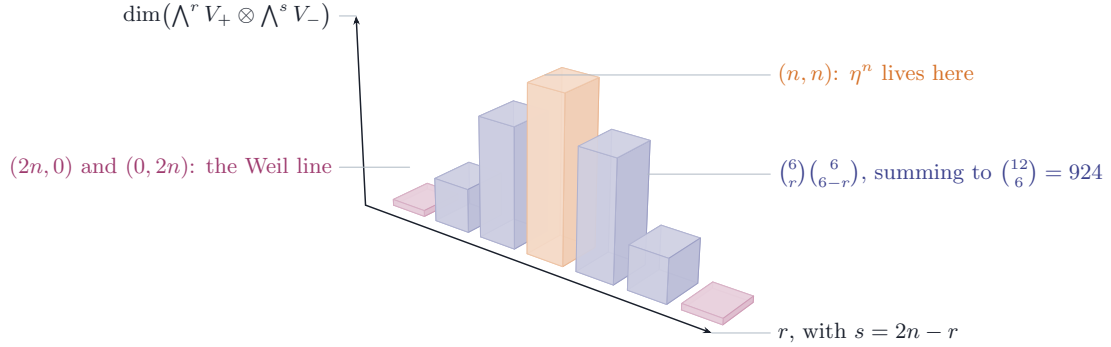


FIGURE 3. The  $K^\times$ -grading of  $H^{2n}(A, \mathbb{C})$  at  $n = 3$ , with each summand  $\bigwedge^r V_+ \otimes \bigwedge^s V_-$  of (15) drawn at its true dimension  $\binom{2n}{r} \binom{2n}{s}$ ; the seven bars sum to  $\binom{12}{6} = 924$ . The two extreme cells,  $(2n, 0)$  and  $(0, 2n)$ , are where the Weil line sits, and the middle cell  $(n, n)$  is where  $\eta^n$  sits. They are different cells, which is the whole of Theorem 1.3: no equivariant construction can move a class from the one to the other.

## 6. THE COHOMOLOGICAL FOURIER–MUKAI TRANSFORM

The divisor obstruction rests on one computation: the cohomological Fourier–Mukai transform sends each power of the polarisation class to a rational multiple of a power of the polarisation class on the other side, and to nothing else.<sup>16</sup> We prove this in closed form. The computation is universal, in the sense that it takes place in a cohomology ring whose isomorphism type is determined by the dimension alone, so a single calculation settles it for every principally polarised abelian variety at once.

### 6.1. Grothendieck–Riemann–Roch on an abelian variety.

**Proposition 6.1.** *Let  $A$  be an abelian variety of dimension  $g$  and  $\mathcal{G}$  an object of  $D^b(A^\vee)$ . Then*

$$\mathrm{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\mathrm{ch}(\mathcal{G})),$$

where the transform on the right is the cohomological one,  $\Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot \mathrm{ch}(\mathcal{P}))$ .

<sup>16</sup>The Todd class of an abelian variety is trivial, since the tangent bundle is trivial, so Grothendieck–Riemann–Roch for a morphism of abelian varieties reduces to the statement that the Chern character commutes with pushforward. That is the one simplification which makes the closed form below a finite computation rather than an estimate, and it is the reason this paper can be precise where the general case would not be.

*Proof.* Grothendieck–Riemann–Roch for the proper morphism  $p: A \times A^\vee \rightarrow A$  reads

$$\mathrm{ch}(Rp_*\mathcal{K}) \cdot \mathrm{td}(T_A) = p_*(\mathrm{ch}(\mathcal{K}) \cdot \mathrm{td}(T_{A \times A^\vee}))$$

for  $\mathcal{K} \in D^b(A \times A^\vee)$ . The tangent bundle of an abelian variety is trivial, so  $\mathrm{td}(T_A) = 1$  and  $\mathrm{td}(T_{A \times A^\vee}) = 1$ . Taking  $\mathcal{K} = Lq^*\mathcal{G} \otimes^L \mathcal{P}$  and using multiplicativity of the Chern character together with  $\mathrm{ch}(Lq^*\mathcal{G}) = q^*\mathrm{ch}(\mathcal{G})$  gives the statement.  $\blacksquare$

## 6.2. The universal model.

*Notation 6.2.* Let  $A$  be a principally polarised abelian variety of dimension  $g$  with polarisation class  $\theta$ , and put  $V = H^1(A, \mathbb{Q})$ , a  $\mathbb{Q}$ -vector space of dimension  $2g$  carrying the symplectic form attached to  $\theta$ . Choose a symplectic basis  $e_1, \dots, e_{2g}$  of  $V$ , so that

$$\theta = \sum_{j=1}^g e_j \wedge e_{g+j}. \quad (16)$$

The principal polarisation identifies  $A^\vee$  with  $A$  and  $H^1(A^\vee, \mathbb{Q})$  with the dual  $V^\vee$ ; let  $f_1, \dots, f_{2g}$  be the basis dual to  $e_1, \dots, e_{2g}$ , so that

$$\theta' = \sum_{j=1}^g f_j \wedge f_{g+j}. \quad (17)$$

Then  $H^*(A \times A^\vee, \mathbb{Q}) = \wedge^*(V \oplus V^\vee)$ .

**Lemma 6.3** (The Poincaré class). *In the model of Notation 6.2,*

$$\ell := c_1(\mathcal{P}) = \sum_{i=1}^{2g} e_i \wedge f_i \in V \otimes V^\vee \subset \wedge^2(V \oplus V^\vee), \quad (18)$$

the canonical element of  $V \otimes V^\vee = \mathrm{Hom}(V, V)$  corresponding to the identity, and  $\mathrm{ch}(\mathcal{P}) = e^\ell$ .

*Proof.* The Poincaré bundle is normalised to be trivial on  $A \times \{0\}$  and on  $\{0\} \times A^\vee$ , so  $c_1(\mathcal{P})$  has no component in  $\wedge^2 V$  or in  $\wedge^2 V^\vee$  and lies in the middle summand  $V \otimes V^\vee$  of  $\wedge^2(V \oplus V^\vee) = \wedge^2 V \oplus (V \otimes V^\vee) \oplus \wedge^2 V^\vee$ . The defining property of  $\mathcal{P}$  is that its restriction to  $A \times \{\lambda\}$  is the line bundle  $\lambda$ ; on cohomology this says that the induced map  $H^1(A^\vee, \mathbb{Q}) \rightarrow H^1(A, \mathbb{Q})^\vee$  obtained by contracting  $c_1(\mathcal{P})$  is the canonical identification. That is exactly the statement that the element of  $\mathrm{Hom}(V, V)$  determined by  $c_1(\mathcal{P})$  is the identity, which is (18). Since  $\mathcal{P}$  is a line bundle,  $\mathrm{ch}(\mathcal{P}) = e^{c_1(\mathcal{P})}$ .  $\blacksquare$

*Remark 6.4.* Every principally polarised abelian variety of dimension  $g$  gives the same data in Notation 6.2: a symplectic space of dimension  $2g$  over  $\mathbb{Q}$ , its dual, the symplectic form, and the canonical element. The constants computed below therefore depend only on  $g$  and  $k$  and not on  $A$ . This is why a single computation suffices.

## 6.3. The closed form.

**Theorem 6.5** (Transform of the polarisation powers). *Let  $A$  be a principally polarised abelian variety of dimension  $g$ . Then for every  $0 \leq k \leq g$ ,*

$$\Phi_{\mathcal{P}}(\theta'^k) = (-1)^{g(g+1)/2+k} \frac{k!}{(g-k)!} \theta^{g-k}, \quad (19)$$

and  $\Phi_{\mathcal{P}}(\theta'^k) = 0$  for  $k > g$ . In particular  $\Phi_{\mathcal{P}}(\mathbb{Q}[\theta']) = \mathbb{Q}[\theta]$ .

*Proof.* Work in the model of [Notation 6.2](#). By definition  $\Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot e^\ell)$ , and  $p_*$  is the operation of extracting the coefficient of the top exterior form  $f_1 \wedge \cdots \wedge f_{2g}$  in the  $V^\vee$  variables.

The classes  $e_i \wedge f_i$  have even degree, so they commute, and they have square zero. Hence for every  $m$ ,

$$\frac{\ell^m}{m!} = \sum_{\substack{S \subseteq \{1, \dots, 2g\} \\ |S|=m}} \prod_{i \in S} e_i \wedge f_i, \quad (20)$$

the product being taken in increasing order of  $i$ . Likewise, from [\(17\)](#),

$$\frac{\theta'^k}{k!} = \sum_{\substack{T \subseteq \{1, \dots, g\} \\ |T|=k}} \prod_{j \in T} f_j \wedge f_{g+j}, \quad \frac{\theta^{g-k}}{(g-k)!} = \sum_{\substack{U \subseteq \{1, \dots, g\} \\ |U|=g-k}} \prod_{j \in U} e_j \wedge e_{g+j}. \quad (21)$$

Fix  $k$ . In the product  $\theta'^k \cdot e^\ell$ , the terms that survive  $p_*$  are those of total degree  $2g$  in the  $f$  variables. A term of [\(21\)](#) indexed by  $T$  carries the  $2k$  variables  $\{f_j, f_{g+j} : j \in T\}$ , and a term of [\(20\)](#) indexed by  $S$  carries the  $|S|$  variables  $\{f_i : i \in S\}$ . For the product to be nonzero the two index sets must be disjoint, and for the total  $f$ -degree to be  $2g$  we need

$$S = \{1, \dots, 2g\} \setminus (T \cup (g+T)), \quad |S| = 2g - 2k.$$

So  $S$  is determined by  $T$ , only the term  $\ell^{2g-2k}/(2g-2k)!$  of  $e^\ell$  contributes, and the surviving terms are in bijection with the  $k$ -subsets  $T$  of  $\{1, \dots, g\}$ . Writing  $U = \{1, \dots, g\} \setminus T$  we have  $S = U \cup (g+U)$ , so the  $V$ -part of the term indexed by  $T$  is  $\prod_{i \in U \cup (g+U)} e_i$ , which up to sign is the term of [\(21\)](#) indexed by  $U$ . The bijection  $T \leftrightarrow U$  is exactly the one between the terms of  $\theta'^k/k!$  and those of  $\theta^{g-k}/(g-k)!$ .

It remains to compute the sign  $\sigma$  with which each term contributes, and to check that it is the same for all  $T$ . Both are permutation computations in the ordered word

$$\left( \prod_{j \in T} f_j f_{g+j} \right) \cdot \left( \prod_{i \in S} e_i f_i \right)$$

against the target word  $(\prod_{j \in U} e_j e_{g+j}) \cdot (f_1 \cdots f_{2g})$ . Both words use the same multiset of generators, and the permutation relating them is a product of transpositions whose parity depends only on the cardinalities  $|T| = k$  and  $|U| = g - k$ , not on which elements  $T$  contains, because relabelling the  $g$  symplectic pairs by a permutation  $\pi$  of  $\{1, \dots, g\}$  carries the word for  $T$  to the word for  $\pi(T)$  and multiplies both words by the same sign. Hence  $\sigma = \sigma(g, k)$ , and evaluating it at  $T = \{1, \dots, k\}$  gives

$$\sigma(g, k) = (-1)^{g(g+1)/2 + k}. \quad (22)$$

Assembling, and accounting for the factorials in [\(21\)](#),

$$\Phi_{\mathcal{P}}(\theta'^k) = k! \cdot \sigma(g, k) \cdot \frac{\theta^{g-k}}{(g-k)!},$$

which is [\(19\)](#). For  $k > g$  we have  $\theta'^k = 0$  because  $\theta'$  is a symplectic form on a space of dimension  $2g$ . The last assertion follows since the constants in [\(19\)](#) are nonzero for every  $0 \leq k \leq g$ .  $\blacksquare$

*Remark 6.6* (On the sign). The identity [\(22\)](#) is a finite permutation count for each pair  $(g, k)$ . It is verified independently, together with the uniformity of the sign over all  $T$ , for every  $g \leq 12$  and every  $k$  in [Appendix D](#); the full expansion of the exterior algebra, which is the more expensive check and confirms the whole of [\(19\)](#) including the factorials, is

carried out there for  $g \leq 4$ . For even  $g = 2n$ , which is the case relevant to Weil type, (22) reads  $\sigma(2n, k) = (-1)^{n+k}$ .

**Corollary 6.7** (Stability of the divisor line). *Let  $A$  be a principally polarised abelian variety and let  $x \in H^*(A^\vee, \mathbb{Q})$  lie in the subring  $\mathbb{Q}[\theta']$  generated by  $\theta'$ . Then  $\Phi_{\mathcal{P}}(x) \in \mathbb{Q}[\theta]$ .*

*Proof.*  $\mathbb{Q}[\theta']$  is spanned by the powers  $\theta'^k$ , and Theorem 6.5 sends each to a multiple of a power of  $\theta$ ; extend by linearity.  $\blacksquare$

**Remark 6.8** (Non-principal polarisations). If  $\theta$  is a polarisation of degree  $\delta^2$  rather than a principal one, the isogeny  $\phi_\theta: A \rightarrow A^\vee$  has degree  $\delta^2$  and one has  $\phi_\theta^* \theta' = \delta \theta$  for a suitable normalisation of  $\theta'$ . The argument of Theorem 6.5 goes through with  $\theta$  replaced by  $\delta^{-1} \phi_\theta^* \theta'$  and the constants changed by powers of  $\delta$ , all of which are nonzero rationals. Since all statements below concern only membership in  $\mathbb{Q}[\theta]$ , and not the values of the constants, nothing changes. We therefore state the results for principal polarisations and use them in general.

## 7. THE DIVISOR OBSTRUCTION

### 7.1. Statement and proof.

**Lemma 7.1** (Restriction from an ambient projective space). *Let  $Y$  be a smooth projective variety, let  $i: Y \rightarrow \mathbb{P}^{N-1}$  be a morphism defined by a linear system, let  $h \in H^2(\mathbb{P}^{N-1}, \mathbb{Q})$  be the hyperplane class, and let  $\mathcal{F} \in D^b(\mathbb{P}^{N-1})$ . Then*

$$\mathrm{ch}(Li^* \mathcal{F}) \in \mathbb{Q}[i^* h] \subset H^*(Y, \mathbb{Q}).$$

*Proof.* The Chern character commutes with derived pullback:  $\mathrm{ch}(Li^* \mathcal{F}) = i^* \mathrm{ch}(\mathcal{F})$ . This is standard for a locally free sheaf, where it follows from naturality of Chern classes, and it extends to  $D^b(\mathbb{P}^{N-1})$  because every object there has a finite resolution by direct sums of line bundles  $\mathcal{O}(m)$ , by Beilinson's theorem, and the Chern character is additive on distinguished triangles.<sup>17</sup> The cohomology ring of  $\mathbb{P}^{N-1}$  is  $\mathbb{Q}[h]/(h^N)$ , so  $\mathrm{ch}(\mathcal{F})$  is a polynomial in  $h$  with rational coefficients, and applying  $i^*$ , a ring homomorphism, gives a polynomial in  $i^* h$ .  $\blacksquare$

**Lemma 7.2** (The hyperplane class of a linear system). *Let  $A$  be an abelian variety and  $i: A^\vee \rightarrow \mathbb{P}^{N-1}$  a morphism defined by a linear system  $|M|$  with  $M \in \mathrm{Pic}(A^\vee)$ . Then  $i^* h = c_1(M) \in \mathrm{NS}(A^\vee) \otimes \mathbb{Q}$ .*

*Proof.* A morphism to  $\mathbb{P}^{N-1}$  defined by a linear system  $|M|$ , or by a subsystem of it, satisfies  $i^* \mathcal{O}(1) \cong M$  by construction, whence  $i^* h = c_1(i^* \mathcal{O}(1)) = c_1(M)$ .  $\blacksquare$

**Theorem 7.3** (Theorem 1.1). *Let  $A$  be a principally polarised complex abelian variety of dimension  $g$  with  $\mathrm{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q} \cdot \theta'$ , let  $i: A^\vee \rightarrow \mathbb{P}^{N-1}$  be any morphism defined by a linear system, and let  $\mathcal{F}$  be any object of  $D^b(\mathbb{P}^{N-1})$ . Put  $\mathcal{E} = \Phi_{\mathcal{P}}(Li^* \mathcal{F})$ . Then*

$$\mathrm{ch}_k(\mathcal{E}) \in \mathbb{Q} \cdot \theta^k \quad \text{for every } k \geq 0.$$

<sup>17</sup>Beilinson's theorem gives more than is used here: every object of  $D^b(\mathbb{P}^{N-1})$  is quasi-isomorphic to a finite complex whose terms are direct sums of  $\mathcal{O}(-m)$  with  $0 \leq m \leq N-1$ . The proof needs only that  $K_0(\mathbb{P}^{N-1})$  is generated by the classes of line bundles, which already follows from the Koszul resolutions of the structure sheaves of linear subspaces; Remark 20.3 records which other ambient varieties behave in the same way.

*Proof.* By Lemma 7.1,  $\text{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[i^*h]$ , and by Lemma 7.2,  $i^*h = c_1(M)$  for the line bundle  $M$  defining  $i$ . By hypothesis  $\text{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$ , so  $c_1(M) = m\theta'$  for some  $m \in \mathbb{Q}$ , and therefore  $\text{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[\theta']$ . By Corollary 6.7,  $\Phi_{\mathcal{P}}$  carries  $\mathbb{Q}[\theta']$  into  $\mathbb{Q}[\theta]$ , so  $\text{ch}(\mathcal{E}) \in \mathbb{Q}[\theta]$  by Proposition 6.1. Taking the component in  $H^{2k}$  and using that  $\mathbb{Q}[\theta] \cap H^{2k}(A, \mathbb{Q}) = \mathbb{Q}\theta^k$  gives the statement. ■

*Remark 7.4* (On the hypothesis on NS). The hypothesis  $\text{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$  holds at the very general point of any positive-dimensional family of abelian varieties of Weil type with  $\text{End}^0(A) = \iota(K)$ , which is the situation in which the Hodge conjecture for Weil classes is open; see Lemma 7.5. Without it the conclusion must be stated as  $\text{ch}_k(\mathcal{E}) \in D^{2k}(A^\vee)$  transported to  $A$ , that is,  $\text{ch}_k(\mathcal{E})$  lies in the image under  $\Phi_{\mathcal{P}}$  of the divisor subring of  $A^\vee$ . Theorem 1.3 covers the general case directly and without any hypothesis on NS, which is one reason to have two proofs.

**Lemma 7.5.** *Let  $\mathcal{A} \rightarrow S$  be the universal family of abelian varieties of  $(K, -1, n)$ -Weil type over the corresponding moduli space, with  $n \geq 2$ . At a very general point  $s \in S$  one has  $\text{End}^0(A_s) = \iota(K)$  and  $\text{NS}(A_s) \otimes \mathbb{Q} = \mathbb{Q}\eta_s$ , of dimension one.*

*Proof.* The locus in  $S$  where  $\text{End}^0$  is strictly larger than  $\iota(K)$  is a countable union of proper closed subvarieties, since each extra endomorphism imposes a nontrivial algebraic condition on the period point and there are countably many possible endomorphism algebras; its complement is the very general locus. At such a point the Hodge group is the full special unitary group  $\text{SU}(V, \psi)$  of the  $K$ -hermitian form  $\psi$  attached to the polarisation, by Weil's computation of the Hodge group of a general abelian variety of Weil type. The space  $\text{Hdg}^1(A_s) = \text{NS}(A_s) \otimes \mathbb{Q}$  is the space of Hodge-group invariants in  $H^2(A_s, \mathbb{Q})$ . By Proposition 5.1 in degree two,  $H^2 \otimes \mathbb{C} = \wedge^2 V_+ \oplus (V_+ \otimes V_-) \oplus \wedge^2 V_-$ , and  $\text{SU}(V, \psi)$  acts on  $V_+ \otimes V_-$  with a one-dimensional space of invariants, spanned by  $\psi$  itself, and on  $\wedge^2 V_\pm$  with no invariants for  $2n \geq 3$ , since  $\wedge^2$  of the standard representation of  $\text{SL}_{2n}$  has no invariant vector unless  $2n = 2$ . Hence the invariants are one-dimensional and spanned by  $\eta_s$ . ■

## 7.2. Consequences for Weil classes.

*Proof of Corollary 1.2.* Let  $g = 2n$  and let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $\text{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$ . By Theorem 7.3,  $\text{ch}_n(\mathcal{E}) \in \mathbb{Q} \cdot \theta^n = \mathbb{Q} \cdot \eta^n \subset D^{2n}(A)$ . By Proposition 3.10,  $W(A) \cap D^{2n}(A) = 0$ . So if  $\text{ch}_n(\mathcal{E})$  is a Weil class then it lies in  $W(A) \cap D^{2n}(A) = 0$ . ■

*Second proof of Corollary 1.2, via characters.* By Lemma 5.2 the class  $\eta^n$  is a  $K^\times$ -eigenspace for the character  $N^n$ , hence so is every element of  $\mathbb{Q}\eta^n$ . By Theorem 7.3,  $\text{ch}_n(\mathcal{E}) \in \mathbb{Q}\eta^n$ , so  $\text{ch}_n(\mathcal{E})$  is an eigenspace for  $N^n$ . By Corollary 5.6 it is a Weil class only if it vanishes. ■

The second proof is longer as written but it is the one that generalises, because it does not use the hypothesis on NS, only the equivariance. We record the version that does not mention the ambient space at all.

**Theorem 7.6** (Equivariance suffices). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and let  $\mathcal{E} \in D^b(A)$  be an object such that for every  $\tau$  in the order  $\iota^{-1}(\text{End}(A)) \subset K$  one has*

$$\iota(\tau)^* \text{ch}_n(\mathcal{E}) = N_{K/\mathbb{Q}}(\tau)^n \text{ch}_n(\mathcal{E}).$$

*Then  $\text{ch}_n(\mathcal{E})$  is a Weil class only if  $\text{ch}_n(\mathcal{E}) = 0$ .*

*Proof.* The hypothesis says  $\text{ch}_n(\mathcal{E})$  lies in the isotypic component of  $N^n$  for the action of the order, and this component is the same as the one for the full group  $K^\times$  because the order spans  $K$  over  $\mathbb{Q}$  and the action is  $\mathbb{Q}$ -linear. Apply [Corollary 5.6](#). ■

*Remark 7.7* (What the two hypotheses cost). [Theorem 7.3](#) asks that the object be restricted from an ambient projective space and imposes a condition on  $\text{NS}(A^\vee)$ . [Theorem 7.6](#) asks only that the resulting class be equivariant with the norm character. The second hypothesis is weaker and is satisfied automatically by any construction whose input is  $K$ -invariant, which is the case for every secant construction we are aware of, since secant varieties of a  $K$ -invariant subvariety are  $K$ -invariant. The reason to state both is that the first is checkable by inspection of the construction and the second requires an equivariance computation.

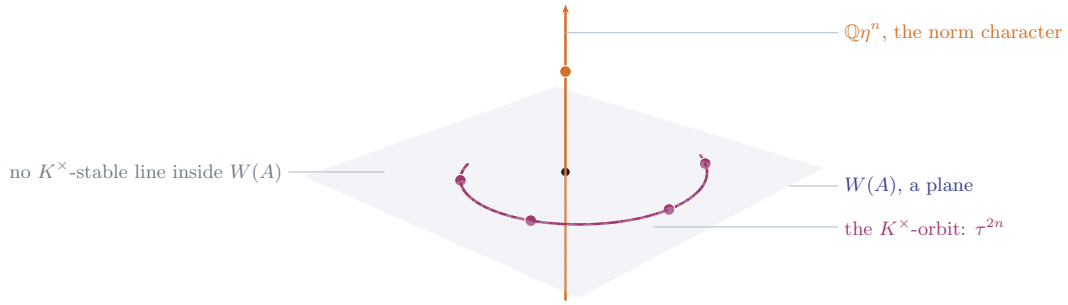


FIGURE 4. The linear algebra behind [Theorem 1.1](#), drawn inside  $H^{2n}(A, \mathbb{Q})$ . The Weil line  $W(A)$  is a plane, free of rank one over  $K$  and of rank two over  $\mathbb{Q}$ , and  $K^\times$  acts on it through  $\tau^{2n}$ , so the orbit of any nonzero class is a spiral that spans the plane and lies on no line in it. The transverse axis carries  $\eta^n$ , on which  $K^\times$  acts through the norm. A construction returning a single eigenclass returns a line, and by [Proposition 13.1](#) no line inside  $W(A)$  is stable.

## 8. THE NUMERICAL COUNT GIVES NO OBSTRUCTION

An earlier version of part of this material asserted a numerical obstruction to secant constructions, on the ground that the Euler characteristic which the dimension count demands lies below the value at which a linear system on an abelian variety becomes free of base points.<sup>18</sup> That assertion is wrong, and since it is the kind of error that is easy to repeat we set it out and correct it here. The corrected conclusion is that the numerical count obstructs nothing; the obstructions in this paper are the two of [Theorems 1.1](#) and [1.3](#), and neither uses any count.

<sup>18</sup>The content of this section was asserted in the opposite direction in the earlier version cited on the title page, where the two thresholds were presented as necessary conditions and used to rule out a range of constructions. They are sufficient and not necessary, and the two counterexamples below are elementary. We have kept the section rather than deleting it, because an error that was believed is worth printing next to its correction.

### 8.1. The count.

**Setup 8.1** (The secant numerology). Let  $A$  be of  $(K, -1, n)$ -Weil type, so  $\dim A = \dim A^\vee = 2n$ . Suppose  $A^\vee$  is embedded in  $\mathbb{P}^{N-1}$  by a complete linear system  $|M|$  with  $M$  ample, so that  $N = h^0(M) = \chi(M)$ . The  $n$ -th secant variety of a nondegenerate  $2n$ -fold has expected dimension

$$\dim \operatorname{Sec}^n(A^\vee) = n(2n+1) - 1 = 2n^2 + n - 1,$$

and a construction that intersects  $\operatorname{Sec}^n(A^\vee)$  with  $A^\vee$  in the expected dimension inside  $\mathbb{P}^{N-1}$  obtains

$$\dim(\operatorname{Sec}^n(A^\vee) \cap A^\vee) = (2n^2 + n - 1) + 2n - (N - 1) = 2n^2 + 3n - N.$$

Asking for this to equal  $n$ , which is what produces a class of the required codimension, forces

$$N = \chi(M) = 2n^2 + 2n. \quad (23)$$

**Lemma 8.2.** *For every  $n \geq 2$ ,*

$$2n^2 + 2n < 2^{2n} < 3^{2n},$$

*and the ratio  $2^{2n}/(2n^2 + 2n)$  tends to infinity.*

*Proof.* At  $n = 2$  the left inequality reads  $12 < 16$ . For  $n \geq 2$  the function  $4^n/(2n^2 + 2n)$  has ratio of consecutive terms  $4(2n^2 + 2n)/(2(n+1)^2 + 2(n+1)) = 4n/(n+2) > 1$  for  $n > 2/3$ , so the quotient is increasing from a value exceeding 1 at  $n = 2$ ; both claims follow.  $\blacksquare$

**8.2. Why this is not an obstruction.** The theorem of Lefschetz says that for  $M$  ample on an abelian variety,  $M^{\otimes 2}$  is free of base points and  $M^{\otimes 3}$  is very ample. If  $M = P^{\otimes 2}$  with  $P$  ample on a variety of dimension  $m$ , then  $\chi(M) = 2^m \chi(P) \geq 2^m$ , and similarly  $\chi \geq 3^m$  in the very ample case. It is tempting to read these as thresholds: no base-point-free system below  $2^m$ , no very ample system below  $3^m$ . Both readings invert a sufficient condition into a necessary one, and both are false already for abelian surfaces.

**Proposition 8.3** (The thresholds are not necessary). *Let  $S$  be a general complex abelian surface, so  $m = 2$ .*

- (i) *A polarisation of type  $(1, 3)$  on  $S$  has  $\chi = 3 < 4 = 2^2$  and its complete linear system is free of base points; the associated morphism  $S \rightarrow \mathbb{P}^2$  is a six-sheeted covering.*
- (ii) *A polarisation of type  $(1, 5)$  on  $S$  has  $\chi = 5 < 9 = 3^2$  and is very ample; the image is a Horrocks–Mumford abelian surface of degree 10 in  $\mathbb{P}^4$  [HM73].*

*In particular neither  $2^m$  nor  $3^m$  is a lower bound for the Euler characteristic of a base-point-free, respectively very ample, ample line bundle on an abelian variety of dimension  $m$ .*

*Proof.* Both statements are in the classification of  $(1, d)$ -polarised abelian surfaces; see Gross and Popescu [GP98, §1.2], where the base locus of  $|L|$  is determined for each  $d$ : it is empty for  $d \geq 3$ , consists of four points for  $d = 2$ , and  $|L|$  is very ample for  $d \geq 5$  on a general surface, the case  $d = 5$  being the Horrocks–Mumford embedding of [HM73]. The Euler characteristics are  $\chi = d$  by Riemann–Roch on an abelian surface.  $\blacksquare$

**Corollary 8.4.** *Setup 8.1 imposes no contradiction. The requirement  $\chi(M) = 2n^2 + 2n$  of (23) is compatible with  $M$  being very ample for all  $n$  for which a polarisation of that Euler characteristic and suitable type exists on a  $2n$ -dimensional abelian variety; at  $n = 2$  the*

requirement  $\chi = 12$  on an abelian fourfold is met by polarisations of type  $(1, 1, 2, 6)$  and  $(1, 1, 1, 12)$ , and nothing in the numerology excludes very ampleness for either.

*Remark 8.5* (What we retract). The claim withdrawn here is the assertion that the polarisation required by (23) admits no base-point-free linear system. It appeared in the supplementary material of the earlier version cited on the title page, together with a table of values of  $2n^2 + 2n$ ,  $2^{2n}$  and  $3^{2n}$ . The table is arithmetically correct and Lemma 8.2 reproduces it; the inference drawn from it is not. We are grateful to have found this before it propagated, and we note that the obstructions this paper does prove, Theorems 1.1 and 1.3, are of a different character: they bound where a class can lie in a decomposition of cohomology, and no choice of polarisation, ambient space or object affects them.

## 9. THE SEMIREGULARITY TARGET

Markman's construction at  $n = 2$  proceeds by deforming a sheaf across a family and controlling the obstruction by a semiregularity argument in the sense of Bloch and of Buchweitz and Flenner [Blo72, BF03]. It is sometimes said that this route is blocked for  $n \geq 4$  by a dimension count, the target of the semiregularity map being too large. We compute the target and show that the count says nothing of the kind. The section is included because the belief that the semiregularity route is numerically blocked, combined with the belief that the secant route is numerically blocked, leaves the impression that the difficulty at  $n \geq 4$  is arithmetic. It is not: it is Theorems 1.1 and 1.3.

**Lemma 9.1** (Hodge numbers of an abelian variety). *For an abelian variety  $A$  of dimension  $g$  one has  $h^{p,q}(A) = \binom{g}{p} \binom{g}{q}$ .*

*Proof.*  $H^{p,q}(A) = \wedge^p H^{1,0} \otimes \wedge^q H^{0,1}$  by (1) and the compatibility of the Hodge decomposition with the exterior algebra structure, and  $\dim H^{1,0} = \dim H^{0,1} = g$ . ■

**Proposition 9.2** (The semiregularity target). *Let  $A$  be an abelian variety of dimension  $g = 2n$ . The target of the semiregularity map, namely  $\bigoplus_q H^{q+2}(A, \Omega_A^q)$ , has dimension*

$$\sum_{q \geq 0} h^{q,q+2}(A) = \binom{4n}{2n-2}.$$

*Proof.* By Lemma 9.1 with  $g = 2n$ ,

$$\sum_{q \geq 0} h^{q,q+2} = \sum_{q \geq 0} \binom{2n}{q} \binom{2n}{q+2} = \sum_{q \geq 0} \binom{2n}{q} \binom{2n}{2n-q-2} = \binom{4n}{2n-2},$$

the last step being the Vandermonde convolution. ■

**Corollary 9.3** (The count does not bind). *The target dimension  $\binom{4n}{2n-2}$  takes the values 28, 495, 8008, 125970, 1961256, 30421755 for  $n = 2, \dots, 7$ , growing by a factor exceeding ten at each step. Any source whose dimension grows polynomially in  $n$ , for instance the quadratic quantity  $\binom{2n}{2} = n(2n-1)$  which takes the values 6, 15, 28, 45, 66, 91 over the same range, is eventually negligible against it. In particular a comparison of the two dimensions cannot obstruct the existence of a semiregular sheaf for large  $n$ ; the inequality it produces points the other way.*

*Remark 9.4* (Consistency at  $n = 2$ ). At  $n = 2$  the target is 28 and the sheaf produced by Markman has  $\dim \text{Ext}^2 = 12$ , comfortably inside it, which is consistent with the construction succeeding there. Nothing in the numbers changes character between  $n = 2$

and  $n = 4$ . Whatever prevents the construction from being carried out for larger  $n$ , it is not this comparison.

*Remark 9.5* (What semiregularity would still need). We emphasise that [Corollary 9.3](#) is not an argument that the semiregularity route works. It removes one purported obstruction and leaves the real requirement in place, which is the construction of a sheaf on  $A$  whose Chern character has a nonzero component along the Weil line. By [Theorem 1.3](#) such a sheaf cannot be  $K$ -equivariant with the norm character, so it cannot be produced by any functorial construction from  $K$ -invariant data on  $A^\vee$ ; Markman's sheaf at  $n = 2$  evades this because it is not so produced. That is the sense in which the problem for  $n \geq 3$  is a problem about breaking equivariance, and not about dimensions.

**9.1. The annihilator of the Weil line, and objects of rank zero.** [Remark 9.5](#) says that the remaining requirement is a sheaf whose Chern character has a component along the Weil line. The most economical way to arrange that is to make the Chern character lie *entirely* on the Weil line: to take an object of rank zero, with every  $\text{ch}_k$  vanishing except  $\text{ch}_n$ , which is what the construction of [Section 14](#) does. We show here that this economy is fatal to semiregularity, for a reason that has nothing to do with how the object is built and everything to do with where its Chern character sits.<sup>19</sup>

**Lemma 9.6** (The annihilator of the Weil line in  $H^{0,2}$ ). *Let  $A$  be of  $(K, -1, n)$ -Weil type and let  $\omega \in W(A) \otimes \mathbb{C}$  be any nonzero element of the Weil line. Then the multiplication map*

$$H^{0,2}(A) \longrightarrow H^{n,n+2}(A), \quad z \longmapsto z \cdot \omega,$$

*has kernel of dimension exactly  $n^2$ . Its kernel is spanned by the products  $u \wedge v$  with  $u \in V_+^{0,1}$  and  $v \in V_-^{0,1}$ , and it is the same subspace for every  $\omega$  on the line.*

*Proof.* Write  $\omega = a\alpha_+ + b\alpha_-$  with  $\alpha_\pm$  generating  $\bigwedge^{2n} V_\pm$  and  $(a, b) \neq (0, 0)$ . Since  $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$  with each summand of dimension  $n$ , the space  $H^{0,2} = \bigwedge^2 H^{0,1}$  decomposes as

$$H^{0,2} = \bigwedge^2 V_+^{0,1} \oplus (V_+^{0,1} \otimes V_-^{0,1}) \oplus \bigwedge^2 V_-^{0,1},$$

of dimensions  $\binom{n}{2}$ ,  $n^2$  and  $\binom{n}{2}$ . A mixed product  $u \wedge v$  with  $u \in V_+^{0,1}$ ,  $v \in V_-^{0,1}$  satisfies  $u \wedge v \wedge \alpha_+ = 0$  because  $u$  already occurs among the  $2n$  factors of  $\alpha_+$ , and  $u \wedge v \wedge \alpha_- = 0$  for the same reason with  $v$ . So the middle summand lies in the kernel.

Conversely let  $z = z_{++} + z_{+-} + z_{--}$  be in the kernel. The product  $z \cdot \omega$  has the component  $a z_{--} \wedge \alpha_+$  in  $\bigwedge^2 V_-^{0,1} \otimes \bigwedge^{2n} V_+$  and the component  $b z_{++} \wedge \alpha_-$  in  $\bigwedge^2 V_+^{0,1} \otimes \bigwedge^{2n} V_-$ , and

<sup>19</sup>We recall what the map is and what it is for, since the rest of this subsection turns on it. For a coherent sheaf  $\mathcal{E}$  on a smooth projective variety  $A$  the semiregularity map is defined through the Atiyah class  $\text{at}(\mathcal{E}) \in \text{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega_A^1)$ : a class  $\xi \in \text{Ext}^2(\mathcal{E}, \mathcal{E})$  is sent to the trace of  $\xi \cdot \exp(\text{at}(\mathcal{E}))$ , and the component of that trace in  $H^q(A, \Omega_A^{q+2}) = H^{q,q+2}(A)$  is the  $q$ -th coordinate of  $\sigma_{\mathcal{E}}(\xi)$ . Bloch introduced the map for local complete intersections [[Blo72](#)], and Buchweitz and Flenner extended it to arbitrary coherent sheaves and perfect complexes [[BF03](#)]. Its use is this. If  $A$  moves in a family  $\mathcal{A} \rightarrow S$ , the obstruction to extending  $\mathcal{E}$  to first order along  $v \in T_s S$  is a class  $\text{ob}_v(\mathcal{E}) \in \text{Ext}^2(\mathcal{E}, \mathcal{E})$ , and the theorem of Buchweitz and Flenner identifies  $\sigma_{\mathcal{E}}(\text{ob}_v(\mathcal{E}))$  with the derivative of  $\text{ch}(\mathcal{E})$  in the direction  $v$ , projected off the diagonal of the Hodge decomposition. So if  $\text{ch}(\mathcal{E})$  stays of Hodge type along  $S$  then  $\sigma_{\mathcal{E}}(\text{ob}_v(\mathcal{E})) = 0$ , and if  $\sigma_{\mathcal{E}}$  is also injective then  $\text{ob}_v(\mathcal{E}) = 0$  and  $\mathcal{E}$  deforms. Injectivity is what converts a statement about the Chern character into a statement about the sheaf, and it is the one hypothesis in the argument that sees more than  $\text{ch}(\mathcal{E})$ . Two special cases are all that is needed below: for a line bundle  $L$ ,  $\text{Ext}^2(L, L) = H^2(\mathcal{O}_A) = H^{0,2}(A)$  and  $\sigma_L$  is multiplication by  $e^{c_1(L)}$ ; and for a direct sum,  $\text{Ext}^2$  is the sum of the  $\text{Ext}^2(\mathcal{E}_i, \mathcal{E}_j)$  and  $\sigma$  acts on each diagonal block by multiplication by the Chern character of that summand. Both follow at once from the definition by the trace, and both are what items (XIII) and (XIV) of [Appendix D](#) implement.

these two summands of  $\bigwedge^{2n+2} V_{\mathbb{C}}$  are independent of each other and of everything else in  $z \cdot \omega$ . Each of the maps  $z_{--} \mapsto z_{--} \wedge \alpha_+$  and  $z_{++} \mapsto z_{++} \wedge \alpha_-$  is injective, since wedging a nonzero element of  $\bigwedge^2 V_-^{0,1}$  with a basis of  $V_+$  produces a nonzero element of  $\bigwedge^{2n+2}(V_+ \oplus V_-)$ . Hence if  $a \neq 0$  then  $z_{--} = 0$ , and if  $b \neq 0$  then  $z_{++} = 0$ . When both  $a$  and  $b$  are nonzero the kernel is exactly the middle summand. When one of them vanishes, say  $b = 0$ , the class  $\omega = a \alpha_+$  is not real and lies in the complexified Weil line; the kernel is then  $V_+^{0,1} \otimes V_-^{0,1} \oplus \bigwedge^2 V_+^{0,1}$ , of dimension  $n^2 + \binom{n}{2}$ . For the rational generators of [Theorem 4.9](#), where  $a$  and  $b$  are both nonzero, the kernel has dimension exactly  $n^2$ . ■

The dimension  $n^2$  was also computed directly in the exterior algebra, modulo a prime, for both rational generators of the Weil line at  $n = 2, 3, 4$ , and agrees ([Appendix D](#), item (XIV)).

**Theorem 9.7** (Objects of rank zero are not semiregular). *Let  $A$  be of  $(K, -1, n)$ -Weil type. Let  $\mathcal{E}_1$  and  $\mathcal{E}_2$  be coherent sheaves on  $A$  such that*

$$\mathrm{ch}(\mathcal{E}_1) - \mathrm{ch}(\mathcal{E}_2) = c\omega, \quad c \neq 0, \quad \omega \in W(A),$$

*and such that  $\mathrm{Ext}^2(\mathcal{E}_1, \mathcal{E}_2) = \mathrm{Ext}^2(\mathcal{E}_2, \mathcal{E}_1) = 0$ . Put  $\mathcal{E} = \mathcal{E}_1 \oplus \mathcal{E}_2$ . Then the semiregularity map*

$$\sigma_{\mathcal{E}}: \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \bigoplus_{q \geq 0} H^{q, q+2}(A)$$

*has kernel of dimension at least  $n^2$ . In particular  $\mathcal{E}$  is not semiregular, for any  $n \geq 1$ .*

*Proof.* The vanishing of the off-diagonal  $\mathrm{Ext}^2$  groups gives  $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) = \mathrm{Ext}^2(\mathcal{E}_1, \mathcal{E}_1) \oplus \mathrm{Ext}^2(\mathcal{E}_2, \mathcal{E}_2)$ , and each summand contains  $H^2(\mathcal{O}_A) \cdot \mathrm{id}_{\mathcal{E}_i}$  through the  $\mathcal{O}_A$ -module structure, the map  $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}_i}$  being injective for  $\mathcal{E}_i \neq 0$ .<sup>20</sup> For  $z \in H^2(\mathcal{O}_A) = H^{0,2}(A)$  consider the element  $\xi_z = z \cdot \mathrm{id}_{\mathcal{E}_1} - z \cdot \mathrm{id}_{\mathcal{E}_2}$ . The semiregularity map is compatible with direct sums and on the summand  $H^{0,2} \cdot \mathrm{id}_{\mathcal{E}_i}$  it is multiplication by the Chern character of  $\mathcal{E}_i$ , so

$$\sigma_{\mathcal{E}}(\xi_z) = z \cdot \mathrm{ch}(\mathcal{E}_1) - z \cdot \mathrm{ch}(\mathcal{E}_2) = cz \cdot \omega.$$

By [Lemma 9.6](#) this vanishes for every  $z$  in an  $n^2$ -dimensional subspace of  $H^{0,2}$ , and the corresponding  $\xi_z$  are nonzero and linearly independent. Hence  $\dim \ker \sigma_{\mathcal{E}} \geq n^2$ . ■

**Corollary 9.8** (The construction of [Section 14](#)). *Let  $L_1, \dots, L_m$  be line bundles on  $A$  with  $\sum_i c_1(L_i) = 0$  such that the virtual class  $\sum_i ([L_i] - [L_i^{-1}])$  has Chern character on the Weil line and every difference  $c_1(L_i) \pm c_1(L_j)$ ,  $i \neq j$ , is nondegenerate of index different from 2. Then the sheaf  $\bigoplus_i (L_i \oplus L_i^{-1})$  has semiregularity kernel of dimension at least  $n^2$ . At  $n = 3$  the explicit object of [Section 14](#) has kernel of dimension exactly  $15 = n^2 + n(n-1)$ , of which the  $n^2 = 9$  dimensions on the odd side are the ones supplied by [Theorem 9.7](#); the remaining  $n(n-1) = 6$  come from the even side and are specific to the line-bundle structure.*

<sup>20</sup>The map  $H^2(\mathcal{O}_A) \rightarrow \mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ ,  $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}}$ , is the edge map of the local-to-global spectral sequence, and when  $\mathrm{rk} \mathcal{E} \neq 0$  it is split by the trace up to the factor  $\mathrm{rk} \mathcal{E}$ . For a direct sum of line bundles every summand has rank one, so the map is injective on each block with no hypothesis, and the element  $\xi_z$  of the proof, which is  $z \cdot \mathrm{id}$  on the first block and  $-z \cdot \mathrm{id}$  on the second, is nonzero for every nonzero  $z$ . This is the only place the direct-sum structure is used: it is what allows an element of  $\mathrm{Ext}^2$  whose image under  $\sigma$  is  $z \cdot (\mathrm{ch} \mathcal{E}_1 - \mathrm{ch} \mathcal{E}_2)$  rather than  $z \cdot (\mathrm{ch} \mathcal{E}_1 + \mathrm{ch} \mathcal{E}_2)$ . For a single indecomposable sheaf of rank zero the same conclusion holds by the first sentence of the proof applied to  $z \cdot \mathrm{id}_{\mathcal{E}}$ , provided that element is nonzero, which is the case whenever the restriction of  $H^{0,2}(A)$  to the support of  $\mathcal{E}$  is injective; it fails for a skyscraper sheaf, whose support is a point, and the theorem is stated for direct sums to keep that hypothesis in view.

*Proof.* Take  $\mathcal{E}_1 = \bigoplus L_i$  and  $\mathcal{E}_2 = \bigoplus L_i^{-1}$ . The index hypothesis gives  $H^2(L_i^{\pm 1} \otimes L_j^{\mp 1}) = 0$  for  $i \neq j$  and hence the required  $\text{Ext}^2$  vanishing, and the Chern character hypothesis is the one of [Theorem 9.7](#). The exact decomposition of the kernel at  $n = 3$  into an odd part of dimension 9 and an even part of dimension 6 is a computation, recorded in [Appendix D](#): since  $e^\lambda$  and  $e^{-\lambda}$  combine into  $\cosh \lambda$  and  $\sinh \lambda$ , which occupy disjoint degrees, the map splits by parity, and the odd part is exactly the annihilator of [Lemma 9.6](#) tensored with the sign vector.  $\blacksquare$

*Remark 9.9* (Every object in the family, and what the theorem says). The same computation was run over every configuration  $\lambda_i = s_i \gamma + c_i \ell$  with  $\sum s_i = \sum c_i = 0$  in a box of side 7 whose virtual Chern character lies on the Weil line: eighteen distinct configurations, every one with kernel exactly 15. Adding an  $\eta$ -component produced no Weil objects at all in a box of side 3. The deficiency is not a defect of the particular six line bundles chosen; it is the same for every member of the family, and [Theorem 9.7](#) says why.

What the theorem removes is precisely the economy described at the start of this subsection. An object realising a Weil class through semiregularity cannot have its Chern character concentrated on the Weil line. It must carry a nonzero rank, or nonzero Chern character in other degrees, so that  $z \mapsto z \cdot \text{ch}(\mathcal{E})$  is injective on  $H^{0,2}$ ; the degree-two component of that product is  $\text{rk}(\mathcal{E}) \cdot z$ , and rank zero is exactly the case in which it vanishes. Markman's sheaves have nonzero rank, and this is not incidental to their success: the class he shows to be deformation-invariant is the normalised character  $\exp(-c_1(\mathcal{E})/\text{rk} \mathcal{E}) \text{ch}(\mathcal{E})$ , which divides by the rank, and the objects at  $n = 3$  are reflexive sheaves [[Mar25a](#), [Mar25b](#)]. So the fourth ingredient of [Remark 14.17](#) is narrowed once more: not a difference of two sheaves whose classes cancel outside the Weil line, but a single sheaf of positive rank whose Chern character has a Weil component alongside the rest.

The hypothesis of [Theorem 9.7](#) is stated for a direct sum because that is where the element  $\xi_z$  is visibly nonzero. The mechanism behind it is more general, and it reaches the object that [Section 12](#) uses.

**Proposition 9.10** (Concentrated Chern characters are never semiregular). *Let  $A$  be of  $(K, -1, n)$ -Weil type and let  $E$  be a nonzero perfect complex on  $A$  with  $\text{ch}_k(E) = 0$  for  $k \neq n$  and  $\text{ch}_n(E) = N\omega$  for some  $N \neq 0$  and some nonzero  $\omega \in W(A)$ . If the natural map*

$$H^2(\mathcal{O}_A) \longrightarrow \text{Ext}^2(E, E), \quad z \longmapsto z \cdot \text{id}_E, \quad (24)$$

*is injective, then  $\dim \ker \sigma_E \geq n^2$ , so  $E$  is not semiregular. The map (24) is injective whenever  $E$  is represented by a virtual sum of line bundles, in particular at every base point of [Corollary 15.4](#), [Theorem 15.25](#) and [Theorem 15.10](#).*

*Proof.* For any perfect complex the semiregularity map satisfies  $\sigma_E(z \cdot \text{id}_E) = z \cdot \text{ch}(E)$ , since the trace of  $z \cdot \text{id}_E \circ \text{At}(E)^q$  is  $z$  times the degree- $q$  component of  $\text{ch}(E)$ . Hence for  $z$  in the annihilator of  $\omega$  in  $H^{0,2}(A)$ ,

$$\sigma_E(z \cdot \text{id}_E) = N z \cdot \omega = 0,$$

and that annihilator has dimension exactly  $n^2$  by [Lemma 9.6](#). If (24) is injective the classes  $z \cdot \text{id}_E$  are independent, which gives the bound.

For the last sentence, write  $E$  in  $K_0(A)$  as  $\sum_i \varepsilon_i [L_i]$  with  $\varepsilon_i = \pm 1$  and  $L_i$  line bundles, realised as a direct sum of the  $L_i$  and of shifts of the remaining ones. The map (24) is then  $z \mapsto (z, \dots, z)$  into  $\bigoplus_i \text{Ext}^2(L_i, L_i) = \bigoplus_i H^2(\mathcal{O}_A)$ , which is injective. At a split member  $\omega_1$  is a polynomial in  $\beta, \hat{\beta}, \ell$  by [Theorem 15.2\(iii\)](#), at a quaternionic member it is a polynomial in divisor classes by [Theorem 15.25](#), and at a product it is one by [Theorem 15.10](#); in

each case the Chern character is a ring expression in classes of line bundles, so such a representation exists.  $\blacksquare$

**Proposition 9.11** (The class forces the size of the object). *Let  $A$  be of  $(K, -1, n)$ -Weil type, let  $\omega$  be a nonzero element of the Weil line and let  $E$  be a perfect complex on  $A$  with*

$$\mathrm{ch}(E) = r + N\omega, \quad r \in H^0(A, \mathbb{Q}), \quad N \neq 0,$$

*and no other component. Then*

$$\chi(E, E) = (-1)^n N^2 \int_A \omega^2,$$

*independently of  $r$ , and consequently*

$$\sum_i \dim \mathrm{Ext}^i(E, E) \geq N^2 \left| \int_A \omega^2 \right| > 0.$$

*Proof.* The Todd class of an abelian variety is 1, so Riemann–Roch for the Ext algebra reads  $\chi(E, E) = \int_A \mathrm{ch}(E)^\vee \mathrm{ch}(E)$  with  $\mathrm{ch}_k^\vee = (-1)^k \mathrm{ch}_k$ . Expanding,

$$\mathrm{ch}(E)^\vee \mathrm{ch}(E) = r^2 + (1 + (-1)^n) r N \omega + (-1)^n N^2 \omega^2,$$

and of the three terms only the last has degree  $4n$ , so only it survives the integral; the rank enters solely through  $r^2$ , which sits in degree zero. For the inequality,  $\chi(E, E) = \sum_i (-1)^i \dim \mathrm{Ext}^i(E, E)$ , so the total dimension is at least  $|\chi(E, E)|$ . Finally  $\int_A \omega^2 \neq 0$ : writing  $\omega = \alpha_+ + \alpha_-$  with  $\alpha_\pm$  spanning  $\bigwedge^{2n} V_\pm$ , each of  $\alpha_\pm$  is a top form on its own eigenspace and squares to zero, so  $\omega^2 = 2\alpha_+ \alpha_-$ , which is a nonzero multiple of the fundamental class. The computation is item (XVIII) of [Appendix D](#).  $\blacksquare$

*Remark 9.12* (The constant, and what it is not). The bound is on the whole Ext algebra and is insensitive to the rank, which is the point: enlarging the rank buys no room. The constant  $|\int \omega^2|$  depends on the normalisation of  $\omega$ ; measured against  $\eta$  it picks up factors of  $d$  and of the polarisation type, so no dimension-free growth rate should be read off it. What is intrinsic, and what the bound uses, is only that the Weil class is not isotropic. Together with [Proposition 9.10](#) this says that an object carrying the class is pinned from two sides at once: its semiregularity map has a kernel of dimension at least  $n^2$  unless  $H^2(\mathcal{O}_A)$  acts on it degenerately, and its Ext algebra has dimension at least  $N^2 |\int \omega^2|$  whatever its rank.

*Remark 9.13* (What this does to the semiregularity reduction). [Theorem 16.28](#) closes the family under the hypothesis that  $\sigma_E$  is injective for the complex  $E$  of [Proposition 16.26](#), whose Chern character is  $N\omega_1$  in degree  $n$  and zero elsewhere. That is exactly the hypothesis of [Proposition 9.10](#), so at every base point this paper constructs the hypothesis of [Theorem 16.28](#) fails for that complex, and fails by the fixed margin  $n^2$ .

Two things keep this from being a refutation of [Theorem 16.28](#) itself. The hypothesis is about a chosen perfect complex representing a class in  $K_0(A)_\mathbb{Q}$ , and distinct representatives of one class have distinct  $\mathrm{Ext}^2$  groups, so the theorem is not void; what is ruled out is the representative the construction supplies. And [Proposition 9.10](#) needs (24) to be injective, which is a genuine hypothesis for an object of rank zero: it fails, for instance, for a sheaf supported at a point. So a representative escaping [Proposition 9.10](#) must receive  $H^2(\mathcal{O}_A)$  non-injectively, which for a rank-zero object means its support is degenerate. We record the situation as it stands: of the two reductions of [Section 12](#), the degree bound of [Theorem 16.9](#) is untouched by anything here, and the semiregularity reduction of [Theorem 16.28](#) is unavailable for every object we can presently write down.

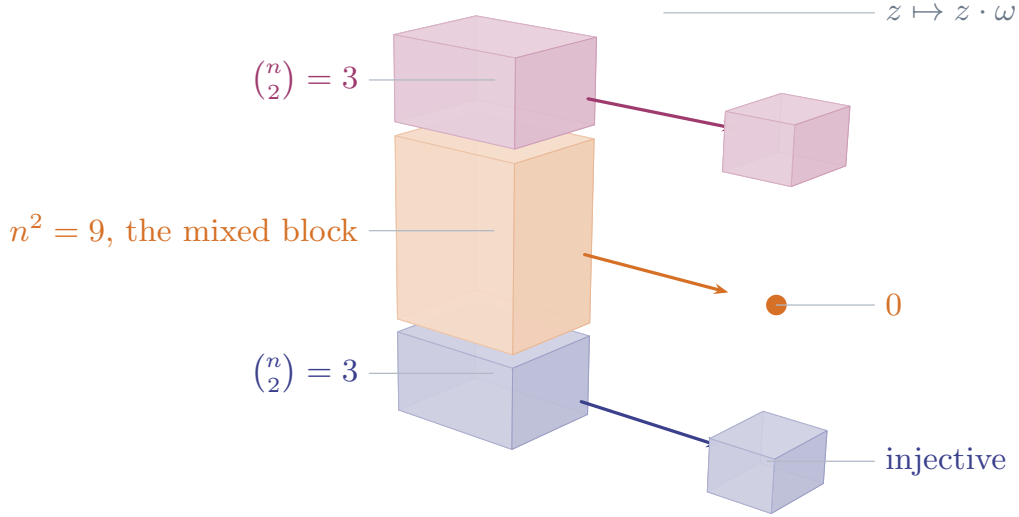


FIGURE 5. The structure behind Lemma 9.6, drawn at  $n = 3$ .  $H^{0,2}(A) = \wedge^2 H^{0,1}$  splits into three blocks by how many factors come from  $V_+^{0,1}$  and how many from  $V_-^{0,1}$ . The two outer blocks, of dimension  $\binom{n}{2} = 3$  each, are carried injectively by multiplication with the Weil class. The middle block, of dimension  $n^2 = 9$  and spanned by the mixed products, goes to zero, because each such product repeats a factor of  $\alpha_+$  and a factor of  $\alpha_-$ . That  $n^2$  is the deficit in Proposition 9.10.

**9.2. The annihilator is the tangent space to the family.** The number  $n^2$  in Lemma 9.6 is the dimension of the Weil family. That is not a coincidence, and this subsection proves it is not: the polarisation identifies the annihilator with the tangent space to the family, canonically and in every dimension. Two consequences follow. The first is that the deficiency of the semiregularity map at an object of rank zero is not a numerical accident to be repaired by a better choice of object; it is the family itself, seen inside  $H^{0,2}$ . The second is that the first-order Hodge locus of the Weil line is exactly the Weil family, so the directions the semiregularity map cannot see are precisely the directions along which the class stays Hodge and would have to be carried.

Write  $V_\pm$  for the two eigenspaces of  $K$  on  $H^1(A, \mathbb{C})$ , each of dimension  $2n$ , and

$$V_\pm^{1,0} = V_\pm \cap H^{1,0}, \quad V_\pm^{0,1} = V_\pm \cap H^{0,1},$$

each of dimension  $n$  by Lemma 3.5. Let  $E$  denote the polarisation form on  $H^1(A, \mathbb{Q})$ , extended to  $H^1(A, \mathbb{C})$ . We use throughout that  $E$  vanishes on  $V_+ \times V_+$  and on  $V_- \times V_-$  and pairs  $V_+$  with  $V_-$  perfectly, which is condition (iii) of Definition 3.1 read on the eigenspaces, and that  $E$  vanishes on  $H^{1,0} \times H^{1,0}$  and on  $H^{0,1} \times H^{0,1}$ , which is the first Riemann bilinear relation.<sup>21</sup>

<sup>21</sup>The first of these is the computation in the proof of Proposition B.8: for  $x, y \in V_+$  one has  $E(\sqrt{-d}x, y) = \sqrt{-d}E(x, y)$  from the eigenvalue and  $E(\sqrt{-d}x, y) = -E(x, \sqrt{-d}y) = -\sqrt{-d}E(x, y)$  from Definition 3.1(iii), so  $E(x, y) = 0$ ; the same on  $V_-$ , and nondegeneracy of  $E$  then forces the pairing between  $V_+$  and  $V_-$  to be perfect. Together the two vanishing statements say that of the four blocks of  $E$  on  $(V_+^{1,0} \oplus V_+^{0,1}) \times (V_-^{1,0} \oplus V_-^{0,1})$  only two are nonzero, namely  $V_+^{1,0} \times V_-^{0,1}$  and  $V_+^{0,1} \times V_-^{1,0}$ , and each of those two is a perfect pairing of  $n$ -dimensional spaces. That is the only input to Theorem 9.17.

A first-order deformation of  $A$  is a class in  $H^1(A, T_A)$ , which for an abelian variety is canonically  $\text{Hom}(H^{1,0}, H^{0,1})$ , since the tangent bundle is trivial and  $H^{1,0}$  is the dual of its space of sections. Such a  $v$  is extended to  $H^1(A, \mathbb{C})$  by zero on  $H^{0,1}$  and acts on  $H^k(A, \mathbb{C}) = \bigwedge^k H^1(A, \mathbb{C})$  as the derivation it generates; we write  $v \lrcorner \xi$  for that action. A Hodge class  $\xi$  of type  $(p, p)$  remains of type  $(p, p)$  to first order along  $v$  exactly when  $v \lrcorner \xi = 0$ . We call  $v$  *polarised* when  $E(vx, y) + E(x, vy) = 0$  for all  $x, y \in H^{1,0}$ , which is the condition that the deformation preserve the polarisation, and the polarised  $v$  form a space of dimension  $n(2n + 1) = \dim \mathcal{A}_{2n}$ .

**Proposition 9.14** (The first-order Hodge locus of the Weil line). *Let  $A$  be of  $(K, -1, n)$ -Weil type and let  $v$  be a polarised first-order deformation of  $A$ . The following are equivalent.*

- (i)  $v \lrcorner \omega = 0$  for every  $\omega$  in the Weil line.
- (ii)  $v$  commutes with the action of  $K$ .
- (iii)  $v(V_+^{1,0}) \subseteq V_+^{0,1}$  and  $v(V_-^{1,0}) \subseteq V_-^{0,1}$ .

The space of such  $v$  has dimension  $n^2$ , and under [Proposition B.8](#) it is the tangent space at  $[A]$  to the Weil family  $\mathcal{D}_{n,n}$ . Already one of the two rational generators of the Weil line cuts it out.

*Proof.* The equivalence of (ii) and (iii) is immediate:  $K$  acts on  $V_{\pm}$  by the two scalars  $\pm\sqrt{-d}$ , so commuting with  $K$  is preserving  $V_+$  and  $V_-$ , and the extension of  $v$  kills  $H^{0,1}$ , so the content is where it sends  $V_{\pm}^{1,0}$ .

For (i) and (iii), let  $\alpha_+$  generate  $\bigwedge^{2n} V_+$  and write  $\alpha_+ = e_1 \wedge \cdots \wedge e_{2n}$  for a basis  $e_1, \dots, e_{2n}$  of  $V_+$  adapted to  $V_+ = V_+^{1,0} \oplus V_+^{0,1}$ . As a derivation,

$$v \lrcorner \alpha_+ = \sum_{j=1}^{2n} e_1 \wedge \cdots \wedge v(e_j) \wedge \cdots \wedge e_{2n}.$$

Split  $v(e_j)$  along  $H^1(A, \mathbb{C}) = V_+ \oplus V_-$ . The  $V_+$  components contribute  $\text{tr}(\pi_+ \circ v|_{V_+}) \alpha_+$ , and that trace vanishes: the map  $\pi_+ \circ v|_{V_+}$  kills  $V_+^{0,1}$  and carries  $V_+^{1,0}$  into  $V_+^{0,1}$ , so in the adapted basis it is strictly block off-diagonal and has zero diagonal. Writing  $v(e_j)_- = \sum_k c_{jk} f_k$  for a basis  $f_1, \dots, f_{2n}$  of  $V_-$ , the  $V_-$  components contribute

$$\sum_{j,k} \pm c_{jk} e_1 \wedge \cdots \wedge \widehat{e_j} \wedge \cdots \wedge e_{2n} \wedge f_k,$$

and the  $4n^2$  monomials appearing here are distinct members of the standard basis of  $\bigwedge^{2n} H^1(A, \mathbb{C})$ , hence linearly independent. So  $v \lrcorner \alpha_+ = 0$  if and only if every  $c_{jk}$  vanishes, which is  $v(V_+) \subseteq V_+$ . The same argument on  $\alpha_-$  gives  $v \lrcorner \alpha_- = 0$  if and only if  $v(V_-) \subseteq V_-$ . The Weil line spans  $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$  over  $\mathbb{C}$  by [Proposition 3.7](#), and each of the two rational generators of [Theorem 4.9](#) has both components nonzero, so a single generator already forces both conditions. This is (i) if and only if (iii).

For the dimension, let  $v$  satisfy (iii) and write  $v_+$  and  $v_-$  for its two blocks. If  $x, y$  both lie in  $V_+^{1,0}$  then  $v_+x$  and  $y$  both lie in  $V_+$ , so both terms of the polarised condition vanish, and likewise if both lie in  $V_-^{1,0}$ . For  $x \in V_+^{1,0}$  and  $y \in V_-^{1,0}$  the condition reads  $E(v_+x, y) = -E(x, v_-y)$ . The left side is determined by  $v_+$ , and the pairing between  $V_+^{1,0}$  and  $V_-^{0,1}$  is perfect, so this determines  $v_-y$  for each  $y$  and hence determines  $v_-$  from  $v_+$ . So the polarised deformations satisfying (iii) are a space of dimension  $\dim \text{Hom}(V_+^{1,0}, V_+^{0,1}) = n^2$ . By [Proposition B.8](#) the family  $\mathcal{D}_{n,n}$  is an open subset of the Grassmannian of  $n$ -planes in  $V_+$ , whose tangent space at  $V_+^{1,0}$  is  $\text{Hom}(V_+^{1,0}, V_+/V_+^{1,0}) = \text{Hom}(V_+^{1,0}, V_+^{0,1})$ , and the identification just made is that isomorphism.  $\blacksquare$

**Proposition 9.15** (Infinitesimal rigidity of divisor classes). *Keep the notation above and write a class  $\delta \in H^{1,1}$  in the four blocks*

$$a \text{ on } P \otimes \bar{P}, \quad b \text{ on } P' \otimes \bar{P}', \quad c \text{ on } P \otimes \bar{P}', \quad e \text{ on } P' \otimes \bar{P},$$

*the first two carrying the norm character of  $K^\times$  and the last two the characters  $\tau^2$  and  $\bar{\tau}^2$ ; reality makes  $a$  and  $b$  anti-hermitian and ties  $e$  to  $-\bar{c}^T$ . Identify the tangent space to  $\mathcal{D}_{n,n}$  with the  $n \times n$  matrices  $V$  as in [Theorem 9.17](#). Then contraction with a tangent vector is*

$$\Phi_\delta(V) = (Va + b^T V, \text{alt}(Vc), \text{alt}(-V^T e)), \quad (25)$$

*with values in  $\bar{P}' \otimes \bar{P} \oplus \wedge^2 \bar{P}' \oplus \wedge^2 \bar{P} = H^{0,2}$ , and  $\ker \Phi_\delta$  is the tangent space to the locus where  $\delta$  stays of type  $(1,1)$ . Consequently:*

- (i)  $\Phi_\delta = 0$  if and only if  $\delta$  is a real multiple of  $\eta$ . The rigid classes form a real line, which is the infinitesimal form of the statement that  $\text{NS}(A)_\mathbb{Q}$  has rank one at the general member.
- (ii) For the norm-character class with block  $a$  and  $b = c = e = 0$ ,

$$\dim \ker \Phi_\delta = n(n - \text{rank } a),$$

*so such a class cuts a locus of codimension  $n \text{rank } a$ , at least  $n$ .*

- (iii) A general class has  $\ker \Phi_\delta = 0$ : it rigidifies the family to a point.

*Proof.* For (25), a polarised deformation commuting with  $K$  sends  $P$  into  $\bar{P}'$  and  $P'$  into  $\bar{P}$ , and the symmetry  $E(vx, y) = E(x, vy)$  forces the second matrix to be  $-V^T$ , which is [Theorem 9.17](#). Applying  $v$  to the  $(1, 0)$  slot of each block and collecting terms in  $H^{0,2}$  gives the three displayed components; the sign in the first comes from  $\bar{P} \wedge \bar{P}' = -\bar{P}' \wedge \bar{P}$ .

(i) If  $\Phi_\delta = 0$  then  $\text{alt}(Vc) = 0$  for every  $V$ , which for  $n \geq 2$  forces  $c = 0$ , and likewise  $e = 0$ . The first component gives  $Va = -b^T V$  for every  $V$ ; taking  $V$  to run over the matrix units shows  $a$  is a scalar  $\lambda$  and  $b^T = -\lambda$ . Anti-hermitian scalars are the purely imaginary ones, so the solutions form a real line. It contains  $\eta$ , whose blocks are  $a = i$ ,  $b = -i$ , and the two therefore agree.

(ii) With  $b = c = e = 0$  the condition is  $Va = 0$ , that is, the rows of  $V$  annihilate the column space of  $a$ ; the solution space has dimension  $n(n - \text{rank } a)$ .

(iii) Immediate from (i) and (ii), and checked on samples.

All three, together with the value  $n^2$  for the dimension of the tangent space, are verified over  $\mathbb{Q}(i)$  in exact arithmetic for  $n = 2, 3, 4$  in item (XIX) of [Appendix D](#); at  $n = 4$  the kernels in (ii) are 12, 8, 4, 0 for ranks 1 to 4. ■

*Remark 9.16* (Why the count matters here). [Proposition 9.15\(ii\)](#) is the reason the divisorial supply of [Theorem 15.16](#) cannot be enlarged by a better choice of class. Any norm-character class other than a multiple of  $\eta$  has  $\text{rank } a \geq 1$  and so cuts codimension at least  $n$ ; the locus it defines is proper, and by (i) no class except the polarisation avoids cutting at all. So each new class buys a subvariety, never an open set, which is [Proposition 15.18](#) again with a dimension attached.

**Theorem 9.17** (The annihilator is the tangent space). *Let  $A$  be of  $(K, -1, n)$ -Weil type and let  $\omega$  be a rational generator of the Weil line. The polarisation induces an isomorphism*

$$T_{[A]} \mathcal{D}_{n,n} \xrightarrow{\sim} \text{Ann}_{H^{0,2}(A)}(\omega),$$

---

<sup>22</sup>The cone is drawn with its rulings so that it reads as the ruled surface it is rather than as a shaded volume. The two planes meet only at the origin, so the fibres are disjoint away from it and the strip they sweep is a cone; that is why the block is a tensor product and not a direct sum.

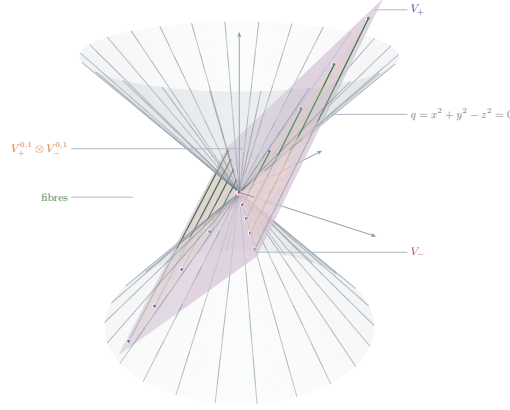


FIGURE 6. The geometry behind Lemma 9.6, drawn for the real shadow of the hermitian form. The ruled surface is the null cone of  $H$ ; the two transverse planes are the conjugate isotropic subspaces  $V_+$  and  $V_-$ , which conjugation interchanges, so neither is defined over  $\mathbb{Q}$  and only their sum is. The segments are the fibres joining a vector of one to its conjugate in the other, and the strip they sweep is the mixed block  $V_+^{0,1} \otimes V_-^{0,1}$ , of dimension  $n^2$ , which multiplication by the Weil class sends to zero and which is the deficit in Proposition 9.10. <sup>22</sup>

canonical in the sense that it depends on no choice of basis and on no choice of  $\omega$  on the line. Explicitly, if  $p_1, \dots, p_n$  is a basis of  $V_+^{1,0}$  and  $q_1, \dots, q_n$  the basis of  $V_-^{0,1}$  dual to it under  $E$ , then  $\varphi \in \text{Hom}(V_+^{1,0}, V_+^{0,1})$  corresponds to

$$\sum_{a=1}^n q_a \wedge \varphi(p_a) \in V_-^{0,1} \otimes V_+^{0,1} \subset \wedge^2 H^{0,1} = H^{0,2}(A).$$

*Proof.* The form  $E$  pairs  $V_+$  with  $V_-$  perfectly and vanishes on  $V_+^{1,0} \times V_-^{1,0}$ , the latter because both spaces lie in  $H^{1,0}$ . Hence  $E$  restricted to  $V_+^{1,0} \times V_-$  kills  $V_-^{1,0}$  and descends to a pairing

$$V_+^{1,0} \times V_-^{0,1} \longrightarrow \mathbb{C},$$

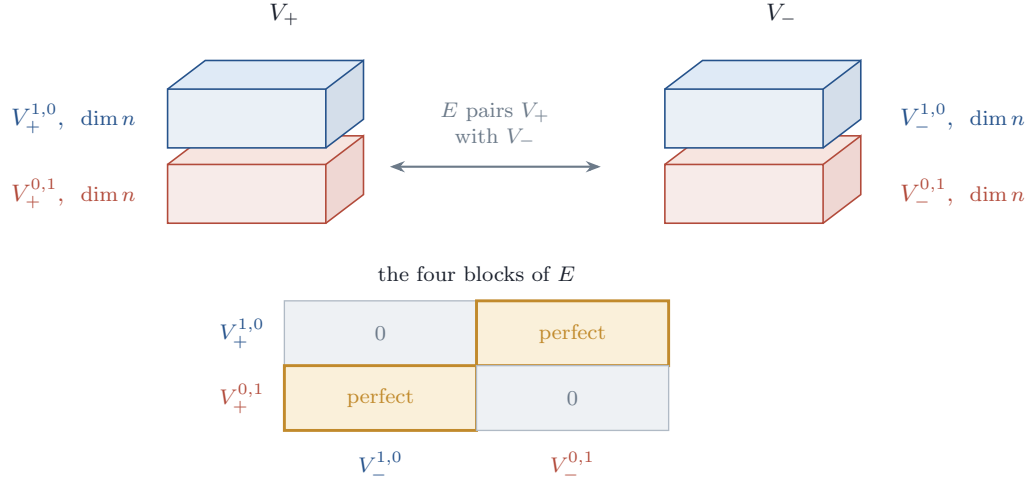
which is perfect, both spaces having dimension  $n$  and the total pairing being nondegenerate. This gives a canonical isomorphism  $(V_+^{1,0})^* \cong V_-^{0,1}$ , and therefore

$$T_{[A]} \mathcal{D}_{n,n} = \text{Hom}(V_+^{1,0}, V_+^{0,1}) = (V_+^{1,0})^* \otimes V_+^{0,1} \cong V_-^{0,1} \otimes V_+^{0,1},$$

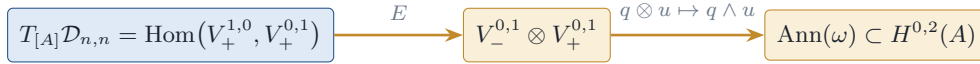
the first equality being the tangent space computation at the end of the proof of Proposition 9.14. Since  $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$ , the map  $q \otimes u \mapsto q \wedge u$  identifies  $V_-^{0,1} \otimes V_+^{0,1}$  with the middle summand of  $\wedge^2 H^{0,1}$ , and that summand is the annihilator by Lemma 9.6. The displayed formula is the composite of these isomorphisms written in a dual pair of bases, so it is independent of the basis chosen; and Lemma 9.6 gives the same subspace for every  $\omega$  on the line, so the isomorphism does not depend on  $\omega$  either.  $\blacksquare$

**Corollary 9.18** (The kernel contains the tangent space to the family). *Let  $A$  be of  $(K, -1, n)$ -Weil type and let  $E$  be a nonzero perfect complex on  $A$  satisfying the hypotheses of Proposition 9.10, that is, with Chern character concentrated on the Weil line and with*

$K$  splits  $H^1(A, \mathbb{C})$  into two halves, and each half splits by Hodge type.



The two zeros are the first Riemann relation,  $E = 0$  on  $H^{1,0} \times H^{1,0}$  and on  $H^{0,1} \times H^{0,1}$ . The upper right block is the one that moves the tangent space.



Both spaces have dimension  $n^2$ . The composite depends on no basis and on no choice of  $\omega$  on the line.

---

$n = 2$ :  $\dim T = \dim \text{Ann} = 4$      $n = 3$ : 9     $n = 4$ : 16  
 checked exactly over  $\mathbb{Q}(i)$ , item (XVI) of the verification

FIGURE 7. **Theorem 9.17.** The polarisation has only two nonzero blocks on  $V_+ \times V_-$ : it pairs  $V_+^{1,0}$  with  $V_-^{0,1}$  and  $V_+^{0,1}$  with  $V_-^{1,0}$ , each perfectly, and the other two blocks vanish by the first Riemann relation. The upper right block turns the tangent space  $\text{Hom}(V_+^{1,0}, V_+^{0,1})$  to the Weil family into  $V_-^{0,1} \otimes V_+^{0,1}$ , which is the middle block of  $\bigwedge^2 H^{0,1}$  drawn in Figure 5 and the annihilator of the Weil line. The two  $n^2$  dimensional spaces are therefore the same subspace of  $H^{0,2}$ , so the directions the semiregularity map cannot see are the directions of the family along which the class stays Hodge.

(24) *injective. Then  $\ker \sigma_E$  contains a subspace canonically isomorphic to  $T_{[A]}\mathcal{D}_{n,n}$ , namely the image of the composite*

$$T_{[A]}\mathcal{D}_{n,n} \cong \operatorname{Ann}_{H^{0,2}}(\omega) \hookrightarrow H^2(\mathcal{O}_A) \xrightarrow{z \mapsto z \cdot \operatorname{id}_E} \operatorname{Ext}^2(E, E).$$

*By Proposition 9.14 that subspace is also the tangent space to the first-order Hodge locus of the class  $\operatorname{ch}_n(E)$ .*

*Proof.* Immediate from Theorem 9.17, Proposition 9.10 and Proposition 9.14. ■

**Theorem 9.19** (The weakened criterion is unavailable for concentrated Chern characters). *Let  $A$  be of  $(K, -1, n)$ -Weil type and let  $E$  be a nonzero perfect complex on  $A$  satisfying the hypotheses of Proposition 9.10. Let*

$$\operatorname{ev}_E: HH^2(A) \longrightarrow \operatorname{Ext}^2(E, E)$$

*be the evaluation on  $E$  of the natural transformations  $\operatorname{id} \rightarrow \operatorname{id}[2]$  of  $D^b(A)$ , as in [Mar25b, Section 11.4]. Then*

$$\operatorname{ev}_E(H^2(\mathcal{O}_A)) \cap \ker \sigma_E = \Delta := \{ z \cdot \operatorname{id}_E : z \in \operatorname{Ann}_{H^{0,2}}(\omega) \},$$

*a subspace of dimension  $n^2$ . In particular  $\sigma_E$  is not injective on the image of  $\operatorname{ev}_E$ , so  $E$  satisfies neither the hypothesis of [Mar25b, Lemma 11.3] nor the weakened hypothesis of [Mar25b, Question 11.4].*

*Proof.* The Hochschild cohomology  $HH^*(A) = \operatorname{Ext}_{A \times A}^*(\mathcal{O}_\Delta, \mathcal{O}_\Delta)$  is the algebra of natural transformations of the identity functor, and the Hochschild–Kostant–Rosenberg isomorphism [Cal05, BF08] writes  $HH^2(A) \cong H^2(\mathcal{O}_A) \oplus H^1(A, T_A) \oplus H^0(A, \wedge^2 T_A)$ , with no correction from the Todd class because  $T_A$  is trivial. The first summand is the image of  $H^2(\mathcal{O}_A) = \operatorname{Ext}^2(\mathcal{O}_A, \mathcal{O}_A)$  under pullback to  $A \times A$ , and the natural transformation it defines sends a complex  $F$  to the composite  $F = F \otimes \mathcal{O}_A \xrightarrow{\operatorname{id} \otimes z} F \otimes \mathcal{O}_A[2] = F[2]$ , that is to  $z \cdot \operatorname{id}_F$ ; this is the statement that the action of  $HH^*(A)$  on  $\operatorname{Ext}^*(F, F)$  restricts on  $H^*(\mathcal{O}_A)$  to the  $\mathcal{O}_A$ -module structure.<sup>23</sup> Hence  $\operatorname{ev}_E(z) = z \cdot \operatorname{id}_E$  for  $z \in H^2(\mathcal{O}_A)$ , and  $\sigma_E(z \cdot \operatorname{id}_E) = z \cdot \operatorname{ch}(E) = N z \cdot \omega$  by the proof of Proposition 9.10, which vanishes exactly for  $z \in \operatorname{Ann}_{H^{0,2}}(\omega)$ . So  $\operatorname{ev}_E(H^2(\mathcal{O}_A)) \cap \ker \sigma_E = \Delta$ , and  $\dim \Delta = \dim \operatorname{Ann}_{H^{0,2}}(\omega) = n^2$  by Lemma 9.6 and the injectivity of (24). Since  $\Delta \neq 0$ , the restriction of  $\sigma_E$  to the image of  $\operatorname{ev}_E$  has a nonzero kernel. The hypothesis of [Mar25b, Lemma 11.3] is that  $\ker \operatorname{ev}_E$  equals the kernel of contraction with  $\operatorname{ch}(E)$  and that  $\operatorname{ev}_E$  is surjective; the second condition would make  $\sigma_E$  injective on all of  $\operatorname{Ext}^2(E, E)$ , contradicting Proposition 9.10, and the weakened hypothesis of [Mar25b, Question 11.4] is injectivity of  $\sigma_E$  on the image, which has just been shown to fail. ■

**Theorem 9.20** (A line bundle summand off the polarisation line obstructs). *Let  $A$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$  and let  $T = T_{[A]}\mathcal{D}_{n,n}$ . Let  $E$  be a perfect complex on  $A$  with a direct summand  $L[p]$ ,  $L$  a line bundle, and suppose that  $\operatorname{ch}(E)$  remains of Hodge*

<sup>23</sup>Concretely: a class  $z \in H^2(\mathcal{O}_A)$  is represented by a 2-extension of  $\mathcal{O}_A$  by itself, its pullback  $\pi_1^* z$  to  $A \times A$  is a 2-extension of  $\mathcal{O}_{A \times A}$  by itself, and tensoring with  $\mathcal{O}_\Delta$  gives a 2-extension of  $\mathcal{O}_\Delta$  by itself, which is the element of  $HH^2(A)$  in question. The Fourier–Mukai transform with kernel  $\mathcal{O}_\Delta$  is the identity, and the transform with kernel the extension is the extension of the identity by itself twisted by  $z$ , so its value on  $F$  is  $z \cdot \operatorname{id}_F$ . Markman uses exactly this identification when he writes the left vertical arrow of the diagram in the proof of his Lemma 11.3 as HKR followed by contraction with  $\operatorname{ch}(E)$ : on the summand  $H^2(\mathcal{O}_A)$  contraction is the cup product  $z \mapsto z \cdot \operatorname{ch}(E)$ , which is  $\sigma_E(z \cdot \operatorname{id}_E)$  by the computation in the proof of Theorem 9.7.

type to first order along  $\mathcal{D}_{n,n}$ , that is  $v \lrcorner \text{ch}(E) = 0$  for every  $v \in T$ . If  $c_1(L)$  is not a real multiple of  $\eta$ , then there is  $v \in T$  with

$$\text{ev}_E(v) \neq 0 \quad \text{and} \quad \sigma_E(\text{ev}_E(v)) = 0.$$

Consequently  $E$  is not semiregular, does not satisfy the weakened hypothesis of [Mar25b, Question 11.4], and is obstructed to first order along  $v$ . The same holds for a twisted complex, with  $c_1(L)$  replaced by the twisted first Chern class of the summand, which is  $c_1(L)$  shifted by the class common to all summands.

*Proof.* The class  $c_1(L)$  is a real class of type  $(1,1)$  and not a real multiple of  $\eta$ , so by Proposition 9.15(i) the contraction  $\Phi_{c_1(L)}: T \rightarrow H^{0,2}(A)$  is not zero: there is  $v \in T$  with  $v \lrcorner c_1(L) \neq 0$ . On the summand  $H^1(A, T_A)$  of  $HH^2(A)$  the evaluation map is contraction with the Atiyah class,  $\text{ev}_E(v) = v \lrcorner \text{At}(E)$ ; this is the top row of the diagram of Buchweitz and Flenner reproduced as [Mar25b, (2.1)], and its identification with the Hochschild evaluation is [BF08], used by Markman in the proof of his Lemma 11.3. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of  $L[p]$  is  $c_1(L)$ , so the component of  $\text{ev}_E(v)$  in the summand  $\text{Ext}^2(L[p], L[p]) = H^{0,2}(A)$  of  $\text{Ext}^2(E, E)$  is  $v \lrcorner c_1(L) \neq 0$ . On the other hand the same diagram gives  $\sigma_E(\text{ev}_E(v)) = v \lrcorner \text{ch}(E)$ , which vanishes by hypothesis. So  $\text{ev}_E(v)$  is a nonzero element of  $\ker \sigma_E \cap \text{im } \text{ev}_E$ ; this is the failure of both hypotheses, and  $v \lrcorner \text{At}(E) \neq 0$  is the statement that  $E$  does not lift to a first-order deformation of the pair along  $v$ , as recalled in [Mar25b, Section 2]. Twisting by a class  $B$  replaces the Atiyah class of every rank one summand by its shift by  $B$  and  $\text{ch}(E)$  by the twisted Chern character, and nothing else in the argument changes. ■

*Remark 9.21* (What this measures, and what it does not settle). The semiregularity criterion converts a statement about the Chern character into a statement about the object, and it does so by requiring  $\sigma_E$  to be injective. Corollary 9.18 says that for an object whose Chern character is concentrated on the Weil line the requirement fails, and that the failure is measured by the tangent space to the very family across which the class is to be transported, in every dimension and with no hypothesis beyond (24). That is a statement about the size and the position of the kernel; by itself it is not a statement that the obstruction classes are nonzero, and it does not show that the Weil class fails to deform. For the explicit object of Proposition 18.14 the obstruction classes are computed in Proposition 18.16: they are nonzero in every direction of the family normal to the split locus, they lie in  $\ker \sigma_E$ , and they meet the subspace of Corollary 9.18 only in zero. The Weil class deforms along the family, by Proposition 9.14; that object does not.

Nor does it decide the weaker criterion in the form that matters. Markman asks, in [Mar25b, Question 11.4], whether the semiregularity theorem survives when  $\sigma_E$  is required to be injective only on the image of the evaluation map  $\text{ev}_E$  rather than on all of  $\text{Ext}^2(E, E)$ , and records in his Section 12 that an affirmative answer would carry the method to higher dimensions and to fields of larger degree. Theorem 9.19 says that the classes of Corollary 9.18 lie in that image, so for concentrated Chern characters the weakened hypothesis fails as well, in every dimension, by the same  $n^2$  dimensions; and Theorem 9.20 says that it fails for every object with a line bundle summand off the polarisation line, whatever the rank, at every member of the family, which by Corollary 18.22 removes every sum of line bundles from consideration at once. For the explicit split object of Proposition 18.14 the failure is complete: Proposition 18.18 shows that every one of the fifteen dimensions of  $\ker \sigma_E$  is the value of  $\text{ev}_E$  on a Hochschild class that already preserves  $\text{ch}(E)$  to first order. What is not decided, and what the question is really about, is whether an object of positive rank can satisfy the weakened hypothesis without satisfying the original one. Markman's own

sheaves are of positive rank and are not of the concentrated shape, which is [Remark 9.9](#); he avoids the question for them by descending to a quotient on which  $\sigma_E$  is injective on an invariant subspace containing the image of  $\text{ev}_E$  [[Mar25b](#), Proposition 11.5], and he proposes in his Section 12 a rank one complex on a principally polarised abelian fourfold, built from  $(d+9)/2$  translates of the theta divisor, as a candidate for the weakened criterion in dimension eight, with the verification still to be done. [Theorem 18.34](#) carries out that verification and finds that the candidate fails, for every discriminant, at every isolated double point of its support.

Voisin's account of the present state of the subject [[Voi26](#)] makes the same point qualitatively and for the variational Hodge conjecture at large, that semiregular objects are hard to construct and too restricted to carry a general argument. [Corollary 9.18](#) is that observation in quantitative form for one family of classes: the restriction is exactly  $n^2$  dimensions, and those dimensions are the family. Totaro's account [[Tot26](#)] is more optimistic about pushing Markman's method further, and records that it needed the semiregularity theorem for twisted sheaves, which Pridham supplied [[Pri24](#)].

## 10. THE HYPERPLANE OBSTRUCTION

We turn to the second attack, induction on the dimension of the variety through hyperplane sections.<sup>24</sup> The ingredients used in such arguments are all theorems: the Lefschetz hyperplane theorem, hard Lefschetz, the algebraicity of the Hodge locus [[CDK95](#)], and the properness of relative Hilbert schemes. The question is whether they compose into an induction, and this section shows that they do not, for a reason that is elementary and that applies to every argument of the shape, independently of how the family is chosen.

Throughout,  $X$  is a smooth complex projective variety of dimension  $d$ ,  $j: Y \hookrightarrow X$  is a smooth hypersurface in the linear system of an ample divisor, and  $L = c_1(\mathcal{O}_X(Y)) \in H^2(X, \mathbb{Q})$ .

### 10.1. Two classical inputs.

**Theorem 10.1** (Lefschetz hyperplane theorem). *The restriction  $j^*: H^k(X, \mathbb{Q}) \rightarrow H^k(Y, \mathbb{Q})$  is an isomorphism for  $k \leq d-2$  and is injective for  $k = d-1$ .*

**Lemma 10.2** (Gysin and restriction). *For every  $k$  and every  $x \in H^k(X, \mathbb{Q})$ ,*

$$j_* j^* x = x \cup L.$$

*Moreover  $j_*: H^k(Y, \mathbb{Q}) \rightarrow H^{k+2}(X, \mathbb{Q})$  is injective for  $k \geq d$ .*

*Proof.* The first identity is the projection formula together with  $j_* 1 = [Y] = L$ . For the second, Poincaré duality on  $X$  and on  $Y$  identifies  $j_*: H^k(Y) \rightarrow H^{k+2}(X)$  with the dual of

$$j^*: H^{2d-k-2}(X) \longrightarrow H^{2(d-1)-k}(Y) = H^{2d-k-2}(Y),$$

so  $j_*$  is injective on  $H^k(Y)$  exactly when this  $j^*$  is surjective. By [Theorem 10.1](#) that  $j^*$  is surjective whenever  $2d-k-2 \leq d-2$ , that is whenever  $k \geq d$ . ■

<sup>24</sup>Lefschetz's theorem on hyperplane sections is quoted here in its topological form. The proof by Morse theory, due to Andreotti and Frankel, gives it for any smooth affine variety and is the shortest route; the statement we use is the consequence that  $H^k(X, \mathbb{Q}) \rightarrow H^k(Y, \mathbb{Q})$  is an isomorphism for  $k < \dim Y$  and injective for  $k = \dim Y$ .

### 10.2. The middle degree.

*Proof of Theorem 1.4.* Let  $d = 2p$  and let  $\alpha \in H^{2p}(X, \mathbb{Q})$  be primitive, which in the middle degree means  $L\alpha = 0$ , since  $H_{\text{prim}}^d(X, \mathbb{Q}) = \ker(L: H^d(X, \mathbb{Q}) \rightarrow H^{d+2}(X, \mathbb{Q}))$ . By Lemma 10.2,

$$j_*(j^*\alpha) = \alpha \cup L = 0.$$

The class  $j^*\alpha$  lies in  $H^d(Y, \mathbb{Q})$  and  $d \geq d$ , so Lemma 10.2 says that  $j_*$  is injective there. Hence  $j^*\alpha = 0$ .  $\blacksquare$

**Corollary 10.3** (Blindness in the middle degree). *Let  $d = 2p$  and let  $\alpha, \beta \in H^{2p}(X, \mathbb{Q})$  be primitive Hodge classes. Then  $j^*\alpha = j^*\beta = 0$  for every smooth ample hypersurface  $Y$ . In particular no property of the restricted classes distinguishes  $\alpha$  from  $\beta$ , and no statement about Hodge classes on hypersurfaces of  $X$ , however strong, carries any information about the primitive middle cohomology of  $X$ .*

*Remark 10.4* (The size of what is lost). The primitive middle cohomology is not a small correction term. For a smooth hypersurface of degree  $m$  in  $\mathbb{P}^{d+1}$  its dimension grows like  $m^{d+1}$ , and for a general such hypersurface with  $d = 2p$  the Hodge classes in it are exactly what the Hodge conjecture is about. Corollary 10.3 says that the whole of it is invisible to hyperplane restriction.

**10.3. Below the middle degree.** Below the middle degree the restriction map is injective, indeed an isomorphism, so one can transport the class downwards. The difficulty moves to the return trip.

**Definition 10.5** (Lefschetz standard conjecture). For a smooth projective variety  $X$  of dimension  $d$  with ample class  $L$ , hard Lefschetz gives isomorphisms  $L^{d-k}: H^k(X, \mathbb{Q}) \rightarrow H^{2d-k}(X, \mathbb{Q})$ , and makes  $H^*(X, \mathbb{Q})$  a representation of  $\mathfrak{sl}_2$  in which  $L$  raises the degree by 2 and  $H$  acts on  $H^k$  by  $k - d$ . Let  $\Lambda$  denote the unique endomorphism of degree  $-2$  with  $[L, \Lambda] = H$ . The Lefschetz standard conjecture  $B(X)$  asserts that  $\Lambda$  is induced by an algebraic cycle on  $X \times X$ ; this is equivalent to Kleiman's formulations [Kle68, Kle94].

**Lemma 10.6** ( $\Lambda$  undoes  $L$  on primitive classes). *Let  $\alpha \in H^k(X, \mathbb{Q})$  be primitive with  $k \leq d$ . Then  $\Lambda L\alpha = (d - k)\alpha$ .*

*Proof.* This is the  $\mathfrak{sl}_2$  relation  $[L, \Lambda] = H$  of Hodge theory, where  $H$  acts on  $H^k$  by the scalar  $k - d$ , together with  $\Lambda\alpha = 0$  for  $\alpha$  primitive. Explicitly,  $\Lambda L\alpha = L\Lambda\alpha - [L, \Lambda]\alpha = 0 - (k - d)\alpha = (d - k)\alpha$ .  $\blacksquare$

*Proof of Theorem 1.5.* Let  $2p \leq d - 2$ , so  $j^*: H^{2p}(X, \mathbb{Q}) \rightarrow H^{2p}(Y, \mathbb{Q})$  is an isomorphism by Theorem 10.1. If  $j^*\alpha = \text{cl}(W)$  for some  $W \in \text{CH}^p(Y)_{\mathbb{Q}}$ , then  $j_*W \in \text{CH}^{p+1}(X)_{\mathbb{Q}}$  and

$$\text{cl}(j_*W) = j_*j^*\alpha = L\alpha$$

by Lemma 10.2, so  $L\alpha$  is algebraic.

For the second statement, suppose  $B(X)$  holds and  $\alpha$  is primitive of degree  $2p < d$ . Then  $\Lambda$  is induced by an algebraic correspondence, hence carries algebraic classes to algebraic classes, and  $\alpha = (d - 2p)^{-1}\Lambda(L\alpha)$  by Lemma 10.6, so  $\alpha$  is algebraic. Conversely, suppose that for every smooth projective  $X$  and every Hodge class  $\alpha$  the algebraicity of  $L\alpha$  implies that of  $\alpha$ . Applying this in every degree gives that  $\Lambda$  preserves the subspace of algebraic classes in every degree, which by the standard equivalences for the Lefschetz conjecture, as set out by Kleiman [Kle68, Kle94], is  $B(X)$ .  $\blacksquare$

*Proof of Corollary 1.6.* Suppose the argument uses only  $j^*$  and  $j_*$  to relate classes on  $X$  to classes on smooth ample hypersurfaces  $Y \subset X$ . If  $2p = d$  and  $\alpha$  is primitive, then  $j^*\alpha = 0$  by Theorem 1.4, and  $j_*$  applied to any cycle on  $Y$  produces a class in the image of  $j_*$ , which by Lemma 10.2 and  $L\alpha = 0$  cannot be  $\alpha$  unless  $\alpha = 0$ ; so no conclusion about  $\alpha$  is available. Otherwise the best that  $j_*j^*$  produces is  $L\alpha$ , and Theorem 1.5 identifies the passage from  $L\alpha$  to  $\alpha$  with  $B(X)$ . ■

**10.4. Diagnosis of a specific inductive scheme.** We record where an induction of this shape breaks in practice, because the break is usually not where one expects and is easy to miss.

*Remark 10.7* (The dimension bookkeeping). A common formulation asks the reader to place  $X$  in a Lefschetz pencil by choosing a generic pencil  $\{X_t\}_{t \in \mathbb{P}^1}$  of hyperplane sections of  $X$  and then to set  $X = X_{t_0}$  for some  $t_0$ . The two requirements are incompatible: the members of a pencil of hyperplane sections of  $X$  have dimension  $d - 1$ , so none of them is  $X$ . An induction that reads  $\dim X_t = d - 1 < d$  in its inductive step, and  $X_{t_0} = X$  when it transports the resulting cycle back, is using both readings of the same symbol.

Resolving the ambiguity in either direction is what Theorems 1.4 and 1.5 address. If one insists that  $X$  be a member of the pencil, then the pencil lives in an ambient variety of dimension  $d + 1$  and every member has dimension  $d$ ; the inductive parameter does not decrease and the induction is circular. If instead one keeps  $\dim X_t = d - 1$ , then the class  $\alpha$  lives on  $X$  and not on the members, the only available transports are  $j^*$  and  $j_*$ , and Corollary 1.6 applies.

*Remark 10.8* (Pointwise existence is not a family). A second gap appears in arguments that produce cycles fibrewise and then assemble them. Suppose an inductive hypothesis furnishes, for each  $t$  in the base  $S$  of a family, a cycle  $Z_t$  with a prescribed class. It is not legitimate to conclude that the Hilbert polynomial  $t \mapsto P_{Z_t}$  is constructible, or semicontinuous, or takes finitely many values. Those are theorems about a flat family, and the input here is a bare existence statement at each point with no coherence between points. Without constructibility there is no section of a relative Hilbert scheme over a dense open subset, and therefore nothing for the valuative criterion of properness to extend. The relative Hilbert scheme is indeed proper over  $S$ , and that is exactly why the step looks safe; properness extends a section that exists over a punctured curve, and supplies no section where none is given.

## 11. THE KUGA–SATAKE ROUTE AND WHERE IT STOPS

Theorem 1.4 closes induction on dimension through hyperplane sections. One classical construction does reach a middle degree Hodge class without any such induction, and we set it out in full, both because it shows what a successful route looks like and because its boundary is numerical and can be located exactly. The construction is Kuga and Satake’s [KS67], in the form given by Deligne [Del72]. This section builds it, proves that Theorem 1.4 has no bearing on it, and then shows that for abelian varieties of Weil type it is available precisely when  $n = 1$ , which is the one case in which the Weil line consists of divisor classes and there is nothing to prove.

The arithmetic here is checked in `kugasatake.py`, item (VIII) of Appendix D.<sup>25</sup>

<sup>25</sup>Kuga and Satake’s paper is four pages and constructs the abelian variety; the Hodge-theoretic properties that make it useful, in particular the embedding of Proposition 11.7, are Deligne’s, and

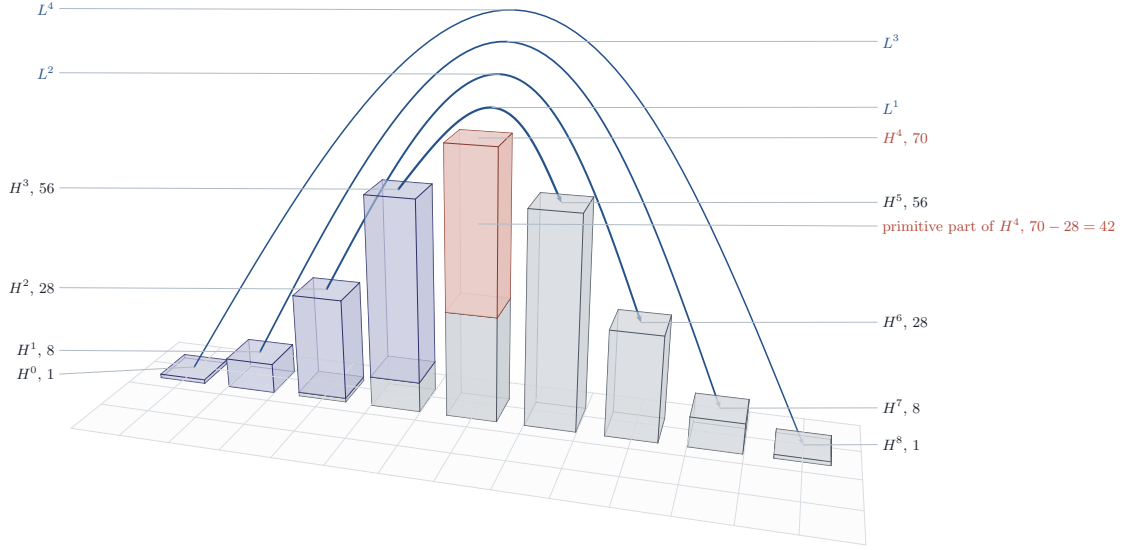


FIGURE 8. The Betti numbers  $\binom{8}{k}$  of an abelian fourfold, drawn as prisms, each split into the image of the Lefschetz operator, in slate, and the primitive part above it, with the hard Lefschetz isomorphisms  $L^{4-k}: H^k \rightarrow H^{8-k}$  drawn as arcs at increasing depth. Below the middle degree the map  $j^*$  to a smooth ample hypersurface is injective and the class descends, but the return trip  $j_*j^*$  multiplies by  $L$ , so recovering the class is the Lefschetz standard conjecture. Above the middle one is reduced to the lower half by an arc. The clay block is the primitive middle cohomology, of dimension  $70 - 28 = 42$ ; it is where the Hodge conjecture lives, it is annihilated outright by [Theorem 1.4](#), and it is the one place no arc and no restriction reaches.

### 11.1. The construction.

**Setup 11.1** (K3 type). Let  $T$  be a  $\mathbb{Q}$ -vector space of dimension  $b$  with a nondegenerate symmetric form  $q$ , and let  $T$  carry a Hodge structure of weight two,  $T_{\mathbb{C}} = T^{2,0} \oplus T^{1,1} \oplus T^{0,2}$ , polarised by  $q$ , with

$$\dim_{\mathbb{C}} T^{2,0} = 1.$$

Such a structure is said to be of *K3 type*. Write  $T^{2,0} = \mathbb{C}\sigma$ ; the Hodge–Riemann relations give  $q(\sigma, \sigma) = 0$  and  $q(\sigma, \bar{\sigma}) > 0$ , so the real plane  $P = \langle \Re\sigma, \Im\sigma \rangle \subset T_{\mathbb{R}}$  is positive definite and  $q$  has signature  $(2, b-2)$  on  $T_{\mathbb{R}}$ . Normalise  $\sigma$  so that  $e_1 = \Re\sigma$  and  $e_2 = \Im\sigma$  are orthonormal.

**Definition 11.2** (Clifford algebra).  $C(T, q)$  is the quotient of the tensor algebra on  $T$  by the ideal generated by  $v \otimes v - q(v)$  for  $v \in T$ . It is  $\mathbb{Z}/2$ -graded,  $C = C^+ \oplus C^-$ , by the parity of the word length.

**Lemma 11.3.** *For an orthogonal basis  $f_1, \dots, f_b$  of  $T$  the products  $f_S$  over subsets  $S \subseteq \{1, \dots, b\}$ , taken in increasing order, form a basis of  $C(T, q)$ . Hence  $\dim_{\mathbb{Q}} C = 2^b$  and  $\dim_{\mathbb{Q}} C^{\pm} = 2^{b-1}$ , and  $C^- \cdot C^+ \cdot C^- \subseteq C^+$ .*

Rapoport’s complement to that paper supplies the comparison with the Shimura variety. Satake had already set up the Clifford algebra families in [\[Sat66\]](#), which is where the Riemann form comes from.

*Proof.* The relations  $f_i^2 = q(f_i)$  and  $f_i f_j = -f_j f_i$  for  $i \neq j$ , the latter obtained by expanding  $(f_i + f_j)^2 = q(f_i) + q(f_j)$ , let one rewrite any word in the  $f_i$  as a scalar times an increasing product, so the  $f_S$  span; they are independent by the standard filtration argument comparing  $C$  with  $\bigwedge^* T$ . The dimension counts are  $\sum_k \binom{b}{k} = 2^b$  and  $\sum_{k \text{ even}} \binom{b}{k} = 2^{b-1}$ . The last statement is the grading: odd plus even plus odd is even. ■

**Lemma 11.4** (The complex structure). *With  $e_1, e_2$  as in Setup 11.1, put  $J = e_1 e_2 \in C^+$ . Then  $J^2 = -1$ , and left multiplication by  $J$  is an  $\mathbb{R}$ -linear operator on  $C^+ \otimes_{\mathbb{Q}} \mathbb{R}$  whose square is  $-\text{id}$ .*

*Proof.*  $J^2 = e_1 e_2 e_1 e_2 = -e_1 e_1 e_2 e_2 = -q(e_1)q(e_2) = -1$ , using anticommutativity of  $e_1$  and  $e_2$  and then  $e_i^2 = q(e_i) = 1$ . Left multiplication by an element of  $C^+$  preserves  $C^+$ , and the square of left multiplication by  $J$  is left multiplication by  $J^2 = -1$ . ■

**Definition 11.5** (Kuga–Satake variety). Fix a lattice  $C_{\mathbb{Z}}^+ \subset C^+$  stable under multiplication. The complex torus

$$\text{KS}(T) = (C^+ \otimes_{\mathbb{Q}} \mathbb{R}, J) / C_{\mathbb{Z}}^+$$

has dimension  $\frac{1}{2} \dim_{\mathbb{Q}} C^+ = 2^{b-2}$ . Satake’s construction of a Riemann form on  $C^+$  compatible with  $J$  [Sat66] makes it an abelian variety, and by construction

$$H^1(\text{KS}(T), \mathbb{Q}) = C^+(T)$$

as a Hodge structure of weight one, the Hodge decomposition being the eigenspace decomposition of left multiplication by  $J$ .

**Example 11.6.** For a projective K3 surface,  $T$  is the transcendental lattice inside  $H^2(X, \mathbb{Q})$ ; taking all of  $H^2$  with  $b = 22$  gives  $\dim \text{KS} = 2^{20} = 1,048,576$ . At the other end,  $b = 3$  gives  $\dim \text{KS} = 2$ , an abelian surface; this is the case relevant below.

## 11.2. What the construction transports.

**Proposition 11.7** (The embedding). *Fix  $v_0 \in T$  with  $q(v_0) \neq 0$ . The map*

$$\Psi: T \longrightarrow \text{End}_{\mathbb{Q}}(C^+), \quad \Psi(v)(x) = v x v_0,$$

*is injective and is a morphism of Hodge structures  $T(1) \rightarrow \text{End}(H^1(\text{KS}(T), \mathbb{Q}))$ , both sides having weight zero. Consequently a Hodge class in  $T$ , that is a rational class of type  $(1, 1)$ , is carried to a Hodge class of type  $(0, 0)$  in  $\text{End}(H^1(\text{KS}(T), \mathbb{Q}))$ , that is to an element of  $\text{End}^0(\text{KS}(T))$ .*

*Proof.* The map lands in  $\text{End}(C^+)$  by the last part of Lemma 11.3, since  $v, v_0 \in T \subset C^-$ . It is injective because  $\Psi(v)(v_0^{-1}) = v$ , where  $v_0^{-1} = v_0/q(v_0)$ . That it respects the Hodge structures is Deligne’s computation [Del72, §4]: the Hodge structure on  $\text{End}(C^+)$  is induced by  $J$  acting on both sides, and  $\Psi$  intertwines the weight-two structure on  $T$ , twisted once, with it. The last sentence is the definition of a morphism of Hodge structures together with the theorem of Lefschetz on  $(1, 1)$  classes in the form that says a Hodge class of type  $(0, 0)$  in  $\text{End}(H^1(B, \mathbb{Q}))$  of an abelian variety  $B$  is an element of  $\text{End}^0(B)$ . ■

**Remark 11.8** (What is and is not known about the transport). The graph of an element of  $\text{End}^0(\text{KS}(T))$  is an algebraic cycle on  $\text{KS}(T) \times \text{KS}(T)$ , so the image of a Hodge class under  $\Psi$  is algebraic. Carrying that conclusion back to the variety  $X$  from which  $T$  came requires the correspondence  $\Psi$  itself to be induced by an algebraic cycle on  $X \times \text{KS}(T) \times \text{KS}(T)$ , and that is what is not known in general. What is known is that  $\Psi$  is absolute Hodge, by Deligne’s theorem for abelian varieties together with the analysis of the Kuga–Satake

correspondence, and that it is motivated in André's sense [And96a]; van Geemen [vG00] surveys the cases where algebraicity has been established. So the Kuga–Satake route converts the Hodge conjecture for  $T$  into the algebraicity of one specific correspondence, which is progress of a definite kind and is not a proof.

### 11.3. The hyperplane obstruction does not apply.

**Proposition 11.9.** *Theorems 1.4 and 1.5 do not constrain the Kuga–Satake route.*

*Proof.* Both theorems are statements about the pair of maps  $j^*$  and  $j_*$  attached to the inclusion  $j: Y \hookrightarrow X$  of a smooth member of an ample linear system, and both proofs use two properties of that pair which no other correspondence has: the Lefschetz hyperplane theorem, which makes  $j^*$  an isomorphism below the middle degree and hence forces the vanishing of Lemma 10.2 in the primitive middle degree; and the identity  $j_*j^* = \cdot\eta$ , which is what makes the return trip multiply by the polarisation rather than recover the class. The Kuga–Satake correspondence is a cycle class on a triple product of varieties of unrelated dimensions,  $\dim \text{KS}(T) = 2^{b-2}$  against  $\dim X$ , and it satisfies neither property; in particular the composite of  $\Psi$  with its transpose is multiplication by  $q(v_0)$  on  $T$  and not by a power of any polarisation of  $X$ . So neither hypothesis of Theorem 1.4 is met. ■

This is the content already asserted in Remark 20.11. The point of isolating it is that it identifies what an escape from Theorem 1.4 must look like: a correspondence whose transpose composite is not the Lefschetz operator. Section 13 makes the analogous identification for Theorems 1.1 and 1.3.

**11.4. Where the route stops for Weil type.** The limitation of the Kuga–Satake construction is the hypothesis  $\dim T^{2,0} = 1$ , and for abelian varieties of Weil type the possible values of  $\dim T^{2,0}$  can be listed completely.

**Proposition 11.10** (The pieces of  $H^2$ ). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type. In the character grading of Proposition 5.1,*

$$H^2(A, \mathbb{C}) = \wedge^2 V_+ \oplus (V_+ \otimes V_-) \oplus \wedge^2 V_-, \quad (26)$$

*the middle summand is the isotypic component of the norm character and is defined over  $\mathbb{Q}$ , and the outer two are interchanged by conjugation so their sum is defined over  $\mathbb{Q}$ . The Hodge numbers are*

$$(h^{2,0}, h^{1,1}, h^{0,2}) = \begin{cases} \left( \binom{n}{2}, n^2, \binom{n}{2} \right) & \text{on } \wedge^2 V_{\pm}, \\ (n^2, 2n^2, n^2) & \text{on } V_+ \otimes V_-, \end{cases}$$

*and the total is  $2\binom{2n}{2} + 4n^2 = \binom{4n}{2} = \dim H^2(A, \mathbb{Q})$ .*

*Proof.* The splitting is (15) at  $k = 2$ . The characters are  $\tau^2$ ,  $\tau\bar{\tau} = N(\tau)$  and  $\bar{\tau}^2$ , so the middle summand is the norm-isotypic component, and it is self-conjugate, hence defined over  $\mathbb{Q}$ ; the outer two are exchanged. For the Hodge numbers use (3) and Lemma 3.5:  $V_{\pm}^{1,0}$  and  $V_{\pm}^{0,1}$  have dimension  $n$  each, so  $(\wedge^2 V_+)^{2,0} = \wedge^2 V_+^{1,0}$  has dimension  $\binom{n}{2}$  and  $(\wedge^2 V_+)^{1,1} = V_+^{1,0} \otimes V_+^{0,1}$  has dimension  $n^2$ , while  $(V_+ \otimes V_-)^{2,0} = V_+^{1,0} \otimes V_-^{1,0}$  has dimension  $n^2$  and the  $(1,1)$  part has dimension  $2n^2$ . The total is  $2(2\binom{n}{2} + n^2) + 4n^2 = 2(2n^2 - n) + 4n^2 = 8n^2 - 2n = \binom{4n}{2}$ . ■

**Proposition 11.11** (No K3-type piece beyond  $n = 1$ ). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $\text{Hg}(A) = \text{SU}(V, H)$ . Then  $H^2(A, \mathbb{Q})$  decomposes into exactly three rational*

sub-Hodge structures that cannot be decomposed further, namely

$$\mathbb{Q}\eta, \quad (V_+ \otimes V_-)_{\mathbb{Q}} \ominus \mathbb{Q}\eta, \quad (\wedge^2 V_+ \oplus \wedge^2 V_-)_{\mathbb{Q}},$$

with  $h^{2,0}$  equal to 0,  $n^2$  and  $n(n-1)$  respectively. Consequently  $H^2(A, \mathbb{Q})$  contains a rational sub-Hodge structure of K3 type if and only if  $n = 1$ .

*Proof.* A rational sub-Hodge structure  $W \subseteq H^2(A, \mathbb{Q})$  is stable under  $\mathrm{Hg}(A)$ : the orthogonal projector onto  $W$  for the polarisation is a Hodge class in  $\mathrm{End}(H^2)$ , hence is fixed by  $\mathrm{Hg}(A)$ , hence  $W$  is a  $\mathrm{Hg}(A)$ -submodule. Conversely a  $\mathrm{Hg}(A)$ -submodule is a sub-Hodge structure, because the Hodge decomposition is the eigenspace decomposition of an element of  $\mathrm{Hg}(A)(\mathbb{R})$ . So the rational sub-Hodge structures are the  $\mathbb{Q}$ -subrepresentations of  $\mathrm{SU}(V, H)$ .

After complexification  $\mathrm{SU}(V, H)$  becomes  $\mathrm{SL}(V_+)$ , acting on  $V_+$  tautologically and on  $V_- \cong V_+^{\vee}$  dually. In (26) the outer summands  $\wedge^2 V_+$  and  $\wedge^2 V_-$  are irreducible, being fundamental representations, and the middle summand is  $V_+ \otimes V_+^{\vee} = \mathrm{End}(V_+) = \mathfrak{sl}(V_+) \oplus \mathbb{C}$ , the sum of the adjoint representation and the trivial one. This gives four complex irreducible pieces; conjugation fixes the two coming from the middle and exchanges the outer two, so over  $\mathbb{Q}$  they group into the three listed structures, and no further splitting is possible.

The trivial summand is spanned by  $\eta$  by Lemma 5.2, and has  $h^{2,0} = 0$  since  $\eta$  is of type  $(1, 1)$ . The adjoint summand has  $h^{2,0} = n^2$ , since removing the one-dimensional trivial piece from  $V_+ \otimes V_-$  removes a  $(1, 1)$  class only. The outer structure has  $h^{2,0} = 2\binom{n}{2} = n(n-1)$  by Proposition 11.10.

For a sum of these to have  $h^{2,0} = 1$  we need one of  $n^2$ ,  $n(n-1)$ ,  $n^2 + n(n-1) = 2n^2 - n$  to equal 1, using  $h^{2,0}(\mathbb{Q}\eta) = 0$ . The equation  $n(n-1) = 1$  has no positive integer solution, and each of  $n^2 = 1$  and  $2n^2 - n = 1$  gives  $n = 1$ . At  $n = 1$  the adjoint summand indeed has  $h^{2,0} = 1$  and the outer one is zero. ■

**Corollary 11.12** (The route is available exactly where it is not needed). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ . The Kuga–Satake construction applies to a rational sub-Hodge structure of  $H^2(A, \mathbb{Q})$  if and only if  $n = 1$ ; and for  $n = 1$  the Weil line lies in  $H^2(A, \mathbb{Q})$  and consists of divisor classes, hence of algebraic classes by the Lefschetz theorem on  $(1, 1)$  classes, so nothing needs transporting.*

*Proof.* The first clause is Proposition 11.11 with Setup 11.1. For the second,  $2n = 2$ , so  $W(A) \subseteq H^2(A, \mathbb{Q}) \cap H^{1,1}(A)$ , which is  $\mathrm{NS}(A) \otimes \mathbb{Q}$ . ■

*Remark 11.13* (Why there is no higher analogue on the shelf). For  $n \geq 2$  the Weil line sits in  $H^{2n}(A, \mathbb{Q})$  with  $2n \geq 4$ , and the Clifford construction has no counterpart in weight greater than two: it turns a quadratic form into an algebra, and a Hodge structure of weight  $w$  carries a polarisation that is symmetric only for  $w$  even and has no reason to produce a Clifford algebra with the right complex structure. One might instead apply the construction to a sub-Hodge structure of  $H^2$  and hope to recover the degree  $2n$  class from the resulting abelian variety; Proposition 11.11 rules that out by showing the hypothesis fails for every  $n \geq 2$ . What remains conceivable is a construction of Kuga–Satake shape for the weight  $2n$  structure directly, producing an auxiliary variety on which the Weil line becomes a divisor class or an endomorphism. We do not know of one, and we note that Theorem 1.3 does not obstruct it: such a construction is a correspondence and need not produce a  $K^{\times}$ -eigenclass. This is the same gap as in Remark 20.4, seen from the other side.

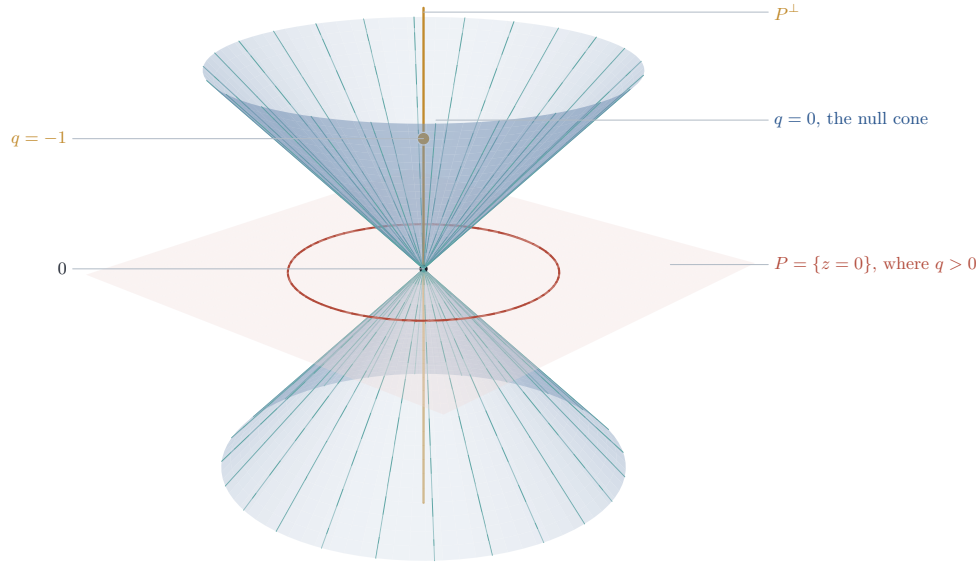


FIGURE 9. The geometry the Kuga–Satake construction starts from, in the smallest case  $b = 3$ . The quadratic form is  $q = x^2 + y^2 - z^2$  of signature  $(2, 1)$ , with its null cone; the horizontal plane  $P = \{z = 0\}$  is positive definite and meets the cone only at the origin. The construction needs such a plane, and it gets one from [Setup 11.1](#):  $P = \langle \Re\sigma, \Im\sigma \rangle$  for the period  $\sigma$  spanning  $T^{2,0}$ . An orthonormal basis  $e_1, e_2$  of  $P$  gives  $J = e_1 e_2$  with  $J^2 = -1$ , by [Lemma 11.4](#), and  $J$  is the complex structure on  $\text{KS}(T)$ . Positive definite planes correspond to the points of one sheet of the hyperboloid  $q = -1$ , which is the period domain. The hypothesis that makes all of this available is  $\dim T^{2,0} = 1$ , and by [Proposition 11.11](#) an abelian variety of Weil type supplies it only when  $n = 1$ .

## 12. WHAT IS LEFT

This section assembles the obstructions into a statement about where the Hodge conjecture can still be attacked. The statement is an equivalence and not a reduction, in the sense that neither side is easier than the other; its value is that it names the single place where a new idea has to enter, and [Theorem 1.4](#) says that the classical inductive handle is blind exactly there.

### 12.1. Reduction to primitive classes.

**Proposition 12.1** (Lefschetz decomposition of Hodge classes). *Let  $X$  be smooth projective of dimension  $d$  with ample class  $L$ . Every  $\alpha \in \text{Hdg}^p(X)$  can be written*

$$\alpha = \sum_{r \geq \max(0, p-d+p)} L^r \beta_r, \quad \beta_r \in \text{Hdg}^{p-r}(X) \cap H_{\text{prim}}^{2p-2r}(X, \mathbb{Q}),$$

*and the  $\beta_r$  are uniquely determined. Consequently the Hodge conjecture holds for  $X$  if and only if every primitive rational Hodge class on  $X$  is algebraic.*<sup>26</sup>

<sup>26</sup>The rationality of  $L$  is what makes the primitive parts rational. For a real Kähler class that is not a multiple of a rational one the primitive subspaces are real but not rational, and a rational Hodge class need not decompose into rational primitive pieces.

*Proof.* The Lefschetz decomposition  $H^k(X, \mathbb{Q}) = \bigoplus_r L^r H_{\text{prim}}^{k-2r}(X, \mathbb{Q})$  is a decomposition of rational Hodge structures, because  $L$  is of type  $(1, 1)$  and the primitive parts are kernels of morphisms of Hodge structures. A Hodge class therefore has Hodge classes as its components. For the last statement, if each primitive  $\beta_r$  is algebraic then so is  $L^r \beta_r$ , since  $L$  is algebraic and algebraic classes form a subring, and hence so is  $\alpha$ ; the converse is trivial. ■

## 12.2. The equivalence.

**Theorem 12.2.** *The following are equivalent.*

- (i) *The Hodge conjecture holds for every smooth complex projective variety.*
- (ii) *The Lefschetz standard conjecture  $B(X)$  holds for every smooth complex projective variety  $X$ , and every primitive rational Hodge class of middle degree on every smooth complex projective variety of even dimension is algebraic.*

*Proof.* (i) implies (ii). The second half of (ii) is a special case of (i). For  $B(X)$ , the Hodge conjecture for  $X \times X$  implies that the Künneth components of the class of the diagonal that realise  $\Lambda$  are algebraic:  $\Lambda$  is a morphism of Hodge structures of bidegree  $(-1, -1)$ , hence is given by a Hodge class on  $X \times X$  by the standard correspondence between morphisms of Hodge structures and Hodge classes on the product, and (i) applied to  $X \times X$  makes that class algebraic. This is Kleiman's argument [Kle68, §2].

(ii) implies (i). We argue by induction on  $d = \dim X$ . For  $d \leq 1$  the Hodge conjecture is classical. Let  $d \geq 2$  and assume the conjecture for all smooth projective varieties of dimension less than  $d$ . By Proposition 12.1 it suffices to prove that a primitive Hodge class  $\alpha \in H^{2p}(X, \mathbb{Q})$  is algebraic, and by hard Lefschetz we may assume  $2p \leq d$ .

If  $2p = d$ , then  $\alpha$  is a primitive Hodge class of middle degree on a variety of even dimension, so it is algebraic by the second half of (ii).

If  $2p < d$ , choose a smooth hypersurface  $Y \in |mH|$  for an ample  $H$  and  $m \gg 0$ , which exists by Bertini, and put  $L = c_1(\mathcal{O}_X(Y))$ . Since  $2p \leq d - 2$ , Theorem 10.1 makes  $j^*: H^{2p}(X, \mathbb{Q}) \rightarrow H^{2p}(Y, \mathbb{Q})$  an isomorphism, and  $j^*\alpha$  is a rational Hodge class on  $Y$  because  $j^*$  is a morphism of Hodge structures. As  $\dim Y = d - 1 < d$ , the inductive hypothesis makes  $j^*\alpha$  algebraic, say  $j^*\alpha = \text{cl}(W)$ . Then  $\text{cl}(j_*W) = j_*j^*\alpha = L\alpha$  by Lemma 10.2, so  $L\alpha$  is algebraic. By  $B(X)$  the operator  $\Lambda$  carries algebraic classes to algebraic classes, and by Lemma 10.6,  $\alpha = (d - 2p)^{-1}\Lambda(L\alpha)$ , which is therefore algebraic. Note  $d - 2p \geq 2$ , so the scalar is nonzero. ■

*Remark 12.3* (This is not a simplification). Both halves of (ii) are open, and  $B(X)$  is itself a consequence of (i), so Theorem 12.2 does not reduce the problem to something smaller. What it does is locate the difficulty. The induction in the proof of (ii) implies (i) runs smoothly through every degree except the middle one, where it stops and hands the work to the hypothesis; and Theorem 1.4 says that the step it uses below the middle degree, restriction to a hypersurface, returns zero at the middle degree, so the induction cannot be extended to cover it by any refinement of the same step.

**Corollary 12.4** (Where a new idea must enter). *Any proof of the Hodge conjecture must, at some point, produce an algebraic cycle representing a primitive Hodge class of middle degree on a variety of even dimension, by a method that does not factor through restriction to hyperplane sections of that variety.*

*Proof.* Immediate from Theorem 12.2 and Corollary 10.3. ■

*Remark 12.5* (Methods that do not factor through hyperplane sections). [Corollary 12.4](#) is not a statement that the conjecture is unapproachable, and we list the known methods that are outside its scope, since they are where the successful cases come from. The Kuga–Satake construction replaces a weight-two Hodge structure of K3 type by the  $H^1$  of an abelian variety, and the class it produces is of type  $(1, 1)$ , hence algebraic by Lefschetz; this is not a restriction to a subvariety and is untouched. Degenerations and limits of Hodge structures move the variety rather than cutting it. Correspondences from other varieties, of which the Kuga–Satake construction is one instance and Markman’s construction another, are also outside the scope. Finally, arguments through the coniveau filtration, which show that a class supported on a proper closed subvariety is controlled by that subvariety, are compatible with everything here: they lower the dimension legitimately, because the class is actually supported on the smaller variety, which is exactly what a primitive coniveau-zero class is not.

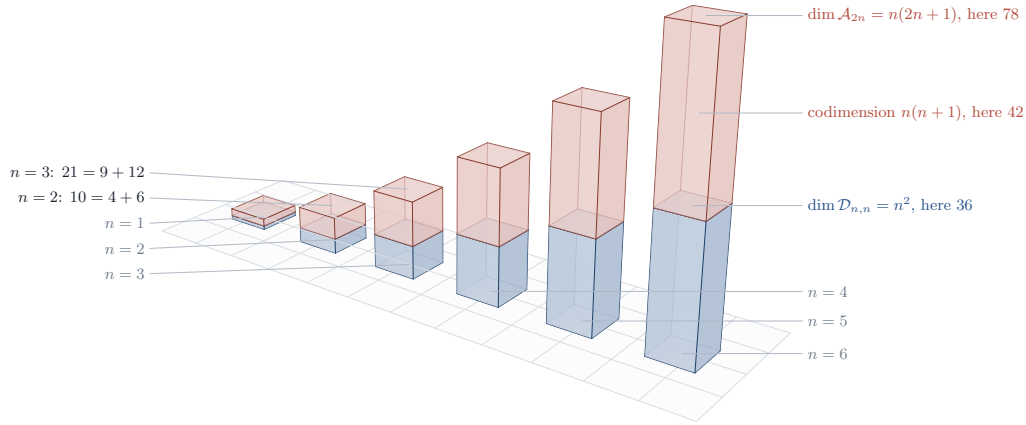


FIGURE 10. The parameter count behind [Theorem 12.2](#). For  $n = 1, \dots, 6$  each prism has the height  $n(2n + 1)$  of the moduli of principally polarised abelian  $2n$ -folds; its lower part, of height  $n^2$ , is the Weil family of [Proposition B.8](#), and its upper part is the codimension  $n(n + 1)$ . The gap widens quadratically, so the Weil locus is thin, and by [Corollary B.9](#) it is an algebraic subvariety. Both facts matter for what is left: the locus one would have to deform along is small and is a variety, and neither of those things produces a cycle on it.

### 13. WHAT A CONSTRUCTION MUST BREAK, AND HOW MARKMAN BREAKS IT

[Theorem 1.3](#) says that a  $K^\times$ -eigenclass for the norm character has no component along the Weil line.<sup>27</sup> It is natural to ask what the contrapositive demands of a construction that does reach the Weil line, and the answer is sharper than one might expect: the demand is not on the size of the object, nor on the ambient geometry, nor on any dimension count, but

<sup>27</sup>The three hypotheses below are not independent in spirit. A construction that imports its input from projective space will normally be equivariant for whatever acts on the input, and an equivariant construction will normally return a single eigenclass, so the three tend to hold or fail together. What [Proposition 13.5](#) records is that Markman’s construction fails all three at once, and that no two of them can be dropped in isolation.

on the way the object transforms. This section states that demand, and then shows that the one construction known to meet it, Markman's, meets it in a specific and reproducible way. The section is the most useful one in the paper for anybody who wants to build something new, so we have tried to make the requirement concrete rather than merely negative.

### 13.1. The orbit criterion.

**Proposition 13.1** (Weil classes generate a plane). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and let  $\alpha \in W(A)$  be nonzero. Then*

$$\text{span}_{\mathbb{Q}}\{\iota(\tau)^*\alpha : \tau \in K^\times\} = W(A),$$

*a two-dimensional  $\mathbb{Q}$ -vector space, and as a representation of  $K^\times$  it is irreducible over  $\mathbb{Q}$ .*

*Proof.* By Proposition 3.7(i) the space  $W(A)$  is free of rank one over  $K$ , and  $\iota(\tau)^*$  acts on it as multiplication by the scalar  $\tau^{2n} \in K$ . Hence the span in question is  $S \cdot \alpha$ , where  $S = \text{span}_{\mathbb{Q}}\{\tau^{2n} : \tau \in K^\times\} \subseteq K$ . Taking  $\tau = 1$  shows  $1 \in S$ . By the argument in the proof of Lemma 5.4 the map  $c(\tau) = \tau/\bar{\tau}$  has infinite image while  $\mu(K)$  is finite, so there is  $\tau_0$  with  $(\tau_0/\bar{\tau}_0)^{2n} \neq 1$ , that is with  $\tau_0^{2n} \neq \bar{\tau}_0^{2n}$ , that is with  $\tau_0^{2n} \notin \mathbb{Q}$ . Then  $S \supseteq \mathbb{Q} + \mathbb{Q}\tau_0^{2n} = K$ , so  $S = K$  and the span is  $K\alpha = W(A)$ .

Irreducibility: a proper nonzero  $K^\times$ -stable  $\mathbb{Q}$ -subspace would be a line, on which  $K^\times$  would act by a  $\mathbb{Q}$ -valued character, forcing  $\tau^{2n} \in \mathbb{Q}$  for all  $\tau$ , which we have just excluded.  $\blacksquare$

**Corollary 13.2** (The orbit criterion). *Let  $\mathcal{E}$  be an object of  $D^b(A)$ , or  $Z$  a cycle in  $\text{CH}^n(A)_{\mathbb{Q}}$ , whose class  $\text{ch}_n(\mathcal{E})$ , respectively  $\text{cl}(Z)$ , is a nonzero Weil class. Then there is no function  $\chi: K^\times \rightarrow \mathbb{Q}$  with*

$$\iota(\tau)^* \text{ch}_n(\mathcal{E}) = \chi(\tau) \text{ch}_n(\mathcal{E}) \quad \text{for all } \tau \in K^\times.$$

*Equivalently, the  $K^\times$ -orbit of the class spans a plane and not a line.*

*Proof.* Such a  $\chi$  would make  $\mathbb{Q} \cdot \text{ch}_n(\mathcal{E})$  a  $K^\times$ -stable line inside  $W(A)$ , contradicting Proposition 13.1.  $\blacksquare$

Corollary 13.2 is the precise sense in which a construction must break equivariance. Here is what it forbids. Suppose a construction assigns to the data  $(A, \iota, \eta)$  an object  $\mathcal{E}(A, \iota, \eta)$  in a way that is natural, meaning that an isomorphism of the data induces an isomorphism of the objects. The endomorphism  $\iota(\tau)$  is, for  $\tau$  in the order, an isogeny of  $A$  commuting with  $\iota$  and multiplying  $\eta$  by  $N(\tau)$ ; naturality then forces  $\iota(\tau)^*\mathcal{E} \cong \mathcal{E}$  up to the normalisation that the polarisation change imposes, and the induced relation on  $\text{ch}_n$  is precisely of the forbidden form  $\chi(\tau) \text{ch}_n(\mathcal{E})$  with  $\chi$  a power of the norm. So:

*No construction that is natural in the  $K$ -action, taken alone, can produce a Weil class. The construction must depend on a choice that  $\text{Gal}(K/\mathbb{Q})$  moves.*

The rest of this section shows what such a choice looks like in practice.

**13.2. The two conjugate isotropic subspaces.** The choice that  $\text{Gal}(K/\mathbb{Q})$  moves is visible already in Section 5 and we name it here. The splitting  $V_{\mathbb{C}} = V_+ \oplus V_-$  of (2) is defined over  $K$  and not over  $\mathbb{Q}$ : conjugation interchanges the two summands. Consequently  $\bigwedge^{2n} V_+$  and  $\bigwedge^{2n} V_-$  are individually defined over  $K$ , and only their sum is rational. A class in  $W(A)$  is of the form  $w + \bar{w}$  with  $w \in \bigwedge^{2n} V_+$ , and the two summands are the two conjugate halves.

**Definition 13.3** (Conjugate pair data). By *conjugate pair data* for  $(A, \iota, \eta)$  we mean an ordered choice of one of the two summands  $V_{\pm}$ , equivalently an ordered choice of one of the two embeddings  $K \hookrightarrow \mathbb{C}$ , equivalently a choice of one of the two points of a  $\text{Gal}(K/\mathbb{Q})$ -orbit of size two attached to the  $K$ -structure.

**Proposition 13.4** (Corestriction produces the plane). *Let  $w \in \wedge^{2n} V_+$  be nonzero. Then  $w + \bar{w}$  and  $\sqrt{-d}(w - \bar{w})$  lie in  $W(A)$  and span it over  $\mathbb{Q}$ . In particular, a rational class in the Weil line is exactly a  $\text{Gal}(K/\mathbb{Q})$ -corestriction of a class in one conjugate half, and any construction producing such a class must produce, in effect, a pair of conjugate inputs and combine them.*

*Proof.* Both displayed elements are fixed by conjugation, hence rational, and both lie in  $\wedge^{2n} V_+ \oplus \wedge^{2n} V_-$  by construction, hence in  $W(A)$  by [Definition 3.6](#). They are linearly independent over  $\mathbb{Q}$  because  $w$  and  $\bar{w}$  are linearly independent over  $\mathbb{C}$ , and  $W(A)$  has dimension two by [Proposition 3.7\(i\)](#). ■

This is the design principle. A construction reaching the Weil line has to be, in a suitable sense, *bilinear in the two conjugate halves*: it must take one input attached to  $V_+$  and one attached to  $V_-$  and pair them. A construction fed a single  $K$ -invariant input cannot do this, which is [Corollary 13.2](#) again.

**13.3. How Markman’s construction meets the criterion.** Markman’s work [[Mar25a](#), [Mar25b](#)] is the only construction we know of that produces Weil classes, and it meets the criterion exactly in the manner just described. We set out the relevant structure; the details are in [[Mar25b](#)], and we claim no part of them.

The construction does not begin on  $A$  or on  $A^{\vee}$ . It begins on an abelian  $n$ -fold  $X$ , that is, on a variety of half the dimension of  $A$ . Inside  $H^{\text{ev}}(X, \mathbb{Q})$  one selects a plane  $P$  spanned by Hodge classes, and one requires that  $P$  meet the spinor variety of even pure spinors in two points. Those two points are conjugate over the imaginary quadratic field  $K$ ; they correspond to two maximal isotropic subspaces  $W_1$  and  $W_2$  that are interchanged by  $\text{Gal}(K/\mathbb{Q})$ . This is the conjugate pair data of [Definition 13.3](#), realised concretely.

To each of the two points one attaches a *secant sheaf* on  $X$ : a coherent sheaf whose Chern character is the corresponding point of  $P$ . In the worked case where  $X$  is the Jacobian of a curve of genus three, the secant sheaf is  $\mathcal{I}_{\cup C_i}(\Theta)$ , the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi image of the curve. Call the two sheaves  $\mathcal{F}_1$  and  $\mathcal{F}_2$ .

The abelian variety of Weil type is then  $A = X \times \widehat{X}$ , of dimension  $2n$ , with its  $K$ -action and polarisation inherited from  $P$ . The object that carries the Weil class is

$$E = \Phi(\mathcal{F}_1^{\vee} \boxtimes \mathcal{F}_2), \quad (27)$$

where  $\Phi: D^b(X \times X) \rightarrow D^b(X \times \widehat{X})$  is Orlov’s derived equivalence. One then shows that the characteristic class  $\kappa(E)$  remains of Hodge type under every deformation of  $X \times \widehat{X}$  as a polarised abelian  $2n$ -fold of Weil type, which is what makes the class algebraic on the whole family. The conclusion in [[Mar25b](#)] is the algebraicity of the Weil classes for abelian sixfolds of Weil type of discriminant  $-1$ , and, through Schoen’s degeneration argument, for all abelian fourfolds of Weil type, whence the Hodge conjecture for abelian fourfolds.

**Proposition 13.5** (Which hypotheses are escaped). *The object (27) satisfies none of the hypotheses of [Theorem 1.1](#) and does not satisfy the hypothesis of [Theorem 7.6](#). Specifically:*

- (i) The source is not a projective space. *The input  $\mathcal{F}_1^{\vee} \boxtimes \mathcal{F}_2$  lives on  $X \times X$ , an abelian variety. The proof of [Theorem 1.1](#) used that  $H^*(\mathbb{P}^{N-1}, \mathbb{Q})$  is generated in*

degree two, so that  $\text{ch}$  of any object is a polynomial in the hyperplane class. The cohomology of  $X \times X$  is not generated in degree two: it contains Künneth classes of every degree, and  $\text{ch}(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$  is not confined to any divisor subring.

- (ii) The transform is not  $\Phi_{\mathcal{P}}$  on  $A \times A^\vee$ . The relevant equivalence is between  $D^b(X \times X)$  and  $D^b(X \times \widehat{X})$ , and [Theorem 6.5](#), which computes the transform with Poincaré kernel on powers of the polarisation of  $A$ , says nothing about it.
- (iii) The construction is not  $K$ -equivariant. The two tensor factors of (27) carry the two conjugate points of  $P \cap (\text{spinor variety})$ , one each. Interchanging them is the action of the nontrivial element of  $\text{Gal}(K/\mathbb{Q})$ , and it exchanges  $E$  with a different object. Hence  $\text{ch}_n(E)$  is not a  $K^\times$ -eigenclass for the norm, and [Corollary 13.2](#) is satisfied rather than violated.

*Proof.* (i) and (ii) are a comparison of hypotheses and require no argument beyond reading (27). For (iii), if  $\text{ch}_n(E)$  were an eigenclass for the norm then by [Corollary 5.6](#) it would have zero component along the Weil line, whereas Markman establishes that it spans one [\[Mar25b\]](#). ■

*Remark 13.6* (The lesson for the secant idea). We are explicit here, because the point is easy to miss and is the reason this section exists. The idea of using secant geometry to produce Weil classes is not what fails. Markman’s construction is a secant construction, and it works. What fails is siting the secant geometry on the dual  $A^\vee$  of the  $2n$ -fold inside an ambient projective space, and transporting by the Poincaré kernel. That siting has three defects at once, by [Proposition 13.5](#): it puts the input where the cohomology is generated by a divisor, it uses the transform for which [Theorem 6.5](#) applies, and it feeds the construction  $K$ -invariant data, since a secant variety of a  $K$ -invariant subvariety is  $K$ -invariant. Moving the secant geometry down to an abelian  $n$ -fold repairs all three, and it does so because the halving of the dimension is what makes room for the two conjugate inputs.

**13.4. What would be needed above the known range.** We can now state precisely what a construction for the cases that remain open would have to supply. The known range is  $n = 2$ , all discriminants, and  $n = 3$  with discriminant  $-1$ ; see [\[Mar25b\]](#) and, for a different route at discriminant 1, [\[FF26\]](#).

**Question 13.7.** Fix  $n \geq 4$  and an imaginary quadratic field  $K$ . Does there exist an abelian  $n$ -fold  $X$ , a plane  $P \subset H^{\text{ev}}(X, \mathbb{Q})$  spanned by Hodge classes meeting the spinor variety in two points conjugate over  $K$ , secant sheaves  $\mathcal{F}_1, \mathcal{F}_2$  on  $X$  realising those two points, and a deformation argument showing that  $\kappa(\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2))$  remains of Hodge type across the moduli of polarised abelian  $2n$ -folds of Weil type with multiplication by  $K$ ?

Each of the four ingredients is a concrete demand and none is obviously out of reach; what is absent at present is the analogue of the genus three Jacobian, namely a family of abelian  $n$ -folds in which the plane  $P$  and the two secant sheaves can be written down. By [Corollary 13.2](#) any successful answer must, however it is phrased, be bilinear in a pair of conjugate inputs, because that is the only way a rational class can acquire a  $K^\times$ -orbit of dimension two. We offer [Question 13.7](#) as the sharpest form we can give of what is left on the abelian side, and we note that it is a question about writing down one sheaf on one abelian variety, not a question about a general mechanism.

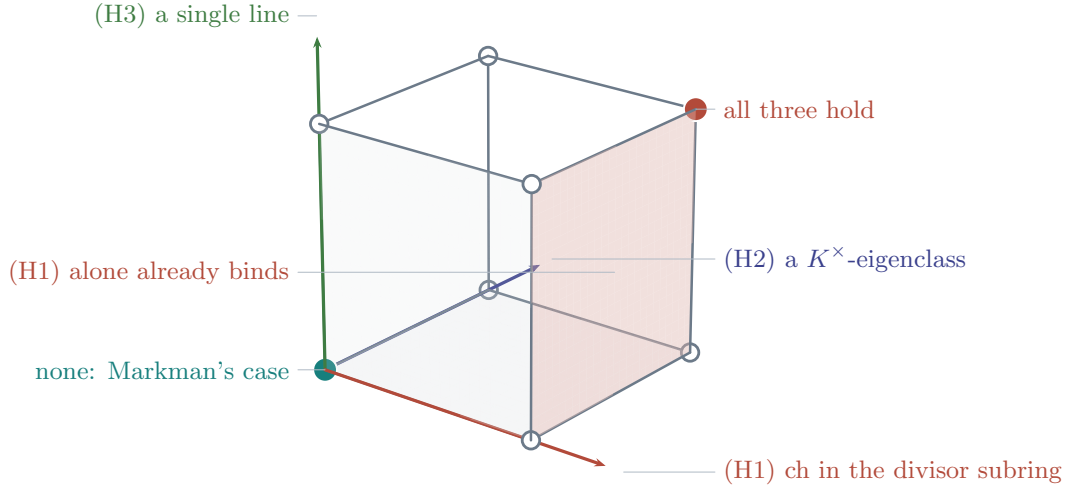


FIGURE 11. The eight positions a secant construction can occupy, drawn as the Boolean lattice of subsets of the three hypotheses isolated in [Proposition 13.5](#): (H1) the Chern character of the input lies in the subring generated by divisor classes, (H2) the output is a  $K^\times$ -eigenclass, (H3) the output spans a single line rather than a plane. A construction satisfying all three sits at the top vertex, where [Theorem 1.1](#) and [Theorem 1.3](#) both apply; the shaded face is where (H1) alone already binds. Markman's construction sits at the bottom vertex, satisfying none of the three, which is why nothing here bears on it, and the Weil line needs a plane by [Proposition 13.1](#).

#### 14. AN EXPLICIT SECANT PLANE IN EVERY DIMENSION

[Question 13.7](#) asks for four things: an abelian  $n$ -fold  $X$ , a plane  $P \subset H^{\text{ev}}(X, \mathbb{Q})$  of Hodge classes meeting the spinor variety in two points conjugate over  $K$ , a pair of secant sheaves realising the plane, and a deformation argument. This section supplies the first two in closed form, for every  $n \geq 2$  and every imaginary quadratic  $K$ , and reduces the third to an explicit equation on a Chern character. The fourth is untouched and is the hardest of the four; we say so at the end.

The framework is Markman's: the Weil structure on  $X \times \widehat{X}$  obtained by embedding  $K$  into  $\text{End}^0(X \times \widehat{X})$ , and the notion of a  $K$ -secant sheaf whose Chern character lies in a plane meeting the spinor variety at two conjugate points, are introduced in [\[Mar25a, Mar25b\]](#), and the deformation step that finishes the argument at  $n = 3$  is his. What we do here is write the first two ingredients out in closed form, uniformly in  $n$ ,  $K$  and the discriminant, so that the third becomes a finite system; we claim nothing beyond that. What follows is elementary once set up, and we were surprised that it comes out in closed form. The reason it does is that the two conjugate isotropic subspaces demanded by [Corollary 13.2](#) can be produced by a single element of the Néron–Severi group, and that the pure spinor of the graph of a two-form is the exponential of that form.

##### 14.1. The Weil structure.

**Setup 14.1.** Let  $X$  be a complex abelian variety of dimension  $n$  with  $H_1(X, \mathbb{Q}) = U$  of dimension  $2n$  and complex structure  $J$  on  $U \otimes \mathbb{R}$ . Let  $d$  be a squarefree positive integer,  $s = \sqrt{-d}$  and  $K = \mathbb{Q}(s)$ . Let  $\beta \in \text{NS}(X) \otimes \mathbb{Q}$  be nondegenerate, regarded as an alternating

form on  $U$ , so that the Riemann relation  $J^\top \beta J = \beta$  holds, and write  $\phi_\beta: X \rightarrow \hat{X}$  for the associated isogeny. Put

$$A = X \times \hat{X}, \quad \dim A = 2n, \quad H_1(A, \mathbb{Q}) = U \oplus U^\vee,$$

and let

$$M = \begin{pmatrix} 0 & -\beta^{-1} \\ d\beta & 0 \end{pmatrix} \in \text{End}^0(A), \quad (28)$$

acting on  $H_1(A, \mathbb{Q}) = U \oplus U^\vee$  by  $(u, \phi) \mapsto (-\beta^{-1}\phi, d\beta u)$ .<sup>28</sup>

**Theorem 14.2** (An explicit Weil structure). *In Setup 14.1:*

- (i)  $M^2 = -d$ , so  $\iota(s) = M$  defines an embedding  $\iota: K \hookrightarrow \text{End}^0(A)$ ;
- (ii)  $M$  commutes with the complex structure of  $A$ , and this holds exactly because  $\beta$  satisfies the Riemann relation, that is because  $\beta \in \text{NS}(X) \otimes \mathbb{Q}$ ;
- (iii)  $M$  acts on  $H^{1,0}(A)$  with eigenvalues  $s$  and  $-s$ , each of multiplicity  $n$ , so  $(A, \iota, \eta)$  is of  $(K, -1, n)$ -Weil type for a suitable polarisation  $\eta$ ;
- (iv) the eigenspaces  $W_\pm \subset H_1(A, \mathbb{C})$  of  $M$  for  $\pm s$  are maximal isotropic for the canonical pairing on  $U \oplus U^\vee$ , are interchanged by complex conjugation, and are the graphs of the two-forms  $\mp s\beta$ .

*Proof.* (i) Block multiplication gives  $M^2 = \text{diag}(-\beta^{-1}d\beta, -d\beta\beta^{-1}) = -d \cdot \text{id}$ .

(ii) The complex structure of  $A = X \times \hat{X}$  is  $J_A = \text{diag}(J, -J^\top)$ . The relation  $MJ_A = J_AM$  amounts to the two identities  $d\beta J = -dJ^\top \beta$  and  $-\beta^{-1}(-J^\top) = J(-\beta^{-1})$ , and each is equivalent to  $\beta J = -J^\top \beta$ . That is exactly the Riemann relation: from  $J^\top \beta J = \beta$  and  $J^2 = -1$  one gets  $J^\top \beta = \beta J^{-1} = -\beta J$ , as required. Conversely  $\beta J = -J^\top \beta$  gives  $J^\top \beta J = -\beta J J = \beta$ .

(iii)  $M$  is block anti-diagonal and  $H^{1,0}(A) = H^{1,0}(X) \oplus H^{1,0}(\hat{X})$  with both summands of dimension  $n$ , and  $M$  exchanges them. If  $M(x, y) = \lambda(x, y)$  then  $-\beta^{-1}y = \lambda x$  and  $d\beta x = \lambda y$ , so  $y = d\beta x / \lambda$  and, substituting,  $-dx / \lambda = \lambda x$ , that is  $\lambda^2 = -d$ . For each of the two values of  $\lambda$  the assignment  $x \mapsto (x, d\beta x / \lambda)$  is an isomorphism from  $H^{1,0}(X)$  onto the  $\lambda$ -eigenspace, which is therefore of dimension  $n$ . That is the balanced signature of Definition 3.1(ii). For (iii) of that definition one takes for  $\eta$  the polarisation of  $A$  obtained from the canonical principal polarisation of  $X \times \hat{X}$  twisted so that the Rosati involution sends  $M$  to  $-M$ ; this is possible because  $\beta$  is symmetric under the Rosati involution of  $X$ , being a class in NS.

(iv) The same computation identifies  $W_\lambda = \{(x, d\beta x / \lambda)\}$ , the graph of the form  $d\beta / \lambda$ , and  $d\beta / \lambda = d\beta / (\pm s) = \mp s\beta$  since  $d/s = -s$ . The canonical pairing on  $U \oplus U^\vee$  is  $\langle (x, \xi), (x', \xi') \rangle = \xi'(x) + \xi(x')$ , and on the graph it gives  $d\beta(x', x) / \lambda + d\beta(x, x') / \lambda = 0$  because  $\beta$  is alternating; so  $W_\lambda$  is isotropic, and it has dimension  $2n$ , half of  $\dim(U \oplus U^\vee) = 4n$ , hence is maximal. Conjugation sends  $s$  to  $-s$  and so interchanges the two. ■

<sup>28</sup>The placement of the factor  $d$  is a convention, and it has to be made once and kept. The endomorphism  $\begin{pmatrix} 0 & -d\beta^{-1} \\ \beta & 0 \end{pmatrix}$  also squares to  $-d$  and also commutes with the complex structure, and it defines a Weil structure on the same  $A$ ; but it is a different embedding of  $K$ , its eigenspaces are the graphs of  $\mp s\beta/d$  rather than of  $\mp s\beta$ , its polarisation is  $\beta + d\hat{\beta}$  rather than  $d\beta + \hat{\beta}$ , and the class in  $\bigwedge^2 V_+ \oplus \bigwedge^2 V_-$  it produces is  $\beta - d\hat{\beta}$  rather than  $\gamma = d\beta - \hat{\beta}$ . The convention of (28) is the one whose action on cohomology is  $x \mapsto -Bx$ ,  $\xi \mapsto dB^{-1}\xi$  in the proof of Theorem 15.2, and it is the one every script uses; item (XI) of Appendix D checks the three eigenvalues that pin it down, and item (XXV) shows what goes wrong if the two are mixed: the space of polarised deformations killing the Weil line then has dimension  $n(n+1)/2$  instead of  $n^2$ . For  $d = 1$  the two conventions coincide.

*Remark 14.3.* [Theorem 14.2](#) says that abelian  $2n$ -folds of Weil type of the split form  $X \times \hat{X}$  exist for every  $n$ , every imaginary quadratic  $K$  and every polarised abelian  $n$ -fold  $X$ , and that the Weil structure is written down by one element of  $\text{NS}(X)$ . These are not general members of the moduli of Weil type, and the Weil classes on them are of interest precisely because Markman's deformation argument propagates algebraicity from them to the general member; see [Remark 14.17](#) below.

**14.2. The plane.** Write  $t = \beta \in \text{NS}(X) \otimes \mathbb{Q}$ , so that by [Theorem 14.2\(iv\)](#) the two conjugate maximal isotropic subspaces are the graphs of  $\mp s t$ ; the letter  $t$  is kept because everything below holds for any nondegenerate  $t \in \text{NS}(X) \otimes \mathbb{Q}$ . The pure spinor attached to the graph of a two-form  $\omega$  is  $e^\omega$ , so the two conjugate points of the even spinor variety determined by the Weil structure are  $[e^{-st}]$  and  $[e^{st}]$ .

**Definition 14.4** (The secant plane). With  $t$  and  $d$  as above, put

$$u_t = \sum_{j \geq 0} \frac{(-d)^j}{(2j)!} t^{2j}, \quad v_t = \sum_{j \geq 0} \frac{(-d)^j}{(2j+1)!} t^{2j+1}, \quad (29)$$

both sums being finite because  $t^k = 0$  for  $k > n$ , and let  $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t \subset H^{\text{ev}}(X, \mathbb{Q})$ .

**Theorem 14.5** (The plane is an explicit rational secant). *With notation as above:*

- (i)  $u_t$  and  $v_t$  are rational Hodge classes, and  $e^{\pm st} = u_t \pm s v_t$ ;
- (ii)  $P_t$  is a two-dimensional  $\mathbb{Q}$ -subspace of  $H^{\text{ev}}(X, \mathbb{Q})$  spanned by Hodge classes, and  $\mathbb{P}(P_t)$  is a line defined over  $\mathbb{Q}$  passing through the two conjugate points  $[e^{\pm st}]$  of the even spinor variety;
- (iii) that line is not contained in the spinor variety, so it meets it in exactly those two points.

*Proof.* (i) Each  $t^k$  is a rational Hodge class because  $t \in \text{NS}(X) \otimes \mathbb{Q}$  is of type  $(1, 1)$  and the algebraic classes form a subring. Separating  $e^{st} = \sum_k s^k t^k / k!$  into even and odd powers of  $s$  and using  $s^{2j} = (-d)^j$ ,  $s^{2j+1} = (-d)^j s$  gives  $e^{st} = u_t + s v_t$ , and conjugating gives  $e^{-st} = u_t - s v_t$ .

(ii) The two elements are linearly independent over  $\mathbb{Q}$  because  $u_t$  has nonzero component in  $H^0$  and  $v_t$  does not, while  $v_t \neq 0$  since  $t \neq 0$ . From (i), both  $e^{st}$  and  $e^{-st}$  lie in  $P_t \otimes K$ , so  $\mathbb{P}(P_t)$  passes through the two points.

(iii) The even spinor variety is cut out by quadrics, by Cartan's description of pure spinors, so a line either lies inside it or meets it in a subscheme of length at most two. The point  $[u_t]$  of  $\mathbb{P}(P_t)$  is not a pure spinor: a pure spinor with nonzero component in  $H^0$  is a scalar multiple of  $e^\omega$  for a two-form  $\omega$ , and  $u_t$  has zero component in  $H^2$  while its component in  $H^4$  is  $-\frac{d}{2}t^2 \neq 0$ , whereas  $e^\omega$  with vanishing  $H^2$  part is 1. So the line is not contained in the spinor variety, and the two points of (ii) exhaust the intersection. ■

**Example 14.6.** For  $n = 4$  and  $d = 1$ ,  $u_t = 1 - \frac{1}{2}t^2 + \frac{1}{24}t^4$  and  $v_t = t - \frac{1}{6}t^3$ . For  $n = 4$  and  $d = 3$ ,  $u_t = 1 - \frac{3}{2}t^2 + \frac{3}{8}t^4$  and  $v_t = t - \frac{1}{2}t^3$ . Both are visibly rational and visibly in the subring generated by  $t$ .

**14.3. The Chern character equation.** We can now say exactly what a secant sheaf has to be. The following is the substance of this section: the four demands of [Question 13.7](#) collapse, on the sheaf side, to one equation with all invariants computed.

**Theorem 14.7** (The invariants of a secant sheaf). *Let  $X$ ,  $t$ ,  $d$  be as above and let  $\mathcal{F}$  be a coherent sheaf on  $X$  with  $\text{ch}(\mathcal{F}) \in P_t$ , say  $\text{ch}(\mathcal{F}) = a u_t + b v_t$  with  $a, b \in \mathbb{Q}$ . Then*

$$\text{rk } \mathcal{F} = a, \quad c_1(\mathcal{F}) = b t, \quad \text{ch}_2(\mathcal{F}) = -\frac{ad}{2} t^2, \quad c_2(\mathcal{F}) = \frac{ad+b^2}{2} t^2,$$

and more generally  $\text{ch}_k(\mathcal{F})$  is the degree  $k$  term of (29) scaled by  $a$  or  $b$  according to the parity of  $k$ . The Bogomolov discriminant is

$$\Delta(\mathcal{F}) = 2a c_2(\mathcal{F}) - (a-1)c_1(\mathcal{F})^2 = N_{K/\mathbb{Q}}(b+as) \cdot t^2. \quad (30)$$

In particular  $\Delta(\mathcal{F}) \cdot H^{n-2} > 0$  for every ample  $H$ , so the Bogomolov inequality places no obstruction on the existence of  $\mathcal{F}$ .

*Proof.* The first four identities are read off from (29) degree by degree, using  $\text{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$  to convert  $\text{ch}_2$  into  $c_2$ . For (30),

$$2a \cdot \frac{ad+b^2}{2}t^2 - (a-1)b^2t^2 = (a^2d + ab^2 - ab^2 + b^2)t^2 = (a^2d + b^2)t^2,$$

and  $N_{K/\mathbb{Q}}(b+as) = (b+as)(b-as) = b^2 + a^2d$ . Positivity of the norm on  $K^\times$  and of  $t^2H^{n-2}$  for  $t$  in the positive cone give the last assertion.  $\blacksquare$

*Remark 14.8* (The norm form as the discriminant). Identity (30) is the reason the construction has a chance at all. The discriminant of the sheaf is the norm form of  $K$  evaluated at the coordinates of its Chern character in the plane. The two conjugate points of the spinor variety are the two isotropic directions of that norm form after extension to  $K$ , and the rational points of the plane are exactly the non-isotropic ones. A sheaf therefore sits at a rational point of a conic whose isotropic directions carry the Weil structure, which is the precise sense in which the conjugate pair of Corollary 13.2 is built into the invariants rather than imposed on them.

**Corollary 14.9** (The case of an abelian surface). *Let  $n = 2$ , so  $X$  is an abelian surface and  $A = X \times \widehat{X}$  an abelian fourfold of Weil type. A sheaf with  $\text{ch}(\mathcal{F}) \in P_t$  has Mukai vector  $v(\mathcal{F}) = (a, bt, -\frac{ad}{2}t^2)$  and Mukai self-pairing*

$$\langle v(\mathcal{F}), v(\mathcal{F}) \rangle = N_{K/\mathbb{Q}}(b+as) \cdot t^2 > 0.$$

*Proof.* On an abelian surface the Mukai pairing of  $v = (r, c, e)$  with itself is  $c^2 - 2re$ , which here is  $b^2t^2 - 2a(-\frac{ad}{2}t^2) = (b^2 + a^2d)t^2$ .  $\blacksquare$

Corollary 14.9 is consistent with the known range: the positivity of the Mukai self-pairing is exactly the numerical condition under which moduli of stable sheaves on an abelian surface with a given Mukai vector are non-empty, and  $n = 2$  is settled.

**14.4. The genus three case, and what the field predicts.** At  $n = 3$  the equation of Theorem 14.7 pins down the secant sheaf far more tightly, and it reproduces the shape of the object Markman uses. We record the computation because it is a check on the whole framework.

**Proposition 14.10** (Rank one at  $n = 3$ ). *Let  $X$  be the Jacobian of a smooth curve of genus three, let  $\Theta$  be the theta divisor and  $C \subset X$  the Abel–Jacobi curve, so that  $\Theta^2/2 = [C]$ . Take  $t = \Theta$  and  $a = 1$  in Theorem 14.7. Then a rank one torsion-free sheaf  $\mathcal{F}$  with  $\text{ch}(\mathcal{F}) \in P_\Theta$  and  $c_1(\mathcal{F}) = b\Theta$  is of the form  $\mathcal{F} = \mathcal{I}_Z(b\Theta)$  with*

$$[Z] = \frac{b^2 + d}{2} \Theta^2 = \frac{b^2 + d}{b^2} \cdot b^2 [C].$$

*If  $Z$  is a disjoint union of translates of  $C$ , the number of translates is  $m = (b^2 + d)/b^2$ , which is a positive integer precisely when  $b^2 \mid d$ ; for  $b = 1$ ,*

$$m = d + 1.$$

*Proof.* A rank one torsion-free sheaf with  $c_1 = b\Theta$  on a smooth projective threefold is  $\mathcal{I}_Z(b\Theta)$  for a subscheme  $Z$  of codimension at least two, and  $\text{ch}(\mathcal{I}_Z(b\Theta)) = e^{b\Theta} - [Z] - (\text{terms in } H^6)$ . Comparing the components in  $H^4$  with [Theorem 14.7](#) at  $a = 1$ , which gives  $\text{ch}_2 = -\frac{d}{2}\Theta^2$ , yields  $\frac{b^2}{2}\Theta^2 - [Z] = -\frac{d}{2}\Theta^2$ , that is  $[Z] = \frac{b^2+d}{2}\Theta^2$ . The Poincaré formula  $\Theta^{g-1}/(g-1)! = [C]$  at  $g = 3$  reads  $\Theta^2/2 = [C]$ , whence  $[Z] = (b^2 + d)[C]$ , and if  $Z$  is  $m$  translates of  $C$  then  $m = b^2 + d$  when  $b = 1$  and  $m = (b^2 + d)/b^2$  after dividing by the class of  $b^2$  translates in general.  $\blacksquare$

| $d$         | 1 | 2 | 3 | 5 | 7 | 11 |
|-------------|---|---|---|---|---|----|
| $m = d + 1$ | 2 | 3 | 4 | 6 | 8 | 12 |

TABLE 2. The number of translates of the Abel–Jacobi curve required at  $n = 3$ ,  $b = 1$ , as a function of the field  $K = \mathbb{Q}(\sqrt{-d})$ , by [Proposition 14.10](#).

*Remark 14.11.* The sheaf used in [\[Mar25b\]](#) on a genus three Jacobian is  $\mathcal{I}_{\cup C_i}(\Theta)$ , the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi curve. That is exactly the shape [Proposition 14.10](#) forces, with the number of translates determined by the field. We take this as evidence that the framework of this section is the right one and not merely a consistent one.

**14.5. The third ingredient, in every dimension.** [Theorem 14.7](#) says what a secant sheaf must look like. It does not say that one exists, and at  $n \geq 4$  that is the question the section has so far left open. We settle it here, and the answer is that the object exists on every polarised abelian variety, in every dimension, for every imaginary quadratic field, and that it can be assembled out of powers of a single line bundle.

Two observations do the work. The first is that the object wanted is a rational point of a plane, and a plane carries no preferred scale. If a positive multiple of a rational point of  $P_t$  is realised by a sheaf, then a rational point of  $P_t$  is realised by a sheaf, and since the Hodge conjecture is a statement about classes with rational coefficients, replacing  $\omega$  by  $M\omega$  costs nothing. The second is that once one is free to clear denominators, the Chern characters of the powers of one line bundle already span the subring in which the secant plane lives, because the matrix expressing them in the basis  $t^k/k!$  is a Vandermonde matrix on distinct nodes.

What is missing is only the passage from a class in  $K_0$  to a sheaf, and for that the sign of the rank is the whole of the hypothesis.

**Lemma 14.12** (Positive rank is enough). *Let  $Y$  be a smooth projective variety over a field and let  $\alpha \in K_0(Y)$  satisfy  $\text{rk } \alpha > 0$ . Then  $\alpha = [\mathcal{F}]$  for some coherent sheaf  $\mathcal{F}$  on  $Y$ .*

*Proof.* Write  $m = \dim Y$  and  $a = \text{rk } \alpha \geq 1$ , and let  $F^c K_0(Y)$  be the topological filtration, the subgroup generated by the classes  $[\mathcal{O}_Z]$  of closed integral subschemes  $Z \subseteq Y$  of codimension at least  $c$ . Since  $Y$  is smooth, every coherent sheaf has a finite locally free resolution, so  $K_0(Y) = K'_0(Y)$  and the filtration is exhaustive with  $F^{m+1} = 0$ ; moreover  $F^c/F^{c+1}$  maps onto  $\text{CH}^c(Y)$  by sending the class of a sheaf supported in codimension  $c$  to its associated cycle, and  $F^1$  is the kernel of the rank while  $F^1/F^2$  is  $\text{Pic}(Y)$  by the first Chern class [\[Ful98, §15.1 and Ex. 15.1.5\]](#). We construct coherent sheaves  $\mathcal{F}_c$  of rank  $a$  with  $\alpha - [\mathcal{F}_c] \in F^c$ , by induction on  $c$ .

Start at  $c = 2$ . Let  $L$  be a line bundle with  $c_1(L) = c_1(\alpha)$  and set  $\mathcal{F}_2 = L \oplus \mathcal{O}_Y^{\oplus(a-1)}$ . Then  $\alpha - [\mathcal{F}_2]$  has rank zero and first Chern class zero, hence lies in  $F^2$ .

For the inductive step let  $\delta = \alpha - [\mathcal{F}_c] \in F^c$  with  $c \geq 2$ . Modulo  $F^{c+1}$  the class  $\delta$  is a finite sum  $\sum_j m_j [\mathcal{O}_{Z_j}]$  with  $m_j \in \mathbb{Z}$  and  $\text{codim } Z_j = c$ . The terms with  $m_j > 0$  are removed by replacing  $\mathcal{F}_c$  with  $\mathcal{F}_c \oplus \bigoplus_{m_j > 0} \mathcal{O}_{Z_j}^{\oplus m_j}$ , which changes neither the rank nor the codimension of the remaining difference. For a term with  $m_j < 0$  we must subtract, and here the rank is used. The restriction  $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$  is a quotient of  $\mathcal{F}_c$  and has rank  $a \geq 1$  at the generic point of the integral scheme  $Z_j$ ; choosing a subsheaf of generic corank one and dividing by it gives a surjection  $\mathcal{F}_c \twoheadrightarrow \mathcal{Q}$  with  $\mathcal{Q}$  of rank one on  $Z_j$ . Then  $[\mathcal{Q}] \in F^c$  with associated cycle  $[Z_j]$ , so  $[\mathcal{Q}] - [\mathcal{O}_{Z_j}] \in F^{c+1}$ . The kernel of that surjection is a coherent sheaf, still of rank  $a$  because  $\mathcal{Q}$  is supported in positive codimension, and its class is  $[\mathcal{F}_c] - [\mathcal{Q}]$ . Performing this  $|m_j|$  times, and once for each  $j$ , produces  $\mathcal{F}_{c+1}$  with  $\alpha - [\mathcal{F}_{c+1}] \in F^{c+1}$  and  $\text{rk } \mathcal{F}_{c+1} = a$ . At  $c = m + 1$  the difference lies in  $F^{m+1} = 0$ , so  $\alpha = [\mathcal{F}_{m+1}]$ .  $\blacksquare$

**Theorem 14.13** (Secant sheaves exist in every dimension). *Let  $X$  be an abelian variety of dimension  $n$ , let  $t \in \text{NS}(X) \otimes \mathbb{Q}$  be a polarisation and let  $d > 0$ . Let  $p$  be a rational point of the secant plane  $P_t$  whose  $u_t$  coordinate is positive. Then there are a positive integer  $M$ , an integer  $D > 0$  with  $Dt \in \text{NS}(X)$ , integers  $n_0, \dots, n_n$  and a coherent sheaf  $\mathcal{F}$  on  $X$  such that, setting*

$$\alpha = \sum_{j=0}^n n_j [\mathcal{O}_X(jDt)] \in K_0(X), \quad (31)$$

one has  $\alpha = [\mathcal{F}]$  in  $K_0(X)$ ,  $\text{ch}(\mathcal{F}) = Mp$  and  $\text{rk } \mathcal{F} = Ma > 0$ , where  $a$  is the  $u_t$  coordinate of  $p$ . The integers  $M$  and  $n_j$  depend only on  $n$ ,  $d$  and the coordinates of  $p$ , and are obtained by inverting one Vandermonde matrix. Moreover  $Mp$  is again a rational point of  $P_t$ ; applying the statement to two linearly independent rational points of  $P_t$  produces a pair of coherent sheaves whose Chern characters span  $P_t$ .

*Proof.* Scaling  $p$  by a positive integer changes neither the hypothesis nor the conclusion, so we may assume  $p = au_t + bv_t$  with  $a, b \in \mathbb{Z}$  and  $a > 0$ . Choose  $D > 0$  with  $Dt \in \text{NS}(X)$ , so that  $\mathcal{O}_X(jDt)$  is a line bundle for every  $j \in \mathbb{Z}$ .

Since  $t$  is a polarisation,  $t^n \neq 0$  and  $t^{n+1} = 0$  in  $H^{\text{ev}}(X, \mathbb{Q})$ , so the  $n+1$  classes  $t^k/k!$  with  $0 \leq k \leq n$  are a basis of the subring  $\mathbb{Q}[t]$ . In that basis

$$p = \sum_{k=0}^n c_k \frac{t^k}{k!}, \quad c_k = \begin{cases} a(-d)^{k/2}, & k \text{ even}, \\ b(-d)^{(k-1)/2}, & k \text{ odd}, \end{cases}$$

by Proposition 15.26(i), and all the  $c_k$  are integers. On the other side,

$$\text{ch}(\mathcal{O}_X(jDt)) = e^{jDt} = \sum_{k=0}^n (jD)^k \frac{t^k}{k!},$$

so in the same basis the  $n+1$  vectors  $\text{ch}(\mathcal{O}_X(jDt))$  for  $j = 0, \dots, n$  are the columns of the Vandermonde matrix  $((jD)^k)_{0 \leq j, k \leq n}$  on the  $n+1$  distinct nodes  $0, D, 2D, \dots, nD$ . Its determinant is  $\prod_{0 \leq i < j \leq n} (jD - iD) = D^{n(n+1)/2} \prod_{0 \leq i < j \leq n} (j - i) \neq 0$ , so the system

$$\sum_{j=0}^n x_j (jD)^k = c_k \quad (0 \leq k \leq n)$$

has a unique solution  $x \in \mathbb{Q}^{n+1}$ . Let  $M$  be the least common multiple of the denominators of the  $x_j$  and put  $n_j = Mx_j \in \mathbb{Z}$ . With  $\alpha$  as in (31),

$$\mathrm{ch}(\alpha) = \sum_j n_j e^{jDt} = \sum_{k=0}^n \left( \sum_j Mx_j (jD)^k \right) \frac{t^k}{k!} = M \sum_{k=0}^n c_k \frac{t^k}{k!} = Mp.$$

In particular  $\mathrm{rk} \alpha = \mathrm{ch}_0(\alpha) = Mc_0 = Ma > 0$ , so [Lemma 14.12](#) gives a coherent sheaf  $\mathcal{F}$  with  $[\mathcal{F}] = \alpha$ . The Chern character is a function of the class in  $K_0(X)$ , whence  $\mathrm{ch}(\mathcal{F}) = Mp$  and  $\mathrm{rk} \mathcal{F} = Ma$ . Finally  $Mp = (Ma)u_t + (Mb)v_t$  is a rational point of  $P_t$ , and two linearly independent rational points stay linearly independent after scaling, which gives the last assertion.  $\blacksquare$

*Remark 14.14* (The multiple is real, and it is the factorials). The integer  $M$  is not an artifact of the proof. The lattice spanned by the Chern characters of the powers of a single line bundle is spanned equally well by the classes  $(e^t - 1)^k$  for  $0 \leq k \leq n$ , because the  $(n+1)$ st finite difference of a polynomial of degree  $n$  vanishes; and  $(e^t - 1)^k = t^k + \dots$  has leading term  $t^k$ , not  $t^k/k!$ . The secant plane, by contrast, is spanned by classes whose coordinates in the basis  $t^k/k!$  are integers. The index between the two lattices is what  $M$  measures, and it is created by the factorials in  $\mathrm{ch}$ . Item (XXI) of [Appendix D](#) computes the least  $M$  exactly for  $1 \leq n \leq 10$ , every squarefree  $d \leq 11$  and a range of  $a$  and  $b$ , together with the witness (31) in each case. At  $n = 4$ ,  $d = 3$ ,  $a = b = 1$  the least multiple is  $M = 6$  and the witness is

$$\alpha = -23 [\mathcal{O}_X] + 66 [\mathcal{O}_X(t)] - 54 [\mathcal{O}_X(2t)] + 20 [\mathcal{O}_X(3t)] - 3 [\mathcal{O}_X(4t)],$$

of rank  $6 = Ma$ ; at  $n = 8$  the largest multiple over the range tested is 2016, and  $M = 1$  occurs in none of the seventy two cases tested once  $n \geq 7$ . So the freedom to rescale is used, and used essentially.

The same computation checks, independently and in the exterior algebra of a symplectic lattice, that  $\theta^k/k!$  has integer coefficients for a principal polarisation. That is the reason the coordinates  $c_k$  are integers to begin with, and it is Poincaré's formula in the form that does not need a curve.

*Remark 14.15* (What this closes, and what it does not). [Theorem 14.13](#) settles the third of the four demands of [Question 13.7](#), for every  $n \geq 2$ , every imaginary quadratic field and every polarised abelian  $n$ -fold, and it does so with line bundles alone. Three qualifications belong with it, and they are not small.

First, the theorem produces a sheaf and nothing more. It says nothing about stability, about the rank being one, about the Ext groups, or about semiregularity, and the construction of [Lemma 14.12](#) is not designed to give any of them: the sheaf is assembled from line bundles and from quotients supported on subvarieties, and its only controlled invariant is its class. [Proposition 9.11](#) applies to it as to any other object carrying the class, and bounds  $\sum_i \dim \mathrm{Ext}^i(\mathcal{F}, \mathcal{F})$  from below.

This does not collide with [Theorem 18.5](#), for the following reason. That theorem concerns a perfect complex on the  $2n$ -fold  $A$  whose Chern character is an effective combination  $\sum_i m_i e^{\lambda_i}$  with every  $m_i$  positive, and it rules such a complex out because the norm form of  $K$  is positive definite. The class (31) lives on the  $n$ -fold  $X$ , not on  $A$ , and its coefficients alternate in sign; the witness at  $n = 4$  has three negative ones. Positivity is exactly what is given up, and [Lemma 14.12](#) is what replaces it, by requiring only that the total rank be positive.

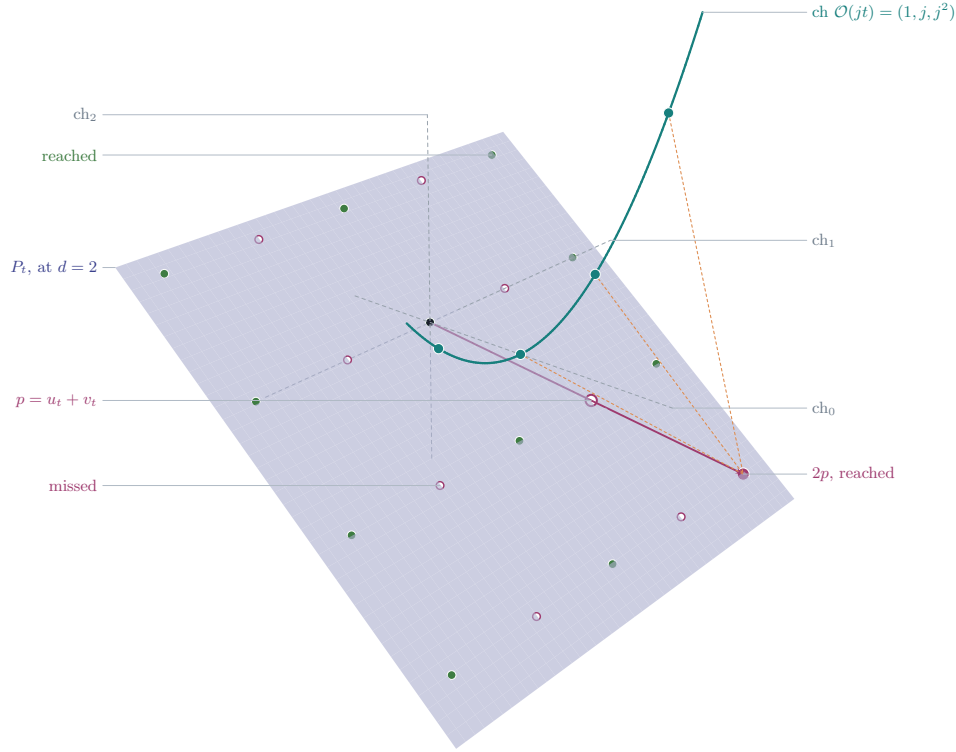


FIGURE 12. [Theorem 14.13](#) at  $n = 2$  and  $d = 2$ , in the coordinates  $(ch_0, ch_1, ch_2)$  of the basis  $1, t, t^2/2$ , the third axis at two fifths scale. The characters of the powers of one line bundle lie on the moment curve  $j \mapsto (1, j, j^2)$ , and the Vandermonde determinant says that any three of its points are affinely independent. The integral points of  $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t$  split by the criterion  $ch_2 \equiv ch_1 \pmod{2}$  into those the line bundles reach, drawn filled, and those they miss, drawn open. Doubling  $p = u_t + v_t$  lands on the lattice, at the witness  $-3[\mathcal{O}_X] + 8[\mathcal{O}_X(t)] - 3[\mathcal{O}_X(2t)]$  of the three dashed leaders.

Second, the rank is large. [Proposition 14.10](#) at  $n = 3$  produces a rank one sheaf, which is what Markman's argument uses and what [Remark 14.11](#) matches against the literature. The sheaf of [Theorem 14.13](#) has rank  $Ma$ , and  $M$  grows quickly with  $n$ . Whether the rank can be brought down to one at  $n \geq 4$ , equivalently whether the lattice generated by the classes  $t^k/k!$  is realised by algebraic cycles on a general abelian  $n$ -fold rather than only after clearing denominators, is a separate and open question; on a Jacobian it is answered by the Abel–Jacobi cycles and Poincaré's formula, and  $M = 1$  can be taken there.

Third, the fourth ingredient is untouched, and it is the one that matters. A pair of secant sheaves gives a class on  $X \times \hat{X}$ ; making that class remain of Hodge type across the moduli of polarised abelian  $2n$ -folds of Weil type is a variational statement, and nothing here bears on it. [Remark 14.17](#) says what that step requires and stands unchanged. We therefore do not claim the Hodge conjecture for abelian  $2n$ -folds of Weil type at any  $n \geq 4$ ; what has changed is that three of the four ingredients of [Question 13.7](#) are now theorems rather than questions.

#### 14.6. What is supplied and what is not.

**Theorem 14.16** (Three of the four ingredients, in general). *For every  $n \geq 2$ , every squarefree  $d > 0$  and every polarised abelian  $n$ -fold  $X$  carrying a nondegenerate  $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$ , the data required by [Question 13.7](#) exist and are explicit in the first three items:*

- (a) *the abelian  $2n$ -fold of Weil type is  $A = X \times \hat{X}$  with the  $K$ -action (28), by [Theorem 14.2](#);*
- (b) *the plane is  $P_t$  of [Definition 14.4](#) with  $t = \beta$ , it is spanned by Hodge classes and it meets the spinor variety in exactly the two conjugate points  $[e^{\pm st}]$ , by [Theorem 14.5](#);*
- (c) *a pair of coherent sheaves  $\mathcal{F}_1, \mathcal{F}_2$  on  $X$  whose Chern characters are linearly independent rational points of  $P_t$  exists, by [Theorem 14.13](#), and each may be taken to have the class of an explicit integer combination of powers of one line bundle.*

*The invariants of any such sheaf are forced by the explicit system of [Theorem 14.7](#), which imposes  $n - 1$  linear conditions on a Chern character already constrained to  $\mathbb{Q}[t]$ , and are unobstructed numerically by (30).*

*Proof.* (a), (b) and (c) are the cited theorems. For the count in the last sentence, a Chern character in the subring  $\mathbb{Q}[t] \subset H^{\mathrm{ev}}(X, \mathbb{Q})$  is a polynomial of degree at most  $n$  in  $t$ , so lies in a space of dimension  $n + 1$ , while  $P_t$  has dimension two. ■

*Remark 14.17* (What is not supplied). The fourth ingredient of [Question 13.7](#) is untouched by anything in this section, and it is the hardest. Granting the pair of secant sheaves  $\mathcal{F}_1, \mathcal{F}_2$  on  $X$  that [Theorem 14.13](#) now supplies, one must still show that the characteristic class of  $\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$  remains of Hodge type under every deformation of  $X \times \hat{X}$  as a polarised abelian  $2n$ -fold of Weil type, and that the object deforms with the variety, as a twisted sheaf to first order. That step is carried out by Markman at  $n = 3$  [[Mar25b](#)], and it rests on a semiregularity analysis. What we add to it is negative: [Theorem 18.29](#) identifies exactly which Hochschild classes the weakened form of that analysis must kill for a secant object, and [Theorem 18.34](#) shows that the only candidate proposed above  $n = 3$  does not kill them. Without the step one obtains algebraicity only on the split locus  $X \times \hat{X}$ , where the Weil classes are of much less interest, since the general abelian  $2n$ -fold of Weil type is simple and is not of that form.

We therefore do not claim a proof of the Hodge conjecture for abelian  $2n$ -folds of Weil type at any  $n \geq 4$ , and nothing in this section should be read as one. What we claim is that three of the four ingredients are now written down for every  $n$ , the third of them with an explicit witness in line bundles, and that the fourth is isolated.

## 15. BASE POINTS IN EVERY WEIL FAMILY

**15.1. The split member, in closed form.** Before turning to what is missing, we record what the family gives outright.<sup>29</sup> On the split member the Weil classes are not merely known to be algebraic; they can be written down as polynomials in three divisor classes, and the polynomial is the same for every  $n$  and every  $d$ .

*Notation 15.1.* On  $A = X \times \hat{X}$  write

$$\beta \in H^2(A, \mathbb{Q}), \quad \hat{\beta} \in H^2(A, \mathbb{Q}), \quad \ell = c_1(\mathcal{P}) \in H^2(A, \mathbb{Q})$$

<sup>29</sup>The split locus is thin: a general abelian  $2n$ -fold of Weil type is simple, so it is not isogenous to a product at all, and the members  $X \times \hat{X}$  form a subvariety of  $\mathcal{D}_{n,n}$  of positive codimension. That is exactly why [Corollary 15.4](#) is a base case and not a solution, and why [Question 15.13](#) is about propagation.

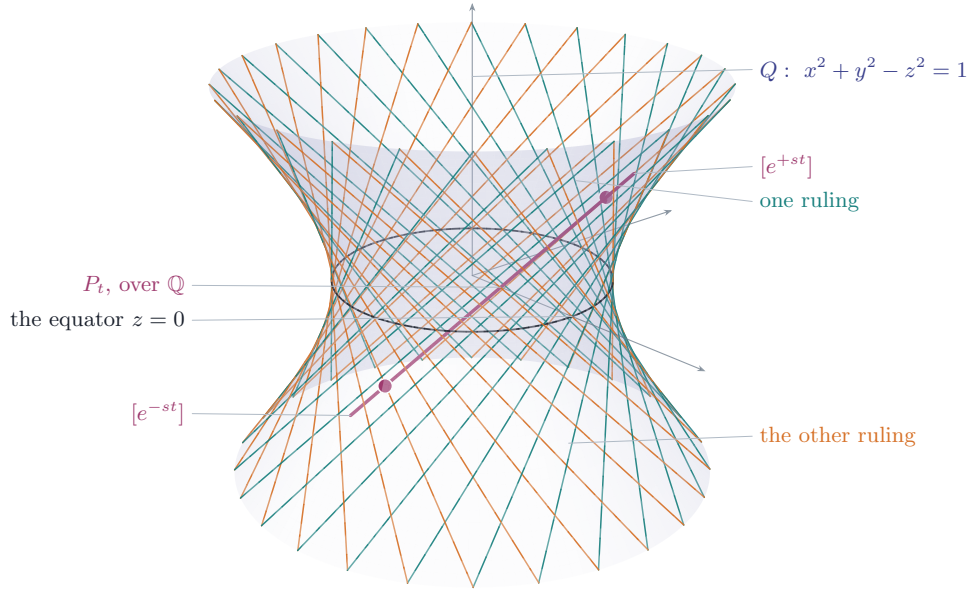


FIGURE 13. The geometry of [Theorem 14.5](#), drawn in the smallest case in which the even spinor variety is a surface, where it is a smooth quadric with its two rulings. The line  $P_t$  of [Definition 14.4](#) is rational, and it meets the variety in exactly the two points  $[e^{\pm st}]$ , which are conjugate over  $K$  and not individually rational. That is the configuration the Weil line demands: a class defined over  $\mathbb{Q}$  cannot pick out one of the two points, so the construction must carry the pair, and the pair spans a plane rather than a line. [Proposition 13.1](#) is the same statement on the cohomological side, and [Figure 11](#) places it among the hypotheses a construction can satisfy.

for the pullback of the polarisation of  $X$ , the pullback of the dual polarisation of  $\widehat{X}$ , and the class of the Poincaré bundle. All three are divisor classes, so every polynomial in them is algebraic. Put

$$\gamma = d\beta - \widehat{\beta}. \quad (32)$$

**Theorem 15.2** (The Weil line of the split member). *Let  $A = X \times \widehat{X}$  carry the  $K$ -action (28) determined by a principal  $\beta$ , and let  $V_{\pm} \subset H^1(A, \mathbb{C})$  be the eigenspaces of  $\iota(\sqrt{-d})$ . Then*

- (i)  $\gamma \mp \sqrt{-d}\ell$  lies in  $\wedge^2 V_{\pm}$ , so it is an eigenclass for the character  $\tau \mapsto \tau^2$  on  $H^2(A, \mathbb{C})$  respectively for its conjugate;
- (ii) the generators of  $\wedge^{2n} V_{\pm}$  are

$$\omega_{\pm} = \frac{(-1)^{n(n-1)/2}}{n! d^n} (\gamma \mp \sqrt{-d}\ell)^n; \quad (33)$$

- (iii) consequently the Weil line of  $A$  is

$$W(A) = \left\langle \Re(\gamma - \sqrt{-d}\ell)^n, \Im(\gamma - \sqrt{-d}\ell)^n \right\rangle_{\mathbb{Q}},$$

and every Weil class on  $A$  is a polynomial with rational coefficients in the three divisor classes  $\beta, \widehat{\beta}, \ell$ , hence is algebraic.

*Proof.* Write  $U = H^1(X, \mathbb{Q})$  with basis  $x_1, \dots, x_{2n}$  chosen so that  $\beta = \sum_{j \leq n} x_j \wedge x_{n+j}$ , let  $\xi_1, \dots, \xi_{2n}$  be the dual basis of  $H^1(\hat{X}, \mathbb{Q}) = U^\vee$ , and let  $B: U \rightarrow U^\vee$  be the isomorphism induced by  $\beta$ , so that  $Bx_j = \xi_{n+j}$  and  $Bx_{n+j} = -\xi_j$ . The endomorphism (28) acts on homology, and its transpose acts on  $H^1(A, \mathbb{Q}) = U \oplus U^\vee$  by  $u \mapsto -Bu$  and  $\phi \mapsto d B^{-1}\phi$ ;<sup>30</sup> its square is  $-d$ ; so, writing  $\delta = \sqrt{-d}$ ,

$$V_\pm = \left\{ u \pm \frac{\delta}{d} Bu : u \in U_{\mathbb{C}} \right\}, \quad (34)$$

since  $M^*(u \pm \frac{\delta}{d} Bu) = -Bu \pm \delta u = \pm \delta(u \pm \frac{\delta}{d} Bu)$ .

(i) Take the sum over  $j \leq n$  of the wedges of the two eigenvectors attached to  $x_j$  and to  $x_{n+j}$ :

$$\sum_{j=1}^n \left( x_j \pm \frac{\delta}{d} Bx_j \right) \wedge \left( x_{n+j} \pm \frac{\delta}{d} Bx_{n+j} \right) = \beta \pm \frac{\delta}{d} \Sigma - \frac{1}{d} \hat{\beta},$$

using  $\delta^2 = -d$ , where  $\Sigma = \sum_j (x_j \wedge Bx_{n+j} + Bx_j \wedge x_{n+j})$  and  $\sum_j Bx_j \wedge Bx_{n+j} = \sum_j \xi_{n+j} \wedge (-\xi_j) = \hat{\beta}$ . Now

$$\Sigma = \sum_{j=1}^n (x_j \wedge (-\xi_j) + \xi_{n+j} \wedge x_{n+j}) = - \sum_{i=1}^{2n} x_i \wedge \xi_i = -\ell.$$

So the displayed sum is  $\frac{1}{d}(d\beta - \hat{\beta} \mp \delta\ell) = \frac{1}{d}(\gamma \mp \delta\ell)$ . It is a sum of wedges of pairs of vectors of  $V_\pm$ , hence lies in  $\wedge^2 V_\pm$ , which proves (i); the statement about characters is [Proposition 5.1](#) at  $(r, s) = (2, 0)$  and  $(0, 2)$ .

(ii) By (i) the  $n$ -th power  $(\gamma \mp \delta\ell)^n$  lies in  $\wedge^{2n} V_\pm$ , which is a line by [Proposition 3.7](#). It is nonzero: expanding  $\gamma \mp \delta\ell = d\beta - \hat{\beta} \mp \delta\ell$ , the only term of the  $n$ -th power lying in  $\wedge^{2n} U$  is  $d^n \beta^n$ , and  $\beta^n \neq 0$  because  $\beta$  is nondegenerate; the remaining terms all involve at least one  $\xi$ , so no cancellation can occur. Hence  $(\gamma \mp \delta\ell)^n$  spans  $\wedge^{2n} V_\pm$  and is a scalar multiple of  $\omega_\pm = \wedge_{i=1}^{2n} (x_i \pm \frac{\delta}{d} Bx_i)$ . To find the scalar, compare the components in  $\wedge^{2n} U$ : that of  $\omega_\pm$  is  $x_1 \wedge \dots \wedge x_{2n}$ , while

$$d^n \beta^n = d^n n! \bigwedge_{j=1}^n (x_j \wedge x_{n+j}) = d^n n! (-1)^{n(n-1)/2} x_1 \wedge \dots \wedge x_{2n},$$

the sign being that of the permutation carrying  $(1, n+1, 2, n+2, \dots, n, 2n)$  to  $(1, 2, \dots, 2n)$ . Inverting gives (33).

(iii) The two displayed classes are the real and imaginary parts of  $(\gamma - \delta\ell)^n$ , which up to the nonzero rational factor in (33) are  $\omega_- + \omega_+$  and  $(\omega_- - \omega_+)/\delta$ ; by [Proposition 3.7](#)(i) these span  $W(A)$  over  $\mathbb{Q}$ . Expanding  $(\gamma - \delta\ell)^n$  by the binomial theorem and separating even from odd powers of  $\delta$  writes each of them as a rational polynomial in  $\beta, \hat{\beta}, \ell$ , because  $\delta^{2m} = (-d)^m$  and  $\delta^{2m+1} = (-d)^m \delta$ . Products of divisor classes are algebraic, so both are algebraic.  $\blacksquare$

**Example 15.3.** At  $n = 2$  the formula reads

$$\omega_1 = d\ell^2 - \gamma^2, \quad \omega_2 = 2\gamma\ell,$$

<sup>30</sup>In the basis  $u_1, \dots, u_{2n}$  of  $H_1(X, \mathbb{Q})$  dual to the  $x_i$ , the map  $\beta: H_1(X) \rightarrow H^1(X)$  sends  $u_j \mapsto x_{n+j}$  and  $u_{n+j} \mapsto -x_j$ , so (28) sends  $u_j \mapsto dx_{n+j}$ ,  $u_{n+j} \mapsto -dx_j$ ,  $x_{n+j} \mapsto -u_j$  and  $x_j \mapsto u_{n+j}$ ; pairing with the dual basis gives the stated action on  $H^1(A, \mathbb{Q})$ , where  $\xi_i$  is the class dual to  $x_i$  regarded as an element of  $H_1(\hat{X}, \mathbb{Q}) = H^1(X, \mathbb{Q})$ .

and at  $n = 3$ ,

$$\omega_1 = d\gamma\ell^2 - \frac{1}{3}\gamma^3, \quad \omega_2 = \gamma^2\ell - \frac{1}{3}d\ell^3,$$

with  $\gamma = d\beta - \widehat{\beta}$  throughout.<sup>31</sup> Both cases, and the expansion for every  $n \leq 5$  and several  $d$ , are confirmed in `split_locus.py`; see item (X) of [Appendix D](#).

**Corollary 15.4** (Algebraicity on the split locus). *For every  $n \geq 1$ , every squarefree  $d > 0$  and every principally polarised abelian  $n$ -fold  $X$ , the Weil classes of  $A = X \times \widehat{X}$  with the  $K$ -action (28) are algebraic, and an algebraic representative is given by (33) as an explicit rational combination of intersections of three divisors.*

*Proof.* [Theorem 15.2](#)(iii), together with the Lefschetz theorem on  $(1, 1)$  classes, which makes  $\beta, \widehat{\beta}, \ell$  algebraic, and the fact that the algebraic classes form a subring of  $H^*(A, \mathbb{Q})$ . ■

*Remark 15.5.* [Corollary 15.4](#) does not contradict [Theorem 1.3](#). At the split point  $\mathrm{NS}(A) \otimes \mathbb{Q}$  contains the three independent classes  $\beta, \widehat{\beta}, \ell$ , so the hypothesis  $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$  of [Proposition 3.10](#) fails, and this is the situation of [Remark 20.5](#). What [Theorem 1.3](#) forbids is a  $K^\times$ -eigenclass for the norm character meeting the Weil line, and  $\gamma \mp \sqrt{-d}\ell$  is an eigenclass for  $\tau^2$  and for  $\bar{\tau}^2$ , not for the norm. Indeed (33) is the first place in this paper where a class in the right eigenspace is exhibited at all, and it is exhibited only because the split member has enough divisors to build one.

**15.2. Products, and a base case in every dimension.** The split member is one way to produce a member with enough divisors. There is a second, and it behaves well under the one operation the Weil line respects. Throughout this subsection  $K$  is fixed and all Weil structures are for that  $K$ .

**Lemma 15.6** (Products of Weil type). *Let  $(A_1, \iota_1, \eta_1)$  be of  $(K, -1, n_1)$ -Weil type and  $(A_2, \iota_2, \eta_2)$  of  $(K, -1, n_2)$ -Weil type. Then  $A = A_1 \times A_2$ , with the diagonal action  $\iota = \iota_1 \times \iota_2$  and the product polarisation  $\eta = p_1^*\eta_1 + p_2^*\eta_2$ , is of  $(K, -1, n_1 + n_2)$ -Weil type.*

*Proof.* Write  $V_i = H^1(A_i, \mathbb{Q})$ , so  $V = H^1(A, \mathbb{Q}) = V_1 \oplus V_2$  as  $K$ -vector spaces and  $\dim_K V = 2n_1 + 2n_2$ . The Rosati involution of the product polarisation is the product of the two Rosati involutions, so it restricts on  $K$  to conjugation, which is the sign  $-1$  of [Definition 3.1](#). For the multiplicities, the Hodge decomposition of  $V$  is the direct sum of those of  $V_1$  and  $V_2$ , and the eigenspace decomposition is likewise  $V_\pm = V_{1,\pm} \oplus V_{2,\pm}$ . Hence

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_{1,+}^{1,0} + \dim_{\mathbb{C}} V_{2,+}^{1,0} = n_1 + n_2,$$

and the same for the other three, which is [Lemma 3.5](#) for  $n = n_1 + n_2$ . ■

The Weil line is the top exterior power over  $K$ , and a top exterior power of a direct sum is the tensor product of the top exterior powers. That is the whole content of the next theorem; what has to be checked is what it does to the rational generators, since those are defined through the real structure and not over  $K$ .

By [Proposition 3.7](#)(i) and (iii) the Weil line  $W(A)$  is free of rank one over  $K$ , and the basis  $\omega_1, \omega_2$  of [Theorem 4.9](#) identifies  $W(A)$  with  $K$  itself by  $a\omega_1 + b\omega_2 \mapsto a + b\sqrt{-d}$ . Write  $\omega(A) = \omega_1(A) + \sqrt{-d}\omega_2(A)$  for the resulting  $K$ -valued invariant.

<sup>31</sup>An earlier version printed  $\gamma\ell^2 - \frac{1}{3}\gamma^3$  for  $\omega_1$  at  $n = 3$ , which lies in the Weil plane only when  $d = 1$ ; the displayed formulas are now checked as printed, for several  $d$ , by item (X).

**Theorem 15.7** (The Weil class is multiplicative). *With the notation of Lemma 15.6, and cup product on  $H^*(A, \mathbb{Q})$  taken after pullback along the two projections,*

$$\begin{aligned}\omega_1(A) &= \frac{1}{2}(\omega_1(A_1)\omega_1(A_2) - \omega_2(A_1)\omega_2(A_2)), \\ \omega_2(A) &= \frac{1}{2}(\omega_1(A_1)\omega_2(A_2) + \omega_2(A_1)\omega_1(A_2)),\end{aligned}\tag{35}$$

*that is  $\omega(A_1)\omega(A_2) = 2\omega(A_1 \times A_2)$  in the  $K$ -valued notation. For  $k$  factors,  $\prod_j \omega(A_j) = 2^{k-1}\omega(A_1 \times \cdots \times A_k)$ .*

*Proof.* Let  $\alpha_{\pm}^i$  generate  $\bigwedge^{2n_i} V_{i,\pm}$ . Since  $V_{\pm} = V_{1,\pm} \oplus V_{2,\pm}$ ,

$$\bigwedge^{2n} V_+ = \bigwedge^{2n_1} V_{1,+} \otimes \bigwedge^{2n_2} V_{2,+},$$

so  $\alpha_+ = \alpha_+^1 \wedge \alpha_+^2$  generates it, and likewise  $\alpha_- = \alpha_-^1 \wedge \alpha_-^2$ ; the classes  $\alpha_{\pm}^i$  have even degree  $2n_i$ , so no sign enters when the two blocks of generators are interleaved. By Theorem 4.9(iii),  $\omega_1 = \alpha_+ + \alpha_-$  and  $\omega_2 = \sqrt{-d}(\alpha_+ - \alpha_-)/(-d)$  up to the normalisation fixed there, so it suffices to expand. Writing  $\omega_1^i$  and  $\omega_2^i$  for the generators of the factors,

$$\omega_1^1 \omega_1^2 - \omega_2^1 \omega_2^2 = (\alpha_+^1 + \alpha_-^1)(\alpha_+^2 + \alpha_-^2) + (\alpha_+^1 - \alpha_-^1)(\alpha_+^2 - \alpha_-^2) = 2(\alpha_+^1 \alpha_+^2 + \alpha_-^1 \alpha_-^2) = 2\omega_1(A),$$

the cross terms cancelling in pairs, and the same computation with one factor conjugated gives the second identity of (35). In the  $K$ -valued notation the two identities together say  $\omega(A_1)\omega(A_2) = 2\omega(A)$ , since the map  $a\omega_1 + b\omega_2 \mapsto a + b\sqrt{-d}$  carries the right-hand sides of (35) to the real and imaginary parts of the product. The statement for  $k$  factors follows by induction, each application contributing one factor of 2. The identities are verified in exact arithmetic over  $\mathbb{Q}(\sqrt{-d})$  for all  $n_1, n_2 \leq 3$  and for the four-factor products listed in Appendix D, item (XV). ■

**Corollary 15.8** (Algebraicity propagates along products). *If the Weil classes of  $A_1$  and of  $A_2$  are algebraic, then so are those of  $A_1 \times A_2$ , and an algebraic representative is the rational combination of external products given by (35).*

*Proof.* Algebraic classes are closed under pullback along the projections and under cup product, and (35) expresses  $\omega_1(A)$  and  $\omega_2(A)$  in those operations applied to the  $\omega_j(A_i)$ . ■

The base case costs nothing.

**Lemma 15.9** (Dimension two). *Let  $A_0$  be of  $(K, -1, 1)$ -Weil type, so  $\dim A_0 = 2$ . Then  $W(A_0) \subset H^2(A_0, \mathbb{Q})$  and both generators are algebraic.*

*Proof.* The Weil classes lie in  $H^{2n} = H^2$  and are of type  $(1, 1)$  by Proposition 3.7(ii), so they are algebraic by the Lefschetz theorem on  $(1, 1)$  classes. No hypothesis on  $A_0$  beyond the Weil structure is used. ■

**Theorem 15.10** (An algebraic Weil class in every dimension). *Let  $n \geq 1$ , let  $d > 0$  be squarefree, and let  $A_1, \dots, A_n$  be abelian surfaces of  $(K, -1, 1)$ -Weil type. Then  $A = A_1 \times \cdots \times A_n$  is of  $(K, -1, n)$ -Weil type and its Weil classes are algebraic; explicitly, by Theorem 15.7 and Lemma 15.9 they are  $2^{-(n-1)}$  times the real and imaginary parts of  $\prod_{j=1}^n (\omega_1(A_j) + \sqrt{-d}\omega_2(A_j))$ , a polynomial with integer coefficients in  $2n$  divisor classes.*

*Proof.* Lemma 15.6 gives the Weil structure, Lemma 15.9 gives algebraicity of each factor's classes, and Corollary 15.8 applied  $n - 1$  times gives the conclusion; the displayed form is Theorem 15.7 for  $k = n$  factors. Each  $\omega_j(A_i)$  is a divisor class by Lemma 15.9, so the product is a polynomial in  $2n$  of them. ■

*Remark 15.11* (What this locus is, and what it is not). [Theorem 15.10](#) is unconditional and holds at every  $n$ , and like [Corollary 15.4](#) and [Theorem 15.31](#) it produces an algebraic Weil class in every dimension while quoting nothing beyond the Lefschetz theorem on  $(1, 1)$  classes. It should not be mistaken for more than it is. The family of polarised abelian  $2n$ -folds of  $(K, -1, n)$ -Weil type has dimension  $n^2$ , and the locus of products of  $n$  surfaces has dimension  $n$ , so for  $n \geq 2$  it is a proper closed subvariety, and a smaller one than the quaternionic locus of [Theorem 15.25](#), whose dimension is  $n(n+1)/2$ . A general member of the family is simple, so it is not isogenous to a product at all, and neither [Theorem 15.10](#) nor [Corollary 15.4](#) touches it. What the product construction adds is a second family of base points, defined for every  $n$  and every  $d$ , at which the class is not merely known to be algebraic but is written down; and it adds the observation that the algebraicity locus is closed under the product operation, so that any future case  $n_0$  propagates to every multiple of  $n_0$  and to every sum of solved cases. Combined with Markman’s theorem at  $n = 2$  [[Mar25a](#), [Mar25b](#)] this gives, at every  $n$ , algebraicity on the locus of products of abelian fourfolds and surfaces of Weil type, of dimension  $2n$  when  $n$  is even. The general member remains outside every one of these loci, and by [Theorem 15.14](#) the gap between them and the general member is exactly the propagation statement that this paper does not prove.

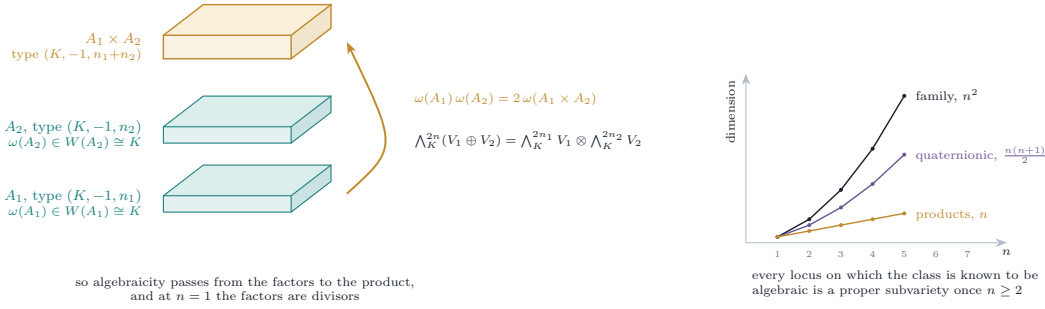


FIGURE 14. [Theorem 15.7](#) and its reach. Left: the Weil line of each factor is a  $K$ -line, the Weil line of the product is the tensor product of the two over  $K$ , and in the  $K$ -valued notation the class is multiplicative up to the factor 2. Algebraicity therefore passes from the factors to the product, and at  $n = 1$  the factors are divisor classes, so a product of  $n$  abelian surfaces of Weil type carries an explicit algebraic Weil class in every dimension. Right: the dimensions of the loci on which the class is known to be algebraic, against the dimension  $n^2$  of the family. Both the quaternionic locus and the product locus are proper subvarieties once  $n \geq 2$ , and a general member lies on neither.

**15.3. The residual statement.** Everything above lets the remaining question be stated in one sentence with no auxiliary hypotheses, and we state it, because a problem in that form is worth more than a programme.

**Setup 15.12** (The family). Fix  $n \geq 2$  and a squarefree  $d > 0$ . Let  $\mathcal{D}_{n,n}$  be the period domain of [Proposition B.8](#) and let

$$\pi: \mathcal{A} \longrightarrow \mathcal{D}_{n,n}$$

be the family of polarised abelian  $2n$ -folds of  $(K, -1, n)$ -Weil type it parametrises, with  $\omega$  the flat section of  $R^{2n}\pi_*\mathbb{Q}$  given fibrewise by the class  $\omega_1$  of (11). Let  $\mathcal{D}^{\text{spl}} \subset \mathcal{D}_{n,n}$  be the locus of split members  $X \times \widehat{X}$  carrying the action (28).

**Question 15.13** (The residual statement). Is  $\omega_s$  algebraic for every  $s \in \mathcal{D}_{n,n}$ ?

**Theorem 15.14** (The reduction). *Fix  $n \geq 2$  and a squarefree  $d > 0$ , and keep the notation of Setup 15.12.*

- (a) *If Question 15.13 has a positive answer, then the Hodge conjecture holds, in every degree, for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with  $\text{Hg}(A) = \text{SU}(V, H)$ .*
- (b) *Conversely, if the Hodge conjecture holds for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type, then Question 15.13 has a positive answer.*
- (c)  *$\mathcal{D}^{\text{spl}}$  is nonempty,  $\omega_s$  is algebraic at every  $s \in \mathcal{D}^{\text{spl}}$  with the explicit representative (33), and  $\mathcal{D}_{n,n}$  is connected.*

So Question 15.13 is the variational Hodge conjecture for one flat class on one connected family, it is necessary as well as sufficient, and its base case is supplied. Theorem 15.31 supplies one in every family, for every discriminant, so the hypothesis of nonemptiness is never the obstacle.

*Proof.* (a) Let  $A$  be of  $(K, -1, n)$ -Weil type with  $\text{Hg}(A) = \text{SU}(V, H)$ . By Proposition B.8 its Hodge structure is a point  $s$  of  $\mathcal{D}_{n,n}$ , so  $A$  is a fibre of  $\pi$ , and the Weil line of  $A$  is generated over  $K$  by  $\omega_s$ . A positive answer makes  $\omega_s$  algebraic, hence makes the whole Weil line algebraic by the argument of Corollary 4.15, and by Theorem 4.14 the Weil line together with the powers of  $\eta$  exhausts  $\text{Hdg}^*(A)$ .

(b) For every  $s$  the class  $\omega_s$  is a Hodge class on the fibre  $A_s$ , by Proposition 3.7(ii), and  $A_s$  is of  $(K, -1, n)$ -Weil type by construction. So the Hodge conjecture for  $A_s$  gives its algebraicity.

(c)  $\mathcal{D}^{\text{spl}}$  is nonempty because the construction (28) applies to any principally polarised abelian  $n$ -fold and produces a point of  $\mathcal{D}_{n,n}$  by Theorem 14.2; algebraicity there is Corollary 15.4; and  $\mathcal{D}_{n,n}$  is connected because it is a bounded symmetric domain, by Proposition B.8. ■

*Remark 15.15* (Why this is the precise target). Three things have been removed from the problem by the preceding sections, and we list what is left after each.

Theorem 4.14 removes the question of how many classes are at issue: one, in one degree, on each fibre. Theorem 1.1 and Theorem 1.3 remove two families of would-be constructions, so no effort need be spent on them. Theorem 15.2 removes the base case: the class is algebraic somewhere in the family, with a representative one can write on a single line.

What is left is propagation, and it is exactly Grothendieck's variational Hodge conjecture for the pair  $(\pi, \omega)$  of Setup 15.12. That statement is open in general; what Theorem 15.14 adds is that for the Hodge conjecture on abelian varieties of Weil type it is not merely sufficient but necessary, so no other route can avoid an equivalent difficulty. Section 16.1 takes the propagation apart. The passage from a formal deformation to an algebraic one, which is usually the awkward half, is supplied there by properness of the Hilbert scheme, and what survives is the infinitesimal half at a single point, or else a bound on the size of the cycles already constructed.

**15.4. The divisorial criterion, and exactly what it reaches.** Theorem 15.2 is a computation on one member of the family. The mechanism behind it is not special to that member, and isolating the mechanism turns the base case into a criterion that can be

tested at any point. The criterion, and the precise determination of where it holds, is the content of this subsection.

**Theorem 15.16** (The divisorial criterion). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type and suppose there exists  $\Theta \in \Lambda^2 V_-$  with*

- (i)  $\Theta$  nondegenerate as an alternating form on  $V_-^\vee$ , equivalently  $\Theta^n \neq 0$ ;
- (ii)  $\Theta$  of Hodge type  $(1, 1)$ ;
- (iii)  $\Theta + \bar{\Theta}$  rational, that is in  $H^2(A, \mathbb{Q})$ .

*Then the Weil classes of  $A$  are algebraic. Explicitly, writing  $g = \Theta + \bar{\Theta}$  and  $\lambda = (\Theta - \bar{\Theta})/\sqrt{-d}$ , both  $g$  and  $\lambda$  are divisor classes and*

$$W(A) = \left\langle \Re(g + \sqrt{-d}\lambda)^n, \Im(g + \sqrt{-d}\lambda)^n \right\rangle_{\mathbb{Q}}, \quad (36)$$

*so every Weil class is a rational polynomial in two divisor classes.*

*Proof.* Conjugation interchanges  $\Lambda^2 V_-$  and  $\Lambda^2 V_+$ , so  $\bar{\Theta} \in \Lambda^2 V_+$ , and both  $g$  and  $\lambda$  are fixed by conjugation, hence real. Rationality of  $g$  is (iii); rationality of  $\lambda$  needs an argument, and it is the following. Fix  $\tau = a + b\sqrt{-d} \in K^\times$  with  $ab \neq 0$ . By Proposition 5.1 the operator  $\iota(\tau)^*$  acts on  $\Lambda^2 V_+$  by  $\tau^2 = a^2 - db^2 + 2ab\sqrt{-d}$  and on  $\Lambda^2 V_-$  by  $\bar{\tau}^2$ , so

$$J := \frac{\iota(\tau)^* - (a^2 - db^2)}{2ab}$$

is a  $\mathbb{Q}$ -rational endomorphism of  $H^2(A, \mathbb{Q})$  preserving  $R$  and acting on  $\Lambda^2 V_\pm$  by  $\pm\sqrt{-d}$ . Applying it to  $g = \Theta + \bar{\Theta}$  gives  $J(g) = -\sqrt{-d}\Theta + \sqrt{-d}\bar{\Theta} = -\sqrt{-d} \cdot \sqrt{-d}\lambda = d\lambda$ , so  $\lambda = J(g)/d$  is rational. Both  $g$  and  $\lambda$  are of type  $(1, 1)$  by (ii) and conjugation symmetry, so both lie in  $\text{NS}(A) \otimes \mathbb{Q}$  by the Lefschetz theorem on  $(1, 1)$  classes, and both are algebraic.

By (i) the form  $\Theta$  on the  $2n$ -dimensional space  $V_-$  has nonzero Pfaffian, so  $\Theta^n$  is a nonzero element of the line  $\Lambda^{2n} V_-$ , and it therefore spans it; conjugating,  $\bar{\Theta}^n$  spans  $\Lambda^{2n} V_+$ . Hence  $\Theta^n + \bar{\Theta}^n$  and  $(\Theta^n - \bar{\Theta}^n)/\sqrt{-d}$  span  $W(A)$  over  $\mathbb{Q}$ , by Proposition 3.7(i). Finally  $2^n \Theta^n = (g + \sqrt{-d}\lambda)^n$ , and expanding by the binomial theorem with  $(\sqrt{-d})^{2m} = (-d)^m$  writes the real and imaginary parts as rational polynomials in  $g$  and  $\lambda$ . Products of divisor classes are algebraic. ■

**Corollary 15.17.** *The split member of Theorem 15.2 satisfies the criterion, with  $\Theta = \gamma - \sqrt{-d}\ell$  and  $\gamma = d\beta - \hat{\beta}$ ; the two divisor classes of (36) are  $2\gamma$  and  $-2\ell$ .*

*Proof.* Theorem 15.2(i) gives  $\Theta \in \Lambda^2 V_+$  for the sign  $\gamma - \sqrt{-d}\ell$ , and the conjugate lies in  $\Lambda^2 V_-$ ;  $\gamma$  and  $\ell$  are divisor classes, hence of type  $(1, 1)$ , so  $\Theta$  is of type  $(1, 1)$  and  $\Theta + \bar{\Theta} = 2\gamma$  is rational. Nondegeneracy is the nonvanishing of  $\Theta^n$ , proved in Theorem 15.2(ii). ■

So the base case is one instance of a general criterion. The next two results say exactly how far the criterion reaches, and both are negative in a way that is worth knowing.

**Proposition 15.18** (The criterion fails at a general point). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$  and  $\text{Hg}(A) = \text{SU}(V, H)$ . Put*

$$R = (\Lambda^2 V_+ \oplus \Lambda^2 V_-) \cap H^2(A, \mathbb{Q}).$$

*Then  $R \cap \text{NS}(A)_\mathbb{Q} = 0$ , and no  $\Theta$  as in Theorem 15.16 exists.*

*Proof.* By Proposition 11.11 the space  $R$  is a rational sub-Hodge structure of  $H^2(A, \mathbb{Q})$  admitting no proper nonzero rational sub-Hodge structure, and by Proposition 11.10 it has  $h^{2,0} = 2\binom{n}{2} = n(n-1)$ , which is positive for  $n \geq 2$ . The intersection  $R \cap \text{NS}(A)_\mathbb{Q}$  is

a rational sub-Hodge structure of  $R$ , so it is 0 or all of  $R$ ; it is of pure type  $(1, 1)$ , and  $R$  is not, so it is 0. A  $\Theta$  as in [Theorem 15.16](#) would give the nonzero element  $\Theta + \bar{\Theta}$  of  $R \cap \text{NS}(A)_{\mathbb{Q}}$ .  $\blacksquare$

**Proposition 15.19** (What the split locus is). *Fix  $n \geq 1$  and a squarefree  $d > 0$ , and let  $A = X \times \hat{X}$  carry the  $K$ -action (28) determined by a nondegenerate  $\beta \in \text{NS}(X) \otimes \mathbb{Q}$  of elementary divisors  $(d_1, \dots, d_n)$ .*

- (i) *Let  $B$  be the Gram matrix of  $\beta$  in a basis of  $U$ , and let  $\hat{\beta}$  be the form on  $U^{\vee}$  with Gram matrix  $-B^{-1}$  in the dual basis, which for principal  $\beta$  is the class of [Notation 15.1](#). The polarisation of  $A$  is  $\eta = \beta + \hat{\beta}/d$ , a positive rational multiple of  $d\beta + \hat{\beta}$ ; its Gram matrix on  $H_1(A, \mathbb{Q}) = U \oplus U^{\vee}$  is the block form  $Q = B \oplus (-B^{-1}/d)$ , and  $M$  is anti-self-adjoint for it.*
- (ii) *In the  $K$ -basis  $u_1, \dots, u_{2n}$  of  $H_1(A, \mathbb{Q})$  the hermitian form of [Lemma 4.5](#) has Gram matrix  $\sqrt{-d}B$ , so*

$$\det H = (-d)^n \det B = (-d)^n (d_1 \cdots d_n)^2,$$

*and  $(-1)^n \det H = d^n (d_1 \cdots d_n)^2$  is a norm from  $K$ . Every split member therefore lies in the Weil family of trivial discriminant, whatever  $\beta$  is chosen.*

- (iii) *The split members form a locus of dimension  $n(n+1)/2$  inside  $\mathcal{D}_{n,n}$ , of codimension  $n(n-1)/2$ , which is zero only for  $n=1$  and is a divisor for  $n=2$ .*
- (iv) *For  $X$  with  $\text{Hg}(X) = \text{Sp}(H^1(X, \mathbb{Q}))$  one has  $\text{NS}(A)_{\mathbb{Q}} = \langle \beta, \hat{\beta}, \ell \rangle$ , of rank three, and in degree  $2n$  the monomials in those three classes span the whole space of Hodge classes of  $A$ .*

*Proof.* (i) In the chosen basis of  $U$  and the dual basis of  $U^{\vee}$  the map  $\beta: U \rightarrow U^{\vee}$  has matrix  $B^{\top} = -B$  and its inverse has matrix  $-B^{-1}$ , so (28) becomes  $M = \begin{pmatrix} 0 & B^{-1} \\ -dB & 0 \end{pmatrix}$ , with  $M^2 = -d$  and  $M^{\top} = \begin{pmatrix} 0 & dB \\ -B^{-1} & 0 \end{pmatrix}$ , using  $B^{\top} = -B$  and  $(B^{-1})^{\top} = -B^{-1}$ . For a block form  $Q = \begin{pmatrix} P & \Lambda \\ -\Lambda^{\top} & S \end{pmatrix}$  the Rosati relation  $M^{\top}Q + QM = 0$  reads  $B\Lambda^{\top} + \Lambda B = 0$ ,  $dB S + P B^{-1} = 0$ ,  $B^{-1}P + d S B = 0$  and  $B^{-1}\Lambda + \Lambda^{\top} B^{-1} = 0$ . Taking  $\Lambda = 0$  and  $P = B$  forces  $S = -B^{-1}/d$  from the second relation, and the third then reads  $-I + I = 0$ . So  $Q = B \oplus (-B^{-1}/d)$ , which is the Gram matrix of  $\beta + \hat{\beta}/d$ . For principal  $\beta$  one may take  $B = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ , so that  $-B^{-1} = B$  and  $\hat{\beta} = \sum_j \xi_j \wedge \xi_{n+j}$  as in [Notation 15.1](#). The computation is repeated entry by entry for  $n \leq 4$  in item (XI) of [Appendix D](#), together with the check that the induced action on  $H^2(A, \mathbb{Q})$  multiplies  $d\beta + \hat{\beta}$  by  $d$  and  $\gamma, \ell$  by  $-d$ , and that  $\beta + d\hat{\beta}$  is not an eigenclass when  $d > 1$ .

(ii) The set  $u_1, \dots, u_{2n}$  is a  $K$ -basis because  $MU = d\beta(U) = U^{\vee}$ , and  $H(u_i, u_j) = Q(Mu_i, u_j) + \sqrt{-d}Q(u_i, u_j)$  by (10); the first term vanishes since  $Mu_i \in U^{\vee}$ ,  $u_j \in U$  and  $Q$  has no cross block by (i), and the second is  $\sqrt{-d}\beta_{ij}$ . Hence the Gram matrix is  $\sqrt{-d}B$  and  $\det H = (\sqrt{-d})^{2n} \det B = (-d)^n \det B$ . The determinant of an alternating matrix is the square of its Pfaffian, here  $(d_1 \cdots d_n)^2$ . Now  $d = N_{K/\mathbb{Q}}(\sqrt{-d})$  is a norm and every square is a norm, and the norms form a subgroup of  $\mathbb{Q}^{\times}$ , so  $d^n (d_1 \cdots d_n)^2$  is a norm.

(iii) A split member is determined by  $X$  together with  $\beta$ , and the polarised moduli of  $X$  has dimension  $\dim \mathcal{A}_n = n(n+1)/2$  while the type of  $\beta$  is discrete. By [Proposition B.8](#) the ambient domain has dimension  $n^2$ , and  $n^2 - n(n+1)/2 = n(n-1)/2$ .

(iv) The Hodge classes of  $A = X \times \hat{X}$  are the invariants of  $\text{Hg}(A) = \text{Hg}(X) = \text{Sp}(U)$  acting on  $\Lambda^*(U \oplus U^{\vee})$ . In degree two the invariants are the three lines spanned by  $\beta, \hat{\beta}$  and  $\ell$ , one from each Kunneth summand  $\Lambda^2 U, U \otimes U^{\vee}$  and  $\Lambda^2 U^{\vee}$ . In degree  $2n$  the dimension

of the invariant space is computed from the primitive decomposition  $\bigwedge^a U = \bigoplus_j \bigwedge_0^{a-2j} U$  together with Schur's lemma, and it agrees with the rank of the set of monomials  $\beta^i \hat{\beta}^j \ell^k$  with  $i + j + k = n$ ; both numbers are computed for  $n \leq 3$  in item (XI) of [Appendix D](#) and agree. ■

*Remark 15.20* (Where this leaves the residual statement). Putting [Proposition 15.19\(ii\)](#) beside [Theorem 15.14](#) changes the shape of what is left. The Weil families for a fixed  $K$  and fixed  $n$  are classified by the discriminant of the hermitian form, an element of  $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$ , which is an infinite group. The split construction of [Section 15.1](#) produces a base point in exactly one of them, the one with trivial discriminant; products of abelian surfaces of Weil type ([Theorem 15.10](#), [Lemma 15.30](#)) reach every discriminant, but only on a locus of dimension  $n$ . So [Question 15.13](#) is really two questions, and they are of different kinds.

- (R1) *Trivial discriminant.* A base point exists, with the explicit cycle (33), and what is missing is propagation off a locus of codimension  $n(n-1)/2$ . At  $n = 2$  that locus is a divisor and the propagation is known, by Markman [[Mar25a](#), [Mar25b](#)] and by Floccari and Fu [[FF26](#)]; combined with [Theorem 15.14\(a\)](#) this recovers the Hodge conjecture for abelian fourfolds of Weil type of trivial discriminant with full Hodge group. At  $n = 3$  it is settled as well, by Markman's theorem on the sixfolds of trivial discriminant [[Mar25a](#), [Mar25b](#)], for every  $K$ . For  $n \geq 4$  it is open, and by [Proposition 20.13](#) it is then the only question left in the imaginary quadratic case.
- (R2) *Nontrivial discriminant.* The construction of this subsection never lands there, by [Proposition 15.19\(ii\)](#), so it supplies no base point; and by [Proposition 15.18](#) no divisorial argument can supply one at a general member. A base point must come from a special member at which  $R$  acquires Hodge classes.

Question (R2) is settled in [Section 15.5](#): the special members required are those with quaternionic multiplication extending  $K$ , they exist in every family, and on them [Theorem 15.16](#) applies. After that only (R1) remains, now with a base point in every family.

*Remark 15.21* (The split locus is larger than a Hodge locus has to be). A component of the Noether–Lefschetz locus of a weight two variation has codimension at most  $h^{2,0}$ , here  $n(n-1)$ . The split locus has codimension  $n(n-1)/2$  by [Proposition 15.19\(iii\)](#), exactly half of that, so it is a component of twice the expected dimension. That is a reflection of [Proposition 15.19\(iv\)](#): at a split point  $R$  acquires two independent Hodge classes at once, namely  $\gamma$  and  $\ell$ , and each of them costs  $h^{2,0}/2$  conditions rather than  $h^{2,0}$ . Whether there are components of the expected codimension carrying only one class, and whether the criterion of [Theorem 15.16](#) can be met on them, is the concrete form of question (R2).

**15.5. A base point in every family.** [Proposition 15.19\(ii\)](#) says the split construction reaches one discriminant class. [Theorem 15.16](#) does not care where its  $\Theta$  comes from, and the source used below is an endomorphism rather than a pair of divisors chosen by hand. It reaches every class, and with it question (R2) of [Remark 15.20](#) is settled.

**Definition 15.22** (Discriminant). For  $(A, \iota, \eta)$  of  $(K, -1, n)$ -Weil type with hermitian form  $H$  as in [Lemma 4.5](#), the *discriminant* is the class of  $(-1)^n \det H$  in  $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$ . The sign makes the representative positive, because  $H$  has signature  $(n, n)$ ; and the class is well defined because changing the  $K$ -basis multiplies  $\det H$  by a norm. Hermitian forms of rank  $2n$  over  $K$  with signature  $(n, n)$  are classified by this class, so it labels the Weil families.

The mechanism behind [Theorem 15.16](#) can be named. A class in  $R$  is the same thing as an endomorphism that conjugates  $K$ .

**Lemma 15.23** (What a class in  $R$  is). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type. The map  $\mu \mapsto g_\mu$ ,  $g_\mu(x, y) = \eta(\mu x, y)$ , is a bijection between*

- (a) *the elements  $\mu \in \text{End}^0(A)$  with  $\mu^\dagger = \mu$  and  $\mu \iota(\tau) = \iota(\bar{\tau}) \mu$  for all  $\tau \in K$ , and*
- (b) *the elements of  $R \cap \text{NS}(A)_\mathbb{Q}$ .*

*Under it,  $\mu$  invertible corresponds to  $g_\mu$  nondegenerate, hence to a nondegenerate  $\Theta$ . If such a  $\mu \neq 0$  exists and the centraliser of  $\iota(K)$  in  $\text{End}^0(A)$  is  $\iota(K)$ , then  $\mu^2 \in \mathbb{Q}^\times$  and  $\iota(K) + \iota(K)\mu$  is a quaternion algebra  $(-d, \mu^2)_\mathbb{Q}$  inside  $\text{End}^0(A)$ .*

*Proof.* A class in  $\text{NS}(A)_\mathbb{Q}$  is a symmetric homomorphism  $A \rightarrow A^\vee$  up to isogeny, that is  $\eta\mu$  with  $\mu^\dagger = \mu$ . The condition  $g_\mu(\iota(\tau)x, y) = g_\mu(x, \iota(\tau)y)$ , which says that  $g_\mu$  is  $K$ -bilinear and hence lies in  $R = \bigwedge_K^2 V$ , reads  $\eta(\mu\iota(\tau)x, y) = \eta(\mu x, \iota(\tau)y) = \eta(\iota(\tau)^\dagger \mu x, y) = \eta(\iota(\bar{\tau})\mu x, y)$ , using [Definition 3.1](#)(iii); so it is exactly  $\mu\iota(\tau) = \iota(\bar{\tau})\mu$ . For the last sentence,  $\mu^2$  commutes with  $\iota(K)$ , hence lies in  $\iota(K)$  by hypothesis, and  $\mu\mu^2\mu^{-1} = \mu^2$  forces it to be fixed by conjugation, so  $\mu^2 \in \mathbb{Q}^\times$ . Then  $\iota(\sqrt{-d})$  and  $\mu$  generate the quaternion algebra with the stated invariants.  $\blacksquare$

So the criterion asks for quaternionic multiplication extending  $K$ . Such abelian varieties exist in abundance, and their invariants can be computed.

**Setup 15.24** (The quaternionic model). Let  $b \in \mathbb{Q}_{>0}$  and let  $B = (-d, b)_\mathbb{Q}$  be the quaternion algebra with basis  $1, i, j, ij$  and  $i^2 = -d, j^2 = b, ij = -ji$ . Let  $V = B^n$  as a left  $B$ -module, of dimension  $4n$  over  $\mathbb{Q}$ , let  $K = \mathbb{Q}(i) \subset B$  act through  $i$ , and fix  $a_1, \dots, a_n \in \mathbb{Q}^\times$  and

$$E(x, y) = \sum_{k=1}^n a_k \text{trd}(\overline{x_k} i y_k), \quad (37)$$

where  $\text{trd}$  is the reduced trace and  $\bar{\phantom{x}}$  the canonical involution of  $B$ .

**Theorem 15.25** (Weil type from quaternionic multiplication). *With the data of [Setup 15.24](#):*

- (i)  *$E$  is alternating and nondegenerate, and its Rosati involution is  $\beta \mapsto i^{-1}\bar{\beta}i$ , which restricts to complex conjugation on  $K$  and fixes  $j$ ; it is positive, the form  $\text{trd}(\beta\beta^\dagger)$  being diagonal with entries  $2, 2d, 2b, 2db$  on the basis;*
- (ii) *in the  $K$ -basis  $e_1, e_1j, \dots, e_n, e_nj$  of  $V$  the hermitian form of [Lemma 4.5](#) is*

$$H = \text{diag}(2a_1d, -2a_1bd, \dots, 2a_nd, -2a_nbd),$$

*of signature  $(n, n)$ ;*

- (iii) *there is a complex structure on  $V_\mathbb{R}$  commuting with  $B$  and polarised by  $E$ , and for every such the resulting abelian variety  $A$  is of  $(K, -1, n)$ -Weil type, the balanced signature being forced and not imposed;*
- (iv) *the discriminant of  $A$  is the class of  $b^n$ ;*
- (v)  *$\mu =$  left multiplication by  $j$  satisfies [Lemma 15.23](#)(a), so  $g = g_\mu$  lies in  $R \cap \text{NS}(A)_\mathbb{Q}$  and is nondegenerate, the divisorial criterion [Theorem 15.16](#) applies, and the Weil classes of  $A$  are algebraic.*

*Proof.* (i)  $\text{trd}(z) = \text{trd}(\bar{z})$  and  $\overline{\bar{x}iy} = \bar{y}i\bar{x} = -\bar{y}ix$  give  $E(y, x) = -E(x, y)$ . Nondegeneracy is the nondegeneracy of the reduced trace pairing. For the Rosati,  $E(\beta x, y) = E(x, \beta^\dagger y)$  for all  $x, y$  reads  $\bar{\beta}i = i\beta^\dagger$ , that is  $\beta^\dagger = i^{-1}\bar{\beta}i$ . Then  $i^\dagger = -i$  and, since  $(-j)i = ij$ , also

$j^\dagger = i^{-1}(ij) = j$ . Positivity is the displayed diagonal, which is positive because  $d > 0$  and  $b > 0$ .

(ii)  $V = Ke_k \oplus Ke_k j$  over all  $k$ , since  $B = K \oplus Kj$ . By (10),  $H(x, y) = E(ix, y) + \sqrt{-d} E(x, y)$ . On one factor,  $E(ix, y) = a \operatorname{trd}(\bar{i}x i y) = -a \operatorname{trd}(\bar{x} i^2 y) = ad \operatorname{trd}(\bar{x} y)$ , so

$$H(x, y) = a[d \operatorname{trd}(\bar{x} y) + \sqrt{-d} \operatorname{trd}(\bar{x} i y)].$$

Now  $\operatorname{trd}(1) = 2$ ,  $\operatorname{trd}(i) = \operatorname{trd}(j) = \operatorname{trd}(ij) = 0$ ,  $\bar{j}j = -b$  and  $\bar{j}ij = ib$ , which give  $H(1, 1) = 2ad$ ,  $H(j, j) = -2abd$  and  $H(1, j) = 0$ . Since  $b > 0$  the two entries of each block have opposite signs, so the signature is  $(n, n)$ .

(iii) A complex structure commuting with  $B$  and compatible with  $E$  is the same as a subspace  $V^{1,0} \subset V_{\mathbb{C}}$ , stable under  $B$ , with  $V_{\mathbb{C}} = V^{1,0} \oplus \bar{V}^{1,0}$  and the two Riemann relations. Being  $K$ -stable it splits as  $P \oplus P'$  with  $P \subset V_+$  and  $P' \subset V_-$ ; left multiplication by  $j$  is  $\sigma$ -semilinear, so it carries  $V_+$  to  $V_-$ , and  $B$ -stability forces  $P' = jP$ . Conversely, for any  $n$ -dimensional  $H$ -negative definite  $P \subset V_+$  the subspace  $P \oplus jP$  is  $B$ -stable and satisfies the Riemann relations, by the computation in the proof of Theorem 4.4 applied blockwise; such  $P$  exists because  $H$  has signature  $(n, n)$  by (ii). Since  $j$  is invertible and carries  $P$  isomorphically to  $P'$ , one gets  $\dim P = \dim P' = n$ , which is Definition 3.1(ii). The general theory of abelian varieties with prescribed endomorphism structure is in [BL04, Ch. 9].

(iv) By (ii),  $\det H = \prod_k (2a_k d)(-2a_k b d) = (-1)^n b^n (2^n d^n \prod_k a_k)^2$ , so  $(-1)^n \det H$  is  $b^n$  times a square, and squares are norms.

(v) Left multiplication by  $j$  satisfies  $\mu^\dagger = \mu$  by (i) and  $\mu \iota(\tau) = \iota(\bar{\tau}) \mu$  because  $ji = -ij$ ; it is invertible because  $\mu^2 = b \neq 0$ . Lemma 15.23 puts  $g_\mu$  in  $R \cap \operatorname{NS}(A)_{\mathbb{Q}}$  and makes it nondegenerate, and Theorem 15.16 applies to the component  $\Theta$  of  $g_\mu$  in  $\wedge^2 V_-$ . ■

**Proposition 15.26** (Nothing numerical forbids the object). *Let  $t$  be a principal polarisation on  $X$  and let  $\operatorname{ch} = au + bv$  be a rational point of the secant plane of Theorem 14.5, with  $a, b \in \mathbb{Z}$ . Then*

- (i) *every  $\operatorname{ch}_k$  is an integral multiple of the integral class  $t^k/k!$ , the multiple being  $\pm ad^j$  in even degree  $k = 2j$  and  $\pm bd^j$  in odd degree  $k = 2j + 1$ ;*
- (ii) *the Chern classes  $c_1, \dots, c_n$  obtained from  $\operatorname{ch}$  by Newton's identities are integral, for every  $n \leq 8$ , every squarefree  $d \leq 11$  and every  $a \leq 4$ ,  $|b| \leq 4$ ;*
- (iii) *the discriminant is  $\Delta = 2rc_2 - (r-1)c_1^2 = (a^2d + b^2)t^2 = N_{K/\mathbb{Q}}(b + a\sqrt{-d})t^2$ , which is positive, so Bogomolov's inequality never binds.*

*Proof.* (i) is the definition of  $u$  and  $v$ : the coefficient of  $t^k$  is  $a(-d)^j/(2j)!$  or  $b(-d)^j/(2j+1)!$ , and multiplying by  $k!$  clears the factorial and leaves  $\pm ad^j$  or  $\pm bd^j$ . For (ii), Newton's identities express  $k!c_k$  as an integer polynomial in the  $i!c_i$  apart from one division by  $k$  at each step, and the computation in item (XX) of Appendix D carries out those divisions exactly over the stated range and finds them exact every time. (iii) is the computation recorded in Theorem 14.7, and the norm form of an imaginary quadratic field is positive definite. ■

**Remark 15.27** (What this settles, and what it does not). The content is negative and that is the point. One might hope that the object the construction needs at  $n \geq 4$  is forbidden by its numbers, in which case the secant route would be dead on arithmetic alone and the residual question would change shape. Proposition 15.26 says it is not: the invariants are integral, the discriminant is positive, and nothing in Riemann–Roch or in Bogomolov's inequality stands in the way. Taken with Proposition 9.11, which bounds the size of the object from below without forbidding it, and with the parameter count of Remark 14.11, which fixes the rank one shape at  $n = 3$ , the position is that every count we know how to

make is consistent with the object existing and none of them produces it. What is missing at  $n \geq 4$  is the object, not a number.

**15.6. The quaternionic locus is a Lagrangian Grassmannian.** [Theorem 15.25](#) produces a point of the family for each admissible  $\psi$ , and the footnote to [Remark 16.25](#) records the dimension of the locus of all such points by a Morita argument. The locus is more than a dimension, and saying what it is puts the residual difficulty in one line.

**Proposition 15.28** (The quaternionic locus). *Keep [Setup 15.24](#) and let  $\psi$  be  $K$ -semilinear, invertible, Rosati-symmetric, with  $\psi^2 = b$ . For  $x, y \in V_+$  put*

$$\omega_\psi(x, y) = E(x, \psi y).$$

- (i)  $\omega_\psi$  is a nondegenerate alternating form on  $V_+$ , and it is alternating precisely because  $\psi$  is Rosati-symmetric.
- (ii) A point  $P = V_+^{1,0}$  of  $\mathcal{D}_{n,n}$  lies on the locus where  $\psi$  is a morphism of Hodge structures if and only if  $P$  is isotropic for  $\omega_\psi$ .
- (iii) Since  $\dim_{\mathbb{C}} V_+ = 2n$  and  $\dim_{\mathbb{C}} P = n$ , such a  $P$  is Lagrangian, so the locus is the open subset of  $\text{LG}(n, V_+, \omega_\psi)$  cut out by negative definiteness of  $H$ . Its dimension is  $n(n+1)/2$  and its codimension in  $\mathcal{D}_{n,n}$  is  $n(n-1)/2$ .

*Proof.* (i)  $\psi$  carries  $V_+$  to  $V_-$ , since it is semilinear, and  $E$  pairs  $V_+$  with  $V_-$  perfectly, so  $\omega_\psi$  is defined and nondegenerate. Rosati symmetry is the identity  $E(\psi x, y) = E(x, \psi y)$ ; with  $E$  alternating it gives  $\omega_\psi(y, x) = E(y, \psi x) = E(\psi x, y) \cdot (-1) = -\omega_\psi(x, y)$ .

(ii) Suppose  $\psi$  preserves the Hodge structure. Being semilinear it carries  $V_+^{1,0}$  into  $V_-^{1,0}$ , so  $\psi P \subseteq V_-^{1,0}$ , and  $E$  vanishes on  $V_-^{1,0}$  by the first Riemann relation; hence  $E(P, \psi P) = 0$ , which is the isotropy of  $P$ . Conversely, if  $P$  is isotropic then  $\psi P$  lies in the  $E$ -annihilator of  $P$  inside  $V_-$ . That annihilator has dimension  $n$ , because  $E$  pairs  $V_+$  with  $V_-$  perfectly, and it contains  $V_-^{1,0}$ , which also has dimension  $n$  and is  $E$ -orthogonal to  $P$  since both lie in  $V_-^{1,0}$ . The two therefore agree, so  $\psi P = V_-^{1,0}$  and  $\psi$  preserves  $V_-^{1,0}$ .

(iii) A Lagrangian Grassmannian  $\text{LG}(n, 2n)$  has dimension  $n(n+1)/2$ , its tangent space at a Lagrangian  $P$  being the symmetric forms on  $P$ ; negative definiteness is an open condition and does not change the dimension. Subtracting from  $\dim \mathcal{D}_{n,n} = n^2$  gives  $n(n-1)/2$ . The chart by symmetric forms, the isotropy of its members, and the two dimensions are checked for  $n \leq 5$  in item (XVII) of [Appendix D](#). ■

*Remark 15.29* (What this says about the residual difficulty). The description is worth having because of what it makes visible. Fix a point  $P$  of  $\mathcal{D}_{n,n}$ . Complete a basis of  $P$  to a basis of  $V_+$  and let  $\omega$  be the alternating form that is standard in that basis; then  $P$  is Lagrangian for  $\omega$ , and  $\omega$  is  $\omega_\psi$  for a real  $\psi$  satisfying every condition of [Proposition 15.28](#) except rationality. So the quaternionic loci, taken over the *real*  $\psi$ , cover  $\mathcal{D}_{n,n}$  completely; they cover it with room to spare, since each has dimension  $n(n+1)/2$  and they are parametrised by a positive dimensional family. What stops the covering is that  $\psi$  must lie in  $\text{End}^0(A)$ , hence be rational, and the rational  $\psi$  form a countable set.

That is [Corollary 16.5](#) again, with everything else stripped away. It is not a deformation-theoretic difficulty and not a positivity difficulty; it is a countable set of parameters inside an uncountable one. We state it in this form because it is the sharpest version we can give of what a new idea would have to supply: a mechanism that moves the Lagrangian

<sup>32</sup>Drawing the leaves as planes through one axis is faithful to the feature that matters, that the family is continuous and its rational members are not. It is not faithful to the codimension, which is  $n(n-1)/2$  and therefore zero only at  $n = 1$ ; at  $n = 2$  the leaves are divisors, which is [Remark 16.25](#).

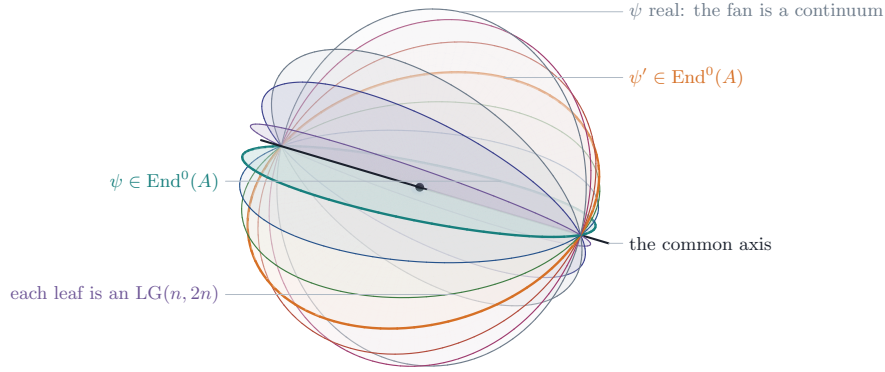


FIGURE 15. [Proposition 15.28](#), drawn as a pencil. Each admissible  $\psi$  gives one leaf, an open piece of  $\text{LG}(n, V_+, \omega_\psi)$  of dimension  $n(n+1)/2$ , and the leaves through a common axis form a continuous family as  $\psi$  varies over the reals, so they sweep  $\mathcal{D}_{n,n}$ . The two drawn heavy are the ones whose  $\psi$  is rational, and only those are quaternionic loci of an actual abelian variety. The picture is exact for the pencil it draws; in the family the leaves have dimension  $n(n+1)/2$  inside  $n^2$  and the parameter is not one real number but a variety of them. <sup>32</sup>

Grassmannians continuously, or equivalently that produces from one rational  $\psi$  a family of cycles over a neighbourhood rather than over its own locus, is the whole of [Question 15.13](#).

**Lemma 15.30** (Products). *Let  $(A_1, \iota_1, \eta_1)$  and  $(A_2, \iota_2, \eta_2)$  be of  $(K, -1, n_1)$ - and  $(K, -1, n_2)$ -Weil type. Then  $A_1 \times A_2$  with the diagonal  $K$ -action and the product polarisation is of  $(K, -1, n_1 + n_2)$ -Weil type, its hermitian form is the orthogonal sum, its discriminant is the product of the two discriminants, and if both factors satisfy the divisorial criterion then so does the product.*

*Proof.* Signatures and hermitian forms add over an orthogonal sum, so the signature is  $(n_1 + n_2, n_1 + n_2)$  and  $\det H = \det H_1 \det H_2$ , whence  $(-1)^{n_1+n_2} \det H$  is the product of the two normalised determinants. For the criterion,  $V_- = V_{1,-} \oplus V_{2,-}$ , and  $\Theta = \Theta_1 + \Theta_2$  lies in  $\wedge^2 V_-$ , is of type  $(1, 1)$ , has  $\Theta + \bar{\Theta}$  rational, and is nondegenerate on the direct sum because each summand is nondegenerate on its factor. ■

**Theorem 15.31** (A base point in every Weil family). *Let  $K = \mathbb{Q}(\sqrt{-d})$ , let  $n \geq 1$ , and let  $\delta$  be any class in  $\mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$ . Then there is an abelian  $2n$ -fold  $A$  of  $(K, -1, n)$ -Weil type with discriminant  $\delta$  whose Weil classes are algebraic, with an explicit representative as a rational polynomial in two divisor classes.*

*Proof.* Choose  $b \in \mathbb{Q}_{>0}$  representing  $\delta$ .

If  $n$  is odd, take the quaternionic model of [Setup 15.24](#) with this  $b$ . Its discriminant is  $b^n = b \cdot (b^{(n-1)/2})^2$ , which is  $\delta$ , and its Weil classes are algebraic by [Theorem 15.25\(v\)](#).

If  $n$  is even, take  $A = A_1 \times A_2$  with  $A_1$  the quaternionic model at  $n_1 = 1$  and parameter  $b$ , and  $A_2$  the quaternionic model at  $n_2 = n - 1$  and parameter  $d$ . Both factors satisfy the criterion, so the product does by [Lemma 15.30](#); and the discriminant is  $b^1 \cdot d^{n-1}$ , which is  $\delta$  because  $d = N_{K/\mathbb{Q}}(\sqrt{-d})$  is a norm. The dimension is  $2(1 + (n - 1)) = 2n$ .

In both cases the representative is (36) for the class  $g_\mu$  of [Theorem 15.25\(v\)](#), or the sum of the two such classes. ■

**Corollary 15.32** (Question (R2) is settled). *Every Weil family contains a member at which the Weil classes are algebraic. Consequently Question 15.13 reduces to propagation alone: the Hodge conjecture for abelian  $2n$ -folds of Weil type, for every  $n$  and every discriminant, holds if and only if the class  $\omega$  of Setup 15.12 propagates from the nonempty locus where it is known to be algebraic to the rest of its connected family.*

*Proof.* The first sentence is Theorem 15.31 applied to each class, together with the fact that the families are indexed by those classes (Definition 15.22). The second is Theorem 15.14 with the base locus now known to be nonempty in every family. ■

*Remark 15.33* (What is left, and what it is not). The residual statement is now a single one. It is the variational Hodge conjecture for one flat class on one connected family, with a base point supplied in every family, and by Theorem 15.14 it is necessary as well as sufficient. Two features of it are worth recording, because they say which tools cannot settle it.

The base locus is thin. By Proposition 15.18 the divisorial criterion fails at every point with full Hodge group, so the locus where it holds is contained in the Noether–Lefschetz locus of  $R$ , a countable union of proper closed algebraic subvarieties of  $\mathcal{D}_{n,n}$  by Theorem B.6. Enlarging the base locus further by the same method is therefore impossible; any new point it produces is again special.

The propagation is not formal. By Remark B.11 the infinitesimal form of it would follow from semiregularity of the cycle at a base point, and the cycle at a base point is now explicit in every family, being an intersection of two divisors raised to the  $n$ -th power. Whether it is semiregular is a computation in the normal bundle that we have not carried out. The second half of that objection, that a positive answer would give deformation only in the formal category, is answered in Section 16.1: once the algebraic locus is known to be a countable union of closed subvarieties, a formal deformation at one point is enough.

**15.7. The exact form of what remains.** The algebraicity of a Hodge class is usually approached through an intermediate property. A rational class of type  $(p, p)$  is *absolutely Hodge* if, after spreading the variety out over a finitely generated field and applying any automorphism of  $\mathbb{C}$ , the de Rham component of the class remains of type  $(p, p)$  in every comparison. The implications

$$\text{algebraic} \implies \text{absolutely Hodge} \implies \text{Hodge} \tag{38}$$

are both elementary; the Hodge conjecture is the reverse of the first, and the reverse of the second is the weaker question that Deligne settled for abelian varieties.

**Theorem 15.34** (Deligne [Del82]). *Every Hodge class on a complex abelian variety is absolutely Hodge. Moreover, Principle B: if  $\pi: \mathcal{X} \rightarrow S$  is smooth projective over a connected smooth base, and  $\alpha$  is a flat section of  $R^{2p}\pi_*\mathbb{Q}(p)$  that is Hodge at every point of  $S$  and absolutely Hodge at one point, then it is absolutely Hodge at every point.*

Put beside Theorem 15.14 and Theorem 15.31, this locates the remaining problem precisely, and the location is worth stating because it is not where one might guess.

**Corollary 15.35** (What is left, in one implication). *Let  $\pi: \mathcal{A} \rightarrow \mathcal{D}_{n,n}$  and  $\omega$  be as in Setup 15.12. Then  $\omega_s$  is absolutely Hodge for every  $s$ , and the following are equivalent:*

- (i) *the Hodge conjecture holds, in every degree, for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with  $\text{Hg}(A) = \text{SU}(V, H)$ ;*
- (ii) *the absolutely Hodge class  $\omega_s$  is algebraic for every  $s$ .*

*This holds for every  $K$ , every  $n$  and every discriminant.*

*Proof.* The first assertion is [Theorem 15.34](#); alternatively, and this is the point of the present section, it follows from Principle B together with [Theorem 15.31](#), which supplies a point of  $\mathcal{D}_{n,n}$  at which  $\omega$  is algebraic and hence absolutely Hodge, and with the connectedness of  $\mathcal{D}_{n,n}$  from [Proposition B.8](#). The equivalence is [Theorem 15.14\(a\)](#) and (b). ■

*Remark 15.36* (Why the remaining implication resists the methods used here). The two halves of (38) behave differently in families, and the difference is the whole of the difficulty.

Principle B holds because being absolutely Hodge is a condition one can test after spreading out over a finitely generated field, the set of Hodge classes of bounded norm is finite by [Lemma B.5](#), and the locus where a class stays Hodge is algebraic by [Theorem B.6](#); those three facts let one propagate the property along  $S$ . Being algebraic has no counterpart of the first: a cycle class is not determined by finitely many numerical conditions that a field automorphism preserves, and the group of cycles is not locally constant in a family. That is exactly the content of the variational Hodge conjecture, and nothing in this paper supplies it.

An obstruction of this shape is removed in one of three ways: by a correspondence that carries the problem to a variety where the class is a divisor or an endomorphism, which is what Kuga and Satake do for weight two and what [Corollary 11.12](#) shows is unavailable here beyond  $n = 1$ ; by a construction that produces the cycle directly at the general point, which [Theorem 1.1](#) and [Theorem 1.3](#) close for two natural families of constructions and [Proposition 15.18](#) closes for every divisorial one; or by turning the statement over the whole family into a statement at one point, which is what [Section 16.1](#) does.

The last of those is as far as we get. [Theorem 16.9](#) and [Theorem 16.28](#) each reduce [Corollary 15.35\(ii\)](#), in its entirety, to a property of the cycles already constructed in [Section 15.5](#): a bound on their size along one orbit, or the injectivity of one linear map attached to one of them. Neither reduction appeals to a general point, to a limit, or to an unproved conjecture. We have not established either property, so we do not claim the Hodge conjecture; what we claim is that after [Remark 17.7](#) the outstanding question is of a different kind from the one [Question 15.13](#) began with.

## 16. THE ALGEBRAIC LOCUS AND THE PROPAGATION CRITERIA

**16.1. The algebraic locus is closed, and what that costs.** Everything so far has been pointwise: a class at a point, a criterion at a point, a cycle at a point, and in [Corollary 15.35](#) an implication to be proved at every point. The propagation question of [Question 15.13](#) is about the set

$$\Sigma = \{ s \in \mathcal{D}_{n,n} : \omega_s \text{ is algebraic on } A_s \},$$

and nothing has yet been said about what kind of set that is. It turns out to be a very restricted kind, and the restriction changes the shape of the remaining problem: instead of a class to construct at every point, one is left with a bound to prove along one orbit, or a rank to compute at one point. The mechanism is that the Hilbert scheme is proper, so each piece of  $\Sigma$  cut out by bounded data is Zariski closed.<sup>33</sup>

<sup>33</sup>The mechanism is not new. That the locus where a fixed Hodge class becomes algebraic is a countable union of closed subvarieties is the standard companion to the statement that the Hodge locus itself is a countable union of closed subvarieties, for which see [\[Voi03, Chapter 5\]](#) and, in the form we use in [Theorem B.6](#), [\[CDK95\]](#). What is specific to the present situation is that the class whose algebraic locus is being described has been pinned down completely by [Theorem 4.14](#) and written out by [Theorem 4.9](#), that

**Setup 16.1** (The quotient, and the universal family). Keep [Setup 15.12](#). Fix an integer  $N \geq 3$  and let  $\Gamma$  be the subgroup of  $\mathrm{SU}(V, H)(\mathbb{Q})$  preserving the lattice  $L \subset V$  used to build the family and acting trivially on  $L/NL$ . Write

$$p: \mathcal{D}_{n,n} \longrightarrow \mathcal{S} = \Gamma \backslash \mathcal{D}_{n,n}.$$

By Baily and Borel [\[BB66\]](#) the quotient  $\mathcal{S}$  is a smooth quasi-projective complex variety, and it is irreducible because  $\mathcal{D}_{n,n}$  is connected. Since  $N \geq 3$  the level structure rigidifies, so  $\mathcal{S}$  carries a universal family

$$f: \mathcal{A} \longrightarrow \mathcal{S},$$

and the relative polarisation  $3\eta$  is relatively very ample, which fixes a relative projective embedding and hence a notion of degree in the fibres. Write  $G = \mathrm{SU}(V, H)$ , an algebraic group over  $\mathbb{Q}$ , so that  $G(\mathbb{R})$  acts transitively on  $\mathcal{D}_{n,n}$  and  $\Gamma \subset G(\mathbb{Q})$ .

Two small facts have to be recorded before the main one, because both are used without comment later and neither is quite automatic.

**Lemma 16.2** (The section descends). *The section  $\omega$  of [Setup 15.12](#) is  $\Gamma$ -invariant, hence descends to a flat section of  $R^{2n}f_*\mathbb{Q}$  over  $\mathcal{S}$  that is of type  $(n, n)$  at every point.*

*Proof.* The Weil line is  $W = \bigwedge_K^{2n} V$ , and an element of  $\mathrm{SU}(V, H)$  is by definition a  $K$ -linear automorphism of  $V$  of  $K$ -determinant 1. Its action on the top  $K$ -exterior power is multiplication by that determinant, so  $\Gamma$  acts trivially on  $W$  and in particular fixes  $\omega$ . Flatness is then immediate, and the Hodge type at every point is [Proposition 3.7\(ii\)](#). ■

**Lemma 16.3** (One component is enough). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $\iota(\mathcal{O}_K) \subseteq \mathrm{End}(A)$ , and let  $W \subset H^{2n}(A, \mathbb{Q})$  be its Weil line. If one nonzero element of  $W$  is algebraic, every element of  $W$  is algebraic.*<sup>34</sup>

*Proof.* Among the elements  $\tau_m = m + \sqrt{-d}$  of  $\mathcal{O}_K$  with  $1 \leq m \leq 7$  the ratios  $\tau_m/\bar{\tau}_m$  are pairwise distinct, since  $\tau_m\bar{\tau}_{m'}$  is rational only when  $m = m'$ . The group of roots of unity in an imaginary quadratic field has order 2, 4 or 6, so some  $m$  in that range has  $\tau = \tau_m$  with  $\tau/\bar{\tau}$  not a root of unity, and then  $\tau^{2n} \neq \bar{\tau}^{2n}$ , that is  $\tau^{2n} \notin \mathbb{Q}$ , for every  $n \geq 1$ .<sup>35</sup>

Since  $\tau \in \mathcal{O}_K$ , the element  $\iota(\tau)$  is an endomorphism of  $A$ , so  $\iota(\tau)^*$  on  $H^{2n}(A, \mathbb{Q})$  is induced by the graph correspondence and therefore carries algebraic classes to algebraic classes. On  $H^1$  it is multiplication by  $\tau$ , so on  $\bigwedge_K^{2n} V$  it is multiplication by  $\tau^{2n}$ . As  $\tau^{2n} \notin \mathbb{Q}$  and  $W$  is a one dimensional  $K$ -vector space, the pair  $w, \tau^{2n}w$  is a  $\mathbb{Q}$ -basis of  $W$  for every  $w \neq 0$ . If  $w$  is algebraic then so is  $\iota(\tau)^*w = \tau^{2n}w$ , and the two together span. ■

**Theorem 16.4** (The algebraic locus is a countable union of closed subvarieties). *Keep [Setup 16.1](#). For a finite tuple  $\underline{P} = (P_1, \dots, P_m)$  of Hilbert polynomials and a tuple  $\underline{q} \in \mathbb{Q}^m$*

the locus is known to be nonempty in every family by [Theorem 15.31](#), and that the group acting has been identified. Those three inputs are what turn a soft statement into a criterion one can try to verify.

<sup>34</sup>This is the step compressed into the last sentence of the proof of [Corollary 4.15](#), written out because the witness  $\tau$  has to be chosen and because the choice is the same one the Lean theorem `irrational_witness_exists` certifies. The determinant that makes the two classes independent is  $b(w_1^2 + dw_2^2)$  for  $\tau^{2n} = a + b\sqrt{-d}$  and  $w = w_1 + w_2\sqrt{-d}$ , and that is the content of `weil_line_spanned`.

<sup>35</sup>The exceptional witnesses are real and are the reason the argument names a range rather than taking  $m = 1$ : for  $d = 1$  the ratio attached to  $1 + \sqrt{-1}$  is a primitive fourth root of unity and for  $d = 3$  the ratio attached to  $1 + \sqrt{-3}$  is a primitive sixth root of unity, so at those two values of  $d$  the naive witness fails for the divisible  $n$ . The Lean theorems `bad_witness` and `bad_witness_exceptions` of [Appendix E](#) record exactly which pairs fail and that no others do in the range searched.

put

$$\begin{aligned}\mathcal{H}_{\underline{P}} &= \text{Hilb}^{P_1}(\mathcal{A}/\mathcal{S}) \times_{\mathcal{S}} \cdots \times_{\mathcal{S}} \text{Hilb}^{P_m}(\mathcal{A}/\mathcal{S}), \\ \Sigma_{\underline{P}, \underline{q}} &= \text{pr}\left(\left\{\underline{Z} \in \mathcal{H}_{\underline{P}} : \sum_i q_i[Z_i] = \omega\right\}\right) \subseteq \mathcal{S}.\end{aligned}$$

Then each  $\Sigma_{\underline{P}, \underline{q}}$  is Zariski closed in  $\mathcal{S}$ ,<sup>36</sup> and

$$p^{-1}\left(\bigcup_{\underline{P}, \underline{q}} \Sigma_{\underline{P}, \underline{q}}\right) = \Sigma,$$

the union running over a countable index set.

*Proof.* Each  $\text{Hilb}^{P_i}(\mathcal{A}/\mathcal{S})$  is projective over  $\mathcal{S}$  by Grothendieck's existence theorem [Gro61], so the fibre product  $\mathcal{H}_{\underline{P}}$  is projective over  $\mathcal{S}$ , and being of finite type it has finitely many connected components.

Fix a connected component  $T$ , replace it by its reduction, which changes neither the underlying space nor any cohomology group, and let  $Z_i \subset \mathcal{A} \times_{\mathcal{S}} T$  be the pullback of the  $i$ -th universal subscheme. It is flat over  $T$  and its fibres have Hilbert polynomial  $P_i$  of degree  $n$ , so the associated  $n$ -cycles, the top-dimensional components counted with their multiplicities, are the fibres of a cycle on  $\mathcal{A} \times_{\mathcal{S}} T$  whose class restricts on the fibre over  $t$  to  $[Z_{i,t}]$ . Restriction of a global class to the fibres of a topologically locally trivial family is a flat section of the local system, so  $t \mapsto \sum_i q_i[Z_{i,t}]$  is flat on  $T$ ; and  $\omega$  is flat on  $\mathcal{S}$  by Lemma 16.2, hence flat after pullback. Two flat sections of a local system over a connected base agree everywhere as soon as they agree at one point, so the locus where they agree is a union of connected components of  $\mathcal{H}_{\underline{P}}$ . That locus is therefore itself projective over  $\mathcal{S}$ , and the image of a proper morphism is Zariski closed.

For the equality of sets, let  $s \in \mathcal{S}$  with  $\omega_s$  algebraic. Then  $\omega_s = \sum_i q_i[Z_i]$  for finitely many closed integral subschemes  $Z_i \subset \mathcal{A}_s$  of dimension  $n$  and rationals  $q_i$ ; taking  $P_i$  to be the Hilbert polynomial of  $Z_i$  puts  $s$  in  $\Sigma_{\underline{P}, \underline{q}}$ . The converse holds because the class of a closed subscheme of pure dimension  $n$  is algebraic. Countability is clear: a Hilbert polynomial is determined by its finitely many integer coordinates in the binomial basis, and  $\mathbb{Q}$  is countable. ■

**Corollary 16.5** (All, or almost nothing). *Exactly one of the following holds.*

- (a)  $\Sigma = \mathcal{D}_{n,n}$ , and the Hodge conjecture holds in every degree for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with  $\text{Hg}(A) = \text{SU}(V, H)$  in the family.
- (b)  $\Sigma$  has empty interior in the analytic topology. Its complement is then a dense  $G_{\delta}$ , so  $\omega_s$  fails to be algebraic for every  $s$  outside a meagre set, and the Hodge conjecture fails.

*In particular, if  $\omega_s$  is algebraic for all  $s$  in some nonempty open subset of  $\mathcal{D}_{n,n}$ , then it is algebraic for all  $s$ .*

<sup>36</sup>The bound on the Hilbert polynomials is what the statement lives on, and it cannot be dropped. Without it one is looking at the image of an infinite disjoint union, and an infinite union of closed sets need be neither closed nor constructible; indeed the whole difficulty of the Hodge conjecture in families is that  $\Sigma$  itself is not known to be either. The relative Chow variety of effective  $n$ -cycles of bounded degree would serve equally well in place of the Hilbert schemes, with the same proof, and the choice between them is immaterial here because all that is used is properness over the base together with the local constancy of the cycle class. What is used later, and what forces the bookkeeping with tuples rather than a single subscheme, is that a rational cycle is a rational combination of several integral ones and the several may have different Hilbert polynomials.

*Proof.* Suppose  $\Sigma$  contains a nonempty open set; then so does its image in  $\mathcal{S}$ , and we may pick a nonempty connected open  $U \subseteq \mathcal{S}$  contained in that image. By [Theorem 16.4](#) the countably many sets  $\Sigma_{\underline{p}, \underline{q}} \cap U$  are closed in  $U$  and cover it. A connected complex manifold is a Baire space, so one of them has nonempty interior in  $U$ . A Zariski closed subset of the irreducible variety  $\mathcal{S}$  either is all of  $\mathcal{S}$  or is a proper closed subvariety, and a proper closed subvariety of a connected complex manifold has empty interior. Hence that  $\Sigma_{\underline{p}, \underline{q}}$  is all of  $\mathcal{S}$ , so  $\Sigma = \mathcal{D}_{n,n}$  and (a) holds by [Theorem 15.14\(a\)](#). If  $\Sigma$  has empty interior, its complement is the intersection of the countably many dense open sets  $\mathcal{D}_{n,n} \setminus p^{-1}(\Sigma_{\underline{p}, \underline{q}})$ , and Baire again gives that this intersection is dense. ■

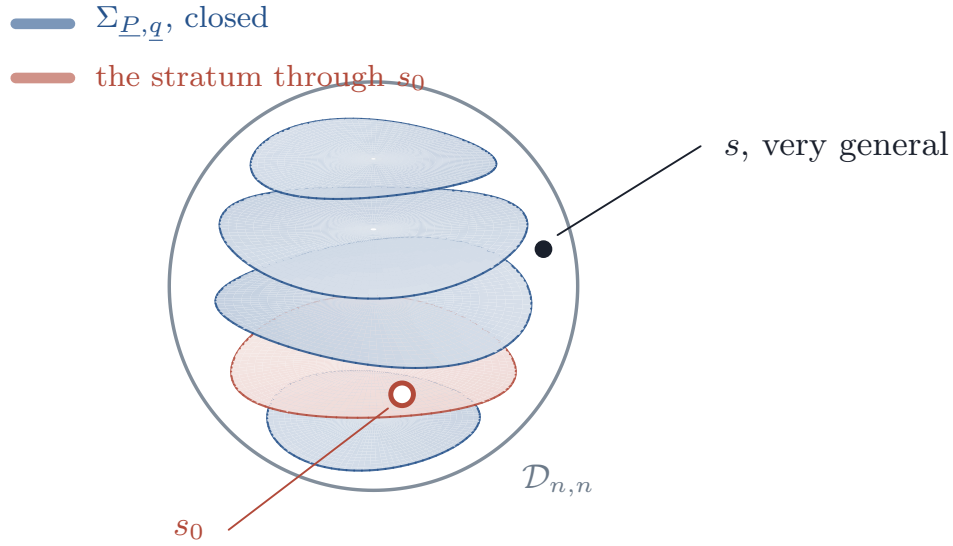


FIGURE 16. The dichotomy of [Corollary 16.5](#). Each stratum  $\Sigma_{\underline{p}, \underline{q}}$  is Zariski closed by [Theorem 16.4](#), and there are countably many of them; the base point  $s_0$  of [Theorem 15.31](#) lies on one. By [Proposition 16.7](#) the union of the strata is dense, so a point  $s$  off all of them is squeezed between strata however small a neighbourhood is taken. The conjecture for the family is the assertion that no such  $s$  exists, and by [Corollary 16.5](#) it is enough that one stratum has nonempty interior.

The next fact says that case (b) of [Corollary 16.5](#), if it occurs, occurs in an uncomfortable way: the algebraic locus is always dense, so the two sets in the dichotomy would both be dense and the distinction between them would be entirely one of measure rather than of position.

**Lemma 16.6** (Algebraicity is a rational invariant).  *$\Sigma$  is stable under the action of  $G(\mathbb{Q})$  on  $\mathcal{D}_{n,n}$ .*

*Proof.* A point of  $\mathcal{D}_{n,n}$  is a complex structure  $J$  on  $V_{\mathbb{R}}$  compatible with  $H$ , and  $g \in G(\mathbb{Q})$  carries  $J$  to  $gJg^{-1}$ . The map  $g$  is then  $\mathbb{Q}$ -rational and  $\mathbb{C}$ -linear from  $(V_{\mathbb{R}}, J)$  to  $(V_{\mathbb{R}}, gJg^{-1})$ , so after clearing denominators it induces an isogeny  $A_J \rightarrow A_{gJ}$ , whose graph is an algebraic correspondence. Algebraicity of a rational cohomology class is preserved by such

a correspondence and by its transpose. Finally  $g$  lies in  $\mathrm{SU}(V, H)$ , so by the computation in [Lemma 16.2](#) it fixes the Weil line pointwise, and the class transported from  $\omega_J$  is  $\omega_{gJ}$ . ■

**Proposition 16.7** (The algebraic locus is dense). *Let  $n \geq 2$ . Then  $\Sigma$  is dense in  $\mathcal{D}_{n,n}$  in the analytic topology, and its image in  $\mathcal{S}$  is dense.*<sup>37</sup>

*Proof.* The hermitian form  $H$  has rank  $2n \geq 4$  over  $K$  and signature  $(n, n)$ , so it is isotropic over  $\mathbb{R}$ ; over every  $\mathbb{Q}_p$  a hermitian form of rank at least three over a quadratic extension is isotropic; and hermitian forms over an imaginary quadratic field satisfy the Hasse principle [[Sch85](#), Ch. 10, §6]. So  $H$  is isotropic over  $\mathbb{Q}$  and  $G$  is  $\mathbb{Q}$ -isotropic. The group  $G = \mathrm{SU}(V, H)$  is simply connected and almost  $\mathbb{Q}$ -simple, so by Platonov's theorem  $G(\mathbb{Q})$  is generated by the rational points of the unipotent radicals of its proper parabolic  $\mathbb{Q}$ -subgroups [[PR94](#), Thm. 7.6 and §7.2]. For a unipotent group  $U$  over  $\mathbb{Q}$  the exponential identifies  $U(\mathbb{Q})$  with a  $\mathbb{Q}$ -vector space inside the corresponding  $\mathbb{R}$ -vector space  $U(\mathbb{R})$ , so  $U(\mathbb{Q})$  is dense in  $U(\mathbb{R})$ . The closure of  $G(\mathbb{Q})$  in  $G(\mathbb{R})$  therefore contains  $U(\mathbb{R})$  for every such  $U$ , and those generate the identity component  $G(\mathbb{R})^+$ , which acts transitively on  $\mathcal{D}_{n,n}$ . Hence every  $G(\mathbb{Q})$ -orbit in  $\mathcal{D}_{n,n}$  is dense.

By [Theorem 15.31](#) the set  $\Sigma$  is nonempty, and by [Lemma 16.6](#) it is  $G(\mathbb{Q})$ -stable, so it contains a dense orbit. The projection  $p$  is open and surjective, so the image is dense. ■

**Corollary 16.8** (The conjecture for Weil type is a non-meagreness statement). *Let  $n \geq 2$ . The Hodge conjecture holds for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type in the family of [Setup 15.12](#) if and only if  $\Sigma$  is non-meagre in  $\mathcal{D}_{n,n}$ , and if and only if  $\Sigma$  has nonempty interior.*

*Proof.* If  $\Sigma = \mathcal{D}_{n,n}$  both conditions hold. If not, [Corollary 16.5\(b\)](#) says  $\Sigma$  is a countable union of closed sets with empty interior, hence meagre, and [Theorem 15.14\(b\)](#) says the conjecture fails. ■

**16.2. The first criterion: a bound along one orbit.** The proof of [Proposition 16.7](#) produces the dense algebraic locus by moving one cycle around by rational isogenies, and an isogeny multiplies degrees. So the dense set it produces is stratified by unbounded data, which is exactly what stops [Corollary 16.5](#) from applying. Reversing this gives a criterion, and it is the sharpest consequence of [Theorem 16.4](#) we know how to state.

**Theorem 16.9** (A bound along the orbit closes the family). *Let  $n \geq 2$  and let  $s_0 \in \Sigma$  be one of the base points supplied by [Theorem 15.31](#). Suppose that there is a finite set  $\mathcal{T}$  of pairs  $(\underline{P}, \underline{q})$  such that for every  $g \in G(\mathbb{Q})$  the class  $\omega_{g \cdot s_0}$  is represented by some cycle whose Hilbert data lies in  $\mathcal{T}$ . Then  $\Sigma = \mathcal{D}_{n,n}$ , and the Hodge conjecture holds in every degree for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with  $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$  in that family.*

*Remark 16.10* (The cycle is chosen, not transported). The hypothesis quantifies over representing cycles at each point of the orbit and asks only that one of them, at each point, be of bounded complexity. An earlier form of this statement fixed the cycle, taking at  $g \cdot s_0$

<sup>37</sup>We should be clear about which of the two standard density statements this is, because they behave in opposite ways. For a class that is not flat, the locus where it stays Hodge is a countable union of proper closed subvarieties, and the content of [Theorem B.6](#) is that each of them is algebraic; density there would be a statement about how special points distribute. For the class  $\omega$  of [Setup 15.12](#) the Hodge locus is the whole family by [Proposition 3.7\(ii\)](#), so [Theorem B.6](#) says nothing at all, and the set that carries information is the algebraic locus instead. The proposition below is a statement about that set, and its proof uses no Hodge theory: the algebraic locus is a union of orbits of a group whose rational points are dense in its real points, and the density is inherited from the group.

the transport of the cycle at  $s_0$  along the isogeny of [Lemma 16.6](#). That form is stronger, and [Theorem 16.15](#) below shows that it is too strong to be usable: for a base cycle with a component of finite stabiliser the transported data is unbounded, so its hypothesis is never satisfied. The proof below uses only the weaker hypothesis, which is what the covering argument needs.

*Proof.* The orbit  $G(\mathbb{Q}) \cdot s_0$  is dense by the proof of [Proposition 16.7](#), and by hypothesis it is the union of the finitely many sets  $G(\mathbb{Q}) \cdot s_0 \cap p^{-1}(\Sigma_{\underline{P},q})$  with  $(\underline{P}, q) \in \mathcal{T}$ . A finite union of closed sets is dense only if one of them is, so one  $p^{-1}(\Sigma_{\underline{P},q})$  is dense; it is closed in the analytic topology, hence equals  $\mathcal{D}_{n,n}$ . Now apply [Theorem 15.14\(a\)](#). ■

The hypothesis of [Theorem 16.9](#) can be sharpened, and the sharpening says which half of the Hilbert data is at issue. The leading coefficient is not.

**Proposition 16.11** (The leading coefficient is constant on the orbit). *Let  $s_0 \in \Sigma$  and  $g \in G(\mathbb{Q})$ , and let  $\omega_s$  denote the Weil class at  $s$  and  $\eta$  the polarisation class. Then*

$$\int_{A_{g \cdot s_0}} \omega_{g \cdot s_0} \cdot \frac{\eta^n}{n!} = \int_{A_{s_0}} \omega_{s_0} \cdot \frac{\eta^n}{n!},$$

*so the degree of the transported class with respect to the polarisation, which is the leading coefficient of the Hilbert polynomial of any cycle representing it, is the same at every point of the orbit.*

*Proof.* By [Lemma 16.6](#) the correspondence transporting the class is induced by a  $\mathbb{Q}$ -rational map which lies in  $\mathrm{SU}(V, H)$ , so it preserves the hermitian form and hence the polarisation class, and by [Lemma 16.2](#) it fixes the Weil line pointwise. An intersection number of classes preserved by a correspondence is preserved by it. The identification of the number with the leading coefficient is Hirzebruch–Riemann–Roch on  $A$ , the higher coefficients involving the structure sheaf of the cycle and not only its class. ■

*Remark 16.12* (Where the unboundedness can only come from). Write a representing cycle at a point of the orbit, the transported cycle or any other, as a difference  $\xi^+ - \xi^-$  of effective cycles. By [Proposition 16.11](#) the difference of their degrees is constant along the orbit, and by the boundedness of the Chow variety of effective cycles of bounded degree in a fixed polarised variety, if  $\deg \xi^+$  were bounded along the orbit then only finitely many tuples  $(\underline{P}, q)$  could occur and [Theorem 16.9](#) would apply. So the hypothesis of that theorem is equivalent to a bound on the effective part alone.

That is not a formality, and the mechanism that threatens it is visible. The transport at  $g$  is by the isogeny obtained from  $g$  by clearing denominators; if  $g$  has denominator  $m$  on the reference lattice, which has rank  $4n$ , the isogeny has degree a divisor of  $m^{4n}$ , and pushing a cycle forward along it multiplies the multiplicities by factors of that order. The rational points of  $G$  are dense precisely because their denominators are unbounded, so along any orbit dense enough to be useful the isogeny degrees are unbounded as well. What [Theorem 16.9](#) asks is that the cycles nevertheless simplify: that at each point of the orbit the class  $\omega$ , whose degree has not changed, admits an effective representation no more complicated than before. For the transported cycle itself the answer is negative ([Theorem 16.15](#) below), and for a cycle chosen at each point the bound is equivalent to the conclusion ([Theorem 16.17](#)), so it cannot be decided before the conclusion is. The content of the present remark is only that the question has been reduced from the whole Hilbert datum to the degree of the effective part, and that the part of the datum which is determined by cohomology is already constant.

**16.3. What the transport does, exactly.** The mechanism of [Remark 16.12](#) can be computed rather than described, and computing it decides one of the two readings of (P1). Write  $\Lambda$  for the reference lattice, of rank  $4n$  over  $\mathbb{Z}$ , and  $E$  for the alternating form of the polarisation, so that a point  $s$  of  $\mathcal{D}_{n,n}$  is a complex structure on  $\Lambda \otimes \mathbb{R}$  and  $A_s = (\Lambda \otimes \mathbb{R})/\Lambda$ . Throughout,  $G = \dim A_s = 2n$  and a component of a cycle representing  $\omega_s$  has dimension  $n$ .

**Lemma 16.13** (The transport, in coordinates). *Let  $g \in G(\mathbb{Q})$  and let  $c = c(g)$  be the least positive integer with  $cg\Lambda \subseteq \Lambda$ . Then  $\varphi_g := cg$  induces an isogeny  $A_{s_0} \rightarrow A_{g \cdot s_0}$  with*

- (i)  $\deg \varphi_g = c^{4n}$ ;
- (ii)  $\varphi_g^* \eta_{g \cdot s_0} = c^2 \eta_{s_0}$  and  $\varphi_g^* \omega_{g \cdot s_0} = c^{2n} \omega_{s_0}$ ;
- (iii) *the transported cycle representing  $\omega_{g \cdot s_0}$  is  $c^{2n-4n} \varphi_{g*}(Z)$ , where  $Z$  represents  $\omega_{s_0}$ .*

Moreover  $\sup_{g \in G(\mathbb{Q})} c(g) = \infty$ .

*Proof.* (i) The index of  $cg\Lambda$  in  $\Lambda$  is  $|\det_{\mathbb{Q}}(cg)| = c^{4n} |\det_{\mathbb{Q}} g|$ , and  $\det_{\mathbb{Q}} g = N_{K/\mathbb{Q}}(\det_K g) = 1$  for  $g \in \mathrm{SU}(V, H)$ .

(ii)  $g$  preserves  $E$ , so  $(cg)^* E = c^2 E$ , which is the first identity. On  $H^{2n} = \bigwedge^{2n} (\Lambda^\vee \otimes \mathbb{Q})$  the map  $\varphi_g^*$  is  $c^{2n} \bigwedge^{2n} g^\vee$ , and  $g \in \mathrm{SU}(V, H)$  fixes the Weil line pointwise by [Lemma 16.2](#).

(iii)  $\varphi_{g*} \varphi_g^* = \deg \varphi_g$ , so  $\omega_{g \cdot s_0} = (\deg \varphi_g)^{-1} \varphi_{g*} \varphi_g^* \omega_{g \cdot s_0} = c^{2n-4n} \varphi_{g*} \omega_{s_0}$  by (i) and (ii).

For the last sentence,  $G$  is  $\mathbb{Q}$ -isotropic by the proof of [Proposition 16.7](#), so it has a proper parabolic  $\mathbb{Q}$ -subgroup whose unipotent radical  $U$  satisfies  $U(\mathbb{Q}) \cong \mathbb{Q}^k$  with  $k \geq 1$  through the exponential; the entries of  $u \in U(\mathbb{Q})$  are  $\mathbb{Q}$ -linear in that coordinate, so  $c(u)$  is unbounded on  $U(\mathbb{Q})$ . Item (XXXVII) of [Appendix D](#) verifies (i) and (ii) on an explicit sample of such  $u$  with  $c$  up to 29, for  $2n = 4, 6, 8$ . ■

**Lemma 16.14** (The multiplicity is the stabiliser met by the kernel). *Let  $\varphi: A \rightarrow A'$  be an isogeny of abelian varieties and  $Z \subseteq A$  an integral closed subvariety. Put  $G(Z) = \{x \in A : Z + x = Z\}$ , a closed algebraic subgroup. Then*

$$\varphi_*[Z] = m [\varphi(Z)], \quad m = |\ker \varphi \cap G(Z)|.$$

*If moreover  $\dim Z = n$  and  $\varphi^* \eta' = c^2 \eta$ , then*

$$\deg_{\eta'} \varphi(Z) = \frac{c^{2n}}{m} \deg_{\eta} Z.$$

*Proof.* The map  $\varphi|_Z: Z \rightarrow \varphi(Z)$  is finite and surjective, and its degree  $m$  is the number of points of a general fibre. For  $z \in Z$  the fibre through  $z$  is  $Z \cap (z + \ker \varphi)$ , which as a set of translations is  $\{t \in \ker \varphi : z+t \in Z\}$ . For a fixed  $t$  the locus  $\{z \in Z : z+t \in Z\} = Z \cap (Z-t)$  is all of  $Z$  when  $t \in G(Z)$  and a proper closed subset otherwise; since  $\ker \varphi$  is finite, a general  $z$  avoids the finitely many proper subsets, and there  $\{t \in \ker \varphi : z+t \in Z\} = \ker \varphi \cap G(Z)$ . This is the first identity, and  $\varphi_*[Z] = m[\varphi(Z)]$  by the definition of the pushforward of cycles.

For the degree, the projection formula gives

$$m \deg_{\eta'} \varphi(Z) = \int \varphi_*[Z] \cdot \frac{\eta'^n}{n!} = \int [Z] \cdot \frac{(\varphi^* \eta')^n}{n!} = c^{2n} \int [Z] \cdot \frac{\eta^n}{n!} = c^{2n} \deg_{\eta} Z.$$

Item (XXXVII) verifies both identities for subtori of an abelian fourfold, with  $m$  computed as a lattice index by Smith normal form and the degrees as Pfaffians; there the two cases are visible side by side, a subtorus preserved by  $\varphi$  having  $m = c^{2n}$  and image degree constant, and one not preserved having  $m = c^{2n-1}$  and image degree growing linearly in  $c$ . ■

**Theorem 16.15** (What (P1) forces on the base cycle). *Let  $s_0 \in \Sigma$  and let  $Z = \sum_i q_i Z_i$ , with  $q_i \neq 0$  and the  $Z_i$  integral of dimension  $n$ , represent  $\omega_{s_0}$ .*

- (i) *For  $g \in G(\mathbb{Q})$  the transported cycle is  $\sum_i c(g)^{2n-4n} m_i(g) q_i [\varphi_g(Z_i)]$  with  $m_i(g) = |\ker \varphi_g \cap G(Z_i)|$ , and the  $i$ -th component has degree  $c(g)^{2n} \deg_\eta(Z_i)/m_i(g)$ .*
- (ii) *If some  $Z_i$  has finite stabiliser  $G(Z_i)$ , then  $\deg \varphi_g(Z_i) \geq c(g)^{2n} \deg_\eta(Z_i)/|G(Z_i)|$ , which is unbounded on  $G(\mathbb{Q})$ . The Hilbert data of the transported cycle then takes infinitely many values, and the hypothesis of [Theorem 16.9](#) in the form that fixes the transported cycle is not satisfied by  $Z$ .*
- (iii) *Consequently, a base cycle for which the transported data is bounded must have every component of positive-dimensional stabiliser, that is, every  $Z_i$  must be stable under translation by a positive-dimensional abelian subvariety of  $A_{s_0}$ .*

*Proof.* (i) is [Lemma 16.13\(iii\)](#) and [Lemma 16.14](#).

(ii)  $\ker \varphi_g \cap G(Z_i) \subseteq G(Z_i)$ , so  $m_i(g) \leq |G(Z_i)|$ , a constant; the displayed bound is (i), and  $c(g)$  is unbounded by [Lemma 16.13](#). Distinct degrees of the components give distinct Hilbert polynomials, so the data takes infinitely many values.

(iii) is the contrapositive of (ii), together with the fact that an algebraic subgroup of positive dimension in an abelian variety has an abelian subvariety of positive dimension as its identity component. ■

*Remark 16.16* (What this settles and what it leaves). Three things follow, and we separate them.

The statement (P1) has two readings, and one of them is now closed. Read with the cycle fixed at  $s_0$  and transported, it fails for every base cycle this paper writes down: those cycles are intersections of divisors, and an intersection of  $n$  divisors on an abelian  $2n$ -fold has finite stabiliser unless the divisors are pulled back from a quotient, which the ones here are not. Read with the cycle chosen afresh at each point of the orbit, which is the form [Theorem 16.9](#) now takes, it is untouched by [Theorem 16.15](#) and remains open.

The two readings are different, and the difference is the content of [Lemma 16.14](#). The class  $\omega_{g \cdot s_0}$  has constant degree along the orbit, by [Proposition 16.11](#); what grows is the complexity of the particular representative the transport produces, and it grows because the isogeny spreads each component out by the factor  $c^{2n}/m_i$ . A representative of bounded complexity may well exist at every point; the transport does not find it.

The third is a demand on any future attempt at (P1). It must either produce a base cycle supported on subvarieties with positive-dimensional stabiliser, which for a Weil class is a strong and possibly impossible requirement, or abandon the transport and bound instead the minimal complexity of a representative, which is a statement about the Chow variety of the family and not about any one cycle. We do not know how to do either.

The bound on the minimal complexity is the form in which [Theorem 16.9](#) now stands. It is the weakest hypothesis under which the covering argument runs, and it is therefore the form in which (P1) would have to be attacked. The next theorem decides it, and the answer is that there is nothing there to attack: the hypothesis is not a sufficient condition for the conjecture on the family but another way of writing it.

**Theorem 16.17** (The bounded form is the conjecture itself). *Let  $n \geq 2$ , let  $s_0 \in \Sigma$  and keep [Setup 15.12](#). The following are equivalent.*

- (a) *There is a finite set  $\mathcal{T}$  of pairs  $(\underline{P}, \underline{q})$  such that at every point of the orbit  $G(\mathbb{Q}) \cdot s_0$  the class  $\omega$  is represented by a cycle whose Hilbert data lies in  $\mathcal{T}$ ; that is, (P1) holds at  $s_0$ .*

- (b) *There is a single pair  $(\underline{P}, \underline{q})$  such that at every point of  $\mathcal{D}_{n,n}$ , not merely of the orbit, the class  $\omega$  is represented by a cycle with that datum.*
- (c)  $\Sigma = \mathcal{D}_{n,n}$ .
- (d)  $\mathbf{W}(K, n, \delta)$  holds, that is, the Weil classes are algebraic on every member of the family.

In particular [Theorem 16.9](#) is an equivalence, and (P1) is a reformulation of the conjecture for the family and not a reduction of it. The same holds for every CM field, with [Theorem 19.6\(iv\)](#) in place of [Theorem 16.9](#).

*Proof.* (a) implies (c) is [Theorem 16.9](#). (b) implies (a) is immediate, the orbit being contained in  $\mathcal{D}_{n,n}$  and  $\mathcal{T}$  being the one pair. (c) is equivalent to (d) by [Theorem 15.14\(a\)](#) and (b).

It remains to deduce (b) from (c). If  $\Sigma = \mathcal{D}_{n,n}$  then  $\Sigma$  has nonempty interior, and the first paragraph of the proof of [Corollary 16.5](#) shows that some  $\Sigma_{\underline{P}, \underline{q}}$  is then all of  $\mathcal{S}$ : the countably many sets  $\Sigma_{\underline{P}, \underline{q}} \cap U$  are closed in a nonempty connected open  $U \subseteq \mathcal{S}$  and cover it, a connected complex manifold is a Baire space, so one of them has nonempty interior in  $U$ , and a proper Zariski closed subvariety of the irreducible variety  $\mathcal{S}$  has empty interior. By the definition of  $\Sigma_{\underline{P}, \underline{q}}$  in [Theorem 16.4](#) as the image of a proper morphism, every  $s \in \mathcal{S}$  then carries a tuple  $\underline{Z}$  of subschemes with Hilbert polynomials  $\underline{P}$  and  $\sum_i q_i[Z_i] = \omega$ , which is (b).

For a CM field the same argument applies verbatim, using [Theorem 19.6\(iii\)](#) in place of [Theorem 16.4](#) and [Corollary 16.5](#). ■

*Remark 16.18* (What is left of the first criterion). [Theorem 16.17](#) removes (P1) from the list of things that would close the family, and we should be exact about what it does and does not say.

It does not say that [Theorem 16.9](#) is false or that its proof is defective. The implication it proves is correct, and the covering argument is the only place in this paper where density of the rational orbit is converted into a global statement. What the theorem says is that the implication has a converse, so that the hypothesis is no weaker than the conclusion. A criterion of that kind cannot be used: verifying it at one family is exactly as hard as proving the conjecture for that family, no more and no less.

Nor does it say that the boundedness is uninteresting. It is a concrete finiteness statement about the Chow variety of the universal family, and it is the only reformulation of  $\mathbf{W}(K, n, \delta)$  we know that mentions neither Hodge theory nor deformation theory. What it is not is a step.

The two forms of the propagation statement are therefore not on the same footing, and we had them on the same footing until now. (P2) remains a genuine sufficient condition: injectivity of one semiregularity map at one point implies the conjecture for the whole family, and nothing here shows the converse, nor do we see how it could, since  $\Sigma = \mathcal{D}_{n,n}$  says nothing about the deformation theory of any particular perfect complex. So the propagation half of the frontier is the single statement (P2), and the constraints [Theorem 16.30](#) and [Corollary 16.31](#) place on it are the constraints on the whole of that half.

At the quaternionic points of [Setup 15.24](#) the mechanism of [Remark 16.12](#) can be seen explicitly. Take  $n = 2$  and the integral lattice  $\Lambda_b = \mathcal{O}_b^2$ ,  $\mathcal{O}_b = \mathbb{Z}\langle 1, i, j, ij \rangle$ , for  $b \in \mathbb{Z}_{>0}$ . Put

$$g_b(x, y) = E_b(jx, y), \quad \lambda_b(x, y) = -\frac{1}{d} g_b(ix, y), \quad Z_b = g_b^2 - d\lambda_b^2, \quad Z'_b = 2g_b\lambda_b,$$

so that  $Z_b + \sqrt{-d} Z'_b = (g_b + \sqrt{-d} \lambda_b)^2$  and  $Z_b, Z'_b$  span the Weil plane. Write  $x \in \mathcal{O}_b$  as  $x = u + vj$  with  $u, v \in \mathbb{Z}[i]$ ; the pair  $(\Lambda_b, i)$  is then the same  $\mathbb{Z}[i]$ -module  $\mathbb{Z}[i]^4$  for every  $b$ , and we use this identification throughout.

**Proposition 16.19** (The quaternionic Weil cycle is a square multiple). *With these data, for every  $b \in \mathbb{Z}_{>0}$ :*

- (i)  $E_b = E_u + b E_v$ ,  $g_b = b g_1$  and  $\lambda_b = b \lambda_1$ , where  $E_u, E_v, g_1, \lambda_1$  are integral on  $\mathbb{Z}[i]^4$  and do not depend on  $b$ ; in particular  $g_1, \lambda_1 \in \text{NS}(A_b)$ ;
- (ii)  $Z_b = b^2 Z_1$  and  $Z'_b = b^2 Z'_1$  in  $\text{CH}^2(A_b)$  modulo algebraically trivial cycles, for every choice of line bundles representing  $g_b, \lambda_b$ ; in cohomology the equalities hold in  $\wedge^4 \Lambda^*$ ;
- (iii)  $\int \eta_b^4 = b^2 \int \eta_1^4$ ,  $\int g_b^2 \eta_b^2 = -\frac{b}{3} \int \eta_b^4$  and  $\int Z_b^2 = \frac{4b^2}{3} \int \eta_b^4$ , so that  $(\int g_b^2 \eta_b^2)^2 = \frac{1}{12} \int Z_b^2 \int \eta_b^4$ ;
- (iv) for  $\alpha \in K^\times$  the element  $\alpha j$  generates the same algebra at the same point, and replacing  $j$  by  $\alpha j$  multiplies the two ratios in (iii) by  $N(\alpha)$  and  $N(\alpha)^2$ .

*Proof.* For  $x = u + vj$  one has  $\bar{x} = \bar{u} - vj$ , and the relations  $ju = \bar{u}j$ ,  $j^2 = b$  give  $\bar{x}iy \equiv \bar{u}iu' + bviv'$  modulo  $Kj$ . Since  $\text{trd}$  vanishes on  $Kj$  and restricts to  $\text{tr}_{K/\mathbb{Q}}$  on  $K$ ,

$$E_b(x, y) = \sum_k a_k (\text{tr}(i \bar{u}_k u'_k) + b \text{tr}(i v_k \bar{v}'_k)).$$

From  $jx = b\bar{v} + \bar{u}j$  one reads off  $g_b(x, y) = b \sum_k a_k (\text{tr}(i v_k u'_k) + \text{tr}(i \bar{u}_k \bar{v}'_k))$ , which is  $b g_1$ , and  $\lambda_b = b \lambda_1$  follows. The traces take integral values on  $\mathbb{Z}[i]$ , and  $g_1 = b^{-1} g_b$  is a rational multiple of a Hodge class, which proves (i). Then  $g_b + \sqrt{-d} \lambda_b = b(g_1 + \sqrt{-d} \lambda_1)$  and (ii) holds in cohomology. Choosing  $L, M$  with  $c_1(L) = g_1$ ,  $c_1(M) = \lambda_1$ , the cycle  $c_1(L)^2 - d c_1(M)^2$  equals  $b^2(c_1(L)^2 - d c_1(M)^2)$  in  $\text{CH}^2(A_b)$ ; another choice changes both sides by algebraically trivial cycles.

For (iii),  $E_u$  and  $E_v$  live on complementary halves  $U, V'$  of  $\Lambda \otimes \mathbb{Q}$  of rank four, so  $\eta_u^3 = \eta_v^3 = 0$  and  $\int \eta_b^4 = 6b^2 \int \eta_u^2 \eta_v^2$ . The form  $g_1$  pairs  $U$  with  $V'$ , so  $g_1^2 \in \wedge^2 U^* \otimes \wedge^2 V'^*$ , and only the mixed term survives in  $\int g_b^2 \eta_b^2 = 2b^3 \int g_1^2 \eta_u \eta_v$ ; likewise  $\int Z_b^2 = b^4 \int Z_1^2$ . Hence the ratios  $\int g_b^2 \eta_b^2 / \int \eta_b^4$  and  $\int Z_b^2 / \int \eta_b^4$  are  $b$  and  $b^2$  times their values at  $b = 1$ , which are  $-1/3$  and  $4/3$  by direct evaluation. For (iv),  $(\alpha j)^2 = N(\alpha)b$  and  $g_{\alpha j} + \sqrt{-d} \lambda_{\alpha j}$  is  $\alpha$  or  $\bar{\alpha}$  times  $g_j + \sqrt{-d} \lambda_j$ . ■

Item (XXIII) of [Appendix D](#) confirms (i) to (iv) in exact arithmetic for  $d \in \{1, 3\}$ , two weight vectors and fifteen values of  $b$  up to 101. It also computes the integral Weil lattice  $W_{\mathbb{Z}} = (\mathbb{Q}Z_b + \mathbb{Q}Z'_b) \cap \wedge^4 \Lambda^*$  and the sublattice spanned inside it by products of two integral divisor classes: for  $d \in \{1, 2, 3, 7\}$ , three weight vectors and seven values of  $b$ , the Gram matrix of  $W_{\mathbb{Z}}$  is  $\text{diag}(8d, 8d^2)$  and the index is  $2(a_1 a_2)^2$ , both independent of  $b$ .

*Remark 16.20* (The multiple sits in the polarisation). At these points the Weil class carries no multiple that grows with  $b$  once it is written in integral terms: by [Proposition 16.19\(ii\)](#) it is  $b^2 Z_1$ , with  $Z_1$  one integral algebraic cycle built from the two fixed divisor classes  $g_1, \lambda_1$ . The factor that the orbit argument would otherwise have to absorb into the coefficients  $\underline{q}$  of [Theorem 16.9](#) is therefore absent. What grows is the polarisation:  $\int \eta_b^4 = b^2 \int \eta_1^4$ , so the points  $A_b$  carry polarisations of unbounded degree, and an isogeny carrying  $\eta_b$  to the polarisation of one fixed family has degree  $\int \eta_b^4 / \int \eta_1^4$ , which grows like  $b^2$ . This is the unbounded isogeny degree of [Remark 16.12](#), located at an explicit family of points, and it is untouched by the integrality of  $Z_1$ .

The base points of [Theorem 15.31](#), and every algebraicity statement for Weil classes proved in this paper, represent  $\omega$  through products of divisor classes. The next result says

that a representation of that kind can hold with bounded degrees only on a nowhere dense set, so the bound asked for in [Theorem 16.9](#) has to come from cycles that are not products of divisors. For a class  $x \in H^2(A_s, \mathbb{R})$  put

$$t(x) = \frac{\int x \eta^{2n-1}}{\int \eta^{2n}}, \quad I(x) = t(x)^2 \int \eta^{2n} - \int (x - t(x)\eta)^2 \eta^{2n-2}.$$

Since  $\mathcal{D}_{n,n}$  is contractible, the local system  $R^2\pi_*\mathbb{Z}$  is trivial; write  $H_{\mathbb{Z}}^2$  for its fibre, identified with  $H^2(A_s, \mathbb{Z})$  at every  $s$ . Both  $t$  and  $I$  are polynomial functions on  $H_{\mathbb{Z}}^2$  and do not depend on  $s$ .

**Lemma 16.21** (The Hodge norm of a divisor). *Let  $h_s$  be the Hodge norm on  $H^2(A_s, \mathbb{R})$  defined by the polarisation, normalised by  $h_s(\eta) = \int \eta^{2n}$ . Then*

- (i)  $h_s(x) = I(x)$  for every real class  $x$  of type  $(1, 1)$  at  $s$ ;
- (ii) if  $D$  is an effective divisor on  $A_s$  then  $I([D]) \leq 2(\int [D] \eta^{2n-1})^2 / \int \eta^{2n}$ .

*Proof.* Write  $x = t\eta + x_0$  with  $t = t(x)$ , so that  $x_0$  is primitive. The Lefschetz decomposition is orthogonal for  $h_s$ , and on primitive classes  $h_s(x_0) = -\int x_0 \wedge C_s x_0 \wedge \eta^{2n-2}$ , where  $C_s$  is the Weil operator, positive by the Hodge–Riemann relations. For  $x_0$  of type  $(1, 1)$ ,  $C_s x_0 = x_0$ , which gives (i). An effective divisor on an abelian variety is nef, since a translate of it meets any given curve properly, so  $\int [D]^2 \eta^{2n-2} \geq 0$ . Expanding,  $\int [D]^2 \eta^{2n-2} = t^2 \int \eta^{2n} + \int x_0^2 \eta^{2n-2}$ , hence  $-\int x_0^2 \eta^{2n-2} \leq t^2 \int \eta^{2n}$  and  $I([D]) \leq 2t^2 \int \eta^{2n}$ , which is (ii). ■

**Proposition 16.22** (Divisor products of bounded degree). *In [Setup 15.12](#), fix  $C > 0$  and let  $\Sigma_C \subset \mathcal{D}_{n,n}$  be the set of points  $s$  at which  $\omega_s$  is a  $\mathbb{Q}$ -linear combination of classes  $[D_1] \cdots [D_n]$  with every  $D_i$  an effective divisor on  $A_s$  of degree  $\int [D_i] \eta^{2n-1} \leq C$ . Then  $\Sigma_C$  is contained in a locally finite union of proper closed analytic subsets of  $\mathcal{D}_{n,n}$ . In particular  $\Sigma_C$  is nowhere dense, and on any dense subset of  $\mathcal{D}_{n,n}$  at which  $\omega$  is a combination of products of effective divisors the degrees of those divisors are unbounded on every nonempty open set.*

*Proof.* For  $x \in H_{\mathbb{Z}}^2$  let  $\text{Hdg}(x) \subset \mathcal{D}_{n,n}$  be the closed analytic set where  $x$  is of type  $(1, 1)$ . If  $x \notin \mathbb{Q}\eta$  it is proper, because  $\text{NS}(A_s) \otimes \mathbb{Q} = \mathbb{Q}\eta$  at a very general point by [Lemma 7.5](#). Let  $s \in \Sigma_C$ . Not every  $[D_i]$  in the representation lies in  $\mathbb{Q}\eta$ , since  $\mathbb{Q}\eta^n$  and the Weil line carry different characters by [Lemma 5.4](#). So  $s \in \text{Hdg}(x)$  for some  $x$  in

$$X_C = \{x \in H_{\mathbb{Z}}^2 \setminus \mathbb{Q}\eta : I(x) \leq 2C^2 / \int \eta^{2n}\},$$

by [Lemma 16.21\(ii\)](#). Now let  $U \subset \mathcal{D}_{n,n}$  be open with compact closure and  $s_0 \in U$ . The Hodge norm depends continuously on the point, so  $h_{s_0} \leq c_U h_s$  for some  $c_U$  and all  $s \in U$ . If  $\text{Hdg}(x)$  meets  $U$  at  $s$  and  $x \in X_C$ , then by [Lemma 16.21\(i\)](#)  $h_{s_0}(x) \leq c_U h_s(x) = c_U I(x) \leq 2c_U C^2 / \int \eta^{2n}$ , and only finitely many integral classes satisfy this. So the sets  $\text{Hdg}(x)$ ,  $x \in X_C$ , form a locally finite family of proper closed analytic subsets. Their union is closed with empty interior in the connected manifold  $\mathcal{D}_{n,n}$ , and it contains  $\Sigma_C$ . ■

*Remark 16.23* (What the proposition closes). The loci on which this paper proves  $\omega$  algebraic are the split, quaternionic and product loci and their translates under  $G(\mathbb{Q})$ , and on each of them  $\omega$  is a rational polynomial in divisor classes, since an isogeny carries  $\text{NS} \otimes \mathbb{Q}$  isomorphically. [Proposition 16.22](#) shows that no bounded family of such cycles covers a dense set, so the finiteness in [Theorem 16.9](#) cannot be obtained by bounding the divisors that produce  $\omega$  at the points where it is known to be algebraic. It has to come from the transported cycles of [Lemma 16.6](#) themselves, which are push-forwards along isogenies and not products of divisors on the target. That is the unbounded isogeny

degree of [Remark 16.12](#) and [Remark 16.20](#), now shown to be unavoidable for the divisorial constructions. The argument uses nothing about  $\omega$  beyond the fact that it is not a multiple of  $\eta^n$ , and nothing about the family beyond the generic Picard number, so it applies equally to any variational argument that feeds on divisor products.

At  $n = 2$  the phenomenon can be exhibited on one fixed polarised lattice. Take  $\Lambda = \mathcal{O}^2$  with  $\mathcal{O} = \mathbb{Z}\langle 1, i, j, ij \rangle$ ,  $i^2 = -d$ ,  $j^2 = 1$ , and  $E$  as in (37); this is one member of the family, with its lattice and polarisation fixed. Let  $R = \{x : x(iu, v) = x(u, iv)\} \subset H^2(\Lambda, \mathbb{Q})$  be the space of  $K$ -bilinear classes, where the divisor classes of the quaternionic points live, and for  $x \in R$  let  $\phi_x = E^{-1}x$ , so that  $x(u, v) = E(\phi_x u, v)$ .

**Proposition 16.24** (A pencil of quaternionic loci). *In this setting, for  $d \in \{1, 3\}$ :*

- (i)  $R \cap H^2(\Lambda, \mathbb{Z})$  has rank 12, every  $x \in R$  has  $t(x) = 0$ , and  $I(x) = \frac{1}{24} \text{tr}(\phi_x^2) \int \eta^4$ ; the form  $I$  has signature  $(8, 4)$  on  $R$ ;
- (ii) there are  $x, y \in R \cap H^2(\Lambda, \mathbb{Z})$  such that  $\phi_{x+ky}^2 = \beta(k)$  is a positive rational scalar for every  $k \in \mathbb{Z}$ , with  $\beta(k) = \frac{1}{2} + \frac{1}{2}k + \frac{1}{4}k^2$  for  $d = 1$  and  $\beta(k) = \frac{1}{9} + \frac{2}{9}k^2$  for  $d = 3$ ;
- (iii) for every  $k$  the set  $\text{Hdg}(x + ky)$  is a quaternionic locus with algebra  $(-d, \beta(k))$ , nonempty of dimension 3, and loci with different values of  $\beta$  are not related by any automorphism of  $(\Lambda, E, i)$ ;
- (iv) at a very general point of  $\text{Hdg}(x + ky)$  one has  $\text{NS} \otimes \mathbb{Q} = \mathbb{Q}\eta \oplus \mathbb{Q}x_k \oplus \mathbb{Q}x_k(i \cdot, \cdot)$  with  $x_k = x + ky$ , and for every  $k \in \mathbb{Z}$  the minimum  $\mu_k$  of  $I$  on  $\text{NS} \setminus \mathbb{Q}\eta$  satisfies  $\mu_k \geq \beta(k)/(3N_d^2 \int \eta^4)$ , with  $N_1 = 2$  and  $N_3 = 4$ .

Consequently, at a very general point of  $\text{Hdg}(x + ky)$  every effective divisor whose class is not a multiple of  $\eta$  has degree at least  $\sqrt{\beta(k)}/6/N_d$ , which grows linearly in  $|k|$ .

*Proof.* (i) and (ii) are finite exact computations, item (XXIV) of [Appendix D](#): the trace identity is checked on all pairs of a basis of  $R \cap H^2(\Lambda, \mathbb{Z})$ , which proves it on  $R$  by bilinearity, and in (ii) the three matrices  $\phi_x^2$ ,  $\phi_y^2$  and  $\phi_x\phi_y + \phi_y\phi_x$  are scalar, so  $\phi_{x+ky}^2 = \phi_x^2 + k(\phi_x\phi_y + \phi_y\phi_x) + k^2\phi_y^2$  is scalar for every  $k$ ;  $\beta$  has positive leading coefficient and negative discriminant. For (iii) write  $\phi = \phi_{x+ky}$  and  $\beta = \beta(k)$ . The map  $\phi$  is  $E$ -self-adjoint, anticommutes with  $i$ , and  $\phi^2 = \beta > 0$ , so  $V_{\mathbb{R}} = V_+ \oplus V_-$  with  $V_{\pm}$  the  $\pm\sqrt{\beta}$  eigenspaces,  $E$ -orthogonal to each other, and  $i$  exchanges them. A complex structure  $J$  on  $V_+$  compatible with  $E|_{V_+}$ , extended to  $V_- = iV_+$  by  $J(iw) = iJw$ , commutes with  $i$  and  $\phi$  and is compatible with  $E$ , and  $i$  has eigenvalues  $\pm\sqrt{-d}$  each with multiplicity 2 on  $V^{1,0}$ ; so these  $J$  are points of  $\mathcal{D}_{2,2}$  at which  $x + ky$  is of type  $(1, 1)$ , and they form a copy of the Siegel space of  $V_+$ , of dimension 3, in agreement with [Proposition 15.28](#). An automorphism  $\gamma$  of  $(\Lambda, E, i)$  sends  $\phi_x$  to  $\gamma\phi_x\gamma^{-1}$  and so preserves  $\text{tr}(\phi_x^2)$ .

For the first half of (iv), let  $S_k \subset \mathcal{D}_{2,2}$  be the Siegel space just described and  $\mathcal{C}_k \subset \text{Sp}(V_{\mathbb{R}}, E)$  the centraliser of  $i$  and  $\phi$ . An element of  $\mathcal{C}_k$  preserves  $V_+$  and is determined by its restriction there, because it commutes with  $i$ , so  $\mathcal{C}_k \cong \text{Sp}(V_+)$ , acting on  $V_{\mathbb{R}} = V_+ \oplus iV_+$  as  $g \oplus igi^{-1}$ . A real class is of type  $(1, 1)$  at  $J$  exactly when it is invariant under  $J$ . The points of  $S_k$  form one orbit  $\{gJ_0g^{-1} : g \in \mathcal{C}_k\}$  of a fixed point  $J_0$ . Let  $z$  be a real 2-form invariant under every point of  $S_k$ , and write  $\mathfrak{c} = \text{Lie}(\mathcal{C}_k) \cong \mathfrak{sp}(V_+)$ ,  $\mathfrak{k}$  for the centraliser of  $J_0$  in  $\mathfrak{c}$  and  $\mathfrak{p} = [\mathfrak{c}, J_0]$ . For  $X \in \mathfrak{c}$  the curve  $g_t = \exp(tX)$  gives  $(g_tJ_0g_t^{-1})^*z = z$  for all  $t$ , and the derivative at  $t = 0$  says that  $[X, J_0]$  annihilates  $z$ , so  $\mathfrak{p}$  annihilates  $z$ . The annihilator of  $z$  in  $\mathfrak{c}$  is a Lie subalgebra, so it contains  $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$ . Now  $\mathfrak{c} = \mathfrak{k} \oplus \mathfrak{p}$  with  $[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$  and  $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$ , so  $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$  is an ideal of  $\mathfrak{c}$ ; it is nonzero, and  $\mathfrak{sp}_4(\mathbb{R})$  is simple, so

it is all of  $\mathfrak{c}$ .<sup>38</sup> Hence  $z$  is annihilated by  $\mathfrak{c}$  and, since  $\mathrm{Sp}(V_+)$  is connected, invariant under all of  $\mathcal{C}_k$ . So the classes of type  $(1, 1)$  at every point of  $S_k$  are the invariants of  $\mathrm{Sp}(V_+)$  in  $\Lambda^2(V_+ \otimes \mathbb{R}^2) = (\Lambda^2 V_+ \otimes S^2 \mathbb{R}^2) \oplus (S^2 V_+ \otimes \Lambda^2 \mathbb{R}^2)$ , a space of dimension 3 spanned by  $\eta$ ,  $x_k$  and  $x_k(i, \cdot)$ . Item (XXIV) confirms both steps over  $\mathbb{Q}$ : for  $0 \leq k \leq 4$  the centraliser of  $i$  and  $\phi$  in  $\mathfrak{sp}(V, E)$  has dimension 10 and exactly three invariants in  $\Lambda^2 V^*$ ; and at the values of  $k$  where  $\beta(k)$  is a rational square,  $k = -1$  for  $d = 1$  and  $k = -2$  for  $d = 3$ , a rational point  $J_0$  of  $S_k$  is written down,  $\mathfrak{p} = [\mathfrak{c}, J_0]$  has dimension 6, and  $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$  has dimension 10. Every other rational class  $z$  is of type  $(1, 1)$  only on a proper closed analytic subset of the connected manifold  $S_k$ , and a very general point avoids these countably many subsets.

For the bound in (iv), write  $e = \int \eta^4$ ,  $\iota x = x(i, \cdot)$  and  $M_k = (\mathbb{Q}x_k + \mathbb{Q}\iota x_k) \cap H^2(\Lambda, \mathbb{Z})$ . Since  $\phi_{\iota x} = \phi_x i$  and  $i\phi_x = -\phi_x i$ , the identity of (i) gives  $I((a + b\iota)x_k) = (a^2 + db^2)I(x_k)$  and  $I(x_k) = \beta(k)e/3$ . The index of  $\mathbb{Z}x_k + \mathbb{Z}\iota x_k$  in  $M_k$  divides the gcd  $G(k)$  of the  $2 \times 2$  minors of the matrix with rows  $x_k, \iota x_k$ . These minors are integer polynomials of degree at most two in  $k$ , so  $G(k)$  divides every element of the ideal they generate in  $\mathbb{Z}[k]$  that is a constant; item (XXIV) finds such constants, resultants of pairs of minors, whose gcd is  $N_d$ . Hence  $N_d M_k \subset \mathbb{Z}x_k + \mathbb{Z}\iota x_k$ , and every nonzero  $w \in M_k$  has  $I(w) \geq I(x_k)/N_d^2$ . Now let  $y \in \mathrm{NS} \setminus \mathbb{Q}\eta$  and  $z = y - t(y)\eta$ . Then  $0 \neq ez \in M_k$ , because  $ey$  and  $et(y) = \int y\eta^3$  are integral, and  $I(y) = t(y)^2 e + I(z) \geq I(ez)/e^2 \geq \beta(k)/(3N_d^2 e)$ . The final assertion follows from Lemma 16.21(ii). ■

The bound in (iv) is far from sharp. The exact values computed in item (XXIV) for  $0 \leq k \leq 10$  are  $\mu_k = \beta(k) \int \eta^4/3$  or half of it for  $d = 1$ , and  $\mu_k = \beta(k) \int \eta^4/12$  for  $d = 3$ ; the proof loses the factor  $(\int \eta^4)^2$  in passing from  $y$  to its primitive part, and only the growth in  $k$  matters here. Each locus of the pencil carries algebraic Weil classes, by the divisorial criterion, and on the  $k$ -th of them the divisors that produce those classes have degree growing like  $k$ . This is Proposition 16.22 seen on explicit quaternionic loci in one fixed polarised lattice; Figure 17 draws it.

*Remark 16.25* (At  $n = 2$  the hypothesis is weaker still). Let  $n = 2$ , so  $\dim \mathcal{S} = n^2 = 4$ . The quaternionic locus of Setup 15.24 is the symmetric space of  $\mathrm{Sp}(4, \mathbb{R})$  inside  $\mathcal{D}_{2,2}$ ,<sup>39</sup> of dimension  $n(n+1)/2 = 3$ , so it is a divisor. Its  $G(\mathbb{Q})$ -translates are infinitely many pairwise distinct irreducible closed subvarieties of  $\mathcal{S}$  of dimension 3, since a finite union of proper closed analytic subsets cannot be dense. If infinitely many of them carry cycles with one and the same data  $(P, q)$ , then the closed set  $\Sigma_{P,q}$  has an irreducible component containing infinitely many distinct irreducible threefolds, so that component has dimension at least 4 and therefore equals  $\mathcal{S}$ . No density is needed at  $n = 2$ , only infinitude. Combined with the theorem of Moonen and Zarhin [MZ99] that every Hodge class on an abelian fourfold lies in the subalgebra generated by divisor classes and Weil classes, the conclusion

<sup>38</sup>Simplicity of  $\mathfrak{sp}_4(\mathbb{R})$  is the one fact from Lie theory used here, and it is elementary: the complexification  $\mathfrak{sp}_4(\mathbb{C})$  has the irreducible root system  $C_2$ , a complex semisimple Lie algebra with irreducible root system is simple, and an ideal of a real form is an ideal of the complexification after tensoring with  $\mathbb{C}$ , so a real form of a simple complex Lie algebra is simple. Nothing finer than this is needed, in particular no statement about the abstract group  $\mathrm{PSP}_4(\mathbb{R})$ ; and the conclusion  $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}] = \mathfrak{c}$  is also checked by direct computation in item (XXIV), at the two members of the pencil that carry a rational point.

<sup>39</sup>Left multiplication by the quaternion algebra  $B = (-d, b)$  commutes with  $K$ , and  $B \otimes \mathbb{R} \cong M_2(\mathbb{R})$  because  $b > 0$ , so Morita equivalence writes  $V_{\mathbb{R}}$  as  $W \otimes_{\mathbb{R}} \mathbb{R}^2$  with  $W$  symplectic of dimension  $2n$ . The complex structures commuting with  $B$  are then the complex structures on  $W$ , that is the points of the Siegel space  $\mathfrak{H}_n$ , of dimension  $n(n+1)/2$ . The codimension in  $\mathcal{D}_{n,n}$  is  $n^2 - n(n+1)/2 = n(n-1)/2$ , which is the count verified in item (XII) of Appendix D for  $n \leq 10$  and in `quat_locus_codimension` of Appendix E.

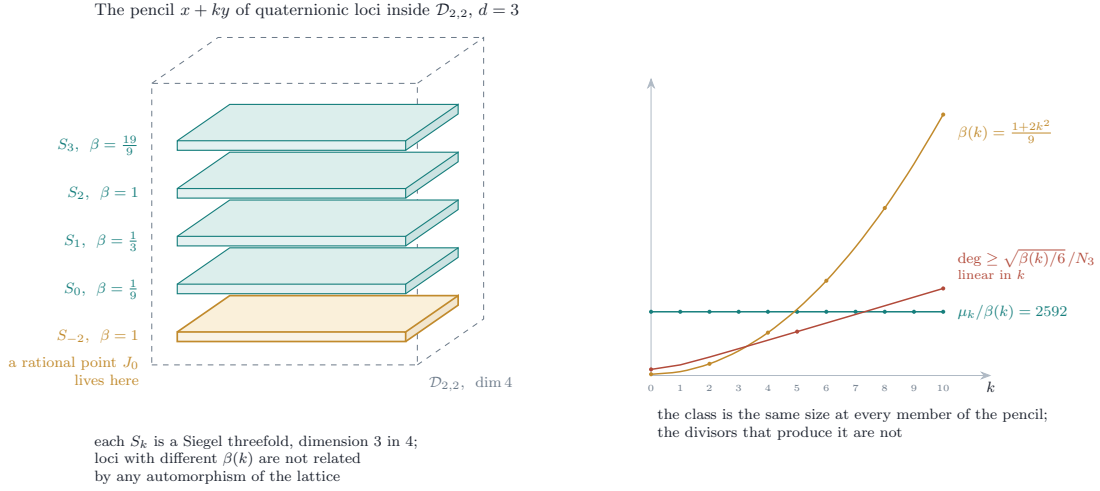


FIGURE 17. **Proposition 16.24** at  $d = 3$ . Left: the members  $S_k$  of the pencil, each a Siegel threefold inside the four-dimensional family, with the member  $k = -2$ , where  $\beta = 1$  and a rational point  $J_0$  can be written down, drawn apart. Right: along the pencil the ratio  $\mu_k/\beta(k)$  is the constant 2592, so the Weil class is the same size at every member, while the least degree of a divisor not proportional to  $\eta$  grows linearly in  $k$ . The class does not change; the divisors that produce it must.

would be the Hodge conjecture for all abelian fourfolds of Weil type, hence for all abelian fourfolds. That conclusion is now Markman's theorem [Mar25a, Corollary 1.6.1], reached by a different route; at  $n = 2$  the argument above would be a second proof, and it is at  $n \geq 3$  that it has something to add.

**16.4. The second criterion: a rank at one point.** The other way to get from a dense algebraic locus to an open one is to make the cycle move, and the classical tool for that is semiregularity. What follows is not a new theorem about semiregularity; it is the observation that after [Theorem 4.14](#), [Theorem 15.31](#) and [Theorem 16.4](#), the semiregularity of a single object at a single point implies the Hodge conjecture for the whole family, with no further propagation argument. The first step is that the object exists, and that its Chern character is as rigid as it can be.

**Proposition 16.26** (The  $K$ -theory class at a base point). *Let  $s_0 \in \Sigma$  be a base point of [Theorem 15.31](#) and write  $A_0 = A_{s_0}$ . There are an integer  $N \geq 1$  and a perfect complex  $E$  on  $A_0$  with*

$$\mathrm{ch}_k(E) = 0 \quad (k \neq n), \quad \mathrm{ch}_n(E) = N \omega_1.$$

*The tuple  $v = \mathrm{ch}(E)$  extends to a flat section of  $\bigoplus_k R^{2k} f_* \mathbb{Q}$  over  $\mathcal{S}$  which is of type  $(k, k)$  at every point.*

*Proof.* By [Theorem 15.31](#) there is  $\xi \in \mathrm{CH}^n(A_0)_{\mathbb{Q}}$  with  $\mathrm{cl}(\xi) = \omega_1$ . For a smooth quasi-projective variety over a field the Chern character  $\mathrm{ch}: K_0(A_0)_{\mathbb{Q}} \rightarrow \mathrm{CH}^*(A_0)_{\mathbb{Q}}$  is a ring isomorphism [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so there is a unique  $u \in K_0(A_0)_{\mathbb{Q}}$  with  $\mathrm{ch}(u) = \xi$ , and its Chern character has no component outside degree  $n$ . Choose  $N$  clearing the denominators of  $u$ ; then  $Nu \in K_0(A_0)$  is a difference of classes of vector bundles and

is therefore the class of a perfect complex  $E$ , with  $\mathrm{ch}(E) = N\xi$ . Passing to cohomology gives the displayed identities.

For the last sentence,  $\mathrm{ch}(E)$  has a single nonzero component, equal to  $N\omega_1$ , and  $\omega$  is a flat section of type  $(n, n)$  everywhere by Lemma 16.2. ■

**Theorem 16.27** (Buchweitz and Flenner [BF03]; Pridham [Pri24]). *Let  $X$  be a smooth projective complex variety,  $E$  a perfect complex on  $X$ , and  $X \hookrightarrow \mathcal{X} \rightarrow \mathrm{Spec} R$  a deformation of  $X$  over an Artin local  $\mathbb{C}$ -algebra along which  $\mathrm{ch}(E)$  stays of Hodge type. If the semiregularity map*

$$\sigma_E: \mathrm{Ext}_X^2(E, E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(X, \Omega_X^q), \quad \sigma_E(\zeta) = \sum_q \frac{1}{q!} \mathrm{tr}(\zeta \circ \mathrm{At}(E)^q),$$

*is injective, then  $E$  extends to a perfect complex on  $\mathcal{X}$ .*

In this form, for perfect complexes and with every obstruction annihilated by  $\sigma_E$  and not only the curvilinear ones, the theorem is proved by Pridham [Pri24], who identifies  $\sigma_E$  with the tangent map of a generalised Abel–Jacobi map on the derived moduli of perfect complexes; for coherent sheaves see also [BLM23]. Perry [Per26] extends it to noncommutative varieties with group actions, which covers twisted derived categories and answers a question of Markman.

**Theorem 16.28** (Semiregularity at one point closes the family). *Let  $n \geq 2$ , let  $s_0$  and  $E$  be as in Proposition 16.26, and suppose the semiregularity map  $\sigma_E$  of Theorem 16.27 is injective.<sup>40</sup> Then  $\Sigma = \mathcal{D}_{n,n}$ , and the Hodge conjecture holds in every degree for every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with  $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$  in that family.*

*Remark 16.29* (The hypothesis is not met by the object above). By Proposition 9.10 the complex of Proposition 16.26 has  $\dim \ker \sigma_E \geq n^2$  at every base point constructed in this paper, so the hypothesis of Theorem 16.28 is not satisfied by it. The theorem is stated because the hypothesis concerns a choice of perfect complex representing a class in  $K_0$ , and a different representative is not excluded; but no representative that is presently available satisfies it. See Remark 9.13, and Theorem 16.30 below for what any representative satisfying it would have to look like.

*Proof.* Let  $\mathfrak{M}$  be the stack of perfect complexes with vanishing negative self-extensions on the fibres of  $f: \mathcal{A} \rightarrow \mathcal{S}$ . It is algebraic and locally of finite presentation over  $\mathcal{S}$  [Lie06], and at a point  $[E]$  over  $s_0$  its deformation theory relative to  $\mathcal{S}$  has tangent space  $\mathrm{Ext}^1(E, E)$  and obstruction space  $\mathrm{Ext}^2(E, E)$ . If  $\mathrm{Ext}^{<0}(E, E) \neq 0$ , which that stack excludes,<sup>41</sup> use instead the analytic Kuranishi germ of pairs of the proof of Theorem 17.5, with the same tangent and obstruction spaces; the argument below then gives a nonempty analytic open set in place of a Zariski open one, which is all that Corollary 16.5 needs.

<sup>40</sup>The hypothesis is not vacuous and it is not free, and the degree-zero component shows both at once. For a line bundle  $L$  one has  $\mathrm{Ext}^2(L, L) = H^2(\mathcal{O}_X)$  and  $\sigma_0 = \mathrm{tr}$  is the identity on it, so  $\sigma_L$  is injective for every line bundle on every variety, and Theorem 16.27 returns the classical fact that a divisor class deforms wherever it stays of type  $(1, 1)$ . The object of Proposition 16.26 has rank zero, so its trace component is not an isomorphism and in general not injective, the source having no reason to be as small as  $h^2(\mathcal{O}_{A_0}) = \binom{2n}{2}$ . The entire hypothesis therefore rests on the components with  $q \geq 1$ , which is exactly the range in which semiregularity stops being formal and starts being a computation.

<sup>41</sup>Lieblisch’s stack parametrises the complexes he calls universally gluable, those with  $\mathrm{Ext}^{<0}(E, E) = 0$ ; without that condition a complex can have automorphisms of negative degree, and the natural moduli object is the derived stack of Toën and Vaquié.

By [Proposition 16.26](#) the Chern character of  $E$  stays of Hodge type along every direction in  $\mathcal{S}$ , because it is a flat section of Hodge classes on the whole base. So [Theorem 16.27](#) applies to every infinitesimal deformation of  $A_0$  inside the family, and injectivity of  $\sigma_E$  kills every obstruction. The morphism  $\mathfrak{M} \rightarrow \mathcal{S}$  is therefore formally smooth at  $[E]$ , and being locally of finite presentation it is smooth at  $[E]$ ; in particular it is dominant near  $[E]$ , so by Chevalley its image contains a nonempty Zariski open  $U \subseteq \mathcal{S}$ .

For  $s \in U$  pick a perfect complex  $E_s$  on  $A_s$  in the fibre. Its Chern character is  $N\omega_1$  in degree  $n$  and zero elsewhere, and Chern characters of perfect complexes are algebraic, so  $N\omega_1$  and hence  $\omega_1$  is algebraic on  $A_s$ ; the rest of the Weil line follows by [Lemma 16.3](#). So  $p^{-1}(U) \subseteq \Sigma$ , which therefore contains a nonempty open set, and [Corollary 16.5](#) gives  $\Sigma = \mathcal{D}_{n,n}$ . Apply [Theorem 15.14\(a\)](#).  $\blacksquare$

**16.5. What the rank at one point forces.** The hypothesis of [Theorem 16.28](#) is a rank, and a rank is not open to inspection until the object is written down. It nevertheless has consequences that can be stated for every representative at once, because the semiregularity map composed with the evaluation map of Hochschild cohomology is contraction into the Chern character ([Setup 18.27](#)), and the Chern character here is a Weil class. The annihilator of a Weil class in  $HT^2$  is large, and injectivity forces the evaluation map to vanish on the whole of it. We compute the annihilator and draw the consequences; they are the exact analogue, for (P2), of what [Theorem 18.34](#) and [Theorem 18.52](#) do for the secant route, and they use the same local products.

**Theorem 16.30** (What injectivity of the semiregularity map forces). *Let  $A$  be an abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type,  $n \geq 2$ , and write  $H^1(A, \mathbb{C}) = V_+ \oplus V_-$  for the eigenspaces of  $K$ ,  $V_{\pm}^{1,0}$ ,  $V_{\pm}^{0,1}$  for their Hodge components and  $T_{\pm} \subset H^0(T_A) = T_0A$  for the eigenspaces of  $K$  on the tangent space, each of dimension  $n$ . Let  $\omega = a\alpha_+ + b\alpha_-$  be a class on the Weil line with  $ab \neq 0$ , for instance any nonzero rational one, and let  $E$  be a nonzero perfect complex on  $A$  with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ , and no other component.*

(i) *The annihilator of  $\text{ch}(E)$  in  $HT^2(A)$  is*

$$\text{Ann}_{HT^2}(\text{ch } E) = (V_+^{0,1} \otimes V_-^{0,1}) \oplus \text{Hom}_K(H^{1,0}, H^{0,1}) \oplus (T_+ \otimes T_-),$$

*of dimension  $n^2 + 2n^2 + n^2 = 4n^2$ : the mixed block of  $H^{0,2}$ , which is the annihilator of [Lemma 9.6](#); the  $K$ -linear first order deformations of  $A$ , which contain the tangent space of the polarised Weil family; and the mixed bivectors.*

- (ii) *If  $\sigma_E$  is injective, then  $\text{ev}_E$  vanishes on this space. Explicitly:  $z \cdot \text{id}_E = 0$  for every  $z \in V_+^{0,1} \otimes V_-^{0,1}$ ;  $E$  extends to first order along every  $K$ -linear deformation of  $A$ ; and the translation classes satisfy  $A_{\theta}A_{\theta'} = 0$  in  $\text{Ext}^2(E, E)$  for every  $\theta \in T_+$  and  $\theta' \in T_-$ .*
- (iii) *Under the same hypothesis the algebra map  $H^*(\mathcal{O}_A) \rightarrow \text{Ext}^*(E, E)$ ,  $z \mapsto z \cdot \text{id}_E$ , kills the ideal generated by the mixed block, which is  $\bigoplus_{a,b \geq 1} \bigwedge^a V_+^{0,1} \otimes \bigwedge^b V_-^{0,1}$ ; so the image of  $H^{0,k}(A)$  has dimension at most  $2\binom{n}{k}$  for  $1 \leq k \leq n$  and is zero for  $k > n$ , and every endomorphism  $f$  of  $E$  has  $\text{tr}(f) = 0$  in  $H^0(\mathcal{O}_A) = \mathbb{C}$ .*
- (iv) *Under the same hypothesis, if at some point  $p$  of its support  $E$  is locally isomorphic to  $i_*F$  with  $F$  locally free of positive rank on a smooth closed subvariety  $Z \ni p$  of codimension at least two, then  $T_+ \subseteq T_pZ$  or  $T_- \subseteq T_pZ$ .*

*Proof.* (i) Write  $H^*(A, \mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$  and  $\alpha_{\pm}$  for a generator of  $\bigwedge^{2n} V_{\pm} = \bigwedge^n V_{\pm}^{1,0} \otimes \bigwedge^n V_{\pm}^{0,1}$ . The three summands of  $HT^2$  act as follows:  $z \in H^{0,2} = \bigwedge^2 H^{0,1}$  by wedge product;

$v \in H^1(T_A) = \text{Hom}(H^{1,0}, H^{0,1})$  as the derivation that replaces one  $(1, 0)$  factor by its image;  $\pi \in H^0(\wedge^2 T_A) = \wedge^2(H^{1,0})^\vee$  by double contraction of  $(1, 0)$  factors. For  $z$ , the computation is [Lemma 9.6](#). For  $v$ , decompose  $\text{Hom}(H^{1,0}, H^{0,1})$  into its four blocks by eigenspaces. A block  $\text{Hom}(V_+^{1,0}, V_+^{0,1})$  replaces a factor of  $\wedge^n V_+^{1,0}$  in  $\alpha_+$  by an element of  $V_+^{0,1}$ , which is already exhausted in  $\alpha_+$ , so it kills  $\alpha_+$ ; it acts trivially on  $\alpha_-$ , which has no  $V_+^{1,0}$  factor. Likewise  $\text{Hom}(V_-^{1,0}, V_-^{0,1})$  kills  $\omega$ . The block  $\text{Hom}(V_+^{1,0}, V_-^{0,1})$  carries  $\alpha_+$  to a nonzero element of  $\wedge^{n-1} V_+^{1,0} \otimes V_-^{0,1} \otimes \wedge^n V_+^{0,1}$  and kills  $\alpha_-$ , and  $\text{Hom}(V_-^{1,0}, V_+^{0,1})$  carries  $\alpha_-$  to a nonzero element of a different summand of  $H^{n-1, n+1}$ ; since  $ab \neq 0$  and the two summands are independent, a general  $v$  kills  $\omega$  if and only if its two mixed blocks vanish, that is, if and only if  $v$  is  $K$ -linear. The space of  $K$ -linear maps has dimension  $2n^2$ , and it contains the  $n^2$ -dimensional tangent space of the polarised family, which is its symmetric part. For  $\pi$ , a mixed bivector  $\theta \wedge \theta'$  with  $\theta \in T_+ = (V_+^{1,0})^\vee$  and  $\theta' \in T_- = (V_-^{1,0})^\vee$  kills  $\alpha_+$  because  $\theta'$  kills every factor of  $\wedge^n V_+^{1,0}$ , and kills  $\alpha_-$  symmetrically; a bivector in  $\wedge^2 T_+$  carries  $\alpha_+$  to a nonzero element of  $\wedge^{n-2} V_+^{1,0} \otimes \wedge^n V_+^{0,1}$  and kills  $\alpha_-$ , and symmetrically for  $\wedge^2 T_-$ . Since the images of the three summands of  $HT^2$  lie in the three different Hodge components  $H^{n, n+2}$ ,  $H^{n-1, n+1}$  and  $H^{n-2, n}$  of  $H^{2n+2}(A, \mathbb{C})$ , the annihilator is the direct sum of the three annihilators computed, of dimensions  $n^2$ ,  $2n^2$  and  $n^2$ . Item (XXXV) of [Appendix D](#) carries out the whole computation in the exterior algebra for  $n = 2$  and  $n = 3$ .

(ii) By the commutative square  $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$  of [Setup 18.27](#),  $\sigma_E$  vanishes on  $\text{ev}_E(\text{Ann}_{HT^2}(\text{ch } E))$ ; if  $\sigma_E$  is injective,  $\text{ev}_E$  vanishes there. The three explicit statements are (52) on the three summands of (i): on  $H^{0,2}$  the evaluation map is  $z \mapsto z \cdot \text{id}_E$ ; on  $H^1(T_A)$  it is the obstruction map  $v \mapsto v \lrcorner \text{At}(E)$ , whose vanishing on  $v$  is the extension of  $E$  to first order along  $v$  [[Tod09](#)]; and on a bivector  $\theta \wedge \theta'$  it is  $\frac{1}{2}(\theta \wedge \theta') \lrcorner \text{At}(E)^2 = A_\theta A_{\theta'}$ , as in the proof of [Theorem 18.34](#).

(iii) The map  $z \mapsto z \cdot \text{id}_E$  is a homomorphism of graded algebras from  $H^*(\mathcal{O}_A) = \wedge^* H^{0,1}$  to the Yoneda algebra, because the classes from  $H^*(A, \mathbb{C})$  are graded central there, so its kernel is an ideal, and by (ii) it contains the mixed block  $V_+^{0,1} \otimes V_-^{0,1}$ . The ideal generated by the mixed block in  $\wedge^*(V_+^{0,1} \oplus V_-^{0,1}) = \wedge^* V_+^{0,1} \otimes \wedge^* V_-^{0,1}$  is the sum of the summands  $\wedge^a V_+^{0,1} \otimes \wedge^b V_-^{0,1}$  with  $a, b \geq 1$ , since each such summand is a multiple of the mixed block and the remaining summands,  $a = 0$  or  $b = 0$ , meet the ideal in zero; the quotient in degree  $k \geq 1$  is  $\wedge^k V_+^{0,1} \oplus \wedge^k V_-^{0,1}$ , of dimension  $2\binom{n}{k}$ , and zero for  $k > n$ . In particular the top class of  $H^{2n}(\mathcal{O}_A)$ , which is mixed for  $n \geq 1$ , acts by zero. Serre duality on  $A$ , whose canonical bundle is trivial, is the perfect pairing  $\text{Ext}^{2n}(E, E) \times \text{Hom}(E, E) \rightarrow H^{2n}(\mathcal{O}_A) \cong \mathbb{C}$ ,  $(\xi, f) \mapsto \text{tr}(\xi \circ f)$ ; for  $\xi = w \cdot \text{id}_E$  with  $w$  the top class this is  $w \cdot \text{tr}(f)$ , where  $\text{tr}(f) \in H^0(\mathcal{O}_A) = \mathbb{C}$ , so  $w \cdot \text{id}_E = 0$  forces  $\text{tr}(f) = 0$  for every  $f$ .

(iv) Let  $\mathcal{O}$  be the local ring of  $A$  at  $p$  and choose coordinates with  $Z = \{x_1 = \cdots = x_c = 0\}$ ,  $c \geq 2$ . Locally  $E \cong (\mathcal{O}/J)^{\oplus r}$  with  $J = (x_1, \dots, x_c)$ , a regular sequence, so  $\text{Ext}_{\mathcal{O}}^*(\mathcal{O}/J, \mathcal{O}/J) = \wedge_{\mathcal{O}/J}^* N$  as a graded algebra, where  $N = \text{Hom}(J/J^2, \mathcal{O}/J)$  is the normal module with basis the classes  $\partial_1, \dots, \partial_c$  dual to  $x_1, \dots, x_c$ ; the  $\text{Ext}^1$  is computed on the Koszul resolution exactly as in the proof of [Lemma 18.33](#), and the product is the wedge because the Yoneda algebra of a Koszul complex is the exterior algebra on the dual of the generators. For  $E$  of rank  $r$  the local Yoneda algebra is  $\wedge^* N \otimes M_r(\mathcal{O}/J)$ . By [Lemma 18.32](#) the germ of  $A_\theta$  at  $p$  is the jet class along  $\theta$ ; on the Koszul complex the jet class of a derivation  $D$  is the chain map  $-D(d)$ , and for  $D = \theta = \sum_i c^i \partial_i$  its component in  $\text{Ext}^1 = N \otimes M_r(\mathcal{O}/J)$  is  $\sum_{i \leq c} (c^i \bmod J) \partial_i \otimes \text{id}$ , the class of  $\theta$  modulo  $T_Z$ . Multiplicativity of the germ map

then gives the germ of  $A_\theta A_{\theta'}$  as  $(\theta \bmod T_Z) \wedge (\theta' \bmod T_Z) \otimes \text{id}$  in  $\bigwedge^2 N \otimes M_r(\mathcal{O}/J)$ , whose value at  $p$  is  $\bar{\theta}(p) \wedge \bar{\theta}'(p) \in \bigwedge^2(T_p A/T_p Z)$ . If  $A_\theta A_{\theta'} = 0$  for all  $\theta \in T_+$ ,  $\theta' \in T_-$ , then  $\bar{a} \wedge \bar{b} = 0$  for all  $\bar{a}$  in the image  $I_+$  of  $T_+$  and  $\bar{b}$  in the image  $I_-$  of  $T_-$  in  $N_p = T_p A/T_p Z$ , and  $I_+ + I_- = N_p$  since  $T_+ \oplus T_- = T_p A$ . If both  $I_\pm$  were nonzero, every nonzero  $\bar{b} \in I_-$  would be proportional to every nonzero  $\bar{a} \in I_+$ , so  $I_+ = I_-$  would be a line and  $N_p = I_+ + I_-$  of dimension one, against  $c \geq 2$ . Hence  $I_+ = 0$  or  $I_- = 0$ , which is the claim. The two pieces of linear algebra, the shape of the ideal in (iii) and the dichotomy just used, are checked in item (XXXV). ■

**Corollary 16.31** (Where (P2) can and cannot be met). *Let  $A$  be as in Theorem 16.30 and suppose that no proper abelian subvariety  $B \subset A$  has  $T_0 B \supseteq T_+$  or  $T_0 B \supseteq T_-$ ; equivalently, that  $A$  has no  $K$ -stable abelian subvariety on which  $K$  acts through a single embedding, which holds whenever  $A$  is simple, at every split member  $X \times \hat{X}$  with  $X$  simple of endomorphism algebra  $\mathbb{Q}$ , and at a general point of the quaternionic locus of Setup 15.24. Let  $E$  be a perfect complex on  $A$  with  $\text{ch}(E) = N\omega$  as above and  $\sigma_E$  injective. Then no irreducible component  $Z_0$  of the support of  $E$  of codimension at least two has a dense open subset along which  $E$  is a locally free sheaf of positive rank on  $Z_0$ . In particular  $E$  is not a sheaf, and along every component of its support of codimension at least two it has, generically, either at least two nonvanishing cohomology sheaves or a cohomology sheaf that is not locally free on the reduced support.*

*Proof.* Suppose  $Z_0$  is such a component, with  $U \subseteq Z_0^{\text{sm}}$  dense open and  $E|_U$  a locally free sheaf of positive rank on  $U$ . By Theorem 16.30(iv),  $U = U_+ \cup U_-$  with  $U_\pm = \{p \in U : T_\pm \subseteq T_p Z_0\}$ , both closed in  $U$ ; as  $U$  is irreducible,  $U = U_+$  say. Then every constant vector field  $\theta \in T_+$  is tangent to  $Z_0$  along  $U$ , hence along  $Z_0^{\text{sm}}$ . The set  $S = \{(p, t) \in Z_0 \times \mathbb{C} : p + \exp(t\theta) \in Z_0\}$  is a closed analytic subset of the irreducible space  $Z_0 \times \mathbb{C}$ , and it contains an open neighbourhood of  $Z_0^{\text{sm}} \times \{0\}$ , because the local flow of the restriction of  $\theta$  to the submanifold  $Z_0^{\text{sm}}$  exists on such a neighbourhood; so  $S = Z_0 \times \mathbb{C}$ , and  $Z_0 + \exp(t\theta) \subseteq Z_0$ , hence  $= Z_0$ , for every  $t$ . Let  $G = \{x \in A : Z_0 + x = Z_0\}$ , a closed algebraic subgroup of  $A$ , and  $B$  its identity component, an abelian subvariety. The connected set  $\exp(T_+)$  contains 0 and lies in  $G$ , hence in  $B$ , so  $T_+ \subseteq T_0 B$ . By hypothesis  $B = A$ , so  $Z_0 = A$ , against the codimension. If  $E$  were a sheaf, its support would have codimension exactly  $n \geq 2$  and along a dense open set of every component it would be locally free of positive rank on the reduced support, which is excluded; the last sentence is the negation of the same condition.

For the equivalence in the hypothesis, let  $\sqrt{-d}$  act on  $A$  through an order of  $K$ . If  $B \neq A$  has  $T_0 B \supseteq T_+$ , then  $H_1(B, \mathbb{Q}) \cap \sqrt{-d} H_1(B, \mathbb{Q})$  is a  $K$ -stable sub-Hodge structure, hence  $H_1(C, \mathbb{Q})$  for a  $K$ -stable abelian subvariety  $C \subseteq B$ , and its real points  $T_0 B \cap \sqrt{-d} T_0 B$  still contain  $T_+$ , on which  $\sqrt{-d}$  acts by a scalar. The  $K$ -stable complement  $C'$  of  $C$  for the polarisation is nonzero, since  $C \subseteq B \neq A$ , and  $T_0 C'$  is a  $K$ -stable complement of  $T_0 C \supseteq T_+$ , so  $T_0 C' \subseteq T_-$  and  $K$  acts on  $C'$  through a single embedding. Conversely, if a nonzero  $K$ -stable  $C' \neq A$  has  $T_0 C' \subseteq T_-$ , its  $K$ -stable complement  $B$  is proper with  $T_0 B \supseteq T_+$ . For the three cases: a simple  $A$  has no proper abelian subvariety; for  $X \times \hat{X}$  with  $X$  simple and  $\text{End}^0(X) = \mathbb{Q}$  the sub-Hodge structures of  $H_1(X) \otimes \mathbb{Q}^2$  are  $H_1(X) \otimes W$  with  $W \subseteq \mathbb{Q}^2$ , and  $W$  is  $K$ -stable only for  $W = 0$  or  $\mathbb{Q}^2$ ,  $K$  acting on  $\mathbb{Q}^2$  through a quadratic field; and at a general quaternionic member  $A$  is either simple or, when the quaternion algebra of Setup 15.24 splits, isogenous to  $C \times C$  with  $C$  a general principally polarised abelian  $n$ -fold, whose sub-Hodge structures are again  $H_1(C) \otimes W$  with the same conclusion. ■

**16.6. The whole Hochschild action, and the two branches.** [Theorem 16.30](#) computes the annihilator degree by degree and reads off what the vanishing of the evaluation map on it means for  $E$ . The annihilator has a structure that the degree-by-degree description hides, and bringing it out costs nothing and gives more: it is the mixed part of a splitting of  $HH^1(A)$  into two halves, one for each of the two conjugate pieces of the Weil class. Everything in this subsection is about the Hochschild cohomology of  $A$  alone, and the consequences for  $E$  follow from it formally.

**Setup 16.32** (Hochschild cohomology in these terms). For an abelian variety  $A$  the Hochschild–Kostant–Rosenberg isomorphism identifies

$$HH^*(A) \cong HT^*(A) = \bigoplus_{p,q} H^p(\wedge^q T_A) = \wedge^*(H^{0,1}(A) \oplus H^0(T_A)),$$

as a graded ring, the grading being  $p+q$ ; the Todd class of  $A$  is trivial, so the twist by  $\mathrm{td}^{1/2}$  in [\[Cal05\]](#), [\[CRV12\]](#) is the identity and the identification is a ring isomorphism on the nose. Write  $HH^1(A) = H^{0,1}(A) \oplus H^0(T_A)$ , of dimension  $4n$ , so that  $HH^*(A)$  is the exterior algebra on it, of dimension  $2^{4n}$ . It acts on  $H^*(A, \mathbb{C}) = \wedge^*(H^{1,0} \oplus H^{0,1})$  by wedging with the classes of  $H^{0,1}$  and contracting the classes of  $H^{1,0}$  against  $H^0(T_A)$ ; the two families of operators involve disjoint generators, so they pairwise anticommute and the action is an action of the exterior algebra. In degree two this is the operator  $x_\perp$  of [Setup 18.27](#), and  $\mathrm{ev}_E: HH^*(A) \rightarrow \mathrm{Ext}^*(E, E)$  is a homomorphism of graded rings into the graded centre, with  $\sigma_E \circ \mathrm{ev}_E = (\cdot)_\perp \mathrm{ch}(E)$  [\[BF08, Section 6\]](#), [\[Cal05\]](#), [\[CRV12\]](#).

**Theorem 16.33** (The annihilator is the mixed part of a splitting). *Let  $A$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$ , and let  $\omega = a\alpha_+ + b\alpha_-$  lie on the Weil line with  $ab \neq 0$ . Put*

$$P = \mathrm{Ann}_{HH^1}(\alpha_+) = V_+^{0,1} \oplus T_-, \quad Q = \mathrm{Ann}_{HH^1}(\alpha_-) = V_-^{0,1} \oplus T_+.$$

- (i)  $\dim P = \dim Q = 2n$  and  $HH^1(A) = P \oplus Q$ ; in particular  $\mathrm{Ann}_{HH^1}(\omega) = 0$ .
- (ii)  $\mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$ , of dimension  $4n^2$ . This is the space computed in [Theorem 16.30\(i\)](#), written as the mixed part of the splitting.
- (iii) Contraction with  $\alpha_-$  is injective on  $\wedge^* P$ , and contraction with  $\alpha_+$  is injective on  $\wedge^* Q$ ; the two images meet in exactly one line, in degree  $2n$ .
- (iv) For every  $k \geq 1$ ,

$$\mathrm{Ann}_{HH^k}(\omega) = \left( \bigoplus_{\substack{i+j=k \\ i,j \geq 1}} \wedge^i P \wedge \wedge^j Q \right) \oplus \epsilon_k,$$

where  $\epsilon_k = 0$  for  $k \neq 2n$  and  $\epsilon_{2n}$  is the line of (iii). So  $\dim \mathrm{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$  for  $k \geq 1$ , while  $\mathrm{Ann}_{HH^0}(\omega) = 0$ , and the annihilator is the ideal generated by its degree-two part together with that one line.

*Proof.* Choose bases  $x_1, \dots, x_n$  of  $V_+^{1,0}$  and  $y_1, \dots, y_n$  of  $V_-^{1,0}$ , and bases  $\bar{x}_i$  of  $V_+^{0,1}$  and  $\bar{y}_j$  of  $V_-^{0,1}$  (the bar is notation only: complex conjugation carries  $V_+^{1,0}$  to  $V_-^{0,1}$ ), so that  $\alpha_+ = x_1 \cdots x_n \bar{x}_1 \cdots \bar{x}_n$  and  $\alpha_- = y_1 \cdots y_n \bar{y}_1 \cdots \bar{y}_n$ , and write  $\partial_{x_i}, \partial_{y_j}$  for the dual basis of  $H^0(T_A) = T_+ \oplus T_-$ .

(i) Of the  $4n$  basis operators,  $\bar{x}_i \wedge$  kills  $\alpha_+$  because  $\bar{x}_i$  already occurs in it, and  $\partial_{y_j} \lrcorner$  kills  $\alpha_+$  because no  $y$  occurs in it; so  $P \supseteq V_+^{0,1} \oplus T_-$ . Conversely  $\bar{y}_j \wedge \alpha_+ \neq 0$  and  $\partial_{x_i} \lrcorner \alpha_+ \neq 0$ , and these images lie in different summands of  $H^*$ , so no nonzero combination of them kills  $\alpha_+$ :  $P = V_+^{0,1} \oplus T_-$ , of dimension  $2n$ . The computation for  $Q$  is the same with the roles of

the two eigenspaces exchanged, and  $P \cap Q = 0$  with  $\dim P + \dim Q = 4n$ , so  $HH^1 = P \oplus Q$ . A class killing  $\omega$  in degree one kills  $\alpha_+$  and  $\alpha_-$  separately, the two images lying in different summands and  $ab \neq 0$ , so it lies in  $P \cap Q = 0$ .

(iii) A monomial  $\bar{x}_I \partial_{y_J} \in \Lambda^* P$  contracts  $\alpha_-$  to  $\pm \bar{x}_I y_{[n] \setminus J} \bar{y}_{[n]}$ , and these are linearly independent for distinct pairs  $(I, J)$ , which is the injectivity; the statement for  $Q$  is symmetric. For the intersection, the image of  $\Lambda^* P$  consists of classes with no  $x$  and at most  $n$  factors  $\bar{x}$ , and the image of  $\Lambda^* Q$  of classes with  $n$  factors  $\bar{x}$  and  $n - \#\{\partial_x\}$  factors  $x$ . A common class therefore has no  $x$  and all  $n$  factors  $\bar{x}$  and, by the same count in the other eigenspace, no  $y$  and all  $n$  factors  $\bar{y}$ : it is a multiple of  $\bar{x}_{[n]} \bar{y}_{[n]}$ , and it comes from  $\pi_P = \bar{x}_{[n]} \partial_{y_{[n]}}$  and  $\pi_Q = \bar{y}_{[n]} \partial_{x_{[n]}}$ , both of degree  $2n$ . So the intersection is the single line spanned by that class.

(ii) and (iv) Decompose  $\Lambda^k HH^1 = \bigoplus_{i+j=k} \Lambda^i P \wedge \Lambda^j Q$ . A summand with  $i, j \geq 1$  annihilates  $\alpha_+$ , because it has a factor in  $P$  and the action is by an algebra, and annihilates  $\alpha_-$  for the same reason with  $Q$ ; so it lies in the annihilator of  $\omega$ . On the remaining two summands,  $\xi_P \in \Lambda^k P$  has  $\xi_P \lrcorner \omega = b \xi_P \lrcorner \alpha_-$  and  $\xi_Q \in \Lambda^k Q$  has  $\xi_Q \lrcorner \omega = a \xi_Q \lrcorner \alpha_+$ , so by (iii) a combination  $\xi_P + \xi_Q$  annihilates  $\omega$  if and only if  $b \xi_P \lrcorner \alpha_- = -a \xi_Q \lrcorner \alpha_+$ , which by (iii) happens only for  $k = 2n$  and then only on the line  $b^{-1} \pi_P - a^{-1} \pi_Q$  up to scale. Taking  $k = 2$  and  $n \geq 2$ , so that  $2 \neq 2n$ , gives (ii), the dimension being  $\dim P \cdot \dim Q = 4n^2$ ; the general count is  $\binom{4n}{k} - \binom{2n}{k} - \binom{2n}{k}$  plus the exceptional line. Item (XXXVIII) of [Appendix D](#) carries out the whole computation in the exterior algebra over  $\mathbb{Q}$  for  $n = 1$  and  $n = 2$  in every degree, for two choices of  $(a, b)$ , and for  $n = 3$  in degrees up to three. ■

**Corollary 16.34** (What the criterion forces on the Hochschild action). *Let  $A$  and  $\omega$  be as above and let  $E$  be a nonzero perfect complex on  $A$  with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ .*

- (i) *Unconditionally,  $\dim \text{Ext}^2(E, E) \geq 2n(2n - 1)$ .*
- (ii) *If  $\sigma_E$  is injective then  $\text{ev}_E$  vanishes on  $P \wedge Q$ , hence on the ideal it generates, and the ring homomorphism  $\text{ev}_E: HH^*(A) \rightarrow \text{Ext}^*(E, E)$  factors through*

$$R = \Lambda^* P \oplus_{\mathbb{C}} \Lambda^* Q, \quad \dim_{\mathbb{C}} R = 2^{2n+1} - 1,$$

*the two exterior algebras glued along their scalars, inside  $HH^*(A)$  of dimension  $2^{4n}$ . At  $n = 2$  that is 31 out of 256.*

- (iii) *Granting the compatibility  $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$  in degree  $k$  [[Cal05](#)], [[CRV12](#)], the induced map on  $R$  is injective away from the line of [Theorem 16.33\(iii\)](#), so*

$$\dim \text{Ext}^k(E, E) \geq 2 \binom{2n}{k} \quad (1 \leq k < 2n), \quad \sum_{k \geq 1} \dim \text{Ext}^k(E, E) \geq 2^{2n+1} - 3.$$

*Proof.* (i) By [Setup 16.32](#),  $\ker \text{ev}_E \subseteq \text{Ann}_{HH^2}(\text{ch } E)$  in degree two, since  $\sigma_E(\text{ev}_E(\xi)) = \xi \lrcorner \text{ch}(E)$  and the right side is nonzero off the annihilator. Hence  $\dim \text{Ext}^2(E, E) \geq \dim HH^2(A) - \dim \text{Ann}_{HH^2}(\omega) = \binom{4n}{2} - 4n^2 = 2n(2n - 1)$  by [Theorem 16.33\(ii\)](#). No hypothesis on  $\sigma_E$  is used.

(ii)  $\text{ev}_E$  vanishes on the annihilator by [Theorem 16.30\(ii\)](#), its kernel is an ideal because it is a ring map, and the ideal generated by  $P \wedge Q$  in  $\Lambda^*(P \oplus Q) = \Lambda^* P \otimes \Lambda^* Q$  is the sum of the summands with  $i, j \geq 1$ , whose complement is  $\Lambda^* P \oplus_{\mathbb{C}} \Lambda^* Q$  of dimension  $2 \cdot 2^{2n} - 1$ .

(iii) With the compatibility in degree  $k$ , the argument of (i) gives  $\ker \text{ev}_E \subseteq \text{Ann}_{HH^k}(\omega)$ , and by [Theorem 16.33\(iv\)](#) that annihilator meets  $\Lambda^k P \oplus \Lambda^k Q$  only in the exceptional line. So the map induced on  $R$  has kernel contained in that one line and its image has dimension at least  $\dim R - 1 = 2^{2n+1} - 2$ . One of those dimensions sits in degree zero, where the image is  $\mathbb{C} \cdot \text{id}_E$ , nonzero because  $E$  is; the rest is distributed over the positive degrees,

which is the second display, and in degree  $k$  the image contains the  $2\binom{2n}{k}$  dimensions of  $\wedge^k P \oplus \wedge^k Q$  for  $k < 2n$ , which is the first.  $\blacksquare$

*Remark 16.35* (The two branches, and what they are not). The ring  $R$  of [Corollary 16.34\(ii\)](#) is the ring of functions on two odd affine spaces of dimension  $2n$  crossing at the origin. Read that way, the statement is that if  $E$  satisfies the criterion then the deformations of  $D^b(A)$  along which its Hochschild action survives form two branches,  $P$  and  $Q$ , meeting only at the trivial deformation; and the two branches are the annihilators of the two conjugate pieces  $\alpha_+$  and  $\alpha_-$  of the Weil class. The criterion says that  $E$  sees the two halves of the class separately and their interaction not at all.

Two things this is not. The splitting  $HH^1 = P \oplus Q$  is not the splitting into eigenspaces of  $K$ : the eigenspaces are  $V_+^{0,1} \oplus T_+$  and  $V_-^{0,1} \oplus T_-$ , and  $P$  and  $Q$  cross them. It is the splitting into the annihilators of  $\alpha_+$  and of  $\alpha_-$ , and it exists because the Weil class is a sum of two pieces each of which is a top form on an eigenspace. And the conclusion is not a contradiction:  $R$  is a perfectly good ring, the bound of (iii) is consistent with  $\chi(E, E)$  computed in [Proposition 9.11](#), since  $\sum_k (-1)^k 2\binom{2n}{k} = 0$ , and nothing here forbids an  $E$  of the required shape. What the corollary does is say how large such an  $E$  must be and how little of the Hochschild cohomology it may see. We have not constructed one, and we do not claim that one exists.

**16.7. The criterion as a dimension, and where the object can live.** The same square gives two more things, and between them they change what one would look for. The first converts the rank of  $\sigma_E$  into a single dimension, which is the form in which a candidate can actually be tested. The second says where the support of  $E$  may lie, and it excludes the place where anyone would look first, namely a cycle representing the Weil class.

**Theorem 16.36** (The criterion is a dimension). *Let  $A$ ,  $\omega$  and  $E$  be as in [Corollary 16.34](#), so that  $\text{ch}(E) = N\omega$  with  $N \neq 0$  and no other component.*

- (i) *If  $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$ , the least value that [Corollary 16.34\(i\)](#) allows, then  $\text{ev}_E: HH^2(A) \rightarrow \text{Ext}^2(E, E)$  is surjective with kernel exactly  $P \wedge Q$ , the map  $\sigma_E$  is injective, and every class of  $\text{Ext}^2(E, E)$  is a sum of products  $\text{ev}_E(p)\text{ev}_E(p') + \text{ev}_E(q)\text{ev}_E(q')$  with  $p, p' \in P$  and  $q, q' \in Q$ , hence lies in the graded centre of  $\text{Ext}^*(E, E)$ .*
- (ii) *Conversely, if  $\sigma_E$  is injective, then  $\text{ev}_E(HH^2(A))$  has dimension exactly  $2n(2n - 1)$ , and  $\sigma_E$  carries it isomorphically onto  $HH^2(A) \lrcorner \omega$ .*
- (iii) *Consequently, if at one base point  $s_0$  of a family some perfect complex  $E$  with  $\text{ch}(E) = N\omega_1$  has  $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$ , then  $\Sigma = \mathcal{D}_{n,n}$ , and the Hodge conjecture holds in every degree for every member of that family with  $\text{Hg}(A) = \text{SU}(V, H)$ .*

*Proof.* (i) The square of [Setup 16.32](#) gives  $\ker \text{ev}_E \subseteq \text{Ann}_{HH^2}(\omega) = P \wedge Q$  in degree two, the equality being [Theorem 16.33\(ii\)](#). Hence  $\text{rank } \text{ev}_E \geq \binom{4n}{2} - 4n^2 = 2n(2n - 1) = \dim \text{Ext}^2(E, E)$ , so  $\text{ev}_E$  is onto and its kernel, of dimension  $4n^2$  and contained in  $P \wedge Q$ , is all of  $P \wedge Q$ . Every class of  $\text{Ext}^2(E, E)$  is then  $\text{ev}_E(\xi)$  for some  $\xi$ , and  $\sigma_E(\text{ev}_E(\xi)) = \xi \lrcorner \omega$  vanishes only for  $\xi \in P \wedge Q = \ker \text{ev}_E$ ; so  $\sigma_E$  is injective. Since  $HH^2(A) = \wedge^2 P \oplus (P \wedge Q) \oplus \wedge^2 Q$  and  $\text{ev}_E$  is a ring map into the graded centre, the image is  $\text{ev}_E(\wedge^2 P) + \text{ev}_E(\wedge^2 Q)$ , which is the description of the classes.

(ii) By [Theorem 16.30\(ii\)](#) the kernel of  $\text{ev}_E$  in degree two contains  $P \wedge Q$ , and by the argument of (i) it is contained in it, so it equals  $P \wedge Q$  and the image has dimension

$2n(2n - 1)$ . On that image  $\sigma_E \circ \text{ev}_E$  is contraction into  $\omega$ , whose kernel is exactly  $P \wedge Q$ , so  $\sigma_E$  maps the image isomorphically onto  $HH^2(A) \lrcorner \omega$ .

(iii) By (i) the map  $\sigma_E$  is injective, and [Theorem 16.28](#) applies.  $\blacksquare$

[Theorem 16.36](#)(iii) is a sufficient condition for  $\mathbf{W}(K, n, \delta)$  that mentions no map at all: one object, at one point, and one number.<sup>42</sup> At  $n = 2$  the number is 12, at  $n = 3$  it is 30, and the two computations behind it are item (XXXIX) of [Appendix D](#). Part (ii) says that nothing is lost in reading the criterion this way on the central part of  $\text{Ext}^2$ ; whether an injective  $\sigma_E$  forces the whole of  $\text{Ext}^2(E, E)$  to be central is a statement about  $\text{Ext}^{2n-2}$ , by Serre duality, which the square in degree two does not reach, and nothing below uses it.

The second consequence rests on a local fact about complexes supported at a point, which we state and prove first.

**Lemma 16.37** (Two jet classes on a complex of finite length). *Let  $R = \mathbb{C}[[x_1, \dots, x_c]]$  with  $c \geq 2$ , let  $M$  be a bounded complex of finitely generated free  $R$ -modules whose cohomology has finite length, and put  $\chi(M) = \sum_i (-1)^i \ell(H^i(M))$ . For  $a \in \mathbb{C}^c$  let  $J(a) \in \text{Ext}_R^1(M, M)$  be the jet class of  $M$  along the constant vector field  $\sum_i a_i \partial_i$ , represented by the chain map  $-\sum_i a_i \partial_i(d)$  as in the proof of [Theorem 16.30](#)(iv). If  $\chi(M) \neq 0$  and  $a, b$  are linearly independent, then  $J(a)J(b) \neq 0$  in  $\text{Ext}_R^2(M, M)$ .*

*Proof.* Let  $T$  be an abelian variety of dimension  $c$  and  $t \in T$  a point. Linear coordinates on the universal cover identify the completed local ring  $\hat{\mathcal{O}}_{T,t}$  with  $R$  in such a way that the translation invariant vector fields become the constant fields  $\sum_i a_i \partial_i$ .<sup>43</sup> A complex of  $R$ -modules with finite length cohomology is the same thing as a complex of  $\mathcal{O}_{T,t}$ -modules with finite length cohomology, since both are modules over  $\mathcal{O}_{T,t}/\mathfrak{m}^N$  for  $N$  large, so  $M$  defines a perfect complex  $M_T$  on  $T$  supported at  $t$ . Its local  $\mathcal{E}xt$  sheaves are skyscrapers at  $t$ , so the local to global spectral sequence gives  $\text{Ext}_T^*(M_T, M_T) = \text{Ext}_R^*(M, M)$ , and by [Lemma 18.32](#) the translation classes  $A_a, A_b$  of  $M_T$  restrict to  $J(a), J(b)$ . Apply the square of [Setup 18.27](#) on  $T$  to the bivector  $a \wedge b$ :

$$\sigma_{M_T}(A_a A_b) = (a \wedge b) \lrcorner \text{ch}(M_T).$$

A sheaf of length  $\ell$  supported at a point has Chern character  $\ell \cdot [\text{pt}]$ , and Chern characters are additive over the cohomology of a complex, so  $\text{ch}(M_T) = \chi(M) [\text{pt}]$ . The class of a point is a nonzero multiple of  $dz_1 \cdots dz_c d\bar{z}_1 \cdots d\bar{z}_c$ , and contracting it with  $a \wedge b$  removes two holomorphic factors and leaves a nonzero class of type  $(c - 2, c)$ , because  $a$  and  $b$  are independent. So the right side is nonzero when  $\chi(M) \neq 0$ , hence so is  $A_a A_b$ , and its restriction  $J(a)J(b)$  is nonzero.  $\blacksquare$

The hypothesis on  $\chi(M)$  cannot be dropped, and here is why.<sup>44</sup> Item (XL) of [Appendix D](#) takes the Koszul resolution  $K$  of the residue field  $k$  of  $\mathbb{Q}[x_1, x_2]$  and forms the cone  $C$

<sup>42</sup>The dimension of  $\text{Ext}^2(E, E)$  is computable in a way the rank of  $\sigma_E$  is not. For a complex built from sheaves whose cohomology is known, the local to global spectral sequence  $E_2^{p,q} = H^p(A, \mathcal{E}xt^q(E, E)) \Rightarrow \text{Ext}^{p+q}(E, E)$  reduces the count to the cohomology of the local  $\mathcal{E}xt$  sheaves, and on an abelian variety the cohomology of a sheaf supported on an abelian subvariety, or of a homogeneous bundle, is a question of linear algebra. The rank of  $\sigma_E$ , by contrast, requires the trace of the exponential of the Atiyah class on every class of  $\text{Ext}^2$ , which is a computation with the object and not only with its invariants.

<sup>43</sup>The exponential map of  $T$  is a local biholomorphism from  $T_t T = \mathbb{C}^c$ , and it carries the constant vector fields on  $\mathbb{C}^c$  to the translation invariant ones. Its germ at the origin identifies  $\hat{\mathcal{O}}_{T,t}$  with  $\mathbb{C}[[x_1, \dots, x_c]]$ , and the identification is compatible with derivations. Nothing about  $T$  beyond its being a compact torus carrying a projective structure is used.

<sup>44</sup>The Euler characteristic is the only additive invariant here: the Grothendieck group of complexes with finite length cohomology over a regular local ring is  $\mathbb{Z}$ , generated by the residue field, since every module

of the class  $J(\partial_2)J(\partial_1)$ , viewed as a chain map  $K \rightarrow K[2]$ . The complex  $C$  has two cohomology modules, each isomorphic to  $k$ , so  $\chi(C) = 0$ , and on  $C$  the product  $J(\partial_2)J(\partial_1)$  is null homotopic: the gluing has killed it. On the split complex  $k \oplus k[1]$ , whose Euler characteristic is also zero, the same product is not null homotopic. The same item tests the lemma on eighteen finite length modules in two and three variables, among them powers of the maximal ideal, monomial and non-monomial complete intersections, a Gorenstein algebra of socle degree two and three modules that are not cyclic, and on three complexes with several cohomology modules, for every pair of coordinate directions and for a random pair; in every case with  $\chi \neq 0$  the product is nonzero.

**Theorem 16.38** (Where an object meeting the criterion can live). *Let  $A$  be an abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with  $n \geq 2$ , let  $\omega = a\alpha_+ + b\alpha_-$  with  $ab \neq 0$ , and let  $E$  be a perfect complex on  $A$  with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ , no other component, and  $\sigma_E$  injective. For an irreducible component  $Z$  of the support of  $E$  write*

$$m_Z(E) = \sum_i (-1)^i \ell_{\mathcal{O}_{A,Z}}(\mathcal{H}^i(E)_Z)$$

for the multiplicity of  $Z$  in the cycle of  $E$ .

- (i) *If  $Z$  has codimension at least two and  $m_Z(E) \neq 0$ , then  $Z$  is a union of cosets of an abelian subvariety  $B \subsetneq A$  with  $T_0B \supseteq T_+$  or  $T_0B \supseteq T_-$ .*
- (ii) *If  $A$  has no proper abelian subvariety  $B$  with  $T_0B \supseteq T_+$  or  $T_0B \supseteq T_-$ , which by [Corollary 16.31](#) holds when  $A$  is simple, at every split member  $X \times \bar{X}$  with  $X$  simple and  $\text{End}^0(X) = \mathbb{Q}$ , and at a general point of the quaternionic locus, then  $m_Z(E) = 0$  for every component  $Z$  of codimension at least two, and the cycle of  $E$  is a divisor.*
- (iii) *At every member of the family, whatever its endomorphisms, the support of  $E$  has an irreducible component of codimension less than  $n$ .*

*In particular no perfect complex whose support has codimension at least  $n$ , at any member of any Weil family, satisfies the criterion of [Theorem 16.28](#). This includes every complex assembled from the structure sheaves of the components of a cycle representing  $\omega$ , the base cycle of [Theorem 15.31](#) among them.*

*Proof.* (i) Let  $c \geq 2$  be the codimension of  $Z$  and let  $U \subseteq Z$  be the dense open set of smooth points of  $Z_{\text{red}}$  that lie on no other component of the support. Fix  $\theta \in T_+$  and  $\theta' \in T_-$ ; by [Theorem 16.30\(ii\)](#),  $A_\theta A_{\theta'} = 0$  in  $\text{Ext}^2(E, E)$ . Suppose that at some  $p \in U$  the images  $\bar{\theta}, \bar{\theta}'$  of  $\theta, \theta'$  in  $N_p = T_p A / T_p Z$  are linearly independent, and complete them by images of constant fields  $\nu_3, \dots, \nu_c$  to a basis of  $N_p$ . In linear coordinates on the universal cover the constant fields are coordinate derivations, and the germ  $S$  at  $p$  of the translate of the span of  $\theta, \theta', \nu_3, \dots, \nu_c$  is a smooth germ of dimension  $c$ , transverse to  $Z$  at  $p$ , to which  $\theta$  and  $\theta'$  are tangent. Let  $j: S \rightarrow A$  be the inclusion and  $M = Lj^*E$  on  $\hat{\mathcal{O}}_{S,p} \cong \mathbb{C}[[x_1, \dots, x_c]]$ . Derived restriction is a ring map  $\text{Ext}^*(E, E) \rightarrow \text{Ext}^*(M, M)$ , and it carries the germ of  $A_\theta$  to the jet class of  $M$  along  $\theta|_S$ , because the Atiyah class is functorial for pullback and  $\theta$  is tangent to  $S$  [[BF03](#), Section 3]. So  $J(\theta)J(\theta') = 0$  on  $M$ . Near  $p$  the support of  $E$  is  $Z$ , and  $Z \cap S = \{p\}$ , so  $M$  has finite length cohomology; its Euler characteristic is the intersection multiplicity of the cycle of  $E$  with  $S$  at  $p$ , which,  $S$  being transverse to  $Z$  at a smooth point lying on no other component, is  $m_Z(E)$ , by Serre's formula for

---

of finite length has a composition series with that one factor. The lemma says that the product of two independent jet classes detects the generator. What it cannot detect is a class of Euler characteristic zero, and the example in the text shows that it does not.

intersection multiplicities applied to each cohomology sheaf and summed with signs [Ser00, Chapter V].<sup>45</sup> By Lemma 16.37,  $m_Z(E) = 0$ .

So if  $m_Z(E) \neq 0$ , then at every  $p \in U$  and for all  $\theta \in T_+$ ,  $\theta' \in T_-$  the images satisfy  $\bar{\theta} \wedge \bar{\theta}' = 0$  in  $\wedge^2 N_p$ . As in the proof of Theorem 16.30(iv), the images  $I_{\pm}$  of  $T_{\pm}$  in  $N_p$  span  $N_p$ , which has dimension  $c \geq 2$ , so one of them is zero:  $T_+ \subseteq T_p Z$  or  $T_- \subseteq T_p Z$ . The two conditions cut out closed subsets of the irreducible set  $U$  whose union is  $U$ , so one of them, say the first, holds on all of  $U$ , and the argument of Corollary 16.31 shows that  $Z$  is stable under translation by the identity component  $B$  of its stabiliser, an abelian subvariety with  $T_0 B \supseteq T_+$ . Since  $Z \neq A$ ,  $B \neq A$ .

(ii) is (i) when no such  $B$  exists.

(iii) Suppose every component of the support has codimension at least  $n$ . Then  $\text{ch}_k(E) = 0$  for  $k < n$  and  $\text{ch}_n(E) = \sum_Z m_Z(E)[Z]$ , the sum over the components of codimension exactly  $n$ ; this is Riemann–Roch without denominators in its simplest case [Ful98, Chapter 15]. By (i), each  $Z$  with  $m_Z(E) \neq 0$  is  $\pi^{-1}(\pi(Z))$  for the quotient  $\pi: A \rightarrow A/B$  by an abelian subvariety  $B$  with  $T_0 B \supseteq T_+$  or  $T_0 B \supseteq T_-$ , so  $[Z] = \pi^*[\pi(Z)]$  lies in the subring of  $H^*(A, \mathbb{C})$  generated by  $H^1(A/B, \mathbb{C})$ , which consists of the classes vanishing on  $T_0 B$  and its conjugate. When  $T_0 B \supseteq T_+$ , a  $(1, 0)$ -form vanishing on  $T_+$  lies in  $V_-^{1,0}$ , and a  $(0, 1)$ -form vanishing on the conjugate of  $T_+$  lies in  $V_+^{0,1}$ , because complex conjugation exchanges the two eigenvalues of  $K$ ;<sup>46</sup> so the subring lies in  $\wedge^*(V_-^{1,0} \oplus V_+^{0,1})$ . When  $T_0 B \supseteq T_-$  it lies in  $\wedge^*(V_+^{1,0} \oplus V_-^{0,1})$ . Choose bases of the four pieces. Both exterior algebras are spanned by monomials in these bases, so their sum is too, and  $\alpha_+$ , a single monomial in  $n$  generators of  $V_+^{1,0}$  and  $n$  of  $V_+^{0,1}$ , lies in neither; nor does  $\alpha_-$ . As  $ab \neq 0$ , the class  $N\omega$  has nonzero coordinates on both and is not in the sum, which contradicts  $\text{ch}_n(E) = N\omega$ . So some component has codimension less than  $n$ . The last assertion follows, since a complex assembled from structure sheaves of the components of a cycle of codimension  $n$  is supported on that cycle. ■

*Remark 16.39* (What is left of the search). Put together, Theorem 16.30, Corollary 16.34, Theorem 16.36 and Theorem 16.38 describe an object meeting the criterion at a general member as follows. It has rank zero and first Chern class zero. Its support has a component of codimension less than  $n$ , although its Chern character begins in codimension  $n$ ; at  $n = 2$  that component is a divisor or the whole of  $A$ . Along every component of its support of codimension at least two its cohomology modules have alternating length zero, so it has at least two of them there, glued as in the cone of item (XL). It extends along all  $2n^2$   $K$ -linear deformations of  $A$ , its endomorphisms have trace zero, and its Hochschild action factors through a ring of dimension  $2^{2n+1} - 1$ . The central part of  $\text{Ext}^2(E, E)$ , the image of  $HH^2(A)$ , has dimension exactly  $2n(2n - 1)$ , and if  $\text{Ext}^2(E, E)$  is no larger than its central part, the criterion holds.

<sup>45</sup>Serre's formula expresses  $\sum_i (-1)^i \ell(\text{Tor}_i(F, \mathcal{O}_S))$ , for a module  $F$  and a smooth germ  $S$  meeting its support properly at  $p$ , as a sum over the components  $Z'$  of the support of  $\ell(F_{Z'})$  times the intersection multiplicity of  $Z'$  and  $S$  at  $p$ . For  $S$  transverse to a component smooth at  $p$  the intersection multiplicity is one. For a complex the Tor spectral sequence gives the same formula with the lengths replaced by their alternating sum over the cohomology, which is the definition of  $m_Z(E)$ .

<sup>46</sup>The field  $K$  acts on  $H_1(A, \mathbb{Q})$  by rational endomorphisms, so on the complexification its action commutes with complex conjugation, which therefore carries the eigenspace of  $\sqrt{-d}$  for one embedding of  $K$  to the eigenspace for the other. On the tangent space the conjugate of  $T_+$  is the eigenspace in  $T^{0,1}$  for the second embedding, and it pairs with  $V_-^{0,1}$  and not with  $V_+^{0,1}$ . Item (XXXIX)(D) checks this bookkeeping in an explicit rational model.

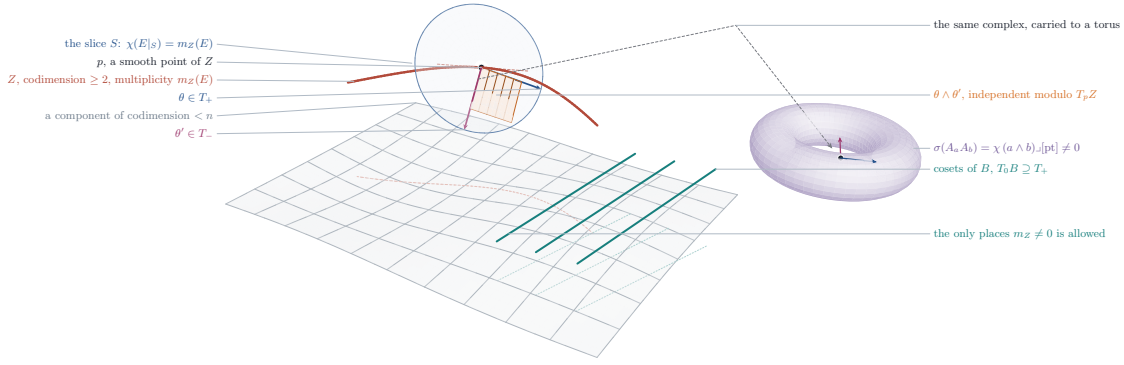


FIGURE 18. The proof of [Theorem 16.38](#). The grey sheet stands for a component of the support of codimension less than  $n$ , which the theorem forces to exist; the clay curve  $Z$  for a component of codimension at least two. At a smooth point  $p$  of  $Z$  the directions  $\theta \in T_+$  and  $\theta' \in T_-$  span the slice  $S$ , transverse to  $Z$ , on which  $E$  restricts to a complex of finite length with Euler characteristic  $m_Z(E)$ ; the amber parallelogram is the bivector  $\theta \wedge \theta'$ , independent modulo  $T_p Z$ . Carried to a torus of the same dimension, the product of the two jet classes is seen by the semiregularity map as  $m_Z(E)$  times the class of a point contracted with the bivector, which is not zero, so  $m_Z(E) = 0$  or the bivector degenerates. The teal lines are cosets of an abelian subvariety  $B$  tangent to  $T_+$ , the only components of codimension at least two on which a nonzero multiplicity survives. The scene is schematic in the dimensions and exact in the incidences.

The Weil class, which lives in codimension  $n$ , therefore has to come from the part of the Chern character that the leading multiplicities do not see: from twists that cancel in rank but not in higher degree. At  $n = 2$  the model is a pair  $\mathcal{O}_D$  and  $\mathcal{O}_D(W)$  on a divisor  $D$ , with  $W$  a divisor on  $D$ , whose difference has Chern character  $-i_*[W]$  plus terms of higher degree and has multiplicity zero along  $D$ . That is a very specific shape, and none of the objects in the literature or in this paper has it. We record it as the form the search now takes, and not as a claim that such an object exists.

**16.8. What the dimension forces on the object.** The numerical criterion of [Theorem 16.36](#) fixes one number,  $\dim \text{Ext}^2(E, E)$ . Together with Serre duality, the lower bounds of [Corollary 16.34](#) in the other degrees and the Hodge–Riemann relations for the Weil class, it fixes a good deal more: the object that carries the class can be taken indecomposable, and in the first open case it has more than one endomorphism. This is where a search has to look.

**Theorem 16.40** (The shape of an object meeting the criterion). *Let  $A$ ,  $\omega$  and  $E$  be as in [Theorem 16.36](#), so that  $\text{ch}(E) = N\omega$  with  $N \neq 0$  and no other component, and suppose that  $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$ . Write  $e_k = \dim \text{Ext}^k(E, E)$ .*

- (i)  $\chi(E, E) = \sum_k (-1)^k e_k = (-1)^n N^2 \int_A \omega^2$ , and this number is positive.
- (ii)  $E \cong E_0 \oplus E'$  with  $E_0$  indecomposable,  $\text{ch}(E_0) = \text{ch}(E)$ ,  $\dim \text{Ext}^2(E_0, E_0) = 2n(2n - 1)$ ,  $\text{ch}(E') = 0$  and  $\text{Ext}^2(E_0, E') = \text{Ext}^2(E', E_0) = \text{Ext}^2(E', E') = 0$ . So  $E_0$  meets the criterion as well.

- (iii) If  $E$  is indecomposable, then  $\text{End}(E) = \mathbb{C} \cdot \text{id} \oplus J$  with  $J$  the nilpotent radical, every  $f \in J$  annihilates  $\text{Ext}^2(E, E)$  and  $\text{Ext}^{2n-2}(E, E)$  under the Yoneda product, and  $\text{ev}_E(\bigwedge^{2n} P) = \text{ev}_E(\bigwedge^{2n} Q)$  is a single line in  $\text{Ext}^{2n}(E, E)$ . If moreover  $e_1 = 4n$ , then  $J$  annihilates  $\text{Ext}^1(E, E)$  as well.
- (iv) If  $n = 2$  or  $n = 3$ , then

$$\dim \text{End}(E) \geq 2 + \frac{1}{2} \chi(E, E) > 2,$$

so  $E$  has at least three linearly independent endomorphisms, and at  $n = 3$  also  $e_1 \leq e_0 + 9$ . In particular no simple object meets the criterion: no stable sheaf, no complex with  $\text{End} = \mathbb{C}$ , and no image of one under an autoequivalence of  $D^b(A)$ ; and an indecomposable object that meets it has a nonzero nilpotent endomorphism. The smallest profiles  $(e_0, \dots, e_{2n})$  that the constraints allow are  $(3, 8, 12, 8, 3)$  at  $n = 2$ , with  $\chi = 2$ , and  $(3, 12, 30, 40, 30, 12, 3)$  and  $(3, 12, 30, 41, 30, 12, 3)$  at  $n = 3$ , with  $\chi = 2$  and  $\chi = 1$ .

Parts (i) and (ii), and part (iv) at  $n = 2$ , use the compatibility  $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$  of [Setup 16.32](#) in degree two only, as [Theorem 16.36](#) does. Part (iii), and part (iv) at  $n = 3$ , use it in the further degrees in which [Corollary 16.34\(iii\)](#) grants it:  $2n - 2$  and  $2n$  for (iii), and three for (iv).

*Proof.* (i) The Todd class of  $A$  is trivial, so the Riemann–Roch theorem gives  $\chi(E, E) = \int_A \text{ch}(E)^\vee \text{ch}(E)$ , and  $\text{ch}(E)^\vee = (-1)^n N \omega$  since  $\omega$  has degree  $2n$ . The polarisation pairs  $V_+$  only with  $V_-$  ([Figure 7](#)), so its class  $\theta$  lies in  $V_+ \otimes V_-$  and  $\theta \wedge \alpha_\pm = 0$ , because  $\alpha_\pm$  spans the top exterior power of  $V_\pm$ . So  $\omega$  is a nonzero rational primitive class of type  $(n, n)$  in the middle degree, and the Hodge–Riemann relations give  $(-1)^{n(2n-1)} \int_A \omega^2 > 0$ , where  $(-1)^{n(2n-1)} = (-1)^n$ . Item (L) of [Appendix D](#) checks the primitivity and the sign of the intersection form on the whole rational Weil plane in the coordinate model, for  $n = 2, 3$  and several discriminants.

(ii) The category  $D^b(A)$  has finite dimensional morphism spaces and split idempotents, so  $E = \bigoplus_i E_i$  with each  $E_i$  indecomposable. The evaluation map and the semiregularity map are compatible with direct sums, and by [Theorem 16.36\(i\)](#) the kernel of  $\text{ev}_E$  in degree two is  $P \wedge Q$ ; so  $\text{ev}_{E_i}(P \wedge Q) = 0$  and, composing with  $\sigma_{E_i}$ ,  $(P \wedge Q) \lrcorner \text{ch}(E_i) = 0$  for every  $i$ . We claim that the only classes of Hodge type killed by  $P \wedge Q$  are those of  $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$ . Write  $H^*(A, \mathbb{C}) = \bigwedge^*(A_+ \oplus B_+ \oplus A_- \oplus B_-)$  with  $A_\pm = V_\pm^{1,0}$  and  $B_\pm = V_\pm^{0,1}$ , and grade it by the four degrees  $(i, j, k, l)$ . As in [Setup 16.32](#),  $P$  acts by wedging with  $B_+$  and contracting  $A_-$ , and  $Q$  by wedging with  $B_-$  and contracting  $A_+$ , so  $P \wedge Q$  is spanned by four kinds of operators, each of which shifts the four degrees by a fixed amount: wedging with  $b_+ \wedge b_-$  kills a component of degree  $(i, j, k, l)$  for all choices exactly when  $j = n$  or  $l = n$ ; wedging with  $b_+$  after contracting  $a_+$ , exactly when  $i = 0$  or  $j = n$ ; contracting  $a_-$  and wedging with  $b_-$ , exactly when  $k = 0$  or  $l = n$ ; and contracting  $a_-$  and  $a_+$ , exactly when  $i = 0$  or  $k = 0$ . Each of these is a statement about a tensor product of maps that are injective unless the degree in question is extreme. A class of type  $(p, p)$  has  $i + k = j + l = p$  in every component. If  $j = n$ , then  $l = n$  would force  $i = k = n$ , against the fourth condition, so  $k = 0$ , hence  $i = n + l$ , so  $i = n$ ,  $l = 0$ : the component lies in  $\bigwedge^n A_+ \otimes \bigwedge^n B_+ = \mathbb{C}\alpha_+$ . If  $j < n$ , then  $l = n$  and  $i = 0$ , hence  $k = n$  and  $j = 0$ : the component lies in  $\mathbb{C}\alpha_-$ . This proves the claim, which item (L) also checks in every degree for  $n = 2, 3$ . Since  $\text{ch}(E_i)$  is rational, it lies on the rational Weil plane, and when it is nonzero [Corollary 16.34\(i\)](#) gives  $\dim \text{Ext}^2(E_i, E_i) \geq 2n(2n - 1)$ , a nonzero rational Weil class having both coefficients  $a$  and  $b$  nonzero. As  $\text{Ext}^2(E, E) = \bigoplus_{i,j} \text{Ext}^2(E_i, E_j)$

has dimension  $2n(2n-1)$  and  $\sum_i \text{ch}(E_i) = N\omega \neq 0$ , exactly one summand  $E_0$  has nonzero Chern character and every other block of  $\text{Ext}^2$  vanishes.

(iii) An indecomposable object of a category with finite dimensional morphism spaces and split idempotents has a local endomorphism ring, so  $\text{End}(E) = \mathbb{C} \cdot \text{id} \oplus J$  with  $J$  nilpotent. By [Theorem 16.36\(i\)](#),  $\text{ev}_E$  maps  $\wedge^2 P \oplus \wedge^2 Q$  isomorphically onto  $\text{Ext}^2(E, E)$ . By Serre duality  $e_{2n-2} = e_2$ , and contraction with  $\omega$  is injective on  $\wedge^{2n-2} P \oplus \wedge^{2n-2} Q$  (item (L)), so, with the compatibility in degree  $2n-2$ ,  $\text{ev}_E$  maps that space isomorphically onto  $\text{Ext}^{2n-2}(E, E)$ . In degree  $2n$  the semiregularity map is the trace, since  $H^{q+2n}(\Omega_A^q) = 0$  for  $q > 0$ , so  $\text{tr ev}_E(\xi)$  is the  $H^{0,2n}$  component of  $\xi \lrcorner \text{ch}(E)$ ; for the generators  $\pi_P, \pi_Q$  of  $\wedge^{2n} P$  and  $\wedge^{2n} Q$  it is  $N_{cP}$  and  $N_{cQ}$  with  $c_P c_Q \neq 0$  (item (L)). For  $f \in \text{End}(E)$  and  $x \in \wedge^2 P \oplus \wedge^2 Q$  write  $f \cdot \text{ev}_E(x) = \text{ev}_E(\varphi_f(x))$ ; then  $f \mapsto \varphi_f$  is a homomorphism of algebras. Let  $x \in \wedge^2 P$  and  $y \in \wedge^{2n-2} P$ , and write  $x \wedge y = \beta(x, y)\pi_P$ . Since  $\text{ev}_E$  is a ring map that kills the ideal generated by  $P \wedge Q$ ,

$$\beta(x, y) \text{tr}(f \text{ev}_E(\pi_P)) = \text{tr}(f \text{ev}_E(x) \text{ev}_E(y)) = \text{tr ev}_E(\varphi_f(x) \wedge y) = N_{cP} \beta(\varphi_f(x)_P, y),$$

and the pairing  $\beta$  is perfect, so  $\varphi_f(x)_P = \mu_f x$  with  $\mu_f = \text{tr}(f \text{ev}_E(\pi_P))/N_{cP}$ . The same computation with  $y \in \wedge^{2n-2} Q$ , where  $x \wedge y = 0$  in the quotient, gives  $\varphi_f(x)_Q = 0$ . So  $f$  acts on  $\text{ev}_E(\wedge^2 P)$  by a scalar  $\mu_f$ , and in the same way on  $\text{ev}_E(\wedge^2 Q)$  by a scalar  $\nu_f$ ;  $\mu$  and  $\nu$  are characters of  $\text{End}(E)$  and vanish on  $J$ . So  $J$  kills  $\text{Ext}^2(E, E)$ , on both sides because  $\text{ev}_E$  lands in the graded centre, and the same argument with the roles of the two degrees exchanged shows that it kills  $\text{Ext}^{2n-2}(E, E)$ . In particular  $\text{tr}(f \text{ev}_E(\pi_P)) = \mu_f N_{cP} = 0$  for  $f \in J$ , so  $\text{ev}_E(\pi_P)$  and, likewise,  $\text{ev}_E(\pi_Q)$  lie in the annihilator of  $J$  under the perfect pairing of  $\text{End}(E)$  with  $\text{Ext}^{2n}(E, E)$ . That annihilator is a line,  $J$  having codimension one, and both elements are nonzero, their traces being  $N_{cP}$  and  $N_{cQ}$ ; so they span the same line. Finally, if  $e_1 = 4n$  then  $\text{Ext}^1(E, E) = \text{ev}_E(HH^1)$ , because  $\text{ev}_E$  is injective on  $HH^1$ , as shown in (iv). For  $f \in J$  and  $h \in HH^1$  write  $f \text{ev}_E(h) = \text{ev}_E(h')$ ; then  $\text{ev}_E(h' \wedge h'') = f \text{ev}_E(h \wedge h'') = 0$  for every  $h'' \in HH^1$ , so  $h' \wedge HH^1 \subseteq P \wedge Q$ , which forces  $h' = 0$  since  $\dim P = \dim Q \geq 2$ .

(iv) By Serre duality  $e_k = e_{2n-k}$ . The map  $\text{ev}_E$  is injective on  $HH^1$  using degree two alone: if  $\text{ev}_E(h) = 0$  then  $\text{ev}_E(h \wedge h') = 0$  for every  $h'$ , so  $h \wedge HH^1 \subseteq \text{Ann}_{HH^2}(\omega) = P \wedge Q$ , and  $h = 0$  as in (iii). So  $e_1 \geq 4n$ , and [Corollary 16.34\(iii\)](#) gives  $e_3 \geq 2\binom{6}{3} = 40$  at  $n = 3$ . At  $n = 2$  the Euler characteristic is  $2e_0 - 2e_1 + 12$ , and at  $n = 3$  it is  $2e_0 - 2e_1 + 60 - e_3$ . So  $2e_0 = \chi + 2e_1 - 12 \geq \chi + 4$  at  $n = 2$ , and  $2e_0 = \chi + 2e_1 + e_3 - 60 \geq \chi + 4$  at  $n = 3$ , where also  $e_1 = e_0 + 30 - \frac{1}{2}(\chi + e_3) \leq e_0 + 9$ . With (i) this gives the bound. An autoequivalence preserves  $\text{End}$ , and a stable sheaf is simple. An indecomposable object with  $\dim \text{End} \geq 3$  has  $J \neq 0$ . The smallest profiles are those with  $e_0 = 3$ , which item (L) lists by exhaustive search over the admissible values, and Section 31 of the Lean certificate checks the inequality  $e_0 \geq 3$  for all natural numbers.  $\blacksquare$

*Remark 16.41* (What the shape does to the search). [Theorem 16.40](#) is a necessary condition and nothing more; it does not say that an object meeting the criterion exists. What it does is remove from the search every object that anyone would construct first. Line bundles, semi-homogeneous bundles, skyscraper sheaves, stable sheaves and their Fourier–Mukai transforms are simple, and direct sums of them are excluded by (ii) together with [Theorem 20.33](#). At  $n = 3$ , the first open case, what remains is an indecomposable object with a local endomorphism algebra of dimension at least three whose radical kills  $\text{Ext}^2$  and  $\text{Ext}^4$  and pairs to zero with the line  $\text{ev}_E(\wedge^6 P)$ , granting the compatibility of [Corollary 16.34\(iii\)](#); the smallest conceivable one has  $\text{Hom}(E, E)$  of dimension three,

$\text{Ext}^1(E, E)$  equal to the image of  $HH^1(A)$ , of dimension twelve and killed by the radical, and  $\text{Ext}^3(E, E)$  of dimension forty or forty-one. Iterated self-extensions and thickenings are the natural objects with nilpotent endomorphisms; an extension of a simple object  $F$  by itself has  $\text{ch} = 2 \text{ch}(F)$ , so  $F$  would have to carry the class already, and the question returns one level down. Nothing in this paper produces an object of the required shape, and the remark is recorded as the form the search now takes. For  $n \geq 4$  the same count only asks for an excess in some even degree other than 2 and  $2n - 2$  (item (L)), and the bound on End disappears.

## 17. THE SEMIREGULARITY CRITERION IN ITS PURE AND CORRECTED FORMS

**17.1. The pure form of the criterion is never met.** So far  $\text{ch}(E) = N\omega$  exactly. If  $\sigma_E$  were injective,  $E$  would extend along every  $K$ -linear deformation of  $A$  (Theorem 16.30(ii)). These reach non-algebraic tori, on which the Weil class is Hodge and the Chern class of nothing. That settles the pure form of the criterion, negatively, in every dimension.

**Setup 17.1** (Weil tori). Let  $A$  be an abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type,  $V = H^1(A, \mathbb{Q})$  with its action of  $K$ , and  $V \otimes \mathbb{C} = V_+ \oplus V_-$  the two eigenspaces, each of dimension  $2n$ . For complementary  $n$ -dimensional subspaces  $X, Y \subset V_+$  the subspace  $X \oplus \bar{Y}$  of  $V \otimes \mathbb{C}$  is the space of holomorphic 1-forms of a complex structure on the real torus underlying  $A$ ; call the resulting complex torus  $T_{X,Y}$ . On  $T_{X,Y}$  the field  $K$  acts with signature  $(n, n)$ , so  $T_{X,Y}$  is a Weil torus in the sense of [Voi02b], and every complex structure on that real torus commuting with  $K$  and of signature  $(n, n)$  arises in this way. The pairs  $(X, Y)$  form a connected complex manifold  $S$  of dimension  $2n^2$ , containing the point of  $A$ ; the polarised Weil family is the submanifold of dimension  $n^2$  on which the polarisation of  $A$  is of type  $(1, 1)$ , and the tangent space of  $S$  is the space  $\text{Hom}_K(H^{1,0}, H^{0,1})$  of  $K$ -linear first order deformations in Theorem 16.30(i). Since  $\alpha_+$  spans  $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$  and  $X$  is holomorphic while  $Y$  is antiholomorphic, the Weil classes are of type  $(n, n)$  on every  $T_{X,Y}$ . We say that a property holds for a very general member of  $S$  if it holds outside a countable union of proper analytic subsets.

**Lemma 17.2** (The Hodge classes of a very general Weil torus). *For a very general member  $T$  of  $S$ ,  $\text{Hdg}^k(T) = 0$  for  $1 \leq k \leq 2n - 1$ ,  $k \neq n$ , and  $\text{Hdg}^n(T)$  is the rational Weil plane.*

*Proof.* A rational class that is of Hodge type at a very general member is of Hodge type on all of  $S$ , because its Hodge locus is analytic and  $S$  is connected. Let  $G = \text{Res}_{K/\mathbb{Q}} \text{GL}_{2n}$  be the group of  $K$ -linear automorphisms of  $V$ . Its real points act transitively on  $S$ , since  $\text{GL}(V_+)$  acts transitively on pairs of complementary subspaces. The Mumford–Tate group  $M$  of a very general member lies in  $G$ , since  $K$  acts by endomorphisms of every member, and it contains the Mumford–Tate group of every member, whose Hodge classes include those of the very general one. So the real Lie algebra of  $M$  contains the complex structure  $J_s$  of every member, the derivative of its Hodge cocharacter, and therefore contains the  $\text{Ad } G(\mathbb{R})$ -span of one  $J_s$ , which is an ideal of  $\mathfrak{g}_{\mathbb{R}} = \mathfrak{gl}_{2n}(\mathbb{C})$  containing a non-central element,  $J_s$  acting on  $V_+$  by  $i$  on  $X$  and  $-i$  on  $Y$ ; hence it contains  $\mathfrak{sl}_{2n}(\mathbb{C})$ , and  $M$  contains  $G^{\text{der}} = \text{Res}_{K/\mathbb{Q}} \text{SL}_{2n}$ . Over  $\mathbb{C}$  this is  $\text{SL}(V_+) \times \text{SL}(V_-)$  acting on  $\bigwedge^*(V_+ \oplus V_-) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$ , whose invariants are spanned by 1,  $\alpha_+$ ,  $\alpha_-$  and  $\alpha_+ \alpha_-$ . Hodge classes are invariant under  $M$ , which gives the statement. Item (LI) of Appendix D confirms it for  $n = 2$  and  $n = 3$  by exhibiting explicit members at which no other rational class of those degrees is of Hodge type. ■

**Lemma 17.3** (A very general Weil torus has no subvarieties). *A very general member  $T$  of  $S$  has no analytic subsets other than finite sets and  $T$  itself.*

*Proof.* Let  $Z \subset T$  be an irreducible analytic subset of codimension  $c$  with  $0 < c < 2n$ . For every Kähler class  $\kappa$  the class  $[Z]$  satisfies  $\int_T [Z] \kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$ , so  $[Z]$  is a nonzero Hodge class of degree  $2c$ . By [Lemma 17.2](#) this forces  $c = n$  and  $[Z]$  into the Weil plane. Take for  $\kappa_0$  the class of a positive definite hermitian form that is block diagonal for the splitting  $H^{1,0} = X \oplus \bar{Y}$ . Its blocks lie in  $X \otimes \bar{X}$  and  $\bar{Y} \otimes Y$ , both inside  $V_+ \otimes V_-$ , so every power of  $\kappa_0$  has a factor from  $V_+$  and a factor from  $V_-$  in each term, and  $\alpha_{\pm} \wedge \kappa_0^n = 0$  because  $\alpha_{\pm}$  spans the top exterior power of  $V_{\pm}$  (item (LI)). So  $\int_T [Z] \kappa_0^n = 0$ , which is impossible. ■

**Proposition 17.4** (Coherent sheaves on a very general Weil torus). *Let  $n \geq 2$  and let  $T$  be a very general member of  $S$ . Every coherent analytic sheaf  $F$  on  $T$  has  $\text{ch}_k(F) = 0$  in  $H^{2k}(T, \mathbb{Q})$  for  $1 \leq k \leq 2n - 1$ . For  $n = 2$  this is Voisin’s theorem that every coherent sheaf on a general Weil torus of dimension four has  $c_2(F) = 0$  [[Voi02b](#), Theorem 10], whose proof we follow.*

*Proof.* Chern classes of coherent sheaves on  $T$  exist in rational Deligne cohomology, extend those of bundles and are additive [[Gri10](#)], so they are Hodge classes. By [Lemma 17.3](#) the torsion of  $F$ , the locus where the torsion-free quotient is not locally free, and the cokernel of the map from that quotient to its reflexive hull are all supported on finite sets, and a sheaf supported on a finite set has Chern character in the top degree only. So it suffices to treat a reflexive sheaf  $G$ . Every coherent sheaf on  $T$  has  $c_1 = 0$ , since  $\text{Hdg}^1(T) = 0$  ([Lemma 17.2](#)); so every torsion-free sheaf has slope zero for every Kähler class, and a torsion-free sheaf is stable for one Kähler class exactly when it has no saturated subsheaf of intermediate rank, which does not depend on the class. A Jordan–Hölder filtration of  $G$  therefore has stable torsion-free factors  $Q_j$ , stable for every Kähler class, and  $\text{ch}(G) = \sum_j \text{ch}(Q_j)$ . By the theorem of Bando and Siu [[BS94](#)] the stable reflexive sheaf  $Q_j^{**}$  carries an admissible Hermite–Einstein metric for every Kähler class  $\kappa$  and satisfies  $\int_T (2rc_2 - (r-1)c_1^2)(Q_j^{**}) \kappa^{2n-2} \geq 0$ , with equality exactly when  $Q_j^{**}$  is a projectively flat vector bundle. Here  $c_1 = 0$  and  $c_2(Q_j^{**}) \in \text{Hdg}^2(T)$ . For  $n \geq 3$  that group is zero; for  $n = 2$  it is the Weil plane, and then  $\int_T c_2(Q_j^{**}) \kappa_0^2 = 0$  for the class  $\kappa_0$  of the proof of [Lemma 17.3](#). In either case equality holds for some  $\kappa$ , so  $Q_j^{**}$  is a projectively flat bundle with  $c_1 = 0$ , hence flat, and its rational Chern classes of positive degree vanish.  $Q_j$  and  $Q_j^{**}$  differ on a finite set, so  $\text{ch}_k(Q_j) = 0$  for  $1 \leq k \leq 2n - 1$ , and the same holds for  $G$  and for  $F$ . ■

**Theorem 17.5** (The pure form of the criterion is never met). *Let  $n \geq 2$ , let  $A$  be an abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type,  $\omega$  a nonzero rational Weil class, and  $E$  a perfect complex on  $A$  with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ , and no other component. Then  $\sigma_E$  is not injective. In particular  $\dim \text{Ext}^2(E, E) > 2n(2n - 1)$ , the hypotheses of [Theorem 16.28](#) and [Theorem 16.36\(iii\)](#) are satisfied by no object at any member of any Weil family of an imaginary quadratic field, and the propagation statement (P2) of [Remark 17.7](#), in the form stated there, is false.*

*Proof.* Suppose  $\sigma_E$  injective. Since  $A$  is projective,  $E$  is isomorphic to a bounded complex  $E^\bullet$  of holomorphic vector bundles. The class  $N\omega$  is of Hodge type on every member of  $S$  ([Setup 17.1](#)), so by the theorem of Buchweitz and Flenner in Pridham’s form ([Theorem 16.27](#), [[BF03](#), [Pri24](#)]), every obstruction to extending  $E$  along an Artinian deformation of  $A$  inside  $S$  vanishes, exactly as in the proof of [Theorem 16.28](#). The deformations of the complex structure inside  $S$  together with those of the holomorphic structures and differentials of  $E^\bullet$  are governed by a differential graded Lie algebra of Dolbeault type with finite

dimensional cohomology, and Kuranishi's method [Kur65], in the form of [GM88, Sections 2–3], gives an analytic germ  $(M, m_0)$  with a morphism  $\pi: (M, m_0) \rightarrow (S, s_0)$  and a complex of holomorphic vector bundles on the pulled back family of tori deforming  $E^\bullet$ , whose completion is a hull of the formal deformation functor. The vanishing of the obstructions says that  $\pi$  is formally smooth, hence smooth, so its image contains a neighbourhood  $U$  of  $s_0$ . Thus every member  $T_s$  with  $s \in U$  carries a bounded complex of holomorphic vector bundles  $E_s^\bullet$  whose Chern character is the parallel transport of  $N\omega$ . Choose  $s \in U$  very general. By Proposition 17.4,  $\text{ch}_n(E_s) = \sum_i (-1)^i \text{ch}_n(E_s^i) = 0$ , while it is the transport of  $N\omega$ , which is not zero. This contradiction proves that  $\sigma_E$  is not injective. The argument uses only that  $\text{ch}(E)$  is of Hodge type on every member of  $S$ , so it applies verbatim when  $\text{ch}(E) - N\omega$  has components in degrees 0 and  $4n$  as well. The remaining statements follow, since  $\dim \text{Ext}^2(E, E) \geq 2n(2n - 1)$  by Corollary 16.34(i), and equality forces  $\sigma_E$  to be injective by Theorem 16.36(i). ■

*Remark 17.6* (What survives, and the corrected statement). Theorem 17.5 turns Theorem 16.30(ii) from a constraint into a contradiction: an object that extends along every  $K$ -linear deformation of  $A$  lands on non-algebraic tori, where the Weil class is Hodge and is the Chern class of nothing. The results of this part that describe the object, Theorem 16.30, Corollary 16.34(ii)–(iii), Theorem 16.36, Theorem 16.38, Theorem 20.33 and Theorem 16.40, remain true as implications, and they are vacuous: no object satisfies their hypotheses. They are kept as the record of what the pure form asked for.

The obstruction does not reach a Chern character with a component that is Hodge on the polarised family and not beyond it, unless a twist removes that component. Call (P2') the statement that at one base point of each Weil family some perfect complex  $E$  with

$$\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k, \quad N \neq 0, \quad c_k \in \mathbb{Q},$$

$\theta$  the polarisation, has injective semiregularity map. The proof of Theorem 16.28 applies to it word for word, with the deformation taken over the polarised family, along which  $\text{ch}(E)$  stays of Hodge type: at a member  $A_s$  it produces  $\text{ch}_n(E_s) = N\omega_1 + c_n \theta^n$  algebraic, and  $\theta^n$  is algebraic, so  $\omega_1$  is. The form above is already the most general one allowed by the subring generated by  $\theta$  and the Weil classes: a Weil class is primitive, so  $\theta\omega_1 = 0$  and that subring is spanned by the powers of  $\theta$  and the Weil plane. Theorem 17.5 excludes directly a Chern character that remains of Hodge type on all of  $S$ , which by Lemma 17.2 is a multiple of the Weil class plus terms in degrees 0 and  $4n$ , and after a line bundle twist the characters  $c e^{t\theta} + N\omega_1 + c' \theta^{2n}$  with  $t\theta$  integral (Theorem 17.9(v)). For every other corrected character with some  $c_k \neq 0$ ,  $1 \leq k \leq 2n - 1$ , the component of the Hodge locus of  $\text{ch}(E)$  through  $A$  is the polarised family itself, whose members are abelian varieties, so nothing forces  $E$  onto non-algebraic tori and the argument cannot be run on  $E$  (Proposition 17.11); the one exception is a ratio at  $n = 2$  at which the locus is five-dimensional (Remark 17.12). The semiregular objects in Markman's proofs for abelian fourfolds and for the sixfolds of trivial discriminant [Mar25a, Mar25b] are not of the pure form either: their Chern characters are not multiples of a Weil class, and they are deformed only along families on which those characters stay of Hodge type. We have not checked that they have exactly the form above, and nothing below depends on it. From here on, in Corollary 20.41, Theorem 20.40 and the frontier, (P2) is read in this corrected form. The annihilator computations above depend on  $\text{ch}(E)$  being a pure Weil class; they are redone for a Chern character with components in every even degree at the end of this subsection (Theorem 17.9, Proposition 17.10, Proposition 17.11).

The same obstruction does not reach the Mumford target of [Section 21.2](#). The exceptional classes there are rigid: their Hodge locus among all complex tori is the Mumford curve ([Theorem 21.21](#), [Proposition 21.24](#)), so a semiregular complex with Chern character  $N\omega$  on the square of a Mumford fourfold would only have to deform along the curve, whose members are all algebraic.

*Remark 17.7* (The two finite statements, and what each of them is). After [Theorem 16.9](#) and [Theorem 16.28](#) the Hodge conjecture for abelian varieties of Weil type follows from either of the following, each of which is a statement about explicit finite data and neither of which mentions a general point. The first is settled by [Theorem 16.17](#), and we record it because it is the form in which the question first presents itself. The second is settled by [Theorem 17.5](#) in the form stated here, and remains open only in the corrected form of [Remark 17.6](#).

- (P1) *A bound.* At every point of the orbit  $G(\mathbb{Q}) \cdot s_0$  the class  $\omega$  is represented by a cycle whose Hilbert data comes from one finite set. This one is settled, and negatively, in the sense that matters: [Theorem 16.15](#) shows that the transport of the base cycle does not satisfy it, for any cycle whose components have finite stabiliser, and [Theorem 16.17](#) shows that read on an arbitrary representative it is equivalent to  $\mathbf{W}(K, n, \delta)$  itself. It is therefore a reformulation of the conjecture for the family, and we record it as such rather than as a criterion.
- (P2) *A rank.* The semiregularity map of a perfect complex with the Chern character of [Proposition 16.26](#) at one base point is injective. In this form the statement is false ([Theorem 17.5](#)), and it survives only in the corrected form of [Remark 17.6](#), where the Chern character may carry powers of the polarisation; what follows records what the first form asked for. Its source is  $\text{Ext}^2(E, E)$  for a single object, and its target has dimension  $\sum_q h^{q, q+2}(A_0) = \binom{4n}{2n-2}$ , computed in [Proposition 9.2](#) and verified in item (V) of [Appendix D](#). So the target is large, which is the direction that helps; what is not known to us is the rank of the map. What is known is what full rank would force ([Theorem 16.30](#)): the evaluation map of Hochschild cohomology must vanish on the  $4n^2$ -dimensional annihilator of the Weil class, so  $E$  must be killed by the mixed block of  $H^{0,2}$ , must extend along every  $K$ -linear deformation, must have all products of transverse translation classes zero, and must have no endomorphism of nonzero trace; and at a base point with no abelian subvariety tangent to an eigenspace it can be a locally free sheaf on no smooth germ of its support of codimension at least two ([Corollary 16.31](#)). The representative that the construction supplies, a virtual sum of line bundles, fails at the first of these ([Proposition 9.10](#)), and any representative that is a sheaf fails at the last. [Theorem 16.33](#) adds the shape of that annihilator, the mixed part of a splitting  $HH^1 = P \oplus Q$  into the annihilators of the two conjugate pieces of the class, and [Corollary 16.34](#) the consequence: the whole Hochschild action of  $E$  factors through a ring of dimension  $2^{2n+1} - 1$  inside one of dimension  $2^{4n}$ , while  $\dim \text{Ext}^2(E, E) \geq 2n(2n - 1)$  for every representative, semiregular or not. [Theorem 16.36](#) turns the rank into that number: equality is enough. And [Theorem 16.38](#) places the support. At every member it has a component of codimension less than  $n$ , so no complex carried by a cycle representing  $\omega$  can serve, and at a general member every component of codimension at least two has multiplicity zero. [Theorem 16.40](#) fixes its shape: an indecomposable summand carries the class and meets the criterion alone, and at  $n = 2$ , and at  $n = 3$  granting the compatibility of [Corollary 16.34](#)(iii), it has at least three independent endomorphisms, so no simple object serves.

Two remarks on what these are not. Neither is the variational Hodge conjecture: (P1) said nothing about deformations at all, and (P2) is the infinitesimal statement at one point with the global propagation already supplied by [Theorem 16.4](#), which is what [Remark B.11](#) said was missing. And neither is implied by the results of [Section 15.5](#): the base points are there, the cycles are there, and both (P1) and (P2) are questions about those same cycles that the construction does not answer. Both are settled in the forms stated here, and neither closes anything: (P1) is equivalent to its conclusion, and (P2) is false and survives only in the corrected form of [Remark 17.6](#), which we have not settled. We do not claim the conjecture.

**17.2. The corrected criterion, computed.** The annihilator computations of this subsection depend on  $\text{ch}(E)$  being a pure Weil class, as [Remark 17.6](#) notes. We now redo them for a Chern character with components in every even degree. In degree two the answer is complete and holds for every  $n$ ; in the other degrees it is a closed formula, proved away from the middle degree and checked in it. The same computation gives the Hodge locus of the character, and with it the list of obstructions of this paper that survive the correction ([Proposition 17.11](#), [Remark 17.13](#)). None of this constructs an object or decides whether one exists. It gives, for each shape of Chern character, the number that  $\dim \text{Ext}^2(E, E)$  has to reach and the exact kernel that  $\text{ev}_E$  must then have, which is the form in which a candidate can be tested.

**Setup 17.8** (Corrected characters). Let  $A$  be an abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type,  $n \geq 2$ , with polarisation  $\theta$ , and let  $\omega = u\alpha_+ + \bar{u}\alpha_-$  be a nonzero real class on the Weil line, in the notation of [Theorem 16.30](#). For complex numbers  $c_0, \dots, c_{2n}$  put

$$p(\theta) = \sum_{k=0}^{2n} c_k \theta^k, \quad \gamma = \omega + p(\theta), \quad \mu_m = m! c_m,$$

and let  $\rho \in \{0, 1, 2, 3\}$  be the rank of the  $(2n-1) \times 3$  Hankel matrix  $H = [\mu_{j+i}]$ ,  $0 \leq j \leq 2n-2$ ,  $0 \leq i \leq 2$ .<sup>47</sup> Write  $r(\gamma)$  for the rank of contraction  $HT^2(A) \rightarrow H^*(A, \mathbb{C})$ ,  $\xi \mapsto \xi \lrcorner \gamma$ . For  $a < n \leq b$ , in the basis of [Theorem 16.33](#) adapted to an eigenbasis  $e_j$  of  $V_+$  and  $f_j$  of  $V_-$  with  $e_j, f_j$  of type  $(1, 0)$  for  $j < n$  and  $j \geq n$  respectively, write  $z_{ab} = w(e_b)w(f_a) \in V_+^{0,1} \otimes V_-^{0,1}$ ,  $v_{ab}^1 = w(e_b)c(e_a)$  and  $v_{ab}^2 = w(f_a)c(f_b)$  in  $H^1(T_A)$ , and  $\pi_{ab} = c(e_a)c(f_b) \in T_+ \otimes T_-$ , where  $w$  is wedge product with a  $(0, 1)$  class and  $c$  contraction of a  $(1, 0)$  class.

For a Chern character,  $\gamma = N\omega_1 + \sum_k c_k \theta^k$  with rational  $c_k$ , and  $u$  is determined by  $N\omega_1$ ; intrinsically,  $|u|^2 = (-1)^n \frac{(2n)!}{2} \int_A \omega^2 / \int_A \theta^{2n}$  in the normalisation of item (LII). The rank  $\rho$  does not change when  $\gamma$  is multiplied by  $e^{t\theta}$ , which acts on the moments by a binomial transform, and  $\omega e^{t\theta} = \omega$  because a Weil class is primitive; so  $\rho$  is an invariant of  $\gamma$  up to twist.

**Theorem 17.9** (The corrected criterion is a number). *In [Setup 17.8](#):*

(i) *For  $n \geq 3$ ,*

$$\text{Ann}_{HT^2}(\gamma) = \bigoplus_{a < n \leq b} \{x z_{ab} + y_1 v_{ab}^1 + y_2 v_{ab}^2 + w \pi_{ab} : (x, i(y_1 - y_2), w) \in \ker H\},$$

*so that  $\dim \text{Ann}_{HT^2}(\gamma) = n^2(4 - \rho)$  and  $r(\gamma) = (4 + \rho)n^2 - 2n$ , whatever  $u \neq 0$  is.*

<sup>47</sup> $H$  is a catalecticant of the moment sequence  $\mu_m = m! c_m$ . A nonzero vector  $(x, \lambda, w)$  in its kernel is a linear recurrence  $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$  of order at most two, so  $\rho \leq 2$  says that the moments satisfy one, and the shapes of [Theorem 17.9\(ii\)](#) are its solutions, read off from the roots of  $wT^2 + \lambda T + x$ .

- (ii)  $\rho = 3$  for a general  $p$ . If  $\rho = 3$ , and either  $n \geq 3$  or  $n = 2$  and  $e = 0$  in the notation of (iii), then  $\text{Ann}_{HT^2}(\gamma)$  is exactly the tangent space  $T_{[A]}\mathcal{D}_{n,n}$  of the polarised Weil family, spanned by the  $n^2$  vectors  $v_{ab}^1 + v_{ab}^2$ , inside  $H^1(T_A)$ , with no component in  $H^{0,2}$  or in  $H^0(\wedge^2 T_A)$ ; and  $r(\gamma) = \binom{4n}{2} - n^2 = 7n^2 - 2n$ . Moreover  $\rho \leq 2$  exactly when  $p$  is one of  $Ae^{s\theta} + Be^{t\theta}$ ,  $(A + B\theta)e^{t\theta}$ ,  $Ae^{t\theta} + B\theta^{2n}$ ,  $A\theta^{2n-1} + B\theta^{2n}$ , and  $\rho \leq 1$  exactly when  $p \in \mathbb{C}e^{t\theta} \cup \mathbb{C}\theta^{2n}$ ; the pure case  $\rho = 0$  gives back  $r = 2n(2n - 1)$  of [Corollary 16.34\(i\)](#).
- (iii) For  $n = 2$ ,  $\dim \text{Ann}_{HT^2}(\gamma) = 4(4 - \rho) + e$  with  $e = \dim \ker((HK)^2 - |u|^2) \leq 2$  and  $K$  the antidiagonal matrix with entries  $1, -2, 1$ ; so  $r(\gamma) \in \{12, 16, 18, 20, 22, 23, 24\}$ , and  $e > 0$  exactly when  $X(X - I)^2 = 4J^2$ , where  $X = |u|^2$  and  $I, J$  are the two invariants of the binary quartic  $\sum_m \binom{4}{m} \mu_m x^{4-m} y^m$ .
- (iv) Let  $E$  be a perfect complex on  $A$  with  $\text{ch}(E) = \gamma$  and  $N \neq 0$ . Then  $\dim \text{Ext}^2(E, E) \geq r(\gamma)$  with no hypothesis on  $\sigma_E$ . If equality holds, then  $\text{ev}_E$  is onto with kernel  $\text{Ann}_{HT^2}(\gamma)$  and  $\sigma_E$  is injective, so that (P2') holds at that point and, by [Remark 17.6](#) and [Theorem 16.28](#), the Hodge conjecture holds for every member of the family with full Hodge group. Conversely, if  $\sigma_E$  is injective then  $\ker \text{ev}_E = \text{Ann}_{HT^2}(\gamma)$ .
- (v) If  $p \in \mathbb{C}e^{t\theta} + \mathbb{C}\theta^{2n}$  with  $t\theta$  the class of a line bundle  $L$ , a set of shapes with  $\rho \leq 2$  that contains  $\mathbb{C}\theta^{2n}$ , then no perfect complex  $E$  with  $\text{ch}(E) = \gamma$  has injective  $\sigma_E$ .

*Proof.* (i) Let the torus  $(\mathbb{C}^\times)^{2n}$  act by  $e_j \mapsto \tau_j e_j$ ,  $f_j \mapsto \tau_j^{-1} f_j$ . It preserves the Hodge decomposition and  $\theta$ , and scales  $\alpha_\pm$  by  $(\prod_j \tau_j)^{\pm 1}$ ; contraction is equivariant. A weight of  $HT^2$  has at most two nonzero coordinates, so for  $n \geq 3$  two such weights never differ by a nonzero multiple of  $(1, \dots, 1)$ , and the images of one weight component in the  $\theta$ -part and in the two  $\alpha$ -parts land in distinct weight spaces. Hence  $\text{Ann}(\gamma) = \text{Ann}(p(\theta)) \cap \text{Ann}(\alpha_+) \cap \text{Ann}(\alpha_-)$ , and the last two meet in  $P \wedge Q$  by [Theorem 16.33\(ii\)](#). The space  $P \wedge Q$  is the sum over  $a < n \leq b$  of the four-dimensional weight spaces spanned by  $z_{ab}, v_{ab}^1, v_{ab}^2, \pi_{ab}$ , and  $\text{Ann}(p(\theta))$  is torus-stable, so the computation splits pair by pair. Put  $m = e_b \wedge f_a$  and  $\theta' = \theta - t_a - t_b$ , where  $t_j = i s_j e_j \wedge f_j$  with  $s_j = \pm 1$  the sign that makes  $\theta = \sum_j t_j$  positive. Since the  $t_j$  are even with  $t_j^2 = 0$ , direct evaluation gives

$$z \lrcorner \theta^k = m \theta'^k, \quad v^1 \lrcorner \theta^k = i k m \theta'^{k-1}, \quad v^2 \lrcorner \theta^k = -i k m \theta'^{k-1}, \quad \pi \lrcorner \theta^k = k(k-1) m \theta'^{k-2},$$

so for  $\xi = xz + y_1 v^1 + y_2 v^2 + w\pi$  and  $\lambda = i(y_1 - y_2)$ ,

$$\xi \lrcorner p(\theta) = m \wedge \sum_{j=0}^{2n-2} \frac{\theta'^j}{j!} (x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2}).$$

The classes  $m \wedge \theta'^j$ ,  $0 \leq j \leq 2n - 2$ , are nonzero and of distinct degrees, so  $\xi$  kills  $p(\theta)$  exactly when  $(x, \lambda, w) \in \ker H$ . The map  $(x, y_1, y_2, w) \mapsto (x, \lambda, w)$  is onto with kernel spanned by  $v^1 + v^2$ , which gives  $1 + (3 - \rho)$  dimensions for each of the  $n^2$  pairs. (ii) The vector  $v_{ab}^1 + v_{ab}^2$  is  $K$ -linear and kills  $\theta$ , and these  $n^2$  vectors span the tangent space of the polarised family by [Proposition 9.14](#); when  $\rho = 3$  they are all of the annihilator. If  $\rho \leq 2$ , a nonzero  $(x, \lambda, w) \in \ker H$  turns the equations  $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$  into a linear recurrence of full rank on  $\mu_0, \dots, \mu_{2n}$ , whose two-dimensional space of solutions is spanned by the listed shapes according to the roots of  $wT^2 + \lambda T + x$ ;  $\rho \leq 1$  is the intersection of two such spaces. (iii) At  $n = 2$  the argument of (i) fails for exactly one pair of weights,  $\epsilon_2 + \epsilon_3$  in  $\wedge^2 P$  and  $-\epsilon_0 - \epsilon_1$  in  $\wedge^2 Q$ , whose difference is  $(1, 1, 1, 1)$ ; these give one coupled block of size eight, whose kernel is isomorphic to  $\ker(X - T)$  with  $T$  conjugate on its image to  $(HK)^2$ , and the characteristic polynomial of  $HK$  is  $y^3 - Iy - 2J$ , which is traceless, so

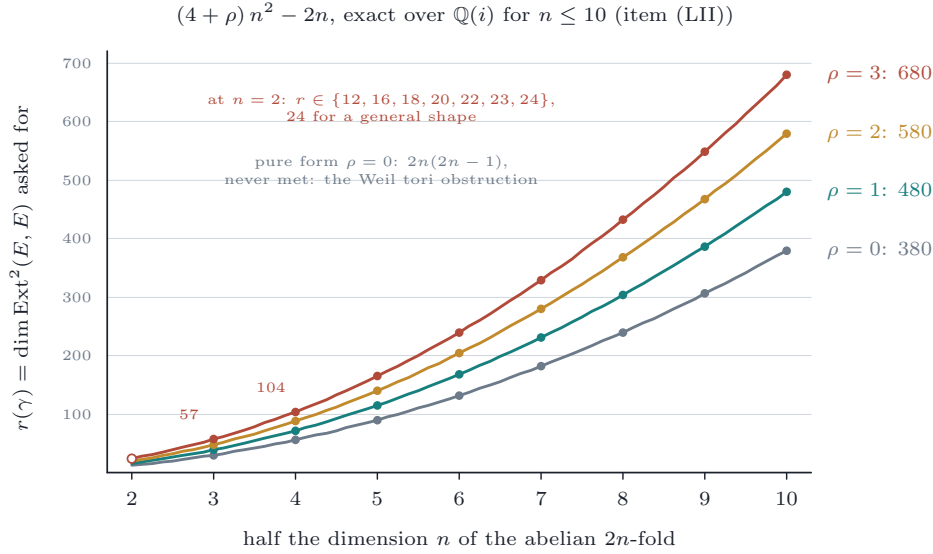


FIGURE 19. The dimension  $r(\gamma) = (4 + \rho)n^2 - 2n$  that  $\text{Ext}^2(E, E)$  must have for a complex with Chern character  $\gamma = N\omega_1 + \sum_k c_k \theta^k$  to meet the corrected criterion, for the four Hankel ranks  $\rho$  of Theorem 17.9; the dots are the values computed exactly over  $\mathbb{Q}(i)$  by item (LII), through  $n = 10$ , where  $HT^2$  has dimension 780. The pure form,  $\rho = 0$ , is the bottom curve  $2n(2n - 1)$ , which no complex attains (Theorem 17.5); a general shape lies on the top curve  $7n^2 - 2n$ . At  $n = 2$  the weight argument fails for one pair of weights and seven values occur.

$e \leq 2$ . (iv) is the argument of Theorem 16.36(i)–(ii) with  $\text{Ann}_{HT^2}(\gamma)$  in place of  $P \wedge Q$ :  $\sigma_E \circ \text{ev}_E$  is contraction into  $\gamma$ , so  $\ker \text{ev}_E \subseteq \text{Ann}_{HT^2}(\gamma)$ . (v) If  $\gamma - \omega$  has components in degrees 0 and  $4n$  only, the proof of Theorem 17.5 applies verbatim. If  $p = c e^{t\theta} + c' \theta^{2n}$  with  $t\theta = c_1(L)$ , then  $\text{ch}(E \otimes L^{-1}) = \gamma e^{-t\theta} = c + \omega + c' \theta^{2n}$ , because  $\omega\theta = 0$  and  $\theta^{2n+1} = 0$ , and  $\sigma_{E \otimes L^{-1}} = e^{-c_1(L)} \cdot \sigma_E$  is injective exactly when  $\sigma_E$  is, so the first case applies to  $E \otimes L^{-1}$ . Item (LII) of Appendix D computes the annihilator structure-free, exactly over  $\mathbb{Q}(i)$ , for  $n \leq 10$ , and checks every statement above. ■

So the number that the corrected criterion asks of one complex is 24 at  $n = 2$ , 57 at  $n = 3$  and 104 at  $n = 4$  for a general shape, against 12, 30 and 56 for the pure form, and injectivity then forces nothing beyond first-order extension along the polarised family, which is what the criterion is for. The larger number is the price of the components  $c_k \theta^k$ , and it is paid in the right place: every direction of  $HT^2$  other than the  $n^2$  directions of the family moves  $\gamma$  off the Hodge classes, so an object meeting the bound has  $\text{ev}_E$  injective on a complement of  $T_{[A]} \mathcal{D}_{n,n}$ . Figure 19 plots the number against  $n$  for the four Hankel ranks.

**Proposition 17.10** (The Hochschild profile of a corrected character). *In Setup 17.8, let  $\rho_k$  be the rank of contraction  $HT^k(A) \rightarrow H^*(A, \mathbb{C})$  into  $\gamma$ , put  $C_l = l! c_l$ , let  $r'$  be the rank of the  $(n + 1) \times (n + 1)$  Hankel matrix  $H' = [C_{p+q}]$ , and  $d = \dim \ker((K'H')^2 - |u|^2)$  with  $K' = \text{antidiag}((-1)^a \binom{n}{a})$ .*

- (i)  $\rho_0 = \rho_{2n} = 1$ ,  $\rho_k = 0$  for  $k > 2n$ , and for  $1 \leq k \leq 2n - 1$ ,  $k \neq n$ ,

$$\rho_k = 2 \binom{2n}{k} + M_k \min(k + 1, 2n + 1 - k, r'), \quad M_k = \sum_{\substack{i+j=k \\ 1 \leq i, j \leq n-1}} \binom{n}{i} \binom{n}{j};$$

in the middle degree the same formula holds with  $d$  subtracted, which is checked for  $n \leq 5$ . In particular  $\rho_k = \rho_{2n-k}$ , and the general profiles are  $(1, 8, 24, 8, 1)$  at  $n = 2$  and  $(1, 12, 57, 112, 57, 12, 1)$  at  $n = 3$ .

- (ii)  $\chi(E, E) = VQ(c) + W$  for a perfect complex with  $\text{ch}(E) = \gamma$ , where  $V = \int_A \theta^{2n} > 0$ ,  $W = (-1)^n \int_A \omega^2 > 0$  and  $Q(c) = \sum_k (-1)^k c_k c_{2n-k}$ ; and  $\chi(E, E)$  is even.
- (iii) Granting the compatibility of [Corollary 16.34\(iii\)](#) in every degree,  $\dim \text{Ext}^k(E, E) \geq \rho_k$ . At  $n = 2$ , at the points with  $d = 1$  one has  $\rho_2 = 23$ , which is odd, so  $\dim \text{Ext}^2(E, E) = r(\gamma)$  cannot hold there, since  $\chi(E, E)$  is even and Serre duality gives  $\chi(E, E) \equiv \dim \text{Ext}^2(E, E) \pmod{2}$ .

*Proof.* (i) Split  $HT^1 = Q \oplus P$  as in [Theorem 16.33](#), so that  $HT^k = \bigoplus_{i+j=k} \wedge^i Q \otimes \wedge^j P$  and  $Q, P$  act on the two tensor factors of  $H^*(A) = H_+ \otimes H_-$  separately. A mixed monomial kills  $\omega$ , and on  $p(\theta)$  the block of mixed monomials with fixed index sets  $I, J$  has entries  $C_{s+t+m+m'}$  depending only on  $s+t$  and  $m+m'$ , so its rank is that of a catalecticant of the binary form  $\sum_k c_k x^k y^{2n-k} / (2n-k)!$ , which is  $\min(k+1, 2n+1-k, r')$  by the classical Hilbert function of an apolar ideal. The pure sources  $\wedge^k Q$  and  $\wedge^k P$  contract  $\alpha_+$  and  $\alpha_-$  injectively into targets that no other source reaches unless  $k = n$ . The symmetry is [Lemma 21.30](#). (ii)  $\text{ch}(E)^\vee$  changes the sign of the components of odd codegree,  $\omega\theta = 0$  kills the cross terms, and Hodge–Riemann gives  $W > 0$ ; parity holds because the Chern character of a perfect complex on a complex torus is integral, and the intersection form on  $\wedge^{2n} H^1(A, \mathbb{Z})$  pairs each monomial only with its complement, so  $\int x^2$  is even. (iii) is the argument of [Corollary 16.34\(iii\)](#). Item (LIII) of [Appendix D](#) checks (i) exactly over  $\mathbb{Q}(i)$  for  $n = 2, 3$  in every degree and for  $n = 4$  in every degree with **-extreme**, and (ii) and the parity at  $n = 2$ . ■

The profile says what the corrected criterion does not constrain. For the pure form, [Theorem 16.40\(iv\)](#) forced  $\dim \text{End}(E) \geq 3$  at  $n = 2$  and  $n = 3$  out of  $\chi(E, E) > 0$ . For a corrected character  $\chi(E, E) = VQ(c) + W$  can have either sign, and when it is small enough the Euler characteristic forces nothing: at  $n = 2$  in the model of `explicit_weil.py` with  $d = 1$ , the character with  $c = (2, 1, 0, 1, -1)$  has  $\chi(E, E) = -136$ , and a simple object is not excluded.

**Proposition 17.11** (The Hodge locus of a corrected character). (i) *Let  $T$  be a complex torus, let  $x$  be a real class of type  $(k, k)$  on  $T$ , and for  $v \in \text{Hom}(H^{1,0}, H^{0,1}) = H^1(T_T)$  let  $J_v$  be the complex structure on the underlying real torus with  $H_v^{1,0} = (1+v)H^{1,0}$ . Then  $x$  is of type  $(k, k)$  for  $J_v$  if and only if  $v \lrcorner x = 0$ . So in this chart the Hodge locus of  $x$  is the linear space  $\text{Ann}_{H^1(T_T)}(x)$ , and it coincides with its first-order locus.*

- (ii) *Let  $J$  be any complex structure on the real torus underlying  $A$ . If  $\theta^k$  is of type  $(k, k)$  for  $J$  for some  $1 \leq k \leq 2n - 1$ , then  $\theta$  is of type  $(1, 1)$  for  $J$ .*
- (iii) *In [Setup 17.8](#) with  $n \geq 3$ , the germ at  $A$  of the Hodge locus of  $\gamma$  among all complex structures is the polarised Weil family  $\mathcal{D}_{n,n}$ , of dimension  $n^2$ , if  $c_k \neq 0$  for some  $1 \leq k \leq 2n - 1$ , and the family of all  $K$ -linear deformations, of dimension  $2n^2$ ,*

otherwise. In the first case the connected component of the Hodge locus through  $A$  is  $\mathcal{D}_{n,n}$ , and all its members are abelian varieties polarised by  $\theta$ .

- (iv) For  $n = 2$  the same holds, with one exception: if  $c_1 = c_3 = 0$  and  $c_2^2 \int_A \theta^4 = \frac{3}{4} \int_A \omega^2$ , the germ has dimension 5. In the rational model of `explicit_weil.py` this is the character  $N\omega_1 \pm \frac{d}{2}N\eta^2 + c_0 + c_4\eta^4$ ; with rational data the ratio is attained exactly in the families of trivial discriminant.

*Proof.* (i) Write  $D_v$  for the derivation of  $\Lambda^* H^1$  extending  $v$ . Since  $v^2 = 0$  on  $H^1$ ,  $\Lambda^*(1 + v) = \exp(D_v)$ , so the Hodge filtration of  $J_v$  is  $F_v^p = \exp(D_v)F^p$ . Now  $\exp(-D_v)x = \sum_m (-1)^m D_v^m x / m!$  with  $D_v^m x$  of type  $(k - m, k + m)$ , so  $x \in F_v^k$  exactly when  $D_v x = v \lrcorner x = 0$ ; and a real class lies in  $F_v^k \cap \overline{F_v^k} = H_v^{k,k}$  as soon as it lies in  $F_v^k$ . (ii) Let  $t \mapsto t_J$  be the circle acting on  $H^{p,q}$  by  $t^{p-q}$ . It acts on  $\Lambda^* H^1(A, \mathbb{R})$  by algebra automorphisms and fixes  $\theta^k$ , so the path  $t \mapsto t_J \theta$  lies in  $\{y \in \Lambda^2 H^1(A, \mathbb{R}) : y^k = \theta^k\}$ . At a nondegenerate  $y$  the differential  $h \mapsto k y^{k-1} h$  of  $y \mapsto y^k$  is injective on  $\Lambda^2$  when  $2 + (k - 1) \leq 2n$ , by the linear hard Lefschetz theorem for the symplectic form  $y$  on a space of dimension  $4n$ ,<sup>48</sup> so near  $\theta$  that set is discrete, the path is constant, and  $\theta$  is of type  $(1, 1)$ . (iii) By (i), applied to each homogeneous component, the Hodge locus of  $\gamma$  in the chart is  $\text{Ann}_{H^1(T_A)}(\gamma)$ , the part of  $\text{Ann}_{HT^2}(\gamma)$  in  $H^1(T_A)$ . In the description of [Theorem 17.9\(i\)](#) that part is  $x = w = 0$  with  $(0, \lambda, 0) \in \ker H$ ; the vector  $(0, 1, 0)$  lies in  $\ker H$  exactly when  $\mu_1 = \dots = \mu_{2n-1} = 0$ , so the part is spanned by the  $n^2$  vectors  $v_{ab}^1 + v_{ab}^2$ , which span  $T_{[A]}\mathcal{D}_{n,n}$ , when some  $c_k$  with  $1 \leq k \leq 2n - 1$  is nonzero, and by all  $v_{ab}^1, v_{ab}^2$ , the  $K$ -linear deformations, otherwise. A germ of submanifold of dimension  $n^2$  inside a linear space of the same dimension is that space. On  $\mathcal{D}_{n,n}$  the class  $\theta$  stays of type  $(1, 1)$  and nondegenerate, so the signature of the hermitian form it defines is constant and every member is polarised by  $\theta$ . The same argument at every member shows that  $\mathcal{D}_{n,n}$  is open in the Hodge locus; it is closed in the space of complex structures, since  $K$ -linearity is a closed condition and a limit of positive hermitian forms that stays nondegenerate stays positive; and it is connected. (iv) At  $n = 2$  the weight argument of [Theorem 17.9\(i\)](#) fails for one pair of weights, and the first-order locus is computed directly: its dimension is 4, 8 when  $c_1 = c_2 = c_3 = 0$ , and 5 exactly at the stated ratio, which in the model reads  $|u| = 4|c_2|$  with  $\int \theta^4$  and  $\int \omega^2$  equal to 24 and  $2|u|^2$  times the same volume. In the rational model  $\int \omega_1^2 = 8d^2$  and  $\int \eta^4 = 24$ , which gives  $c_2 = \pm dN/2$ . For a general rational Weil class  $\omega = x\omega_1 + y\omega_2$  and hermitian form  $\text{diag}(-a_1, -a_2, a_3, a_4)$  one has  $\int \omega^2 = 8d N_{K/\mathbb{Q}}(y + x\sqrt{-d})$  and  $\int \theta^4 = 24 a_1 a_2 a_3 a_4$ , so the ratio asks  $(c_2/N)^2 = d N_{K/\mathbb{Q}}(y + x\sqrt{-d}) / (4a_1 a_2 a_3 a_4)$ , a rational square exactly when  $a_1 a_2 a_3 a_4$  is a norm from  $K$ . Items (LII)(E) and (F) of [Appendix D](#) check the dimensions and the two integrals exactly. ■

So [Theorem 17.5](#) does not extend to the corrected form, and the reason is visible. A semiregular complex deforms along the Hodge locus of its Chern character; for the pure form that locus is the whole  $2n^2$ -dimensional family of  $K$ -linear tori, whose very general member is not algebraic, while for a corrected character with some  $c_k \neq 0$ ,  $1 \leq k \leq 2n - 1$ , it is the polarised family itself, all of whose members are abelian varieties, and there is nothing for the argument through Weil tori to act on. That is a statement about the argument run on  $E$  itself; an autoequivalence can still remove a component, and a line

<sup>48</sup>For a nondegenerate  $y \in \Lambda^2 U^\vee$  on a real space  $U$  of dimension  $2m$ , multiplication by  $y^j$  is injective from  $\Lambda^p$  to  $\Lambda^{p+2j}$  whenever  $p + j \leq m$ , because  $y$  generates an  $\mathfrak{sl}_2$ -action on  $\Lambda^* U^\vee$  in which  $\Lambda^p$  has weight  $p - m$ . Here  $m = 2n$ ,  $p = 2$  and  $j = k - 1$ . The bound is sharp:  $\theta^{2n}$  is a multiple of the volume form for every complex structure.

bundle twist is what reaches the corrected characters of [Theorem 17.9\(v\)](#). The one other place it can reach is the exceptional ratio of (iv), where the Hodge locus leaves the polarised family.

*Remark 17.12* (The exceptional ratio at  $n = 2$ ). At  $n = 2$ , with  $c_1 = c_3 = 0$  and  $c_2^2 \int \theta^4 = \frac{3}{4} \int \omega^2$ , put  $\gamma_2 = \omega + c_2 \theta^2$ . Item (LII)(G) of [Appendix D](#) computes its stabiliser  $\mathfrak{g}$  in  $\mathfrak{gl}(H^1(A, \mathbb{C}))$  exactly: it has dimension 21, Hodge pieces of dimensions 5, 11, 5, commutant on  $H^1$  equal to the scalars, and a real form whose trace form has signature (12, 9); at an ordinary ratio the stabiliser is  $\mathfrak{su}(2, 2)$ , of dimension 15. The scalars do not fix  $\gamma_2$ , so  $\mathfrak{g}$ , acting irreducibly with a nondegenerate trace form, is semisimple; a product of two factors would act on  $H^1 = V_1 \otimes V_2$  with  $\dim V_i \in \{2, 4\}$  and have dimension at most  $3 + 15$ , so  $\mathfrak{g}$  is simple of dimension 21. The simple Lie algebras of dimension 21 are  $\mathfrak{so}(7, \mathbb{C})$  and  $\mathfrak{sp}(6, \mathbb{C})$ , and only the first has an irreducible representation of dimension 8, its spin representation; among its real forms only  $\mathfrak{so}(4, 3)$  has a trace form of signature (12, 9). Its invariants in  $\Lambda^* H^1$  are spanned by 1,  $\gamma_2$  and the class of a point, so  $\gamma_2$  is the invariant 4-form of  $\text{Spin}(4, 3)$ , a split form of the Cayley form.<sup>49</sup> The Hodge locus of  $\gamma$  near  $A$  is the orbit of  $A$  under the connected real group  $G$  of  $\mathfrak{g}$ , since that orbit has dimension  $\dim \mathfrak{g}^{-1,1} = 5$  and lies in the locus; the Hodge group of a very general member  $T$  has Lie algebra an ideal of  $\mathfrak{g}$ , hence all of it, so the Hodge classes of  $T$  are 1,  $\gamma_2$  and the class of a point, and  $\text{NS}(T) = 0$ . The locus reaches tori that are not algebraic, and [Theorem 17.5](#) can be started there. It does not finish: its last step, [Proposition 17.4](#), needs a Kähler class on which the degree-four Chern character integrates to zero, and here

$$\int_A \gamma_2 \kappa_h^2 = \frac{|c_2|}{6} \left( \text{sign}(c_2) \sigma_2(h) + 4 \text{Re}(\phi \det h_{01,23}) \right) \int_A \theta^4, \quad |\phi| = 1,$$

for the Kähler class  $\kappa_h = i \sum h_{jk} z_j \wedge \bar{z}_k$  of a positive hermitian  $h$  in a unitary frame for  $\theta$ , which has the sign of  $c_2$ . Indeed, writing  $h = \sum_j \lambda_j w_j w_j^*$  with  $(w_j)$  orthonormal, Cauchy–Binet gives  $\sigma_2(h) \pm 4 \text{Re}(\phi \det h_{01,23}) = \sum_{j < k} \lambda_j \lambda_k (1 \pm 4 \text{Re}(\phi \xi_{01} \bar{\xi}_{23}))$  with  $\xi = w_j \wedge w_k$  of norm one; the Plücker relation and the inequality of the means give  $|\xi_{01} \bar{\xi}_{23}| \leq \frac{1}{4}$ , so each term is nonnegative, and Parseval gives  $\sum_{j < k} \lambda_j \lambda_k = 6$ , so the sum is at least  $6 \min_j \lambda_j^2 > 0$ . Since  $G$  fixes  $\gamma_2$  and carries Kähler cones to Kähler cones, the same holds on every member of the locus. So the argument excludes at most a torsion-free sheaf with  $c_2 > 0$ , which Bogomolov’s inequality already forbids to be  $\theta$ -semistable, and not a complex. Item (LII)(G) checks the displayed identity at  $c_2 = 1$ ,  $\phi = 1$ , and the constancy of the sign at random Kähler classes, against a change of sign at  $R = 3/16$ . At  $n = 2$  Markman’s theorem settles every Weil fourfold, so this is a statement about the shape of  $(P2')$ , not about the conjecture.

*Remark 17.13* (Which obstructions survive the correction). The obstruction theorems of this section and of [Section 9](#) were proved for a Chern character on the Weil line; for a corrected character they divide as follows. Unchanged, with  $r(\gamma)$  in place of  $2n(2n - 1)$  where a number occurs: [Theorem 9.20](#) and [Corollary 18.22](#), whose only hypothesis is that  $\text{ch}(E)$  stay of Hodge type to first order along  $\mathcal{D}_{n,n}$ ; [Theorem 18.5](#) and [Corollary 18.6](#), since a corrected character is a flat Hodge class in every degree; [Lemma 16.37](#); and the first two parts of [Theorem 16.36](#), which become [Theorem 17.9\(iv\)](#). Unchanged only

<sup>49</sup>In the compact case the stabiliser in  $\text{GL}(8, \mathbb{R})$  of the Cayley 4-form is  $\text{Spin}(7)$ , acting on  $\mathbb{R}^8$  by its spin representation; here the same happens for the real form of signature (4, 3). We know of no earlier appearance of this form among the Hodge classes of abelian fourfolds, but we have not searched the literature on Hodge loci for it systematically.

$m(R) = 1 - \sqrt{3/(4R)}$ : the sign of  $\int \gamma \kappa^2$  is constant on the Kähler cone exactly for  $R \geq 3/4$

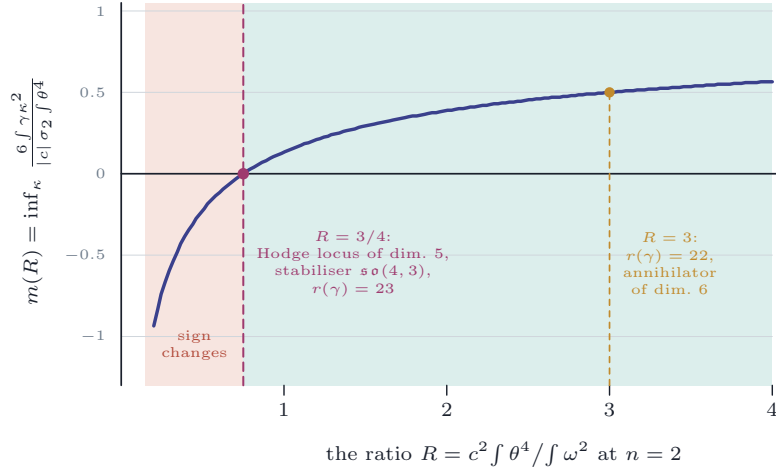


FIGURE 20. The exceptional ratio of Remark 17.12. By the identity displayed there, which is linear in  $\gamma$ , the infimum over the Kähler cone of  $6 \int \gamma \kappa^2 / (|c| \sigma_2(h) \int \theta^4)$  is  $m(R) = 1 - \sqrt{3/(4R)}$ , approached at the boundary of the cone by a positive hermitian form of rank two whose Plücker coordinates satisfy  $|\xi_{01} \xi_{23}| = \frac{1}{4}$ . It is negative exactly for  $R < 3/4$ , and  $R = 3/4$  is also where the Hodge locus of  $\gamma$  jumps from dimension 4 to 5 and its stabiliser becomes  $\mathfrak{so}(4, 3)$ , with  $r(\gamma) = 23$ ; at  $R = 3$  the annihilator has dimension 6 and  $r(\gamma) = 22$ . Items (LII)(D), (E) and (G) check these values.

when  $c_0 = \dots = c_{2n-2} = 0$ , so that  $H^{0,2}$  still kills  $\gamma$ : Theorem 9.7, Proposition 9.10, Corollary 9.18, Theorem 9.19 and Theorem 16.30(iii). Unchanged only when  $\ker H$  contains a vector with  $w \neq 0$ , which needs  $\rho \leq 2$ , the vector giving a compensated class of  $P \wedge Q$  with a component in  $T_+ \otimes T_-$ : Theorem 16.30(iv), Corollary 16.31 and Theorem 16.38. Lost: Theorem 17.5, outside Theorem 17.9(v); the positivity of  $\chi(E, E)$  in Proposition 9.11 and with it Theorem 16.40(i)–(iii), by Proposition 17.10(ii); and Theorem 20.33. So for a general shape,  $\rho = 3$ , what this paper still excludes is a line bundle summand off the polarisation line and a Chern character effective in the exponential basis, and nothing else; Proposition 17.14 below extends the first to summands with a scalar Atiyah class. An object with no such summand, whose Chern character is not effective, and with  $\dim \text{Ext}^2(E, E) = 7n^2 - 2n$ , is not excluded by anything proved here, and none is known; by Lemma 17.16 below it may be taken indecomposable.

**Proposition 17.14** (Summands with a scalar Atiyah class). *Let  $A$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$ , let  $T = T_{[A]} \mathcal{D}_{n,n}$ , and let  $E$  be a perfect complex on  $A$  with  $v_{\perp} \text{ch}(E) = 0$  for every  $v \in T$ . Suppose  $E$  has a direct summand  $F[p]$  with  $\text{At}(F) = \delta \cdot \text{id}_F$ , where  $\delta$  is a real class of type  $(1, 1)$  that is not a real multiple of  $\theta$ , and  $\text{rk } F \neq 0$ . Then there is  $v \in T$  with  $\text{ev}_E(v) \neq 0$  and  $\sigma_E(\text{ev}_E(v)) = 0$ . In particular no finite direct sum of shifted bundles  $\pi_* L$ ,  $\pi$  an isogeny, among them the simple semi-homogeneous bundles, has a corrected character  $\gamma$  with  $N \neq 0$  and injective semiregularity map.*

*Proof.* As in [Theorem 9.20](#),  $\text{ev}_E(v) = v \lrcorner \text{At}(E)$  on  $H^1(T_A)$  and the Atiyah class of a direct sum is block diagonal, so the component of  $\text{ev}_E(v)$  in  $\text{Ext}^2(F, F)$  is  $(v \lrcorner \delta) \cdot \text{id}_F$ ; the map  $z \mapsto z \cdot \text{id}_F$  on  $H^{0,2}$  is injective because  $\text{tr}(z \cdot \text{id}_F) = \text{rk}(F) z$ , and some  $v \in T$  has  $v \lrcorner \delta \neq 0$  by [Proposition 9.15\(i\)](#). On the other hand  $\sigma_E(\text{ev}_E(v)) = v \lrcorner \text{ch}(E) = 0$ . For the last sentence, the proof of [Lemma 18.54\(i\)](#) gives  $\text{At}(\pi_* L) = \delta \cdot \text{id}$  with  $\pi^* \delta = c_1(L)$  for any line bundle  $L$ , and  $\text{ch}(\pi_* L) = \text{rk}(\pi_* L) e^\delta$ ; if every  $\delta$  were a multiple of  $\theta$  the Chern character of the sum would lie in  $\mathbb{Q}[\theta]$ , which does not contain  $\omega$  because  $\omega\theta = 0$  and  $\theta^{n+1} \neq 0$ . ■

[Theorem 16.28](#) asks that  $\sigma_E$  be injective. What its proof uses is that the relative deformations of  $E$  dominate the family, and Pridham's form of [Theorem 16.27](#), in which  $\sigma_E$  annihilates every obstruction, gives that under a weaker hypothesis. It tolerates an object that is obstructed to first order, which is exactly how [Theorem 9.20](#), [Corollary 18.22](#) and [Proposition 17.14](#) fail an object.

**Proposition 17.15** (A flatness form of the criterion). *Let  $n \geq 2$ ,  $s_0 \in \mathcal{D}_{n,n}$ , and let  $E$  be a bounded complex of vector bundles on  $A_{s_0}$  with  $\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$  and  $N \neq 0$ . Let  $K_E$  be the base of a miniversal deformation of  $E$  on the fixed variety  $A_{s_0}$ , put  $e = \dim \text{Ext}^1(E, E)$  and  $k = \dim \ker \sigma_E$ , and grant [Theorem 16.27](#) in Pridham's form, in which  $\sigma_E$  annihilates every obstruction, both on  $A_{s_0}$  and along the Artinian deformations of  $A_{s_0}$  in  $\mathcal{D}_{n,n}$ .*

- (i)  $\dim K_E \geq e - k$ .
- (ii) If  $\dim K_E = e - k$ , then  $\Sigma = \mathcal{D}_{n,n}$ , and the Hodge conjecture holds in every degree for every member of the family with  $\text{Hg}(A) = \text{SU}(V, H)$ .
- (iii)  $\dim K_E = e - k$  as soon as the cone  $\{\xi \in \text{Ext}^1(E, E) : \xi \circ \xi = 0\}$  has codimension  $k$ .

For  $k = 0$  this is [Theorem 16.28](#).

*Proof.* Let  $R$  be the completed local ring of  $\mathcal{D}_{n,n}$  at  $s_0$ , regular of dimension  $n^2$ . By hypothesis  $\ker \sigma_E$  is a complete obstruction space for the deformations of the pair  $(A_s, E)$  over Artinian  $R$ -algebras, whose relative tangent space is  $\text{Ext}^1(E, E)$ ; so this functor has a hull  $P/J$  with  $P = R[[x_1, \dots, x_e]]$  and  $J$  generated by at most  $k$  elements [[FM98](#)], and its fibre over the closed point of  $R$  is a hull of the absolute functor, of dimension  $\dim K_E$ . Krull's height theorem gives  $\dim P/J \geq n^2 + e - k$ , and the same count on the fibre gives (i). If  $\dim K_E = e - k$ , the dimension inequality for a local homomorphism [[Mat86](#), Theorem 15.1] gives  $\dim P/J \leq n^2 + e - k$ , so equality holds,  $J$  has height  $k$  and is generated by a regular sequence,  $P/J$  is Cohen–Macaulay, and it is flat over  $R$  by miracle flatness [[Mat86](#), Theorem 23.1].<sup>50</sup> The analytic Kuranishi germ of pairs, as in the proof of [Theorem 17.5](#), has  $P/J$  as its completion, so it is flat over the germ of  $\mathcal{D}_{n,n}$  at  $s_0$  and therefore open onto a neighbourhood  $U$  of  $s_0$ . Every  $A_s$  with  $s \in U$  then carries a bounded complex of vector bundles, algebraic by GAGA, whose Chern character is the parallel transport of  $\text{ch}(E)$ , so  $N\omega_{1,s}$  differs from an algebraic class by a polynomial in  $\theta$  and is algebraic; [Corollary 16.5](#) gives (ii). For (iii), the quadratic part of the Kuranishi map of  $E$  is  $\xi \mapsto \xi \circ \xi$ , so the tangent cone of  $K_E$  lies in the cone of the hypothesis, and  $\dim K_E \leq e - k$ . Every step uses only deformations along  $\mathcal{D}_{n,n}$ , which are projective, where [Theorem 17.5](#) needs non-projective ones. ■

No object is known that meets (iii) with  $k > 0$ . Among direct sums the flatness form fails for a large class, [Remark 17.25](#) below; for indecomposable objects nothing in this paper

<sup>50</sup>A local homomorphism from a regular local ring to a Cohen–Macaulay local ring is flat exactly when the dimension of the target is the dimension of the source plus that of the fibre; here both sides equal  $n^2 + e - k$ .

excludes it, and the proposition says only that injectivity of  $\sigma_E$  is more than propagation needs.

**17.3. Which objects can meet the corrected criterion.** The corrected criterion asks for one perfect complex. The results below say what such a complex can be made of, for  $K$  imaginary quadratic: a direct sum meets the criterion only through one indecomposable summand (Lemma 17.16); no sum of the objects whose Ext algebra is an exterior algebra meets it (Corollary 17.17); the Orlov products of secant objects meet its twisted form, with  $\dim \text{Ext}^2$  equal to the number of Theorem 17.9, at  $n = 2$  and  $n = 3$  (Theorem 17.19), and cannot meet it at  $n = 4$  (Proposition 17.20); and at  $n = 4$  a combination of natural objects needs at least eight of them, on one conic (Theorem 17.23). None of this is a new case of the Hodge conjecture: the certificates are in families where Markman's theorem already holds, and they are certificates for twisted objects. Item (LV) of Appendix D checks the computations; its scripts are in `code/attack/explicit_objects` and `code/attack/natural_objects`.

**Lemma 17.16** (The summands of an object meeting the criterion). *Let  $A$  be a member of a  $(K, -1, n)$ -Weil family,  $K$  imaginary quadratic and  $n \geq 2$ , let  $T = T_{[A]} \mathcal{D}_{n,n} \subseteq H^1(T_A)$ , and let  $E = \bigoplus_i F_i[k_i]$  be a finite direct sum of perfect complexes with  $v \lrcorner \text{ch}(E) = 0$  for every  $v \in T$ , as holds for every corrected character  $N\omega_1 + \sum_k c_k \theta^k$ . Suppose that  $\sigma_E$  is injective on  $\text{ev}_E(T)$ ; this holds if  $\sigma_E$  is injective, and under the weakened criterion of [Mar25b, Question 11.4].*

- (i) *For every  $i$  and every  $v \in T$ ,  $\text{ev}_{F_i}(v) = 0$  and  $v \lrcorner \text{ch}(F_i) = 0$ , and every  $\text{ch}(F_i)$  lies in  $\bigoplus_k \mathbb{Q} \theta^k \oplus W(A)$ .*
- (ii) *If  $\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$  with  $N \neq 0$ , some  $F_i$  has  $\text{ch}(F_i) = \omega' + \sum_k c'_k \theta^k$  with  $\omega'$  a nonzero rational Weil class, and then  $\dim \text{Ext}^2(F_i, F_i) \geq r(\text{ch } F_i) \geq 2n(2n - 1)$ . If  $\sigma_E$  is injective, so is every  $\sigma_{F_i}$ , and that  $F_i$  meets the corrected criterion on its own.*

*So the corrected criterion is a statement about indecomposable objects.*

*Proof.* For  $v \in T$ ,  $\sigma_E(\text{ev}_E(v)) = v \lrcorner \text{ch}(E) = 0$  by the commutative square of Setup 18.27, so  $\text{ev}_E(v) = 0$ . The Atiyah class is natural, so  $\text{ev}_E(v) = v \lrcorner \text{At}(E)$  is block diagonal with the blocks  $\pm \text{ev}_{F_i}(v)$ , which therefore vanish, and  $v \lrcorner \text{ch}(F_i) = \sigma_{F_i}(\text{ev}_{F_i}(v)) = 0$ . By Proposition 17.11(i) every homogeneous component of  $\text{ch}(F_i)$ , a rational class of type  $(k, k)$ , stays of Hodge type on the germ of  $\mathcal{D}_{n,n}$  at  $A$ , which in the chart of that proposition is the linear space  $T$ ; so it is a Hodge class at a member with Hodge group  $\text{SU}(V, H)$ , and Theorem 4.14 puts it in  $\mathbb{Q} \theta^k$ , or in  $\mathbb{Q} \theta^n \oplus W$  for  $k = n$ . That is (i). For (ii), the Weil parts of the  $\pm \text{ch}(F_i)$  add up to  $N\omega_1 \neq 0$ , so one of them is nonzero; Theorem 17.9 gives  $\dim \text{Ext}^2(F_i, F_i) \geq r(\text{ch } F_i)$  with no hypothesis on  $\sigma$ , and  $r \geq (4 + 0)n^2 - 2n = 2n(2n - 1)$  for  $n \geq 3$ , while at  $n = 2$  the seven values of Figure 19 are at least 12. Injectivity passes to the summands by Lemma 21.35. By the Krull–Schmidt property of  $D^b(A)$  the  $F_i$  may be taken indecomposable. ■

The Weil part  $\omega'$  of the summand need not be a multiple of  $\omega_1$ ; propagating any nonzero rational Weil class propagates the whole Weil plane, through the algebraic action of  $K$ . The lemma is the Weil-family analogue of Corollary 21.36(ii).

**Corollary 17.17** (Sums of objects with exterior Ext algebras). *In Lemma 17.16, let every  $F_i$  have  $\dim \text{Ext}^2(F_i, F_i) < 2n(2n - 1)$ . This holds when  $\text{Ext}^*(F_i, F_i) \cong \bigwedge^* \mathbb{C}^{2n}$ , and so for line bundles, for the sheaves  $\mathcal{O}_B \otimes L$  with  $B \subseteq A$  an abelian subvariety, for simple semi-homogeneous bundles, for skyscraper sheaves, and for the images of all of these under*

autoequivalences of  $D^b(A)$ . Then no such sum with  $\text{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$  and  $N \neq 0$  has  $\sigma_E$  injective on  $\text{ev}_E(T)$ : it fails the corrected criterion and the weakened criterion, some  $v \in T$  obstructs it to first order, and  $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$ .

*Proof.* The first assertion is [Lemma 17.16\(ii\)](#), since  $\binom{2n}{2} = n(2n-1) < 2n(2n-1)$ , and  $\dim \text{Ext}^2(E, E) \geq r(\text{ch } E)$  with equality only if  $\sigma_E$  is injective ([Theorem 17.9](#)). For  $\mathcal{O}_B$  choose a complementary abelian subvariety  $B'$ ; the isogeny  $\mu: B \times B' \rightarrow A$  is étale with group  $G = B \cap B'$ , and  $\mu^* \mathcal{O}_B = \bigoplus_{g \in G} \mathcal{O}_{B \times \{g\}}$  with  $G$  permuting the summands simply transitively, so  $\text{Ext}_A(\mathcal{O}_B, \mathcal{O}_B) = \text{Ext}_{B \times B'}(\mathcal{O}_{B \times 0}, \mathcal{O}_{B \times 0}) = \bigwedge^*(\mathbb{C}^{\dim B} \oplus \mathbb{C}^{\dim B'})$ . For a simple semi-homogeneous bundle  $V$ ,  $V \otimes V^\vee$  is a sum of line bundles in  $\text{Pic}^0$  of which  $\mathcal{O}$  occurs once [[Muk78](#)], so  $\text{Ext}^*(V, V) = H^*(\mathcal{O}_A)$ . Equivalences preserve Ext algebras. ■

With [Lemma 17.16](#) the objects that [Remark 17.13](#) leaves open may be taken indecomposable, and the corollary restores, for corrected characters, the part of [Theorem 20.33](#) that concerns objects with exterior Ext algebras; it extends [Proposition 17.14](#), whose summand has a scalar Atiyah class, to the subtori, the points and the Fourier–Mukai images. In the language of pure spinors, a summand whose character is a rational multiple of the pure spinor of a rational maximal isotropic subspace is excluded unless that character lies in  $\mathbb{Q}[\theta]$ , because the rational maximal isotropic subspaces whose pure spinors are flat over  $\mathcal{D}_{n,n}$  are those with pure spinor 1, the class of a point or  $e^{c\theta}$ .

The secant objects of [Section 14](#) have their own Hochschild profile, which [Theorem 18.29\(i\)](#) and (ii) compute in degrees one and two.

**Proposition 17.18** (The Hochschild profile of a secant class). *Let  $X$  be a principally polarised abelian  $n$ -fold,  $n \geq 2$ , with polarisation  $\theta$ , let  $d \geq 1$ , and let  $v = a u_\theta + b v_\theta$  with  $(a, b) \in \mathbb{Q}^2 \setminus \{0\}$ , in the notation of [\(29\)](#). The rank of contraction  $HT^k(X) \rightarrow H^*(X, \mathbb{C})$ ,  $\xi \mapsto \xi \lrcorner v$ , is 1 for  $k = 0$  and  $k = n$ ,  $2\binom{n}{k}$  for  $1 \leq k \leq n-1$ , and 0 for  $k > n$ . A perfect complex  $F$  on  $X$  with  $\text{ch}(F) = v$  has  $\dim \text{Ext}^k(F, F) \geq 2\binom{n}{k}$  for  $k = 1, 2$ , and*

$$\chi(F, F) = \begin{cases} (-1)^{n/2} 2^{n-1} d^{n/2-1} (a^2 d + b^2) & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases}$$

*Proof.* Put  $s = \sqrt{-d}$  and  $w = \frac{1}{2}(a - bs/d)$ . By [Theorem 14.5\(i\)](#),  $v = w e^{s\theta} + \bar{w} e^{-s\theta}$ , and  $w \neq 0$ . By [Lemma 18.28\(i\)](#),  $\xi \lrcorner e^{\pm s\theta} = y_\pm e^{\pm s\theta}$ , where  $y_\pm \in H^{0,k}(X)$  is the component of  $e^{\pm s\theta} \xi$  in  $\bigwedge^k H^{0,1}$ . The map  $\xi \mapsto (y_+, y_-)$  is onto  $H^{0,k} \oplus H^{0,k}$  for  $k \geq 1$ :  $\xi = z \in \bigwedge^k H^{0,1}$  gives  $(z, z)$ , and  $\xi = z' \otimes \partial$  with  $z' \in \bigwedge^{k-1} H^{0,1}$  gives  $\pm s z' \wedge (\partial \lrcorner \theta)$ , where  $\partial \lrcorner \theta$  runs over  $H^{0,1}$ . If  $w y_+ e^{s\theta} + \bar{w} y_- e^{-s\theta} = 0$ , the components of degrees  $k$  and  $k+2$  give  $w y_+ + \bar{w} y_- = 0$  and  $(w y_+ - \bar{w} y_-) \wedge \theta = 0$ , and for  $k < n$  wedge product with  $\theta = \sum_a p_a \wedge q_a$  is injective on  $\bigwedge^k H^{0,1}$ , so  $y_\pm = 0$ . So the rank is  $2\binom{n}{k}$  for  $1 \leq k < n$ . For  $k = n$ ,  $y \wedge \theta = 0$  for  $y \in H^{0,n}$  and the image is the line spanned by  $w y_+ + \bar{w} y_-$ ; for  $k > n$ ,  $H^{0,k} = 0$ ; and for  $k = 0$  the rank is one. The bounds on  $\text{Ext}^1$  and  $\text{Ext}^2$  follow, as in [Theorem 18.29\(i\)](#) and (ii), from the compatibility of the characteristic morphism with contraction into the Chern character. Finally the Todd class is trivial, so  $\chi(F, F) = \int_X \text{ch}(F)^\vee \text{ch}(F)$  with  $\text{ch}(F)^\vee = w e^{-s\theta} + \bar{w} e^{s\theta}$ , and  $\int_X (e^{2s\theta} + e^{-2s\theta}) = (2s)^n (1 + (-1)^n)$  because  $\int_X \theta^n = n!$ ; with  $|w|^2 = (a^2 d + b^2)/(4d)$  this is the formula. ■

At  $n = 4$  and  $(a, b) = (1, 3)$ , the class of [Proposition 18.25](#), this gives  $\chi(\mathcal{F}, \mathcal{F}) = 8d(d+9)$ , the rank recorded there.

**Theorem 17.19** (Twisted certificates at  $n = 2$  and  $n = 3$ ). *Let  $d \geq 1$ ,  $K = \mathbb{Q}(\sqrt{-d})$ , let  $\Phi$  be Orlov’s equivalence of [Proposition 18.25\(v\)](#), and for a perfect complex  $F$  on a*

principally polarised  $X$  put  $E = \Phi(F \boxtimes F^\vee)$  on  $A = X \times \hat{X}$ , with the  $K$ -action of (28) and the classes  $\eta, \ell$  of the split member.

- (i) Let  $X$  be the Jacobian of a curve  $C$  of genus two,  $C \subset X$  by an Abel–Jacobi map, and  $F = \mathcal{O}_C(p)$  for a point  $p \in C$ . Then  $\text{ch}(F) = v_\theta$ ,  $\dim \text{Ext}^k(F, F) = 1, 4, 1$ , and  $\dim \text{Ext}^k(E, E) = 1, 8, 18, 8, 1$  for  $k = 0, \dots, 4$ , which are the ranks of contraction from  $HT^k(A)$  into  $\text{ch}(E)$ . So  $\dim \text{Ext}^2(E, E) = 18 = r(\text{ch } E)$  and  $\sigma_E$  is injective.
- (ii) Let  $X$  be the Jacobian of a non-hyperelliptic curve  $C$  of genus three,  $i: C^{(2)} \rightarrow X$  the embedding of the theta divisor,  $x$  the class of  $\{p+q : q \in C\}$  and  $\theta$  the restriction of the polarisation, and let  $d = k(k+1)$  with  $k \geq 1$  and  $F = i_*\mathcal{O}_{C^{(2)}}((2k+1)x - k\theta)$ . Then  $\text{ch}(F) = v_\theta$ ,  $\dim \text{Ext}^k(F, F) = 1, 6, 6, 1$ , and  $\dim \text{Ext}^k(E, E) = 1, 12, 48, 74, 48, 12, 1$ . So  $\dim \text{Ext}^2(E, E) = 48 = r(\text{ch } E)$  and  $\sigma_E$  is injective.
- (iii) In both cases  $\text{ch}(E)$  is not killed by  $T = T_{[A]}\mathcal{D}_{n,n}$ , and no line bundle twist of  $E$  has a character killed by  $T$ ; the twisted character  $\kappa = \text{ch}(E)e^{\ell/2}$  is killed by  $T$ , has the form  $N'\omega' + \sum_k c_k\eta^k$  with  $\omega'$  a nonzero rational Weil class, and has the same contraction ranks as  $\text{ch}(E)$ . A class  $B$  with  $\text{ch}(E)e^B$  killed by  $T$  is  $\pm\ell/2$  modulo  $\mathbb{Q}\eta$ . At  $n = 2$ ,  $\kappa$  has  $c_1 = c_3 = 0$  and lies at the exceptional ratio of Remark 17.12, and its annihilator in  $H^1(T_A)$  has dimension 5; at  $n = 3$  that annihilator has dimension  $9 = n^2$ .

*Proof.* (i) By Grothendieck–Riemann–Roch  $\text{ch}(i_*\mathcal{O}_C(p)) = [C] + (1 + 1 - 2)[\text{pt}] = \theta = v_\theta$ , since  $v_\theta = \theta$  when  $n = 2$ . The curve has  $\text{Ext}^1(\mathcal{O}_C, \mathcal{O}_C) = H^1(\mathcal{O}_C) \oplus H^0(N_C)$  with  $N_C = K_C$  by adjunction, of dimension  $2 + 2$ , and  $\text{Ext}^0 = \text{Ext}^2 = \mathbb{C}$  by Serre duality; twisting by  $\mathcal{O}_C(p)$  changes nothing. The Ext algebra of  $F^\vee$  is that of  $F$ ,  $\Phi$  is an equivalence, and by the Künneth formula  $\text{Ext}^*(E, E) = \text{Ext}^*(F, F) \otimes \text{Ext}^*(F^\vee, F^\vee)$ , which gives  $1, 8, 18, 8, 1$ . By Proposition 17.18 and the Künneth decomposition  $HT^2(X \times X) = HT^2 \otimes 1 \oplus HT^1 \otimes HT^1 \oplus 1 \otimes HT^2$  the rank of contraction into  $\text{ch}(F) \boxtimes \text{ch}(F^\vee)$  is  $1 + 4 \cdot 4 + 1 = 18$ , and it is transported by  $\Phi$  as in the proof of Theorem 18.29(iv). The other degrees are the same count. Equality in degree two forces  $\sigma_E$  injective, as in Theorem 17.9. (ii) On  $C^{(2)}$  one has  $x^2 = 1$ ,  $x\theta = 3$ ,  $\theta^2 = 6$  and  $K_{C^{(2)}} = \theta$ ,  $i_*x = \frac{1}{2}\Theta^2$  and  $i_*\theta = \Theta^2$ , so Riemann–Roch gives  $\text{ch}_2(F) = 0$  and  $\text{ch}_3(F) = -k(k+1)[\text{pt}]$ , that is  $\text{ch}(F) = \Theta - d[\text{pt}] = v_\theta$ . The local-to-global spectral sequence, with  $h^p(\mathcal{O}_{C^{(2)}}) = 1, 3, 3$  and  $h^p(K_{C^{(2)}}) = 3, 3, 1$ , gives  $\dim \text{Ext}^1, \dim \text{Ext}^2 \leq 6$ , and Proposition 17.18 gives  $\geq 6$ ;  $\text{Ext}^0 = \mathbb{C}$  and Serre duality gives  $\text{Ext}^3 = \mathbb{C}$ . The rest is as in (i), with  $r(\text{ch } E) = 6n^2 - 2n = 48$  by Theorem 18.29(iv). (iii) is computed exactly in item (LV), over the function field  $\mathbb{Q}(d)$  for  $n \leq 5$  and in all the conventions for  $\Phi$ ; that  $r(\kappa) = r(\text{ch } E)$  is Lemma 18.28(i). For the uniqueness of  $B$ : the coefficient of the classes  $x_j\xi_j$  in  $\frac{1}{2}\ell + c\eta$  is  $\frac{1}{2}$  for every  $c$ , so no integral class is congruent to  $\pm\frac{1}{2}\ell$  modulo  $\mathbb{Q}\eta$ , on any  $X$ . ■

These are the numerical form of the criterion met with equality, so the injectivity of  $\sigma_E$  comes out of a dimension count and does not need [Mar25b, Question 11.4]. They certify the twisted object only: the character  $\kappa$  stays of Hodge type along  $\mathcal{D}_{n,n}$ , while  $\text{ch}(E)$  does so only on a locus of dimension 3 at  $n = 2$  and 6 at  $n = 3$ , so propagation needs the extension of Theorem 16.27 to twisted objects, which Perry proves [Per26], and an argument extracting untwisted cycles, as in [Mar25b, Section 2]. The families concerned, of trivial discriminant with  $n = 2$  and  $n = 3$ , are Markman’s.

**Proposition 17.20** (No Orlov product at  $n = 4$ ). *Let  $X$  be a principally polarised abelian fourfold,  $d \geq 1$ , and  $F_1, F_2$  perfect complexes on  $X$  with characters in the secant*

plane  $P_\theta \setminus \{0\}$  and  $\text{Ext}^{<0}(F_i, F_i) = 0$ , for instance sheaves. Then  $E = \Phi(F_1 \boxtimes F_2^\vee)$  has  $\dim \text{Ext}^2(E, E) > 88 = r(\text{ch } E)$ .

*Proof.* By the Künneth formula  $\dim \text{Ext}^2(E, E) = e_0 e'_2 + e_1 e'_1 + e_2 e'_0$  with  $e_k = \dim \text{Ext}^k(F_1, F_1)$  and  $e'_k$  likewise for  $F_2$ , and by Proposition 17.18  $e_0 \geq 1$ ,  $e_1 \geq 8$ ,  $e_2 \geq 12$ ; so the sum is at least  $12 + 64 + 12 = 88$ , which is  $r(\text{ch } E)$  by Theorem 18.29(iv). Equality would force  $e_0, e_1, e_2 = 1, 8, 12$ , and with Serre duality  $\chi(F_1, F_1) = 1 - 8 + 12 - 8 + 1 = -2$ , while Proposition 17.18 gives  $\chi(F_1, F_1) = 8d(a^2d + b^2) > 0$ . ■

Without  $\text{Ext}^{<0} = 0$  the count does not close, because negative Ext groups of  $F_i$  contribute products  $e_{-j} e'_{2+j}$  and change the Euler characteristic. For every  $n$  the same count says what a secant object would have to be for its Orlov product to meet the criterion with equality: a complex  $F$  with  $\text{ch}(F) \in P_\theta \setminus \{0\}$ ,  $\text{Ext}^{<0}(F, F) = 0$  and  $\dim \text{Ext}^k(F, F) = 1, 2n, n(n-1)$  for  $k = 0, 1, 2$ . For  $n = 4$  that is excluded above; for  $n$  odd, and for even  $n \geq 6$ , the Euler characteristic leaves room. If such  $F$  existed for a fixed  $d$  and unboundedly many  $n$ , and  $\kappa = \text{ch}(E)e^{\ell/2}$  is killed by  $T$ , which item (LV) checks for  $n \leq 5$ , then with a twisted form of Theorem 16.28 and Proposition 20.13 the Weil classes of every family of  $\mathbb{Q}(\sqrt{-d})$  would be algebraic. That is the secant route of (C1) and (C2) in Section 20.5, with the question of [Mar25b] replaced by a dimension count, and it lies off every minimal route (Theorem 20.40(iv)).

*Remark 17.21 (Untwisting).* The template  $E = \Phi(F_1 \boxtimes F_2^\vee)$  does not give an untwisted object meeting the corrected criterion by the obvious operations. No line bundle twist makes  $\text{ch}(E)$  flat (Theorem 17.19(iii)). At  $n = 2$  the object  $S^2(E[1]) \otimes P$ , for  $P$  a line bundle with  $c_1(P) = \ell$ , is untwisted with flat character, of rank 3 and at the exceptional ratio after a twist by a multiple of  $\eta$ , but its number is  $r = 23$ , which is odd, while Serre duality and the evenness of the Mukai pairing on an abelian variety make  $\dim \text{Ext}^2 \equiv \chi(E, E) \equiv 0 \pmod{2}$ ; so equality is impossible.  $\Lambda^2(E[1]) \otimes P$  has no Weil part. For  $E \otimes V$  with  $V$  a simple semi-homogeneous bundle of slope  $\ell/2$  [Muk78], the character  $\text{rk}(V)\kappa$  is flat, but  $\sigma_{E \otimes V}$  kills the summands  $\text{Ext}^2(E, E \otimes P_\xi)$ ,  $\xi \neq 0$ , of  $\text{Ext}^2(E \otimes V, E \otimes V)$  coming from the decomposition of  $V \otimes V^\vee$ , and at  $n = 2$  these are nonzero; this last point uses the decomposition of  $\text{End } V$  and the way  $\Phi$  intertwines  $\otimes P_\xi$  with translations, which we recall and have not re-derived. None of this shows that the corrected criterion must be met in twisted form; it shows that this template meets it only there.

At  $n = 4$  the natural objects of the split member can be described completely, and the description turns the criterion into a statement about points on a quadric.

**Setup 17.22** (The quadric of Lagrangians). Let  $X$  be a very general principally polarised abelian fourfold,  $U = H^1(X, \mathbb{Q})$ , and  $A = X \times \hat{X}$  with the  $K$ -action of (28), so that  $H^1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^2$  and the rational Hodge ring  $\mathcal{H}_A$  of  $A$  is the ring of  $\text{Sp}(U)$ -invariants in  $\Lambda^*(U \otimes \mathbb{Q}^2)$ , of dimension 55, with 1, 3, 6, 10, 15, 10, 6, 3, 1 in degrees 0, 2, ..., 16. The symmetric pairing on  $H^1(A, \mathbb{Q}) \oplus H_1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^4$  is the product of the symplectic form of  $U$  and a symplectic form on  $\mathbb{Q}^4$ , so the maximal isotropic subspaces stable under  $\text{Sp}(U)$  are the  $U \otimes L$  with  $L$  a Lagrangian plane in  $\mathbb{Q}^4$ . Let  $Q \subset \mathbb{P}^4$  be the Lagrangian Grassmannian  $\text{LG}(2, 4)$ , a smooth quadric threefold in its Plücker embedding. Call a perfect complex on  $A$  *natural* if it is a line bundle, a sheaf  $\mathcal{O}_B \otimes L$  on an abelian subvariety, a point, or the image of one of these under an autoequivalence. Then the character of a natural  $F$  is a multiple of the pure spinor of  $U \otimes L_F$  for a rational point  $L_F \in Q$ , and there is a linear isomorphism  $\Phi_4: H^0(Q, \mathcal{O}(4))^\vee \rightarrow \mathcal{H}_A$  with  $\text{ch}(F) = \Phi_4(\nu_4(L_F))$  up to a rational factor,  $\nu_4$  the fourth Veronese map. Under it the powers of  $\eta$  span the image of a

conic  $C_\eta \subset Q$ , and the Weil plane is the rational part of the line spanned by  $\nu_4(L_+)$  and  $\nu_4(L_-)$  for a conjugate pair of  $K$ -points  $L_\pm$  of  $Q$ ; the line  $l_W = L_+L_-$  meets  $Q$  only in  $L_\pm$ , and the plane of  $C_\eta$  is its polar plane.

The pure spinor description of the characters of natural objects is classical, and we recall it without re-deriving it. Item (LV) computes the dimensions, fits  $\Phi_4$  exactly and checks it on line bundles of every slope, on the abelian subvarieties  $B_{(p,q)} = \{(px, q\varphi x)\}$ , with  $\varphi: X \rightarrow \widehat{X}$  the polarisation, on twisted subvarieties and on points, and computes the positions of  $C_\eta$ ,  $l_W$  and  $L_\pm$ . On the conic of subvarieties the two points  $L_\pm$  have coordinates  $(p : q) = (1 : \mp\sqrt{-d})$ , so that the Weil classes are the values of the octic  $(p, q) \mapsto [B_{(p,q)}]$  at those points.

**Theorem 17.23** (Minimal support at  $n = 4$ ). *In Setup 17.22, let  $F_1, \dots, F_s$  be natural objects whose points  $L_{F_i}$  are pairwise distinct and not on  $C_\eta$ , and suppose that  $\sum_i m_i \text{ch}(F_i) + P = N\omega_1 + \sum_k c_k \eta^k$  with  $m_i \in \mathbb{Q}$ ,  $P \in \mathbb{Q}[\eta]$  and  $N \neq 0$ . Then  $s \geq 8$ . If  $s = 8$ , the eight points lie on one smooth conic  $\Pi \cap Q$  with  $\Pi$  a plane containing  $l_W$ , the conic is disjoint from  $C_\eta$ , and  $\sum_i m_i \text{ch}(F_i) = N\omega_1$  exactly. Conversely any eight rational points of such a conic carry such a relation with every  $m_i \neq 0$ .*

*Proof.* Through  $\Phi_4$  the relation reads  $\sum_i m_i \nu_4(L_{F_i}) + P' = c_+ \nu_4(L_+) + c_- \nu_4(L_-)$  in  $H^0(Q, \mathcal{O}(4))^\vee$ , with  $P'$  in the span of  $\nu_4(C_\eta)$  and  $c_\pm$  conjugate and nonzero because  $N \neq 0$ . A quartic  $f$  vanishing on  $C_\eta$ , at  $L_-$  and at every  $L_{F_i}$  gives  $c_+ f(L_+) = 0$ ; so it suffices to produce such an  $f$  with  $f(L_+) \neq 0$  unless the configuration is as stated. Let  $h_-$  be the linear form  $B(L_-, \cdot)$ , which vanishes on the polar plane of  $l_W$ , hence on  $C_\eta$ , and at  $L_-$ , but not at  $L_+$ , since  $l_W \not\subset Q$ . If the points split into at most three groups  $G_k$  with  $L_+$  outside the span of each, linear forms  $l_k$  vanishing on  $G_k$  and not at  $L_+$  give the quartic  $h_- l_1 l_2 l_3$ , which kills everything but  $c_+$ , so  $c_+ = 0$ . A rational line through  $L_+$  is  $l_W$ , which has no rational point on  $Q$ , so groups of one or two points never contain  $L_+$  in their span, and a triple does only if it lies in a plane through  $l_W$ . For  $s \leq 6$  split into pairs. For  $s = 7$ , either some triple is not in a plane through  $l_W$ , or all seven lie in one such plane  $\Pi$ , hence on  $C'' = \Pi \cap Q$ . That conic is smooth, since a degenerate one is a pair of conjugate lines with at most one rational point, and it misses  $C_\eta$ , since  $\Pi$  meets the plane of  $C_\eta$  in one point  $z$  and  $z \in Q$  would put the lines  $zL_\pm$  in  $Q$ . The quartics vanishing on  $C_\eta$  restrict onto all binary octics on  $C'' \cong \mathbb{P}^1$ , and nine distinct points impose independent conditions on octics, so  $L_\pm$  and the seven points are independent. For  $s = 8$ , if no plane through  $l_W$  holds seven of the points, choose two triples not in such planes and a pair; if exactly seven lie in  $\Pi$ , modify the quartic by a multiple of one vanishing on  $\Pi$  and at the eighth point; if all eight lie on  $C''$ , the ten points  $L_\pm, L_{F_1}, \dots, L_{F_8}$  of the rational normal curve  $\nu_4(C'')$  of degree eight satisfy exactly one relation, with every coefficient nonzero, and it is rational by Galois symmetry. Its part along the span of  $\nu_4(C_\eta)$  vanishes, because quartics restrict onto the sum of the octics on the two disjoint conics, so the relation is pure. ■

**Example 17.24** (Fourteen times a Weil class). Let  $d = 1$  and write  $(\gamma - \sqrt{-1}\ell)^4 = W_1 + \sqrt{-1}W_2$  with  $W_1, W_2$  rational, as in Theorem 15.2(iii). The eight subvarieties  $B_i = B_{(p_i, q_i)}$  with

$$(p_i, q_i) = (1, -3), (1, -2), (1, 2), (1, 3), (2, -1), (2, 1), (3, -1), (3, 1)$$

and  $m = (-1, 16, -16, 1, -16, 16, 1, -1)$  satisfy

$$\sum_i m_i [B_i] = 14W_2,$$

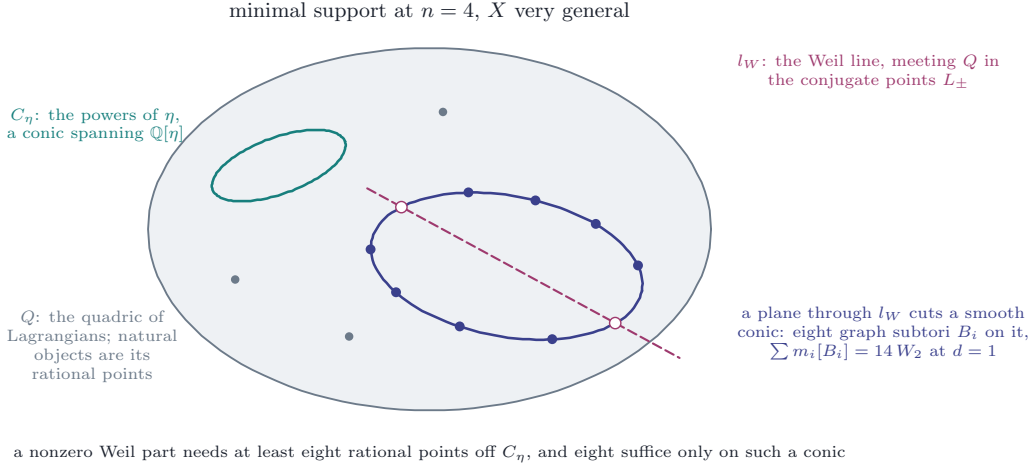


FIGURE 21. **Theorem 17.23.** The quadric  $Q$  of Lagrangians; the rational points are the Chern characters of the natural objects on the split member at  $n = 4$ , up to the Veronese map  $\nu_4$ . The powers of the polarisation fill the conic  $C_\eta$  (teal), and the Weil line is spanned by the conjugate points  $L_\pm$  (open circles) in which the line  $l_W$  (dashed) meets  $Q$ . A combination of natural characters with a nonzero Weil part needs at least eight rational points off  $C_\eta$ , and with eight they lie on one conic cut out by a plane through  $l_W$  (indigo), such as the eight graph subvarieties  $B_i$  of [Example 17.24](#); the grey points are natural objects off any such conic, which no relation of this size can use.

with the sign of  $\varphi$  for which  $B_{(p,q)}$  has conormal forms  $p\xi_j - qx_{n+j}$  and  $p\xi_{n+j} + qx_j$ ; with the opposite sign the right side is  $-14W_2$ . Among the 12870 sets of eight subvarieties with  $|p|, |q| \leq 3$  this one has the least  $\sum_i m_i^2$ , namely 1028, and the least  $\sum_i |m_i|$ , namely 68. In general the coefficients of the relation of [Theorem 17.23](#) for eight subvarieties are proportional to  $((q_i^2 + dp_i^2) \prod_{j \neq i} (p_i q_j - p_j q_i))^{-1}$ . The sum  $E = \bigoplus_i \mathcal{O}_{B_i}^{\oplus |m_i|}[k_i] \oplus \mathcal{O}(\eta) \oplus \mathcal{O}(2\eta) \oplus \mathcal{O}(3\eta)$ , with  $k_i = 0$  for  $m_i > 0$  and 1 otherwise, has  $\text{ch}(E) = 14W_2 + \sum_k c_k \eta^k$  with  $k! c_k = 1 + 2^k + 3^k$ , so  $\rho = 3$  and  $r = 104$ , while  $\dim \text{Ext}^2(E, E) = 28868$ ; by [Corollary 17.17](#) it is far from the criterion, and every direct sum of natural objects with  $N \neq 0$  has  $\dim \text{Ext}^2 \geq 8 \cdot 28 = 224$ .

*Remark 17.25* (The flatness form for direct sums). [Lemma 17.16](#) does not apply to the flatness form of [Proposition 17.15](#), but a direct sum fails that form too when its cross-extensions cannot interact. Let  $E = \bigoplus_i (\mathcal{O}_{B_i} \otimes L_i)^{\oplus m_i}[k_i]$  with the  $B_i$  pairwise distinct abelian subvarieties of dimension  $n$  meeting pairwise in finite sets, such as the  $B_{(p,q)}$ . The extensions between different summands are concentrated in degree  $n$ , so an arrow of  $\text{Ext}^1$  between summands raises the shift by  $n - 1$ , and a chain of  $a \geq 1$  arrows lands in  $\text{Ext}^{2+a(n-1)} \neq \text{Ext}^2$ . Every term of the Maurer–Cartan equation that involves a cross-extension therefore lands in a zero group, the base  $K_E$  is the product of the varieties of commuting  $2n$ -tuples of  $m_i \times m_i$  matrices with the space of cross-extensions, and a count of dimensions, using the simultaneously diagonalisable component, gives  $\dim K_E > e - k$ ; so (ii) and (iii) of [Proposition 17.15](#) fail. We give this in outline only; item (LV) checks the counts on the sum of [Example 17.24](#). Indecomposable objects, among them iterated

extensions of at least eight natural blocks, and sums whose cross-extensions compose to nonzero classes are not covered.

## 18. SECANT OBJECTS ON THE SPLIT MEMBER

**18.1. What the object must be, and what it cannot be.** [Theorem 16.28](#) asks for one perfect complex at one base point. The cheapest candidate is a sum of line bundles on a split member, because there the Weil classes are already polynomials in  $\beta$ ,  $\hat{\beta}$  and  $\ell$  by [Theorem 15.2](#), so the Chern character one wants is manifestly in reach. This subsection settles that candidate. It does not exist, and the obstruction is not semiregularity: it is the Chern character condition, and it fails because the norm form of  $K$  is positive definite.<sup>51</sup>

**Lemma 18.1** (The semiregularity map of a split object). *Let  $Y$  be a smooth projective complex variety and let*

$$E = \bigoplus_{i=1}^s L_i[p_i]^{\oplus m_i}, \quad \lambda_i = c_1(L_i),$$

with the  $L_i$  line bundles, the  $\lambda_i$  pairwise distinct and  $p_i \in \mathbb{Z}$ . Then for  $\zeta = (\zeta_{ij}) \in \text{Ext}^2(E, E)$ ,

$$\sigma_E(\zeta)_q = \frac{1}{q!} \sum_{i=1}^s (-1)^{p_i} \text{tr}(\zeta_{ii}) \lambda_i^q, \quad (39)$$

so  $\sigma_E$  annihilates  $\text{Ext}^2(L_i[p_i], L_j[p_j])$  for  $i \neq j$  and the traceless part of each diagonal block. Consequently  $\sigma_E$  is injective if and only if all three of the following hold:

- (i)  $m_i = 1$  for every  $i$ ;
- (ii)  $H^{2+p_i-p_j}(Y, L_j L_i^{-1}) = 0$  for all  $i \neq j$ ;
- (iii) the map

$$\Phi: H^2(\mathcal{O}_Y)^{\oplus s} \longrightarrow H^*(Y, \mathbb{C}), \quad (\zeta_i) \longmapsto \sum_i e^{\lambda_i} \zeta_i,$$

is injective.

*Proof.* The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of a line bundle is its first Chern class, so  $\text{At}(E)$  is diagonal with entries  $\lambda_i$  and  $\text{At}(E)^q$  is diagonal with entries  $\lambda_i^q$ . The matrix product  $\zeta \circ \text{At}(E)^q$  therefore has  $(i, j)$  entry  $\zeta_{ij} \lambda_i^q$ , and the trace keeps only the diagonal; a shift by  $p$  multiplies the trace by  $(-1)^p$ . That is (39), and it exhibits the two spaces the map kills.

For the equivalence, the off-diagonal part of  $\text{Ext}^2(E, E)$  is  $\bigoplus_{i \neq j} \text{Ext}^2(L_i[p_i], L_j[p_j]) = \bigoplus_{i \neq j} H^{2+p_i-p_j}(L_j L_i^{-1})$ , which must vanish, giving (ii); the diagonal block at  $i$  is  $H^2(\mathcal{O}_Y) \otimes \text{End}(\mathbb{C}^{m_i})$  and only its trace survives, so  $m_i = 1$ , giving (i). What is left is  $\bigoplus_i H^2(\mathcal{O}_Y)$ , and by (39) an element  $(\zeta_i)$  of it dies exactly when  $\sum_i \lambda_i^q \zeta_i = 0$  for every  $q \geq 0$ , which is (iii) after absorbing the signs  $(-1)^{p_i}$  into the  $\zeta_i$  and the factorials into the grading. ■

<sup>51</sup>The norm form has now closed three different doors in this paper, by three different mechanisms, and they should be kept apart. [Theorem 1.3](#) uses the norm character of  $K^\times$  and an equivariance hypothesis. [Proposition 15.18](#) uses the fact that the part  $R$  of  $H^2$  on which  $K^\times$  acts through  $\tau^2$  and  $\bar{\tau}^2$  has no Hodge classes at a general point, which is a statement about the Hodge group. [Theorem 18.5](#) below uses neither: it uses that  $N_{K/\mathbb{Q}}$  is a positive definite quadratic form on  $K$  and that the multiplicities of a filtration are positive integers. The three are independent, and only the third is available at a special point, which is exactly where the object of [Proposition 16.26](#) has to live.

Condition (ii) is cheap, because the shifts  $p_i$  are at our disposal: only their parities are fixed by the Chern character, and a line bundle whose first Chern class is nondegenerate of index  $k$  has cohomology in degree  $k$  alone, so one may choose the  $p_i$  to avoid every index that occurs. Condition (iii) is a rank. Condition (i) is free. What is not free is the Chern character itself, and that is what the rest of this subsection is about.

*Notation 18.2* (Coordinates on the divisor classes). Let  $A_0 = X \times \widehat{X}$  be a split member,  $n \geq 2$ , and keep  $\gamma = d\beta - \widehat{\beta}$  from [Notation 15.1](#). Write

$$\theta^\pm = \gamma \mp \sqrt{-d}\ell \in \Lambda^2 V_\pm, \quad R_0 = \mathbb{Q}\gamma \oplus \mathbb{Q}\ell \subset R.$$

Every  $\theta \in R_0$  is  $\theta = u\theta^+ + \bar{u}\theta^-$  for a unique  $u \in K$ , and in coordinates  $u = g + h\sqrt{-d}$  corresponds to  $\theta = 2g\gamma + 2dh\ell$ . By [Proposition 15.19\(iv\)](#) the group  $\text{NS}(A_0)_\mathbb{Q}$  at a split member with  $X$  general is  $\mathbb{Q}\eta \oplus R_0$ , so every divisor class is  $e\eta + \theta$  with  $e \in \mathbb{Q}$  and  $\theta \in R_0$ , and we write  $u(\lambda) \in K$  for the coordinate of its  $R_0$ -part.

**Lemma 18.3** (A basis in degree four). *For  $n \geq 2$  the six products  $\beta^2, \widehat{\beta}^2, \ell^2, \beta\widehat{\beta}, \beta\ell, \widehat{\beta}\ell$  are linearly independent in  $H^4(A_0, \mathbb{Q})$ . Equivalently the six classes  $\eta^2, \eta\theta^+, \eta\theta^-, (\theta^+)^2, (\theta^-)^2, \theta^+\theta^-$  are a basis of the degree four part of the subring generated by the divisor classes.*

*Proof.* Grade  $H^*(A_0, \mathbb{Q}) = \Lambda^* U \otimes \Lambda^* U^\vee$  by bidegree. Then  $\beta$  has bidegree  $(2, 0)$ ,  $\widehat{\beta}$  has  $(0, 2)$  and  $\ell$  has  $(1, 1)$ , so the six products have bidegrees  $(4, 0), (0, 4), (2, 2), (2, 2), (3, 1), (1, 3)$ . All are distinct except for the pair  $\beta\widehat{\beta}$  and  $\ell^2$ , and those two are independent in  $\Lambda^2 U \otimes \Lambda^2 U^\vee$  because  $\beta\widehat{\beta} = \beta \otimes \widehat{\beta}$  is decomposable there while  $\ell^2 = -2 \sum_{i < j} (x_i x_j) \otimes (\xi_i \xi_j)$  has all  $\binom{2n}{2}$  terms and  $\binom{2n}{2} > n$  for  $n \geq 2$ . The second description follows because the change of basis between the two sets of six classes is invertible over  $K$ :  $\eta = d\beta + \widehat{\beta}$  and  $\gamma = d\beta - \widehat{\beta}$  span the same plane as  $\beta$  and  $\widehat{\beta}$  since  $\det \begin{pmatrix} d & 1 \\ d & -1 \end{pmatrix} = -2d \neq 0$ , and  $\theta^\pm$  span the same plane as  $\gamma$  and  $\ell$  over  $\mathbb{C}$ . ■

The next lemma says the same thing at every abelian variety of Weil type, not only at a split member, and it says it without coordinates. It is the fact the positivity argument really turns on.

**Lemma 18.4** ( $\theta^+\theta^-$  is never proportional to  $\eta^2$ ). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$ , and let  $0 \neq \theta^\pm \in \Lambda^2 V_\pm$ . Then  $\theta^+\theta^-$  is not a scalar multiple of  $\eta^2$  in  $H^4(A, \mathbb{C})$ .*

*Proof.* Both classes lie in the summand  $\Lambda^2 V_+ \otimes \Lambda^2 V_-$  of  $H^4(A, \mathbb{C})$ , the isotypic component for the character  $N^2$ , so the question is one of tensor rank there. The polarisation class  $\eta$  lies in  $V_+ \otimes V_-$  and is the hermitian form of [Proposition 3.7](#), hence nondegenerate: in dual bases  $\eta = \sum_{a=1}^{2n} v_a \wedge w_a$  with  $v_a$  a basis of  $V_+$  and  $w_a$  a basis of  $V_-$ . Squaring,

$$\eta^2 = \sum_{a,b} (v_a \wedge w_a)(v_b \wedge w_b) = -2 \sum_{a < b} (v_a v_b) \otimes (w_a w_b),$$

which is diagonal in the induced bases of  $\Lambda^2 V_+$  and  $\Lambda^2 V_-$  and therefore has tensor rank exactly  $\binom{2n}{2}$ . The class  $\theta^+\theta^-$  is, up to sign,  $\theta^+ \otimes \theta^-$ , of tensor rank one. For  $n \geq 2$  one has  $\binom{2n}{2} \geq 6 > 1$ , so the two cannot be proportional. ■

**Theorem 18.5** (The positivity obstruction). *Let  $(A, \iota, \eta)$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$ , and suppose that the only divisor classes on which  $K^\times$  acts through the norm character are the multiples of  $\eta$ , that is*

$$\text{NS}(A)_\mathbb{Q} \cap (V_+ \otimes V_-) = \mathbb{Q}\eta. \quad (40)$$

Let  $E$  be a perfect complex on  $A$  whose Chern character is effective in the exponential basis, meaning

$$\mathrm{ch}(E) = \sum_{i=1}^s m_i e^{\lambda_i}, \quad m_i \in \mathbb{Q}_{>0}, \quad \lambda_i \in \mathrm{NS}(A)_{\mathbb{Q}}, \quad (41)$$

and write  $\lambda_i = e_i \eta + \theta_i$  with  $\theta_i$  in the part  $R$  of  $H^2(A, \mathbb{Q})$  carrying the characters  $\tau^2$  and  $\bar{\tau}^2$ , and  $u_i \in K$  for the  $\theta^+$ -coordinate of  $\theta_i$ . If  $\mathrm{ch}_2(E)$  is a flat Hodge class, then

$$\sum_{i=1}^s m_i N_{K/\mathbb{Q}}(u_i) = 0, \quad (42)$$

and since  $N_{K/\mathbb{Q}}$  is positive definite and every  $m_i$  is positive,  $u_i = 0$  for every  $i$ . Hence every  $\lambda_i$  is a rational multiple of  $\eta$ , and  $\mathrm{ch}_k(E) \in \mathbb{Q} \eta^k$  for every  $k$ . In particular  $\mathrm{ch}_n(E)$  has no component on the Weil line.

*Proof.* From (41),  $\mathrm{ch}_2(E) = \frac{1}{2} \sum_i m_i \lambda_i^2$ . Decompose  $H^4(A, \mathbb{C})$  into  $K^\times$ -isotypic pieces and keep the one for  $N^2$ , namely  $\wedge^2 V_+ \otimes \wedge^2 V_-$ . Writing  $\theta_i = u_i \theta^+ + \bar{u}_i \theta^-$  for a fixed choice of generators  $\theta^\pm$  of  $\wedge^2 V_\pm$ , one has

$$\lambda_i^2 = e_i^2 \eta^2 + 2e_i \eta \theta_i + u_i^2 (\theta^+)^2 + \bar{u}_i^2 (\theta^-)^2 + 2N_{K/\mathbb{Q}}(u_i) \theta^+ \theta^-,$$

and of these five terms only the first and the last are in the  $N^2$  piece:  $\eta \theta_i$  carries  $N\tau^2$  and  $N\bar{\tau}^2$ , and  $(\theta^\pm)^2$  carry  $\tau^4$  and  $\bar{\tau}^4$ , all four distinct from  $N^2$ . Hypothesis (40) is what makes the decomposition  $\lambda_i = e_i \eta + \theta_i$  available with  $e_i$  a scalar. So the  $N^2$ -part of  $\mathrm{ch}_2(E)$  is

$$\frac{1}{2} \left( \sum_i m_i e_i^2 \right) \eta^2 + \left( \sum_i m_i N_{K/\mathbb{Q}}(u_i) \right) \theta^+ \theta^-.$$

Now compare with the flat Hodge classes of degree four. For  $n \geq 3$  they are  $\mathbb{Q} \eta^2$  by Theorem 4.14. For  $n = 2$  they are  $\mathbb{Q} \eta^2 \oplus W(A)$ , and  $W(A)$  spans the summands for  $\tau^4$  and  $\bar{\tau}^4$ , which are not  $N^2$ . In both cases the  $N^2$ -part of a flat Hodge class of degree four is a multiple of  $\eta^2$ . By Lemma 18.4 the class  $\theta^+ \theta^-$  is not such a multiple, so its coefficient must vanish, which is (42).

Finally  $N_{K/\mathbb{Q}}(g + h\sqrt{-d}) = g^2 + dh^2$  with  $d > 0$ , so  $N_{K/\mathbb{Q}}(u) \geq 0$  with equality only at  $u = 0$ , and a sum of nonnegative terms with positive weights vanishes only when every term does. Then  $\mathrm{ch}_k(E) = \frac{1}{k!} (\sum_i m_i e_i^k) \eta^k$ , and  $\eta^k$  is not on the Weil line by Theorem 4.14. ■

**Corollary 18.6** (The cheapest objects do not exist). *Let  $A$  satisfy (40), which holds at every split member of Theorem 15.2 and at every quaternionic base point of Theorem 15.25.<sup>52</sup> Then Theorem 16.28 cannot be applied at  $A$  with any of the following:*

- (a) *a coherent sheaf admitting a filtration by line bundles, in particular any direct sum of line bundles;*
- (b) *a simple semi-homogeneous vector bundle in the sense of Mukai [Muk78], since such a bundle of rank  $r$  has  $\mathrm{ch}(E) = r e^{c_1(E)/r}$ ;*
- (c) *any iterated extension of the objects in (a) and (b).*

*In each case the failure is not of the semiregularity hypothesis but of the Chern character hypothesis of Proposition 16.26: no such  $E$  has a Chern character which is a flat section of Hodge classes with a nonzero Weil component.*

<sup>52</sup>At a split member  $\mathrm{NS}(A)_{\mathbb{Q}}$  has rank three with basis  $\beta, \hat{\beta}, \ell$  by Proposition 15.19(iv), and of these  $\eta$  spans the norm-character part while  $\gamma$  and  $\ell$  span  $R$ . At a quaternionic point  $\mathrm{NS}(A)_{\mathbb{Q}}$  is the space of Rosati-symmetric elements of  $B = (-d, b)_{\mathbb{Q}}$ , which is spanned by  $1, j$  and  $ij$ ; the first gives  $\eta$  and the other two are semilinear over  $K$ , hence lie in  $R$ . In both cases the norm-character part of  $\mathrm{NS}(A)_{\mathbb{Q}}$  is exactly  $\mathbb{Q} \eta$ .

*Proof.* Each of (a), (b) and (c) has a Chern character of the shape (41) with positive multiplicities, by additivity of  $\text{ch}$  along a filtration and by Mukai's computation of the Chern character of a simple semi-homogeneous bundle. Apply Theorem 18.5. ■

*Remark 18.7* (Where the theorem has force, and why it does not contradict Markman). At a very general member of the family  $\text{NS}(A)_{\mathbb{Q}} = \mathbb{Q}\eta$ , so Theorem 18.5 is true for the empty reason that there are no  $\theta_i$  to begin with. Its content is at the special members, which is exactly where Theorem 16.28 needs its object to live and where the base points of Theorem 15.31 are. That is the opposite of the usual situation in this paper, where the special members are the easy ones.

The theorem also does not touch the objects of Markman [Mar25a, Mar25b]. Those are sheaves whose class in  $K_0$  is not a positive combination of exponentials of divisor classes, so (41) fails for them at the first step; and they live at very general points, where as just noted the statement is vacuous anyway. What the theorem does is remove the class of objects one would reach for first, and it removes it for a reason which is visible before any cohomology is computed.

*Proof.* Immediate from Theorem 18.5, since the conclusion there contradicts the requirement that  $\text{ch}_n(E)$  have a nonzero component on the Weil line. ■

*Remark 18.8* (What is left of the candidate, and what it must look like). Theorem 18.5 uses positivity of the multiplicities and nothing else, so it says exactly where to go next: the object must be a complex whose class in  $K_0$  is not effective, that is one in which the line bundles appear with multiplicities of both signs. Such objects exist. Writing  $E = F \oplus G[1]$  with  $F$  and  $G$  sums of line bundles, the class in  $K_0$  is  $[F] - [G]$  and (42) becomes  $\sum_{L_i \subset F} N(u_i) = \sum_{L_j \subset G} N(u_j)$ , which is satisfiable; and Lemma 18.1 still applies verbatim, with the shifts absorbing condition (ii).

Two further facts, both computational, say how little room is left at  $n = 2$ , and how much more there is beyond it. First, the map  $\Phi$  of Lemma 18.1(iii) is injective at  $n = 2$  for  $s \leq 3$  and, for a general choice of  $\lambda_1, \dots, \lambda_4$ , has rank 22 rather than 24; so at  $n = 2$  no more than three line bundles may occur at all. This is special to  $n = 2$ : at  $n = 3$  the same map is injective for a general choice of up to ten classes, and the dimension count permits up to 33. Second, an exhaustive search over the classes  $a\beta + b\hat{\beta} + c\ell$  with  $|a|, |b|, |c| \leq 3$ , for  $d = 1$  and  $d = 2$ , finds no signed tuple of at most three line bundles whose Chern character is a flat section of Hodge classes with a nonzero Weil component; the tuples of at most four that do satisfy the Chern character condition with a nonzero Weil component, 888 of them at  $d = 1$  and 30 at  $d = 2$ , all have exactly four terms, and all of them fail Lemma 18.1(iii): the kernel has dimension 2, 3, 4 or 6 at  $d = 1$  and 3 or 6 at  $d = 2$ , never 0. Item (XIII) of Appendix D carries both computations.

So the object of Theorem 16.28, if it exists at a split member, is not a sum of line bundles in any arrangement of signs and shifts: it must have  $\text{Ext}^2$  that the trace does not see, that is, it must be indecomposable in the derived category. That is a sharper description of the remaining construction than Remark 17.7 gives, and it is as far as we have been able to push it.

**18.2. The object, written down, and what it costs.** Proposition 16.26 produced the object of Theorem 16.28 by an existence argument: the Chern character is an isomorphism onto the Chow ring with rational coefficients, so a  $K$ -theory class with the right character exists. That is enough for the theorem and tells one nothing about the object. This subsection writes it down. The answer turns the question into a Diophantine one, and the Diophantine problem has a classical name.

**Setup 18.9** (Objects supported on  $R$ ). Let  $A$  be of  $(K, -1, n)$ -Weil type,  $n \geq 2$ , satisfying (40), and keep  $\theta^\pm$  from Notation 18.2. For  $u \in K$  with  $\theta(u) = u\theta^+ + \bar{u}\theta^- \in \text{NS}(A)_\mathbb{Q}$ , and for multiplicities  $m_i \in \mathbb{Z}$ , consider

$$v = \sum_{i=1}^s m_i e^{\theta(u_i)}, \quad T_{q,r} = \sum_{i=1}^s m_i u_i^q \bar{u}_i^r \quad (q, r \geq 0).$$

**Proposition 18.10** (The Chern character in terms of the  $u_i$ ). In Setup 18.9,

$$v_k = \sum_{\substack{q+r=k \\ 0 \leq q, r \leq n}} \frac{T_{q,r}}{q! r!} (\theta^+)^q (\theta^-)^r, \quad (43)$$

because  $(\theta^\pm)^q = 0$  for  $q > n$ . Consequently  $v$  is a flat section of Hodge classes with a nonzero Weil component if

$$T_{q,r} = 0 \text{ for all } 0 \leq q, r \leq n \text{ with } (q, r) \notin \{(0, 0), (n, 0), (0, n)\}, \quad T_{n,0} \neq 0, \quad (44)$$

and then  $v_0 = T_{0,0} = \sum_i m_i$ ,  $v_k = 0$  for  $0 < k < 2n$  with  $k \neq n$ , and

$$v_n = \frac{1}{n!} (T_{n,0} (\theta^+)^n + \bar{T}_{n,0} (\theta^-)^n),$$

a nonzero rational multiple of a generator of the Weil line, by Theorem 15.2.

*Proof.* Expand  $\theta(u_i)^k$  by the binomial theorem, divide by  $k!$  and collect; the vanishing of  $(\theta^\pm)^q$  for  $q > n$  is because  $\theta^\pm \in \wedge^2 V_\pm$  and  $\dim_\mathbb{C} V_\pm = 2n$ . Under (44) the only surviving terms of (43) are the ones listed, and  $(\theta^+)^n$  spans  $\wedge^{2n} V_+$ , which is the Weil line over  $\mathbb{C}$ . ■

So the search for the object of Theorem 16.28 is the search for a finitely supported integer-valued function on  $\mathcal{O}_K$  that annihilates every monomial  $u^q \bar{u}^r$  with  $q, r \leq n$  except three of them. Two things follow at once, and they pull in opposite directions.

**Lemma 18.11** (Equal norms force the Prouhet–Tarry–Escott condition). Suppose all the  $u_i$  in Setup 18.9 have the same norm  $N = N_{K/\mathbb{Q}}(u_i) > 0$ . Then (44) holds if and only if

$$\sum_i m_i u_i^k = 0 \text{ for } k = 0, 1, \dots, n-1, \quad \sum_i m_i u_i^n \neq 0. \quad (45)$$

With  $m_i = \pm 1$  this says that the  $u_i$  split into two multisets on the circle  $N = N$  whose power sums agree through degree  $n-1$  and differ at degree  $n$ : a Prouhet–Tarry–Escott pair of degree  $n-1$ , constrained to a norm circle of  $\mathcal{O}_K$ .

*Proof.* For  $q \geq r$  one has  $u^q \bar{u}^r = N^r u^{q-r}$ , so  $T_{q,r} = N^r \sum_i m_i u_i^{q-r}$ , and the pairs  $(q, r)$  with  $q, r \leq n$  realise every difference  $k = q - r$  from 0 to  $n$ . The excluded pairs  $(0, 0)$  and  $(n, 0)$  are the ones with  $r = 0$  and  $k = 0$  or  $k = n$ ; but  $k = 0$  also occurs at  $(1, 1), \dots, (n, n)$ , which are not excluded, so  $\sum_i m_i = 0$  is required after all. The remaining differences  $k = 1, \dots, n-1$  each occur at some non-excluded pair, and  $k = n$  occurs only at  $(n, 0)$ . ■

**Theorem 18.12** (Equal norms cannot be semiregular). Let  $E = \bigoplus_i L_i[p_i]$  realise (45) with all norms equal. Then  $\text{ch}_0(E) = 0$ , and the semiregularity map  $\sigma_E$  has a kernel of dimension at least  $n^2$ .

*Proof.* By Lemma 18.11 the case  $k = 0$  of (45) gives  $\sum_i m_i = 0$ , which is  $\text{ch}_0(E)$ . For the kernel, take  $\zeta_i = m_i \zeta$  for a single  $\zeta \in H^2(\mathcal{O}_A)$ ; then by Lemma 18.1(iii)

$$\Phi((m_i \zeta)_i) = \left( \sum_i m_i e^{\lambda_i} \right) \zeta = \text{ch}(E) \zeta,$$

which by [Proposition 18.10](#) is a multiple of  $\omega \zeta$  with  $\omega$  a generator of the Weil line. Now  $W(A)_{\mathbb{C}}$  is spanned by  $\bigwedge^{2n} V_+$  and  $\bigwedge^{2n} V_-$ , and the Weil type condition says  $\dim V_{\pm}^{0,1} = n$ ; writing  $H^2(\mathcal{O}_A) = \bigwedge^2(V_+^{0,1} \oplus V_-^{0,1})$ , the product of a generator of  $\bigwedge^{2n} V_+$  with  $\zeta$  kills the components of  $\zeta$  in  $\bigwedge^2 V_+^{0,1}$  and in  $V_+^{0,1} \otimes V_-^{0,1}$ , and likewise with the roles exchanged. So  $\omega \zeta = 0$  for every  $\zeta$  in  $V_+^{0,1} \otimes V_-^{0,1}$ , a space of dimension  $n^2$ , and all of those lie in the kernel of  $\Phi$ .  $\blacksquare$

**Corollary 18.13** (At least  $n+1$  norms). *If  $E$  as in [Setup 18.9](#) satisfies (44) and  $\text{ch}_0(E) \neq 0$ , then the  $u_i$  take at least  $n+1$  distinct norms.*

*Proof.* Let  $N_1, \dots, N_\nu$  be the distinct norms and  $M_j$  the sum of the  $m_i$  over the level  $N(u_i) = N_j$ . The conditions  $T_{r,r} = 0$  for  $r = 1, \dots, n$  read  $\sum_j M_j N_j^r = 0$  for  $r = 1, \dots, n$ . The matrix  $(N_j^r)_{1 \leq r \leq n, 1 \leq j \leq \nu}$  has rank  $\min(n, \nu)$  because the  $N_j$  are distinct and nonzero, so  $\nu \leq n$  forces every  $M_j = 0$  and hence  $\text{ch}_0(E) = \sum_j M_j = 0$ .  $\blacksquare$

*The smallest solution, and what it does.* At  $n = 3$  and  $K = \mathbb{Q}(\sqrt{-3})$  the Prouhet–Tarry–Escott condition of [Lemma 18.11](#) has a solution of striking simplicity: the norm circle  $N = 1$  carries the six roots of unity of  $K$ , and the two cosets of the cube roots of  $-1$  and of  $+1$  are a Prouhet–Tarry–Escott pair of degree two.<sup>53</sup>

**Proposition 18.14** (The object at  $n = 3$ , written out). *Let  $K = \mathbb{Q}(\sqrt{-3})$ ,  $n = 3$ , and let  $A_0 = X \times \widehat{X}$  be a split member. In the basis  $\beta, \widehat{\beta}, \ell$  of  $\text{NS}(A_0)_{\mathbb{Q}}$  put*

$$\lambda_1 = (3, -1, 3), \quad \lambda_2 = (-6, 2, 0), \quad \lambda_3 = (3, -1, -3),$$

*and  $\lambda_{3+i} = -\lambda_i$  for  $i = 1, 2, 3$ , with  $m_i = +1$  for  $i \leq 3$  and  $m_i = -1$  for  $i \geq 4$ . Then the six classes are distinct and integral, the perfect complex*

$$E = (L_1 \oplus L_2 \oplus L_3) \oplus (L_4 \oplus L_5 \oplus L_6)[3]$$

*has*

$$\text{ch}_k(E) = 0 \quad (k \neq 3), \quad \text{ch}_3(E) = 12\omega_1,$$

*and every difference  $\lambda_j - \lambda_i$  with  $i \neq j$  is nondegenerate of index exactly 3, so that conditions (i) and (ii) of [Lemma 18.1](#) hold for this shift. Condition (iii) fails: the map  $\Phi$  has rank 75 instead of 90, with kernel of dimension 15 =  $\dim H^2(\mathcal{O}_{A_0})$ .*

*Proof.* Everything asserted is a finite exact computation, carried out in item (XIII) of [Appendix D](#). The classes are the images under  $\theta(u) = 2g\gamma + 2dh\ell$ ,  $u = g + h\sqrt{-3}$ , of the six roots of unity of  $K$  scaled by 2; the Chern character is [Proposition 18.10](#) with the Prouhet–Tarry–Escott data above; the index computation and the rank of  $\Phi$  are the two remaining items. For the shift, condition (ii) asks that  $2 + p_i - p_j \neq 3$  whenever  $i \neq j$ , that is  $p_i - p_j \neq 1$ ; within each of the two blocks the difference is 0, and between them it is  $\pm 3$ .  $\blacksquare$

**Remark 18.15** (What this settles and what it does not). Three things are settled. The object of [Proposition 16.26](#) is not an abstraction: at  $n = 3$  over  $\mathbb{Q}(\sqrt{-3})$  it is six line bundles and a shift, and its Chern character is exactly 12 times a Weil class with every other graded piece zero. The construction is completely general, and [Lemma 18.11](#) identifies the

<sup>53</sup>Explicitly, with  $\zeta$  a primitive sixth root of unity, the cube roots of  $-1$  are  $\zeta, \zeta^3, \zeta^5$  and those of  $+1$  are  $1, \zeta^2, \zeta^4$ ; each triple sums to zero and has zero sum of squares, while the sums of cubes are  $-3$  and  $+3$ . A search over the norm circles of  $\mathcal{O}_K$  with  $N \leq 200$  and  $|g|, |h| \leq 25$ , for  $K = \mathbb{Q}(\sqrt{-d})$  with  $d = 1, 2, 3, 7$ , finds this solution and its conjugate at  $n = 3$ , one further pair at  $d = 1$  and  $n = 4$ , and nothing else; item (XIII) of [Appendix D](#) records the search.

constraint as a Prouhet–Tarry–Escott problem on a norm circle, which is a question of a kind that can be attacked. And [Corollary 18.13](#) says what any semiregular solution must look like: the  $u_i$  must take at least  $n + 1$  distinct norms, because otherwise the rank is zero and [Theorem 18.12](#) produces a kernel of dimension  $n^2$ .

The remaining question was the existence of a solution with  $n + 1$  norms. The level sums are then forced up to scale, and they compete with the supply of elements of  $\mathcal{O}_K$  at each level; [Theorem 18.20](#) carries out that competition in a range, and [Corollary 18.22](#) then shows that no solution in any range could have served, because a line bundle summand off the polarisation line is obstructed along the family whatever the norms.

*The obstruction classes of the explicit object.* [Corollary 9.18](#) and [Remark 9.21](#) say that the kernel of  $\sigma_E$  contains a copy of the tangent space of the Weil family, and that this is a statement about where the semiregularity map is blind, not a statement that the object is obstructed. For the explicit object the obstruction classes can be written down, and they are not zero. The computation is elementary and we record it in general.

Recall the conventions of [Section 9.2](#): a first-order deformation of  $A$  is  $v \in \text{Hom}(H^{1,0}, H^{0,1}) = H^1(A, T_A)$ , extended by zero on  $H^{0,1}$  and acting on  $H^*(A, \mathbb{C})$  as a derivation  $v \lrcorner$ ; a class  $\xi$  of type  $(p, p)$  stays of type  $(p, p)$  to first order along  $v$  exactly when  $v \lrcorner \xi = 0$ . For a line bundle  $L$  with  $\lambda = c_1(L)$  the Atiyah class is  $\lambda \in H^1(\Omega_A^1)$ , its contraction with  $v$  is  $v \lrcorner \lambda \in H^2(\mathcal{O}_A) = H^{0,2}(A)$ , and this is the obstruction to deforming  $L$  along  $v$ ; a shift does not change it. For  $E = \bigoplus_i L_i[p_i]$  the Atiyah class is diagonal with entries  $\lambda_i$ , as in the proof of [Lemma 18.1](#), so the obstruction to deforming  $E$  along  $v$  is the diagonal element

$$\text{ob}_E(v) = (v \lrcorner \lambda_i)_i \in \bigoplus_i H^{0,2}(A) \subset \text{Ext}^2(E, E). \quad (46)$$

**Proposition 18.16** (The obstruction map of a split object). *Let  $A_0 = X \times \widehat{X}$  be a split member with  $\beta$  principal and  $X$  general, so that  $\text{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \widehat{\beta}, \ell \rangle$  by [Proposition 15.19\(iv\)](#). Let  $T = T_{[A_0]} \mathcal{D}_{n,n}$  be the space of polarised first-order deformations commuting with  $K$ , of dimension  $n^2$  by [Proposition 9.14](#), and put*

$$T_{\text{spl}} = \{ v \in \text{Hom}(H^{1,0}, H^{0,1}) : v \lrcorner \beta = v \lrcorner \widehat{\beta} = v \lrcorner \ell = 0 \}.$$

*Let  $E = \bigoplus_{i=1}^s L_i[p_i]$  with  $\lambda_i = c_1(L_i)$  and let  $\text{ob}_E: T \rightarrow \bigoplus_i H^{0,2}(A_0)$  be the map (46) restricted to  $T$ .*

- (i)  $T_{\text{spl}}$  is the tangent space to the split locus: it has dimension  $n(n + 1)/2$ , it is contained in  $T$ , and it consists of the deformations of  $X \times \widehat{X}$  induced by the polarised deformations of  $(X, \beta)$ .
- (ii) If the classes  $\lambda_i$  together with  $\eta$  span  $\text{NS}(A_0)_{\mathbb{Q}}$ , then  $\ker \text{ob}_E = T_{\text{spl}}$  and  $\text{rank } \text{ob}_E = n(n - 1)/2$ , the codimension of the split locus in  $\mathcal{D}_{n,n}$ .
- (iii) If  $\text{ch}_k(E) \in \mathbb{Q}\eta^k$  for  $k \neq n$  and  $\text{ch}_n(E) \in \mathbb{Q}\eta^n \oplus W(A_0)$ , then  $\sigma_E \circ \text{ob}_E = 0$  on  $T$ : for every  $v \in T$ ,

$$\sigma_E(\text{ob}_E(v)) = v \lrcorner \text{ch}(E) = 0. \quad (47)$$

- (iv) For the object of [Proposition 18.14](#), at  $n = 3$  and  $K = \mathbb{Q}(\sqrt{-3})$ :  $\ker \text{ob}_E = T_{\text{spl}}$  has dimension 6 and  $\text{ob}_E$  has rank 3; the image of  $\text{ob}_E$  lies in  $\ker \sigma_E$ ; the subspace  $\Delta = \{z \cdot \text{id}_E : z \in \text{Ann}_{H^{0,2}}(\omega_1)\}$  of [Corollary 9.18](#) has dimension 9 and meets the image of  $\text{ob}_E$  only in zero, so that  $\Delta \oplus \text{im } \text{ob}_E$  is a twelve-dimensional subspace of the fifteen-dimensional kernel of  $\sigma_E$ ; and  $\text{ob}_E(v)$  has components  $v \lrcorner \lambda_i$  on  $L_i$  and  $-v \lrcorner \lambda_i$  on  $L_i^{-1}[3]$ , so it lies on the line of vectors of the shape  $(z, \dots, z)$  only when

it vanishes. In particular  $E$  has a nonzero first-order obstruction along every  $v \in T$  not tangent to the split locus, and no element  $z \cdot \text{id}_E$  of  $\text{Ext}^2(E, E)$  cancels it.

*Proof.* (i) Let  $P = H^{1,0}(X) \subset H^1(X, \mathbb{C}) = U_{\mathbb{C}}^{\vee}$  with basis  $p_1, \dots, p_n$ , so that  $p_1, \dots, p_n, \bar{p}_1, \dots, \bar{p}_n$  is a basis of  $U_{\mathbb{C}}^{\vee}$ , and let  $q_a, \bar{q}_a$  be the dual basis of  $U_{\mathbb{C}} = H^1(\hat{X}, \mathbb{C})$ ; the dual of a conjugate basis is the conjugate of the dual basis, so the notation is consistent. The Poincaré class is the identity element of  $U^{\vee} \otimes U$ ,

$$\ell = \sum_{a=1}^n (p_a \wedge q_a + \bar{p}_a \wedge \bar{q}_a),$$

and it is of type  $(1, 1)$ , which forces  $H^{1,0}(\hat{X}) = \langle \bar{q}_a \rangle$  and  $H^{0,1}(\hat{X}) = \langle q_a \rangle$ : the Hodge structure of  $\hat{X}$  is the dual one. Since  $\beta$  is of type  $(1, 1)$  and nondegenerate,  $\beta = \sum_{a,b} h_{ab} p_a \wedge \bar{p}_b$  with  $h \in \text{GL}_n(\mathbb{C})$ ; and since  $\beta$  is principal with Gram matrix  $B$  in a symplectic basis of  $U$  and  $\hat{\beta}$  has Gram matrix  $B = -B^{-1}$  in the dual basis of  $U^{\vee}$ , the Gram matrix of  $\hat{\beta}$  in the basis  $p_a, \bar{p}_a$  is minus the inverse of the Gram matrix  $\begin{pmatrix} 0 & h \\ -h^{\top} & 0 \end{pmatrix}$  of  $\beta$  in the basis  $q_a, \bar{q}_a$ , which gives  $\hat{\beta} = -\sum_{a,b} (h^{-1})_{ab} \bar{q}_a \wedge q_b$ .

Now  $H^{1,0}(A_0) = \langle p_a \rangle \oplus \langle \bar{q}_a \rangle$  and  $H^{0,1}(A_0) = \langle \bar{p}_a \rangle \oplus \langle q_a \rangle$ , so a deformation  $v$  is four  $n \times n$  matrices,

$$v(p_a) = \sum_c (A_{ca} \bar{p}_c + C_{ca} q_c), \quad v(\bar{q}_a) = \sum_c (B_{ca} \bar{p}_c + D_{ca} q_c),$$

and  $v$  kills  $\bar{p}_a$  and  $q_a$ . Contracting,

$$v \lrcorner \ell = \sum_{a,c} (A_{ca} \bar{p}_c \wedge q_a + C_{ca} q_c \wedge q_a + B_{ca} \bar{p}_a \wedge \bar{p}_c + D_{ca} \bar{p}_a \wedge q_c),$$

and reading off the three kinds of monomials,  $v \lrcorner \ell = 0$  if and only if  $D = -A^{\top}$  and  $B, C$  are symmetric. Next

$$v \lrcorner \beta = \sum_{a,b,c} h_{ab} (A_{ca} \bar{p}_c \wedge \bar{p}_b + C_{ca} q_c \wedge \bar{p}_b),$$

so  $v \lrcorner \beta = 0$  if and only if  $Ch = 0$ , that is  $C = 0$ , and  $Ah$  is symmetric. Likewise  $v \lrcorner \hat{\beta} = 0$  if and only if  $B = 0$  and  $Dh^{-1}$  is symmetric. Given  $D = -A^{\top}$ , the last condition reads  $A^{\top} h^{-1} = h^{-\top} A$ , which after multiplication by  $h^{\top}$  on the left and by  $h$  on the right is  $h^{\top} A^{\top} = Ah$ , the symmetry of  $Ah$  again. Hence

$$T_{\text{spl}} = \{(A, B, C, D) = (A, 0, 0, -A^{\top}) : Ah \text{ symmetric}\},$$

of dimension  $n(n+1)/2$ . It is the tangent space to the split locus: the same computation on  $X$  alone shows that the polarised first-order deformations of  $(X, \beta)$  are the matrices  $A$  with  $Ah$  symmetric, the space  $T_{[X]} \mathcal{A}_n$  of dimension  $n(n+1)/2$ ; the deformation of  $\hat{X}$  induced by  $A$  is the dual one,  $-A^{\top}$ ; and the map  $A \mapsto (A, 0, 0, -A^{\top})$  is injective. Finally  $T_{\text{spl}} \subseteq T$ : for  $v \in T_{\text{spl}}$ ,  $v \lrcorner \eta = 0$  because  $\eta \in \langle \beta, \hat{\beta} \rangle$ , so  $v$  is polarised, and  $v \lrcorner \omega_1 = v \lrcorner \omega_2 = 0$  because  $\omega_1$  and  $\omega_2$  are polynomials in  $\beta, \hat{\beta}, \ell$  by [Theorem 15.2](#) and  $v \lrcorner$  is a derivation; by [Proposition 9.14](#) such a  $v$  commutes with  $K$ .

(ii) Every  $v \in T$  has  $v \lrcorner \eta = 0$ . If the  $\lambda_i$  and  $\eta$  span  $\text{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \hat{\beta}, \ell \rangle$ , then  $v \lrcorner \lambda_i = 0$  for all  $i$  is equivalent, for  $v \in T$ , to  $v \lrcorner \beta = v \lrcorner \hat{\beta} = v \lrcorner \ell = 0$ , so  $\ker \text{ob}_E = T \cap T_{\text{spl}} = T_{\text{spl}}$  by (i), and the rank is  $n^2 - n(n+1)/2 = n(n-1)/2$ , which is the codimension of [Proposition 15.19\(iii\)](#).

(iii) By (39),  $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_{ii} e^{\lambda_i}$  for a diagonal  $\zeta$ . Since  $v_\perp$  is a derivation and  $v_\perp \lambda_i$  has even degree,  $v_\perp \lambda_i^q = q \lambda_i^{q-1} (v_\perp \lambda_i)$  and so  $v_\perp e^{\lambda_i} = (v_\perp \lambda_i) e^{\lambda_i}$ . Therefore

$$\sigma_E(\text{ob}_E(v)) = \sum_i (-1)^{p_i} (v_\perp \lambda_i) e^{\lambda_i} = v_\perp \sum_i (-1)^{p_i} e^{\lambda_i} = v_\perp \text{ch}(E).$$

For  $v \in T$  one has  $v_\perp \eta^k = k \eta^{k-1} (v_\perp \eta) = 0$ , and  $v_\perp \omega = 0$  for every  $\omega \in W(A_0)$  by Proposition 9.14; under the hypothesis on  $\text{ch}(E)$  this gives (47).

(iv) The three classes  $\lambda_1 = \gamma + 3\ell$ ,  $\lambda_2 = -2\gamma$ ,  $\lambda_3 = \gamma - 3\ell$  span  $\langle \gamma, \ell \rangle$ , and  $\langle \eta, \gamma \rangle = \langle \beta, \hat{\beta} \rangle$  by the determinant in the proof of Lemma 18.3, so (ii) applies and gives the kernel and the rank; Proposition 18.14 gives the hypothesis of (iii), so the image lies in  $\ker \sigma_E$ . The subspace  $\Delta$  has dimension  $\dim \text{Ann}_{H^{0,2}}(\omega_1) = n^2 = 9$  by Lemma 9.6. The components of  $\text{ob}_E(v)$  on  $L_i$  and on  $L_i^{-1}[3]$  are  $v_\perp \lambda_i$  and  $v_\perp (-\lambda_i) = -v_\perp \lambda_i$ , so an element of the image of the shape  $(z, \dots, z)$  has  $z = -z$ , hence is zero; this is  $\Delta \cap \text{im ob}_E = 0$ , and with  $\dim \ker \sigma_E = 15$  from Proposition 18.14 the dimension count follows. Every number in (iv) is also computed directly in item (XXV) of Appendix D, in the model of item (XIII). ■

*Remark 18.17* (What the obstruction classes say). Proposition 18.16 sharpens Remark 9.21 in one direction and closes a door in another. The direction: for the explicit object the obstruction classes along the family are nonzero, and they are nonzero exactly in the  $n(n-1)/2$  directions normal to the split locus, which are the directions in which the Weil class has to be transported. The door: those classes lie in the kernel of  $\sigma_E$  by (47), and (47) holds for every object whose Chern character is a flat section of Hodge classes along the family, so the semiregularity map cannot detect the obstruction of any such object. The obstruction is therefore genuine, and it is invisible to the criterion. For the explicit object the situation is the worst possible: the twelve dimensions  $\Delta \oplus \text{im ob}_E$  of the kernel are accounted for by the two mechanisms of Corollary 9.18 and (46), the remaining three lie on the even side of Corollary 9.8 as well, and none of the fifteen is an artefact. The classes of the shape  $z \cdot \text{id}_E$  are the ones a change of the twisting class would contribute, one and the same  $z$  to every rank one summand; the last assertion of (iv) says that no such change removes the obstruction. So the explicit object does not move off the split locus to first order, as a complex and as a twisted complex, and the question of Question 15.13 for it is not one of verifying a hypothesis but of replacing the object.

The positive content is the recipe. By (ii), a split object whose classes span  $\text{NS}(A_0)_\mathbb{Q}$  with  $\eta$  has obstruction map of rank  $n(n-1)/2$  against the family, and the spanning hypothesis is more than is needed: in the model of item (XXV) a single class  $\theta \in \text{NS}(A_0)_\mathbb{Q}$  not proportional to  $\eta$  already has  $\{v \in T : v_\perp \theta = 0\} = T_{\text{spl}}$ , for each of the seven classes tested at  $(n, d) = (2, 1), (2, 3), (3, 3)$ . So every split object built from line bundles other than powers of  $\eta$  is obstructed in exactly the normal directions to the split locus, and Theorem 18.5 says that an object built from powers of  $\eta$  carries no Weil class. What survives is the statement of Remark 9.9: the object must have positive rank.

*The evaluation map of the explicit object.* Theorem 9.19 places the copy  $\Delta$  of the tangent space inside the image of the evaluation map. For a split object the whole of  $\text{ev}_E$  can be written down, and for the explicit object the entire kernel of  $\sigma_E$  turns out to lie in its image. We identify  $HH^2(A_0)$  with

$$HT^2(A_0) = H^{0,2}(A_0) \oplus \text{Hom}(H^{1,0}, H^{0,1}) \oplus \wedge^2 (H^{1,0})^\vee \quad (48)$$

by the Hochschild–Kostant–Rosenberg isomorphism, the three summands being  $H^2(\mathcal{O})$ ,  $H^1(T)$  and  $H^0(\wedge^2 T)$  for the trivial tangent bundle. An element  $x = (z, v, \pi)$  acts on

$H^*(A_0, \mathbb{C})$  by  $z \wedge$ , by the derivation  $v \lrcorner$  of [Section 9.2](#), and by the double contraction  $\pi \lrcorner$ , which removes two  $(1, 0)$  slots; we write  $x \lrcorner \xi$  for the sum of the three.<sup>54</sup>

**Proposition 18.18** (The evaluation map of a split object). *Let  $A_0$  be a split member with  $X$  general and let  $E = \bigoplus_{i=1}^s L_i[p_i]$  with  $\lambda_i = c_1(L_i)$  and all  $\text{Ext}^2(L_i[p_i], L_j[p_j])$ ,  $i \neq j$ , zero, so that  $\text{Ext}^2(E, E) = \bigoplus_i H^{0,2}(A_0)$ .*

(i) *Under (48),  $\text{ev}_E$  is diagonal:*

$$\text{ev}_E(z, v, \pi)_i = z + v \lrcorner \lambda_i + \frac{1}{2} \pi \lrcorner \lambda_i^2 \in H^{0,2}(A_0) = \text{Ext}^2(L_i[p_i], L_i[p_i]). \quad (49)$$

(ii) *For every  $x \in HT^2(A_0)$ ,*

$$\sigma_E(\text{ev}_E(x)) = x \lrcorner \text{ch}(E), \quad (50)$$

*and consequently  $\ker \sigma_E \cap \text{im } \text{ev}_E = \text{ev}_E(\{x : x \lrcorner \text{ch}(E) = 0\})$ . The weakened hypothesis of [\[Mar25b, Question 11.4\]](#) holds for  $E$  if and only if  $\text{ev}_E$  kills every  $x$  that preserves  $\text{ch}(E)$  to first order.*

(iii)  *$\dim HT^2(A_0) = 2\binom{2n}{2} + 4n^2$  and  $\dim \text{Ext}^2(E, E) = s\binom{2n}{2}$ ; for  $s \geq 5$  and every  $n \geq 2$  the second exceeds the first, so  $\text{ev}_E$  is not surjective and [\[Mar25b, Lemma 11.3\]](#) cannot apply to a sum of five or more line bundles.*

(iv) *For the object of [Proposition 18.14](#), with  $\lambda_i = u_i \theta^+ + \bar{u}_i \theta^-$  as in [Notation 18.2](#): the space  $\{x : x \lrcorner \text{ch}(E) = 0\}$  is  $\text{Ann}_{H^{0,2}(\omega_1)} \oplus \{v : v \lrcorner \omega_1 = 0\} \oplus \{\pi : \pi \lrcorner \omega_1 = 0\}$ , of dimensions 9, 18 and 9; the first and the third summands are both carried by  $\text{ev}_E$  onto  $\Delta$ ; the second is carried onto*

$$\Sigma = \{ (u_i a + \bar{u}_i b)_{i=1}^6 : a \in \wedge^2 V_+^{0,1}, b \in \wedge^2 V_-^{0,1} \}, \quad \dim \Sigma = n(n-1) = 6,$$

*which contains the image of the obstruction map of [Proposition 18.16](#);  $\Delta \cap \Sigma = 0$ ; and*

$$\ker \sigma_E = \Delta \oplus \Sigma \subseteq \text{im } \text{ev}_E.$$

*So every direction in which  $\sigma_E$  is blind is the value of  $\text{ev}_E$  on a Hochschild class that preserves  $\text{ch}(E)$  to first order, and the weakened hypothesis fails for  $E$  by all fifteen dimensions. The rank of  $\text{ev}_E$  is 45. The same containment  $\ker \sigma_E \subseteq \text{im } \text{ev}_E$ , with  $\dim \ker \sigma_E = 15$ , holds for each of the eighteen configurations of [Remark 9.9](#).*

*Proof.* (i) The evaluation map is the action of  $HH^*(A_0)$  on  $\text{Ext}^*(E, E)$ , which under HKR is contraction of a polyvector in  $H^q(\wedge^p T)$  with  $\frac{1}{p!} \text{At}(E)^p$  [\[BF08\]](#); for  $p = 0$  this is  $z \cdot \text{id}_E$  as in the proof of [Theorem 9.19](#), for  $p = 1$  it is  $v \lrcorner \text{At}(E)$ , the map (46), and for  $p = 2$  it is  $\frac{1}{2} \pi \lrcorner \text{At}(E)^2$ . The Atiyah class of  $E$  is diagonal with entries  $\lambda_i$ , so all three are diagonal with the entries displayed.

(ii) By (39),  $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_i e^{\lambda_i}$  for diagonal  $\zeta$ . The three factors in (49) are exactly the scalars by which  $z \wedge$ ,  $v \lrcorner$  and  $\pi \lrcorner$  act on  $e^{\lambda_i}$ : the first trivially, the second because  $v \lrcorner$  is a derivation and  $v \lrcorner \lambda_i$  has even degree, the third by the footnote above. Hence

<sup>54</sup>The double contraction is a second order operator. For a class  $\lambda$  of type  $(1, 1)$ , which has a single  $(1, 0)$  slot,  $\pi \lrcorner \lambda = 0$  and  $\pi \lrcorner \lambda^k = \binom{k}{2} \lambda^{k-2} (\pi \lrcorner \lambda^2)$ , because  $\pi \lrcorner$  is a composite of two interior products, each of which is an antiderivation, and the only nonzero terms are those in which the two interior products hit two different factors. Summing over  $k$  with the factorials gives  $\pi \lrcorner e^\lambda = \frac{1}{2} (\pi \lrcorner \lambda^2) e^\lambda$ , exactly parallel to  $v \lrcorner e^\lambda = (v \lrcorner \lambda) e^\lambda$  for the first order operator  $v \lrcorner$ . This is the whole content of the compatibility (50) below: the semiregularity map of a line bundle is multiplication by  $e^\lambda$ , and every summand of  $HT^2$  acts on  $e^\lambda$  by a scalar-valued factor which is precisely the value of  $\text{ev}_E$ . The same identity, for arbitrary  $E$ , is the commutative square of Buchweitz and Flenner [\[BF03, Corollary 4.3\]](#) and its Hochschild form in [\[Mar25b, Lemma 11.3\]](#); on an abelian variety the Todd class is trivial and no correction to the HKR map is needed [\[Cal05, BF08\]](#).

$\sigma_E(\text{ev}_E(x)) = \sum_i (-1)^{p_i} x \lrcorner e^{\lambda_i} = x \lrcorner \text{ch}(E)$ . The description of  $\ker \sigma_E \cap \text{im } \text{ev}_E$  follows:  $\text{ev}_E(x)$  lies in  $\ker \sigma_E$  if and only if  $x \lrcorner \text{ch}(E) = 0$ .

(iii) The three summands of (48) have dimensions  $\binom{2n}{2}$ ,  $4n^2$  and  $\binom{2n}{2}$ . The inequality  $s \binom{2n}{2} > 2 \binom{2n}{2} + 4n^2$  is  $(s-2)(2n-1) > 4n$ , and  $4n/(2n-1) \leq 8/3$  for  $n \geq 2$ , so it holds for  $s \geq 5$ ; it is checked for  $2 \leq n \leq 30$  and  $5 \leq s \leq 40$  in [Appendix E](#).

(iv) Since  $\text{ch}(E) = 12\omega_1$  is of type  $(3, 3)$ , the three summands of  $x \lrcorner \text{ch}(E)$  have types  $(3, 5)$ ,  $(2, 4)$  and  $(1, 3)$ , so the kernel is the direct sum of the three kernels. The first is  $\text{Ann}_{H^{0,2}}(\omega_1)$ , of dimension  $n^2 = 9$  by [Lemma 9.6](#). The second is  $\text{Hom}(V_+^{1,0}, V_+^{0,1}) \oplus \text{Hom}(V_-^{1,0}, V_-^{0,1})$ , of dimension  $2n^2 = 18$ , by the proof of [Proposition 9.14](#), whose argument for (i)  $\Leftrightarrow$  (iii) does not use the polarisation. For the third, decompose  $\Lambda^2(H^{1,0})^\vee$  as  $\Lambda^2(V_+^{1,0})^\vee \oplus (V_+^{1,0})^\vee \otimes (V_-^{1,0})^\vee \oplus \Lambda^2(V_-^{1,0})^\vee$ ; here  $\omega_1$  is a combination of  $(\theta^+)^3$  and  $(\theta^-)^3$  with both coefficients nonzero,  $(\theta^\pm)^3$  spans  $\Lambda^3 V_\pm^{1,0} \otimes \Lambda^3 V_\pm^{0,1}$ , the outer blocks of  $\pi$  act injectively on the respective factor  $\Lambda^3 V_\pm^{1,0}$  and land in different summands of  $H^{1,3}$ , and the middle block kills both; so the third kernel is the middle block, of dimension  $n^2 = 9$ .

At a split point  $\theta^\pm$  are of type  $(1, 1)$ , so  $\theta^\pm \in V_\pm^{1,0} \otimes V_\pm^{0,1}$ , and they are nondegenerate pairings because  $(\theta^\pm)^3 \neq 0$ . Write  $\lambda_i^2 = u_i^2(\theta^+)^2 + \bar{u}_i^2(\theta^-)^2 + 2N(u_i)\theta^+\theta^-$ . A middle block  $\pi$  kills  $(\theta^\pm)^2$ , which have no  $V_\mp^{1,0}$  slot, so  $\frac{1}{2}\pi \lrcorner \lambda_i^2 = N(u_i)\pi \lrcorner (\theta^+\theta^-)$ ; the six  $u_i$  of [Proposition 18.14](#) are  $\pm\zeta_6, \pm 1, \pm\zeta_6^{-1}$ , all of norm one, so this value is independent of  $i$  and  $\text{ev}_E(\pi) = z_\pi \cdot \text{id}_E$  with  $z_\pi = \pi \lrcorner (\theta^+\theta^-)$ . The map  $\pi \mapsto z_\pi$  from the middle block to  $V_+^{0,1} \otimes V_-^{0,1}$  is the tensor product of the two nondegenerate pairings, hence an isomorphism onto the middle block of  $H^{0,2}$ , which is  $\text{Ann}_{H^{0,2}}(\omega_1)$  by [Lemma 9.6](#). So  $\text{ev}_E$  carries the third kernel onto  $\Delta$ , as it carries the first by [Theorem 9.19](#).

For  $v = (v_+, v_-)$  in the second kernel,  $v \lrcorner \lambda_i = u_i v_+ \lrcorner \theta^+ + \bar{u}_i v_- \lrcorner \theta^-$  with  $a = v_+ \lrcorner \theta^+ \in \Lambda^2 V_+^{0,1}$  and  $b = v_- \lrcorner \theta^- \in \Lambda^2 V_-^{0,1}$ ; as  $v_\pm$  ranges over  $\text{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$ ,  $a$  and  $b$  range over the whole of  $\Lambda^2 V_\pm^{0,1}$  because the pairings  $\theta^\pm$  are nondegenerate, which is the surjectivity of  $\text{alt}(Vc)$  in (25). So the image is  $\Sigma$ . The map  $(a, b) \mapsto (u_i a + \bar{u}_i b)_i$  is injective, because  $(u_1, \bar{u}_1)$  and  $(u_2, \bar{u}_2)$  are linearly independent in  $\mathbb{C}^2$ ,  $u_1/\bar{u}_1 = \zeta_6^2$  differing from  $u_2/\bar{u}_2 = 1$ ; hence  $\dim \Sigma = 2 \binom{n}{2} = n(n-1) = 6$ . The obstruction map of [Proposition 18.16](#) is the restriction to the polarised  $K$ -commuting  $v$ , so its image lies in  $\Sigma$ . An element of  $\Sigma$  of the shape  $(z, \dots, z)$  has  $u_i a + \bar{u}_i b = z = -z$  on comparing  $i$  with  $3+i$ , so  $z = 0$  and then  $a = b = 0$ :  $\Delta \cap \Sigma = 0$ . By (ii),  $\Delta \oplus \Sigma \subseteq \ker \sigma_E \cap \text{im } \text{ev}_E$ , and its dimension  $9 + 6 = 15$  equals  $\dim \ker \sigma_E$  by [Proposition 18.14](#), so the three spaces coincide. The rank of  $\text{ev}_E$  and the statement for the eighteen configurations are the computations of item (XXVI) of [Appendix D](#), which also recomputes every dimension in this proof in the model of item (XIII).  $\blacksquare$

*Remark 18.19* (Question 11.4 for split objects, and what is left of it). For a split object the two criteria stand or fall together, and for the explicit object both fall completely. [Theorem 9.19](#) makes the first  $n^2$  dimensions of the failure uniform over every concentrated object in every dimension; [Proposition 18.18](#)(iv) accounts for the remaining  $n(n-1)$  at  $n = 3$  by the deformations of the two halves  $V_\pm$  of the Hodge structure separately, and the equality  $\ker \sigma_E = \Delta \oplus \Sigma$  says that the two mechanisms exhaust the kernel: nothing in it is accidental. Nothing here touches the case the question was asked for. A sheaf of positive rank has  $z \cdot \text{ch}(E) \neq 0$  for every  $z \neq 0$ , so its  $\Delta$  is zero, and whether  $\text{ev}_E$  kills the deformations that preserve  $\kappa(E)$  is a statement about the sheaf and not about its Chern character. Markman's candidate in his Section 12, a rank one complex on a principally polarised abelian fourfold with Chern character on the secant  $\langle e^{\pm\sqrt{-d}\Theta} \rangle$ , is exactly such a

The evaluation map carries the first-order symmetries of  $\text{ch}(E)$  onto the whole kernel of  $\sigma_E$ .

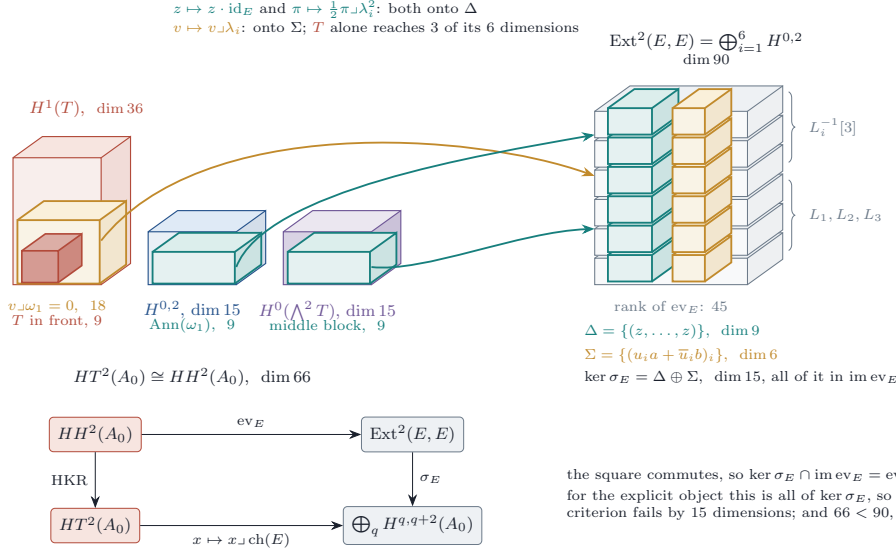


FIGURE 22. **Proposition 18.18** for the explicit object at  $n = 3$ . Left: the three summands of  $HT^2(A_0)$ , drawn with heights proportional to their dimensions, and inside each the part that preserves  $\text{ch}(E)$  to first order: the annihilator of  $\omega_1$ , the deformations killing  $\omega_1$  with the polarised  $K$ -commuting tangent space  $T$  in front, and the middle block of bivectors. Right:  $\text{Ext}^2(E, E)$  as six stacked copies of  $H^{0,2}$ , one per summand of  $E$ , and inside it the kernel of  $\sigma_E$ : the column  $\Delta$ , one and the same class in every layer, and the column  $\Sigma$ , whose entries depend on the summand through  $u_i$ . The three arrows are the evaluation map on the three kernels; the two teal ones both land on  $\Delta$ , the ochre one on  $\Sigma$ , and together they cover the kernel. Below, the commutative square that makes this a statement about the weakened criterion of [Mar25b, Question 11.4].

sheaf, and he records that the verification remains to be done. **Theorem 18.29** reduces that verification to one identity on the fourfold, and **Theorem 18.34** shows that the identity fails: the candidate does not satisfy the weakened criterion either.

**Theorem 18.20** (Where a semiregular split object can live). *Keep [Setup 18.9](#) and suppose  $E$  satisfies (44) with  $\text{ch}_0(E) \neq 0$ , so that the  $u_i$  take  $\nu \geq n + 1$  distinct norms by [Corollary 18.13](#). Write  $N_1 < \dots < N_\nu$  for those norms,  $M_j$  for the sum of the  $m_i$  at level  $N_j$ , and  $r_K(N) = \#\{u \in \mathcal{O}_K : N_{K/\mathbb{Q}}(u) = N\}$ .*

- (i) *If  $\nu = n + 1$  then, with  $P = \prod_k N_k$ ,  $S = \text{ch}_0(E)$  and  $D_j = N_j \prod_{k \neq j} (N_j - N_k)$ ,*

$$M_j = (-1)^n \frac{SP}{D_j}, \quad (51)$$

*and the least  $S > 0$  making every  $M_j$  an integer is  $S_{\min} = \text{lcm}_j(|D_j|/\text{gcd}(|D_j|, P))$ . Multiplicity one, [Lemma 18.1\(i\)](#), forces  $|M_j| \leq r_K(N_j)$  for every  $j$ .*

- (ii) *For  $4 \leq n \leq 8$ , every  $d \in \{1, 2, 3, 7, 11, 19\}$  and every choice of norms below the bounds recorded in item (XXII) of [Appendix D](#), no set of  $n + 1$  norms and no set of*

$n + 2$  norms satisfies those constraints. In that range no split object of the shape of [Setup 18.9](#) is semiregular, and the reason is arithmetic rather than geometric: the least integral scale outruns the supply of elements at the levels it demands.

- (iii) At  $n = 3$  the constraints of (i) can be met. With norms at most 200 there are exactly eleven sets of four norms that meet them, all of them at  $d = 1$  or  $d = 2$ , and after dividing out the common factor each has one of the three shapes  $(1, 2, 3, 4)$ ,  $(1, 2, 4, 5)$  and  $(1, 4, 9, 10)$ . The smallest is  $d = 2$  at  $(9, 18, 27, 36)$ , where  $S = 1$ ,  $|M| = (4, 6, 4, 1)$  and  $\text{ch}_0(E) = 1$ .
- (iv) For all eleven, the full system (44) has no solution: no assignment of signs to the elements of  $\mathcal{O}_K$  at those norms, with the level sums of (51), satisfies the off-diagonal conditions. In particular no split object of the shape of [Setup 18.9](#) with norms at most 200 is semiregular at  $n = 3$ .

*Proof.* (i) The conditions  $T_{r,r} = 0$  for  $r = 1, \dots, n$  say that the linear functional  $f \mapsto \sum_j M_j f(N_j)$  kills  $x, x^2, \dots, x^n$ , hence sends every polynomial  $f$  of degree at most  $n$  to  $S f(0)$ . Applying it to  $f(x) = \prod_{k \neq j} (x - N_k)$ , which has degree  $n$ , gives

$$M_j \prod_{k \neq j} (N_j - N_k) = S (-1)^n \prod_{k \neq j} N_k,$$

and multiplying by  $N_j$  turns this into (51). The value of  $S_{\min}$  is the definition of a least common multiple, and the bound on  $|M_j|$  holds because each  $u$  of norm  $N_j$  contributes at most one term  $m_i = \pm 1$  to  $M_j$ .

(ii), (iii) and (iv) are the computations of item (XXII) of [Appendix D](#), carried out in exact integer arithmetic. For (ii) and (iii) the script enumerates every set of  $n + 1$  norms represented by  $\mathcal{O}_K$  below the bound, forms  $D_j$  and  $S_{\min}$ , and tests  $|M_j| \leq r_K(N_j)$ ; at  $\nu = n + 2$  the solution space of the diagonal conditions is two dimensional and its integer points inside the same box are swept by running two coordinates over their own ranges. For (iv) the off-diagonal conditions are, after grouping by norm,

$$\sum_j N_j^r A_{1,j} = 0 \quad (r = 0, 1, 2), \quad \sum_j N_j^r A_{2,j} = 0 \quad (r = 0, 1), \quad A_{k,j} = \sum_{N(u)=N_j} m_u u^k,$$

which are additive over the levels; the four levels are therefore split into one half that is tabulated and one that is streamed against the table, so the enumeration is complete and not a sample. The four level sets attacked directly are  $d = 2$  at  $(9, 18, 27, 36)$ ,  $(27, 54, 81, 108)$  and  $(33, 66, 99, 132)$  and  $d = 1$  at  $(5, 20, 45, 50)$ ; the other four of the eight are their images under multiplication by an element of norm 2 or 4, which is a bijection on the relevant norm levels because the element counts agree, so a solution for one would give a solution for the other. The remaining three,  $d = 1$  at  $(13, 52, 117, 130)$ ,  $(17, 68, 153, 170)$  and  $(25, 50, 100, 125)$ , have levels of 8, 8, 8, 16 and 12, 12, 12, 16 elements. For them the scale is  $S = \pm S_{\min}$ , since  $|S| \geq 2$  violates the bound  $|M_j| \leq r_K(N_j)$  and  $S \mapsto -S$  is the symmetry  $m \mapsto -m$ . The first three levels are streamed against the fourth, matched through a linear hash of the ten integer coordinates of the off-diagonal conditions, and every match is checked on all ten coordinates exactly. A solution has all ten coordinates zero and cannot be missed, so the search is complete; it covers  $4.29 \times 10^{14}$  sign patterns in the largest case and finds none. ■

*Remark 18.21* (What the range means). The dimension that matters is  $n \geq 4$ , since  $n = 2$  and  $n = 3$  are settled by [\[Mar25a, Mar25b\]](#) and [\[FF26\]](#). There [Theorem 18.20\(ii\)](#) says that the split route closes before any geometry is used: the level sums cannot exist, so there is nothing to test for semiregularity. At  $n = 3$ , where the geometry does produce an object,

the level sums do exist and the off-diagonal conditions are what remove them, which is a different mechanism and a reminder that (ii) is a statement about a range and not a theorem for all  $n$ . What both cases share is that the obstruction is arithmetic: it lives in the norm form of  $K$  and in the supply of integers of a given norm, not in the cohomology of  $A$ .

The range is now beside the point, because a second mechanism, this one in the cohomology of  $A$ , closes the split route for every  $n$  and every set of norms at once.

**Corollary 18.22** (No sum of line bundles runs either argument). *Let  $A$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$ , and let  $E = \bigoplus_i L_i[p_i]^{\oplus m_i}$  be a sum of shifted line bundles, twisted by a common class or not, whose (twisted) Chern character remains of Hodge type to first order along  $\mathcal{D}_{n,n}$  and has a nonzero Weil component. Then  $\sigma_E$  is not injective on the image of  $\text{ev}_E$ :  $E$  is neither semiregular nor semiregular in the weakened sense of [Mar25b, Question 11.4], and it is obstructed to first order along some direction of the Weil family. In particular every object of Setup 18.9 satisfying (44), for every  $n \geq 2$ , every field  $K$ , every rank and every set of norms, fails both criteria: the conclusion of Theorem 18.20, that no split object of that shape is semiregular, holds without any bound on the norms, in every dimension, and for the weakened criterion as well.*

*Proof.* Write  $\lambda_i$  for the twisted first Chern classes of the summands. If every  $\lambda_i$  were a real multiple of  $\eta$ , it would be a rational multiple, both classes being rational, and the twisted Chern character  $\sum_i m_i e^{\lambda_i}$  would be a polynomial in  $\eta$ ; its component in degree  $2n$  would then lie in  $\mathbb{Q}\eta^n$ , which meets the Weil line in zero by Proposition 5.1, against the hypothesis. So some  $\lambda_i$  is not a real multiple of  $\eta$ , and Theorem 9.20 applies to the summand  $L_i[p_i]$ . For an object of Setup 18.9 the classes are  $\lambda_i = \theta(u_i) \in R_0$ , and  $R_0 \cap \mathbb{Q}\eta = 0$ ; no twist is needed because  $c_1(E) = \theta(T_{1,0}) = 0$  by (44), as  $(1, 0)$  is not an excluded pair for  $n \geq 2$ ; and (44) gives a flat Chern character with a nonzero Weil component by Proposition 18.10. ■

*Remark 18.23* (What is closed and what is not). Corollary 18.22 ends the search that Theorem 18.20 began: there is no sum of line bundles, at any member of any Weil family in any dimension, that could run the semiregularity argument in either form, and the arithmetic of Theorem 18.20 is now a description of how the level sums fail rather than the reason the route fails. The reason is Proposition 9.15(i): the only divisor class that stays of type  $(1, 1)$  across the whole family is the polarisation, so every line bundle summand off that line is obstructed in some direction of the family, and the semiregularity map, whose composite with the obstruction is contraction with the flat class  $\text{ch}(E)$ , cannot see it. What is not closed is the object the argument needs: one with no rank one summand off the polarisation line, of positive rank, indecomposable enough that its Atiyah class contracted with every direction of  $T$  vanishes. Markman's reflexive sheaves at  $n = 3$  are of that kind, and nothing here says that their analogues at  $n \geq 4$  do not exist.

*Markman's candidate in dimension eight.* There is one concrete candidate of that kind above  $n = 3$ , proposed by Markman in [Mar25b, Section 12] with the remark that it is yet to be checked against the weakened criterion. Its Chern character is a finite computation in the ring  $\mathbb{Q}[\Theta]/(\Theta^5)$ , and we carry it out, because it places the candidate exactly at a rational point of the secant plane of Definition 14.4 and reduces what has to be checked to a single condition; that condition is then decided, in the negative, in Theorem 18.34.

**Setup 18.24** (Markman's object). Let  $(X, \Theta)$  be a principally polarised abelian fourfold,  $d \geq 3$  odd and squarefree,  $K = \mathbb{Q}(\sqrt{-d})$  and  $N = (d + 9)/2$ . Let  $D_i$ ,  $i \in \mathbb{Z}/N\mathbb{Z}$ , be  $N$  translates of  $\Theta$  in general position: any two, three or four of them meet transversally,

in a surface, a curve and  $\Theta^4 = 24$  points, and any five have empty intersection. Put  $Z_i = D_i \cap D_{i+1}$ ,  $Z = \bigcup_i Z_i$ , let  $C_i = Z_i \cap Z_{i+1} = D_i \cap D_{i+1} \cap D_{i+2}$  be the curves along which consecutive components meet, and let  $\nu: \tilde{Z} \rightarrow Z$  be the partial normalisation of  $Z$  at  $m = (d+9)(2d-1)$  of its isolated points of self-intersection, the points of  $Z_i \cap Z_j$  for  $i, j$  at cyclic distance at least three. Put

$$\mathcal{F} = [\mathcal{O}_X \rightarrow \nu_* \mathcal{O}_{\tilde{Z}}] \otimes \mathcal{O}_X(3\Theta),$$

the complex placed in degrees 0 and 1.

**Proposition 18.25** (The candidate sits at the point  $(1, 3)$  of the secant plane). *In Setup 18.24, with  $t = \Theta$  and  $u_t, v_t$  as in (29) for  $n = 4$ :*

- (i)  $\chi(\mathcal{O}_{Z_i}) = 14$ ,  $\chi(\mathcal{O}_{C_i}) = -36$ , the isolated self-intersection points of  $Z$  number  $12N(N-5) = (d+9)(3d-3)$ , and  $\chi(\mathcal{O}_{\tilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(27-d)$ ;
- (ii)  $\text{ch}(\nu_* \mathcal{O}_{\tilde{Z}}) = N\Theta^2 - 2N\Theta^3 + \chi(\mathcal{O}_{\tilde{Z}}) [\text{pt}]$ ;
- (iii)  $\text{ch}(\mathcal{F}) = u_t + 3v_t$ , that is

$$\text{ch}(\mathcal{F}) = 1 + 3\Theta - \frac{d}{2}\Theta^2 - \frac{d}{2}\Theta^3 + \frac{d^2}{24}\Theta^4,$$

the rational point  $(a, b) = (1, 3)$  of the secant plane  $P_\Theta$ ; with the twist  $\mathcal{O}_X(\Theta)$  in place of  $\mathcal{O}_X(3\Theta)$  the degree two component would be  $-\frac{d+8}{2}\Theta^2$  instead, so the twist in [Mar25b, Section 12] is to be read as  $3\Theta$ ;

- (iv)  $N_{K/\mathbb{Q}}(3 + \sqrt{-d}) = d + 9 = 2N$  is the number of divisors, and by Theorem 14.7 the Bogomolov discriminant of  $\mathcal{F}$  is  $\Delta(\mathcal{F}) = (d+9)\Theta^2$ ;
- (v) the objects  $\Phi(\mathcal{F} \boxtimes \mathcal{F})$  and  $\Phi(\mathcal{F} \boxtimes \mathcal{F}^\vee)$  on  $X \times \hat{X}$  have ranks  $\int_X \text{ch}(\mathcal{F})^2 = 8d(d-9)$  and  $\int_X \text{ch}(\mathcal{F}) \text{ch}(\mathcal{F}^\vee) = 8d(d+9)$ , up to sign, both nonzero for squarefree  $d$ ; the same formula gives  $-8q$  for the pair of secant sheaves of [Mar25b, Section 11] on a threefold, which is his rank  $8q$  up to the shift.

Consequently  $\mathcal{F} \boxtimes \mathcal{F}$  satisfies conditions (1), (2b) and (2c) of Markman's strategy [Mar25b, Section 4]: the transform has nonzero rank, its normalised Chern character is invariant under the special Mumford–Tate group by [Mar25b, Corollary 9.2], and its Weil component is nonzero by [Mar25b, Proposition 10.2], the relevant coefficient being  $N(c)$  with  $c = \frac{1}{2}(1 + 3/\sqrt{-d}) \neq 0$ . What remains is condition (2a) in its weakened form: that  $\text{ev}_E$  kills every class in  $H^1(T) \oplus H^0(\wedge^2 T)$  that preserves  $\kappa(E)$  to first order, for  $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$ .

*Proof.* On a principally polarised abelian fourfold  $\chi(\mathcal{O}_X(k\Theta)) = k^4$  and  $\Theta^4 = 24$ . The Koszul complex of  $r$  transversal translates of  $\Theta$  gives  $\text{ch}$  of the structure sheaf of their intersection as  $(1 - e^{-\Theta})^r$ , so  $\text{ch}(\mathcal{O}_{Z_i}) = \Theta^2 - \Theta^3 + \frac{7}{12}\Theta^4$ ,  $\text{ch}(\mathcal{O}_{C_i}) = \Theta^3 - \frac{3}{2}\Theta^4$  and  $\text{ch}$  of four divisors is  $\Theta^4$ ; integrating,  $\chi(\mathcal{O}_{Z_i}) = 14$ ,  $\chi(\mathcal{O}_{C_i}) = -36$  and 24 points, as also follows from  $0 - 2 \cdot 1 + 16$ ,  $0 - 3 + 48 - 81$  and  $0 - 4 + 96 - 324 + 256$ .

(i), (ii). At every point  $Z$  is locally a union of coordinate subspaces of  $\mathbb{C}^4$ : along  $C_i$  two surfaces  $\{x_1 = x_2 = 0\} \cup \{x_2 = x_3 = 0\}$ , at a point of  $C_i \cap C_{i+1}$  three of them, at an isolated point  $\{x_1 = x_2 = 0\} \cup \{x_3 = x_4 = 0\}$ . The ideals of such an arrangement are monomial and distribute over sums, so the Mayer–Vietoris complex

$$0 \rightarrow \mathcal{O}_Z \rightarrow \bigoplus_i \mathcal{O}_{Z_i} \rightarrow \bigoplus_{i < j} \mathcal{O}_{Z_i \cap Z_j} \rightarrow \bigoplus_{i < j < k} \mathcal{O}_{Z_i \cap Z_j \cap Z_k} \rightarrow \cdots$$

with scheme-theoretic intersections is exact. Pairs at cyclic distance one meet in  $C_i$ ,  $N$  of them; pairs at distance at least two meet in the 24 points of four divisors,  $N(N-3)/2$  of them; the only triples with nonempty intersection are the consecutive ones  $Z_i \cap Z_{i+1} \cap Z_{i+2} =$

$C_i \cap C_{i+1}$ , 24 points each,  $N$  of them; and no four components meet, since that would involve at least five divisors. Hence

$$\mathrm{ch}(\mathcal{O}_Z) = N \mathrm{ch}(\mathcal{O}_{Z_i}) - N \mathrm{ch}(\mathcal{O}_{C_i}) - \frac{N(N-3)}{2} \Theta^4 + N \Theta^4,$$

whose degree two part is  $N\Theta^2$ , degree three part  $-N\Theta^3 - N\Theta^3 = -2N\Theta^3$ , and degree four part  $\chi(\mathcal{O}_Z)[\mathrm{pt}]$  with  $\chi(\mathcal{O}_Z) = 14N + 36N - 12N(N-3) + 24N = 110N - 12N^2$ . The isolated singular points are those of pairs at distance at least three, of which there are  $N(N-5)/2$ , giving  $12N(N-5) = 3(d+9)(d-1)$  points; the pairs at distance two lie on the double curves and are not isolated. At an isolated point the two branches meet transversally, and separating them gives  $0 \rightarrow \mathcal{O}_Z \rightarrow \nu_* \mathcal{O}_{\tilde{Z}} \rightarrow k(p) \rightarrow 0$ , so each of the  $m$  normalisations raises  $\chi$  by one and leaves the lower degrees of  $\mathrm{ch}$  unchanged. With  $N = (d+9)/2$ ,  $\chi(\mathcal{O}_{\tilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(55 - 3(d+9) + 2d-1) = (d+9)(27-d)$ .

(iii)  $\mathrm{ch}(\mathcal{F}) = e^{3\Theta}(1 - \mathrm{ch}(\nu_* \mathcal{O}_{\tilde{Z}}))$ . The degree two part is  $(\frac{9}{2} - N)\Theta^2 = -\frac{d}{2}\Theta^2$ , which is where  $N = (d+9)/2$  enters; with the twist  $k\Theta$  it would be  $(\frac{k^2}{2} - N)\Theta^2$ , equal to  $-\frac{d}{2}\Theta^2$  exactly for  $k^2 = 9$ , and  $k = -3$  is excluded by the degree three part. The degree three part is  $(\frac{27}{6} - 3N + 2N)\Theta^3 = (\frac{9}{2} - N)\Theta^3 = -\frac{d}{2}\Theta^3$ , and the degree four part is  $(\frac{81}{24} - \frac{9N}{2} + 6N - \frac{\chi(\mathcal{O}_{\tilde{Z}})}{24})\Theta^4 = \frac{81+36N-\chi(\mathcal{O}_{\tilde{Z}})}{24}\Theta^4 = \frac{d^2}{24}\Theta^4$  by (i). Comparing with (29) at  $n = 4$ ,  $u_t = 1 - \frac{d}{2}t^2 + \frac{d^2}{24}t^4$  and  $v_t = t - \frac{d}{6}t^3$ , gives  $\mathrm{ch}(\mathcal{F}) = u_t + 3v_t$ .

(iv)  $N(3 + \sqrt{-d}) = 9 + d$ , and (30) with  $(a, b) = (1, 3)$ .

(v) Orlov's equivalence is  $\Phi = (\mathrm{id} \times \Phi_P)^{-1} \circ \mu^*$  with  $\mu(x, y) = (x + y, y)$ , so the restriction of  $\Phi(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$  to  $\{x\} \times \hat{X}$  is the Fourier–Mukai transform of  $\tau_x^* \mathcal{F}_1 \otimes \mathcal{F}_2$ , whose derived fibre at  $L \in \hat{X}$  is  $R\Gamma(X, \tau_x^* \mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L^{\pm 1})$ ; the rank of a perfect complex is the Euler characteristic of its derived fibre, and  $\chi(\tau_x^* \mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L) = \int_X \mathrm{ch}(\mathcal{F}_1) \mathrm{ch}(\mathcal{F}_2)$  by Riemann–Roch, translations and  $L \in \mathrm{Pic}^0$  acting trivially on cohomology. With  $\mathrm{ch}(\mathcal{F}^\vee) = u_t - 3v_t$ , the involution  $\tau$  changing the sign of the odd degrees,  $\int u_t^2 = 8d^2$ ,  $\int u_t v_t = 0$  and  $\int v_t^2 = -8d$  give the two values. On a threefold  $\int (u_t + v_t)^2 = 2 \int u_t v_t = -8q$ .

The last paragraph: (2b) is the  $\mathrm{Spin}(V_{\mathbb{Q}})_B$ -invariance of  $\kappa(E)$ , which [Mar25b, Corollary 9.2] derives from  $\mathrm{ch}(\mathcal{F}) \in B$  and the nonvanishing of the rank; (2c) is [Mar25b, Proposition 10.2], which for  $K$  imaginary quadratic reduces to the nonvanishing of the component of  $\mathrm{ch}(\mathcal{F}) \otimes \mathrm{ch}(\mathcal{F})$  in  $\ell_W \otimes \ell_{\bar{W}}$ , and writing  $\mathrm{ch}(\mathcal{F}) = c\ell_W + \bar{c}\ell_{\bar{W}}$  with  $\ell_W = e^{\sqrt{-d}\Theta}$  gives that coefficient as  $c\bar{c}$ . Every number in (i) to (v) is recomputed for every odd squarefree  $d \leq 101$  in item (XXVII) of Appendix D. ■

*Remark 18.26* (The one condition left). [Proposition 18.25](#) does not verify Markman's candidate; it verifies everything about it that is a statement about classes, and it shows that the one remaining statement is about the object. That statement is, for  $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$  on the eightfold  $X \times \hat{X}$  of Weil type with  $\kappa(E) = \mathrm{ch}(E)e^{-c_1(E)/\mathrm{rk} E}$ : every  $x \in H^1(T) \oplus H^0(\wedge^2 T)$  with  $x \lrcorner \kappa(E) = 0$  satisfies  $\mathrm{ev}_E(x) = 0$  in  $\mathrm{Ext}^2(E, E)$ . The  $H^2(\mathcal{O})$  summand imposes nothing, because  $\mathrm{rk} E \neq 0$  makes  $z \cdot \kappa(E) \neq 0$  for  $z \neq 0$ . By [Theorem 9.20](#) the condition fails as soon as  $E$  has a rank one direct summand whose class is not a multiple of the polarisation; so the first thing to know about  $E$  is whether it is indecomposable. Markman's object at  $n = 3$  is reflexive of rank  $8q$  away from a divisor and he proves the condition by descending to a quotient on which  $\sigma_E$  is injective on an invariant subspace [Mar25b, Proposition 11.5]. Whether the same can be done for  $\Phi(\mathcal{F} \boxtimes \mathcal{F})$  is a question about Ext groups of one sheaf on one eightfold, not about any Hodge class. [Theorem 18.29](#) below moves it from the eightfold to the fourfold and from the object  $E$  to  $\mathcal{F}$  itself: it is equivalent

to the single identity (56) in  $\text{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2)$ , the other half of the condition holding by construction; and Theorem 18.34 shows that the identity fails, so the answer is no.

*The one condition, reduced to the factor.* The condition left in Remark 18.26 is about an object on an eightfold. It is equivalent to a condition on the factor  $\mathcal{F}$  on the fourfold, and on the fourfold it splits into two, one of which holds for Markman's object because the object is constructed uniformly over the moduli of principally polarised fourfolds. The reduction is not special to that object: it works for every secant object on every abelian  $n$ -fold with  $n \geq 3$ , and its ingredients are three computations in the Hodge algebra of one abelian variety, which is why they can be checked by machine. We fix the conventions first.

**Setup 18.27** (The weakened criterion for one object). Let  $Y$  be an abelian variety and  $F$  a nonzero perfect complex on  $Y$ . Write  $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\wedge^2 T_Y)$  as in (48), with  $x = (z, v, \pi)$  acting on  $H^*(Y, \mathbb{C})$  by  $x \lrcorner$ , and likewise  $HT^1(Y) = H^{0,1}(Y) \oplus H^0(T_Y)$ , with  $(z, \theta)$  acting by  $z \wedge$  and by the interior product  $\theta \lrcorner$ . Normalise the Atiyah class  $\text{At}(F) \in \text{Ext}^1(F, F \otimes \Omega_Y^1)$  so that  $\text{ch}(F) = \text{tr exp At}(F)$ ; for a line bundle it is then  $c_1(L)$ , as in (46).<sup>55</sup> The evaluation map of Theorem 9.19 is, under HKR,

$$\text{ev}_F(z, v, \pi) = z \cdot \text{id}_F + v \lrcorner \text{At}(F) + \frac{1}{2} \pi \lrcorner \text{At}(F)^2 \in \text{Ext}^2(F, F), \quad (52)$$

the contraction of the polyvector with the exponential Atiyah class [Hua21, Theorem A], [BF08]; its restriction to  $H^1(T_Y)$  is the obstruction map, whose kernel is the space of first order deformations of  $Y$  to which  $F$  extends [Tod09], [Mar25a, Section 8.3]. Put

$$\text{Ann}_{HT^2}(\text{ch } F) = \{x \in HT^2(Y) : x \lrcorner \text{ch}(F) = 0\},$$

the Hochschild classes preserving  $\text{ch}(F)$  to first order. We say that  $F$  *satisfies the weakened criterion* if  $\text{ev}_F$  vanishes on  $\text{Ann}_{HT^2}(\text{ch } F)$ . By the commutative square  $\sigma_F \circ \text{ev}_F = (\cdot) \lrcorner \text{ch}(F)$  of [BF08, Proposition 6.2.1 and Corollary 6.3.2], in the form quoted in [Mar25b, Lemma 11.3],

$$\text{ev}_F(\text{Ann}_{HT^2}(\text{ch } F)) = \ker \sigma_F \cap \text{im } \text{ev}_F,$$

so the weakened criterion is exactly the hypothesis of [Mar25b, Question 11.4], that  $\sigma_F$  restricts to the image of  $\text{ev}_F$  as an injective map; Proposition 18.18(ii) is the case of a split object.

**Lemma 18.28** (The  $B$ -field transform). *Let  $B \in H^{1,1}(Y)$ . For  $x = (z, v, \pi) \in HT^2(Y)$  put*

$$e^B x = (z + v \lrcorner B + \frac{1}{2} \pi \lrcorner B^2, v + \pi \lrcorner B, \pi), \quad (53)$$

where  $\pi \lrcorner B \in \text{Hom}(H^{1,0}, H^{0,1}) = H^1(T_Y)$  is the contraction of one slot of  $\pi$  with  $B$ : for  $\pi = \partial_a \wedge \partial_b$  it sends  $p_b$  to  $\partial_a \lrcorner B$ ,  $p_a$  to  $-\partial_b \lrcorner B$  and the other basis vectors to zero. Then  $e^B$  is a linear automorphism of  $HT^2(Y)$  with inverse  $e^{-B}$ , and:

- (i)  $x \lrcorner (w e^B) = ((e^B x) \lrcorner w) e^B$  for every even class  $w \in H^{\text{ev}}(Y, \mathbb{C})$ ;
- (ii) for a line bundle  $L$  with  $c_1(L) = B$ ,  $\text{ev}_{F \otimes L}(x) = \text{ev}_F(e^B x)$  and  $\text{Ann}_{HT^2}(\text{ch}(F \otimes L)) = e^{-B} \text{Ann}_{HT^2}(\text{ch } F)$ ; hence  $F$  satisfies the weakened criterion if and only if  $F \otimes L$  does.

The computation in (ii) is formal: it uses only that the Atiyah class of the twisted object is  $\text{At}(F) + B \cdot \text{id}_F$  and its Chern character  $\text{ch}(F)e^B$ . It therefore applies verbatim to the twisted sheaves of [Mar25b, Section 2], with  $B$  rational: for  $F$  of nonzero rank  $r$ , the

<sup>55</sup>This is the convention of [Tod09] and of [Mar25a, footnote 14]; Buchweitz and Flenner use the opposite sign, which changes  $\text{At}(F)$  to  $-\text{At}(F)$  and leaves  $\text{At}(F)^2$  and every statement below unchanged except for the sign of the middle term of  $\text{ev}_F$ , which is irrelevant to the vanishing statements.

criterion for the twisted object  $F \otimes L^{-1}$  with class  $\kappa(F) = \text{ch}(F)e^{-c_1(F)/r}$  is the same condition as the criterion for  $F$  with class  $\text{ch}(F)$ .

*Proof.* Everything takes place in a graded algebra on which the interior products  $\iota_a = \partial_a \lrcorner$  act as odd derivations, and the only properties used are that  $w$  is even, that  $B$  is even and central with  $\iota_a \iota_b B = 0$ , which holds for  $B$  of type  $(1, 1)$  and  $\partial_a, \partial_b$  of type  $(1, 0)$ , and that the odd scalar classes  $\iota_a B$  anticommute with every odd element. In  $H^*(Y, \mathbb{C})$  this is clear; the same argument applies in  $\bigoplus_{p,q} \text{Ext}^p(F, F \otimes \Omega_Y^q) = \text{Ext}^*(F, F) \otimes \bigwedge^* H^{1,0}$ , on which the polyvectors act through the second factor by odd derivations, because the classes coming from  $H^*(Y, \mathbb{C})$  are graded central in the Yoneda algebra. First  $\iota_a(B^k) = k(\iota_a B)B^{k-1}$ , since  $\iota_a B$  is odd and  $B$  is even, so  $\iota_a e^B = (\iota_a B)e^B$ . For  $w$  even this gives  $v \lrcorner (we^B) = (v \lrcorner w)e^B + w \sum_a v(p_a)(\iota_a B)e^B = (v \lrcorner w + (v \lrcorner B)w)e^B$ , and  $z(we^B) = (zw)e^B$ . For the bivector, with  $\pi \lrcorner = \iota_a \iota_b$ ,

$$\iota_b(we^B) = (\iota_b w)e^B + w(\iota_b B)e^B,$$

and applying  $\iota_a$ , using that  $\iota_b w$  and  $\iota_b B$  are odd,

$$\iota_a \iota_b(we^B) = [\iota_a \iota_b w + (\iota_a B)\iota_b w - (\iota_b B)\iota_a w + w((\iota_a \iota_b B) - (\iota_b B)(\iota_a B))]e^B.$$

The middle two terms are  $(\pi \lrcorner B) \lrcorner w$  for the element  $\pi \lrcorner B$  of  $\text{Hom}(H^{1,0}, H^{0,1})$  displayed in the statement, acting as a derivation, and the last bracket is  $\frac{1}{2}\iota_a \iota_b(B^2) = \iota_a((\iota_b B)B)$ . Summing the three actions gives (i). Applying (i) with  $B$  replaced by  $-B$  and then by  $B$ , and using  $(\pi \lrcorner B) \lrcorner B = 2(\iota_a B)(\iota_b B) = \pi \lrcorner B^2$  when  $\iota_a \iota_b B = 0$ , shows  $e^{-B}e^B = \text{id}$ .

(ii) The Atiyah class of  $F \otimes L$  is  $\text{At}(F) + B \cdot \text{id}_F$ , the Atiyah class being additive on tensor products [BF03, Section 3], and the two summands commute, so  $\exp \text{At}(F \otimes L) = \exp \text{At}(F) \cdot e^B$ . By (i) in the algebra  $\text{Ext}^*(F, F) \otimes \bigwedge^* H^{1,0}$ , with  $w = \exp \text{At}(F)$ ,

$$x \lrcorner \exp \text{At}(F \otimes L) = ((e^B x) \lrcorner \exp \text{At}(F))e^B,$$

and the component in  $\text{Ext}^2(F, F) \otimes \bigwedge^0$  of the right hand side is that of  $(e^B x) \lrcorner \exp \text{At}(F)$ , because  $e^B = 1 + B + \dots$  raises the exterior degree. This is  $\text{ev}_{F \otimes L}(x) = \text{ev}_F(e^B x)$ . By (i) in  $H^*(Y, \mathbb{C})$  with  $w = \text{ch}(F)$ ,  $x \lrcorner \text{ch}(F \otimes L) = 0$  if and only if  $(e^B x) \lrcorner \text{ch}(F) = 0$ , which is the statement on the annihilators. Hence  $\text{ev}_{F \otimes L}(\text{Ann}_{HT^2}(\text{ch}(F \otimes L))) = \text{ev}_F(\text{Ann}_{HT^2}(\text{ch}(F)))$ . The identity (i) is checked on random data in item (XXVIII) of Appendix D. ■

**Theorem 18.29** (Reduction to the factor). *Let  $(X, t)$  be a polarised abelian variety of dimension  $n \geq 3$ , let  $d \geq 1$ , and let  $F$  be a perfect complex on  $X$  with  $\text{ch}(F) = a u_t + b v_t$  for some  $(a, b) \in \mathbb{Q}^2 \setminus \{0\}$ , in the notation of (29). Then:*

- (i) *No nonzero class in  $HT^1(X)$  preserves  $\text{ch}(F)$ : the map  $HT^1(X) \rightarrow H^*(X, \mathbb{C})$ ,  $y \mapsto y \lrcorner \text{ch}(F)$ , is injective. Its two lowest components form the matrix  $\begin{pmatrix} a & b \\ b & -ad \end{pmatrix}$  of determinant  $-(a^2 d + b^2) = -N_{K/\mathbb{Q}}(b + a\sqrt{-d}) \neq 0$ .*
- (ii) *The classes in  $HT^2(X)$  preserving  $\text{ch}(F)$  to first order are*

$$\text{Ann}_{HT^2}(\text{ch } F) = \{(0, v, 0) : v \lrcorner t = 0\} \oplus \{(\frac{d}{2} \pi \lrcorner t^2, 0, \pi) : \pi \in H^0(\bigwedge^2 T_X)\}, \quad (54)$$

*the polarised first order deformations of  $(X, t)$ , of dimension  $n(n+1)/2$ , and the Poisson directions each compensated by a class in  $H^{0,2}$ , of dimension  $n(n-1)/2$ ; the total is  $n^2 = \dim \mathcal{D}_{n,n}$ , and the space does not depend on  $(a, b)$ .*

- (iii)  *$F$  satisfies the weakened criterion if and only if*
  - (a)  *$v \lrcorner \text{At}(F) = 0$  in  $\text{Ext}^2(F, F)$  for every  $v \in H^1(T_X)$  with  $v \lrcorner t = 0$ , that is,  $F$  extends to first order along every polarised deformation of  $(X, t)$ ; and*
  - (b)  *$\text{At}(F)^2 = -dt^2 \cdot \text{id}_F$  in  $\text{Ext}^2(F, F \otimes \Omega_X^2)$ .*

The trace of (b) is automatic:  $\text{tr At}(F)^2 = 2 \text{ch}_2(F) = -ad t^2 = \text{tr}(-dt^2 \cdot \text{id}_F)$ . So (b) says that the trace free part of the square of the Atiyah class vanishes.

- (iv) Let  $F_1, F_2$  be two such complexes,  $G = F_1 \boxtimes F_2$  on  $X \times X$  and  $E = \Phi(G)$  on  $X \times \widehat{X}$ , with  $\Phi$  Orlov's equivalence as in [Proposition 18.25\(v\)](#). Then  $\text{Ann}_{HT^2}(\text{ch } G) = \text{Ann}_{HT^2}(\text{ch } F_1) \otimes 1 \oplus 1 \otimes \text{Ann}_{HT^2}(\text{ch } F_2)$ , of dimension  $2n^2$ , and

$E$  satisfies the weakened criterion  $\iff G$  does  $\iff F_1$  and  $F_2$  do.

*Proof.* Choose a basis  $p_1, \dots, p_n$  of  $H^{1,0}(X)$  and a basis  $q_1, \dots, q_n$  of  $H^{0,1}(X)$  with  $t = \sum_a p_a \wedge q_a$ , which is possible because  $t$  is a positive class of type  $(1, 1)$ ; let  $\partial_a$  be the dual basis of  $T$ , so that  $\iota_a p_b = \delta_{ab}$  and  $\iota_a q_b = 0$ . Write  $\text{ch}(F) = \sum_k c_k t^k$  with  $c_k k! = a(-d)^{k/2}$  for  $k$  even and  $c_k k! = b(-d)^{(k-1)/2}$  for  $k$  odd, so that

$$c_{k+1} (k+1)! = -d c_{k-1} (k-1)! \quad (k \geq 1). \quad (55)$$

Two facts about the Hodge algebra are used throughout: for  $v \in H^1(T)$  acting as a derivation,  $v \lrcorner t^k = k (v \lrcorner t) t^{k-1}$ ; and for a bivector  $\pi$  of type  $(2, 0)$  in the tangent directions,  $\pi \lrcorner t = 0$  because  $t$  has a single  $(1, 0)$  slot in each term, so  $\pi \lrcorner t^k = \binom{k}{2} (\pi \lrcorner t^2) t^{k-2}$  by the footnote to [\(48\)](#). Finally, multiplication by  $t$  is injective on  $H^{0,1}$  when  $n \geq 2$  and on  $H^{0,2}$  when  $n \geq 3$ :  $q_a \wedge t$  contains the monomial  $p_c q_c q_a$  for any  $c \neq a$ , and  $q_a q_b \wedge t$  contains  $p_c q_c q_a q_b$  for any  $c \notin \{a, b\}$ , and distinct monomials are independent.

(i) For  $y = (z, \theta)$  the class  $y \lrcorner \text{ch}(F)$  has components of type  $(k-1, k)$  equal to  $c_{k-1} z t^{k-1} + c_k \theta \lrcorner t^k = (c_{k-1} z + k c_k (\theta \lrcorner t)) t^{k-1}$ . For  $k = 1$  this is  $az + b(\theta \lrcorner t)$  and for  $k = 2$  it is  $(bz - ad(\theta \lrcorner t))t$ , by [\(55\)](#). Both  $z$  and  $\theta \lrcorner t = \sum_a \theta^a q_a$  lie in  $H^{0,1}$ , on which  $t$  acts injectively, so  $y \lrcorner \text{ch}(F) = 0$  forces  $(z, \theta \lrcorner t)$  into the kernel of the displayed matrix, whose determinant is  $-a^2 d - b^2 < 0$ . Hence  $z = 0$  and  $\theta \lrcorner t = 0$ , and the latter gives  $\theta = 0$  because  $t$  is nondegenerate.

(ii) For  $x = (z, v, \pi)$  the component of  $x \lrcorner \text{ch}(F)$  of type  $(k-1, k+1)$  is

$$c_{k-1} z t^{k-1} + c_k v \lrcorner t^k + c_{k+1} \pi \lrcorner t^{k+1} = \left( c_{k-1} z + k c_k W + \binom{k+1}{2} c_{k+1} P \right) t^{k-1},$$

with  $W = v \lrcorner t$  and  $P = \pi \lrcorner t^2$ , both in  $H^{0,2}$ . By [\(55\)](#),  $\binom{k+1}{2} c_{k+1} = -\frac{d}{2} c_{k-1}$ , so the bracket is  $c_{k-1} Z + k c_k W$  with  $Z = z - \frac{d}{2} P$ . For  $k = 1$  and  $k = 2$  this gives  $aZ + bW$  and  $(bZ - adW)t$ , and  $t$  is injective on  $H^{0,2}$  because  $n \geq 3$ ; the same determinant forces  $Z = W = 0$ . Conversely  $Z = W = 0$  kills every component. So  $\text{Ann}_{HT^2}(\text{ch } F)$  is the set of  $(z, v, \pi)$  with  $v \lrcorner t = 0$  and  $z = \frac{d}{2} \pi \lrcorner t^2$ , which is [\(54\)](#); the two summands are independent because the first has  $\pi = 0$  and the second is the graph of a linear map from the  $\pi$  component. A  $v \in \text{Hom}(H^{1,0}, H^{0,1})$  has  $v \lrcorner t = \sum_a v(p_a) \wedge q_a$ , which vanishes exactly when the matrix of  $v$  is symmetric; this is the tangent space of the polarised deformations, of dimension  $n(n+1)/2$ , as in [Proposition 18.16\(i\)](#). The bivectors contribute  $\dim \wedge^2 T = n(n-1)/2$ . Nothing in the description involves  $(a, b)$ .

(iii) By (ii) and [\(52\)](#),  $\text{ev}_F$  vanishes on  $\text{Ann}_{HT^2}(\text{ch } F)$  if and only if it vanishes on both summands. On the first it is  $v \mapsto v \lrcorner \text{At}(F)$ , which is (a). On the second it is

$$\frac{d}{2} (\pi \lrcorner t^2) \cdot \text{id}_F + \frac{1}{2} \pi \lrcorner \text{At}(F)^2 = \frac{1}{2} \pi \lrcorner (\text{At}(F)^2 + dt^2 \cdot \text{id}_F),$$

since  $\pi \lrcorner (t^2 \cdot \text{id}_F) = (\pi \lrcorner t^2) \cdot \text{id}_F$ . Now  $\Omega_X^2 = \mathcal{O}_X \otimes \wedge^2 H^{1,0}$ , so  $\text{Ext}^2(F, F \otimes \Omega_X^2) = \text{Ext}^2(F, F) \otimes \wedge^2 H^{1,0}$ , and the contractions with all  $\pi \in \wedge^2 (H^{1,0})^\vee$  vanish on an element of this space if and only if the element is zero, the pairing being perfect. This is (b). For the trace,  $\text{ch}(F) = \text{tr exp At}(F)$  gives  $\text{ch}_2(F) = \frac{1}{2} \text{tr At}(F)^2$ , and  $\text{ch}_2(F) = c_2 t^2 = -\frac{ad}{2} t^2$ , while  $\text{tr}(\text{id}_F) = \text{rk } F = c_0 = a$ .

(iv) The Hochschild cohomology of a product is the tensor product,  $HT^2(X \times X) = HT^2 \otimes HT^0 \oplus HT^1 \otimes HT^1 \oplus HT^0 \otimes HT^2$ , the Atiyah class of  $G$  is  $\text{At}(F_1) \boxtimes 1 + 1 \boxtimes \text{At}(F_2)$  [BF03, Proposition 3.14], [Mar25a, (8.4.1)], so  $\exp \text{At}(G) = \exp \text{At}(F_1) \boxtimes \exp \text{At}(F_2)$ , and  $\text{Ext}^2(G, G)$  is the corresponding Künneth sum of  $\text{Ext}^i(F_1, F_1) \otimes \text{Ext}^j(F_2, F_2)$ . A polyvector on a factor contracts only the forms of that factor, so for  $x_1 \in HT^i(X)$  and  $x_2 \in HT^j(X)$  with  $i + j = 2$ ,

$$(x_1 \otimes x_2) \lrcorner \text{ch}(G) = \pm (x_1 \lrcorner \text{ch}(F_1)) \otimes (x_2 \lrcorner \text{ch}(F_2)), \quad \text{ev}_G(x_1 \otimes x_2) = \pm \text{ev}_{F_1}(x_1) \otimes \text{ev}_{F_2}(x_2),$$

with  $\text{ev}(1) = \text{id}$  on  $HT^0$  and Koszul signs that play no role; this is the decomposition [Mar25a, (8.4.2), (8.4.3)]. Let  $x = x_{20} + x_{11} + x_{02}$  be a class in  $\text{Ann}_{HT^2}(\text{ch } G)$  according to the three summands. Since  $\text{ch}(F_i)$  is even,  $x_{11} \lrcorner \text{ch}(G)$  lies in  $H^{\text{odd}} \otimes H^{\text{odd}}$  while the other two terms lie in  $H^{\text{ev}} \otimes H^{\text{ev}}$ , so  $x_{11} \lrcorner \text{ch}(G) = 0$  separately; and  $x_{11} \lrcorner \text{ch}(G)$  is the image of  $x_{11}$  under the tensor product of the two maps of (i), which is injective, so  $x_{11} = 0$ . For the even part write  $x_{20} = x_1 \otimes 1$  and  $x_{02} = 1 \otimes x_2$ ; then  $(x_1 \lrcorner \text{ch}(F_1)) \otimes \text{ch}(F_2) + \text{ch}(F_1) \otimes (x_2 \lrcorner \text{ch}(F_2)) = 0$ . Applying a linear form on the second factor taking the value 1 on  $\text{ch}(F_2)$  shows  $x_1 \lrcorner \text{ch}(F_1) = \lambda \text{ch}(F_1)$  for a scalar  $\lambda$ , and then  $x_2 \lrcorner \text{ch}(F_2) = -\lambda \text{ch}(F_2)$ . But every component of  $x_1 \lrcorner \text{ch}(F_1)$  has Hodge type  $(p, p+2)$  while  $\text{ch}(F_1) \neq 0$  has type  $(p, p)$ ; so  $\lambda = 0$  and  $x_i \in \text{Ann}_{HT^2}(\text{ch } F_i)$ . This proves the description of  $\text{Ann}_{HT^2}(\text{ch } G)$  and its dimension  $2n^2$  by (ii). On it  $\text{ev}_G(x_1 \otimes 1 + 1 \otimes x_2) = \pm \text{ev}_{F_1}(x_1) \otimes \text{id}_{F_2} \pm \text{id}_{F_1} \otimes \text{ev}_{F_2}(x_2)$ , whose two terms lie in different Künneth summands and vanish exactly when  $\text{ev}_{F_i}(x_i) = 0$ , as  $\text{id}_{F_i} \neq 0$ . So  $G$  satisfies the criterion if and only if  $F_1$  and  $F_2$  do.

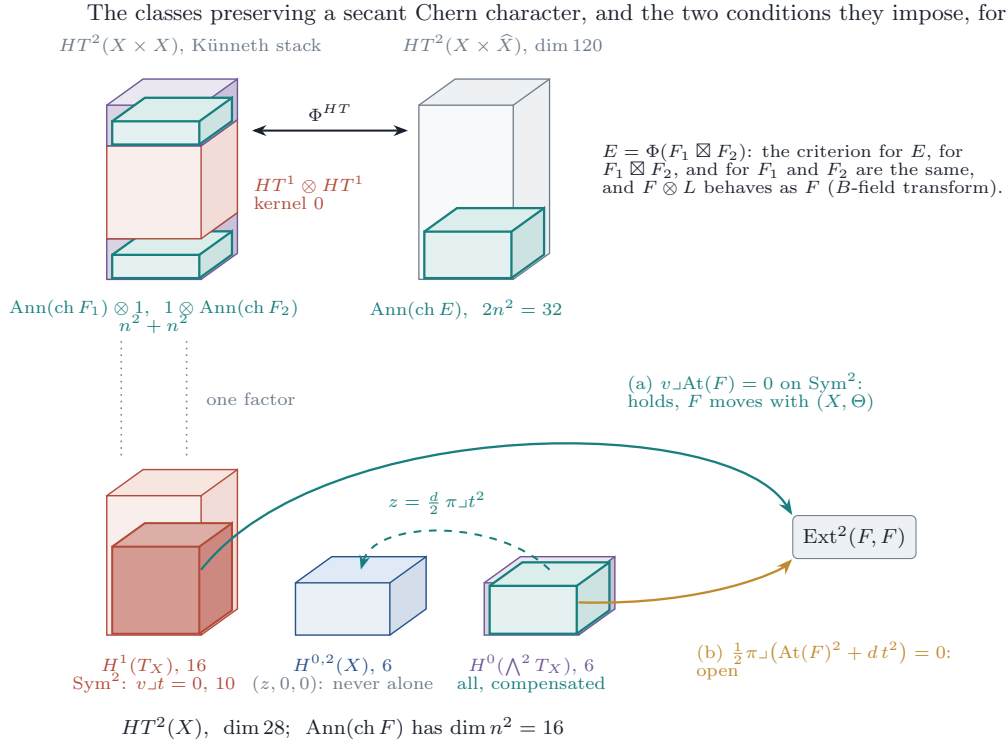
For the transport under  $\Phi$  we use the commutative diagram of [Mar25a, Lemma 8.3.10]: an equivalence  $\Phi: D^b(A) \rightarrow D^b(B)$  of derived categories of abelian varieties induces isomorphisms  $\Phi^{HT}: HT^2(A) \rightarrow HT^2(B)$  and  $\Phi^H: H^*(A, \mathbb{C}) \rightarrow H^*(B, \mathbb{C})$  with  $\Phi^H(x \lrcorner \text{ch } G) = \Phi^{HT}(x) \lrcorner \text{ch } \Phi(G)$  [CRV12], and  $\Phi \circ \text{ev}_G = \text{ev}_{\Phi(G)} \circ \Phi^{HT}$  on  $\text{Hom}(G, G[2]) \rightarrow \text{Hom}(\Phi G, \Phi G[2])$  [Hua21, Theorem A]; on abelian varieties the Todd class is trivial, so there is no Duflo correction [Mar25a, Remark 8.3.11]. The first square gives  $\Phi^{HT}(\text{Ann}_{HT^2}(\text{ch } G)) = \text{Ann}_{HT^2}(\text{ch } E)$  and the second  $\text{ev}_E(\text{Ann}_{HT^2}(\text{ch } E)) = \Phi(\text{ev}_G(\text{Ann}_{HT^2}(\text{ch } G)))$ , with  $\Phi$  an isomorphism on  $\text{Ext}^2$ . So  $E$  satisfies the criterion if and only if  $G$  does. Items (i) and (ii) are recomputed in the model of Appendix D, item (XXVIII), for  $n = 3, 4, 5$  and several  $(d, a, b)$ , together with the controls that the bare Poisson class  $(0, 0, \pi)$  and the class compensated with the opposite sign do not preserve  $\text{ch}(F)$ . ■

**Corollary 18.30** (The one condition, on the fourfold). *In Setup 18.24 let  $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$ . Then  $E$  satisfies the weakened criterion of [Mar25b, Question 11.4] if and only if*

$$\text{At}(\mathcal{F})^2 = -d \Theta^2 \cdot \text{id}_{\mathcal{F}} \quad \text{in } \text{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2). \quad (56)$$

*Equivalently, writing  $\text{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$  with  $A_a \in \text{Ext}^1(\mathcal{F}, \mathcal{F})$  in a basis  $p_a$  of  $H^{1,0}(X)$ , all the commutators  $A_a A_b - A_b A_a \in \text{Ext}^2(\mathcal{F}, \mathcal{F})$  are the scalars  $\pm 2d(q_a \wedge q_b) \cdot \text{id}_{\mathcal{F}}$  prescribed by  $\Theta^2$ , the sign being fixed by the Koszul rule.*

*Proof.* By Proposition 18.25(iii),  $\text{ch}(\mathcal{F}) = u_{\Theta} + 3v_{\Theta}$ , so Theorem 18.29 applies with  $n = 4$ ,  $(a, b) = (1, 3)$  and  $F_1 = F_2 = \mathcal{F}$ , and  $E$  satisfies the criterion if and only if  $\mathcal{F}$  satisfies (a) and (b). It remains to see that (a) holds. The data of Setup 18.24 are a principally polarised fourfold  $(X, \Theta)$  and  $N$  points  $x_i \in X$  with  $D_i = \tau_{x_i}^* \Theta$ ; the conditions imposed on the  $D_i$ , that the intersections of two, three and four of them be smooth of the expected codimension and that five of them be disjoint, are open in the parameter space of such data over the local universal deformation  $S$  of  $(X, \Theta)$ , by the openness of smoothness for



$2n^2$  classes on the eightfold,  $n^2$  on each factor:  $n(n+1)/2$  honest deformations of  $(X, t)$  and  $n(n-1)/2$  Poisson directions with a gerbe compensation. The weakened criterion for  $\Phi(F \boxtimes F)$  is (a) and (b), and (b) is  $\text{At}(F)^2 = -d t^2 \cdot \text{id}_F$  in  $\text{Ext}^2(F, F \otimes \Omega^2)$ : the trace free part of the square of the Atiyah class vanishes.

FIGURE 23. [Theorem 18.29](#) for  $n = 4$ . Top: the Künneth stack of  $HT^2(X \times X)$ , whose middle summand  $HT^1 \otimes HT^1$  meets the kernel in zero by (i), so that the classes preserving  $\text{ch}(F_1 \boxtimes F_2)$  are two copies of the kernel on one factor; across Orlov's isomorphism,  $HT^2$  of the eightfold with the  $2n^2$  classes preserving  $\text{ch}(E)$ . Bottom:  $HT^2(X)$  as three slabs with heights proportional to their dimensions, and inside them the  $n^2$  classes of (54): the polarised deformations in front of  $H^1(T_X)$ , and the whole of  $H^0(\wedge^2 T_X)$ , each bivector tilted into  $H^{0,2}$  by its compensation  $\frac{d}{2} \pi \lrcorner t^2$ ; nothing in  $H^{0,2}$  alone preserves  $\text{ch}(F)$ . On the right, the two conditions that the evaluation map imposes on the two parts, (a) closed by the construction of  $\mathcal{F}$  and (b) the identity (56).

a proper flat family and the properness of the projection; and the isolated points of  $Z$ , being transversal intersections of four divisors, form a finite étale cover of the open set where the conditions hold, so the  $m$  points to be normalised can be followed locally. The partial normalisation separates two smooth branches at each of them and commutes with base change. So the construction produces a perfect complex  $\mathcal{F}_S$  on the universal family over a neighbourhood of the base point in  $S$ , whose fibre at the base point is  $\mathcal{F}$  and whose base has Kodaira–Spencer image the whole space of polarised deformations, of dimension  $10 = n(n+1)/2$ . The obstruction  $v \lrcorner \text{At}(\mathcal{F})$  to extending  $\mathcal{F}$  along  $v$  vanishes for every  $v$  along which  $\mathcal{F}$  does extend [Tod09], which is (a).

For the reformulation,  $\Omega_X^1 = \mathcal{O}_X \otimes H^{1,0}$  gives  $\text{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$ , and  $\text{At}(\mathcal{F})^2 = -\sum_{a<b} (A_a A_b - A_b A_a) \otimes p_a \wedge p_b$  by the Koszul rule, while  $\Theta^2 = -2 \sum_{a<b} (p_a \wedge p_b)(q_a \wedge q_b)$  for  $\Theta = \sum_a p_a \wedge q_a$ . Comparing coefficients of  $p_a \wedge p_b$  gives the commutator relations. ■

*Remark 18.31* (What the reduction closes, and what it does not). Three things are closed. First, the eightfold has disappeared: the condition of Remark 18.26, on an object of rank  $8d(d-9)$  on  $X \times \hat{X}$  of Weil type, is equivalent to one identity (56) for a rank one complex on the fourfold, in the Yoneda algebra  $\text{Ext}^*(\mathcal{F}, \mathcal{F})$ ; the dimension of the space of classes that has to be killed drops from  $2n^2$  to  $n^2$  and, by (a), to  $n(n-1)/2 = 6$ ; at  $n = 3$  the count  $n^2 = 9$  is the nine-dimensional kernel of [Mar25a, Corollaries 8.3.12 and 8.5.2], now written out for every secant object in every dimension. Second, the half of that space made of honest deformations imposes nothing, because Markman’s construction is uniform over the moduli of principally polarised fourfolds. Third, the two forms of the criterion, for  $E$  with  $\text{ch}(E)$  and for the twisted object with  $\kappa(E)$ , are the same by Lemma 18.28, so nothing is lost by working with the untwisted object.

The identity itself is a Heisenberg relation: the components of the Atiyah class in the  $n$  tangent directions must commute up to the scalars prescribed by  $\Theta^2$ , and the only part of it that the Chern character forces is the trace. It is exactly the statement that  $\mathcal{F}$  deforms, as an object of the deformed category, along the Poisson directions of  $X$  compensated by the gerbe classes  $\frac{d}{2} \pi \lrcorner \Theta^2$ , the noncommutative first order deformations of  $X$  with a gerby correction in the language of [Mar25a, Section 8.5]. It holds in the one case where the method is known to work: for Markman’s sheaves  $\mathcal{F}_1 = I_{\bigcup C_i}(\Theta)$  on the Jacobian threefold of [Mar25b, Section 11.1], with  $\text{ch} = u_\Theta + v_\Theta$  and  $d = q$ , the kernel of the obstruction map equals the annihilator of the Chern character [Mar25a, Corollary 8.3.12], which is stronger than the weakened criterion, so by Theorem 18.29(iii) these sheaves satisfy  $\text{At}(\mathcal{F}_1)^2 = -q \Theta^2 \cdot \text{id}$  on the threefold. For the rank one complex of Setup 18.24 on the fourfold the identity is a computation of Yoneda products in  $\text{Ext}^*(\mathcal{F}, \mathcal{F})$ , and it can be made local: the next paragraphs carry it out at one point of the support and find that it fails.

*The identity fails at an isolated double point.* The identity (56) can be tested locally. The evaluation map is an algebra homomorphism [Mar25a, (8.3.3)], so on a bivector  $\theta \wedge \theta'$  it is the Yoneda product of the classes

$$A_\theta = \text{ev}_{\mathcal{F}}(\theta) = \theta \lrcorner \text{At}(\mathcal{F}) \in \text{Ext}^1(\mathcal{F}, \mathcal{F}), \quad \theta \in H^0(T_X),$$

which are the first order deformations of  $\mathcal{F}$  along the translations of  $X$ ; and the product of two global classes restricts to the product of their germs. At a point where the support of  $\mathcal{F}$  has two smooth branches of codimension two meeting transversally, the germs multiply nontrivially, and since the compensating classes  $z \cdot \text{id}_{\mathcal{F}}$  have zero germs, the identity fails. Markman’s construction has such points, and it cannot avoid them.

**Lemma 18.32** (Localisation). *Let  $F$  be a perfect complex on a smooth projective variety  $Y$ , let  $\mathcal{E}xt^m(F, F)$  be the sheaf associated with  $U \mapsto \mathrm{Hom}_{D(U)}(F|_U, F|_U[m])$ , and let  $e: \mathrm{Ext}^m(F, F) \rightarrow H^0(Y, \mathcal{E}xt^m(F, F))$  send a morphism  $F \rightarrow F[m]$  to its germs. Then  $e$  is compatible with Yoneda products;  $e(z \cdot \mathrm{id}_F) = 0$  for  $z \in H^m(\mathcal{O}_Y)$ ,  $m \geq 1$ ; and for  $\theta \in H^0(T_Y)$  the germ of  $A_\theta$  at  $p$  is the class of the jet extension  $0 \rightarrow F_p \rightarrow J_\theta(F_p) \rightarrow F_p \rightarrow 0$  of the germ  $F_p$  along the derivation  $\theta$  of  $\mathcal{O}_{Y,p}$ , which is  $\mathcal{O}_{Y,p}$ -linear in  $\theta$ .*

*Proof.* Composition of morphisms commutes with restriction to open sets, which is the first assertion. For the second,  $z \cdot \mathrm{id}_F$  restricted to an affine open  $U$  is  $z|_U \cdot \mathrm{id}_{F|_U}$ , and  $z|_U \in H^m(U, \mathcal{O}_U) = 0$ . For the third, the Atiyah class of  $F$  is the class of the first jet sequence  $0 \rightarrow F \otimes \Omega_Y^1 \rightarrow J^1(F) \rightarrow F \rightarrow 0$  [BF03, Section 3], its restriction to  $U$  is the jet sequence of  $F|_U$ , and contracting with  $\theta$  is pushing out along  $\mathrm{id} \otimes \theta: F \otimes \Omega^1 \rightarrow F$ , which produces the extension  $J_\theta(F)$  whose underlying sheaf is  $F \oplus F$  with  $f \cdot (s, s') = (fs + \theta(f)s', fs')$ . Linearity in  $\theta$  is linearity of the contraction. ■

**Lemma 18.33** (Two branches through a point). *Let  $\mathcal{O}$  be the local ring of  $\mathbb{C}^4$  at the origin or its completion  $\mathbb{C}[[x_1, \dots, x_4]]$ ; let  $P_1 = \{x_1 = x_2 = 0\}$  and  $P_2 = \{x_3 = x_4 = 0\}$ , two smooth surfaces meeting transversally at the origin; put  $N_1 = \langle \partial_1, \partial_2 \rangle$  and  $N_2 = \langle \partial_3, \partial_4 \rangle$ , the normal spaces of  $P_1$  and of  $P_2$  at the origin, so that  $\wedge^2 T_0 = \wedge^2 N_1 \oplus N_1 \otimes N_2 \oplus \wedge^2 N_2$ . For a germ of vector field  $\theta$  and a perfect complex  $F$  of  $\mathcal{O}$ -modules write  $a_\theta \in \mathrm{Ext}_{\mathcal{O}}^1(F, F)$  for the class of the jet extension along  $\theta$ .*

- (i) *For  $F = I = I_{P_1} \cap I_{P_2}$ , the ideal of the union,  $\mathrm{Ext}_{\mathcal{O}}^2(I, I) \cong N_1 \otimes N_2$ , a vector space of dimension four on which the maximal ideal acts trivially, and  $a_\theta a_{\theta'}$  is, up to a fixed nonzero scalar, the component of  $\theta(0) \wedge \theta'(0)$  in  $N_1 \otimes N_2$ . In particular  $a_\theta a_{\theta'} \neq 0$  unless  $\theta(0) \wedge \theta'(0) \in \wedge^2 N_1 \oplus \wedge^2 N_2$ .*
- (ii) *For  $F = K = [\mathcal{O} \rightarrow \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$ , the complex in degrees 0 and 1 whose cohomology is  $I$  in degree 0 and the residue field  $k$  in degree 1,  $a_\theta a_{\theta'} \neq 0$  unless  $\theta(0) \wedge \theta'(0) \in N_1 \otimes N_2$ .*

*In both cases the bilinear map  $(\theta, \theta') \mapsto a_\theta a_{\theta'}$  is nonzero, and it depends only on  $\theta(0) \wedge \theta'(0)$ .*

*Proof.* Write  $A = \mathbb{C}[x_1, x_2]$ ,  $B = \mathbb{C}[x_3, x_4]$ ,  $R = A \otimes_{\mathbb{C}} B$ , with maximal ideals  $\mathfrak{m}_A, \mathfrak{m}_B$  at the origin, and  $k = \mathbb{C}$ . The ideal of the union is  $I = \mathfrak{m}_A R \cap \mathfrak{m}_B R = \mathfrak{m}_A \otimes_{\mathbb{C}} \mathfrak{m}_B$ , because the two ideals are generated by disjoint sets of variables. Everything below is homogeneous, the Ext groups in degree two are of finite length, and finite length modules and jet classes are unchanged by the flat base change from  $R$  to  $\mathcal{O}$ ; so we compute over  $R$ . For finitely generated modules  $M, M'$  over  $A$  and  $N, N'$  over  $B$ , the tensor product of free resolutions resolves  $M \otimes_{\mathbb{C}} N$  and the Künneth formula gives

$$\mathrm{Ext}_R^n(M \otimes N, M' \otimes N') = \bigoplus_{i+j=n} \mathrm{Ext}_A^i(M, M') \otimes_{\mathbb{C}} \mathrm{Ext}_B^j(N, N'), \quad (57)$$

compatibly with Yoneda products up to the Koszul sign, since a product of chain maps on the tensor product of resolutions is the tensor product of the products.

The building block is one plane. The Koszul complex  $0 \rightarrow A \xrightarrow{(-x_2, x_1)^t} A^2 \xrightarrow{(x_1, x_2)} \mathfrak{m}_A \rightarrow 0$  is a free resolution, so  $\mathfrak{m}_A$  has projective dimension one and  $\mathrm{Ext}_A^i(\mathfrak{m}_A, -) = 0$  for  $i \geq 2$ ; and  $\mathrm{Ext}_A^1(\mathfrak{m}_A, \mathfrak{m}_A) = \mathrm{coker}(\mathrm{Hom}(A^2, \mathfrak{m}_A) \rightarrow \mathrm{Hom}(A, \mathfrak{m}_A)) = \mathfrak{m}_A / \mathfrak{m}_A^2$ , a two-dimensional space killed by  $\mathfrak{m}_A$ . On a complex of free modules with differential  $d$  the jet extension along a derivation  $D$  is the complex with terms  $P_i \oplus P_i$  and differential  $\begin{pmatrix} d & -D(d) \\ 0 & d \end{pmatrix}$ ,  $D(d)$  the entrywise derivative, because the twisted module  $J_D(P_i)$  is free on the elements  $(e, 0)$  and  $(0, e)$  and  $d(0, e) = \sum_l d_{le}(0, e_l) - \sum_l D(d_{le})(e_l, 0)$ ; so the jet class is represented by the

degree  $-1$  chain map  $-D(d)$ . For  $D = \partial_i$  on the Koszul complex this is  $e \mapsto \delta_{i2}f_1 - \delta_{i1}f_2$  on the generator  $e$  of the last term, whose image in  $\mathfrak{m}_A/\mathfrak{m}_A^2$  is  $\delta_{i2}x_1 - \delta_{i1}x_2$ :

$$\alpha_1 := a_{\partial_1} = -[x_2], \quad \alpha_2 := a_{\partial_2} = [x_1],$$

a basis of  $\text{Ext}_A^1(\mathfrak{m}_A, \mathfrak{m}_A)$ , which we identify with  $N_1$  by  $\partial_i \mapsto \alpha_i$ . Similarly  $\beta_3, \beta_4$  form a basis of  $\text{Ext}_B^1(\mathfrak{m}_B, \mathfrak{m}_B)$ , identified with  $N_2$ .

(i) By (57) with  $M = M' = \mathfrak{m}_A$ ,  $N = N' = \mathfrak{m}_B$ ,

$$\text{Ext}_R^2(I, I) = \text{Ext}_A^1 \otimes \text{Ext}_B^1 = N_1 \otimes N_2,$$

the other two summands vanishing by the projective dimension, and it is killed by  $\mathfrak{m}_A$  and  $\mathfrak{m}_B$ . The derivation  $\partial_1$  acts on the factor  $A$ , so the jet extension of  $I = \mathfrak{m}_A \otimes \mathfrak{m}_B$  along it is  $J_{\partial_1}(\mathfrak{m}_A) \otimes \mathfrak{m}_B$  and  $a_{\partial_1} = \alpha_1 \otimes 1$ ; likewise  $a_{\partial_2} = \alpha_2 \otimes 1$ ,  $a_{\partial_3} = 1 \otimes \beta_3$ ,  $a_{\partial_4} = 1 \otimes \beta_4$ . Hence  $a_{\partial_i}a_{\partial_j} = \pm\alpha_i \otimes \beta_j$  for  $i \leq 2 < j$ , the four of them a basis of  $N_1 \otimes N_2$ , while  $a_{\partial_1}a_{\partial_2} = \alpha_1\alpha_2 \otimes 1 = 0$  and  $a_{\partial_3}a_{\partial_4} = 0$  because  $\text{Ext}_A^2(\mathfrak{m}_A, \mathfrak{m}_A) = 0$ . For germs  $\theta = \sum_i c^i \partial_i$  and  $\theta' = \sum_j c'^j \partial_j$  the linearity of Lemma 18.32 and the triviality of the module structure give  $a_{\theta}a_{\theta'} = \sum_{i,j} c^i(0)c'^j(0)a_{\partial_i}a_{\partial_j}$ , the image of  $\theta(0) \wedge \theta'(0)$  under the projection of  $\wedge^2 T_0$  onto  $N_1 \otimes N_2$  followed by the isomorphism  $\partial_i \otimes \partial_j \mapsto \pm\alpha_i \otimes \beta_j$ .

(ii) Let  $Q = \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}$ . The map  $\mathcal{O} \rightarrow Q$  has kernel  $I$  and cokernel  $\mathcal{O}/(I_{P_1} + I_{P_2}) = k$ , so  $K$  sits in an exact triangle  $I \xrightarrow{\iota} K \xrightarrow{\pi} k[-1] \xrightarrow{\varepsilon} I[1]$ . The jet extension along  $\theta$  is an exact functor, natural in its argument, so the jet classes commute with  $\iota$  and  $\pi$ ; and for the residue field  $a_{\theta}^k$  depends only on  $\theta(0)$ , because  $\text{Ext}^1(k, k)$  is killed by  $\mathfrak{m}$  and the class is linear in  $\theta$ . Therefore

$$\pi[2] \circ a_{\theta}^K a_{\theta'}^K = a_{\theta}^k a_{\theta'}^k \circ \pi = \pi^*(a_{\theta(0)}^k a_{\theta'(0)}^k), \quad \pi^*: \text{Ext}^2(k, k) \rightarrow \text{Ext}^1(K, k),$$

and it suffices to show that the right hand side is nonzero. The long exact sequence of the triangle for  $\text{Hom}(-, k)$  reads

$$\text{Hom}(I, k) \xrightarrow{\varepsilon^*} \text{Ext}^2(k, k) \xrightarrow{\pi^*} \text{Ext}^1(K, k),$$

so  $\ker \pi^* = \text{im } \varepsilon^*$ . Now  $k = k_A \otimes k_B$  and, by (57),  $\text{Ext}^*(k, k)$  is the tensor product of  $\text{Ext}_A^*(k, k)$  and  $\text{Ext}_B^*(k, k)$ . On the Koszul resolution  $A \rightarrow A^2 \rightarrow A$  of  $k_A$  the jet chain maps of  $\partial_1, \partial_2$  are  $-D(d)$  as above, and their composite  $A \rightarrow A$  is the scalar  $-1$ , whose class in  $\text{Ext}_A^2(k, k) = \text{Hom}(A, k) = k$  is nonzero; the degree one classes  $\gamma_1, \gamma_2$  of  $\partial_1, \partial_2$  are the basis  $-e_1^*, -e_2^*$  of  $\text{Ext}_A^1(k, k) = \text{Hom}(A^2, k)$ , and their product spans  $\text{Ext}_A^2(k, k)$ . With the analogous classes  $\gamma_3, \gamma_4$  over  $B$ , the products  $\gamma_1\gamma_2 \otimes 1$ ,  $\gamma_i \otimes \gamma_j$  ( $i \leq 2 < j$ ) and  $1 \otimes \gamma_3\gamma_4$  form a basis of  $\text{Ext}^2(k, k) = k \oplus (k^2 \otimes k^2) \oplus k$ , so  $\partial_i \wedge \partial_j \mapsto a_{\partial_i}^k a_{\partial_j}^k$  identifies  $\wedge^2(N_1 \oplus N_2)$  with  $\text{Ext}^2(k, k)$ , and  $a_{\theta(0)}^k a_{\theta'(0)}^k$  is the image of  $\theta(0) \wedge \theta'(0)$ . The class  $\varepsilon$  is the Künneth product  $\varepsilon_A \otimes \varepsilon_B$  of the classes of  $0 \rightarrow \mathfrak{m}_A \rightarrow A \rightarrow k \rightarrow 0$  and its  $B$ -analogue, because  $K$  is the two-step truncation of the tensor product of the complexes  $[A \rightarrow k_A]$  and  $[B \rightarrow k_B]$  and  $\varepsilon$  is the inclusion of the omitted term  $k_A \otimes k_B$ . Hence  $\varepsilon^*$  maps  $\text{Hom}(I, k) = \text{Hom}(\mathfrak{m}_A, k) \otimes \text{Hom}(\mathfrak{m}_B, k)$  into  $\text{Ext}_A^1(k, k) \otimes \text{Ext}_B^1(k, k) = N_1 \otimes N_2$ , and onto it, since  $\text{Hom}(\mathfrak{m}_A, k) \rightarrow \text{Ext}_A^1(k, k)$  is an isomorphism: it is the connecting homomorphism of the sequence, injective because  $\text{Hom}(A, k) \rightarrow \text{Hom}(\mathfrak{m}_A, k)$  is zero and surjective because  $\text{Ext}_A^1(A, k) = 0$ . So  $\ker \pi^* = N_1 \otimes N_2$  and  $\pi^*(\theta(0) \wedge \theta'(0)) \neq 0$  exactly when  $\theta(0) \wedge \theta'(0) \notin N_1 \otimes N_2$ .

Both computations are repeated by machine in item (XXIX) of Appendix D, which also finds  $\text{Ext}_{\mathcal{O}}^2(K, K) \cong \wedge^2 N_1 \oplus \wedge^2 N_2$  with  $a_{\theta}a_{\theta'}$  the projection of  $\theta(0) \wedge \theta'(0)$  to it, the exact complement of (i).  $\blacksquare$

**Theorem 18.34** (Markman’s candidate does not satisfy the weakened criterion). *In Setup 18.24, for every odd  $d \geq 3$ , every choice of the translates  $D_i$  in general position and every choice of the  $m$  points to be normalised:*

- (i) *the identity (56) fails: there is a constant bivector  $\pi \in H^0(\wedge^2 T_X)$  with  $\frac{1}{2}\pi \lrcorner (\text{At}(\mathcal{F})^2 + d\Theta^2 \cdot \text{id}_{\mathcal{F}}) \neq 0$  in  $\text{Ext}^2(\mathcal{F}, \mathcal{F})$ ;*
- (ii)  *$\mathcal{F}$  does not satisfy the weakened criterion: the class  $x_{\mathcal{F}} = (\frac{d}{2}\pi \lrcorner \Theta^2, 0, \pi)$  has  $x_{\mathcal{F}} \lrcorner \text{ch}(\mathcal{F}) = 0$  and  $\text{ev}_{\mathcal{F}}(x_{\mathcal{F}}) \neq 0$ ;*
- (iii) *for every perfect complex  $F_2$  on  $X$  with Chern character on the secant plane, in particular for  $F_2 = \mathcal{F}$  and  $F_2 = \mathcal{F}^\vee$ , the object  $E = \Phi(\mathcal{F} \boxtimes F_2)$  on the eightfold  $X \times \hat{X}$  of Weil type does not satisfy the weakened criterion of [Mar25b, Question 11.4]: the class  $x = \Phi^{HT}(x_{\mathcal{F}} \otimes 1)$  satisfies  $x \lrcorner \text{ch}(E) = 0$  and  $\text{ev}_E(x) \neq 0$ , so  $\sigma_E$  is not injective on the image of  $\text{ev}_E$ .*

*Proof.* (i) The surfaces  $Z_i$  and  $Z_j$  at cyclic distance at least three meet in the 24 points of  $D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$ , and there are  $12N(N-5) = (d+9)(3d-3)$  such points, of which  $m = (d+9)(2d-1)$  are normalised; the remaining  $(d+9)(d-2) \geq 12$  are not. Let  $p$  be one of them. The four divisors through  $p$  meet transversally, so their local equations  $x_1, \dots, x_4$  form a regular system of parameters of  $\mathcal{O}_{X,p}$  and identify its completion with  $\mathbb{C}[[x_1, \dots, x_4]]$ , with  $D_i = \{x_1 = 0\}$ ,  $D_{i+1} = \{x_2 = 0\}$ ,  $D_j = \{x_3 = 0\}$ ,  $D_{j+1} = \{x_4 = 0\}$ , hence  $Z_i = P_1$  and  $Z_j = P_2$  in the notation of Lemma 18.33; the completion is flat, so a class whose completed germ is nonzero is nonzero. No other  $D_k$  passes through  $p$ , five divisors having empty intersection, so no other component of  $Z$  does. Near  $p$  the map  $\nu$  is an isomorphism,  $\nu_*\mathcal{O}_{\tilde{Z}} = \mathcal{O}_Z$ , and  $\mathcal{F}$  is the ideal sheaf  $I_Z = I_{P_1} \cap I_{P_2}$  twisted by the locally trivial  $\mathcal{O}_X(3\Theta)$ : the germ  $\mathcal{F}_p$  is the module  $I$  of Lemma 18.33(i).

Let  $\theta, \theta' \in H^0(T_X)$  be constant vector fields whose values at  $p$  are  $\partial_1$  and  $\partial_3$ ; they exist because the constant fields span every tangent space. Put  $\pi = \theta \wedge \theta'$ . Since  $\text{ev}_{\mathcal{F}}$  is an algebra homomorphism,  $\text{ev}_{\mathcal{F}}(\pi) = \frac{1}{2}\pi \lrcorner \text{At}(\mathcal{F})^2 = A_{\theta}A_{\theta'}$ , and by Lemma 18.32 its germ at  $p$  is  $a_{\theta}a_{\theta'}$ , which is nonzero by Lemma 18.33(i),  $\partial_1 \wedge \partial_3$  having nonzero component in  $N_1 \otimes N_2$ . The germ of  $\frac{d}{2}(\pi \lrcorner \Theta^2) \cdot \text{id}_{\mathcal{F}}$  at  $p$  is zero by the same lemma. So the germ at  $p$  of  $\frac{1}{2}\pi \lrcorner (\text{At}(\mathcal{F})^2 + d\Theta^2 \cdot \text{id}_{\mathcal{F}})$  is  $a_{\theta}a_{\theta'} \neq 0$ , and the class itself is nonzero.

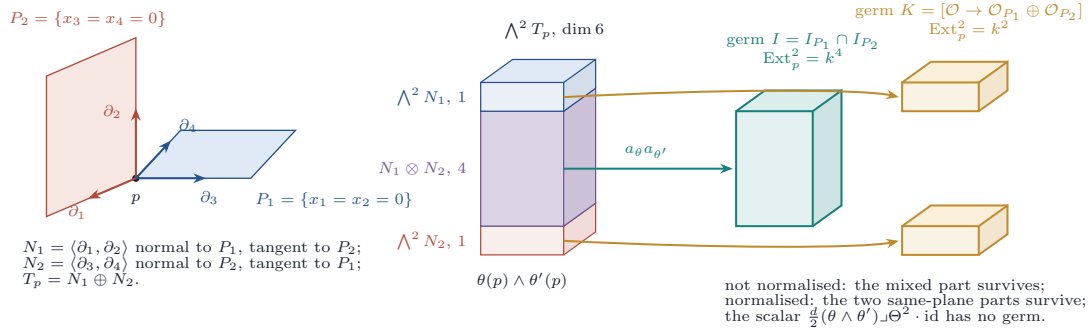
(ii) By Theorem 18.29(ii) the class  $x_{\mathcal{F}}$  preserves  $\text{ch}(\mathcal{F})$ , and by (52)  $\text{ev}_{\mathcal{F}}(x_{\mathcal{F}})$  is the class of (i).

(iii)  $\text{ch}(\mathcal{F}^\vee) = u_{\Theta} - 3v_{\Theta}$  lies on the secant plane. By Theorem 18.29(iv),  $E$  satisfies the criterion only if both factors do, and  $\mathcal{F}$  does not; the displayed class is the transport of  $x_{\mathcal{F}} \otimes 1 \in \text{Ann}_{HT^2}(\text{ch}(\mathcal{F} \boxtimes F_2))$ , on which  $\text{ev}$  is  $\pm \text{ev}_{\mathcal{F}}(x_{\mathcal{F}}) \otimes \text{id}_{F_2} \neq 0$ . The last assertion is Setup 18.27:  $\sigma_E(\text{ev}_E(x)) = x \lrcorner \text{ch}(E) = 0$  with  $\text{ev}_E(x) \neq 0$ . ■

*Remark 18.35* (The normalised points obstruct as well). At a normalised point the germ of  $\mathcal{F}$  is the complex  $K$  of Lemma 18.33(ii), and the same argument with  $\theta(p) = \partial_1$ ,  $\theta'(p) = \partial_2$  produces a nonzero germ. So the failure does not depend on which points are separated, and separating all of them would not help even if it were allowed: normalising every isolated point changes  $\chi(\mathcal{O}_{\tilde{Z}})$  to  $110N - 12N^2 + 12N(N-5) = 50N$ , while the secant plane requires  $(d+9)(27-d) = 72N - 4N^2$ , and the two agree only for  $N = 11/2$ . The two local computations are complementary: at a point of type (i) the obstruction lives in  $N_1 \otimes N_2$ , at a point of type (ii) in  $\wedge^2 N_1 \oplus \wedge^2 N_2$ , and together they exhaust  $\wedge^2 T_p$ .

**Corollary 18.36** (A local obstruction in every dimension). *Let  $(X, t)$  be a polarised abelian variety of dimension  $n \geq 4$  and  $F$  a perfect complex on  $X$  with  $\text{ch}(F) = au_t + bv_t$ ,  $(a, b) \neq (0, 0)$ . Suppose that at some point  $p$  the germ  $F_p$  is isomorphic, in the derived*

At an isolated double point of the support the translation classes multiply to a nonzero germ.



At every isolated double point of  $Z$ , of either kind, some constant bivector  $\pi$  has  $\text{ev}_{\mathcal{F}}(\pi) + \frac{d}{2}(\pi \lrcorner \Theta^2) \text{id} \neq 0$ ; the class  $\pi + \frac{d}{2} \pi \lrcorner \Theta^2$  preserves  $\text{ch}(\mathcal{F})$ , so the weakened criterion fails for  $\mathcal{F}$  and for  $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ .

FIGURE 24. [Lemma 18.33](#) and [Theorem 18.34](#). Left: two smooth surfaces of a fourfold meeting transversally at  $p$ , with the normal spaces  $N_1$  and  $N_2$ , each tangent to the other branch. Centre: the bivectors at  $p$ , split into  $\Lambda^2 N_1$ ,  $N_1 \otimes N_2$  and  $\Lambda^2 N_2$ , with heights proportional to the dimensions. Right: the local  $\text{Ext}^2$  of the germ of  $\mathcal{F}$  at  $p$  in the two cases, and the part of the bivectors that the products of the translation classes carry onto it: the mixed part when the point is not normalised, the two same-plane parts when it is. The compensating scalar has no germ, so in either case the identity (56) fails at  $p$ .

category of  $\mathcal{O}_{X,p}$ -modules, to the ideal of the union of two smooth germs of codimension two meeting transversally along a germ of codimension four, or, when  $n = 4$ , to the complex  $[\mathcal{O} \rightarrow \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$  of two such germs. Then  $F$  does not satisfy the weakened criterion, and neither does  $\Phi(F \boxtimes F_2)$  on  $X \times \hat{X}$  for any secant  $F_2$ . In particular, in every dimension  $n \geq 4$ , the union  $Z$  of the intersections  $D_i \cap D_{i+1}$  of consecutive translates of a polarisation in general position contains, for every pair  $(i, j)$  of nonconsecutive indices, a subvariety  $Z_i \cap Z_j$  of codimension four along which  $Z$  has two transversal branches; any secant object of the form  $[\mathcal{O}_X \rightarrow \nu_* \mathcal{O}_{\tilde{Z}}] \otimes L$  in which  $\nu$  is an isomorphism at a general point of one such  $Z_i \cap Z_j$  fails the criterion.

*Proof.* In formal coordinates  $x_1, \dots, x_n$  at  $p$  with the two branches  $\{x_1 = x_2 = 0\}$  and  $\{x_3 = x_4 = 0\}$ , which exist because the four local equations of the branches form part of a regular system of parameters, the ideal is  $I \otimes_{\mathbb{C}} \mathbb{C}[x_5, \dots, x_n]$  with  $I$  as in [Lemma 18.33](#), and (57) with the third factor gives  $\text{Ext}^2(I_p, I_p) = N_1 \otimes N_2 \otimes \mathcal{O}_{W,p}$ ,  $W$  the intersection of the branches. The jet classes along  $\partial_5, \dots, \partial_n$  vanish, these derivations acting on a free factor, and the products of the classes along  $\partial_1, \dots, \partial_4$  are as in the lemma; reducing modulo the maximal ideal of  $\mathcal{O}_{W,p}$  shows that  $a_\theta a_{\theta'} \neq 0$  whenever  $\theta(p) \wedge \theta'(p)$  has a nonzero component in  $N_1 \otimes N_2$ . The rest of the argument is that of [Theorem 18.34](#), using [Theorem 18.29](#) for  $n \geq 3$ . For the last assertion,  $Z_i \cap Z_j = D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$  is a transversal intersection of four divisors, of codimension four, and at a general point of it no fifth divisor passes, so  $Z$  has exactly the two branches  $Z_i$  and  $Z_j$  there. ■

*Remark 18.37* (What this settles). [Theorem 18.34](#) answers the question left open in [\[Mar25b, Section 12\]](#): the secant objects proposed there do not satisfy the weaker criterion of Question 11.4, for any discriminant. The reason is local and geometric. Question 11.4 asks whether the semiregularity map may be required to be injective only on the image of the evaluation map; for a secant object that image contains, by [Theorem 18.29](#), the products  $A_\theta A_{\theta'}$  of the translation classes shifted by the scalars  $\frac{d}{2}(\theta \wedge \theta') \lrcorner \Theta^2$ , and the semiregularity map kills them all. For the criterion to hold these shifted products must vanish, and they cannot: at an isolated double point of the support, the translation classes of the two branches multiply to a nonzero local class, while a scalar has no local class at all. What survives of the fourth ingredient of [Remark 14.17](#) is therefore sharper than before. A secant object of positive rank on a fourfold that satisfies the weakened criterion must have, at every point, germs whose translation classes multiply to zero in the local  $\text{Ext}^2$ ; by [Lemma 18.33](#) this excludes every support with two transversal branches of codimension two through a point, whether or not the branches are separated, and by [Corollary 18.36](#) it excludes the chain of consecutive intersections of translates in every dimension. An object supported on a smooth surface, or on surfaces meeting only along curves, is not excluded by this argument, but no such object with Chern character on the secant plane is known.

**18.3. Smooth support, and the four discriminants.** [Theorem 18.34](#) and [Corollary 18.36](#) remove every secant object whose support carries two transversal branches of codimension two through a point. What they leave is a support that is smooth, or that meets itself only along curves, and [Remark 18.37](#) closes with the remark that no such object is known. This subsection says why the local argument cannot reach that case, and then determines what the case can be. The answer is narrow: on the fourfold the invariants of the support are forced outright, and the discriminant is confined to a finite list.

The first point is that at a smooth point there is nothing local left to compute.

**Proposition 18.38** (No local obstruction at a smooth point). *Let  $Y$  be a smooth variety with  $\omega_Y \cong \mathcal{O}_Y$  and let  $S \subset Y$  be smooth of codimension two. Then*

$$\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y, \quad \mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S) = N_{S/Y}, \quad \mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) = 0 \quad (q \geq 2),$$

*and the same holds for  $\mathcal{I}_S \otimes L$  for any line bundle  $L$ . Consequently, for  $F = \mathcal{I}_S \otimes L$  both sides of (56) have vanishing germs in  $\mathcal{E}xt^2$ , and [Lemma 18.32](#) decides nothing.*

*Proof.* Twisting by  $L$  changes neither side, so take  $F = \mathcal{I}_S$ . The sheaf  $\mathcal{I}_S$  is torsion free of rank one and reflexive, whence  $\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y$ . Apply  $\mathcal{H}om(-, \mathcal{I}_S)$  to

$$0 \rightarrow \mathcal{I}_S \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_S \rightarrow 0. \quad (58)$$

Since  $\mathcal{E}xt^q(\mathcal{O}_Y, \mathcal{I}_S) = 0$  for  $q \geq 1$ , the long exact sequence gives

$$\mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) \cong \mathcal{E}xt^{q+1}(\mathcal{O}_S, \mathcal{I}_S) \quad (q \geq 1). \quad (59)$$

Now apply  $\mathcal{H}om(\mathcal{O}_S, -)$  to (58). Because  $S$  is smooth of codimension two, hence a local complete intersection, the Koszul resolution gives  $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_Y) = 0$  for  $k \neq 2$  and  $\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) = \omega_S \otimes \omega_Y^{-1} = \omega_S$ , while  $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_S) = \bigwedge^k N_{S/Y}$  for  $0 \leq k \leq 2$  and zero otherwise. Computing both from the same Koszul complex shows that the map

$$\rho: \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) = \omega_S \longrightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S) = \bigwedge^2 N_{S/Y} = \omega_S$$

induced by  $\mathcal{O}_Y \rightarrow \mathcal{O}_S$  is the identity: in local coordinates in which  $S = \{x_1 = x_2 = 0\}$ , both groups are computed as  $M/(x_1, x_2)M$  for  $M = \mathcal{O}_Y$  and  $M = \mathcal{O}_S$ , and the map is reduction modulo  $(x_1, x_2)$ , which is the identity on  $\mathcal{O}_S$ .

Take  $k = 2$  in the long exact sequence of  $\mathcal{H}om(\mathcal{O}_S, -)$ . Since  $\rho$  is surjective and  $\mathcal{E}xt^3(\mathcal{O}_S, \mathcal{O}_Y) = 0$ , the segment

$$\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) \xrightarrow{\rho} \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S) \rightarrow \mathcal{E}xt^3(\mathcal{O}_S, \mathcal{I}_S) \rightarrow \mathcal{E}xt^3(\mathcal{O}_S, \mathcal{O}_Y) = 0$$

gives  $\mathcal{E}xt^3(\mathcal{O}_S, \mathcal{I}_S) = 0$ , so  $\mathcal{E}xt^2(\mathcal{I}_S, \mathcal{I}_S) = 0$  by (59). For  $k \geq 3$  both  $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_Y)$  and  $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_S)$  vanish, so  $\mathcal{E}xt^{k+1}(\mathcal{O}_S, \mathcal{I}_S) = 0$  and  $\mathcal{E}xt^k(\mathcal{I}_S, \mathcal{I}_S) = 0$ . Taking  $k = 1$  and using that  $\rho$  is injective, the segment

$$0 = \mathcal{E}xt^1(\mathcal{O}_S, \mathcal{O}_Y) \rightarrow \mathcal{E}xt^1(\mathcal{O}_S, \mathcal{O}_S) = N_{S/Y} \rightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{I}_S) \rightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) \xrightarrow{\rho}$$

exhibits  $\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{I}_S) \cong N_{S/Y}$ , which is  $\mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S)$  by (59).

For the last assertion, the right side of (56) has vanishing germs by Lemma 18.32, and the left side lies in  $\text{Ext}^2(F, F \otimes \Omega_Y^2)$  whose sheaf of germs in the  $\mathcal{E}xt^2$  direction is  $\mathcal{E}xt^2(F, F) \otimes \Omega_Y^2 = 0$ .  $\blacksquare$

*Remark 18.39* (Where the obstruction goes instead). Proposition 18.38 is the precise reason the argument of Theorem 18.34 stops at a smooth support, and it also says what replaces it. With  $\mathcal{E}xt^2 = 0$  the local to global spectral sequence for  $\text{Ext}^2(\mathcal{I}_S, \mathcal{I}_S)$  has only two surviving rows,  $H^1(S, N_{S/Y})$  and  $H^2(Y, \mathcal{O}_Y)$ , and they sit in different steps of the filtration: the first is the quotient that records the geometry of  $S$ , the second the subgroup of scalars. A product  $A_\theta A_{\theta'}$  of translation classes therefore has a well defined image in  $H^1(S, N_{S/Y})$ , while the scalar on the right of (56) has image zero there. So the criterion splits into two conditions of different kinds: a vanishing in  $H^1(S, N_{S/Y})$ , which is a statement about the normal bundle and not about any point, and an identity in  $H^2(Y, \mathcal{O}_Y)$ . At a double point the first of these was already inconsistent locally. At a smooth support it is a global question, and we do not decide it. What we can decide is which  $S$  are even available.

The invariants of a smooth support are not free. On an abelian variety the Todd class is trivial, so Grothendieck–Riemann–Roch turns the Chern character of the object into the complete numerical package of the surface, and three classical inequalities then close in on it.

**Theorem 18.40** (The invariants of a smooth support). *Let  $(X, \Theta)$  be a principally polarised abelian fourfold, let  $d > 0$ , let  $b \in \mathbb{Z}_{>0}$ , and let  $\mathcal{F} = \mathcal{I}_S(b\Theta)$  be a coherent sheaf with*

$$\text{ch}(\mathcal{F}) = u_\Theta + b v_\Theta$$

*and  $S \subset X$  smooth. Write  $N = \frac{1}{2}N_{K/\mathbb{Q}}(b + \sqrt{-d}) = \frac{1}{2}(b^2 + d)$ ,  $\iota: S \hookrightarrow X$  and  $\lambda = \iota^*\Theta$ . Then  $S$  is a surface and*

$$\begin{aligned} [S] &= N\Theta^2, & \iota_*K_S &= \frac{4bN}{3}\Theta^3, & \chi(\mathcal{O}_S) &= 4N(2b^2 - N), \\ K_S^2 &= 12N(4b^2 - N), & e(S) &= 12N(4b^2 - 3N), & K_S \cdot \lambda &= 32bN, \end{aligned} \quad (60)$$

*with  $\lambda^2 = 24N$  and  $[S]^2 = 24N^2$ . Moreover  $K_S$  is globally generated, hence nef;  $S$  is minimal of general type; and*

$$\frac{4}{9}b^2 \leq N \leq b^2, \quad \text{equivalently} \quad -\frac{1}{9}b^2 \leq d \leq b^2. \quad (61)$$

*The lower bound is the Hodge index theorem and is implied by  $d > 0$ ; the upper bound is the Bogomolov–Miyaoka–Yau inequality. In terms of  $b$  and  $d$  the five invariants read*

$$\chi(\mathcal{O}_S) = (b^2 + d)(3b^2 - d), \quad K_S^2 = 3(b^2 + d)(7b^2 - d), \quad e(S) = 3(b^2 + d)(5b^2 - 3d), \quad (62)$$

*and the defect in the Bogomolov–Miyaoka–Yau inequality is*

$$3e(S) - K_S^2 = 24(b^4 - d^2), \quad (63)$$

so that the inequality is exactly  $d \leq b^2$  and nothing else. Finally  $K_S$  is never numerically proportional to  $\lambda$ .

*Proof.* Since  $\text{ch}_0(\mathcal{F}) = 1$  and  $c_1(\mathcal{F}) = b\Theta$ , the sheaf  $\mathcal{F}(-b\Theta) = \mathcal{I}_S$  has  $\text{ch}_0 = 1$ ,  $\text{ch}_1 = 0$ , so  $S$  has codimension at least two, and

$$\text{ch}(\mathcal{O}_S) = 1 - \text{ch}(\mathcal{I}_S) = 1 - \text{ch}(\mathcal{F}) e^{-b\Theta}.$$

Expanding the right side in  $\mathbb{Q}[\Theta]/(\Theta^5)$  and collecting gives

$$\text{ch}_2(\mathcal{O}_S) = \frac{b^2+d}{2}\Theta^2 = N\Theta^2, \quad \text{ch}_3(\mathcal{O}_S) = -\frac{2bN}{3}\Theta^3, \quad \text{ch}_4(\mathcal{O}_S) = \frac{3b^4+2db^2-d^2}{24}\Theta^4,$$

which is the computation of item (XXX) of [Appendix D](#). The first identity says  $S$  has codimension exactly two, so it is a surface, with class  $N\Theta^2$ . Because  $T_X$  is trivial,  $\text{td}(T_X) = 1$ , and Grothendieck–Riemann–Roch for  $\iota$  reads  $\text{ch}(\iota_*\mathcal{O}_S) = \iota_* \text{td}(T_S)$ . Comparing degrees three and four,

$$\text{ch}_3(\mathcal{O}_S) = \iota_* \frac{c_1(T_S)}{2} = -\frac{1}{2}\iota_* K_S, \quad \text{ch}_4(\mathcal{O}_S) = \iota_* \frac{c_1^2+c_2}{12} = \chi(\mathcal{O}_S) [\text{pt}],$$

the second by Noether’s formula. With  $\Theta^4 = 24 [\text{pt}]$  these give  $\iota_* K_S = \frac{4bN}{3}\Theta^3$  and, after substituting  $d = 2N - b^2$ ,  $\chi(\mathcal{O}_S) = 3b^4 + 2db^2 - d^2 = 4N(2b^2 - N)$ . Then  $K_S \cdot \lambda = \int_X \iota_* K_S \cdot \Theta = \frac{4bN}{3} \cdot 24 = 32bN$  and  $\lambda^2 = \int_X [S] \cdot \Theta^2 = 24N$ .

For the last two invariants, the normal bundle sequence and the triviality of  $T_X$  give  $c(T_S)c(N_{S/X}) = 1$ , so  $c_1(N_{S/X}) = K_S$  and  $c_2(N_{S/X}) = K_S^2 - e(S)$ ; the self-intersection formula  $[S]^2 = \int_S c_2(N_{S/X})$  then reads

$$24N^2 = K_S^2 - e(S),$$

while Noether’s formula reads  $12\chi(\mathcal{O}_S) = K_S^2 + e(S)$ . Solving the two gives the stated  $K_S^2$  and  $e(S)$ .

The conormal sequence presents  $\Omega_S^1$  as a quotient of  $\Omega_X^1|_S = \mathcal{O}_S^{\oplus 4}$ , so  $\Omega_S^1$  is globally generated, and therefore so is  $K_S = \det \Omega_S^1$ , a quotient of  $\wedge^2 \mathcal{O}_S^{\oplus 4}$ . A globally generated line bundle is nef. A globally generated rank two bundle on a surface has  $c_2 \geq 0$ , the class  $c_2$  being represented by the vanishing scheme of a general section, which is empty or finite; hence

$$e(S) = c_2(\Omega_S^1) = 12N(4b^2 - 3N) \geq 0,$$

so  $N \leq \frac{4}{3}b^2 < 4b^2$  and therefore  $K_S^2 = 12N(4b^2 - N) > 0$ . A nef divisor of positive self-intersection is big, so  $\kappa(S) = 2$  and,  $K_S$  being nef,  $S$  is minimal of general type. The Bogomolov–Miyaoaka–Yau inequality  $c_1^2 \leq 3c_2$  then applies and reads  $K_S^2 \leq 3e(S)$ . Substituting  $N = \frac{1}{2}(b^2 + d)$  in the two expressions gives (62), hence  $3e(S) - K_S^2 = 24(b^4 - d^2)$ , and the inequality is  $d \leq b^2$ , that is  $N \leq b^2$ .

For the lower bound,  $\lambda$  is ample on  $S$ , so the Hodge index theorem gives  $(K_S \cdot \lambda)^2 \geq K_S^2 \lambda^2$ , that is  $(32bN)^2 \geq 12N(4b^2 - N) \cdot 24N$ , which simplifies to  $9N \geq 4b^2$ . Since  $N = \frac{1}{2}(b^2 + d)$  with  $d > 0$ , this holds automatically. Equality in the Hodge index theorem occurs exactly when  $K_S$  is numerically proportional to  $\lambda$ , hence exactly when  $9N = 4b^2$ ; as  $9N = \frac{9}{2}(b^2 + d) > 4b^2$  for  $d > 0$ , that never happens.  $\blacksquare$

**Corollary 18.41** (Four discriminants on the fourfold). *In [Setup 18.24](#), where  $b = 3$  by [Proposition 18.25\(iii\)](#), a secant object  $\mathcal{I}_S(3\Theta)$  with smooth support exists only if  $d \leq 9$ ; since  $d$  is squarefree this leaves  $d \in \{1, 2, 3, 5, 6, 7\}$ , and if in addition  $N = (9 + d)/2$  is an integer, which is what a support assembled from translates of  $\Theta$  requires, only*

$$d \in \{1, 3, 5, 7\}, \quad N \in \{5, 6, 7, 8\}.$$

For these the invariants are forced to the values in [Table 3](#). In particular, for every imaginary quadratic field of discriminant larger than 9 there is no secant object of rank one with smooth support on an abelian fourfold at all.

*Proof.* Immediate from (61) at  $b = 3$ , together with the arithmetic of (60); the table is item (XXX) of [Appendix D](#). The value  $d = 9$  is the equality case of Bogomolov–Miyaoaka–Yau, where  $K_S^2 = 3e(S) = 2916$  and  $S$  would have to be a quotient of the ball; it is excluded here because 9 is not squarefree. ■

| $d$ | $N$ | $\chi(\mathcal{O}_S)$ | $K_S^2$ | $e(S)$ | $K_S \cdot \lambda$ | $\lambda^2$ |
|-----|-----|-----------------------|---------|--------|---------------------|-------------|
| 1   | 5   | 260                   | 1860    | 1260   | 480                 | 120         |
| 3   | 6   | 288                   | 2160    | 1296   | 576                 | 144         |
| 5   | 7   | 308                   | 2436    | 1260   | 672                 | 168         |
| 7   | 8   | 320                   | 2688    | 1152   | 768                 | 192         |

TABLE 3. The only invariants a smooth support can have on an abelian fourfold, by [Theorem 18.40](#) and [Corollary 18.41](#), at  $b = 3$  and with  $N$  an integer. Every entry is forced by  $\text{ch}(\mathcal{F})$  alone; none is a choice. The slope  $K_S^2/\chi(\mathcal{O}_S)$  runs from 93/13 at  $d = 1$  to 42/5 at  $d = 7$ , and reaches the Bogomolov–Miyaoaka–Yau ceiling of 9 only at the excluded value  $d = 9$ .

*Remark 18.42* (What is left of the smooth case). Three things have changed and one has not.

The local argument is out of reach here, and [Proposition 18.38](#) says so for a structural reason rather than as a failure of effort: the sheaf  $\mathcal{E}xt^2$  of a smooth ideal vanishes, so both sides of (56) are locally zero and the germ comparison that decides [Theorem 18.34](#) has nothing to compare.

The surface is pinned. Its class, its canonical class, its Euler characteristic, its Euler number and its two intersection numbers against the polarisation are all determined by  $d$  alone, with no freedom, and they are compatible with existence only for  $d \leq 9$ . That is the content of [Corollary 18.41](#), and it is a genuine restriction: the secant construction with smooth support is not available for all but finitely many imaginary quadratic fields, whatever geometry one is willing to invent.

The residual question is now a single existence statement. For  $d \in \{1, 3, 5, 7\}$ , does a principally polarised abelian fourfold carry a smooth surface  $S$  with the invariants of [Table 3](#), and if so does the image of  $A_\theta A_{\theta'}$  in  $H^1(S, N_{S/X})$  vanish for all  $\theta, \theta' \in H^0(T_X)$ ? The first half is a question about surfaces in abelian fourfolds with prescribed Chern numbers, of the kind that is answered by construction or by a classification argument; the second is a question about one cohomology group of one normal bundle. Neither is settled here.

What has not changed is the standing of the conjecture. Even a positive answer to both would produce Weil classes on abelian eightfolds for at most four discriminants, and [Corollary 15.35](#) would still be open for every other discriminant, for every  $n \geq 5$ , and for every field of degree larger than two.

**18.4. The obstruction is a condition on depth.** [Proposition 18.38](#) says that at a smooth point of the support there is nothing local to compare. That is a special case of something much more general, and identifying the general case tells us exactly which

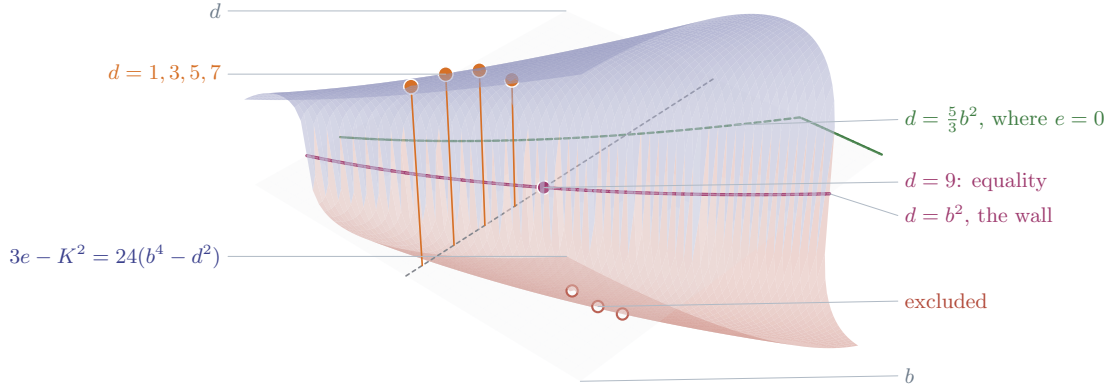


FIGURE 25. [Theorem 18.40](#) over the quarter plane of  $(b, d)$ . The height of the surface is the defect  $3e(S) - K_S^2 = 24(b^4 - d^2)$  in the Bogomolov–Miyaoka–Yau inequality for the surface the theorem attaches to the pair, compressed by a signed cube root so that four orders of magnitude fit one frame; the compression is odd and monotone, so the zero locus and the sign are the true ones. The surface is drawn indigo where the defect is positive and clay where it is negative, and it crosses zero along  $d = b^2$ , which is therefore the whole content of the inequality. The weaker wall  $d = \frac{5}{3}b^2$ , where  $e(S)$  itself changes sign, is drawn dashed and lies above it. On the line  $b = 3$ , which is the value [Proposition 18.25\(iii\)](#) forces on the fourfold, the four squarefree discriminants  $d = 1, 3, 5, 7$  with  $N$  an integer sit above the wall and are marked with their drop lines;  $d = 9$  sits exactly on it, the equality case, and  $d = 11, 13, 15$  sit below and are excluded.

supports [Theorem 18.34](#) and [Corollary 18.36](#) remove and which they cannot touch. The dividing line is not smoothness, and it is not being a local complete intersection. It is depth.

**Theorem 18.43** (Cohen–Macaulay supports carry no local obstruction). *Let  $Y$  be a smooth variety with  $\omega_Y \cong \mathcal{O}_Y$ , let  $Z \subset Y$  be a closed subscheme of codimension two, and let  $p \in Z$ . If the local ring  $\mathcal{O}_{Z,p}$  is Cohen–Macaulay, then*

$$\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = 0 \quad \text{for every } q \geq 2,$$

*and the same holds for  $\mathcal{I}_Z \otimes L$  for every line bundle  $L$ . Consequently both sides of (56) have vanishing germ at  $p$ , and [Lemma 18.32](#) decides nothing there.*

*Proof.* Twisting by  $L$  changes no  $\mathcal{E}xt$  sheaf, so take  $F = \mathcal{I}_Z$ . Write  $A = \mathcal{O}_{Y,p}$ , a regular local ring of dimension  $m$ , and  $I = \mathcal{I}_{Z,p}$ . By hypothesis  $A/I$  is Cohen–Macaulay of dimension  $m - 2$ , so  $\text{depth } A/I = m - 2$ , and the Auslander–Buchsbaum formula gives

$$\text{pd}_A(A/I) = m - \text{depth}_A(A/I) = 2.$$

From the exact sequence  $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$  with  $A$  free it follows that  $\text{pd}_A I = 1$ . A module of projective dimension one has a free resolution of length one, so  $\text{Ext}_A^q(I, M) = 0$  for every  $A$ -module  $M$  and every  $q \geq 2$ ; taking  $M = I$  gives the vanishing. The last assertion is [Lemma 18.32](#): the right side of (56) is  $z \cdot \text{id}_F$  with  $z \in H^2(\mathcal{O}_Y)$  and has vanishing germs, and the left side lies in a sheaf whose  $\mathcal{E}xt^2$  factor is zero. ■

*Remark 18.44* (The hypothesis is sharp, and it is the one that fails). Two observations make [Theorem 18.43](#) the right statement.

The first is that Cohen–Macaulayness is strictly weaker than the conditions one might reach for instead. Item (XXXI) of [Appendix D](#) computes  $\mathcal{E}xt^2(\mathcal{I}_Z, \mathcal{I}_Z)$  for ten codimension two germs in  $\mathbb{C}^4$ . It vanishes for the smooth germ  $(x_1, x_2)$ ; for two planes meeting along a line,  $(x_1, x_2x_3)$ , and for three planes meeting along a line,  $(x_1, x_2x_3x_4)$ ; for the double plane  $(x_1^2, x_2)$  and the quadric cone  $(x_1, x_2^2 - x_3x_4)$ ; and also for the fat plane  $(x_1, x_2)^2$  and for the cone over the twisted cubic, the ideal of  $2 \times 2$  minors of  $\begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \end{pmatrix}$ , neither of which is a local complete intersection, both needing three generators in codimension two. So reducedness, irreducibility and the complete intersection property are all irrelevant; the projective dimension is what matters, and by Auslander–Buchsbaum that is depth.

The second is that the germs where it does not vanish are exactly the ones the construction of [Setup 18.24](#) produces. The same computation finds  $\mathcal{E}xt^2 \neq 0$ , with nonzero products of translation classes, for two planes meeting at a point, for three planes meeting pairwise at a point, and for a plane with an embedded point; the projective dimensions there are 3, 3 and 4. A scheme whose local ring is Cohen–Macaulay is connected in codimension one at that point, by Hartshorne’s connectedness theorem, so a support with two components through  $p$  meeting only at  $p$  can never be Cohen–Macaulay there. That is precisely the configuration [Lemma 18.33](#) analyses.

**Corollary 18.45** (What the local argument removes, in final form). *Let  $(X, t)$  be a polarised abelian variety of dimension  $n \geq 4$  and  $F$  a perfect complex with  $\mathrm{ch}(F) = au_t + bv_t$ ,  $(a, b) \neq (0, 0)$ , whose germ at each point is the germ of  $\mathcal{I}_Z \otimes L$  for a codimension two subscheme  $Z$ . Then exactly one of the following holds.*

- (i)  *$Z$  fails to be Cohen–Macaulay at some point  $p$  of one of the types of [Remark 18.44](#), and then  $F$  does not satisfy the weakened criterion, nor does  $\Phi(F \boxtimes F_2)$  for any secant  $F_2$ , by [Theorem 18.34](#) and [Corollary 18.36](#).*
- (ii)  *$Z$  is Cohen–Macaulay, and then there is no local obstruction at any point whatever. The criterion is then a single global condition: the image of  $A_\theta A_{\theta'}$  in  $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$  must vanish for all  $\theta, \theta' \in H^0(T_X)$ , the identity in  $H^2(\mathcal{O}_X)$  being all that the scalar on the right of (56) can meet.*

*In case (ii),  $\mathcal{I}_Z$  has projective dimension one at every point, so it is locally the ideal of maximal minors of a Hilbert–Burch matrix, and when  $Z$  is in addition smooth  $\mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z) = N_{Z/X}$  and the invariants of  $Z$  are those of [Theorem 18.40](#).*

*Proof.* The two cases are exclusive and exhaustive by definition. In case (i) the local ring is not Cohen–Macaulay and the cited results apply, the germs being those of [Lemma 18.33](#) and [Corollary 18.36](#). In case (ii) [Theorem 18.43](#) gives  $\mathcal{E}xt^q = 0$  for  $q \geq 2$  at every point, so the local to global spectral sequence for  $\mathrm{Ext}^2(\mathcal{I}_Z, \mathcal{I}_Z)$  has only the two rows  $H^1(\mathcal{E}xt^1)$  and  $H^2(\mathcal{O}_X)$ , and these lie in different steps of the filtration, the second being the subgroup of scalars. The right side of (56) lies in that subgroup, so it has zero image in the quotient  $H^1(\mathcal{E}xt^1)$ , and the criterion forces the image of the left side there to vanish. The projective dimension statement is the proof of [Theorem 18.43](#), and Hilbert–Burch is the structure theorem for modules of projective dimension one over a regular local ring; the last clause is [Proposition 18.38](#) and [Theorem 18.40](#). ■

*Remark 18.46* (Why the normalisation cannot repair the candidate). [Corollary 18.45](#) explains the tension in [Remark 18.35](#) in one word. The partial normalisation performed in [Setup 18.24](#) separates the two branches at an isolated double point, and separating them is exactly what would make the local ring Cohen–Macaulay there. So the construction

is already trying to reach case (ii). What [Remark 18.35](#) shows is that it cannot arrive: normalising every isolated point moves  $\chi(\mathcal{O}_{\tilde{Z}})$  to  $50N$ , while the secant plane demands  $(d+9)(27-d) = 72N - 4N^2$ , and the two agree only at  $N = 11/2$ . Repairing the depth destroys the Chern character, and keeping the Chern character keeps a point where the depth fails. That is the obstruction in its final form, and it is why a construction assembled from intersections of translates of a polarisation cannot be made to work: by [Corollary 18.36](#) such a chain always has a point of the bad type, and by [Remark 18.35](#) it cannot be separated without leaving the plane.

**18.5. Cohen–Macaulay supports, and the two routes that remain.** [Corollary 18.45](#) leaves one class of supports and one condition on them. The class is the Cohen–Macaulay subschemes of codimension two, and for those the proof of [Theorem 18.43](#) gives more than the vanishing it was used for: the ideal sheaf has projective dimension one at every point. That is the Hilbert–Burch shape, and it can be searched.

**Setup 18.47** (A resolution of the support). Let  $(X, \Theta)$  be a principally polarised abelian fourfold with  $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$ , let  $Z \subset X$  be Cohen–Macaulay of codimension two, and suppose  $\mathcal{I}_Z$  admits a global resolution

$$0 \longrightarrow E_1 \xrightarrow{\varphi} E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0, \quad \mathrm{rk} E_0 = r+1, \quad \mathrm{rk} E_1 = r, \quad (64)$$

with  $E_0 = \bigoplus_{i=1}^{r+1} \mathcal{O}(e_i\Theta)$  and  $E_1 = \bigoplus_{j=1}^r \mathcal{O}(f_j\Theta)$  sums of line bundles. Put  $p_i = e_i + b$  and  $q_j = f_j + b$ .

**Lemma 18.48** (The resolution in moments). *In [Setup 18.47](#),  $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_{\Theta} + b v_{\Theta}$  holds if and only if*

$$\sum_{i=1}^{r+1} p_i^k - \sum_{j=1}^r q_j^k = c_k, \quad k = 0, 1, 2, 3, 4, \quad (c_k) = (1, b, -d, -db, d^2). \quad (65)$$

*The equation at  $k = 0$  is the rank count and is automatic. Moreover  $\varphi$  can be injective only if  $\max_i e_i \geq \max_j f_j$ , and  $\mathcal{O}_X$  embeds in  $E_0^{\vee}$  only if  $\min_i e_i \leq 0$ .*

*Proof.* From (64),  $\mathrm{ch}(\mathcal{I}_Z) = \mathrm{ch}(E_0) - \mathrm{ch}(E_1)$ , so  $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = \sum_i e^{p_i\Theta} - \sum_j e^{q_j\Theta}$ . Comparing coefficients of  $\Theta^k/k!$  against  $u_{\Theta} + b v_{\Theta}$ , whose coefficients are the  $c_k$  displayed, gives (65). For the last two statements, on an abelian variety  $\mathrm{Hom}(\mathcal{O}(f\Theta), \mathcal{O}(e\Theta)) = H^0(\mathcal{O}((e-f)\Theta))$  vanishes unless  $e \geq f$ ; if  $\varphi$  is injective then each summand of  $E_1$  must map nonzero to  $E_0$ , which needs  $\max_i e_i \geq f_j$  for every  $j$ . Dualising (64) and using  $\mathcal{I}_Z^{\vee} = \mathcal{O}_X$  gives an injection  $\mathcal{O}_X \hookrightarrow E_0^{\vee}$ , which needs  $-e_i \geq 0$  for some  $i$ .  $\blacksquare$

The case  $r = 1$  is the complete intersections, and it closes outright, for every  $b$  and every  $d$ , by an argument that uses no search.

**Theorem 18.49** (No complete intersection supports a secant object). *Let  $(X, \Theta)$  be a principally polarised abelian fourfold with  $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$ , let  $d > 0$  and  $b \geq 1$ . No complete intersection  $Z = D_1 \cap D_2$  of two divisors satisfies  $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_{\Theta} + b v_{\Theta}$ .*

*Proof.* Such a  $Z$  is [Setup 18.47](#) with  $r = 1$ , the resolution being the Koszul complex of the two equations. Put  $w(x) = x^2 + d$ , which is strictly positive on  $\mathbb{R}$  because  $d > 0$ , and read (65) in the combinations  $c_{k+2} + d c_k$  for  $k = 0, 1, 2$ . Those three numbers are

$$c_2 + d c_0 = -d + d = 0, \quad c_3 + d c_1 = -db + db = 0, \quad c_4 + d c_2 = d^2 - d^2 = 0,$$

so that

$$\sum_i w(p_i) p_i^k = \sum_j w(q_j) q_j^k, \quad k = 0, 1, 2.$$

The two sides are the moments of order 0, 1 and 2 of the positive measures  $\nu_P = \sum_i w(p_i)\delta_{p_i}$  and  $\nu_Q = \sum_j w(q_j)\delta_{q_j}$ , so those measures have the same mass, the same mean and the same variance. At  $r = 1$  the measure  $\nu_Q$  has one atom, hence variance zero, so  $\nu_P$  has variance zero as well, and a positive measure with two atoms has variance zero only if the atoms coincide. So  $p_1 = p_2$ , and then equality of mass and mean forces  $q_1 = p_1$ . The resolution (64) then cancels and  $\mathcal{I}_Z \cong \mathcal{O}(e_1\Theta)$  is a line bundle, which the ideal sheaf of a subscheme of codimension two is not.

Equivalently and by the same computation, the two divisor degrees  $a_1$  and  $a_2$  would have to satisfy  $a_1 + a_2 = \frac{4}{3}b$  and  $(b - a_1)(b - a_2) = \frac{1}{2}(\frac{1}{3}b^2 + d)$ , and the quadratic with that sum and product has discriminant  $-\frac{2}{9}b^2 - 2d < 0$ : the degrees are not even real. ■

For  $r \geq 2$  the moment conditions no longer decide by themselves, and (65) becomes a finite search over the twists. Item (XXXII) of [Appendix D](#) carries it out exactly, by meet in the middle on the four moments, and the outcome is [Table 4](#).

**Theorem 18.50** (The split resolutions, and where they can live). *In [Setup 18.47](#) with  $b = 3$ , over every squarefree  $d < 24$  and every choice of twists in the box recorded in [Table 4](#):*

- (i) *at  $r = 2$  and  $r = 3$  the system (65) has no solution at all, and the same holds for every  $b \leq 6$  over seven fields;*
- (ii) *at  $r = 4, 5, 6$  it has solutions, but only for  $d = 15$  and  $d = 23$ ;*
- (iii) *every  $d$  that survives at any rank exceeds 9.*

*Proof.* The search is item (XXXII) of [Appendix D](#), carried out in exact integer arithmetic; the two conditions of [Lemma 18.48](#) on  $\max_i e_i$  and  $\min_i e_i$  are imposed, and without the first the system acquires solutions that are not resolutions, the smallest being  $p = (-8, -5, 0, 1, 2)$ ,  $q = (-7, -6, -4, 4)$  at  $r = 4$  and  $d = 23$ , where the summand of  $E_1$  with the largest twist admits no nonzero map to  $E_0$ . ■

| $r$ | twists searched       | $d$ surviving | first candidate, as twists of $\Theta$                          |
|-----|-----------------------|---------------|-----------------------------------------------------------------|
| 1   | every $b$ , every $d$ | none          | <a href="#">Theorem 18.49</a>                                   |
| 2   | $[-18, 18]$           | none          |                                                                 |
| 3   | $[-14, 14]$           | none          |                                                                 |
| 4   | $[-11, 11]$           | 23            | $E_0 : (-11, -6, 0, 2, 7), E_1 : (-10, -9, 5, 6)$               |
| 5   | $[-9, 9]$             | 23            | $E_0 : (-8, -4, -3, -2, 1, 6), E_1 : (-7, -6, -6, 4, 5)$        |
| 6   | $[-8, 8]$             | 15, 23        | $E_0 : (-8, -4, -4, -1, -1, 1, 4), E_1 : (-7, -6, -6, 0, 3, 3)$ |

TABLE 4. The split Hilbert–Burch resolutions of [Setup 18.47](#) at  $b = 3$ , over every squarefree  $d < 24$ . Nothing survives below rank four, and what survives above it lives only at  $d = 15$  and  $d = 23$ . Both exceed the bound  $d \leq 9$  of [Corollary 18.41](#), so none of these supports can be smooth.

The search of [Theorem 18.50](#) records where a split resolution can have the right Chern character. It is superseded by a statement that needs no search and no bound on the twists: a split resolution never runs the criterion, at any rank and any discriminant. The reason is a necessary condition that is worth isolating, because it applies to every object and can be tested on any candidate before any global computation is attempted.

**Lemma 18.51** (Morphisms to a line bundle are killed by  $H^2(\mathcal{O})$ ). *Let  $(X, t)$  be a polarised abelian variety of dimension  $n \geq 3$ , let  $F$  be a perfect complex with  $\text{ch}(F) = au_t + bv_t$ ,*

$(a, b) \neq (0, 0)$ , which satisfies the weakened criterion of [Setup 18.27](#), and let  $L$  be a line bundle with  $c_1(L) \in \mathbb{Q}t$ . Then for every  $m \in \mathbb{Z}$ , every morphism  $p: F \rightarrow L[m]$ , every morphism  $q: L[m] \rightarrow F$ , and every  $z \in H^2(\mathcal{O}_X)$ ,

$$z \cdot p = 0 \text{ in } \text{Hom}(F, L[m+2]), \quad z \cdot q = 0 \text{ in } \text{Hom}(L[m], F[2]),$$

where  $z \cdot (\cdot)$  is the  $H^*(\mathcal{O}_X)$ -module structure on morphisms in  $D^b(X)$ . In particular, if  $F = I_Z \otimes L'$  for a closed subscheme  $Z$  and a line bundle  $L'$  with  $c_1(L') \in \mathbb{Q}t$ , then the restriction  $H^{n-2}(\mathcal{O}_X) \rightarrow H^{n-2}(\mathcal{O}_Z)$  is injective.

*Proof.* For a constant bivector  $\pi$  the class  $x_\pi = (\frac{d}{2}\pi \lrcorner t^2, 0, \pi)$  preserves  $\text{ch}(F)$  by [Theorem 18.29\(ii\)](#), so  $\text{ev}_F(x_\pi) = 0$  by hypothesis. The Hochschild action is by natural transformations  $\text{id} \rightarrow \text{id}[2]$  of  $D^b(X)$ , so for  $p: F \rightarrow L[m]$ ,  $\text{ev}_{L[m]}(x_\pi) \circ p = p \circ \text{ev}_F(x_\pi) = 0$ , and  $\text{ev}_{L[m]} = \text{ev}_L[m]$ . By (49),  $\text{ev}_L(x_\pi) = c_\pi \cdot \text{id}_L$  with  $c_\pi = \frac{1}{2}\pi \lrcorner (\lambda^2 + dt^2) \in H^{0,2}(X)$ ,  $\lambda = c_1(L) = et$ , that is  $c_\pi = \frac{e^2+d}{2}\pi \lrcorner t^2$ , and  $\pi \mapsto \pi \lrcorner t^2$  is an isomorphism from  $H^0(\wedge^2 T_X)$  onto  $H^{0,2}(X)$ , as in the proof of [Theorem 18.29\(iii\)](#). Hence  $c_\pi$  runs over all of  $H^2(\mathcal{O}_X)$  as  $\pi$  varies, and  $(c_\pi \cdot \text{id}_L) \circ p = c_\pi \cdot p$ . The statement for  $q$  is the same with the square  $\text{ev}_F(x_\pi) \circ q = q \circ \text{ev}_{L[m]}(x_\pi)$ . For the last assertion apply the first to the inclusion  $p: I_Z \otimes L' \rightarrow L'$ : the sequence  $0 \rightarrow I_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0$  gives an exact sequence

$$\text{Ext}^2(\mathcal{O}_Z, \mathcal{O}_X) \longrightarrow H^2(\mathcal{O}_X) \longrightarrow \text{Ext}^2(I_Z, \mathcal{O}_X),$$

in which the second map is  $z \mapsto z \cdot p$ ; so  $z \cdot p = 0$  for all  $z$  means that the first map is surjective, and by Serre duality with  $\omega_X$  trivial it is the dual of the restriction  $H^{n-2}(\mathcal{O}_X) \rightarrow H^{n-2}(\mathcal{O}_Z)$ .  $\blacksquare$

**Theorem 18.52** (No two-term complex of powers of the polarisation runs the criterion). *Let  $(X, t)$  be a polarised abelian variety of dimension  $n \geq 3$  and  $M$  an ample line bundle with  $c_1(M) \in \mathbb{Q}t$ . Let  $A$  and  $B$  be direct sums of tensor powers  $M^{\otimes e}$ ,  $e \in \mathbb{Z}$ , let  $\varphi: A \rightarrow B$  be any morphism, and let  $F$  be the cone of  $\varphi$ , or any shift of it. If  $F \neq 0$  and  $\text{ch}(F) = au_t + bv_t$  with  $(a, b) \neq (0, 0)$ , then  $F$  does not satisfy the weakened criterion. In particular no object with a resolution  $0 \rightarrow E_1 \rightarrow E_0 \rightarrow F \rightarrow 0$  by sums of powers of  $M$ , whatever the rank, does.*

*Proof.* A pair of summands  $M^{\otimes e}$  of  $A$  and  $M^{\otimes e}$  of  $B$  joined by a nonzero entry of  $\varphi$ , which is then a nonzero scalar, can be cancelled without changing the cone up to isomorphism; do this until no such pair remains. If  $A = 0$  or  $B = 0$  after cancelling,  $F$  is a shifted sum of line bundles,  $\text{ev}_F(x_\pi)$  is diagonal with the nonzero entries  $c_\pi \cdot \text{id}$  of the proof of [Lemma 18.51](#), and the criterion fails as in [Corollary 18.22](#); so suppose both are nonzero and consider the triangle  $A \xrightarrow{\varphi} B \xrightarrow{q} F \rightarrow A[1]$ . Let  $M^{\otimes b_j}$  be a summand of  $B$ ,  $\iota_j$  its inclusion into  $B$ , and  $q_j = q \circ \iota_j: M^{\otimes b_j} \rightarrow F$ . Applying  $\text{Hom}(M^{\otimes b_j}, -)$  to the triangle gives the exact sequence

$$\text{Hom}(M^{\otimes b_j}, A[2]) \xrightarrow{\varphi_*} \text{Hom}(M^{\otimes b_j}, B[2]) \xrightarrow{q_*} \text{Hom}(M^{\otimes b_j}, F[2]),$$

and  $z \cdot q_j = q_*(z \cdot \iota_j)$  for  $z \in H^2(\mathcal{O}_X)$ . Now  $\text{Hom}(M^{\otimes b_j}, B[2]) = \bigoplus_k H^2(M^{\otimes(b_k - b_j)})$ , and  $H^2(M^{\otimes m}) = 0$  for  $m \neq 0$ : for  $m > 0$  by Kodaira vanishing,  $\omega_X$  being trivial, and for  $m < 0$  because a negative line bundle on an abelian variety has cohomology only in degree  $n \geq 3$ . So  $z \cdot \iota_j$  is the element with component  $z$  in the summand  $k = j$  and zero elsewhere. It lies in the image of  $\varphi_*$  only if there is  $\psi \in \text{Hom}(M^{\otimes b_j}, A[2]) = \bigoplus_i H^2(M^{\otimes(a_i - b_j)})$  with  $(\varphi \circ \psi)_j = \sum_i \varphi_{ji} \psi_i = z$ ; but  $\psi_i \neq 0$  forces  $a_i = b_j$ , and for such  $i$  the entry  $\varphi_{ji}$  is a scalar, zero after the cancellation. So  $z \cdot q_j \neq 0$  for every  $z \neq 0$ , which contradicts

**Lemma 18.51.** The last assertion is the case of a resolution, where  $F \cong \text{coker } \varphi$  is the cone of  $\varphi: E_1 \rightarrow E_0$ . ■

The powers of the polarisation are not the only bundles on which the compensated Poisson classes act by scalars. So does every bundle whose Atiyah class is a multiple of the identity, and on an abelian variety these are the semi-homogeneous bundles of Mukai [Muk78]; the argument of Theorem 18.52 goes through for them, with the cohomology of the fibre product of two isogenies in place of the cohomology of a power of  $\Theta$ . We state the result in a form that is self-contained.

**Definition 18.53** (Isogeny blocks). Let  $(X, \Theta)$  be a polarised abelian variety. An *isogeny block* is a vector bundle  $E = \pi_* L$ , where  $\pi: Y \rightarrow X$  is an isogeny and  $L$  is a line bundle on  $Y$  with  $c_1(L) = \nu \pi^* \Theta$  in  $H^2(Y, \mathbb{Q})$ ,  $\nu \in \mathbb{Q}$ ; its *slope* is  $\nu$ , so that  $c_1(E) = \nu \deg(\pi) \Theta$ . The block is *simple* if  $\tau_g^* L \not\cong L$  for every nonzero  $g \in \ker \pi$ . The powers  $\mathcal{O}(e\Theta)$  are the blocks with  $\pi = \text{id}$ .

**Lemma 18.54** (Three properties of a block). *Let  $E = \pi_* L$  be an isogeny block of slope  $\nu$  on an abelian  $n$ -fold.*

- (i)  $\text{At}(E) = \nu \Theta \cdot \text{id}_E$ ; hence for a constant bivector  $\pi$  the compensated class  $x_\pi$  of Theorem 18.29(ii) acts by  $\text{ev}_E(x_\pi) = \frac{\nu^2 + d}{2} (\pi \lrcorner \Theta^2) \cdot \text{id}_E$ , and Lemma 18.51 holds with  $E$  in place of the line bundle  $L$ .
- (ii) For a second block  $E' = \pi'_* L'$  of slope  $\nu'$ ,  $\text{Ext}^q(E', E) = H^q(Z, M)$  with  $Z = Y' \times_X Y$ , a disjoint union of abelian varieties isogenous to  $X$ , and  $M$  a line bundle whose class on each component is  $(\nu - \nu')$  times the pullback of  $\Theta$ . If  $\nu \neq \nu'$  and  $n \geq 3$  then  $\text{Ext}^2(E', E) = 0$ . If  $E$  is simple then  $\text{Ext}^2(E, E) = H^2(\mathcal{O}_X) \cdot \text{id}_E$ .
- (iii) A simple block is  $\mu$ -stable with respect to  $\Theta$ ; hence a nonzero morphism between simple blocks of the same slope is an isomorphism.

*Proof.* (i) Since  $\pi$  is étale,  $\pi^* \Omega_X^1 = \Omega_Y^1$  and  $\pi_*$  is exact, so the first jet sequence of  $\pi_* L$  is the direct image of that of  $L$ , and  $\text{At}(\pi_* L)$  is the image of  $\text{At}(L)$  under  $\pi_*: \text{Ext}_Y^1(L, L \otimes \pi^* \Omega_X^1) \rightarrow \text{Ext}_X^1(\pi_* L, \pi_* L \otimes \Omega_X^1)$ . As  $\text{At}(L) = c_1(L) \cdot \text{id}_L = \nu \pi^* \Theta \cdot \text{id}_L$ , the projection formula gives  $\nu \Theta \cdot \text{id}_{\pi_* L}$ . Then (52) with a scalar Atiyah class gives the value of  $\text{ev}_E(x_\pi)$ , and the proof of Lemma 18.51 used nothing else about  $L$ .

(ii) Adjunction for the finite étale  $\pi'$ , flat base change along the cartesian square with vertices  $Z, Y', Y, X$ , and the projection formula give  $\text{Ext}_X^q(\pi'_* L', \pi_* L) = H^q(Y', L'^{-1} \otimes \pi'^* \pi_* L) = H^q(Z, p'^* L'^{-1} \otimes p^* L)$ , with  $p, p'$  the two projections; the class of  $M = p'^* L'^{-1} \otimes p^* L$  is  $(\nu - \nu')$  times the pullback of  $\Theta$  on every component, which is an abelian variety finite étale over  $Y'$ . A line bundle on an abelian  $n$ -fold whose class is a positive multiple of an ample class has cohomology only in degree 0, and one whose class is a negative multiple only in degree  $n$ , by Kodaira vanishing and Serre duality with trivial canonical bundle; so  $H^2(Z, M) = 0$  when  $\nu \neq \nu'$  and  $n \geq 3$ . When  $E' = E$ , the components of  $Y \times_X Y$  are the graphs of the translations by  $g \in \ker \pi$ , on which  $M$  restricts to  $L^{-1} \otimes \tau_g^* L \in \text{Pic}^0(Y)$ ; a nontrivial element of  $\text{Pic}^0$  has no cohomology, so  $\text{Ext}^q(E, E) = \bigoplus_g H^q(\mathcal{O}_Y)$  over the  $g$  with  $\tau_g^* L \cong L$ , which for a simple block is  $g = 0$  alone, and  $H^q(\mathcal{O}_Y) = H^q(\mathcal{O}_X)$  through  $\pi$ .

(iii) The pullback  $\pi^* E = \bigoplus_{g \in \ker \pi} \tau_g^* L$  is a direct sum of line bundles of one slope, hence  $\mu$ -semistable, and semistability descends along the finite étale  $\pi$ ; so  $E$  is semistable. If  $F \subset E$  is a saturated subsheaf of the same slope, then  $\pi^* F \subset \pi^* E$  is a saturated subsheaf of the same slope of a direct sum of pairwise non-isomorphic line bundles, the  $\tau_g^* L$  being distinct because the block is simple, hence a partial sum  $\bigoplus_{g \in S} \tau_g^* L$ ; it is invariant under the translations by  $\ker \pi$ , which permute the summands transitively, so  $S$  is empty or

everything and  $F = 0$  or  $F = E$ . A nonzero morphism between stable bundles of the same slope is an isomorphism.  $\blacksquare$

**Theorem 18.55** (No two-term complex of simple blocks runs the criterion). *Let  $(X, \Theta)$  be a polarised abelian variety of dimension  $n \geq 3$ , let  $A$  and  $B$  be direct sums of simple isogeny blocks, let  $\varphi: A \rightarrow B$  be any morphism and let  $F$  be the cone of  $\varphi$  or a shift of it. If  $F \neq 0$  and  $\text{ch}(F) = au_\Theta + bv_\Theta$  with  $(a, b) \neq (0, 0)$ , then  $F$  does not satisfy the weakened criterion. In particular no object with a resolution  $0 \rightarrow E_1 \rightarrow E_0 \rightarrow F \rightarrow 0$  by sums of simple isogeny blocks does, whatever the rank.*

*Proof.* Cancel, by Gaussian elimination in the two-term complex, every pair of isomorphic blocks of  $A$  and  $B$  joined by an entry of  $\varphi$  that is an isomorphism; this changes neither the cone nor the block structure of the remaining terms. By Lemma 18.54(iii), every remaining entry between blocks of the same slope is zero. The rest is the proof of Theorem 18.52 with Lemma 18.54 in place of the vanishing of  $H^2(M^{\otimes m})$ : for a block  $E'$  of  $B$  with inclusion  $\iota$  and  $q' = q \circ \iota: E' \rightarrow F$ , the class  $z \cdot \iota \in \text{Hom}(E', B[2])$  has component  $z \cdot \text{id}_{E'} \neq 0$  in  $\text{Ext}^2(E', E') = H^2(\mathcal{O}_X) \cdot \text{id}_{E'}$  by (ii); it lies in the image of  $\varphi_*$  only if  $\sum_i \varphi_{ji} \psi_i = z \cdot \text{id}_{E'}$  for some  $\psi_i \in \text{Ext}^2(E', A_i)$ , and  $\psi_i \neq 0$  forces  $A_i$  to have the slope of  $E'$  by (ii), for which  $\varphi_{ji} = 0$ . So  $z \cdot q' \neq 0$ , against Lemma 18.51 in the form of (i). If  $A$  or  $B$  is empty after cancelling,  $F$  is a shifted sum of blocks, on each of which  $\text{ev}(x_\pi)$  is a nonzero scalar by (i).  $\blacksquare$

*Remark 18.56* (What the blocks are). By Mukai [Muk78] the simple isogeny blocks are exactly the simple semi-homogeneous bundles, the bundles  $E$  with  $\tau_x^* E \cong E \otimes P$  for every  $x$  and some  $P \in \text{Pic}^0(X)$ , and every semi-homogeneous bundle on a variety with  $\text{NS}(X)_\mathbb{Q} = \mathbb{Q}\Theta$  is an iterated extension of such blocks and their twists by  $\text{Pic}^0$ ; we do not use this. What Theorem 18.55 says is that the bundles in a Hilbert–Burch resolution of a candidate cannot all have scalar Atiyah class. The obstruction is the same one every time: the compensated Poisson classes act on a block by the scalar  $\frac{\nu^2+d}{2} \pi \lrcorner \Theta^2$ , which would vanish only for  $\nu^2 = -d$ , and cones of maps between blocks inherit the scalar through the summand inclusions because the  $\text{Ext}^2$  between blocks of different slopes vanishes. A candidate must therefore be built from a bundle whose Atiyah class is not a scalar, that is, one which is not projectively flat.

**Corollary 18.57** (The split shape is closed, and what the two routes must be). *Let  $(X, \Theta)$  be a principally polarised abelian fourfold.*

- (i) *No secant object of rank one whose support has a resolution (64) by sums of powers of  $\Theta$  satisfies the weakened criterion, for any  $r$ , any twists and any  $d$ ; in particular every candidate of Table 4 is excluded, whether or not its matrix degenerates in codimension two.*
- (ii) *If a secant object  $\mathcal{I}_Z(b\Theta)$  satisfies the criterion, then  $Z$  is Cohen–Macaulay,  $\mathcal{I}_Z$  has projective dimension one, its Hilbert–Burch resolution does not split into powers of  $\Theta$ , nor into simple semi-homogeneous bundles, and the restriction  $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_Z)$  is injective. If  $Z$  is smooth then moreover  $d \in \{1, 3, 5, 7\}$ .*
- (iii) *In the range of Theorem 18.50 the two shapes do not meet: a smooth support forces  $d \leq 9$  and a split resolution forces  $d \geq 15$ , so even before (i) no support could be both.*

*Proof.* (i) is Theorem 18.52 with  $M = \mathcal{O}_X(\Theta)$ , the resolution (64) presenting  $\mathcal{I}_Z(b\Theta)$  as the cone of  $\varphi \otimes \mathcal{O}(b\Theta)$ . (ii): Cohen–Macaulay and projective dimension one are Corollary 18.45(ii); non-splitting is (i) and Theorem 18.55; injectivity is the last assertion

of [Lemma 18.51](#); the four discriminants are [Corollary 18.41](#). (iii) is [Corollary 18.41](#) against [Theorem 18.50](#)(iii), the sets  $\{1, 3, 5, 7\}$  and  $\{15, 23\}$  being disjoint. ■

*Remark 18.58* (The route map). Collecting what is now known about the first of the six statements of [Section 20.5](#), the position is this. The local part of the criterion is finished: it is a condition on depth, nothing else ([Theorem 18.43](#), [Corollary 18.45](#)). The support must therefore be Cohen–Macaulay, its ideal has projective dimension one, and there are exactly two shapes left.

The first is a smooth support. Then every invariant is forced, the discriminant is one of four ([Corollary 18.41](#)), and the resolution cannot split, now for every  $d$  and every rank ([Theorem 18.52](#)). What is needed is the existence of such a surface on a principally polarised abelian fourfold, with a Hilbert–Burch matrix over a bundle that is not a sum of line bundles, and then the vanishing of the image of  $A_\theta A_{\theta'}$  in  $H^1(S, N_{S/X})$ . The first half is a question about surfaces with prescribed Chern numbers, of the kind settled by construction or by classification; the second is a question about one cohomology group.

The second is a singular Cohen–Macaulay support. Then the invariants are still forced by (60) with  $c_1(N_{Z/X})$  in place of  $K_Z$ , and by [Corollary 18.57](#)(i) the Hilbert–Burch resolution must not split into powers of  $\Theta$  or into simple semi-homogeneous bundles ([Theorem 18.55](#)): the explicit candidates of [Table 4](#) are excluded, and what is needed is a resolution over a bundle that is not projectively flat, together with the vanishing in  $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ . In both shapes the restriction  $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_Z)$  must be injective ([Lemma 18.51](#)), which for a smooth  $S$  says that the six classes of  $\bigwedge^2 H^1(\mathcal{O}_X)$  remain independent in  $H^2(\mathcal{O}_S)$ .

Both are finite questions in the sense that they are questions about one variety and one sheaf, not about a family or a limit. Neither is settled here, and settling both would give  $\mathbf{W}(K, 4, \delta)$  for the discriminants reached and nothing beyond them.

## 19. WEIL CLASSES OF EVERY CM FIELD

The Weil classes of an imaginary quadratic field are the case  $m = 1$  of a construction that makes sense for every CM field  $F$  of degree  $2m$ , and the question whether those classes are algebraic might seem to lie outside the methods of this paper. It does not. Every theorem of [Section 14](#) that turns the conjecture for a Weil family into a propagation statement is a theorem about a connected period domain, a flat class on it, a group whose rational points act with dense orbits, and a base point where the class is a polynomial in divisor classes; none of the four uses that the field is quadratic. This section carries the four over, and supplies the base point, which is the one input that is new. Nothing here is conditional on a question about semiregularity: the base point is a CM point, and the classes there are products of divisor classes by the Lefschetz theorem on  $(1, 1)$  classes.<sup>56</sup>

### 19.1. The families.

**Setup 19.1** (Weil type for a CM field). Let  $F$  be a CM field of degree  $2m$  with maximal totally real subfield  $F_0$ , and write  $\alpha \mapsto \bar{\alpha}$  for the nontrivial automorphism of  $F/F_0$ , which is complex conjugation under every embedding. The real places of  $F_0$  are  $\tau_1, \dots, \tau_m$ , and each extends to a conjugate pair  $\sigma_\tau, \bar{\sigma}_\tau$  of embeddings of  $F$  into  $\mathbb{C}$ . Let  $V$  be an  $F$ -vector

<sup>56</sup>Nothing is claimed about the whole Hodge ring at a CM point, which can contain classes that are not polynomials in divisor classes. What is used is narrower: the particular Weil classes of [Theorem 19.4](#) are written out as polynomials with rational coefficients in divisor classes, each divisor class is algebraic by the Lefschetz theorem, and products of algebraic classes are algebraic.

space of dimension  $2n$ , let  $H: V \times V \rightarrow F$  be a nondegenerate hermitian form,  $F$ -linear in the first variable, and fix a totally imaginary  $\xi \in F^\times$ , that is  $\bar{\xi} = -\xi$ . Put

$$E(x, y) = \text{Tr}_{F/\mathbb{Q}}(\xi H(x, y)),$$

a nondegenerate alternating  $\mathbb{Q}$ -bilinear form with  $E(\alpha x, y) = E(x, \bar{\alpha}y)$ . For each  $\tau$  let  $V_\tau = V \otimes_{F_0, \tau} \mathbb{R}$ , a vector space of dimension  $2n$  over  $F \otimes_{F_0, \tau} \mathbb{R} \cong \mathbb{C}$ , and let  $(p_\tau, q_\tau)$  be the signature of the hermitian form  $H_\tau$  induced by  $H$  on it. We say that  $(V, H)$  is of  $(F, n)$ -Weil type if  $p_\tau = q_\tau = n$  for every  $\tau$ , and we call the class of  $\det H$  in  $F_0^\times / N_{F/F_0}(F^\times)$  its discriminant  $\delta$ . A polarised abelian variety  $(A, \iota, E)$  of dimension  $2mn$  with  $\iota: F \rightarrow \text{End}^0(A)$  and  $H^1(A, \mathbb{Q}) \cong V$  as  $F$ -modules with polarisation  $E$  is of  $(F, n, \delta)$ -Weil type when  $(V, H)$  is; as in [Lemma 3.2](#) we work throughout with  $V = H^1$ , a complex structure on  $V_\mathbb{R}$  being the same datum as one on the tangent space. The Weil classes of  $A$  are

$$W_F(A) = \bigwedge_F^{2n} V \subseteq \bigwedge_{\mathbb{Q}}^{2n} V = H^{2n}(A, \mathbb{Q}),$$

the  $\mathbb{Q}$ -subspace whose complexification is  $\bigoplus_{\sigma} \bigwedge^{2n} V_{\sigma}$ , where  $V_{\sigma} = V \otimes_{F, \sigma} \mathbb{C}$  runs over the  $2m$  eigenspaces of  $F$  on  $V_{\mathbb{C}}$ ; it is a one-dimensional  $F$ -vector space, of dimension  $2m$  over  $\mathbb{Q}$ . Write

**W** $(F, n, \delta)$ : every abelian variety of  $(F, n, \delta)$ -Weil type has algebraic Weil classes.

The period domain  $\mathcal{D}_F$  is the set of complex structures  $J$  on  $V_{\mathbb{R}}$  commuting with  $F$  and polarised by  $E$ ;  $G_F = \text{Res}_{F_0/\mathbb{Q}} \text{SU}(V, H)$  acts on it;  $\Gamma \subset G_F(\mathbb{Q})$  is an arithmetic subgroup,  $\mathcal{S}_F = \Gamma \backslash \mathcal{D}_F$  the quotient with  $p: \mathcal{D}_F \rightarrow \mathcal{S}_F$ ,  $\pi: \mathcal{A} \rightarrow \mathcal{S}_F$  the universal family, and  $\Sigma_F \subseteq \mathcal{D}_F$  the set of  $s$  at which  $W_F(A_s)$  is algebraic. For  $m = 1$  this is [Setup 15.12](#) and [Setup 16.1](#), with  $\omega_s$  a generator of the line.

*Remark 19.2* (Every such abelian variety is a member). If  $A$  is any abelian variety with a CM field  $F \subseteq \text{End}^0(A)$  such that  $\dim_F H^1(A, \mathbb{Q}) = 2n$  and  $\dim V_{\sigma}^{1,0} = n$  for every embedding  $\sigma$ , then  $A$  is of  $(F, n, \delta)$ -Weil type for some  $\delta$ : the Rosati involution of any polarisation restricts to a positive involution of  $F$ , hence to conjugation, and the argument of [Lemma 4.5](#) writes the polarisation as  $\text{Tr}_{F/\mathbb{Q}}(\xi H)$  for a unique hermitian  $H$ , whose signatures are then forced by the Hodge numbers as in [Proposition 19.3](#)(i). These are exactly the abelian varieties whose Weil classes for  $F$  are Hodge classes, in the sense of [\[MZ99\]](#), so **W** $(F, n, \delta)$  over all  $(F, n, \delta)$  is the statement that all Weil classes of all CM fields are algebraic.

**Proposition 19.3** (The Weil classes of a CM field). *Keep [Setup 19.1](#), with  $(V, H)$  of  $(F, n)$ -Weil type.*

- (i) *A polarised complex structure  $J$  is the same as an  $H_{\tau}$ -orthogonal decomposition  $V_{\tau} = V_{\tau}^{+} \oplus V_{\tau}^{-}$  for every  $\tau$ , with  $H_{\tau}$  definite of the sign of  $\text{Im } \sigma_{\tau}(\xi)$  on  $V_{\tau}^{+}$  and of the opposite sign on  $V_{\tau}^{-}$ ; then  $\dim V_{\sigma_{\tau}}^{1,0} = \dim V_{\tau}^{+} = n$  and  $\dim V_{\sigma_{\tau}}^{1,0} = \dim V_{\tau}^{-} = n$ , so every class in  $W_F(A_J)$  is of type  $(n, n)$ .  $W_F$  is a flat subsystem of  $R^{2n} \pi_* \mathbb{Q}$  of rank  $2m$  on which  $G_F$  acts trivially, and  $\mathcal{D}_F \cong \prod_{\tau} \mathcal{D}_{n,n}$  is connected of dimension  $mn^2$ .*
- (ii) *If one nonzero element of  $W_F(A)$  is algebraic, every element is.*
- (iii) *If  $\text{Hg}(A_s) = G_F$ , the Hodge ring of  $A_s$  is generated by  $\text{NS}(A_s)_{\mathbb{Q}}$  and by  $W_F$ ; so the Hodge conjecture for  $A_s$  in every degree is the algebraicity of  $W_F$ . For  $n \geq 2$  the space  $\text{NS}(A_s)_{\mathbb{Q}}$  is the  $m$ -dimensional space of classes  $\eta_t = [E(t \cdot, \cdot)]$ ,  $t \in F_0$ .*

- (iv) Two members of  $(F, n, \delta)$ -Weil type have isomorphic  $(V, H)$ , so every one of them is a fibre of  $\pi$ .

*Proof.* (i) A complex structure commuting with  $F$  preserves each  $V_\tau$ , since  $F_0 \otimes \mathbb{R}$  acts diagonally, and is  $\mathbb{C}$ -linear on it for the structure  $F \otimes_{F_0, \tau} \mathbb{R} \cong \mathbb{C}$ , so it is  $\pm\sqrt{-1}$  on two complementary  $\mathbb{C}$ -subspaces  $V_\tau^\pm$ . Write  $\sigma_\tau(\xi) = \sqrt{-1} r_\tau$  with  $r_\tau \in \mathbb{R}^\times$ . For  $x \in V_\tau^\epsilon$  and  $y \in V_\tau^{\epsilon'}$  one has  $H_\tau(Jx, Jy) = \epsilon\epsilon' H_\tau(x, y)$ , so  $E(Jx, Jy) = E(x, y)$  forces  $E(x, y) = 0$  for  $\epsilon \neq \epsilon'$ , and replacing  $x$  by  $\sqrt{-1}x$  then forces  $H_\tau(x, y) = 0$ : the decomposition is orthogonal. And  $E(x, Jx) = 2\operatorname{Re}(\sqrt{-1} r_\tau \epsilon \sqrt{-1} H_\tau(x, x)) = 2\epsilon r_\tau H_\tau(x, x)$ , so positivity is the sign condition stated. Conversely such a decomposition defines a polarised  $J$ . Under  $V_\tau \otimes_{\mathbb{R}} \mathbb{C} = V_{\sigma_\tau} \oplus V_{\bar{\sigma}_\tau}$  the space  $V^{1,0}$ , the  $\sqrt{-1}$ -eigenspace of  $J$  on  $V_\mathbb{C}$ , meets  $V_{\sigma_\tau}$  in the image of  $V_\tau^+$  and  $V_{\bar{\sigma}_\tau}$  in the conjugate of the image of  $V_\tau^-$ , which gives the dimensions; since  $H_\tau$  has signature  $(n, n)$  the two spaces have dimension  $n$ . A generator  $\alpha_\sigma$  of  $\Lambda^{2n} V_\sigma$  is then of type  $(n, n)$ . The subspace  $\Lambda_F^{2n} V$  of  $H^{2n}(A_s, \mathbb{Q})$  is defined by the action of  $F$  on the lattice and does not depend on  $s$ , so it is flat;  $\operatorname{SU}(V, H)$  acts on  $\Lambda_F^{2n} V$  by the  $F$ -determinant, which is 1. Finally the choice of  $V_\tau^+$  is a point of the bounded symmetric domain of  $\operatorname{SU}(V_\tau, H_\tau) \cong \operatorname{SU}(n, n)$ , of dimension  $n^2$ , and  $\mathcal{D}_F$  is the product over  $\tau$ .

(ii) For  $\alpha \in \mathcal{O}_F$  the endomorphism  $\iota(\alpha)$  acts on  $\Lambda_F^{2n} V$  by  $\alpha^{2n}$ , through an algebraic correspondence, so the algebraic classes in  $W_F$  form a  $\mathbb{Q}$ -subspace stable under multiplication by every  $\alpha^{2n}$ . In characteristic zero the  $2n$ -th powers of the elements of a field span, over the prime field, the whole image of the  $2n$ -fold multiplication map, which is the field itself; so the  $\mathbb{Q}$ -span of  $\{\alpha^{2n}w\}$  is  $Fw = W_F$  for every  $w \neq 0$ .

(iii) Hodge classes are the invariants of the Hodge group. Over  $\mathbb{C}$  the group is  $\prod_\tau \operatorname{SL}(U_\tau)$  with  $U_\tau = V_{\sigma_\tau}$ , and  $H$  identifies  $V_{\bar{\sigma}_\tau}$  with  $U_\tau^\vee$ , so  $\Lambda^* V_\mathbb{C} = \bigotimes_\tau \Lambda^*(U_\tau \oplus U_\tau^\vee)$  with the  $\tau$ -th factor of the group acting on the  $\tau$ -th tensor factor alone. The invariants of a product group in a tensor product of representations are the tensor product of the invariants, and the invariants of  $\operatorname{SL}(U)$  in  $\Lambda^*(U \oplus U^\vee)$  are generated by the pairing  $\psi \in U \otimes U^\vee$  and the two determinants, by the computation of [Theorem 4.14](#). The classes  $\eta_t$  are Hodge classes, being polarisations of the abelian variety  $A_s$  with the  $F_0$ -twisted form  $E_t = \operatorname{Tr}(\xi t H)$ , their complexification is  $\bigoplus_\tau \mathbb{C}\psi_\tau$ , and the complexification of  $W_F$  contains all the determinants. So the  $\mathbb{Q}$ -algebra generated by  $\operatorname{NS}_\mathbb{Q}$  and  $W_F$  has complexification equal to the ring of invariants, hence equals  $\operatorname{Hdg}^*(A_s)$ . Divisor classes being algebraic, the conjecture reduces to  $W_F$ , and by (ii) to one class. For the last sentence, the invariants in  $\Lambda^2 V_\mathbb{C}$  are  $\bigoplus_\tau \mathbb{C}\psi_\tau$  when  $n \geq 2$ : a summand  $\Lambda^2 U_\tau$  or  $\Lambda^2 U_\tau^\vee$  has an  $\operatorname{SL}(U_\tau)$ -invariant only if  $\dim U_\tau = 2n$  equals 2, and a summand mixing two different  $\tau$  has none.

(iv) Two hermitian forms over  $F/F_0$  of the same rank, the same discriminant and the same signature at every real place of  $F_0$  are isomorphic [[Sch85](#), Ch. 10, §6]. So the datum  $(F, n, \delta)$  determines  $(V, H)$  up to isomorphism, and the Hodge structure of any member is, by (i), a polarised complex structure on that  $V_\mathbb{R}$ , that is a point of  $\mathcal{D}_F$ . ■

## 19.2. A base point in every family.

**Theorem 19.4** (A base point in every family, for every CM field). *Let  $(V, H)$  be of  $(F, n)$ -Weil type with discriminant  $\delta$ . Then  $\mathcal{D}_F$  contains a point  $s_0$  with the following properties.*

- (i)  $A_{s_0}$  is isogenous to  $B_\Phi^n \times B_{\bar{\Phi}}^n$ , where  $B_\Phi$  is an abelian  $m$ -fold with complex multiplication by  $F$  of type  $\Phi = \{\sigma : \operatorname{Im} \sigma(\xi) > 0\}$  and  $B_{\bar{\Phi}}$  its conjugate; there is an  $H$ -orthogonal basis  $e_1, \dots, e_{2n}$  of  $V$  with  $H(e_i, e_i)$  totally positive for  $i \leq n$  and

totally negative for  $i > n$ , and  $Fe_i = H^1(B_\Phi, \mathbb{Q})$  for  $i \leq n$ ,  $Fe_i = H^1(B_{\bar{\Phi}}, \mathbb{Q})$  for  $i > n$ .

- (ii) For  $i \leq n$  put  $i' = i + n$  and let  $D_{ii'} \subseteq H^2(A_{s_0}, \mathbb{Q})$  be the  $F$ -balanced part of  $Fe_i \wedge Fe_{i'}$ , spanned by the classes

$$\delta_i(f) = \sum_k (\omega_k f e_i) \wedge (\omega_k^* e_{i'}), \quad f \in F,$$

for dual  $\mathbb{Q}$ -bases  $(\omega_k)$ ,  $(\omega_k^*)$  of  $F$  with respect to the trace form. Every  $\delta_i(f)$  is a Hodge class of type  $(1, 1)$ , hence a divisor class, and  $D_{ii'}$  is a one-dimensional  $F$ -vector space with  $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$ .

- (iii) The Weil classes are the products

$$w(f) = \sum_{k_1, \dots, k_{n-1}} \delta_1(\omega_{k_1} f) \cdot \delta_2(\omega_{k_1}^* \omega_{k_2}) \cdots \delta_{n-1}(\omega_{k_{n-2}}^* \omega_{k_{n-1}}) \cdot \delta_n(\omega_{k_{n-1}}^*), \quad f \in F,$$

which lie in  $W_F(A_{s_0})$ , span it as  $f$  runs over  $F$ , and are polynomials with rational coefficients in divisor classes. In particular  $W_F(A_{s_0})$  is algebraic, and  $\Sigma_F \neq \emptyset$  for every  $F$ , every  $n \geq 1$  and every  $\delta$ .

*Proof.* (i) Since  $H_\tau$  has signature  $(n, n)$  at every  $\tau$ , the discriminant  $\det H$  has sign  $(-1)^n$  at every real place, so  $b = (-1)^n \det H$  is totally positive, and the diagonal form  $H' = \text{diag}(b, 1, \dots, 1, -1, \dots, -1)$  with  $n$  positive and  $n$  negative entries has the same dimension, the same discriminant and the same signatures as  $H$ . Hermitian forms over  $F/F_0$  are classified by these invariants [Sch85, Ch. 10, §6], so  $(V, H) \cong (F^{2n}, H')$  and the standard basis is the required orthogonal basis  $e_i$ , with  $a_i = H(e_i, e_i)$ . Write  $\sigma_\tau(\xi) = \sqrt{-1} r_\tau$  and define  $J$  on  $V_\tau = \bigoplus_i \mathbb{C} e_i$  to be  $\epsilon_{i,\tau} \sqrt{-1}$  on  $\mathbb{C} e_i$ , with  $\epsilon_{i,\tau} = \text{sign}(r_\tau) \text{sign}(\tau(a_i))$ . The decomposition  $V_\tau^+ = \bigoplus_{\epsilon_{i,\tau}=+} \mathbb{C} e_i$ ,  $V_\tau^- = \bigoplus_{\epsilon_{i,\tau}=-} \mathbb{C} e_i$  is  $H_\tau$ -orthogonal with  $H_\tau$  of sign  $\text{sign}(r_\tau)$  on  $V_\tau^+$ , so Proposition 19.3(i) makes  $J$  a point  $s_0$  of  $\mathcal{D}_F$ . On  $Fe_i \otimes \mathbb{R} = \prod_\tau \mathbb{C} e_i$  the structure  $J$  is multiplication by the element  $(\epsilon_{i,\tau} \sqrt{-1})_\tau$  of  $F \otimes_{\mathbb{Q}} \mathbb{R}$ , so each  $Fe_i$  is a rational sub-Hodge structure of weight one with  $F$ -action, that is the  $H^1$  of an abelian  $m$ -fold with complex multiplication by  $F$ , of CM type  $\Phi_i = \{\sigma : (Fe_i)_\sigma \subseteq V^{1,0}\}$ , which by the computation in Proposition 19.3(i) is  $\{\sigma_\tau^{\epsilon_{i,\tau}}\}$ , where  $\sigma^+ = \sigma$  and  $\sigma^- = \bar{\sigma}$ . For  $i \leq n$  the  $a_i$  are totally positive, so  $\epsilon_{i,\tau} = \text{sign}(r_\tau)$  and  $\Phi_i = \{\sigma : \text{Im } \sigma(\xi) > 0\} = \Phi$ ; for  $i > n$  the signs are reversed and  $\Phi_i = \bar{\Phi}$ . So  $V = \bigoplus_i Fe_i$  splits  $A_{s_0}$  up to isogeny as stated.

(ii) Write  $L_{i,\sigma} = (Fe_i)_\sigma$ , a line, so that  $V_{\mathbb{C}} = \bigoplus_{i,\sigma} L_{i,\sigma}$  and  $L_{i,\sigma}$  has type  $(1, 0)$  exactly when  $\sigma \in \Phi_i$ . For dual bases,  $\sum_k \sigma(\omega_k) \sigma'(\omega_k^*) = 1$  if  $\sigma = \sigma'$  and 0 otherwise, because the matrices  $(\sigma(\omega_k))$  and  $(\sigma(\omega_k^*))$  are inverse transposes by the definition of the trace form. Hence in  $Fe_i \otimes Fe_{i'} \otimes \mathbb{C} = \bigoplus_{\sigma, \sigma'} L_{i,\sigma} \otimes L_{i',\sigma'}$  the class  $\delta_i(f)$  has component  $\sigma(f) \ell_{i,\sigma} \wedge \ell_{i',\sigma}$  on the diagonal  $\sigma = \sigma'$  and zero off it, for suitable generators  $\ell$ ; it is  $F$ -balanced in the sense that  $(\alpha e_i) \wedge e_{i'}$  and  $e_i \wedge (\alpha e_{i'})$  agree on it, which is the identity  $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$ . Since  $\Phi_{i'} = \bar{\Phi}_i$ , exactly one of  $L_{i,\sigma}$ ,  $L_{i',\sigma}$  has type  $(1, 0)$ , so every component, hence  $\delta_i(f)$ , has type  $(1, 1)$ ; and  $\delta_i(f)$  is rational, being a sum of wedges of rational vectors. By the Lefschetz theorem on  $(1, 1)$  classes it is a divisor class. As  $f$  varies the components  $\sigma(f)$  vary independently, so  $D_{ii'} \otimes \mathbb{C} = \bigoplus_{\sigma} L_{i,\sigma} \otimes L_{i',\sigma}$  and  $D_{ii'} \cong F$ .

(iii) The complexification of  $W_F$  is spanned by  $\alpha_\sigma = \pm \prod_{i \leq n} \ell_{i,\sigma} \wedge \ell_{i',\sigma}$ , a generator of  $\bigwedge^{2n} V_\sigma = \bigwedge^{2n} (\bigoplus_i L_{i,\sigma})$ . In the product  $w(f)$ , the component indexed by  $(\sigma_1, \dots, \sigma_n)$ , with  $\sigma_i$  the embedding chosen in the  $i$ -th factor, carries the coefficient  $\sigma_1(f) \prod_{j=1}^{n-1} \sum_{k_j} \sigma_j(\omega_{k_j}) \sigma_{j+1}(\omega_{k_j}^*)$ , which vanishes unless  $\sigma_1 = \dots = \sigma_n$  and then equals  $\sigma_1(f)$ . So  $w(f) = \sum_\sigma \sigma(f) \alpha_\sigma$  up to the sign of the reordering, an element of  $W_F \otimes \mathbb{C}$ .

which is rational because it is a rational polynomial in rational classes; as  $f$  runs over  $F$  the vectors  $(\sigma(f))_\sigma$  span  $\mathbb{C}^{2m}$ , so the  $w(f)$  span  $W_F$ . Each  $w(f)$  is a rational combination of  $n$ -fold products of the divisor classes  $\delta_i(\cdot)$ , and a product of divisor classes is algebraic. Item (XXXVI) of [Appendix D](#) carries out every step of (i), (ii) and (iii) in exact arithmetic for six pairs  $(F, n)$ , with  $F$  one of  $\mathbb{Q}(\zeta_5)$ ,  $\mathbb{Q}(\zeta_7)$ ,  $\mathbb{Q}(\zeta_8)$ ,  $\mathbb{Q}(\zeta_9)$ , so  $m = 2$  and  $m = 3$ , and  $n = 1, 2, 3$ ; for  $F = \mathbb{Q}(\zeta_5)$ , a cyclic quartic CM field with no imaginary quadratic subfield, at  $n = 2$  it also computes the space of rational  $(1, 1)$  classes, of dimension 32, and records that the balanced sum is necessary: the plain product  $\delta_1(1)\delta_2(1)$  is not a Weil class. ■

*Remark 19.5* (The base point for a quadratic field). For  $m = 1$  the point  $s_0$  is a member isogenous to  $E^{2n}$  for an elliptic curve  $E$  with complex multiplication by  $K$ , at which the whole Hodge ring is generated by divisor classes, since the Hodge group is a one-dimensional torus. It is a second base point in every family of [Section 14](#), beside the quaternionic one of [Theorem 15.31](#), and it lies in the product locus of [Theorem 15.10](#). The formula in (iii) is then the explicit form of the class at that point.

### 19.3. The reduction and the propagation.

**Theorem 19.6** (The reduction and the propagation, for every CM field). *Fix  $(F, n, \delta)$  with  $n \geq 2$  and keep [Setup 19.1](#), and let  $\omega_s$  denote a generator of  $W_F(A_s)$  chosen flatly.*

- (i)  $\mathbf{W}(F, n, \delta)$  holds if and only if  $\omega_s$  is algebraic for every  $s \in \mathcal{D}_F$ , and then the Hodge conjecture holds in every degree for every member of the family with  $\mathrm{Hg}(A_s) = G_F$ .
- (ii)  $\Sigma_F$  is stable under  $G_F(\mathbb{Q})$ , every  $G_F(\mathbb{Q})$ -orbit in  $\mathcal{D}_F$  is dense, and  $\Sigma_F$  is dense.
- (iii)  $\Sigma_F = p^{-1}(\bigcup_{\underline{P}, \underline{q}} \Sigma_{\underline{P}, \underline{q}})$  with each  $\Sigma_{\underline{P}, \underline{q}}$  Zariski closed in  $\mathcal{S}_F$ , the union countable; either  $\Sigma_F = \mathcal{D}_F$ , or  $\Sigma_F$  is meagre and the conjecture fails on a dense  $G_\delta$ .
- (iv) If at every point  $s$  of the orbit  $G_F(\mathbb{Q}) \cdot s_0$  the class  $\omega_s$  is represented by a cycle whose Hilbert data comes from one finite set, then  $\Sigma_F = \mathcal{D}_F$ .
- (v) If some perfect complex  $E$  on  $A_{s_0}$  with  $\mathrm{ch}(E) = N\omega_{s_0} + P$ , where  $N \neq 0$  and  $P$  is a polynomial with rational coefficients in the classes  $\eta_t$  of [Proposition 19.3\(iii\)](#), has injective semiregularity map, then  $\Sigma_F = \mathcal{D}_F$ . Complexes with  $\mathrm{ch}(E) = N\omega_{s_0}$  exist.

*Proof.* (i) is [Proposition 19.3\(ii\)](#) and (iii): each member has its Hodge structure at some point of  $\mathcal{D}_F$ , by (i) of that proposition, so the statement  $\mathbf{W}(F, n, \delta)$  is the algebraicity of  $W_F$  at every point, which by (ii) is that of one generator, and by (iii) it gives the conjecture at the members with full Hodge group.

(ii) The stability is the argument of [Lemma 16.6](#):  $g \in G_F(\mathbb{Q})$  carries  $J$  to  $gJg^{-1}$  and induces, after clearing denominators, an isogeny  $A_J \rightarrow A_{gJ}$  whose graph transports algebraic classes to algebraic classes, and  $g$  fixes  $W_F$  pointwise by [Proposition 19.3\(i\)](#). For the density we show that the closure of  $G_F(\mathbb{Q})$  in  $G_F(\mathbb{R}) = \prod_\tau \mathrm{SU}(V_\tau, H_\tau)$  contains the identity component, which acts transitively on  $\mathcal{D}_F$ . The form  $H$  is isotropic over  $F_0$ : it is indefinite at every real place, its signature being  $(n, n)$ ; at every finite place  $v$  its trace form is a quadratic form in  $4n \geq 8$  variables over the local field  $F_{0,v}$ , hence isotropic, and a hermitian form is isotropic as soon as its trace form is [[Sch85](#), Ch. 10, §1]; and hermitian forms over the quadratic extension  $F/F_0$  satisfy the Hasse principle [[Sch85](#), Ch. 10, §6]. Let  $X = \{v \in V : H(v, v) = 0\}$ , the cone of isotropic vectors, a quadric cone in the  $4n$ -dimensional  $F_0$ -space  $V$  with a nonzero  $F_0$ -point; the projective quadric of its lines is  $F_0$ -rational, by projection from that point, so it is  $F_0$ -birational to a projective space and therefore satisfies weak approximation at the archimedean places,  $F_0$  being dense in  $F_0 \otimes \mathbb{R} = \prod_\tau \mathbb{R}$ . Hence the  $F_0$ -points of  $X$  are dense in its points over  $F_0 \otimes \mathbb{R}$ , which are the products  $\prod_\tau X_\tau$  of the isotropic cones of the  $H_\tau$ . For an isotropic

vector  $v \in V$  let  $U_v \subseteq \mathrm{SU}(V, H)$  be the unipotent radical of the stabiliser of the line  $Fv$ , a unipotent  $F_0$ -group; the exponential identifies  $U_v(F_0)$  with an  $F_0$ -vector space and  $U_v(F_0 \otimes \mathbb{R}) = \prod_{\tau} U_{v,\tau}(\mathbb{R})$  with the corresponding real space, so  $U_v(\mathbb{Q}) = U_v(F_0)$  is dense in  $\prod_{\tau} U_{v,\tau}(\mathbb{R})$ . Hence the closure  $L$  of  $G_F(\mathbb{Q})$  contains  $\prod_{\tau} U_{v,\tau}(\mathbb{R})$  for every rational isotropic  $v$ , in particular the factor  $U_{v,\tau}(\mathbb{R})$  for each  $\tau$  separately, and its Lie algebra  $\mathfrak{l}$  contains  $\mathfrak{u}_{v,\tau}$  for every  $\tau$  and every rational isotropic  $v$ . The map  $v \mapsto \mathfrak{u}_{v,\tau}$  is continuous on the cone  $X_{\tau} \setminus 0$  and rational  $v$  are dense in  $\prod_{\tau} X_{\tau}$ , so the closed subspace  $\mathfrak{l} \cap \mathfrak{su}(V_{\tau}, H_{\tau})$  contains  $\mathfrak{u}_{v,\tau}$  for every isotropic  $v \in V_{\tau}$ . The span of these is stable under the adjoint action of  $\mathrm{SU}(V_{\tau}, H_{\tau})$ , which permutes the isotropic lines, so it is a nonzero ideal of the simple Lie algebra  $\mathfrak{su}(n, n)$ , hence everything. Thus  $\mathfrak{l} \supseteq \bigoplus_{\tau} \mathfrak{su}(V_{\tau}, H_{\tau})$ , and  $L \supseteq G_F(\mathbb{R})^+$ . Since  $\Sigma_F \neq \emptyset$  by [Theorem 19.4](#) and is  $G_F(\mathbb{Q})$ -stable, it contains a dense orbit.

(iii) The proofs of [Theorem 16.4](#) and [Corollary 16.5](#) use only that  $\mathcal{S}_F$  is a quasi-projective variety, which is Baily and Borel [[BB66](#)], that the relative Hilbert schemes of  $\mathcal{A}/\mathcal{S}_F$  are projective over it [[Gro61](#)], and that  $W_F$  is a flat subsystem consisting of Hodge classes, which is [Proposition 19.3\(i\)](#); they apply verbatim.

(iv) is the proof of [Theorem 16.9](#), which asks for a cycle chosen at each point ([Remark 16.10](#)): the orbit is dense by (ii), a finite union of the closed sets of (iii) covering it forces one of them to be dense and closed, hence everything; then (i). As for  $K$ , the hypothesis is equivalent to the conclusion ([Theorem 16.17](#)), and the transported cycle does not satisfy it ([Theorem 16.15](#)).

(v) The existence is the proof of [Proposition 16.26](#), using the algebraic cycle of [Theorem 19.4](#) and the isomorphism  $K_0(A)_{\mathbb{Q}} \cong \mathrm{CH}^*(A)_{\mathbb{Q}}$  [[Ful98](#)]; the deformation argument is the proof of [Theorem 16.28](#) for a corrected character, as in [Remark 17.6](#), whose only inputs are the flatness of  $\mathrm{ch}(E)$  as a section of Hodge classes over  $\mathcal{D}_F$ , which holds because  $W_F$  is flat by (i) of [Proposition 19.3](#) and the  $\eta_t$  stay of Hodge type on the whole family, the theorem of Buchweitz and Flenner, and (iii). At a member  $A_s$  reached by the deformation the component of degree  $2n$  of the character is  $N\omega_s + P_{2n}$ , with  $P_{2n}$  a polynomial in divisor classes, so  $\omega_s$  is algebraic. ■

*Remark 19.7* (What is and is not new here). The propagation statements (iv) and (v) are exactly the two forms (P1) and (P2) of [Remark 17.7](#), with  $F$  in place of  $K$ , (v) in the corrected form of [Remark 17.6](#). What this section proves is that they are the whole of the problem for every CM field, and that their hypothesis is never vacuous: the base point exists, its cycle is written down, and the orbit through it is dense. The theorem of [Theorem 16.30](#) on what (v) forces transfers as well, with the annihilator of a class in  $W_F$  computed over the  $2m$  eigenspaces in place of two, and we do not repeat it. Nothing above depends on Question 11.4 of [[Mar25b](#)] or on any secant construction; the route through Weil restriction and CM points is independent of the route through secant sheaves, and reaches every field at once. Markman has since extended the route through secant sheaves to these fields [[Mar25c](#)]; see (C3) in [Section 20.5](#), and [Section 19.4](#) for the smallest case.

**Proposition 19.8** (Composite fields). *Let  $F = KF_0$  with  $K$  imaginary quadratic, and let  $A$  be of  $(F, n, \delta)$ -Weil type. Then  $A$  is of  $(K, -1, mn)$ -Weil type for the restricted action, its Weil classes  $W_K(A) \subseteq H^{2mn}(A, \mathbb{Q})$  lie in the  $\mathbb{Q}$ -span of the  $m$ -fold products of elements of  $W_F(A)$ , and  $W_K(A)$  is algebraic whenever  $W_F(A)$  is. In particular  $\mathbf{W}(F, n, \delta)$  implies the algebraicity of the  $K$ -Weil classes on every member of the  $F$ -family, a closed subvariety of dimension  $mn^2$  of the  $K$ -family of dimension  $m^2n^2$ ; the converse implication is not available, and [Theorem 19.4](#) is not a consequence of [Theorem 15.31](#).*

*Proof.* The eigenspace  $V_{\tau_K}$  of  $K$  on  $V_{\mathbb{C}}$  for the embedding  $\tau_K$  is  $\bigoplus_{\sigma|\tau_K} V_{\sigma}$ , of dimension  $2mn$  with  $\dim V_{\tau_K}^{1,0} = mn$  by [Proposition 19.3\(i\)](#), so  $A$  is of  $K$ -Weil type with balanced signature, and  $\bigwedge^{2mn} V_{\tau_K} = \bigotimes_{\sigma|\tau_K} \bigwedge^{2n} V_{\sigma}$  is spanned by  $\prod_{\sigma|\tau_K} \alpha_{\sigma}$ . So  $W_K \otimes \mathbb{C}$  is contained in the complexification of the  $\mathbb{Q}$ -span  $P$  of the  $m$ -fold products of elements of  $W_F$ , and a rational subspace whose complexification lies in that of  $P$  lies in  $P$ . Products of algebraic classes are algebraic. Item (XXXVI) checks the identity for  $F = \mathbb{Q}(\zeta_8)$  and  $K = \mathbb{Q}(\sqrt{-1})$ . The final sentence is dimension count: for  $m \geq 2$  the  $F$ -family is a proper subvariety, and the  $K$ -Weil classes on it are products that carry no information about their factors. ■

**19.4. The smallest open case: quartic fields and  $n = 2$ .** By [Theorem 20.40\(ii\)](#) the only form of (P2) on a minimal route to the conjecture is the one for the split families of the CM fields of degree at least four, and by [Proposition 20.13\(i\)](#) the Weil classes of any family of such a field in a higher dimension give those of its families at  $n = 2$ . For a quartic field these are families of abelian eightfolds, and they are the smallest families of a field of degree at least four whose Weil classes are not known to be algebraic: we know of no quartic CM field  $F$  and no discriminant  $\delta$  for which  $\mathbf{W}(F, 2, \delta)$  is proved. This subsection determines how far the available mechanisms reach there. Divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, which has codimension two when  $\delta$  is trivial ([Proposition 19.10](#), [Corollary 19.11](#)); for biquadratic  $F$  Markman’s theorem reaches them on a further locus of dimension four ([Proposition 19.12](#)); the Casimir class of the quadratic form on the weight two part  $R_F$  below, the natural class of degree four attached to it, is algebraic exactly when the Weil classes are ([Proposition 19.13](#)); correspondences with abelian varieties whose simple factors have dimension at most seven cannot help at a member with full Hodge group ([Proposition 19.15](#)); and for a general shape the corrected criterion asks  $\dim \operatorname{Ext}^2(E, E) = 112$  of one complex ([Theorem 19.16](#)). None of this is a new case of the Hodge conjecture, and  $\mathbf{W}(F, 2, \delta)$  remains open for every quartic  $F$  and every  $\delta$ . Item (LV) of [Appendix D](#) checks the statements in exact arithmetic; its scripts are in `code/attack/quartic_cm`.

Throughout,  $F$  is a quartic CM field and  $F_0$  its real quadratic subfield, with real places  $\tau_1, \tau_2$ , and  $(V, H)$  is of  $(F, 2)$ -Weil type with discriminant  $\delta$ ; so  $m = n = 2$ , the members are abelian eightfolds, and  $\mathcal{D}_F$  has dimension  $mn^2 = 8$ . By [Proposition 19.3\(i\)](#),  $H^1(A, \mathbb{C}) = \bigoplus_{\sigma} V_{\sigma}$  over the four embeddings  $\sigma$  of  $F$ , with  $\dim V_{\sigma}^{1,0} = \dim V_{\sigma}^{0,1} = 2$ . Put

$$R_F = \bigwedge_F^2 V \subseteq \bigwedge_{\mathbb{Q}}^2 V = H^2(A, \mathbb{Q}),$$

the  $\mathbb{Q}$ -subspace with complexification  $\bigoplus_{\sigma} \bigwedge^2 V_{\sigma}$ ; for  $m = 1$  it is the space  $R$  of [Proposition 15.18](#). It is an  $F$ -vector space of dimension 6, of  $\mathbb{Q}$ -dimension 24, with Hodge numbers  $(4, 16, 4)$ . Fix a generator  $\Omega$  of the  $F$ -line  $W_F = \bigwedge_F^4 V$ , so that  $W_F = \{a\Omega : a \in F\}$ , and define a symmetric  $F$ -bilinear form  $q$  on  $R_F$  by  $x \wedge_F y = q(x, y)\Omega$ . Let  $h$  be the hermitian form on  $R_F$  induced by  $H$ ,  $h(x_1 \wedge x_2, y_1 \wedge y_2) = \det(H(x_i, y_j))$ , and  $h_4$  the one induced on  $\bigwedge_F^4 V$ ; in a basis  $e_1, \dots, e_4$  with  $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$ ,  $h_4(\Omega, \Omega)$  is  $\det H$ , so its class in  $F_0^{\times}/N_{F/F_0}(F^{\times})$  is  $\delta$ . With dual bases  $(\omega_k), (\omega_k^*)$  of  $F$  for the trace form, as in [Theorem 19.4](#), put  $x \star y = \sum_k (\omega_k x) \cup (\omega_k^* y)$  for  $x, y \in R_F$ . Then

$$x \star y = q(x, y)\Omega, \tag{66}$$

because the complexification of  $(\omega_k x) \cup (\omega_k^* y)$  is  $\sum_{\sigma, \sigma'} \sigma(\omega_k) \sigma'(\omega_k^*) x_{\sigma} \wedge y_{\sigma'}$ , the sum over  $k$  of the coefficient is 1 for  $\sigma = \sigma'$  and 0 otherwise, as in the proof of [Theorem 19.4\(ii\)](#), and  $\sum_{\sigma} x_{\sigma} \wedge y_{\sigma}$  is the complexification of  $q(x, y)\Omega$ . Finally let  $\theta_j \in H^{1,1}(A)$  be the component of the polarisation that pairs  $V_{\sigma}$  with  $V_{\bar{\sigma}}$  for the two embeddings  $\sigma, \bar{\sigma}$  over

$\tau_j$ . Then  $\eta_t = \sum_j \tau_j(t) \theta_j$  for the classes  $\eta_t$  of [Proposition 19.3\(iii\)](#), and  $\mathbb{C}[\theta_1, \theta_2]$  is the complexification of the  $\mathbb{Q}$ -algebra generated by the  $\eta_t$ .

**Proposition 19.9** (Hodge classes, and a quaternion algebra acting on  $R_F$ ). *Let  $A$  be a member of an  $(F, 2, \delta)$ -family.*

- (i) *If  $\text{Hg}(A) = G_F$ , the space of Hodge classes of  $A$  has dimension 2 in degree two, spanned by the  $\eta_t$ , and 7 in degree four, spanned by the three products of two  $\eta_t$  and by  $W_F(A)$ , which meets the span of those products in 0.*
- (ii) *The map  $\phi: R_F \rightarrow R_F$  defined by  $q(x, y) = h(x, \phi y)$  satisfies  $\phi(ax) = \bar{a}\phi(x)$  for  $a \in F$  and  $\phi^2 = h_4(\Omega, \Omega)^{-1}$ , and it commutes with  $\text{SU}(V, H)$ . So  $\phi$  and the action of  $F$  are Hodge endomorphisms of  $R_F$  at every member, and the Hodge classes in  $R_F$  form a module over the algebra  $\mathcal{Q}_\delta = F \oplus F\phi$ .*
- (iii)  *$\mathcal{Q}_\delta$  is the cyclic quaternion algebra  $(F/F_0, \delta)$  over  $F_0$ , which is split if and only if  $\delta$  is trivial. If  $\text{Hg}(A) = G_F$ , then  $\mathcal{Q}_\delta$  is the whole algebra of Hodge endomorphisms of  $R_F$ .*
- (iv) *If  $\delta$  is trivial, then  $R_F = W \otimes_{F_0} F$  for a rational sub-Hodge structure  $W$ , stable under  $F_0$ , of dimension 12 and with Hodge numbers  $(2, 8, 2)$ .*

*Proof.* (i) As in the proof of [Proposition 19.3\(iii\)](#), over  $\mathbb{C}$  the group  $G_F$  is  $\text{SL}(U_1) \times \text{SL}(U_2)$  with  $\dim U_j = 4$ , and  $H^1(A, \mathbb{C}) = \bigoplus_j (U_j \oplus U_j^\vee)$ . The invariants of  $\text{SL}(U)$  in  $\bigwedge^a U \otimes \bigwedge^b U^\vee$  form a line if  $a = b$  or  $\{a, b\} = \{0, 4\}$ , and are zero otherwise. In degree two this leaves the bidegree  $(1, 1)$  in one factor: two lines, spanned by  $\theta_1$  and  $\theta_2$ . In degree four it leaves  $(2, 2)$  in one factor or  $(1, 1)$  in both, three lines spanned by  $\theta_1^2$ ,  $\theta_1\theta_2$  and  $\theta_2^2$ , which are nonzero because  $\dim U_j = 4$ ; and  $(4, 0)$  or  $(0, 4)$  in one factor, four lines spanning  $W_F \otimes \mathbb{C}$ . They lie in different summands.

(ii) Choose an  $H$ -orthogonal basis  $e_1, \dots, e_4$  of  $V$  and put  $a_i = H(e_i, e_i) \in F_0$ . Replacing  $\Omega$  by  $\lambda\Omega$  replaces  $\phi$  by  $\bar{\lambda}^{-1}\phi$  and  $h_4(\Omega, \Omega)$  by  $N(\lambda)h_4(\Omega, \Omega)$ , so we may take  $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$ , with  $h_4(\Omega, \Omega) = a_1 a_2 a_3 a_4$ . For  $J = \{j < j'\}$  put  $e_J = e_j \wedge e_{j'}$  and  $a_J = a_j a_{j'}$ , and let  $J^c$  be the complement, with  $e_{J^c} \wedge e_J = \epsilon_J \Omega$ . The  $e_J$  form an  $h$ -orthogonal  $F$ -basis with  $h(e_J, e_J) = a_J$ , and  $q(e_K, e_J)$  is  $\epsilon_J$  for  $K = J^c$  and 0 otherwise, so  $\phi(e_J) = \epsilon_J a_{J^c}^{-1} e_{J^c}$ . Two-forms commute, so  $\epsilon_{J^c} = \epsilon_J$  and  $\phi^2(e_J) = (a_J a_{J^c})^{-1} e_J = h_4(\Omega, \Omega)^{-1} e_J$ . Semilinearity holds because  $q$  is linear and  $h$  antilinear in the second variable. An element of  $\text{SU}(V, H)$  preserves  $h$  and fixes  $\Omega$ , hence preserves  $q$  and commutes with  $\phi$ , and it commutes with  $F$ . The Hodge group of every member lies in  $G_F$ , because the circle defining the Hodge structure preserves  $E$ , commutes with  $F$  and acts on each  $V_\sigma$  with determinant  $z^2 \bar{z}^{-2} = 1$ ; so  $\phi$  and  $F$  commute with it, and they are Hodge endomorphisms.

(iii) The relations  $\phi a = \bar{a}\phi$  and  $\phi^2 = h_4(\Omega, \Omega)^{-1}$  present  $F \oplus F\phi$  as the cyclic algebra  $(F/F_0, h_4(\Omega, \Omega)^{-1})$ , which is  $(F/F_0, \delta)$  because a square in  $F_0$  is a norm; and a cyclic algebra  $(F/F_0, c)$  is split exactly when  $c$  is a norm from  $F$ . It has  $\mathbb{Q}$ -dimension 8, since  $\phi$  is not  $F$ -linear. If  $\text{Hg}(A) = G_F$ , the Hodge endomorphisms of  $R_F$  are the commutant of  $G_F$ . Over  $\mathbb{C}$ ,  $R_F \otimes \mathbb{C} = \bigoplus_j (\bigwedge^2 U_j \oplus \bigwedge^2 U_j^\vee)$ , where  $\bigwedge^2 U_j \cong \bigwedge^2 U_j^\vee$  is irreducible for  $\text{SL}(U_j)$  and trivial for the other factor, so the commutant is  $M_2(\mathbb{C}) \times M_2(\mathbb{C})$ , of dimension 8.

(iv) If  $\delta$  is trivial,  $h_4(\Omega, \Omega) = N(b)$  for some  $b \in F$ , and  $\psi = b\phi$  satisfies  $\psi^2 = b\bar{b}\phi^2 = 1$ . Its fixed space  $W$  is an  $F_0$ -subspace and a sub-Hodge structure,  $\psi$  being a Hodge endomorphism, and an element  $\beta \in F$  with  $\bar{\beta} = -\beta \neq 0$  maps it isomorphically onto the  $(-1)$ -eigenspace, since  $\psi\beta = -\beta\psi$ . So  $R_F = W \oplus \beta W = W \otimes_{F_0} F$ , and  $W$  has half the dimension and half the Hodge numbers of  $R_F$ . Item (LV) checks (i), (ii) and the dimension in (iii) exactly, for several fields and hermitian forms. ■

**Proposition 19.10** (What divisor classes reach). *Let  $A$  be a member of an  $(F, 2, \delta)$ -family.*

- (i) *The following are equivalent: (a)  $W_F(A)$  lies in the subalgebra of  $H^*(A, \mathbb{Q})$  generated by  $\text{NS}(A)_{\mathbb{Q}}$ ; (b)  $W_F(A)$  meets that subalgebra in a nonzero class; (c)  $R_F$  contains a nonzero rational Hodge class. When they hold,  $R_F$  contains a rational Hodge class  $t$  with  $q(t, t) \neq 0$ , and  $W_F(A) = \{(ft) \star t : f \in F\}$ , a space of polynomials with rational coefficients in divisor classes.*
- (ii) *For a nonzero  $\mathcal{Q}_\delta$ -submodule  $M \subseteq R_F$  let  $\mathcal{D}(M) \subseteq \mathcal{D}_F$  be the set of points at which  $M$  consists of Hodge classes, and let  $\text{NL}(R_F) \subseteq \mathcal{D}_F$  be the set of points at which (c) holds. Then  $\text{NL}(R_F) = \bigcup_M \mathcal{D}(M)$ , a countable union. Each nonempty  $\mathcal{D}(M)$  is a closed complex submanifold, isomorphic to a product of two bounded symmetric domains of type IV and dimension  $4 - d$ , where  $d = \dim_F M$ , and  $d = 1$  occurs only if  $\delta$  is trivial. So if  $\delta$  is trivial,  $\text{NL}(R_F)$  is the union of the nonempty  $\mathcal{D}(M)$  with  $d = 1$ , each of dimension exactly 6; if  $\delta$  is not trivial, every  $\mathcal{D}(M)$  has dimension at most 4.*
- (iii)  *$\text{NL}(R_F)$  contains the base point  $s_0$  of Theorem 19.4, and for every  $\delta$  it contains the locus of products  $B_1 \times B_2$  of two abelian fourfolds of  $(F, 1)$ -Weil type, which has dimension 4. The split locus, of members isogenous to  $X \otimes_{\mathcal{O}_{F_0}} \mathcal{O}_F$  with  $X$  an abelian fourfold with  $F_0 \subseteq \text{End}^0(X)$ , has dimension 6, lies in the family of trivial discriminant, and is contained in  $\text{NL}(R_F)$ .*

*Proof.* (i) Clearly (a) implies (b). Let  $F^\times$  act on  $H^*(A, \mathbb{Q})$  by  $\alpha \mapsto \iota(\alpha)^*$ , which on  $\bigotimes_\sigma \wedge^{a_\sigma} V_\sigma$  is the character  $\prod_\sigma \sigma^{a_\sigma}$ . The isotypic parts for the characters  $\sigma^2$  in degree two and  $\sigma^4$  in degree four are  $R_F$  and  $W_F$ . The projections  $p_R$  and  $p_W$  onto them are rational, the two sets of characters being Galois orbits, and they are morphisms of Hodge structures, being  $\mathbb{Q}$ -linear combinations of the  $\iota(\alpha)^*$ . The characters occurring in  $H^2$  are the products  $\rho\rho'$ , and the only way to write  $\sigma^4$  as a product of two of them is  $\sigma^2 \cdot \sigma^2$ , so  $p_W(D \cup D') = q(p_R D, p_R D') \Omega$  for  $D, D' \in H^2(A, \mathbb{Q})$ . If a nonzero  $w \in W_F$  equals  $\sum_i D_i \cup D'_i$  with  $D_i, D'_i \in \text{NS}(A)_{\mathbb{Q}}$ , then  $w = p_W(w) = \sum_i q(p_R D_i, p_R D'_i) \Omega$ , so some  $p_R D_i$  is a nonzero rational Hodge class in  $R_F$ , and (b) implies (c). Conversely, let  $t \in R_F$  be a nonzero rational Hodge class. The part of type  $(1, 1)$  of  $\wedge^2 V_\sigma$  is  $V_\sigma^{1,0} \wedge V_\sigma^{0,1}$ , and  $H$  is definite of opposite signs on the two factors by Proposition 19.3(i), so  $h$  is negative definite on it at each real place, and  $h(t, t) \in F_0$  is totally negative. By Proposition 19.9(ii),  $q(t, \phi t) = h(t, \phi^2 t) = h_4(\Omega, \Omega)^{-1} h(t, t) \neq 0$ , so one of the rational Hodge classes  $t, \phi t, t + \phi t$ , call it  $t'$ , has  $q(t', t') \neq 0$ . Each  $\omega_k f t'$  and each  $\omega_k^* t'$  is a rational Hodge class in  $H^2$ , hence a divisor class, and by (66)  $(f t') \star t' = f q(t', t') \Omega$ , which runs over  $W_F$  as  $f$  runs over  $F$ . So (c) implies (a), with the stated formula.

(ii) The Hodge classes in  $R_F$  form a  $\mathcal{Q}_\delta$ -module, so  $\text{NL}(R_F)$  is the union of the  $\mathcal{D}(M)$ , and there are countably many  $M$ . Fix a real place  $\tau$ , let  $\sigma$  be an embedding over it, and put  $R_\tau = R_F \otimes_{F_0, \tau} \mathbb{R}$ , a complex vector space of dimension 6 through  $F \otimes_{F_0, \tau} \mathbb{R} = \mathbb{C}$ , identified with  $\wedge^2 V_\sigma$ . Since  $\det H$  has sign  $(-1)^n = 1$  at  $\tau$ , the map  $\phi$  induces an antilinear  $\phi_\tau$  on  $R_\tau$  with  $\phi_\tau^2 = \lambda^2$ ,  $\lambda > 0$ . Its eigenspace  $W_\tau = \{x : \phi_\tau x = \lambda x\}$  is a real form of  $R_\tau$  and a real sub-Hodge structure with  $W_\tau \otimes \mathbb{C} \cong \wedge^2 V_\sigma$ , of Hodge numbers  $(1, 4, 1)$ , on which  $q = \lambda h$  is real of signature  $(2, 4)$ , positive on the plane  $\Pi_\tau$  of real classes of type  $(2, 0) + (0, 2)$  and negative on those of type  $(1, 1)$ . The group  $\text{SU}(V_\tau, H_\tau)$  acts on  $W_\tau$  preserving  $q$ , through the isogeny  $\text{SU}(2, 2) \rightarrow \text{SO}(2, 4)^0$ , and the map from  $\mathcal{D}_{2,2}$  to the space of oriented positive planes in  $W_\tau$  that sends a point to  $\Pi_\tau$  is the equivariant isomorphism of the symmetric domains of this group onto one component,  $\mathcal{D}^{\text{IV}}(W_\tau)$ , of dimension 4. For a quadratic space  $P$  of signature  $(2, k)$  write likewise  $\mathcal{D}^{\text{IV}}(P)$ , of dimension  $k$ . Let  $M$  have  $F$ -dimension

$d$ . Then  $M_\tau = M \otimes_{F_0, \tau} \mathbb{R}$  is stable under  $\phi$  and  $F$ , so  $M_\tau = P_\tau \otimes \mathbb{C}$  with  $P_\tau = M_\tau \cap W_\tau$  of real dimension  $d$ , and  $M$  consists of Hodge classes exactly when every  $P_\tau$  is of type  $(1, 1)$ , that is, orthogonal to  $\Pi_\tau$ . So  $\mathcal{D}(M) = \prod_\tau \mathcal{D}^{\text{IV}}(P_\tau^\perp)$ , which is empty unless every  $P_\tau$  is negative definite and then has dimension  $2(4 - d)$ . If  $d = 1$ , then  $M = Fx$  with  $\phi x = bx$  for some  $b \in F$ , and  $\phi^2 x = \bar{b}bx$  gives  $h_4(\Omega, \Omega)^{-1} = N(b)$ , so  $\delta$  is trivial. If  $\delta$  is trivial,  $\mathcal{Q}_\delta \cong M_2(F_0)$ , every nonzero module contains one with  $d = 1$ , and  $\mathcal{D}(M) \subseteq \mathcal{D}(M')$  for  $M' \subseteq M$ .

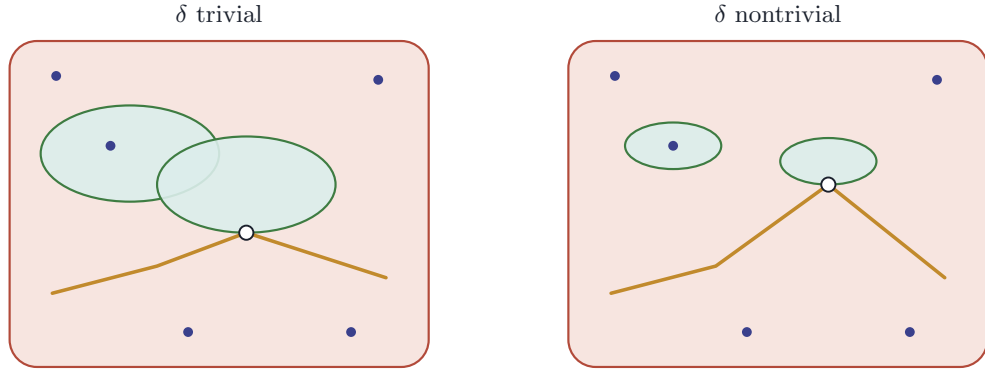
(iii) The classes  $\delta_i(f)$  of [Theorem 19.4\(ii\)](#) have complexification in  $\bigoplus_\sigma \Lambda^2 V_\sigma$ , so they lie in  $R_F$ , and they are Hodge classes at  $s_0$ . For a product,  $V = V_1 \oplus V_2$  with  $H$ -orthogonal  $F$ -planes of signature  $(1, 1)$  at each place, and  $\Lambda_F^2 V_1 \subseteq R_F$  is  $W_F(B_1)$ , of type  $(1, 1)$  by [Proposition 19.3\(i\)](#). Every  $\delta$  is the product of the discriminants of two such planes, for instance of  $\text{diag}(1, -\delta)$  and  $\text{diag}(1, -1)$ , and each factor varies in a family of dimension  $m = 2$ . On the split locus  $H^1(A, \mathbb{Q}) = U \otimes_{F_0} F$ , with  $U = H^1(X, \mathbb{Q})$  and the complex structure of  $X$  on the first factor. The polarisation class  $e$  of  $X$  lies in  $\Lambda_{F_0}^2 U$ , the hermitian form of  $A$  is  $c$  times the sesquilinear extension of the  $F_0$ -bilinear form  $e$ , with  $c \in F$  totally imaginary, and so  $\det H = c^4 \text{Pf}(e)^2$  is a square in  $F_0$  and  $\delta$  is trivial. The class  $e \otimes 1 \in (\Lambda_{F_0}^2 U) \otimes_{F_0} F = R_F$  is fixed by  $\text{Hg}(A) = \text{Hg}(X) \subseteq \text{Res}_{F_0/\mathbb{Q}} \text{Sp}(U, e)$ , so it is a nonzero Hodge class. The polarised abelian fourfolds with real multiplication by  $F_0$  form families of dimension  $[F_0 : \mathbb{Q}] \cdot 3 = 6$ . Item (LV) checks the split locus for three quartic fields. ■

**Corollary 19.11** (A base locus of codimension two). *In every  $(F, 2, \delta)$ -family,  $\Sigma_F \supseteq \text{NL}(R_F)$ , and at every point of  $\text{NL}(R_F)$  the Weil classes are the polynomials in divisor classes of [Proposition 19.10\(i\)](#). If  $\delta$  is trivial,  $\text{NL}(R_F)$  is a countable union of closed complex submanifolds of  $\mathcal{D}_F$  of codimension exactly two, and it contains the split locus; if  $\delta$  is not trivial, they have codimension at least four, and the product loci have codimension four. In both cases  $\text{NL}(R_F)$  contains the base point  $s_0$  of [Theorem 19.4](#) and is stable under  $G_F(\mathbb{Q})$ . At a point outside  $\text{NL}(R_F)$  no nonzero Weil class is a polynomial in divisor classes.*

*Proof.* Everything except the stability is [Proposition 19.10](#). An element  $g \in G_F(\mathbb{Q})$  commutes with  $F$  and with  $\phi$ , so it carries  $\mathcal{Q}_\delta$ -submodules to  $\mathcal{Q}_\delta$ -submodules, and  $g\mathcal{D}(M) = \mathcal{D}(gM)$ . ■

[Theorem 19.4](#) supplied one CM point, whose  $G_F(\mathbb{Q})$ -orbit is dense but countable. When  $\delta$  is trivial, [Corollary 19.11](#) replaces it by a locus of codimension two on which the Weil classes are balanced products of divisor classes, as they are at  $s_0$ ; nothing is claimed about the rest of the Hodge ring at its points. It does not decide  $\mathbf{W}(F, 2, \delta)$ :  $\text{NL}(R_F)$  is a countable union of proper closed submanifolds, hence meagre, and [Theorem 19.6\(iii\)](#) allows  $\Sigma_F$  to be meagre. The last sentence of the corollary is the analogue of [Proposition 15.18](#): no argument through divisor classes reaches a member outside  $\text{NL}(R_F)$ . The codimension is two rather than  $h^{2,0}(R_F) = 4$  (compare [Remark 15.21](#)) because Hodge classes in  $R_F$  come in  $\mathcal{Q}_\delta$ -modules, and a module of  $F$ -dimension one costs one condition at each real place.

**Proposition 19.12** (Scalar extension from an imaginary quadratic subfield). *Let  $F = KF_0$  with  $K$  imaginary quadratic; these are the biquadratic quartic CM fields, and the only ones containing an imaginary quadratic field. Let  $C$  be an abelian fourfold of  $(K, 2, \delta)$ -Weil type and  $B_F = C \otimes_{\mathcal{O}_K} \mathcal{O}_F$ , which is of  $(F, 2, \iota(\delta))$ -Weil type by [Proposition 20.13\(ii\)](#); write  $(V_C, H_C)$  for the hermitian space of  $C$ . For  $f \in F$  let  $g_f \in \text{Hom}(B_F, C) \otimes \mathbb{Q}$  be the element inducing  $v \mapsto v \otimes f$  from  $H^1(C, \mathbb{Q})$  to  $H^1(B_F, \mathbb{Q}) = H^1(C, \mathbb{Q}) \otimes_K F$ .*



clay: the family  $\mathcal{D}_F$ , of dimension 8; the Hodge-generic member is open

grass: the Noether–Lefschetz locus of  $R_F = \bigwedge_F^2 H^1$ , of dimension exactly 6 ( $\delta$  trivial) or at most 4

on it  $W_F$  is a sum of products of divisor classes

ochre: scalar extensions of Weil fourfolds, dimension 4 ( $F$  biquadratic), where Markman’s theorem applies

it meets the grass (ring) where the fourfold is a product of Weil surfaces

indigo: CM points; the base point  $s_0$  of the paper lies on the grass

the quaternion algebra  $(F/F_0, \det H^{-1})$  of  $R_F$  splits exactly when  $\delta$  is trivial

FIGURE 26. Where the Weil classes of an  $(F, 2, \delta)$ -family are known to be algebraic,  $F$  a quartic CM field; each panel is the period domain  $\mathcal{D}_F$ , of dimension 8. The grass regions are components of the Noether–Lefschetz locus  $\text{NL}(R_F)$ , on which  $W_F$  consists of polynomials in divisor classes (Proposition 19.10, Corollary 19.11): of dimension exactly 6 when  $\delta$  is trivial, where the quaternion algebra  $\mathcal{Q}_\delta$  of Proposition 19.9 is split, and at most 4 otherwise. The ochre curve is the locus of scalar extensions of Weil fourfolds, of dimension 4 when  $F$  is biquadratic, on which Markman’s theorem applies (Proposition 19.12); it is not contained in  $\text{NL}(R_F)$  but meets it (ring) where the fourfold is a product of Weil surfaces. The indigo points are CM points; the base point  $s_0$  of Theorem 19.4 lies on  $\text{NL}(R_F)$ . The rest of the domain, the Hodge-generic member included, is open.

- (i)  $W_F(B_F)$  lies in the  $\mathbb{Q}$ -span of the classes  $g_f^* w$ ,  $f \in F$ ,  $w \in W_K(C)$ , which has dimension 10. Hence  $W_F(B_F)$  is algebraic, by Markman’s theorem  $\mathbf{W}(K, 2, \delta)$  [Mar25a] (Proposition 20.12(ii)).
- (ii) The members  $B_F$  form a locus of dimension 4 in the  $(F, 2, \iota(\delta))$ -family. At those with  $\text{Hg}(C) = \text{SU}(V_C, H_C)$  the space  $R_F$  contains no nonzero Hodge class, so this locus is not contained in  $\text{NL}(R_F)$ ; it meets  $\text{NL}(R_F)$ , for instance where  $C = C_1 \times C_2$  with  $C_1, C_2$  abelian surfaces of Weil type, and then  $B_F$  is a product of two abelian fourfolds of  $(F, 1)$ -Weil type.

*Proof.* (i) The map  $v \mapsto v \otimes f$  commutes with the complex structures, that of  $B_F$  being the one of  $C$  tensored with  $F$ , so  $g_f$  exists. Over  $\mathbb{C}$ , for each embedding  $\kappa$  of  $K$  let  $V_\kappa$  be the  $\kappa$ -eigenspace of  $K$  on  $H^1(C, \mathbb{C})$  and  $U_\kappa = F \otimes_{K, \kappa} \mathbb{C} \cong \mathbb{C}^2$ , the sum of the lines  $L_\sigma$ ,  $\sigma|_\kappa$ , on which  $F$  acts through  $\sigma$ . The  $\kappa$ -part of  $H^1(B_F, \mathbb{C})$  is  $V_\kappa \otimes U_\kappa$  and its  $\sigma$ -part is  $V_\kappa \otimes L_\sigma$ , so the  $\sigma$ -component  $\bigwedge^4 V_\kappa \otimes L_\sigma^{\otimes 4}$  of  $W_F(B_F) \otimes \mathbb{C}$  lies in  $\bigwedge^4 V_\kappa \otimes \text{Sym}^4 U_\kappa \subseteq \bigwedge^4 (V_\kappa \otimes U_\kappa)$ .

The map  $g_f^*$  sends a generator  $\alpha_\kappa$  of  $\bigwedge^4 V_\kappa$  to  $\alpha_\kappa \otimes f_\kappa^{\otimes 4}$ , with  $f_\kappa$  the image of  $f$  in  $U_\kappa$ . The fourth powers of the elements of the Zariski dense subset  $F$  of  $U_\kappa$  span  $\text{Sym}^4 U_\kappa$ , so the complexified span of the  $g_f^* w$  is  $\bigoplus_\kappa \bigwedge^4 V_\kappa \otimes \text{Sym}^4 U_\kappa$ , of dimension  $2 \cdot 5$ , and it contains  $W_F(B_F) \otimes \mathbb{C}$ . A rational subspace whose complexification lies in that of a rational subspace lies in it, and pull-backs of algebraic classes are algebraic.

(ii) The map  $C \mapsto B_F$  embeds the four-dimensional domain of the  $K$ -family into  $\mathcal{D}_F$ . As rational Hodge structures  $H^1(B_F, \mathbb{Q}) \cong H^1(C, \mathbb{Q})^2$ , so  $\text{Hg}(B_F) = \text{Hg}(C)$ , and  $R_F \otimes \mathbb{C} = \bigoplus_\sigma \bigwedge^2 V_\sigma \otimes L_\sigma^{\otimes 2}$ , with  $\bigwedge^2 V_\sigma$  irreducible and nontrivial for  $\text{SL}(V_\sigma)$ ; so when  $\text{Hg}(C) = \text{SU}(V_C, H_C)$ ,  $R_F$  has no invariants. For  $C = C_1 \times C_2$  one has  $B_F = (C_1 \otimes_{\mathcal{O}_K} \mathcal{O}_F) \times (C_2 \otimes_{\mathcal{O}_K} \mathcal{O}_F)$ , and [Proposition 19.10\(iii\)](#) applies. Item (LV) checks (i), and that  $R_F$  meets the invariants of  $\text{SU}(V_C, H_C)$  in 0, exactly for three biquadratic fields and four hermitian forms. ■

This is not a new case of the Hodge conjecture.  $B_F$  is isogenous to  $C \times C$ , the classes are Hodge classes there, and when  $\text{Hg}(C) = \text{SU}(V_C, H_C)$  [Proposition 20.25](#) already gives the conjecture for  $C \times C$  from [\[Mar25a\]](#).

**Proposition 19.13** (The Casimir class). *Let  $A$  be a member of an  $(F, 2, \delta)$ -family and  $c \in F^\times$ , and let*

$$C_q \in R_F \otimes_{\mathbb{Q}} R_F \subseteq H^2(A, \mathbb{Q}) \otimes H^2(A, \mathbb{Q}) \subseteq H^4(A \times A, \mathbb{Q})$$

*be the Casimir element  $\sum_j u_j \otimes u_j^*$  of the nondegenerate symmetric form  $\text{Tr}_{F/\mathbb{Q}}(c^{-1}q)$  on  $R_F$ , for a basis  $(u_j)$  and its dual basis  $(u_j^*)$ . Then  $C_q$  is a Hodge class,  $\Delta^* C_q = 6c\Omega$  for the diagonal  $\Delta: A \rightarrow A \times A$ , and  $C_q$  lies in the  $\mathbb{Q}$ -span of the classes  $f^* w$ ,  $w \in W_F(A)$ ,  $f(x, y) = \iota(a)x + \iota(b)y$  with  $a, b \in \mathcal{O}_F$ . Hence  $W_F(A)$  is algebraic if and only if  $C_q$  is.*

*Proof.* For an  $F$ -basis  $(x_I)$  of  $R_F$  and its  $q$ -dual basis  $(x_I^q)$ , the elements  $\omega_k x_I$  and  $c\omega_k^* x_I^q$  form dual bases for  $\text{Tr}_{F/\mathbb{Q}}(c^{-1}q)$ . So by (66)  $\Delta^* C_q = \sum_I x_I \star (c x_I^q) = \sum_I c\Omega = 6c\Omega$ , and if  $C_q$  is algebraic so is  $W_F(A)$ , by [Proposition 19.3\(ii\)](#). Conversely,  $\text{SU}(V, H)$  preserves  $q$ , so  $C_q$  is invariant under the diagonal action of  $G_F$ , which contains  $\text{Hg}(A \times A)$ , and  $C_q$  is a Hodge class. Since  $q$  pairs  $\bigwedge^2 V_\sigma$  only with itself, the complexification of  $C_q$  lies in  $\bigoplus_\sigma \bigwedge^2 V_\sigma \otimes \bigwedge^2 V_\sigma$ , inside the part  $\bigoplus_\sigma \bigwedge^4 (V_\sigma \otimes \mathbb{C}^2)$  of  $H^4(A \times A, \mathbb{C})$  on which  $F^\times$ , acting diagonally, has the characters  $\sigma^4$ . The group  $G_F$  acts on  $V_\sigma$  through  $\text{SL}(V_\sigma)$ , whose invariants in  $\bigwedge^4 (V_\sigma \otimes \mathbb{C}^2)$  are  $\bigwedge^4 V_\sigma \otimes \text{Sym}^4 \mathbb{C}^2$  by Cauchy's formula, so the  $G_F$ -invariants in that part form the rational space  $W_F(A) \otimes_F \text{Sym}_F^4(F^2)$ , of dimension 20. The map  $f^*$  sends  $w$  to  $w \otimes (a, b)^{\otimes 4}$ , and the fourth powers of the elements of  $\mathcal{O}_F^2$  span  $\text{Sym}_F^4(F^2)$ ; so the classes  $f^* w$  span that space, and  $C_q$  is algebraic if  $W_F(A)$  is. Item (LV) checks the three identities exactly for four pairs of field and hermitian form. ■

*Remark 19.14* (What the exceptional isogeny gives). By the proof of [Proposition 19.10\(ii\)](#), at each real place  $R_F$  becomes  $W_\tau \otimes \mathbb{C}$  with  $W_\tau$  of K3 type,  $\text{SU}(V_\tau, H_\tau)$  acting through  $\text{SU}(2, 2) \rightarrow \text{SO}(2, 4)$ , and when  $\delta$  is trivial the  $W_\tau$  are the real components of the rational structure  $W$  of [Proposition 19.9\(iv\)](#). A route of Kuga–Satake type would start from this weight two structure and has to return to degree four, on  $A$  or on  $A \times A$ ; [Proposition 19.13](#) says that the natural class it meets there, the Casimir class of  $q$ , is algebraic exactly when the Weil classes are. So this mechanism restates  $\mathbf{W}(F, 2, \delta)$  at each member and does not reduce it. One can go one step further, which we only sketch. Under the identification  $R_F \otimes R_F \cong \text{End}(R_F)$  given by the intersection pairing,  $C_q$  corresponds to  $\phi$  up to multiplication by an element of  $F$ , provided that pairing restricts to  $R_F$  as  $\text{Tr}_{F/\mathbb{Q}}(ah)$  for some  $a \in F_0$ ; since the Künneth components and the inverse of the Lefschetz operator

of an abelian variety are algebraic [Kle68, §2],  $C_q$  would then be algebraic exactly when  $\phi$  is induced by an algebraic correspondence. We have not checked the form of the pairing, and nothing in this paper uses the sketch.

**Proposition 19.15** (No correspondence from a small abelian variety). *Let  $A$  be a member with  $\mathrm{Hg}(A) = G_F$ , and let  $X$  be an abelian variety such that  $\mathrm{Hg}(X)$  has no normal subgroup isogenous to  $G_F$ ; by the proof, this holds when every simple factor of  $X$  has dimension at most 7, in particular when  $\dim X \leq 7$ , and it holds when  $X$  is of CM type. Then  $\mathrm{Hg}(A \times X) = \mathrm{Hg}(A) \times \mathrm{Hg}(X)$ , and  $\mathrm{Hdg}^*(A^a \times X^b) = \mathrm{Hdg}^*(A^a) \otimes \mathrm{Hdg}^*(X^b)$  for all  $a, b \geq 0$ . So the class of an algebraic cycle  $Z$  on  $A^a \times X^b$  is  $\sum_i \beta_i \otimes \xi_i$  with  $\beta_i \in \mathrm{Hdg}^*(A^a)$  and linearly independent  $\xi_i \in \mathrm{Hdg}^*(X^b)$ ; the correspondence  $Z$  carries  $H^*(X^b, \mathbb{Q})$  into the span of the  $\beta_i$ ; and if the Hodge conjecture holds for  $X^b$ , every  $\beta_i$  is algebraic.*

*Proof.* The group  $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$  is almost simple over  $\mathbb{Q}$ . By Goursat's lemma the kernel  $N$  of the projection  $\mathrm{Hg}(A \times X) \rightarrow \mathrm{Hg}(X)$  maps onto a normal subgroup of  $\mathrm{Hg}(A) = G_F$ . If that subgroup is  $G_F$ , then  $\mathrm{Hg}(A \times X)$  contains  $G_F \times 1$  and is the product. Otherwise  $N$  is finite,  $\mathrm{Hg}(X)$  has a normal subgroup  $H$  isogenous to  $G_F$ , and the cocharacter  $\mu$  defining the Hodge structure of  $X$  projects through  $H \rightarrow G_F$  to that of  $A$ . We show that this is impossible under the hypothesis on the simple factors. Over  $\mathbb{C}$ ,  $H$  is  $\mathrm{SL}_4 \times \mathrm{SL}_4$ , and  $\mu$  projects in each factor to  $\mathrm{diag}(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$  modulo the centre. Since  $\mu$  acts on  $H^1(X, \mathbb{C})$  with two weights differing by one, the weights of its projection spread by at most one on every irreducible constituent for  $H$ ; for the representation of  $\mathrm{SL}_4$  of highest weight  $a\varpi_1 + b\varpi_2 + c\varpi_3$  the spread is  $a + 2b + c$ , and spreads add over the two factors. So every nontrivial constituent is the standard representation of one factor or its dual. These four are permuted transitively by  $\mathrm{Gal}(\mathbb{Q}/\mathbb{Q})$ , as the embeddings of  $F$  are; so a simple factor  $Y$  of  $X$  on whose  $H^1$  the group  $H$  acts nontrivially has  $\dim_{\mathbb{Q}} H^1(Y, \mathbb{Q}) \geq 16$ , that is  $\dim Y \geq 8$ . A torus has no such  $H$ . Next,  $\mathrm{Hg}(A^a \times X^b)$  is  $\mathrm{Hg}(A) \times \mathrm{Hg}(X)$  acting factorwise on  $H^*(A^a) \otimes H^*(X^b)$ , and the invariants of a product of groups in a tensor product of representations of the factors are the tensor product of the invariants. Finally, the intersection pairing is perfect on  $\mathrm{Hdg}^*(X^b)$ , the invariants being orthogonal to the other isotypic parts, so there are  $\xi_j^* \in \mathrm{Hdg}^*(X^b)$  with  $\int \xi_i \xi_j^* = \delta_{ij}$ ; they are algebraic when the conjecture holds for  $X^b$ , and then  $\beta_j = \mathrm{pr}_{A^a*}(Z \cdot \mathrm{pr}_{X^b}^* \xi_j^*)$  is algebraic. ■

So a correspondence with such an  $X$  can produce a Weil class of  $A$  only from an algebraic cycle on some  $A^a \times X^b$  that already yields one on  $A^a$ . At a member with full Hodge group this excludes every abelian variety of dimension at most seven, among them the fourfolds and sixfolds of Weil type of the imaginary quadratic fields on which Markman's theorems are proved, and every CM abelian variety. The same argument, with real points in place of dimensions, excludes rational Hodge structures  $Y$  of K3 type: by Zarhin's theorem [Zar83] the Hodge group of such a  $Y$  is the restriction of scalars of an orthogonal or unitary group over the endomorphism field of its transcendental part, noncompact at exactly one real place, so its real Lie algebra has at most one simple factor isomorphic to  $\mathfrak{su}(2, 2)$ , while that of  $G_F$  has two; hence  $\mathrm{Hg}(A \times Y) = \mathrm{Hg}(A) \times \mathrm{Hg}(Y)$ . This part rests on Zarhin's theorem, which we quote.

**Theorem 19.16** (The corrected criterion for a quartic field at  $n = 2$ ). *Let  $A$  be a member of an  $(F, 2, \delta)$ -family, let  $\omega = \sum_{\sigma} u_{\sigma} \alpha_{\sigma}$ , with  $\alpha_{\sigma}$  spanning  $\wedge^4 V_{\sigma}$  and every  $u_{\sigma} \neq 0$ , as holds for every nonzero rational Weil class, let  $p \in \mathbb{C}[\theta_1, \theta_2]$ , put  $\gamma = \omega + p$ , and let  $r(\gamma)$  be the rank of the contraction  $HT^2(A) \rightarrow H^*(A, \mathbb{C})$ ,  $\xi \mapsto \xi \lrcorner \gamma$ . Write  $T_{\sigma} = (V_{\sigma}^{1,0})^{\vee} \subseteq T_0 A$ .*

- (i)  $\dim HT^2(A) = 28 + 64 + 28 = 120$ , and

$$\text{Ann}_{HT^2}(\omega) = \bigoplus_{\sigma} \text{Hom}(V_{\sigma}^{1,0}, V_{\sigma}^{0,1}) \oplus \bigoplus_{\{\sigma, \sigma'\}, \sigma \neq \sigma'} T_{\sigma} \otimes T_{\sigma'},$$

of dimension  $0 + 16 + 24$ : the  $F$ -linear first order deformations of  $A$  and the bivectors mixing two embeddings. So  $r(\omega) = 80$ . Unlike the case  $m = 1$  of [Theorem 16.30\(i\)](#), no nonzero element of  $H^{0,2}(A)$  kills  $\omega$ .

- (ii)  $r(\gamma) \leq 112$  for every  $p$ . For a general  $p$ ,  $r(\gamma) = 112$ , and whenever  $r(\gamma) = 112$  the annihilator of  $\gamma$  is the tangent space  $T_{[A]}\mathcal{D}_F$  of the family, of dimension 8, inside  $H^1(T_A)$ . The value 112 is also taken at a general element, with nonzero component in  $W_F \otimes \mathbb{C}$ , of the complexified Hodge ring of a member with  $\text{Hg}(A) = G_F$ .
- (iii)  $r(\gamma) \geq 68$  for every  $p$ .
- (iv) At the classes  $\omega$  and coefficients computed in item (LV): for  $p$  a multiple of one monomial  $\theta_1^i \theta_2^j$ ,  $0 \leq i, j \leq 4$ , the rank  $r(\gamma)$  takes the six values 80, 84, 88, 104, 108, 112; for  $p$  a sum of two such terms the values 92 and 96 occur as well, for instance  $r(\omega + \theta_2 + t\theta_1^2) = 92$  and  $r(\omega + \theta_1^2 + t\theta_2^2) = 96$  for each of the seven nonzero values of  $t$  tested. No value below 80 has been found, and whether the bound of (iii) is attained is open.
- (v) Let  $E$  be a perfect complex on  $A$  with  $\text{ch}(E) = N\omega_1 + P$ , where  $N \neq 0$ ,  $\omega_1$  is a nonzero rational Weil class and  $P$  is a polynomial with rational coefficients in the classes  $\eta_t$ . Then  $\dim \text{Ext}^2(E, E) \geq r(\text{ch } E) \geq 68$ . If  $\dim \text{Ext}^2(E, E) = r(\text{ch } E)$ , then  $\text{ev}_E$  is onto with kernel  $\text{Ann}_{HT^2}(\text{ch } E)$  and  $\sigma_E$  is injective, so that  $\mathbf{W}(F, 2, \delta)$  follows by [Theorem 19.6\(v\)](#).

*Proof.* (i)  $HT^2(A) = \bigwedge^2 H^{0,1} \oplus H^{0,1} \otimes T_0 A \oplus \bigwedge^2 T_0 A$ , with  $H^{0,1}$  and  $T_0 A$  of dimension 8. As in the proof of [Theorem 16.30\(i\)](#) the three summands map into different Hodge components, so the annihilator is the sum of the three annihilators; and the torus  $(\mathbb{C}^\times)^4$  acting on each  $V_{\sigma}$  by scalars separates the images further. A block  $V_{\rho}^{0,1} \wedge V_{\rho'}^{0,1}$  of  $\bigwedge^2 H^{0,1}$  kills  $\alpha_{\sigma}$  exactly when  $\sigma \in \{\rho, \rho'\}$ , and otherwise maps injectively into a weight space that no other pair of a block and an embedding reaches; there are four embeddings, so some  $\sigma$  lies outside  $\{\rho, \rho'\}$ , and no nonzero element of  $\bigwedge^2 H^{0,1}$  kills  $\omega$ . A block  $\text{Hom}(V_{\rho}^{1,0}, V_{\rho'}^{0,1})$  of  $H^1(T_A)$  kills  $\alpha_{\sigma}$  for  $\sigma \neq \rho$ ; it kills  $\alpha_{\rho}$  when  $\rho' = \rho$ , because  $V_{\rho}^{0,1}$  is already exhausted in  $\alpha_{\rho}$ , and maps it injectively into its own weight space when  $\rho' \neq \rho$ . A bivector in  $T_{\rho} \otimes T_{\rho'}$  with  $\rho \neq \rho'$  kills every  $\alpha_{\sigma}$ , whose two factors of type  $(1, 0)$  both lie in  $V_{\sigma}^{1,0}$ , and  $\bigwedge^2 T_{\sigma}$  maps  $\alpha_{\sigma}$  to a nonzero multiple of a generator of  $\bigwedge^2 V_{\sigma}^{0,1}$ . The dimensions are 0,  $4 \cdot 4$  and  $6 \cdot 4$ .

(ii) The classes  $\theta_j$  and the rational Weil classes stay of Hodge type along  $\mathcal{D}_F$ , so each is killed by  $T_{[A]}\mathcal{D}_F \subseteq H^1(T_A)$ , as in [Proposition 17.11\(i\)](#); hence  $r(\gamma) \leq 120 - 8$ , and when equality holds the annihilator, of dimension 8, is  $T_{[A]}\mathcal{D}_F$ . Take a basis of  $H^1(A, \mathbb{C})$  adapted to the  $V_{\sigma}$  and to the Hodge decomposition, in which the polarisation pairs the basis of  $V_{\sigma}^{1,0}$  with that of  $V_{\bar{\sigma}}^{0,1}$ . The torus acting diagonally on it contains a subtorus that multiplies each  $\theta_j$  by a scalar and each  $\alpha_{\sigma}$  by an arbitrary scalar, and contraction is equivariant; so  $r(\omega + p)$  depends on  $\omega$  only through a rescaling of the variables of  $p$ , and each statement of the theorem holds for every  $\omega$  once it holds for one. The rank is lower semicontinuous in  $p$ , and item (LV) finds  $r(\gamma) = 112$  exactly at explicit rational  $\omega$  and  $p$ , and at explicit elements of the Hodge ring with nonzero Weil component.

(iii) In such a basis each basis element of  $HT^2$  carries a monomial to a multiple of a monomial. Let  $G$  be the set of monomials that are not of the form  $\xi_0 \lrcorner \mu$  for a basis element  $\xi_0$  and a monomial  $\mu$  occurring in some  $\theta_1^i \theta_2^j$ . The coordinates of  $\xi \lrcorner \gamma$  along  $G$  are those of

$\xi \lrcorner \omega$ , whatever  $p$  is, and the map  $\xi \mapsto (\xi \lrcorner \omega)|_G$  has rank 68, computed exactly in item (LV); the rank of a projection bounds  $r(\gamma)$  from below. The full torus acts on the monomials of  $G$  by scalars and moves any  $\omega$  to any other, so the rank 68 does not depend on  $\omega$ .

(iv) is computed exactly in item (LV), on the 25 monomials and on all 300 pairs of them with seven ratios of coefficients. By the torus of (ii) the value on a monomial of total degree other than two does not depend on its coefficient or on  $\omega$ ; for  $\theta_1^2$ ,  $\theta_1\theta_2$  and  $\theta_2^2$  the ratio of the coefficient to  $\omega$  is an invariant, as at the exceptional ratio of [Remark 17.12](#), and only the ratios computed are asserted.

(v) By the commutative square  $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$  of [Setup 18.27](#), which holds on every abelian variety,  $\dim \text{Ext}^2(E, E) \geq \text{rk } \text{ev}_E \geq r(\text{ch } E)$ , and equality forces  $\text{ev}_E$  to be onto with kernel the annihilator and  $\sigma_E$  to be injective, as in [Theorem 17.9\(iv\)](#); and  $r(\text{ch } E) \geq 68$  by (iii), since the complexification of  $\text{ch}(E)$  is a  $\gamma$  as above with  $\omega = N\omega_1$ . The last implication is [Theorem 19.6\(v\)](#), whose character is exactly of this form. ■

At  $m = 1$  and  $n = 2$  the corresponding numbers are 28, 12 and 24 ([Theorem 17.9](#)); in both cases the general value is the dimension of  $HT^2$  minus that of the family. For the pure shape the argument of [Theorem 16.30\(ii\)](#) and (iv) transfers with the annihilator of (i): an injective  $\sigma_E$  would force  $A_\theta A_{\theta'} = 0$  for  $\theta \in T_\rho$ ,  $\theta' \in T_{\rho'}$ ,  $\rho \neq \rho'$ , and a smooth component of the support of codimension at least two along which  $E$  is locally free of positive rank would be tangent to three of the four  $T_\rho$ . We have not extended [Theorem 17.5](#) to fields of degree four, so we do not know whether the pure shape is excluded here as it is for  $m = 1$ ; nothing below uses it.

*Remark 19.17* (What is missing at  $n = 2$ ). The Weil classes of an  $(F, 2, \delta)$ -family are known to be algebraic at the CM points, on  $\text{NL}(R_F)$ , and, for biquadratic  $F$ , on the scalar-extension loci, in a period domain of dimension 8. Each mechanism examined here either stops on a locus of codimension at least two, or is equivalent to the statement to be proved, or is excluded at a member with full Hodge group. By [Proposition 20.13\(i\)](#),  $\mathbf{W}(F, 3, \delta')$  for a single discriminant  $\delta'$ , the trivial one for instance, would give  $\mathbf{W}(F, 2, \delta)$  for every  $\delta$ ; by [Proposition 20.13\(ii\)](#) the families of  $F$  at  $n = 2$  give nothing new for the imaginary quadratic subfields of  $F$ , if it has any, whose fourfolds are settled by Markman. What is missing is propagation, in the form of [Theorem 19.16\(v\)](#): one perfect complex on one member, with Chern character a nonzero multiple of a rational Weil class plus a polynomial in the  $\eta_t$ , and with  $\dim \text{Ext}^2(E, E) = r(\text{ch } E)$ , which is 112 for a general shape; every such complex has  $\dim \text{Ext}^2(E, E) \geq 68$ . When  $\delta$  is trivial the secant sheaves of [\[Mar25c\]](#), on  $X \times \hat{X}$  with  $X$  an abelian fourfold with real multiplication by  $F_0$ , are natural candidates; we do not know whether their Chern characters have this form, or whether they are semiregular. Even  $\mathbf{W}(F, 2, \delta)$  for every quartic  $F$  and every  $\delta$  would be only the first case of the conclusion of  $\text{P2}_{\text{split}}$ , which concerns every field of degree at least four in infinitely many dimensions, and it would leave (F2) and (F3') of [Corollary 20.41](#) untouched.

## 20. SCOPE

An obstruction theorem earns its keep only if its boundary is drawn exactly. This section states what each result does not cover.

### 20.1. What [Theorem 1.1](#) does not cover.

*Remark 20.1* (Markman's construction). The objects constructed by Markman [\[Mar25a, Mar25b\]](#) are not obtained by restricting from an ambient projective space, and they are

not  $K$ -equivariant. [Section 13](#) analyses exactly which hypotheses they escape and why, and [Proposition 13.5](#) lists the three. Nothing in this paper bears on them, and nothing must: their conclusion is a theorem.

*Remark 20.2* (Twisted objects and gerbes). The hypothesis of [Theorem 1.1](#) is that  $\mathcal{F}$  is an object of  $D^b(\mathbb{P}^{N-1})$ , so a twisted sheaf on a gerbe over  $\mathbb{P}^{N-1}$ , or an object of a derived category of twisted sheaves, is not covered. What is covered, and what the proof actually uses, is that the Chern character of the restricted object lies in the subring of  $H^*(A^\vee, \mathbb{Q})$  generated by divisor classes. Any construction whose input has that property is subject to the conclusion, whatever category it is phrased in.

*Remark 20.3* (Other ambient varieties). Replacing  $\mathbb{P}^{N-1}$  by another ambient variety  $P$  changes the conclusion exactly when  $H^*(P, \mathbb{Q})$  is not generated by divisor classes, since the proof only uses  $\mathrm{ch}(\mathcal{F}) \in D^*(P)$  and the fact that  $i^*$  carries  $D^*(P)$  into  $D^*(A^\vee)$ . For  $P$  a Grassmannian, a flag variety, or any variety whose cohomology is generated in degree two, nothing changes. For  $P$  with exceptional Hodge classes of its own the argument does not apply, but then the problem has been moved rather than solved.

## 20.2. What [Theorem 1.3](#) does not cover.

*Remark 20.4* (Constructions that are not equivariant). [Theorem 1.3](#) constrains eigenclasses. A construction whose output is not a  $K^\times$ -eigenclass is not constrained, and by [Remark 5.7](#) such a construction must break the naturality of the  $K$ -action in an essential way: it must depend on a choice that is not  $K$ -invariant, and its output must have components in both extreme summands of the grading at once. We know of no systematic way to arrange this, and we regard the question of whether one exists as the interesting one left by this paper.

*Remark 20.5* (Abelian varieties that are not very general). At special points of the moduli of Weil type the endomorphism algebra grows, the Néron–Severi group grows with it, and the Weil line can meet the divisor subring. This is the split case, where Weil classes are known to be algebraic by elementary means, and it is consistent with everything here: our statements about  $W(A) \cap D^{2n}(A)$  assume  $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$ , which fails there. [Theorem 1.3](#) itself makes no such assumption and holds at every point, but at special points a class can be a sum of eigenclasses for several characters and the conclusion applies to each summand separately.

## 20.3. What [Corollary 9.18](#) does not cover.

*Remark 20.6* (Objects of positive rank). [Theorem 9.7](#) and [Corollary 9.18](#) both require the Chern character to lie on the Weil line, so that  $z \mapsto z \cdot \mathrm{ch}(\mathcal{E})$  has no degree-two term. An object of nonzero rank has  $\mathrm{rk}(\mathcal{E})z$  in that degree, the map is injective, and nothing here applies. That is the class Markman’s sheaves belong to, and [Remark 9.9](#) records why the distinction is not incidental to his argument.

*Remark 20.7* (The weaker semiregularity hypothesis). What [Corollary 9.18](#) obstructs is the injectivity of  $\sigma_{\mathcal{E}}$  on the whole of  $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ . The variant proposed in [[Mar25b](#), Question 11.4], injectivity on the image of an evaluation map, asks less. What we prove about it is confined to [Theorem 9.19](#), [Proposition 18.18](#), [Theorem 9.20](#) and [Corollary 18.22](#), where it fails, to [Theorem 18.29](#), where for a secant object of positive rank it is shown equivalent to two conditions on the factor, and to [Theorem 18.34](#) and [Corollary 18.36](#), where it fails for Markman’s candidate in dimension eight and for every secant object whose support has two transversal branches of codimension two through a point. We do not decide the

variant for any other object of positive rank, and we do not decide whether an affirmative answer to the question would follow from it.

*Remark 20.8* (Deformation of the class itself). A kernel is not an obstruction class. [Corollary 9.18](#) says that a sufficient criterion for the Weil class to stay algebraic is unavailable at the objects in question; it does not say the class fails to stay algebraic, and no statement in this paper does. [Question 15.13](#) remains open in both directions.

#### 20.4. What [Theorem 1.4](#) does not cover.

*Remark 20.9* (Families that are not linear systems on  $X$ ). [Theorem 1.4](#) is about smooth members of an ample linear system on  $X$ , and about the maps  $j^*$  and  $j_*$  they carry. It does not obstruct an argument that deforms  $X$  itself, that passes to a different variety by a correspondence, or that uses a family in which  $X$  appears as a fibre and the other fibres have the same dimension. In the last case the inductive parameter does not decrease, which is a different difficulty and the one recorded in [Remark 10.7](#).

*Remark 20.10* (Coniveau). A Hodge class of positive coniveau is supported on a proper closed subvariety, and the reduction to that subvariety is legitimate and lowers the dimension. Nothing here obstructs it. The obstruction concerns classes of coniveau zero, which is where the primitive middle cohomology of a general variety sits, and the generalised Hodge conjecture predicts that these are precisely the ones a coniveau argument cannot reach.

*Remark 20.11* (The Kuga–Satake route). For a weight-two Hodge structure of K3 type the Kuga–Satake construction produces an abelian variety  $\text{KS}(T)$  and an injection of  $T$ , twisted once, into the endomorphisms of its first cohomology, carrying the Hodge class in question to a class of type  $(0,0)$ , that is to an endomorphism, whose graph is a cycle. The conclusion is transported back by a correspondence, not by a restriction, so [Theorem 1.4](#) does not apply; this is [Proposition 11.9](#). The route is limited by the requirement  $h^{2,0} = 1$ , and [Proposition 11.11](#) locates that limit exactly for abelian varieties of Weil type: no rational sub-Hodge structure of  $H^2(A, \mathbb{Q})$  is of K3 type unless  $n = 1$ . [Section 11](#) treats the construction in full, including what remains open about the algebraicity of the correspondence itself.

**20.5. What stands between this and the conjecture.** The results above close several routes and narrow one. We set down, once and precisely, what remains, because the remainder is not a single obstruction whose removal would finish the problem. It is a list, and we record it with the exact statement of each entry and the exact place where this paper leaves it; [Proposition 20.13](#) records the one way in which the entries on Weil classes are linked.

Fix an imaginary quadratic field  $K = \mathbb{Q}(\sqrt{-d})$ , an integer  $n \geq 2$ , and a class  $\delta \in \mathbb{Q}_{>0}^\times / \mathbb{N}_{K/\mathbb{Q}}(K^\times)$ , the discriminant of the hermitian form. Write

$\mathbf{W}(K, n, \delta)$ : every abelian  $2n$ -fold of  $(K, -1, n)$ -Weil type with discriminant  $\delta$  has algebraic Weil classes.

**Proposition 20.12** (The Weil statements, and how they are linked). *For each triple  $(K, n, \delta)$  the following hold.*

- (i)  $\mathbf{W}(K, n, \delta)$  is equivalent to the algebraicity of the single class  $\omega_s$  at every point  $s$  of the period domain  $\mathcal{D}_{n,n}$ , and that domain is connected; this is [Corollary 15.35](#).
- (ii)  $\mathbf{W}(K, 3, \delta_0)$  holds for the trivial discriminant  $\delta_0$  [[Mar25a](#), [Mar25b](#)], and with it, by [Proposition 20.13](#),  $\mathbf{W}(K, 2, \delta)$  for every  $K$  and every  $\delta$ , which is how Markman obtains the fourfolds. Nothing else is known.

- (iii) The period domains attached to distinct triples are disjoint, and [Theorem 15.14](#) and [Corollary 15.35](#) are statements about one triple at a time. The triples are nevertheless not independent: by [Proposition 20.13](#),  $\mathbf{W}(K, N, \delta'')$  implies  $\mathbf{W}(K, n, \delta)$  for every  $n < N$  and every  $\delta$ , so the families of trivial discriminant in infinitely many dimensions give every imaginary quadratic family, and the families of a CM field of degree at least four give those of the imaginary quadratic fields it contains.

*Proof.* (i) is [Corollary 15.35](#) together with [Proposition B.8](#). (ii) is the cited work together with [Proposition 20.13\(i\)](#); [Remark 15.20](#) records that the construction of [Section 15.1](#) reaches only  $\delta_0$ . For (iii), the discriminant is a discrete invariant of the hermitian form and is constant on  $\mathcal{D}_{n,n}$  by [Proposition 15.19\(ii\)](#), while  $n$  and  $K$  are fixed by the endomorphism algebra, so the domains are disjoint; the links are [Proposition 20.13](#), which passes between families through a product with a Weil surface or through a scalar extension, and not through a deformation. An earlier version of this proposition asserted that settling one triple leaves every other exactly where it was. That was wrong: it considered only the constructions of this paper, and the product argument of Schoen that Markman uses to pass from sixfolds to fourfolds already links the triples. ■

**Proposition 20.13** (Descent and scalar extension). *Let  $F$  be a CM field of degree  $2m$  and write  $\mathbf{W}(F, n, \delta)$  for the statement that every abelian variety of  $(F, n, \delta)$ -Weil type has algebraic Weil classes, so that  $\mathbf{W}(K, n, \delta)$  is the statement above when  $m = 1$ .*

- (i) Descent. *Let  $B$  be of  $(F, n, \delta)$ -Weil type and  $Y$  of  $(F, 1, \delta')$ -Weil type. Then  $X = B \times Y$ , with the diagonal action of  $F$  and the product polarisation, is of  $(F, n+1, \delta\delta')$ -Weil type, and the correspondence*

$$P(x) = \text{pr}_{B*}(x \cdot \text{pr}_Y^*(\eta_Y^{2m-2} y')),$$

*with  $\eta_Y$  the polarisation of  $Y$  and  $y'$  a nonzero Weil class of  $Y$ , maps the Weil classes of  $X$  isomorphically onto those of  $B$ . Since  $y'$  lies in  $H^2(Y)$  it is algebraic, so  $P$  is algebraic, and every discriminant, a totally positive class in  $F_0^\times / N_{F/F_0}(F^\times)$ , is that of some  $Y$ . Hence  $\mathbf{W}(F, N, \delta'')$  for one  $N$  and one  $\delta''$  implies  $\mathbf{W}(F, n, \delta)$  for every  $n < N$  and every  $\delta$ .*

- (ii) Scalar extension. *Let  $K \subset F$  be CM fields and  $B$  of  $(K, n, \delta)$ -Weil type. Then  $B_F = B \otimes_{\mathcal{O}_K} \mathcal{O}_F$  is of  $(F, n, \iota(\delta))$ -Weil type, and for  $j: B \rightarrow B_F$ ,  $b \mapsto b \otimes 1$ , one has  $j^* \text{Tr}_{F/\mathbb{Q}}(a \det_F) = \text{Tr}_{K/\mathbb{Q}}(\text{Tr}_{F/K}(a) \det_K)$ , so  $j^*$  maps the Weil classes of  $B_F$  onto those of  $B$ . Hence  $\mathbf{W}(F, n, \iota(\delta))$  implies  $\mathbf{W}(K, n, \delta)$ ; every imaginary quadratic  $K$  lies in the quartic CM field  $K(\sqrt{2})$ .*

*Proof.* (i)  $H^1(X) = H^1(B) \oplus H^1(Y)$  as  $F$ -spaces and the hermitian form of  $X$  is the orthogonal sum, of signature  $(n+1, n+1)$  at every real place; its normalised discriminant is the product, and  $H_Y = \text{diag}(1, -t)$  with  $t$  totally positive realises every class. The Weil classes of  $X$  are  $\bigoplus_\sigma \bigwedge^{2n} V_{B,\sigma} \otimes \bigwedge^2 V_{Y,\sigma}$ , because the top exterior power of a direct sum is the tensor product of the top powers. On the  $\sigma$ -component,  $P$  is multiplication by  $\int_Y w_Y^\sigma \eta_Y^{2m-2} y'^{\bar{\sigma}}$ , which is nonzero since  $w_Y^\sigma \wedge y'^{\bar{\sigma}}$  fills the  $(\sigma, \bar{\sigma})$  block and  $\eta_Y$  is nondegenerate on the others; the remaining pairings vanish for weight reasons under the norm-one torus of  $F$ . So  $P$  is rational, diagonal over  $\mathbb{C}$  with nonzero entries, and algebraic, and  $\omega(B)$  is algebraic whenever  $\omega(X)$  is; [Lemma 16.3](#) passes from one Weil class to all of them. (ii)  $H^1(B_F) = H^1(B) \otimes_K F$ , each eigenspace of  $F$  is an eigenspace of  $K$  of signature  $(n, n)$ , the hermitian form is  $H_B \otimes F$  with determinant  $\det H_B$ , and a norm from  $K$  is a norm from  $F$ . The identity for  $j^*$  is the multiplicativity of the determinant under restriction of

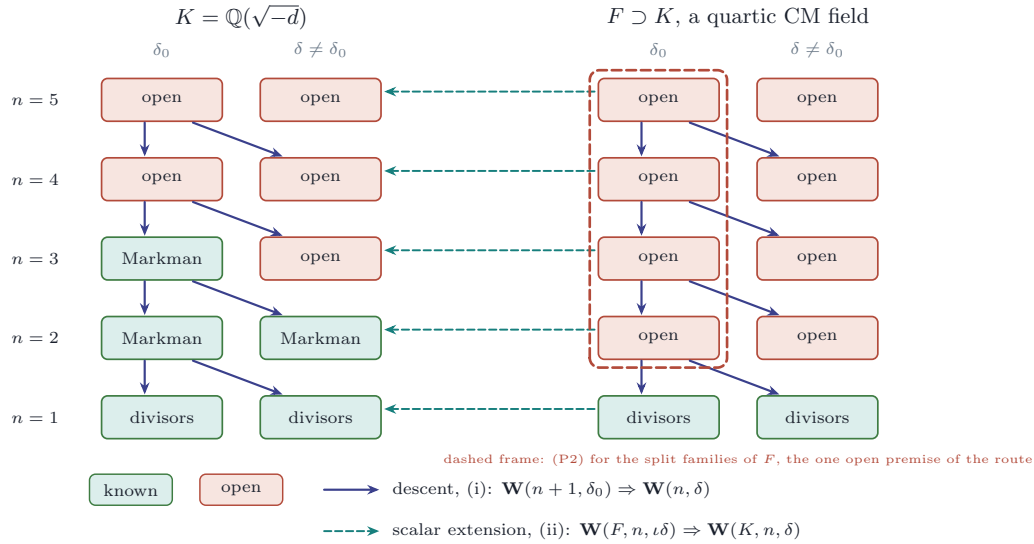


FIGURE 27. The Weil families of an imaginary quadratic field  $K$  and of a quartic CM field  $F \supset K$ , one cell for each  $n$  and each kind of discriminant, grass where the Weil classes are known to be algebraic (divisor classes at  $n = 1$ ; Markman [Mar25a, Mar25b] at  $n = 2$  and at  $n = 3$ ,  $\delta = \delta_0$ ) and clay where they are not. The indigo arrows are the descent of Proposition 20.13(i), from the trivial discriminant in dimension  $n + 1$  to every discriminant in dimension  $n$ ; the dashed teal arrows are the scalar extension of Proposition 20.13(ii). The dashed frame is the propagation statement for the split families of the field of higher degree, which by Theorem 20.40(ii) is the only form of (P2) that any minimal route to the conjecture uses: every clay cell of the left panel lies below it.

scalars, and  $\text{Tr}_{F/K}$  is onto. Item (LIV) of Appendix D checks the linear algebra of both statements exactly, over seven CM fields. ■

The case  $K$  imaginary quadratic and  $N = 3$  of (i) is the degeneration argument of Schoen through which Markman obtains every Weil fourfold, at every discriminant, from the sixfolds of trivial discriminant [Mar25a];<sup>57</sup> what (i) adds is that the same argument works in every dimension and for every CM field. Its consequence for the structure of the problem is not small, and it corrects an earlier version of this section, which asserted that the triples  $(K, n, \delta)$  are separate problems. They are not: the trivial discriminant families in infinitely many dimensions give every imaginary quadratic family, and by (ii) the split families of the CM fields of degree at least four give the imaginary quadratic fields as well. In particular the base points of nontrivial discriminant supplied by Theorem 15.31 are not needed on any route to the conjecture, since the split families already carry base points (33); what Theorem 15.31 shows is that propagation can be started in every family, not that it has to be. Figure 27 draws the web of families that Proposition 20.13 links.

<sup>57</sup>We know Schoen’s argument only through Markman’s account of it. What (i) records is the product and push-forward to which that argument amounts; in the form given here it needs no degeneration at all, only a Weil surface  $Y$  of the complementary discriminant and one of its Weil classes, which is a divisor class.

With that said, here is the list.

- (C1) *One secant object on one fourfold.* There should exist a principally polarised abelian fourfold  $(X, \Theta)$  and a perfect complex  $\mathcal{F}$  on it with  $\text{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$  satisfying (56). By Theorem 18.29 and Corollary 18.30 the transform  $\Phi(\mathcal{F} \boxtimes \mathcal{F})$  would then satisfy the weakened criterion, and by Proposition 18.25 the three conditions of Markman's strategy that are conditions on the Chern character hold at the point  $(1, 3)$  for any such  $\mathcal{F}$ , since they are conditions on  $\text{ch}(\mathcal{F})$  alone. Turning that into algebraicity needs one thing more, and it is not ours to supply. The weakened criterion enters in place of the hypothesis Markman verifies, and whether the semiregularity theorem survives the substitution is exactly [Mar25b, Question 11.4], which is open. So (C1) is a demand on a construction, and an affirmative answer to that question is a separate prerequisite standing beside it; Theorem 20.40 places both. What this paper contributes is the exact form of the demand. The local part of the criterion is a condition on depth and nothing else: if the support is Cohen–Macaulay there is no local obstruction at any point (Theorem 18.43), and if it is not, at a point where two components meet only at that point or where there is an embedded point, the criterion fails outright (Theorem 18.34, Corollary 18.36, Corollary 18.45). The candidate of [Mar25b, Section 12] is of the second kind, and it cannot be moved to the first: separating the branches is what would restore the depth, and Remark 18.46 shows that doing so moves the Chern character off the secant plane. So what is needed is a Cohen–Macaulay support, and for one of those the whole criterion is the single vanishing of the image of  $A_\theta A_{\theta'}$  in  $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ . Two shapes of such a support remain (Corollary 18.57): a smooth one, where the invariants are forced outright and  $d \leq 9$  (Theorem 18.40, Corollary 18.41), and a singular one; in neither may the Hilbert–Burch resolution split into powers of  $\Theta$  or into simple semi-homogeneous bundles (Theorem 18.52, Theorem 18.55), which removes every candidate the search of Theorem 18.50 found, and no complete intersection occurs (Theorem 18.49). Remark 18.58 states the two remaining questions in the form a construction would have to answer.
- (C2) *The same for every  $n$  and every discriminant.* A positive answer to (C1) reaches one triple, and the discriminant of that triple is the trivial one, because the object lives on  $X \times \hat{X}$  and every split member has trivial discriminant by Proposition 15.19(ii). The other triples are reached through Proposition 20.13: a positive answer in infinitely many dimensions gives  $\mathbf{W}(K, n, \delta)$  for every  $n$  and every  $\delta$ , so only uniformity in  $n$  is asked for, and (C1) at  $n = 4$ , with Question 11.4, already gives every sixfold. What no answer of this kind reaches is the CM fields of degree at least four. What this paper contributes: three of the four ingredients of Question 13.7 are supplied for every  $n$  and every  $K$  (Theorem 14.16), including the existence of the secant sheaves themselves (Theorem 14.13); the fourth is untouched (Remark 14.17).
- (C3) *CM fields of degree larger than two.* For a CM field  $F$  of degree  $2m > 2$  acting on an abelian variety, the Weil classes span a space of dimension  $2m$ , and the secant framework of Section 14, which is built on a plane meeting the spinor variety in a conjugate pair, does not apply to them as it stands. Markman has extended it to these fields [Mar25c]: for an abelian variety  $X$  with real multiplication by the totally real subfield  $F_0$  of  $F$  he makes  $F$  act on  $X \times \hat{X}$ , replaces the plane by a rational subspace of dimension  $2^{[F_0:\mathbb{Q}]}$  of the half-spin representation that is secant to the spinor variety, and gives secant sheaves for Jacobians of curves of genus

four with real multiplication by a real quadratic field. The Weil classes of  $F$  on  $X \times \hat{X}$  would be algebraic if such a sheaf were semiregular, and according to the abstract of [Mar25c] that semiregularity has not yet been addressed. We know that work only from its abstract, and nothing in this paper depends on it. The smallest open case, quartic fields at  $n = 2$ , is analysed in Section 19.4. What this paper contributes: everything except the propagation. By Section 19 each such family has a connected period domain, a flat space of Weil classes, a dense set of rational orbits, a base point at a CM member where the classes are written as polynomials in divisor classes (Theorem 19.4), and the two propagation criteria (Theorem 19.6); so the higher CM fields are the same propagation problem as the quadratic ones, and not a separate entry. For a quartic field at  $n = 2$ , Section 19.4 determines how far divisor classes, scalar extension and correspondences reach, and computes the number that the corrected criterion asks of one complex. Appendix C places the imaginary quadratic case inside Albert's classification.

- (C4) *The non-split sixfolds.*  $\mathbf{W}(K, 3, \delta)$  is open for every  $\delta \neq \delta_0$ ; the case  $\delta = \delta_0$  is Markman's [Mar25a]. By Proposition 20.13(i) it would follow from  $\mathbf{W}(K, 4, \delta_0)$ , and by Proposition 20.13(ii) from  $\mathbf{W}(K(\sqrt{2}), 3, \iota(\delta))$ . What this paper contributes: a base point in every family, at a member with quaternionic multiplication (Theorem 15.31); the locus of such members is a Lagrangian Grassmannian of codimension  $n(n-1)/2$  (Proposition 15.28), so propagation off it is needed and is not supplied.
- (C5) *Exceptional classes that are not of Weil type.* On an abelian variety the Hodge ring can contain exceptional classes that the Weil construction does not produce. Through dimension five it does not [MZ99], and in dimension eight it does: the self-product of an abelian fourfold of Mumford type carries two Hodge classes that are neither products of divisor classes nor Weil classes of any quotient (Proposition 20.44). So the Hodge conjecture for abelian varieties does not reduce to Weil classes, and the statement that remains is the algebraicity of these classes themselves. What this paper contributes: the example, made exact, and nothing towards their algebraicity. Every positive statement here is about classes on the Weil line.
- (C6) *Varieties that are not abelian.* No reduction of the Hodge conjecture for arbitrary smooth projective varieties to abelian varieties is known, and none of Hodge-theoretic kind can exist: the  $H^3$  of a very general quintic threefold lies outside the tensor category generated by the cohomology of abelian varieties (Proposition 20.45), so its Hodge structure cannot be realised inside that of an abelian variety by any correspondence. Theorem 1.4 and Theorem 1.5 close the two routes through hyperplane sections that would be the natural candidates. What this paper contributes: those three obstructions, and nothing positive. Nor is this entry the complement of the others. The varieties that are not abelian include  $A \times \mathbb{P}^1$  for every abelian variety  $A$ , and the conjecture for  $A \times \mathbb{P}^1$  gives it for  $A$ , so the conjecture for the varieties that are not abelian is the whole conjecture (Proposition 20.35).

*Remark 20.14* (The shape of the remainder). Two features of this list deserve emphasis, because they are the reason the paper does not claim more than it does.

The narrowing is real but bounded. Corollary 18.41 removes all but four discriminants from the smooth case of (C1), and Corollary 18.36 removes an entire class of supports in every dimension. Those are restrictions on the space of possible proofs, not steps towards one; they say where a construction cannot be, and the set of places it could be remains infinite in every direction except the one that Corollary 18.41 bounds.

The equivalence of [Corollary 15.35](#) is not an obstruction. It says that for a fixed triple the Hodge conjecture in every degree is the algebraicity of one class at every point of one connected domain. That is a restatement of the problem in its smallest form, and a restatement is worth having, but the implication it isolates is the conjecture itself for those varieties, and proving it would close those varieties only.

The third feature is the one a list cannot show, and it is the subject of [Section 20.9](#), once [Section 20.6](#) has supplied the implications of the literature. The entries above are not of equal standing. Some of them lie on every route to the conjecture and some lie on none, and which is which is not a matter of judgement: it is decided by the implications the paper actually proves.

**20.6. The target common to the short routes.** The theorems of this paper are not the only implications into the conjecture. The literature supplies others, and a frontier computed without them overstates what is needed. This subsection proves those implications, or quotes them where they are theorems of the literature, so that the computation of [Section 20.9](#) runs over both. Nothing in it is new as mathematics except the assembly; what is new is that, once assembled, it changes the answer.

**Definition 20.15** (Three statements about all varieties). Here a variety is smooth, projective and defined over  $\mathbb{C}$ , except the base  $S$  in (V).

- (L)  $B(X)$  holds for every variety  $X$  ([Definition 10.5](#)).
- (M) Every Hodge class on every variety  $X$  is *motivated* in the sense of André [[And96b](#), §2.1]: it equals  $\text{pr}_*(\alpha \cup * \beta)$  for some variety  $Y$ , algebraic classes  $\alpha, \beta$  on  $X \times Y$ , the projection  $\text{pr}: X \times Y \rightarrow X$ , and the Lefschetz involution  $*$  of  $H^*(X \times Y, \mathbb{Q})$  with respect to some ample class.
- (V) For every smooth connected complex variety  $S$ , every smooth projective morphism  $f: \mathcal{X} \rightarrow S$  and every global section  $\xi$  of  $R^{2p}f_*\mathbb{Q}$  over  $S^{\text{an}}$ : if  $\xi_s$  is algebraic at one point  $s \in S$ , then  $\xi_t$  is algebraic at every point  $t \in S$ .

The Lefschetz involution of a variety  $Z$  of dimension  $e$  with ample class  $L$  sends  $L^r x$ , for  $x$  primitive of degree  $j$  and  $0 \leq r \leq e - j$ , to  $\pm L^{e-j-r} x$ , with a sign that depends only on  $j$  [[And96b](#)]; the sign plays no role below.

The statement (V) is the variational Hodge conjecture for algebraic classes; it asks nothing about Hodge classes, only that algebraicity, once present, survives deformation.

**Lemma 20.16** (Kleiman). *Let  $Z$  be a variety of dimension  $e$  with ample class  $L$ , and suppose that  $B(Z)$  holds. Then the Künneth projectors  $\pi_k$  of  $H^*(Z, \mathbb{Q})$ , the projectors onto the summands  $L^r P^j$  of the Lefschetz decomposition, and the Lefschetz involution  $*$  are all induced by algebraic cycles on  $Z \times Z$ .*

*Proof.* Operators induced by algebraic cycles on  $Z \times Z$  are closed under composition, sums and rational multiples, because composition of correspondences is [[Kle68](#)]. The operator  $L$  is induced by the image of a divisor of class  $L$  under the diagonal, and  $\Lambda$  by  $B(Z)$ . With  $[L, \Lambda] = H$  and  $H$  acting on  $H^k$  by  $k - e$  (the proof of [Lemma 10.6](#)),  $H = L\Lambda - \Lambda L$  is induced by an algebraic cycle, and since the numbers  $k - e$  for  $0 \leq k \leq 2e$  are distinct,

$$\pi_k = \prod_{i \neq k} \frac{H - (i - e)}{k - i}$$

is as well. Put  $\Omega = H^2 + 2H + 4\Lambda L$ . For  $x$  primitive of degree  $j$  put  $m = e - j \geq 0$ . Then  $\Lambda x = 0$  and  $Hx = -mx$ , and the relation

$$\Lambda L^s x = s(m - s + 1) L^{s-1} x \quad (s \geq 1) \quad (67)$$

follows by induction from  $\Lambda L = L\Lambda - H$ : the case  $s = 1$  is [Lemma 10.6](#), and  $\Lambda L^s x = L\Lambda L^{s-1} x - HL^{s-1} x = ((s-1)(m-s+2) - (2s-2-m))L^{s-1} x = s(m-s+1)L^{s-1} x$ . So  $\Omega x = (m^2 - 2m + 4m)x = m(m+2)x$ , and  $\Omega$  commutes with  $L$ , since  $[L, H] = -2L$  and  $[L, \Lambda L] = HL$  give  $[L, \Omega] = -2(LH + HL) - 4L + 4HL = 0$ ; hence  $\Omega$  acts on  $L^r P^j$  by  $m(m+2)$ . Inside  $H^k$  the summands are  $L^r P^{k-2r}$ , with  $m = e - k + 2r$ , and  $m \mapsto m(m+2)$  is injective on  $m \geq 0$ , so Lagrange interpolation in  $\Omega$  composed with  $\pi_k$  gives the projector  $p_{k,r}$  onto  $L^r P^{k-2r}$  as an algebraic operator. Finally let  $x \in P^j$  and  $0 \leq r \leq m$ . If  $2r \leq m$ , then  $L^{m-2r}(L^r x) = L^{m-r} x$ . If  $2r > m$ , put  $t = 2r - m$ , so  $1 \leq t \leq r$ ; by (67),

$$\Lambda^t L^r x = \prod_{i=0}^{t-1} (r-i)(m-r+i+1) L^{m-r} x,$$

and every factor is at least 1, because  $r-i \geq r-t+1 \geq 1$  and  $m-r+i+1 \geq m-r+1 \geq 1$ . So on  $L^r P^j$  the involution is  $\pm L^{m-2r}$  or a nonzero rational multiple of  $\Lambda^{2r-m}$ , and  $* = \sum_{k,r} c_{k,r} T_{k,r} p_{k,r}$  with rational  $c_{k,r}$  and  $T_{k,r}$  a power of  $L$  or of  $\Lambda$ . Every term is algebraic.  $\blacksquare$

**Proposition 20.17** (The conjecture is (L) and (M)). *The Hodge conjecture for every variety is equivalent to (L) and (M) together.*

*Proof.* Assume the conjecture. Then (L) holds by [Theorem 12.2](#). An algebraic class  $\xi$  on  $X$  of dimension  $d$  is motivated: take  $Y$  a point,  $\alpha = \xi$ , and  $\beta = \varepsilon h^d$  with  $h$  the ample class and  $\varepsilon = \pm 1$  the sign for which  $*(h^d) = \varepsilon$ , which exists because the unit  $1 \in H^0$  is primitive of degree 0; then  $\alpha \cup * \beta = \xi$ . Every Hodge class is algebraic, so (M) holds.

Conversely assume (L) and (M), and let  $\xi$  be a Hodge class on  $X$ . By (M),  $\xi = \text{pr}_*(\alpha \cup * \beta)$  with  $\alpha, \beta$  algebraic on  $X \times Y$ . By (L),  $B(X \times Y)$  holds, so  $*$  is induced by an algebraic cycle ([Lemma 20.16](#)); hence  $*\beta$ , the product  $\alpha \cup * \beta$  and its push-forward  $\xi$  are algebraic. This is André's observation that motivated classes are algebraic as soon as the Lefschetz involutions are [[And96b](#), §2.1].  $\blacksquare$

**Proposition 20.18** (Two sources of (V)). *The Hodge conjecture implies (V), and so does (L).*

*Proof.* Let  $f: \mathcal{X} \rightarrow S$ ,  $\xi$  and  $s$  be as in (V), with  $\xi_s$  algebraic. As  $S$  is smooth and connected it is irreducible, so two affine open subsets meet, and it suffices to treat  $S$  affine, joining  $s$  to an arbitrary  $t$  through a point of the intersection of an affine neighbourhood of each. Then  $\mathcal{X}$  is quasi-projective and, by resolution of singularities, has a smooth projective compactification  $\bar{\mathcal{X}}$ .

Assume the conjecture. Write  $n = 2p$  and  $\rho: H^n(\bar{\mathcal{X}}, \mathbb{Q}) \rightarrow H^0(S^{\text{an}}, R^n f_* \mathbb{Q})$  for restriction; it is surjective by the theorem of the fixed part [[Del71](#), 4.1.1(ii)]. For  $t \in S$  the restriction  $r_t: H^n(\bar{\mathcal{X}}, \mathbb{Q}) \rightarrow H^n(\mathcal{X}_t, \mathbb{Q})$  is  $\rho$  followed by evaluation at  $t$ , and evaluation is injective because  $S$  is connected; so  $K = \ker r_t = \ker \rho$  does not depend on  $t$ . It is a sub-Hodge structure, and the Hodge structure  $H^n(\bar{\mathcal{X}}, \mathbb{Q})$  is polarisable, hence semisimple, so  $H^n(\bar{\mathcal{X}}, \mathbb{Q}) = K \oplus K'$  with  $K'$  a sub-Hodge structure. Choose  $\eta \in K'$  with  $\rho(\eta) = \xi$ . Then  $r_s|_{K'}$  is an injective morphism of Hodge structures and  $r_s(\eta) = \xi_s$  is of type  $(p, p)$ , so every component of  $\eta$  of type other than  $(p, p)$  maps to zero and is zero:  $\eta$  is a Hodge class. So  $\xi_t = r_t(\eta)$  is a Hodge class on  $\mathcal{X}_t$  for every  $t$ , and algebraic by the conjecture.

Assume (L). The algebraic class  $\xi_s$  is motivated (proof of [Proposition 20.17](#)), so  $\xi_t$  is motivated for every  $t$  by André's deformation theorem [[And96b](#), Théorème 0.5], and a motivated class is algebraic under (L) (proof of [Proposition 20.17](#)). ■

**Theorem 20.19** (Deligne, André; Abdulali, André). (V) *implies the Hodge conjecture for every abelian variety, and so does (L).*

*Proof.* Milne calls a class *accessible* if it lies in the least family of rational classes that contains the classes of algebraic cycles, is stable under pull-back by morphisms of varieties, and contains  $t_t$  for every  $t$  whenever  $(t_s)_{s \in S}$  is a global section of  $R^r f_* \mathbb{Q}$  for a smooth projective family over a smooth connected base and contains  $t_s$  for one  $s$ ; and he proves, following Deligne and André, that every Hodge class on an abelian variety is accessible [[Mil20](#), §5.1, Theorem 3]. The algebraic classes contain the classes of algebraic cycles and are stable under pull-back, and under (V) they have the third property. So they contain every accessible class, hence every Hodge class on an abelian variety; this is [[Mil20](#), Remark 2]. The statement for (L) follows by [Proposition 20.18](#), and is the theorem of Abdulali and André in the form of [[Mil20](#), Theorem 4]. ■

*Remark 20.20* (What is known of the three). Each of (L), (M) and (V) is open, and none is proved here in general.  $B(X)$  is known for curves, where it is immediate, and for abelian varieties [[Kle68](#), §2], and [Corollary 20.23](#) below adds the total space of every Mumford family. (M) holds for the Hodge classes on abelian varieties [[And96b](#), Théorème 0.6.2]. (V) holds in degree 2, where by the argument of [Proposition 20.18](#) the class  $\xi_t$  is a Hodge class and the Lefschetz theorem on  $(1, 1)$  classes applies; the one general mechanism for it in higher degree is the semiregularity theorem [[Blo72](#), [BF03](#), [Pri24](#)], which is the mechanism of [Theorem 16.28](#). [Theorem 20.19](#) says what is at stake for this paper: the single statement (V), which the conjecture itself implies, gives the Weil classes of every CM field and the classes beyond the Weil lines at once, by a route that uses none of the theorems proved here.

**20.7. A new case of the Lefschetz standard conjecture.** (L) is known for abelian varieties and open for the varieties fibred in them. This subsection proves it for the total space of an abelian scheme over a curve whose invariant cycles are algebraic, and so for the total space of every Mumford family. The proof assembles the operator  $\Lambda$  from a relative part, which is a Pontryagin product and is algebraic on every abelian scheme, and a part along the base, which is algebraic exactly when the invariant cycles are.

**Proposition 20.21** ( $\Lambda$  as a Pontryagin product). *Let  $A$  be an abelian variety of dimension  $g$ ,  $\ell \in H^2(A, \mathbb{Q})$  the class of an ample line bundle,  $D = \int_A \ell^g / g!$  and  $\gamma = \ell^{g-1} / (g-1)!$ . Let  $\mu: A \times A \rightarrow A$  be the addition and  $x \star y = \mu_*(p_1^* x \cup p_2^* y)$  the Pontryagin product. Then the operator  $\Lambda$  of [Definition 10.5](#) for the class  $\ell$  is*

$$\Lambda(x) = D^{-1} x \star \gamma.$$

*In particular  $\Lambda$  is induced by the algebraic cycle  $D^{-1}(p_1, \mu)_*(p_2^* Z)$  on  $A \times A$ , for any cycle  $Z$  of class  $\gamma$ .*

*Proof.* Put  $V = H_1(A, \mathbb{Q})$ , so that  $H^*(A, \mathbb{Q}) = \bigwedge^* V^\vee$  under cup product and  $H_*(A, \mathbb{Q}) = \bigwedge^* V$  under the Pontryagin product of homology, and let  $\text{PD}(x) = x \cap [A]$ . Cap product makes  $H_*$  a module over  $H^*$ , and for a torus it is contraction; so  $\text{PD}(\ell \cup x) = \ell \cap \text{PD}(x)$  is the contraction  $\iota_\ell$  applied to  $\text{PD}(x)$ . Poincaré duality commutes with  $\mu_*$  and carries  $p_1^* x \cup p_2^* y$  to the cross product of  $\text{PD}(x)$  and  $\text{PD}(y)$ , without sign when  $y$  has even degree, so  $\text{PD}(x \star \gamma) = \text{PD}(x) \cdot \beta$  with  $\beta = \text{PD}(\gamma) \in \bigwedge^2 V$ .

Choose a basis  $x_1, y_1, \dots, x_g, y_g$  of  $V^\vee$  with  $\ell = \sum_i d_i x_i \wedge y_i$ ,  $d_i > 0$ , oriented so that  $\int_A x_1 \wedge y_1 \wedge \dots \wedge x_g \wedge y_g = 1$ ; then  $\int_A \ell^g = g! d_1 \dots d_g$ , so  $D = d_1 \dots d_g$ . Let  $e_i, f_i$  be the dual basis of  $V$ . Then  $\gamma = \sum_j (\prod_{i \neq j} d_i) \prod_{i \neq j} x_i \wedge y_i$  and  $\beta = \varepsilon D \sum_j d_j^{-1} e_j \wedge f_j$  with  $\varepsilon = \pm 1$  independent of  $j$ , since the permutations of the pairs preserve the orientation and exchange the terms. The algebra  $\bigwedge^* V$  is the graded tensor product of the algebras  $\bigwedge^* \langle e_i, f_i \rangle$ , and  $\iota_\ell$  and multiplication by  $D^{-1}\beta$  are sums over  $i$  of even operators on the  $i$ -th factor, namely  $d_i \iota_{x_i \wedge y_i}$  and  $\varepsilon d_i^{-1} e_i \wedge f_i$ . Operators on different factors commute, and on the  $i$ -th factor, with basis  $1, e_i, f_i, e_i \wedge f_i$  and  $\iota_{x_i \wedge y_i}(e_i \wedge f_i) = \sigma$  for a sign  $\sigma$  fixed by the convention for contraction, the commutator  $[\iota_\ell, D^{-1}\beta]$  acts by  $\varepsilon\sigma$  on  $1$ , by  $0$  on  $e_i$  and  $f_i$ , and by  $-\varepsilon\sigma$  on  $e_i \wedge f_i$ , that is by  $\varepsilon\sigma(1-k_i)$  in degree  $k_i$ . Summing over  $i$ ,  $[\iota_\ell, D^{-1}\beta] = \varepsilon\sigma(g-k)$  on  $\bigwedge^k V$ . As PD carries  $H^q$  to  $\bigwedge^{2g-q} V$ , the operator  $\Lambda'(x) = D^{-1}x \star \gamma$  satisfies  $[L, \Lambda'] = \varepsilon\sigma(q-g) = \varepsilon\sigma H$  on  $H^q(A)$ .

The sign is fixed by one value. For  $x$  and  $y$  of complementary degrees,  $x \star y \in H^0(A)$  is  $\int_{A \times A} p_1^* x \cup p_2^* y \cup \mu^*[\text{pt}]$ , and  $\mu^*[\text{pt}]$  is the class of the antidiagonal  $\{(a, -a)\}$ , so  $x \star y = \int_A x \cup [-1]^* y$ ; for  $y = \gamma$ , of even degree, this is  $\int_A x \cup \gamma$ . Hence  $\Lambda'(\ell) = D^{-1} \int_A \ell \cup \gamma = g$ , while  $\Lambda(\ell) = \Lambda(L \cdot 1) = g$  by Lemma 10.6. If  $\varepsilon\sigma = -1$ , then  $-\Lambda'$  would be the operator of Definition 10.5 and  $\Lambda'(\ell) = -g$ . So  $\varepsilon\sigma = 1$ , and  $\Lambda' = \Lambda$  by the uniqueness in Definition 10.5. Finally, the correspondence  $(p_1, \mu)_*(p_2^* Z)$  acts by  $x \mapsto p_{2*}(p_1^* x \cup (p_1, \mu)_* p_2^* [Z]) = \mu_*(p_1^* x \cup p_2^* \gamma)$ , by the projection formula.  $\blacksquare$

**Theorem 20.22** (The Lefschetz operator of an abelian scheme over a curve; cf. Tankeev [Tan03]). *Let  $C$  be a smooth projective connected curve and  $\pi: W \rightarrow C$  an abelian scheme of relative dimension  $g$  with  $W$  projective. Suppose that every class in  $H^*(W_c, \mathbb{Q})$  invariant under the monodromy of  $R^* \pi_* \mathbb{Q}$  is the restriction of an algebraic class on  $W$ . Then there is an ample class  $L_W$  on  $W$  whose operator  $\Lambda$  is induced by an algebraic cycle on  $W \times W$ :  $B(W)$  holds.*

*Proof. Step 1: the decomposition.* Let  $[n]: W \rightarrow W$  be multiplication by  $n$  over  $C$ . It acts on  $R^q \pi_* \mathbb{Q} = \bigwedge^q R^1 \pi_* \mathbb{Q}$  by  $n^q$ . The Leray spectral sequence degenerates [Del71, 4.1.1(i)], and  $[2]^*$  preserves the Leray filtration and acts on its graded pieces  $H^p(C, R^q \pi_* \mathbb{Q})$  by  $2^q$ . So  $H^k(W, \mathbb{Q}) = \bigoplus_q H_{(q)}^k$ , where  $H_{(q)}^k$  is the  $2^q$ -eigenspace of  $[2]^*$  and maps isomorphically onto  $H^{k-q}(C, R^q \pi_* \mathbb{Q})$ ; only  $k - q \in \{0, 1, 2\}$  occurs. The projectors  $P_q = \prod_{q' \neq q} ([2]^* - 2^{q'}) / (2^q - 2^{q'})$  are induced by algebraic cycles, polynomials in the graph of  $[2]$ . As  $[2]^*$  is a ring homomorphism,  $H_{(q)} \cup H_{(q')} \subset H_{(q+q')}$ ; as  $\int_W [2]^* z = 2^{2g} \int_W z$ , the integral vanishes on  $H_{(q)}$  for  $q \neq 2g$ , so Poincaré duality pairs  $H_{(q)}$  with  $H_{(2g-q)}$  only; and  $[2]_* = 2^{2g}([2]^*)^{-1}$  acts on  $H_{(q)}$  by  $2^{2g-q}$ .

*Step 2: the ample class.* Let  $\mathcal{L}$  be a relatively ample line bundle, replaced by  $\mathcal{L} \otimes [-1]^* \mathcal{L}$  so that it is symmetric. Then  $[-1]^*$  fixes  $c_1(\mathcal{L})$  and acts on  $H_{(q)}$  by  $(-1)^q$ , so  $c_1(\mathcal{L})$  has no component in  $H_{(1)}^2$ , while  $H_{(0)}^2 = H^2(C, \mathbb{Q})$  is spanned by the class  $f$  of a fibre. So  $c_1(\mathcal{L}) = \ell + af$  with  $\ell \in H_{(2)}^2$  algebraic and  $a \in \mathbb{Q}$ . Put  $L_W = \ell + mf = c_1(\mathcal{L}) + (m-a)f$  with  $m \in \mathbb{Q}$  so large that  $L_W$  is a positive rational multiple of the class of an ample line bundle, which is possible because  $\mathcal{L}$  is relatively ample. Write  $L_{\text{rel}}$  and  $L_C$  for cup product with  $\ell$  and with  $mf$ , so that  $L_W = L_{\text{rel}} + L_C$ ; on  $H^p(C, R^q \pi_* \mathbb{Q})$  the first raises  $q$  by two and the second raises  $p$  by two.

*Step 3: the relative operator.* Put  $\gamma = \ell^{g-1}/(g-1)!$ , which is algebraic and lies in  $H_{(2g-2)}$ , and  $D = \int_{W_c} \ell^g / g!$ . Let  $\mu, \text{pr}_1, \text{pr}_2: W \times_C W \rightarrow W$  be the addition and the projections, and  $T(x) = \mu_*(\text{pr}_1^* x \cup \text{pr}_2^* \gamma)$ . As in Proposition 20.21,  $T$  is induced by the algebraic cycle  $(\text{pr}_1, \mu)_*(\text{pr}_2^* Z)$  on  $W \times W$ . Since  $\pi \circ \text{pr}_1 = \pi \circ \mu$ , the projection formula

gives  $T(x \cup \pi^*b) = T(x) \cup \pi^*b$ , so  $T$  commutes with  $L_C$ . Since  $[2] \circ \mu = \mu \circ ([2] \times_C [2])$ ,  $[2]_*T(x) = \mu_*(\text{pr}_1^*[2]_*x \cup \text{pr}_2^*[2]_*\gamma)$ , and for  $x \in H_{(q)}$  this is  $2^{2g-q} 2^2 T(x)$ ; so  $T$  maps  $H_{(q)}$  into  $H_{(q-2)}$ . A relative correspondence acts on the Leray spectral sequence through the maps it induces on the local systems, so on  $H^p(C, R^q\pi_*\mathbb{Q})$  the operator  $T$  is  $H^p(C, T_\bullet)$ , with  $T_c$  the Pontryagin product with  $\gamma|_{W_c}$  on the fibre. By [Proposition 20.21](#),  $D^{-1}T_c$  is the operator  $\Lambda$  of the fibre, so  $\Lambda_{\text{rel}} = D^{-1}T$  satisfies  $[L_{\text{rel}}, \Lambda_{\text{rel}}] = H_{\text{rel}}$ , where  $H_{\text{rel}}$  acts on  $H_{(q)}$  by  $q - g$ .

*Step 4: the invariant classes.* Restriction maps  $H^q(W, \mathbb{Q})$  onto the monodromy invariants  $I_q \subset H^q(W_c, \mathbb{Q})$  [[Del71](#), 4.1.1(ii)], and it kills  $H_{(q')}^q$  for  $q' < q$ , which lies in positive Leray degree; so it maps  $H_{(q)}^q$  isomorphically onto  $I_q$ . Every element of  $H_{(q)}^q$  is algebraic: if  $v \in I_q$  is the restriction of an algebraic class  $z$ , then  $P_q z$  is algebraic, lies in  $H_{(q)}^q$  and restricts to  $v$ . The subspace  $I_q$  is a sub-Hodge structure, as the image of a morphism of Hodge structures, and it is stable under cup product with  $\ell_c$  and under the operator  $\Lambda$  of the fibre, which are the restrictions of  $L_{\text{rel}}$  and  $\Lambda_{\text{rel}}$ . The polarisation of  $H^q(W_c, \mathbb{Q})$  attached to  $\ell_c$  has the form  $(v, w) \mapsto \int_{W_c} v \cup S(w)$ , with  $S$  a polynomial in cup product with  $\ell_c$ , the operator  $\Lambda$  and the projectors of the fibre, and its restriction to the sub-Hodge structure  $I_q$  is again a polarisation [[Voi02](#)]. Since  $S(I_q) \subset I_{2g-q}$ , the pairing  $I_q \times I_{2g-q} \rightarrow \mathbb{Q}$  is nondegenerate.

*Step 5: the operator along the base.* For  $y \in H_{(q)}^q$  and  $w \in H_{(2g-q)}^{2g-q}$  one has  $\int_W (y \cup f) \cup w = \int_{W_c} y|_{W_c} \cup w|_{W_c}$ . By Step 4 and Step 1 this shows that  $L_C: H_{(q)}^q \rightarrow H_{(q)}^{q+2}$  is injective, and it is onto because  $\dim H_{(q)}^{q+2} = \dim H_{(2g-q)}^{2g-q} = \dim I_{2g-q} = \dim I_q$ , by Poincaré duality on  $W$  and on the invariants. Choose bases  $(a_j)$  of  $H_{(q)}^q$  and  $(b_k)$  of  $H_{(2g-q)}^{2g-q}$ , all algebraic by Step 4, let  $M_{jk} = \int_W a_j \cup m f \cup b_k$ , which is invertible, and put

$$\Lambda_C(x) = \sum_{j,k} (M^{-1})_{kj} \left( \int_W x \cup b_k \right) a_j \quad (x \in H^{q+2}(W, \mathbb{Q})),$$

for every  $q$ . It is induced by the algebraic cycle  $\sum_{j,k} (M^{-1})_{kj} b_k \times a_j$ . By Step 1,  $\int_W x \cup b_k$  vanishes for  $x \in H_{(q')}^{q+2}$  with  $q' \neq q$ , so  $\Lambda_C$  is zero on  $H^p(C, R^q\pi_*\mathbb{Q})$  for  $p \neq 2$ , and  $\Lambda_C(L_C a_i) = a_i$ . Hence  $[L_C, \Lambda_C]$  is  $-1$  on  $H_{(q)}^q$ ,  $0$  on  $H_{(q)}^{q+1}$  and  $+1$  on  $H_{(q)}^{q+2}$ : it is the operator  $H_C$  that acts on  $H^p(C, R^q\pi_*\mathbb{Q})$  by  $p - 1$ .

*Step 6: the cross relations and the conclusion.*  $[L_C, \Lambda_{\text{rel}}] = 0$  by Step 3. Both  $L_{\text{rel}}\Lambda_C$  and  $\Lambda_C L_{\text{rel}}$  vanish off the summands with  $p = 2$ , and on  $L_C y$  with  $y \in H_{(q)}^q$  both give  $\ell \cup y$ , because  $\ell \cup y \in H_{(q+2)}^{q+2}$  and  $L_C$  commutes with  $L_{\text{rel}}$ ; so  $[L_{\text{rel}}, \Lambda_C] = 0$ . Put  $\Lambda = \Lambda_{\text{rel}} + \Lambda_C$ , of degree  $-2$ . Then

$$[L_W, \Lambda] = [L_{\text{rel}}, \Lambda_{\text{rel}}] + [L_C, \Lambda_C] = H_{\text{rel}} + H_C,$$

which acts on  $H^p(C, R^q\pi_*\mathbb{Q}) \subset H^{p+q}(W)$  by  $(q - g) + (p - 1) = (p + q) - \dim W$ . So  $\Lambda$  is the operator of [Definition 10.5](#) for  $L_W$ , by its uniqueness, and it is induced by an algebraic cycle by Steps 3 and 5. ■

**Corollary 20.23** (Mumford families). *Let  $\pi: W \rightarrow C$  be a non-isotrivial abelian scheme of relative dimension four over a smooth projective connected curve,  $W$  projective, whose very general fibre  $X$  is a fourfold as in [Proposition 20.44](#), with Hodge group the group  $G$  of Mumford's construction [[Mum69](#)], the restriction of scalars from a totally real cubic field  $F$  of the norm one group of a quaternion algebra over  $F$ . Then  $B(W)$  holds. Such families exist, over the compact Shimura curves of [[Mum69](#)].*

*Proof.* By André’s theorem [And92, Theorem 1] the connected monodromy group is a normal subgroup of the derived Mumford–Tate group of  $X$ , which is  $G$ . It is not trivial, because the family is not isotrivial, and  $G$  is  $\mathbb{Q}$ -simple, since over  $\overline{\mathbb{Q}}$  its three factors are indexed by the embeddings of  $F$  and permuted transitively by the Galois group; so it is  $G$ . The monodromy invariants in  $H^*(X, \mathbb{Q})$  are therefore contained in the invariants of  $G$ , which are the Hodge classes of  $X$ , and these are the powers of the polarisation by Proposition 20.44(i). Item (XLII) of Appendix D and the Lean certificate confirm the count: the invariants of  $\mathfrak{sl}_2^3$  in  $\wedge^q(V_1 \otimes V_2 \otimes V_3)$  have dimension 1 for  $q$  even and 0 for  $q$  odd. The powers of  $\ell$  restrict to the powers of the polarisation, so every invariant class is the restriction of an algebraic class, and Theorem 20.22 applies. ■

*Remark 20.24* (What this does and does not reach). The total space  $W$  is a fivefold fibred in abelian fourfolds whose Hodge groups are not defined by endomorphisms; it is not a product and not an abelian variety. For abelian schemes over curves the mechanism of Theorem 20.22 goes back to Tankeev [Tan03], who proves  $B$  for the total space when certain canonically defined Hodge cycles are algebraic; we have not checked whether his hypotheses are visibly met by Mumford families, and make no claim of priority for Corollary 20.23. The hypothesis of Theorem 20.22 is where the difficulty sits. Applied to the fibre square  $W \times_C W \rightarrow C$  the theorem gives  $B(W \times_C W)$  as soon as every invariant class of  $X \times X$  is the restriction of an algebraic class on  $W \times_C W$ ; the invariant classes include the two exceptional classes of Proposition 20.44, which are not known to be algebraic, so this is the smallest case of (F2) in a global form. Theorem 20.22 does not bear on (P2), on the Kuga–Satake class of Theorem 21.4, or on (L) for varieties that are not fibred in abelian varieties over a curve.

**Proposition 20.25** (Powers of a Weil variety; cf. [Mil22]). *Let  $X$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$  and  $\mathrm{Hg}(X) = \mathrm{SU}(V, H)$ . For every  $k \geq 1$  the  $\mathbb{Q}$ -algebra  $\mathrm{Hdg}^*(X^k)$  is generated by the divisor classes of  $X^k$  and the pull-backs  $f_a^* \omega$  of the Weil classes  $\omega$  of  $X$  along the homomorphisms  $f_a = \sum_j a_j \mathrm{pr}_j: X^k \rightarrow X$ ,  $a \in \mathbb{Z}^k$ . Consequently, if the Weil classes of  $X$  are algebraic, the Hodge conjecture holds for every power of  $X$ . By [Mar25a] this is so for every such  $X$  with  $n = 2$ , and with  $n = 3$  and trivial discriminant.*

*Proof.* Write  $H^1(X, \mathbb{C}) = V_+ \oplus V_-$  for the eigenspaces of  $K$  and  $W = V_+$ . The polarisation makes  $V_+$  and  $V_-$  isotropic and pairs them perfectly, so  $V_- = W^\vee$ , and restriction to  $V_+$  identifies  $\mathrm{SU}(V, H)_\mathbb{C}$  with  $\mathrm{SL}(W)$ . As  $\mathrm{Hg}(X^k)$  is  $\mathrm{Hg}(X)$  acting diagonally on  $H^1(X^k, \mathbb{C}) = (W \otimes U) \oplus (W^\vee \otimes U)$ ,  $U = \mathbb{C}^k$ , the Hodge classes of  $X^k$ , tensored with  $\mathbb{C}$ , are the  $\mathrm{SL}(W)$ -invariants in  $\wedge^*((W \otimes U) \oplus (W^\vee \otimes U))$ . That algebra is an  $\mathrm{SL}(W)$ -equivariant quotient of a tensor algebra, with an equivariant section given by antisymmetrisation, so by the first fundamental theorem of invariant theory for  $\mathrm{SL}(W)$  its invariants are generated by the images of the contractions and the determinant tensors:<sup>58</sup> the classes  $D(u, u') = \sum_i (e_i \otimes u) \wedge (e_i^\vee \otimes u')$  and the polarised determinants, which by polarisation are combinations of  $\det_u = \wedge_i (e_i \otimes u)$  and  $\det_u^\vee = \wedge_i (e_i^\vee \otimes u)$ ,  $u, u' \in U$ . For  $2n \geq 3$  the  $D(u, u')$  span the invariants of degree two, which are the divisor classes tensored with  $\mathbb{C}$ . With  $\alpha_\pm$  spanning  $\wedge^{2n} V_\pm$  (Proposition 3.7) one has  $f_a^* \alpha_+ = \det_a$  and  $f_a^* \alpha_- = \det_a^\vee$ , and since  $u \mapsto \det_u$  is polynomial and  $\mathbb{Z}^k$  is Zariski dense in  $U$ , these span the same space as all  $\det_u$ . So the  $\mathbb{Q}$ -algebra generated by  $\mathrm{NS}(X^k)_\mathbb{Q}$  and the  $f_a^* \omega$  is contained in  $\mathrm{Hdg}^*(X^k)$ .

<sup>58</sup>This is Weyl’s first fundamental theorem for  $\mathrm{SL}(W)$  in its tensor form, applied before antisymmetrisation; the exterior algebra needs no separate invariant theory because antisymmetrisation is an equivariant projection with an equivariant section.

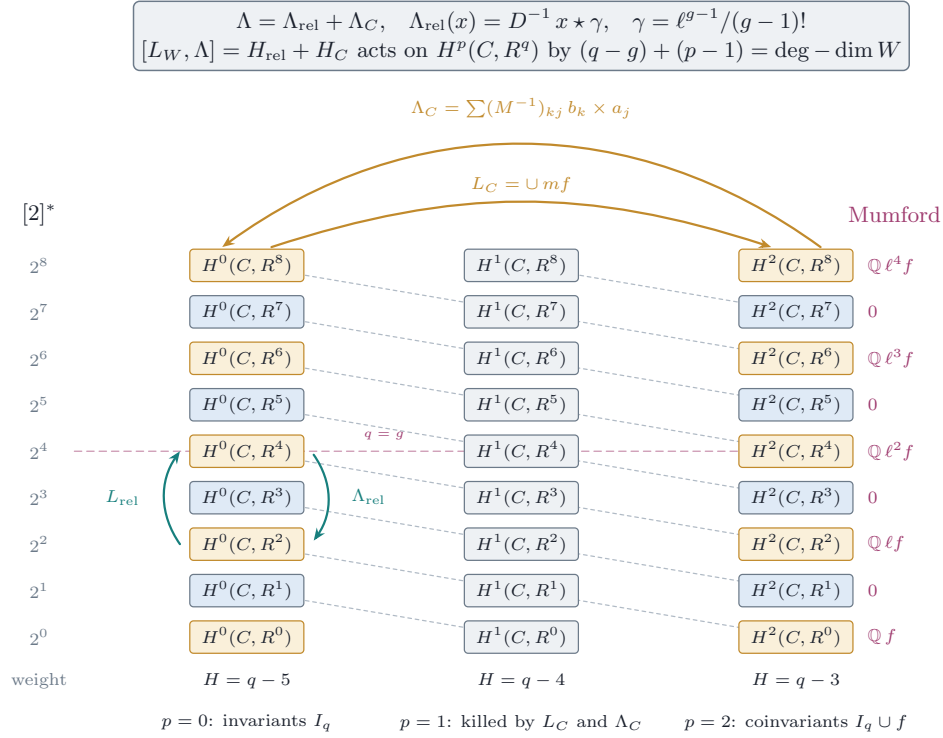


FIGURE 28. The proof of [Theorem 20.22](#) for relative dimension four. The boxes are the summands  $H^p(C, R^q \pi_* \mathbb{Q})$  of  $H^*(W, \mathbb{Q})$ , split off by the eigenvalue  $2^q$  of  $[2]^*$  shown on the left; the dotted diagonals join the summands of one  $H^k(W, \mathbb{Q})$ ,  $k = p + q$ , and the dashed magenta line is the axis  $q = g$  of hard Lefschetz on the fibres. The relative operators  $L_{\text{rel}}$  and  $\Lambda_{\text{rel}}$ , in teal, move within a column;  $\Lambda_{\text{rel}}$  is a Pontryagin product with an algebraic class on every abelian scheme ([Proposition 20.21](#)). The base operators  $L_C$  and  $\Lambda_C$ , in ochre, move along a row between the outer columns and kill the middle one;  $\Lambda_C$  is the displayed cycle built from bases  $(a_j)$ ,  $(b_k)$  of the invariant summands, and is algebraic exactly when the invariant cycles are. Below each column is the weight of  $H = [L_W, \Lambda]$  on it, and the panel at the top is Step 6 of the proof. The ochre boxes carry the invariant cycles and their images under  $L_C$ ; for a Mumford family they are the lines listed in magenta on the right, spanned by  $\ell^i f$  (and by  $\ell^i$  in the left column), and the other boxes of the outer columns are zero ([Corollary 20.23](#)).

and has the same complexification. The generators are algebraic when the Weil classes of  $X$  are, and they are for  $n = 2$  and for  $n = 3$  at trivial discriminant by [Mar25a]. ■

The statement for powers is known (compare [Mil22], and [Li26] for the powers of CM fourfolds); we include the short proof because it is what the next corollary uses.

**Corollary 20.26** (Abelian schemes of small relative dimension). *Let  $\pi: W \rightarrow C$  be an abelian scheme of relative dimension  $g$  over a smooth projective connected curve, with  $W$  projective, and suppose that every monodromy invariant class in  $H^*(W_c, \mathbb{Q})$  is a Hodge class, as happens when the identity component of the Zariski closure of the monodromy group is the Hodge group of a very general fibre.<sup>59</sup>*

- (i) *If  $g \leq 4$ , then  $B(W)$  holds.*
- (ii) *If the fibres are of  $(K, -1, n)$ -Weil type with connected monodromy group  $\mathrm{SU}(V, H)$ , and  $n = 2$ , or  $n = 3$  with trivial discriminant, then  $B$  holds for every fibre power  $W \times_C \cdots \times_C W$ . In particular  $B(W_C)$  of Corollary 20.32 holds in these cases.*

*Proof.* By the theorem of the fixed part the invariant classes form a sub-Hodge structure of every fibre, with a Hodge structure independent of the point, so they are Hodge classes on every fibre. (i) The Hodge conjecture holds for abelian varieties of dimension at most four, by the Lefschetz theorem on  $(1, 1)$  classes and hard Lefschetz up to dimension three and by [Mar25a] in dimension four, so each invariant class is algebraic on every fibre; Lemma 20.28 makes it the restriction of an algebraic class on  $W$ , and Theorem 20.22 applies. (ii) The fibre power is an abelian scheme with projective total space and the monodromy acting diagonally. Its invariant classes are  $\mathrm{SU}(V, H)$ -invariants, hence Hodge classes at every member with  $H_g = \mathrm{SU}(V, H)$ , an uncountable set of points, and there the Hodge conjecture holds for the fibre by Proposition 20.25; Lemma 20.28 and Theorem 20.22 conclude as before. For  $W_C$  the connected monodromy group is  $\mathrm{SU}(V, H)$  when it is Zariski dense, which is the hypothesis of Corollary 20.32(ii). ■

**Proposition 20.27** (The total space of a Mumford family). *Let  $W \rightarrow C$  be as in Corollary 20.23, with  $C$  the compact Shimura curve of [Mum69] of torsion-free level, or a finite étale cover of it. Then the Hodge conjecture holds for the fivefold  $W$ : its Hodge classes of degree  $2p$  are spanned by  $\ell^p$  and  $f\ell^{p-1}$ , with  $f$  the class of a fibre.*

*Proof.* By Step 1 of the proof of Theorem 20.22,  $H^k(W, \mathbb{Q}) = \bigoplus_q H_{(q)}^k$  with  $H_{(q)}^k \cong H^{k-q}(C, R^q\pi_*\mathbb{Q})$ , the summands being sub-Hodge structures cut out by algebraic projectors. The summand with  $k = q$  is the space of invariants, which by the proof of Corollary 20.23 is spanned by the powers of the polarisation, restrictions of the powers of  $\ell$ . The summand with  $k = q + 2$  is  $f \cup H_{(q)}^q$  by Step 5, spanned by the  $f\ell^{q/2}$ . It remains to see that  $H^1(C, R^q\pi_*\mathbb{Q})$  carries no Hodge class. For  $q$  even its weight  $q + 1$  is odd. For  $q$  odd, over  $\mathbb{C}$  the local system splits as a sum of  $S^{k_1}V_1 \otimes U_\lambda \otimes M_\lambda$ , where  $V_1$  is the uniformising variation of the factor of  $G$  split at the real place,  $U_\lambda$  is a unitary variation of type  $(0, 0)$ ,  $M_\lambda$  is constant of type  $(a, a)$  with  $2a + k_1 = q$ , and  $k_1 \equiv q \pmod{2}$  because the central element  $(-1, 1, 1)$  acts on  $V$  by  $-1$ . When  $C$  is étale over the Shimura curve the Kodaira–Spencer

<sup>59</sup>By André’s theorem [And92, Theorem 1] that identity component is always a normal subgroup of the derived Mumford–Tate group of a very general fibre, so the condition holds whenever the family is not isotrivial and the Hodge group of a very general fibre is semisimple and  $\mathbb{Q}$ -simple, as for the Mumford families of Corollary 20.23.

map of  $V_1$  is an isomorphism, and then  $H^1(C, S^{k_1}V_1 \otimes U_\lambda)$  has only the types  $(k_1 + 1, 0)$  and  $(0, k_1 + 1)$  [Zuc79]; after the twist by  $M_\lambda$  these are never balanced, since  $k_1 \geq 1$ .<sup>60</sup> ■

We have not found Proposition 20.27 in the literature, but it is of a kind studied for Kuga fibre varieties and we make no claim of priority. The étale hypothesis cannot be dropped from the argument: over a ramified cover the Kodaira–Spencer map has zeros and balanced types can occur in  $H^1(C, R^q\pi_*\mathbb{Q})$ .

**20.8. The two remaining targets, each reduced to one case of (L).** The machinery of Section 20.7 does more than add a case of (L). For an abelian scheme over a curve it shows that (L) for the total space is exactly the statement that algebraicity propagates along the curve, and the base points of this paper then turn each of the two targets left open in Remark 20.43, one Weil family and the first classes beyond the Weil lines, into (L) for one explicit variety. The reductions are equivalences, so they locate the difficulty rather than remove it. Along the way two things are proved outright: the Hodge conjecture for the self-product of a Mumford fourfold at every CM point, and the impossibility of realising the two branches of Remark 16.35 by two objects.

**Lemma 20.28** (Spreading along a curve). *Let  $\pi: W \rightarrow C$  be a smooth projective morphism onto a smooth projective connected curve,  $W$  projective, and  $\xi$  a global section of  $R^{2p}\pi_*\mathbb{Q}$ . If  $\xi_t$  is algebraic for uncountably many  $t \in C$ , there are an algebraic class  $\tilde{\xi}$  on  $W$  and an integer  $d \geq 1$  with  $\tilde{\xi}|_{W_t} = d\xi_t$  for every  $t \in C$ .*

*Proof.* The proof of Theorem 16.4 applies verbatim with  $(\mathcal{A}, \mathcal{S}, \omega)$  replaced by  $(W, C, \xi)$ : the set of  $t$  at which  $\xi_t$  is algebraic is the union, over countably many tuples of Hilbert polynomials and rational coefficients, of the images of unions  $T$  of connected components of fibre products of relative Hilbert schemes, projective over  $C$ , on which  $\sum_i q_i[Z_i] = \xi$ . Uncountably many  $t$  force one image to be infinite, hence equal to  $C$ . Choose an irreducible component of that  $T$  surjecting onto  $C$ , two of its points over distinct points of  $C$ , and an irreducible curve through them; its normalisation  $C'$  maps finitely onto  $C$ , of some degree  $d$ . Pulling the universal subschemes back to  $W' = W \times_C C'$  gives a cycle  $z = \sum_i q_i[Z_i]$  whose restriction to  $W'_{c'} = W_{g(c')}$  is  $\xi_{g(c')}$  for every  $c' \in C'$ . The projection  $h: W' \rightarrow W$  is finite and flat of degree  $d$ , and by base change  $h_*[z]$  restricts on  $W_t$  to the sum of the restrictions over the points of  $C'$  above  $t$ , counted with multiplicity, which is  $d\xi_t$ . ■

**Theorem 20.29** ((L) for an abelian scheme over a curve is propagation). *Let  $\pi: W \rightarrow C$  be as in Theorem 20.22, but with no hypothesis on its invariant cycles, and keep the notation of its proof. Let  $\Lambda_C$  be the operator that is zero on  $H^p(C, R^q\pi_*\mathbb{Q})$  for  $p \neq 2$  and inverts  $L_C: H_{(q)}^q \rightarrow H_{(q)}^{q+2}$ .*

- (i) *The operator  $\Lambda$  of Definition 10.5 for  $L_W$  is  $\Lambda_{\text{rel}} + \Lambda_C$ . So  $B(W)$  holds for  $L_W$  if and only if  $\Lambda_C$  is induced by an algebraic cycle.*
- (ii) *If  $B(W)$  holds for  $L_W$ , then for every  $c \in C$  and every algebraic class  $y$  on  $W_c$  the class  $\Lambda_C(i_{c*}y)$  is algebraic; up to the factor  $m$  it lies in  $H^0(C, R^q\pi_*\mathbb{Q})$  and restricts on  $W_c$  to the unique monodromy invariant class  $v$  with  $\int_{W_c} v \cup w = \int_{W_c} y \cup w$  for every invariant  $w$  of complementary degree. In particular every global section of  $R^{2p}\pi_*\mathbb{Q}$  that is algebraic at one point is the restriction of an algebraic class on  $W$ , and is algebraic at every point.*

<sup>60</sup>Over a ramified cover the Kodaira–Spencer map of  $V_1$  vanishes at the branch points, the Higgs field of  $S^{k_1}V_1$  is no longer an isomorphism, and the torsion of its cokernel contributes to  $H^1$  in balanced types; this is the only place where the étale hypothesis is used.

- (iii) If at one point  $c_0$  every monodromy invariant class of  $H^*(W_{c_0}, \mathbb{Q})$  is algebraic, then  $B(W)$  holds for  $L_W$  if and only if every monodromy invariant class is the restriction of an algebraic class on  $W$ .

*Proof.* (i) Steps 4 and 5 of the proof of [Theorem 20.22](#) show that  $L_C: H_{(q)}^q \rightarrow H_{(q)}^{q+2}$  is an isomorphism without using the hypothesis on invariant cycles: the nondegeneracy of  $I_q \times I_{2g-q} \rightarrow \mathbb{Q}$  comes from the Hodge–Riemann relations alone. So  $\Lambda_C$  is defined, and Steps 5 and 6, which use only that  $\Lambda_C$  vanishes off  $p = 2$  and inverts  $L_C$ , give  $[L_W, \Lambda_{\text{rel}} + \Lambda_C] = H$ . By the uniqueness in [Definition 10.5](#) this is  $\Lambda$ . As  $\Lambda_{\text{rel}}$  is induced by an algebraic cycle (Step 3),  $\Lambda$  is if and only if  $\Lambda_C$  is.

(ii) The Gysin map  $i_{c*}$  commutes with  $[2]_*$ , which acts on  $H^q(W_c)$  by  $2^{2g-q}$ , so  $i_{c*}y \in H_{(q)}^{q+2}$ , the summand with  $p = 2$ . If  $y$  is algebraic so is  $i_{c*}y$ , and by (i) so is  $\Lambda_C(i_{c*}y)$ . Put  $u = m \Lambda_C(i_{c*}y) \in H_{(q)}^q$ , so  $f \cup u = i_{c*}y$ . For  $w \in H_{(2g-q)}^{2g-q}$  the projection formula gives  $\int_{W_c} u|_{W_c} \cup w|_{W_c} = \int_W (f \cup u) \cup w = \int_W i_{c*}y \cup w = \int_{W_c} y \cup w|_{W_c}$ , and  $w|_{W_c}$  runs over the invariants of complementary degree (Step 4); by the nondegeneracy of Step 4 this determines  $u|_{W_c}$ . If  $y = \xi_c$  for a global section  $\xi$ , then  $v = \xi_c$  has the property, so  $u|_{W_c} = \xi_c$ , and  $u$ , being a global section of  $R^q \pi_* \mathbb{Q}$  equal to  $\xi$  at  $c$ , restricts to  $\xi_t$  at every  $t$ .

(iii) If every invariant class is a restriction of an algebraic class,  $B(W)$  holds by [Theorem 20.22](#). Conversely, under  $B(W)$ , (ii) applied at  $c_0$  to each invariant class makes it the restriction of an algebraic class.  $\blacksquare$

**Theorem 20.30** (The Mumford square at a CM point). *Let  $W \rightarrow C$  be a family as in [Corollary 20.23](#) and  $c \in C$  a CM point. Then every Hodge class on  $X_c \times X_c$  is algebraic. In particular the two exceptional classes of [Proposition 20.44](#) are algebraic on  $X_c \times X_c$ .*

*Proof.* The Hodge group  $T$  of  $X_c$  is a  $\mathbb{Q}$ -torus contained in  $G$ . It lies in a maximal  $\mathbb{Q}$ -torus of  $G = \text{Res}_{F/\mathbb{Q}} \text{SL}_1(D)$ , which is the restriction of scalars of the norm one torus of a quadratic extension of  $F$  inside  $D$ , of rank three, one factor for each embedding of  $F$ . The cocharacter of the Hodge structure lies in the factor of the embedding at which  $D$  splits, and the Galois group permutes the embeddings of  $F$  transitively, so the smallest  $\mathbb{Q}$ -subtorus containing it has rank three:  $T_{\mathbb{C}}$  is a maximal torus of  $G_{\mathbb{C}} \cong \text{SL}_2^3$ . Its eight weights on  $V = H^1(X_c, \mathbb{C}) = V_1 \otimes V_2 \otimes V_3$  are the vectors  $(\pm 1, \pm 1, \pm 1)$ , pairwise distinct. Hence  $\text{End}^0(X_c) \otimes \mathbb{C} = \text{End}_T(V)$  is the algebra of diagonal matrices in a weight basis,  $\text{Hom}(X_c \times X_c, X_c) \otimes \mathbb{C}$  is the space of pairs  $(\varphi, \psi)$  of such matrices acting by  $e_w \mapsto \varphi_w e_w^{(1)} + \psi_w e_w^{(2)}$ , and the Hodge classes of  $X_c$  and of  $X_c \times X_c$ , tensored with  $\mathbb{C}$ , are spanned by the monomials of weight zero.

Item (XLI) of [Appendix D](#) shows that the ring generated by the divisor classes of  $X_c \times X_c$  and by the pull-backs  $f^*(t)$ , for  $f = (\varphi, \psi)$  and  $t$  a Hodge class of  $X_c$  of degree 4 or 6, contains every Hodge class of  $X_c \times X_c$ , in every degree. The map  $f \mapsto f^*(t)$  is polynomial and the rational homomorphisms are Zariski dense in their complexification, so the same span is reached with  $f$  defined over  $\mathbb{Q}$  and  $t$  rational. Divisor classes are algebraic, and the Hodge classes of the abelian fourfold  $X_c$  are algebraic because the Hodge conjecture holds for abelian fourfolds [[Mar25a](#)]; pull-backs and products of algebraic classes are algebraic.  $\blacksquare$

**Corollary 20.31** (The first classes beyond the Weil lines are one case of (L)). *Let  $W \rightarrow C$  be as in [Corollary 20.23](#) and  $W^{(2)} = W \times_C W \rightarrow C$ , an abelian scheme of relative dimension eight with  $W^{(2)}$  projective of dimension nine. The following are equivalent.*

- (a)  $B(W^{(2)})$  holds for the class  $L_{W^{(2)}}$ .

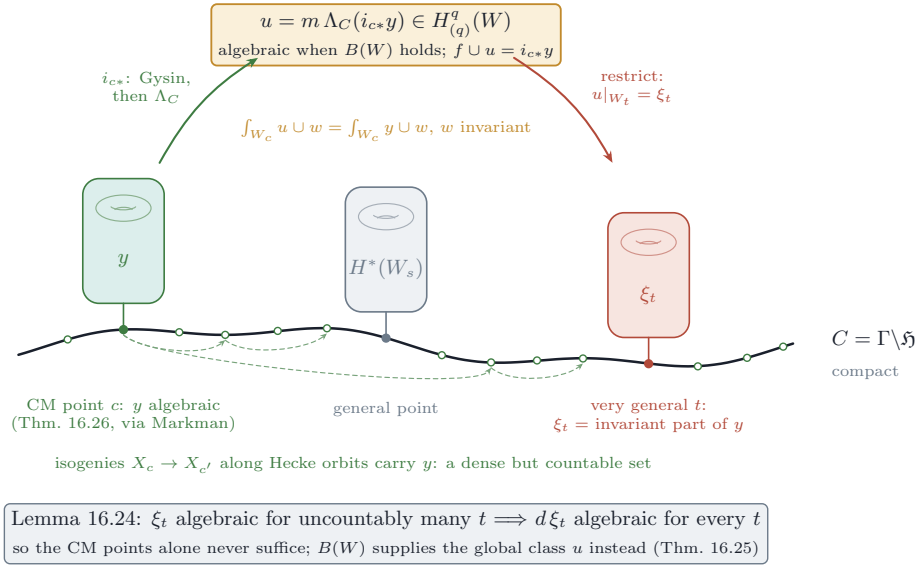


FIGURE 29. [Theorem 20.29\(ii\)](#) along a compact Shimura curve  $C$ . An algebraic class  $y$  on one fibre, at a base point of a Weil family or at a CM point of a Mumford family (hollow dots: the CM points, countable and dense), is pushed into  $W$  by the Gysin map and brought back to the invariant summand by the base operator  $\Lambda_C$ . When  $B(W)$  holds the result  $u$  is an algebraic class on  $W$ , pinned down by the displayed pairing against the invariant classes, and its restriction to a very general fibre is the invariant part of  $y$ . The dashed arcs are isogenies between CM members along Hecke orbits: they carry algebraic classes to algebraic classes but reach only countably many points, and [Lemma 20.28](#), in the box, needs uncountably many. This is how [Corollary 20.32](#) and [Corollary 20.31](#) turn each target into one case of (L).

- (b) *The two exceptional classes of  $X_t \times X_t$  are algebraic for uncountably many  $t \in C$ .*
- (c) *The Hodge conjecture holds for  $X_t \times X_t$  for every  $t \in C$ .*
- (d) *The Hodge conjecture holds for  $W^{(2)}$ ; for this item  $C$  is taken to be the compact Shimura curve of torsion-free level, or a finite étale cover of it.*

*Proof.* As in the proof of [Corollary 20.23](#), the connected monodromy group is  $G$ , so the monodromy invariants in  $H^*(X_t \times X_t, \mathbb{Q})$  are contained in the  $G$ -invariants, which are the Hodge classes of  $X_t \times X_t$  for  $t$  very general; conversely a  $G$ -invariant class is flat, being fixed by the monodromy. Item (XLIII) shows that these invariants form a ring generated by its elements of degree two, the divisor classes, and of degree four, which are the six products of divisor classes and the two exceptional classes.

(c) implies (b) trivially. (b) implies (a): the exceptional classes are restrictions of algebraic classes on  $W^{(2)}$  by [Lemma 20.28](#); so are the invariant divisor classes, which are algebraic on every fibre, by the same lemma; hence so is every invariant class, a polynomial in these, and [Theorem 20.22](#) gives  $B(W^{(2)})$ . (a) implies (c): at a CM point  $c_0$  every Hodge class of  $X_{c_0} \times X_{c_0}$  is algebraic by [Theorem 20.30](#), so by [Theorem 20.29\(iii\)](#) every invariant class is the restriction of an algebraic class, and is algebraic on every fibre. At a CM point

(c) is [Theorem 20.30](#). At any other point  $t$  the Hodge group of  $X_t$  is  $G$ , so its Hodge classes are the invariant ones. Indeed, if the Hodge group  $H_t$  were smaller, the Hodge classes of the powers of  $X_t$  that are not  $G$ -invariant would stay Hodge only on a discrete subset of the period domain, the upper half plane; every point of the orbit of the Hodge structure at  $t$  under  $H_t(\mathbb{R})^+$  carries them, so that orbit is a point, the cocharacter of the Hodge structure is central in  $H_t$ , and  $H_t$ , the smallest  $\mathbb{Q}$ -group containing it, is a torus:  $t$  would be a CM point.

For (d), split  $H^*(W^{(2)}, \mathbb{Q})$  into the summands  $H^p(C, R^q)$ ,  $p \in \{0, 1, 2\}$ , as in the proof of [Proposition 20.27](#). The summands with  $p = 1$  carry no Hodge class, by the argument given there, applied to the fibre square. Those with  $p = 2$  are  $f \cup H_{(q)}^q$ , and  $f \cup u = i_{c*}(u|_{W_c^{(2)}})$  for any  $c$ ; taking  $c$  a CM point, where  $u|_{W_c^{(2)}}$  is a Hodge class and so algebraic by [Theorem 20.30](#), they are algebraic. The summands with  $p = 0$  are the invariants, all of Hodge type. So (d) holds exactly when every invariant class is the restriction of an algebraic class, which by [Theorem 20.29\(iii\)](#), applied at a CM point, is (a).  $\blacksquare$

**Corollary 20.32** (One Weil family is one case of (L)). *Keep [Setup 16.1](#), let  $s_0 \in \mathcal{S}$  be the image of a base point of [Theorem 15.31](#), and let  $t_1 \in \mathcal{S}$  lie outside every proper Zariski closed set  $\Sigma_{\underline{p}, q}$  of [Theorem 16.4](#). Let  $\kappa: C \rightarrow \mathcal{S}$  be a morphism from a smooth projective connected curve whose image contains  $s_0$  and  $t_1$ , and  $W_C = \mathcal{A} \times_{\mathcal{S}} C$ .*

- (i) *If  $B(W_C)$  holds for  $L_{W_C}$ , then  $\Sigma = \mathcal{D}_{n,n}$ , and the Hodge conjecture holds in every degree for every member of the family with  $\text{Hg}(A) = \text{SU}(V, H)$ .*
- (ii) *Conversely, if  $\Sigma = \mathcal{D}_{n,n}$  and the monodromy of  $W_C$  is Zariski dense in  $\text{SU}(V, H)$ , then  $B(W_C)$  holds for  $L_{W_C}$ .*

*Proof.* (i) The class  $\kappa^*\omega$  is a global section of  $R^{2n}\pi_*\mathbb{Q}$  on  $C$ , algebraic at a point over  $s_0$  by [Theorem 15.31](#). By [Theorem 20.29\(ii\)](#) it is algebraic at a point over  $t_1$ , so  $t_1$  lies in some closed set  $\Sigma_{\underline{p}, q}$ , which is therefore all of  $\mathcal{S}$ . [Theorem 16.4](#) gives  $\Sigma = \mathcal{D}_{n,n}$ , and [Theorem 15.14\(a\)](#) the rest. (ii) With dense monodromy the invariants are the  $\text{SU}(V, H)$ -invariants, which are the Hodge classes of a member with  $\text{Hg}(A) = \text{SU}(V, H)$ : the powers of the polarisation and the Weil classes, by [Theorem 4.14](#). All are algebraic on every fibre when  $\Sigma = \mathcal{D}_{n,n}$ , so they are restrictions of algebraic classes by [Lemma 20.28](#), and [Theorem 20.22](#) applies.  $\blacksquare$

Curves  $\kappa$  as in [Corollary 20.32](#) exist: the Baily–Borel compactification of  $\mathcal{S}$  is projective [[BB66](#)], its boundary has codimension  $2n - 1 \geq 3$ , the rational boundary components of the domain of type  $\text{I}_{n,n}$  being of type  $\text{I}_{n-r, n-r}$  with  $r \geq 1$ , and a general complete intersection curve through  $s_0$  and  $t_1$  avoids it.

**Theorem 20.33** (Two branches cannot be two objects). *Let  $A$  be of  $(K, -1, n)$ -Weil type with  $n \geq 2$ ,  $\omega \neq 0$  a rational class on the Weil line, and  $E$  a perfect complex with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ , and  $\sigma_E$  injective, for instance with  $\dim \text{Ext}^2(E, E) = 2n(2n - 1)$ . Let  $E \cong \bigoplus_i F_i[k_i]$  be any decomposition. Then every summand  $F_i$  with  $\text{ch}(F_i) \neq 0$  has a Yoneda product  $\wedge^2 \text{Ext}^1(F_i, F_i) \rightarrow \text{Ext}^2(F_i, F_i)$  that is not injective. In particular  $E$  is not a direct sum of shifts of line bundles, of skyscraper sheaves of points, or of their images under autoequivalences of  $D^b(A)$ , since the Ext algebra of each of these is the exterior algebra on its  $\text{Ext}^1$ .*

*Proof.* By [Theorem 16.36\(ii\)](#) the kernel of  $\text{ev}_E$  in degree two is  $P \wedge Q$ . The characteristic morphism is compatible with the decomposition, so  $\text{ev}_{F_i}(p \wedge q) = \text{ev}_{F_i}(p) \text{ev}_{F_i}(q) = 0$  for all  $p \in P$  and  $q \in Q$ . Suppose the Yoneda product of  $F = F_i$  is injective on  $\wedge^2 \text{Ext}^1$ . Then

the images  $\bar{P}$  and  $\bar{Q}$  of  $P$  and  $Q$  in  $\text{Ext}^1(F, F)$  satisfy  $\bar{p} \wedge \bar{q} = 0$  for all  $\bar{p}, \bar{q}$ , so either  $\bar{P} = 0$ , or  $\bar{Q} = 0$ , or  $\bar{P} = \bar{Q}$  is a line. In the first case  $\text{ev}_F$  kills  $P \wedge HH^1$ , in the second  $Q \wedge HH^1$ , in the third all of  $HH^2 = HH^1 \wedge HH^1$ ; by the compatibility  $\sigma_F \circ \text{ev}_F = (\cdot) \lrcorner \text{ch}(F)$  of [Setup 16.32](#) the same subspace annihilates  $x = \text{ch}(F)$ .

These annihilators are computed in the free module of rank one that  $H^*(A, \mathbb{C})$  is over  $HH^*(A) = \bigwedge^*(P \oplus Q)$ , with generator the holomorphic volume form  $\Omega$ : an element  $y \cdot \Omega$  is killed by  $P \wedge HH^1$  exactly when  $p \wedge y$  has top degree for every  $p \in P$ , that is when  $y \in \det P \wedge \bigwedge^* Q + \bigwedge^{\geq 4n-1}(P \oplus Q)$ . Since  $\det P \cdot \Omega = \pm \alpha_+$ , the first summand gives the classes  $\bigwedge^j Q \cdot \alpha_+$ , of type  $(n-b, n+a)$  with  $a$  wedges by  $V_-^{0,1}$  and  $b$  contractions by  $T_+$ , and the second gives classes of type  $(1, 2n)$ ,  $(0, 2n-1)$  and  $(0, 2n)$ . The components of  $x$  are of type  $(k, k)$ , so  $x \in \mathbb{C}\alpha_+$ . Symmetrically the second case gives  $x \in \mathbb{C}\alpha_-$  and the third  $x = 0$ ; item (XLIII) confirms the three annihilators for  $n \leq 3$ , and so does the Lean certificate. Finally  $x$  is rational, complex conjugation carries  $V_+$  to  $V_-$  and  $H^{1,0}$  to  $H^{0,1}$ , so  $\bar{\alpha}_+ \in \mathbb{C}\alpha_-$ , and a rational multiple of  $\alpha_+$  or of  $\alpha_-$  is zero. So  $\text{ch}(F) = 0$ . As  $\sum_i (-1)^{k_i} \text{ch}(F_i) = N\omega \neq 0$ , some summand has nonzero Chern character, and the first statement follows; the second is the special case in which every summand has an exterior Ext algebra.  $\blacksquare$

*Remark 20.34* (Where the two targets stand). [Corollary 20.32](#) and [Corollary 20.31](#) say that closing one Weil family and closing the first classes beyond the Weil lines are each the Lefschetz standard conjecture for one variety: the total space of the family over a curve through a base point, and the ninefold  $W \times_C W$ . By [Theorem 20.29](#) that case of (L) is in turn the algebraicity of the invariant cycles, so nothing is gained for free: the reduction moves both targets onto the bullseye of [Remark 20.43](#) and shows that they sit there, and no further. Some things are unconditional: the Hodge conjecture for the total space of a Mumford family over its Shimura curve ([Proposition 20.27](#)),  $B$  for every abelian scheme over a curve of relative dimension at most four whose invariant classes are Hodge<sup>61</sup> ([Corollary 20.26](#)), and the following two. The Mumford classes are algebraic at every CM point ([Theorem 20.30](#)), so the first case of (F2) is a propagation problem along one compact Shimura curve, exactly as a Weil family is a propagation problem from its base point. And the object (P2) asks for cannot be assembled from two objects each of which sees one branch ([Theorem 20.33](#)): the two branches of [Remark 16.35](#) must live inside a single summand with a quadratic relation among its first order deformations. Neither target is closed here. For the Mumford target [Section 21.2](#) goes further: it closes off degeneration and deformation into a larger family, reduces semiregularity to one dimension, 119, and specialisation to a degree bound at CM points; and [Section 21.3](#) closes off every construction from line bundles and every twistor deformation of the square, and shows that modulo  $p$  the classes are represented at every closed point of every reduction, so that the target there is a bound on those representatives ([Theorem 21.33](#), [Proposition 21.24](#), [Theorem 21.38](#)).

The logic runs one way only.  $B(W_C)$  and  $B(W \times_C W)$  are two cases of (L), not (L): (L) for every variety implies both, but both together close one Weil family and the first case of (F2) and nothing more. The conjecture needs (L) for every variety together with (M) ([Proposition 20.17](#)), or (F3), which is the conjecture itself ([Proposition 20.35](#)), and no case of (L) proved or reduced in this section bears on (M) or on (F3).

<sup>61</sup>In relative dimension five the same argument goes through as soon as the Hodge conjecture is known for every abelian fivefold; we quote it there, following [MZ99] and [Mar25a], but have not checked the dimension-five statement of [MZ99] ourselves.

**20.9. What would close it, computed rather than listed.** A list of open problems invites a reading it does not deserve, that each entry is a step and that the steps together are a path. The implications proved above, together with those of [Section 20.6](#), decide the question, so we settle it by computation instead of by description. The logical skeleton of the paper and of the literature it quotes is finite: a set of statements, each of them proved here, quoted from the literature, or open, and a set of inference rules, each of them one theorem. Consequences can then be taken.

Such a computation is only as strong as its rule set. A set of statements the rules do not carry to the conjecture is *insufficient* in the sense of the computation, and that means no more than that no rule derives the conjecture from it. An implication that is true and left out makes the computation report a set as necessary when it is not. One such implication is elementary, and it decides the shape of the answer, so we prove it first.

**Proposition 20.35** (The conjecture for the varieties that are not abelian is the conjecture). *Let  $A$  be a complex abelian variety of dimension  $g \geq 1$  and let  $X = A \times \mathbb{P}^1$ . Then  $X$  is a smooth complex projective variety that is not an abelian variety, and if every Hodge class of degree  $2p$  on  $X$  is algebraic, then so is every Hodge class of degree  $2p$  on  $A$ . Consequently the statement*

*the Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety*

*implies the Hodge conjecture for every abelian variety, and it is equivalent to **HC**.*

*Proof.*  $X$  is smooth and projective of dimension  $g + 1$ . By the Künneth formula  $h^{1,0}(X) = h^{1,0}(A) + h^{1,0}(\mathbb{P}^1) = g$ , while an abelian variety of dimension  $g + 1$  has  $h^{1,0} = g + 1$ ; so  $X$  is not an abelian variety. (Equally,  $X$  contains the rational curves  $\{a\} \times \mathbb{P}^1$ , and every morphism from  $\mathbb{P}^1$  to a complex torus is constant, since it lifts to the universal cover  $\mathbb{C}^{g+1}$  [\[BL04, Chapter 1\]](#).)

Let  $\text{pr}: X \rightarrow A$  be the projection and  $D = A \times \{0\}$ , a smooth divisor that  $\text{pr}$  maps isomorphically onto  $A$ , so that  $\text{pr}_*[D] = 1$  in  $H^0(A, \mathbb{Q})$ . Let  $\alpha$  be a Hodge class of degree  $2p$  on  $A$ . Pull-back along a morphism is a morphism of Hodge structures, so  $\text{pr}^*\alpha$  is a Hodge class of degree  $2p$  on  $X$ , and by hypothesis  $\text{pr}^*\alpha = [Z]$  for an algebraic cycle  $Z$  on  $X$  with rational coefficients. The projection formula gives

$$\text{pr}_*([Z] \cup [D]) = \text{pr}_*(\text{pr}^*\alpha \cup [D]) = \alpha \cup \text{pr}_*[D] = \alpha.$$

The cycle class map is compatible with intersection products and with proper push-forward [\[Ful98, Chapter 19\]](#), so the left side is the class of the cycle  $\text{pr}_*(Z \cdot D)$  and  $\alpha$  is algebraic. A point, the abelian variety of dimension zero, satisfies the conjecture trivially.

So the displayed statement gives the conjecture for every abelian variety, and with it the conjecture for every smooth complex projective variety. The converse is immediate: the displayed statement is a special case of **HC**. ■

Since the conjecture for the varieties that are not abelian is the whole conjecture, the statement that belongs on a frontier is weaker: it asks for algebraicity only modulo what abelian varieties supply.

**Definition 20.36** (The conjecture modulo abelian varieties). For a smooth complex projective variety  $X$  and  $p \geq 0$  let  $\text{Ab}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$  be the span of the classes  $\Gamma_*\gamma = \text{pr}_{X*}([\Gamma] \cup \text{pr}_B^*\gamma)$  of degree  $2p$ , where  $B$  runs over the complex abelian varieties, the point included,  $\gamma$  over the Hodge classes on  $B$ , and  $\Gamma$  over the algebraic cycles on  $B \times X$

with rational coefficients. The statement (F3') is: for every  $X$  and every  $p$ ,

$$\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X).$$

Each  $\Gamma_*\gamma$  is a Hodge class, since cup product with an algebraic class and push-forward preserve Hodge classes, so  $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$ ; and taking  $B$  a point shows that  $\mathrm{Ab}^p(X)$  contains every algebraic class. So (F3') says that every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences, which is the form of (F3) that [Remark 20.42](#) asks for.

**Proposition 20.37** (What is known of (F3')).

- (i) **HC** implies (F3'), and (F3') together with the conjecture for abelian varieties implies **HC**.
- (ii) (F3') holds on every abelian variety and on every product  $A \times \mathbb{P}^1$  with  $A$  abelian; the argument of [Proposition 20.35](#) gives nothing for it.
- (iii) Let  $\mathcal{A}$  be the class of smooth complex projective varieties  $X$  for which  $H^*(X, \mathbb{Q})$  is spanned by classes  $\Gamma_*y$  with  $B$  abelian,  $\Gamma$  an algebraic cycle on  $B \times X$  and  $y \in H^*(B, \mathbb{Q})$ . Every  $X \in \mathcal{A}$  satisfies (F3'). The class  $\mathcal{A}$  contains every abelian variety and every smooth projective curve, and it is closed under products and under images of surjective morphisms. So it contains every product of curves, every symmetric product of a curve, and every smooth projective variety onto which a product of curves or an abelian variety maps surjectively.
- (iv) Every  $X$  of dimension  $n$  satisfies (F3') in the degrees  $2p$  with  $p \in \{0, 1, n-1, n\}$ , so every  $X$  of dimension at most three satisfies it in every degree.

*Proof.* (i) If **HC** holds, then  $\mathrm{Hdg}^p(X)$  consists of algebraic classes, which lie in  $\mathrm{Ab}^p(X)$ . Conversely, if the conjecture holds for abelian varieties, every  $\gamma$  in [Definition 20.36](#) is algebraic, so every  $\Gamma_*\gamma$  is, since composition with a correspondence carries algebraic classes to algebraic classes [[Ful98](#), Chapter 16]; then (F3') says that every Hodge class is algebraic.

(ii) On an abelian variety  $A$  take  $B = A$  and  $\Gamma$  the diagonal. On  $X = A \times \mathbb{P}^1$  the Künneth formula gives  $H^*(X, \mathbb{Q}) = \mathrm{pr}^*H^*(A, \mathbb{Q}) \oplus \mathrm{pr}^*H^*(A, \mathbb{Q}) \cup \eta$ , with  $\eta$  the class of a point of  $\mathbb{P}^1$  pulled back to  $X$ , which is algebraic. The first summand is  $\Gamma_*H^*(A, \mathbb{Q})$  for  $\Gamma$  the transpose of the graph of  $\mathrm{pr}$ , and the second is  $\Gamma'_*H^*(A, \mathbb{Q})$  for  $\Gamma' = \Gamma \cdot \mathrm{pr}_X^*\eta$ . So  $X \in \mathcal{A}$ , and (iii) applies.

(iii) Let  $X \in \mathcal{A}$  and fix  $p$ . Since  $H^{2p}(X, \mathbb{Q})$  is finite dimensional, finitely many classes  $\Gamma_{j*}y_j$  span it, and we may take each  $\Gamma_j$  of pure codimension  $c_j$  on  $B_j \times X$  and each  $y_j$  of pure degree  $k_j$ , with  $k_j + 2c_j - 2 \dim B_j = 2p$ . Then  $\Gamma_{j*}: H^{k_j}(B_j, \mathbb{Q}) \rightarrow H^{2p}(X, \mathbb{Q})$  is a morphism of Hodge structures up to a Tate twist, and the sum of these maps is a surjective morphism of polarisable Hodge structures onto  $H^{2p}(X, \mathbb{Q})$ . Polarisable Hodge structures are semisimple, so its kernel has a complement that is a sub-Hodge structure and maps isomorphically onto  $H^{2p}(X, \mathbb{Q})$ ; hence every Hodge class of degree  $2p$  is the image of a Hodge class, that is  $\sum_j \Gamma_{j*}\gamma_j$  with each  $\gamma_j$  a Hodge class on  $B_j$ . So  $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ .

An abelian variety lies in  $\mathcal{A}$  through its diagonal. A curve  $C$  of genus  $g \geq 1$  lies in  $\mathcal{A}$  through an Abel–Jacobi embedding  $\iota: C \rightarrow J(C)$ : the transpose of its graph induces  $\iota^*$ , which is an isomorphism on  $H^0$  and  $H^1$  and sends the class of the theta divisor to  $g$  times the class of a point, so it is surjective; the projective line is the image of an elliptic curve under a double cover, and is covered by the last closure property. For products, if  $H^*(X_i, \mathbb{Q})$  is spanned by the  $\Gamma_{i*}y$ , then by the Künneth formula  $H^*(X_1 \times X_2, \mathbb{Q})$  is spanned by the  $(\Gamma_1 \times \Gamma_2)_*(y_1 \otimes y_2)$ , with  $\Gamma_1 \times \Gamma_2$  an algebraic cycle on  $(B_1 \times B_2) \times (X_1 \times X_2)$ . For images, let  $f: Y \rightarrow X$  be a surjective morphism with  $Y \in \mathcal{A}$ , let  $r = \dim Y - \dim X$  and

let  $h$  be an ample class on  $Y$ . The correspondence  $\Psi = [\Gamma_f] \cdot \text{pr}_Y^* h^r$  on  $Y \times X$  induces  $\Psi_* y = f_*(y \cup h^r)$ , and  $\Psi_* f^* x = x \cup f_* h^r = c x$  with  $c > 0$  the degree of  $h^r$  on a general fibre. So  $\Psi_*$  is surjective, and  $H^*(X, \mathbb{Q})$  is spanned by the classes  $(\Psi \circ \Gamma)_* y$ , with  $\Psi \circ \Gamma$  algebraic [Ful98, Chapter 16]. A symmetric product of a curve is the image of a power of it.

(iv) Every class of degree 0 or  $2n$  is algebraic, and every Hodge class of degree 2 is algebraic by the Lefschetz theorem on  $(1, 1)$  classes. For  $n \geq 2$ , the hard Lefschetz theorem makes cup product with  $h^{n-2}$ , for  $h$  an ample class, an isomorphism  $H^2(X, \mathbb{Q}) \rightarrow H^{2n-2}(X, \mathbb{Q})$  of Hodge structures up to a Tate twist, so every Hodge class of degree  $2n - 2$  is  $h^{n-2}$  times a Hodge class of degree 2, and is algebraic. Algebraic classes lie in  $\text{Ab}^p(X)$ . When  $n \leq 3$  every  $p$  between 0 and  $n$  is one of  $0, 1, n - 1, n$ . ■

*Remark 20.38* (Why (F3') is not closed here). (F3') is the statement that the Hodge conjecture reduces to abelian varieties along algebraic correspondences, and Proposition 20.37 records what is known of such a reduction: the varieties of  $\mathcal{A}$ , whose whole cohomology is reached from abelian varieties by algebraic correspondences, and the degrees in which the conjecture itself is known. The first degree in which it is not known in general is degree four on fourfolds, for fourfolds not known to lie in  $\mathcal{A}$ . The square  $S \times S$  of a K3 surface is the standard example. Its Hodge classes in  $H^2(S) \otimes H^2(S)$  include the classes that induce the Hodge endomorphisms of the transcendental lattice  $T(S)$ , and  $S \times S$  lies in  $\mathcal{A}$  as soon as the Kuga–Satake embedding of  $T(S)$  is induced by an algebraic cycle, the Kuga–Satake Hodge conjecture for  $S$ , which is not known in general: the transpose of such a cycle maps the cohomology of the Kuga–Satake variety onto  $T(S)$ , and the rest of  $H^*(S, \mathbb{Q})$  is algebraic. The rational Hodge isometries are algebraic [Bus19], but a real multiplication is not an isometry, and Remark 21.5 meets exactly this obstacle for the surfaces  $S_\lambda$ .

For varieties whose cohomology is not of abelian type the position is starker. By Proposition 20.45 no correspondence moves the  $H^3$  of a very general quintic threefold into the cohomology of an abelian variety, so such a variety does not lie in  $\mathcal{A}$ . That is no obstruction to (F3') itself, since a Hodge class spans a Hodge structure of Tate type, which is of abelian type, and nothing in Hodge theory forbids it from lying in  $\text{Ab}^p(X)$ . Nor is it a route: to put a Hodge class into  $\text{Ab}^p(X)$  one must construct an algebraic cycle on  $B \times X$  whose class carries it, which is the kind of construction the Hodge conjecture asks for, now on  $B \times X$  instead of on  $X$ . The methods of this paper construct cycles on abelian varieties of Weil type and on nothing else, so they give no case of (F3') outside  $\mathcal{A}$ . We therefore leave it open, and Theorem 20.40 treats it as an open statement.

**Setup 20.39** (The rule set). Item (XXXIII) of Appendix D carries forty-two statements and twenty-two rules; the propagation statement (P2) is carried family by family, as (P2) for the split and for the other families of the imaginary quadratic fields and of the fields of degree at least four. Each statement is marked proved in this paper, quoted from the literature, or open. Each rule is a finite set of premises together with one conclusion, and carries the label of the theorem of this paper that proves it; Table 5 lists the rules whose conclusion is derived rather than assumed. For a set  $T$  of statements write  $\text{Cn}(T)$  for the least set that contains  $T$  and is closed under the rules, which exists because each rule has a single conclusion and the consequence operator is therefore monotone. Write **T** for the set of statements marked proved or quoted, and **HC** for the Hodge conjecture for every smooth complex projective variety. Call a set  $S$  of open statements *sufficient* if **HC**  $\in \text{Cn}(\mathbf{T} \cup S)$ , and *minimal* if no proper subset of  $S$  is sufficient.

**Theorem 20.40** (The closure of what is proved here). *With the notation of Setup 20.39:*

| premises                                                                   | conclusion                                | by                |
|----------------------------------------------------------------------------|-------------------------------------------|-------------------|
| the conjecture for abelian varieties, (F3)                                 | <b>HC</b>                                 | Section 20.5      |
| (F3)                                                                       | abelian varieties                         | Proposition 20.35 |
| the conjecture for abelian varieties, (F3')                                | <b>HC</b>                                 | Proposition 20.37 |
| the Weil classes of every CM field, (F2)                                   | abelian varieties                         | Section 20.5      |
| $K$ imaginary quadratic, the fields of degree at least four                | every CM field                            | Appendix C        |
| $\delta = \delta_0$ , $\delta \neq \delta_0$                               | every discriminant                        | Proposition 20.12 |
| a base point, the reduction, the orbit, (P2) for the family                | that family                               | Theorem 16.28     |
| the same three for a CM field of degree at least four, (P2) for the family | that family                               | Theorem 19.6      |
| its split families, its other families                                     | the fields of degree at least four        | Proposition 19.3  |
| $\delta = \delta_0$                                                        | $\delta \neq \delta_0$                    | Proposition 20.13 |
| the split families of the fields of degree at least four                   | their other families; $\delta = \delta_0$ | Proposition 20.13 |
| a secant object in every dimension, Question 11.4                          | $\delta = \delta_0$                       | Proposition 15.19 |
| a Cohen–Macaulay support, smooth or singular                               | one secant object                         | Corollary 18.57   |
| one secant object, Question 11.4                                           | $\mathbf{W}(K, 4, \delta_0)$              | Corollary 18.30   |
| (L), (M)                                                                   | <b>HC</b>                                 | Proposition 20.17 |
| (L), the deformation theorem for motivated classes                         | (V)                                       | Proposition 20.18 |
| (V), the accessibility of Hodge classes on abelian varieties               | abelian varieties                         | Theorem 20.19     |

TABLE 5. The rules of [Setup 20.39](#) whose conclusion is derived, with the theorem that proves each; the rows for (P2) and for the Cohen–Macaulay support each stand for two rules, the row for descent over the fields of degree at least four for two, descent within a field and scalar extension down to the imaginary quadratic fields, the second row is [Proposition 20.35](#), which an earlier version of the set omitted, the third is [Proposition 20.37\(i\)](#) for the weaker statement (F3') of [Definition 20.36](#), the last three rows are the rules of [Section 20.6](#), one of whose conclusions, (V), is itself open, and the one remaining rule records what the literature supplies on Weil classes. The bounded criterion (P1) is absent: by [Theorem 16.17](#) it is equivalent to its own conclusion, so it is a restatement and not a premise. Item (XXXIII) of [Appendix D](#) carries the whole set as data and checks that every label here is a label of this paper.

- (i) **HC**  $\notin \text{Cn}(\mathbf{T})$ . Nothing in this paper, combined with the theorems it quotes, implies the Hodge conjecture.
- (ii) There are exactly five minimal sufficient sets:

$$\{F3\}, \quad \{L, M\}, \quad \{L, F3'\}, \quad \{V, F3'\}, \quad \{P2_{\text{split}}, F2, F3'\},$$

with (L), (M), (V) as in [Definition 20.15](#), (F2), (F3) as in [Corollary 20.41](#), (F3') as in [Definition 20.36](#), and  $P2_{\text{split}}$  the propagation statement (P2), read in the corrected form of [Remark 17.6](#), for the split families of the CM fields of degree at least four. No minimal set contains (P2) for an imaginary quadratic field or for a family of nontrivial discriminant: [Proposition 20.13](#) derives those. The single open statement (F3) is sufficient, and it is equivalent to **HC** ([Proposition 20.35](#)); no other single open statement is. The sets  $\{L, F3\}$ ,  $\{V, F3\}$  and  $\{P2_{\text{split}}, F2, F3\}$  are sufficient and not minimal. The last of the five is the only one with three elements and the only one whose derivation of **HC** uses a statement proved in this paper.

- (iii) In each of the five, every element is necessary: removing any one leaves **HC** outside the closure.
- (iv) The secant route lies in no minimal sufficient set. Granting both of its open demands, a secant object satisfying (56) in every dimension and an affirmative answer to [[Mar25b](#), Question 11.4], puts  $\mathbf{W}(K, n, \delta_0)$  in the closure, and with it, by [Proposition 20.13](#),  $\mathbf{W}(K, n, \delta)$  for every  $\delta$ ; it leaves the Weil classes of the CM fields of degree at least four outside.
- (v) No statement proved in this paper rests, at any depth, on an open one, and the rule set is acyclic.
- (vi) Every element of a minimal sufficient set other than  $P2_{\text{split}}$  is a consequence of **HC**, and  $\{F3\}$  and  $\{L, M\}$  are each equivalent to it; no implication from **HC** to (P2) is known.
- (vii) An earlier version of this theorem, over a rule set without [Proposition 20.13](#), stated in (iv) that the secant route stops at the trivial discriminant, and listed (P2) for every family in the fifth set. Both came from the omission of the rules of [Proposition 20.13](#), and the five minimal sets are otherwise unchanged.

*Proof.* The consequence operator is monotone, so  $\text{Cn}$  is computed by iterating the one step map to a fixed point, and the set of subsets of the sixteen open statements is finite. Item (XXXIII) of [Appendix D](#) carries out the iteration and the exhaustive search over all  $2^{16}$  subsets of the sixteen open statements in exact arithmetic, and Section 24 of the Lean certificate repeats (i), (ii), (iii) and (iv) in the kernel, over the same rule set under the numbering recorded there, checking in particular all sixteen single open statements and all one hundred and twenty pairs. The content of the statement is the rule set itself, and every rule is one of the theorems named in [Table 5](#). The sufficiency of (F3) alone is the chain  $(F3) \Rightarrow \text{abelian varieties by Proposition 20.35, then abelian varieties} \wedge (F3) \Rightarrow \mathbf{HC}$ ; every sufficient set that contains (F3) and has another element is therefore not minimal, and a sufficient set without (F3) must reach **HC** through one of the two other rules with that conclusion, [Proposition 20.17](#) or [Proposition 20.37](#)(i), the second of which needs the conjecture for abelian varieties, reached by (V), by (L) through (V), or by (P2) with (F2). For (vi): (L) follows from **HC** by [Theorem 12.2](#), (M) by [Proposition 20.17](#), (V) by [Proposition 20.18](#), (F3') by [Proposition 20.37](#)(i), and (F2) and (F3) are special cases of **HC**.

In (iv), every base point the secant route produces is a member  $X \times \widehat{X}$ , whose discriminant is trivial whatever  $\beta$  is chosen ([Proposition 15.19](#)(ii)); the deformation argument stays

inside the family of trivial discriminant, and it is [Proposition 20.13\(i\)](#), not the deformation, that carries the result to the other discriminants. It has no base point over a field of degree at least four, so it reaches none of them. ■

**Corollary 20.41** (The three statements, one of which is the conjecture). *The Hodge conjecture follows from the results of this paper together with the following three statements. No rule of [Setup 20.39](#) derives it from (F1) and (F2) together, and it follows from (F3) alone, which is equivalent to it ([Proposition 20.35](#)); so the list is not a reduction of the conjecture to three smaller problems. With (F3) replaced by the weaker statement (F3') of [Definition 20.36](#), the conjecture modulo abelian varieties, the three statements (F1), (F2), (F3') still suffice, and no rule of [Setup 20.39](#) derives the conjecture from a proper subset of them.*

- (F1) One propagation statement, for every CM field. (P2), in the corrected form of [Remark 17.6](#): at one base point of each Weil family of each CM field some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation has injective semiregularity map. Its first form, with Chern character exactly a multiple of the Weil class, is false for every imaginary quadratic field and every  $n \geq 2$  ([Theorem 17.5](#)): such a complex would have to deform to non-algebraic Weil tori, whose coherent sheaves have no Chern character in degrees 1 to  $2n - 1$ . The corrected form closes  $\mathbf{W}(K, n, \delta)$  for every imaginary quadratic  $K$ , every  $n$  and every  $\delta$  at once, by [Theorem 16.28](#), and  $\mathbf{W}(F, n, \delta)$  for every CM field  $F$  of higher degree, by [Theorem 19.6](#), because those theorems put  $\Sigma = \mathcal{D}$ , so the Weil class is algebraic at every member of the family and not only at the general one, and because [Theorem 15.31](#) and [Theorem 19.4](#) supply the base point in every one of those families. By [Proposition 20.13](#) it is enough to have it for the split families of the CM fields of degree at least four, in infinitely many dimensions. [Theorem 16.30](#), [Theorem 16.38](#), [Theorem 20.33](#), [Theorem 16.36](#) and [Theorem 16.40](#) describe the object of the first form and are vacuous by [Theorem 17.5](#); for the corrected form [Theorem 17.9](#) computes the number that  $\dim \text{Ext}^2(E, E)$  has to reach,  $7n^2 - 2n$  for a general shape. For one family its conclusion is equivalent to the Lefschetz standard conjecture for one variety ([Corollary 20.32](#)). The bounded criterion (P1) is not a second form of the same demand: by [Theorem 16.17](#) it is equivalent to the conclusion it was to imply, and [Theorem 16.15](#) shows that the one cycle it could have been tested on, the transport of the base cycle, does not satisfy it.
- (F2) The classes beyond the Weil lines. Every Hodge class on an abelian variety that lies outside the subring generated by divisor classes and by Weil classes of quotients is algebraic. Through dimension five there is no such class [[MZ99](#)]; in dimension eight there are ([Proposition 20.44](#)), so this is not a reduction of the conjecture for abelian varieties to Weil classes, which is false as a statement about Hodge rings: it is the conjecture for those classes. Its smallest case is brought down to one Kuga–Satake class in [Theorem 21.4](#); it holds at every CM point ([Theorem 20.30](#)), and on all members of a Mumford family it is equivalent to the Lefschetz standard conjecture for one ninefold ([Corollary 20.31](#)). Its classes are rigid along the base curve ([Theorem 21.21](#)), and one complex at one CM point with  $\text{Ext}^2$  of dimension 119 would close it ([Theorem 21.27](#)), as would a bound on the cycles that represent them at the closed points of the reductions of the curve modulo primes ([Theorem 21.38](#)).
- (F3) The varieties that are not abelian. The Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety. This is the Hodge conjecture

itself. The varieties it covers include  $A \times \mathbb{P}^1$  for every abelian variety  $A$ , and the conjecture for  $A \times \mathbb{P}^1$  gives it for  $A$  ([Proposition 20.35](#)); so (F3) implies the conjecture for abelian varieties, hence (F1) and (F2), and hence **HC**. Nor is it a reduction to abelian varieties in the other direction: no realisation of the cohomology of an arbitrary variety inside that of abelian varieties exists ([Proposition 20.45](#)), and [Theorem 1.4](#) and [Theorem 1.5](#) close the two routes through hyperplane sections that would be the natural candidates for one.

*Proof.* The three together are sufficient by the derivation of [Theorem 20.40](#), and (F3) alone by [Proposition 20.35](#). That (F1) and (F2) together are not sufficient in the rule set is [Theorem 20.40\(ii\)](#), since every minimal sufficient set contains (F3), (F3') or (M); the statement about (F1), (F2), (F3') is [Theorem 20.40\(ii\)](#) and (iii) for the last of the five minimal sets. The Weil classes of the CM fields of degree at least four are not a fourth statement: they are derived from (F1) by [Theorem 19.6](#). Nor is the bounded criterion a premise, since by [Theorem 16.17](#) it is a restatement of its own conclusion. ■

*Remark 20.42* (How the earlier count went wrong, and the wording of (F2)). An earlier version of this corollary said that the conjecture follows from no proper subset of the three statements. That was a statement about a rule set that used (F3) only through (abelian varieties)  $\wedge$  (F3)  $\Rightarrow$  **HC** and did not contain the elementary rule (F3)  $\Rightarrow$  (abelian varieties) of [Proposition 20.35](#). The rule set can show that a set suffices; it cannot show that a set is needed, except relative to the implications written into it.

The wording of (F2) has the same feature, one degree at a time. Let  $R(A) \subseteq H^*(A, \mathbb{Q})$  be the subring generated by divisor classes and Weil classes of quotients. Suppose some Hodge class  $\beta$  of degree  $2p$  on  $A$  lies outside  $R(A)$ , and let  $\alpha$  be any Hodge class of degree  $2p$ . If  $\alpha \notin R(A)$ , (F2) makes it algebraic. If  $\alpha \in R(A)$ , then  $\alpha + \beta \notin R(A)$ , because  $R(A)$  is closed under subtraction, so (F2) makes  $\alpha + \beta$  and  $\beta$  algebraic, and with them  $\alpha$ . So in every degree in which (F2) says anything it is the whole conjecture for that degree: on the square of a Mumford fourfold it gives every Hodge class of  $H^4$ , the six products of divisor classes included. Whether (F2) alone gives the conjecture for every abelian variety is not decided here, and nothing depends on it: the rule set does not use such an implication, and adding it would not change [Theorem 20.40\(ii\)](#), since (F3) alone is already sufficient.

The form of (F3) that is not the conjecture itself asks only for algebraicity modulo classes coming from abelian varieties. It is (F3') of [Definition 20.36](#); [Proposition 20.37](#) says what is known of it, and [Remark 20.38](#) why it is not closed.

*Remark 20.43* (The target, and what it would take). With the rules of [Section 20.6](#), [Proposition 20.35](#) and [Proposition 20.37](#) in place, [Theorem 20.40](#) changes the reading of [Corollary 20.41](#) in three ways. First, the route through this paper is a route only with (F3) replaced by (F3'). As stated, its last statement (F3) is the conjecture ([Proposition 20.35](#)), so (F1) and (F2), and everything this paper proves about Weil classes, drop out of every minimal derivation that uses (F3). With (F3') in its place the route  $\{P2_{\text{split}}, F2, F3'\}$  is minimal, and it is not the shortest: (F1) and (F2), which between them give the conjecture for abelian varieties along this route, are replaced on the routes of the literature by the single statement (V), or by (L), which implies it ([Theorem 20.19](#)). Second, every element of every minimal sufficient set except  $P2_{\text{split}}$  is a consequence of the conjecture ([Theorem 20.40\(vi\)](#)), and two of the sets,  $\{F3\}$  and  $\{L, M\}$ , are each equivalent to it. (P2) is a statement about semiregularity maps, and a proof of the conjecture need not prove it; in its first form, with a Chern character that is exactly a Weil class, it is false ([Theorem 17.5](#)), and the route runs through the corrected form of [Remark 17.6](#). Third, every minimal sufficient set

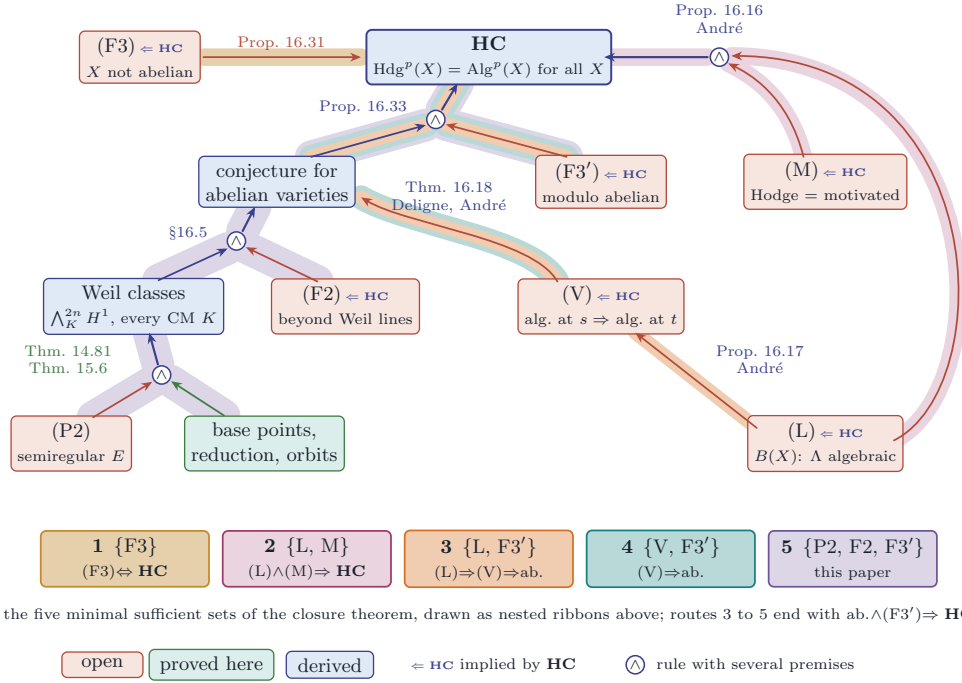


FIGURE 30. The rule set of [Setup 20.39](#) with the rules of [Section 20.6](#), drawn as a hypergraph. A rule with several premises is a junction marked  $\wedge$ , with arrows from its premises and one arrow to its conclusion; each rule carries the result that proves it, and the junction marked §16.5 is the closure map of [Section 20.5](#). Clay marks an open statement, grass what this paper proves, indigo a derived statement, and the tag  $\Leftarrow \mathbf{HC}$  an open statement that the conjecture implies: every clay node except (P2). The arrow from (F3) to the conjecture is [Proposition 20.35](#), and the junction above the conjecture for abelian varieties is [Proposition 20.37\(i\)](#). The five minimal sufficient sets of [Theorem 20.40\(ii\)](#) are drawn as nested ribbons under the arrows they use, widest first, and listed in the cards below: (F3) alone; (L) with (M); (L) through (V) with (F3'); (V) with (F3'); and, on the left, (P2) with (F2) and (F3'). A shared arrow shows the bands of every route through it. The secant route, which reaches the imaginary quadratic families only, is drawn in [Figure 31](#).

contains (F3), (F3') or (M): the varieties that are not abelian are reached only through the conjecture for them, which is the whole conjecture, through the conjecture modulo abelian varieties, which is open (Remark 20.38), or through motivated classes.

So the target at the centre is (L). It lies in two of the five minimal sets, it gives (V) and with it the whole abelian half, the conjecture implies it, and with (M) it is the conjecture. For each  $X$  it asks for one explicit Hodge class on  $X \times X$  to be algebraic, the class inducing  $\Lambda$ . It is open, as are (M), (V) and (F3'), and this paper proves none of them in general: Corollary 20.23 adds the total spaces of Mumford families to the known cases of (L), Proposition 20.37 records the known cases of (F3'), and the routes of the literature use no other theorem of this paper.

*Two of the three are not reductions.* A reader might hope that (F2) and (F3) are reductions in the ordinary sense, statements to the effect that every Hodge class is carried, by some correspondence, to a class on a Weil type variety or on an abelian variety, so that the whole conjecture would follow from the Weil classes alone. Both hopes are refutable, and we refute them, because a frontier that contains a false statement is not a frontier.

**Proposition 20.44** (The Hodge ring is not generated by divisor and Weil classes). *Let  $X$  be a complex abelian fourfold whose Hodge group is a  $\mathbb{Q}$ -form of an almost direct product of three copies of  $\mathrm{SL}_2$ , acting on  $V = H^1(X, \mathbb{Q})$  through the tensor product  $V_1 \otimes V_2 \otimes V_3$  of the three standard representations, with  $\mathrm{End}^0(X) = \mathbb{Q}$ ; such fourfolds exist and form families of dimension one, by Mumford [Mum69], and by [MZ99, 2.5(i)] they are exactly the fourfolds with  $\mathrm{End}^0(X) = \mathbb{Q}$  and Hodge group smaller than  $\mathrm{Sp}_8$ . Then:*

- (i) *the Hodge ring of  $X$  is generated by the polarisation:  $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X) = \dim_{\mathbb{Q}} \mathrm{Hdg}^2(X) = 1$ ;*
- (ii)  *$\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X \times X) = 3$  and  $\dim_{\mathbb{Q}} \mathrm{Hdg}^2(X \times X) = 8$ , while the products of divisor classes span a subspace of dimension 6 in  $H^4(X \times X, \mathbb{Q})$ ;*
- (iii) *the two remaining classes lie in  $H^4(X \times X, \mathbb{Q})$  and are Weil classes of no CM field acting on  $X \times X$ , on a quotient of it, or on an abelian subvariety of it.*

*In particular the statement that every Hodge class on an abelian variety lies in the subring generated by divisor classes and Weil classes of quotients is false in dimension eight, though true through dimension five [MZ99].*

*Proof.* Hodge classes are the invariants of the Hodge group, so  $\mathrm{Hdg}^k(Y) \otimes \mathbb{C}$  is the space of  $\mathrm{Hg}(Y)_{\mathbb{C}}$ -invariants in  $\wedge^{2k} H^1(Y, \mathbb{C})$ , and  $\mathrm{Hg}(X \times X) = \mathrm{Hg}(X)$  acting diagonally on  $V \oplus V$ . Over  $\mathbb{C}$  the group is  $\mathrm{SL}_2^3$  and the weights of  $V$  under its maximal torus are the eight sign vectors  $(\pm 1, \pm 1, \pm 1)$ . Item (XXXV) of Appendix D decomposes every  $\wedge^k V$  by peeling highest weights and finds

$$\begin{aligned} \wedge^2 V &= V(2, 2, 0) \oplus V(2, 0, 2) \oplus V(0, 2, 2) \oplus \mathbf{1}, \\ \wedge^4 V &= V(2, 2, 2) \oplus \bigoplus_{\mathrm{cyc}} V(4, 0, 0) \oplus \bigoplus_{\mathrm{cyc}} V(2, 2, 0) \oplus \mathbf{1}, \end{aligned}$$

each with one trivial summand, which is (i). For (ii),  $\wedge^4(V \oplus V) = \bigoplus_a \wedge^a V \otimes \wedge^{4-a} V$  and the invariants in  $\wedge^a V \otimes \wedge^{4-a} V$  are  $\mathrm{Hom}_{\mathrm{SL}_2^3}(\wedge^a V, \wedge^{4-a} V)$ , since every summand is self-dual; the counts are 1, 1, 4, 1, 1, total 8, the 4 coming from the four distinct irreducible summands of  $\wedge^2 V$ . Likewise the invariants in  $\wedge^2(V \oplus V)$  number  $1 + 1 + 1 = 3$ . The divisor classes of  $X \times X$  are the three symplectic forms  $\psi_{11}, \psi_{12}, \psi_{22}$  on  $V \oplus V$  built from the polarisation  $\psi$  of  $V$ , and the same item multiplies them in the exterior algebra of  $V \oplus V$  and finds their six pairwise products linearly independent; it also counts the  $\mathrm{Sp}_8$ -invariants of  $\wedge^4(V \oplus V)$  by the Weyl character formula and finds six, so the six products are all

the classes a Hodge group equal to  $\mathrm{Sp}_8$  would allow. For (iii), the Weil classes of an abelian variety  $Y$  with respect to a CM field  $F \subseteq \mathrm{End}^0(Y)$  are, by definition, the classes in  $\bigwedge_F^{2k} H^1(Y, \mathbb{Q}) \subseteq H^{2k}(Y, \mathbb{Q})$  with  $2k = \dim_F H^1(Y, \mathbb{Q})$  (Setup 19.1). A CM field acting on  $X \times X$  lies in  $\mathrm{End}^0(X \times X) = M_2(\mathbb{Q})$ , so it is imaginary quadratic,  $2k = 8$ , and its Weil classes lie in  $H^8$ , while the two classes of (ii) have degree four. Every proper abelian subvariety and every proper quotient of  $X \times X$  is isogenous to  $X$ , because  $X$  is simple, and  $\mathrm{End}^0(X) = \mathbb{Q}$  contains no CM field, so  $X$  carries no Weil classes at all. This is the statement of [MZ99, 0.1(iii)] that the exceptional classes of  $X \times X$  are not of Weil type. The last sentence is (ii) and (iii) together with the theorem of [MZ99] for dimension at most five. ■

**Proposition 20.45** (No Hodge-theoretic reduction to abelian varieties). *Let  $Y$  be a very general smooth quintic threefold in  $\mathbb{P}^4$ , and let  $V = H^3(Y, \mathbb{Q})$  with its polarised Hodge structure of weight three. Then  $V$  is not isomorphic to a subquotient of any Hodge structure of the form  $\bigoplus_i T_i$ , where each  $T_i$  is obtained from the  $H^1$  of finitely many abelian varieties by tensor products, duals and Tate twists. In particular the Hodge structure  $H^*(Y, \mathbb{Q})$  is not of abelian type, and no correspondence between  $Y$  and an abelian variety, or a product of abelian varieties, induces an injection of  $V$  into the cohomology of the latter.*

*Proof.* Write  $\mathrm{MT}(W)$  for the Mumford–Tate group of a rational Hodge structure  $W$  and  $\mu_W: \mathbb{G}_m \rightarrow \mathrm{MT}(W)_{\mathbb{C}}$  for its Hodge cocharacter, acting on  $W^{p,q}$  by  $z^p$ . The adjoint action of  $\mu_W$  on  $\mathrm{Lie} \mathrm{MT}(W)_{\mathbb{C}} \subseteq \mathrm{End}(W_{\mathbb{C}})$  has weights among the differences  $p - p'$  of two Hodge indices of  $W$ .

If  $W$  has weight one, these differences lie in  $\{-1, 0, 1\}$ . If  $W'$  is a subquotient of a tensor construction  $T$  on  $W$ , then  $\mathrm{MT}(W')$  is the image of  $\mathrm{MT}(W)$  acting on  $W'$ , and  $\mu_{W'}$  is the image of  $\mu_W$ ; so  $\mathrm{Lie} \mathrm{MT}(W')$  is a quotient of  $\mathrm{Lie} \mathrm{MT}(W)$  as a representation of  $\mathbb{G}_m$  through  $\mu$ , and its weights are a subset of  $\{-1, 0, 1\}$ . The same holds with  $W = \bigoplus H^1(A_i)$ . Hence for every Hodge structure of abelian type the adjoint weights of the Hodge cocharacter lie in  $\{-1, 0, 1\}$ .

For  $V = H^3(Y, \mathbb{Q})$  we show that the adjoint weight 3 occurs. The canonical bundle of a quintic threefold is trivial by adjunction, so  $h^{3,0}(Y) = 1$ ; pick  $0 \neq e \in H^{3,0}$ . The intersection form  $\psi$  on  $V$  is symplectic and pairs  $H^{3,0}$  with  $H^{0,3}$  perfectly, so the rank one endomorphism  $x \mapsto \psi(x, e)e$  of  $V_{\mathbb{C}}$  is nonzero, sends  $H^{0,3}$  to  $H^{3,0}$  and kills the rest, and lies in  $\mathfrak{sp}(V_{\mathbb{C}}, \psi)$  because  $\psi(\psi(x, e)e, y) + \psi(x, \psi(y, e)e) = \psi(x, e)(\psi(e, y) + \psi(y, e)) = 0$ ; its adjoint weight is  $3 - 0 = 3$ . It remains to see that  $\mathfrak{sp}(V_{\mathbb{C}}, \psi) \subseteq \mathrm{Lie} \mathrm{MT}(V)_{\mathbb{C}}$ . Let  $S$  be the parameter space of smooth quintic threefolds, connected and nonsingular, and  $\mathbb{V}$  the polarisable variation of Hodge structure  $R^3$  of the universal family. The connected algebraic monodromy group  $G^0$  of  $\mathbb{V}$  contains the connected monodromy group of any Lefschetz pencil of quintics, that is of hyperplane sections of the fifth Veronese image of  $\mathbb{P}^4$ ; for a Lefschetz pencil of odd fibre dimension the image of the monodromy representation on the  $\ell$ -adic vanishing cohomology  $E \otimes \mathbb{Q}_{\ell}$  is open in  $\mathrm{Sp}(E \otimes \mathbb{Q}_{\ell}, \psi)$  [Del74, Theorem 5.10], so the Zariski closure of the monodromy group over  $\mathbb{Q}$  is  $\mathrm{Sp}(E, \psi)$ ; here  $E = H^3(Y_t, \mathbb{Q})$ , the vanishing cohomology being the orthogonal complement of the image of  $H^3(\mathbb{P}^4, \mathbb{Q}) = 0$ . Hence  $G^0 \supseteq \mathrm{Sp}(V, \psi)$ , and  $G^0 \subseteq \mathrm{Sp}(V, \psi)$  since the monodromy preserves  $\psi$ . At a very general point  $Y$  the group  $G^0$  is a normal subgroup of the derived group of  $\mathrm{MT}(V)$  [And92, Theorem 1], so  $\mathrm{Sp}(V, \psi) \subseteq \mathrm{MT}(V)^{\mathrm{der}} \subseteq \mathrm{Sp}(V, \psi)$ , and the adjoint weight 3 occurs in  $\mathrm{Lie} \mathrm{MT}(V)$ . So  $V$  is not of abelian type, and neither is any Hodge structure containing it as a subquotient, in particular  $H^*(Y, \mathbb{Q})$ . A correspondence inducing an injection of  $V$  into the cohomology of an abelian variety would exhibit  $V$  as a sub-Hodge structure

of a Tate twist of that cohomology, which is of abelian type. Item (XXXV) records the Hodge numbers  $(1, 101, 101, 1)$  from the Jacobian ring of the Fermat quintic, checks the symplectic identity and the weight of the displayed endomorphism in a four-dimensional model with the same weights, and lists the adjoint weights that occur. ■

*Remark 20.46* (What the two propositions do to the frontier). Neither proposition changes [Theorem 20.40](#): the rule set uses (F2) and (F3) as implications, and an implication whose hypothesis is the conjecture for one class of Hodge classes and whose conclusion is the conjecture for a larger class is true whatever the classes are. What does change it is [Proposition 20.35](#): the varieties of (F3) are not a class disjoint from the abelian ones but contain them, through products with  $\mathbb{P}^1$ , so (F3) is the whole conjecture and not its non-abelian half. What they change is how the two statements may be read. Before them one could hope that (F2) and (F3) were structural facts awaiting proof, so that the conjecture would be a theorem about Weil classes and nothing else; [Proposition 20.44](#) shows that the Hodge classes on abelian varieties are not exhausted by divisor and Weil classes once the dimension reaches eight, and [Proposition 20.45](#) that the cohomology of a general variety cannot be moved into that of abelian varieties at all. Each of the two is therefore the Hodge conjecture for a class of varieties on which the methods of this paper, which are methods about Weil lines, have no purchase. The product  $X \times X$  is itself of Weil type for every imaginary quadratic  $K$ , through an embedding  $K \subset M_2(\mathbb{Q}) = \text{End}^0(X \times X)$  whose Rosati involution for the product polarisation is conjugation, with signature  $(4, 4)$ ; its Weil classes lie in  $H^8$ , and its Hodge group is far smaller than  $\text{SU}(4, 4)$ . The two classes in  $H^4$  are of a different kind, and they show that even on a variety of Weil type the Weil line need not be the whole of the exceptional cohomology once the Hodge group is small. They are the smallest instance of (F2), and [Section 21.1](#) identifies them and says what one K3 surface would have to supply for them to be algebraic.

*Remark 20.47* (Where this leaves the secant route). [Theorem 20.40\(iv\)](#) is a statement about the shape of the argument, not about its difficulty, and we say plainly what it does and does not mean.

It does not mean the secant route is worthless. It is the only route on which anything above dimension four has ever been proved, it is the route that settles  $n = 2$  at every discriminant and  $n = 3$  at trivial discriminant, and the reduction of [Theorem 18.29](#) turns its remaining demand into one identity in the Yoneda algebra of a sheaf on a fourfold, which is the form in which a demand can be attacked. What [Corollary 18.45](#), [Theorem 18.49](#) and [Theorem 18.52](#) do to that identity is narrow it to two shapes of support, neither resolved by sums of line bundles, and, in the smooth shape, to four discriminants.

What it does mean is that finishing the secant route would not finish the problem, and would not reduce it either. Through [Proposition 20.13](#) it would give every imaginary quadratic family, the trivial discriminant directly and the others by descent, which is a great deal; an earlier version of this remark said that it stops at the trivial discriminant, and that was an omission of the rule set. It would give nothing over the CM fields of degree at least four, whose split families need (P2) for themselves, and by [Proposition 20.13\(ii\)](#) that statement gives the imaginary quadratic fields as well. Markman's extension of the secant construction to those split families [[Mar25c](#)] is aimed at exactly that statement; its semiregularity step is open, and it is not among the rules of [Setup 20.39](#). So the statement that would have to be proved anyway is the propagation statement for those split families, and once proved it subsumes the secant route entirely.

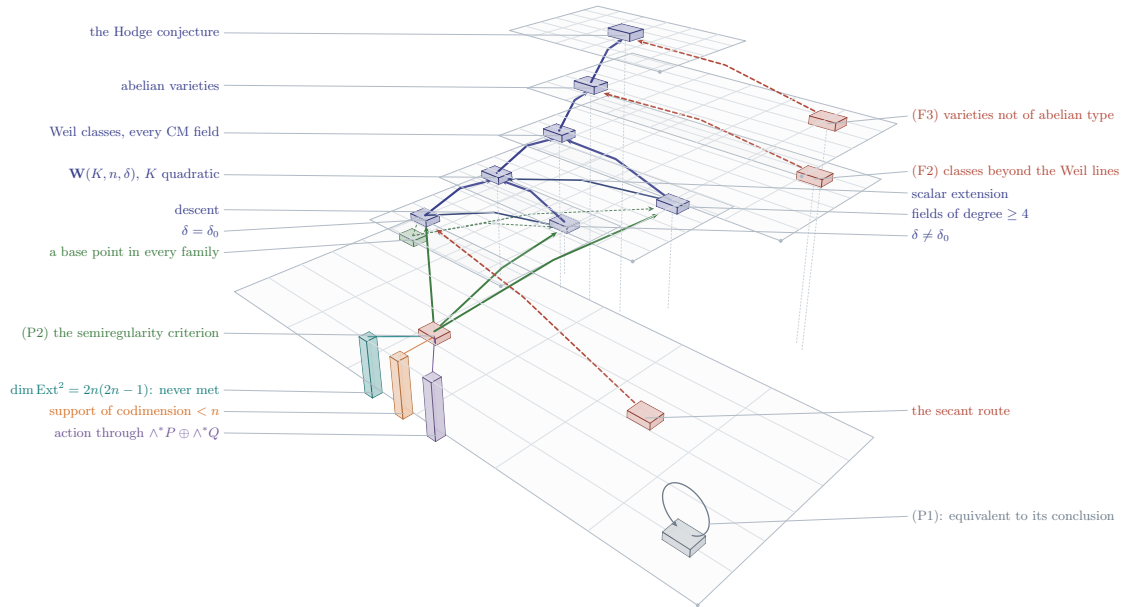


FIGURE 31. The route of this paper in [Theorem 20.40](#), drawn as a stack of plates, one per level of the derivation; the other routes of the literature are drawn in [Figure 30](#). The spine in indigo runs from the Weil families at the bottom to the conjecture at the top. The two statements (F2) and (F3) enter in clay from the right, each at the level where it is needed; (F2) is the conjecture for a class of Hodge classes that no Weil line reaches, not a reduction, and (F3) is the whole conjecture ([Proposition 20.35](#)), so the route drawn is minimal only with (F3) replaced by (F3') of [Definition 20.36](#). From below, the propagation statement (P2) in grass reaches the two discriminant cases of the quadratic fields and the fields of higher degree alike, because [Theorem 15.31](#) and [Theorem 19.4](#) put a base point in every family; it is read in the corrected form of [Remark 17.6](#). The two edges inside the levels are [Proposition 20.13](#): descent from the trivial discriminant to the others, and scalar extension from the fields of degree at least four to the quadratic fields; with them (P2) is needed only for the split families of the fields of higher degree. The three pillars under the grass node are the results of [Theorem 16.36](#), [Theorem 16.38](#) and [Corollary 16.34](#) on the first form, in which the Chern character is exactly a Weil class; by [Theorem 17.5](#) that form is never met, so the number on the first pillar is attained by no complex. The secant route in clay reaches the trivial discriminant, since the member it is built on is split, and through descent the rest of the quadratic case; it has no edge to the fields of higher degree. The bounded criterion (P1) is drawn apart, in slate, closed on itself: it is equivalent to its own conclusion.

It is tempting to read (C1) and (C2) as what remains for the imaginary quadratic case, and the computation shows why that reading is wrong: it puts the demand for a construction where a theorem about propagation belongs. (C1) and (C2) are demands whose satisfaction would be a considerable achievement, would settle the Weil classes of every imaginary quadratic field, and would close nothing that (P2) does not close.

## 20.10. Summary.

Table 6: The routes considered, and the boundary of each result.

| Route                                                                        | Status here                                                                                                                                    | Not covered                                                                                |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Ambient object, restricted to $A^\vee$ , transported by $\Phi_{\mathcal{P}}$ | closed by <a href="#">Theorem 1.1</a>                                                                                                          | ambient varieties with exceptional classes                                                 |
| Any $K$ -equivariant construction with the norm character                    | closed by <a href="#">Theorem 1.3</a>                                                                                                          | constructions that are not eigenclasses                                                    |
| Secant dimension count                                                       | no obstruction, <a href="#">Corollary 8.4</a>                                                                                                  | none                                                                                       |
| Semiregularity dimension count                                               | no obstruction, <a href="#">Corollary 9.3</a>                                                                                                  | none                                                                                       |
| Semiregularity of an object whose Chern character sits on the Weil line      | closed by <a href="#">Theorem 9.7</a> and <a href="#">Corollary 9.18</a> for direct sums, and for every object by <a href="#">Theorem 17.5</a> | objects of positive rank; the weaker hypothesis of <a href="#">[Mar25b, Question 11.4]</a> |
| Induction through hyperplane sections                                        | closed by <a href="#">Theorem 1.4</a> , <a href="#">Theorem 1.5</a>                                                                            | correspondences, degenerations, coniveau                                                   |
| Kuga–Satake, from a weight two structure of K3 type                          | not obstructed, <a href="#">Proposition 11.9</a>                                                                                               | available only at $n = 1$ , <a href="#">Corollary 11.12</a>                                |
| Divisors, on a member with quaternionic multiplication                       | succeeds, <a href="#">Theorem 15.25</a>                                                                                                        | fails at full Hodge group, <a href="#">Proposition 15.18</a>                               |
| Propagation off the base locus                                               | reduced to one bound or one rank, <a href="#">Theorem 16.9</a> , <a href="#">Theorem 16.28</a>                                                 | neither hypothesis proved, <a href="#">Remark 17.7</a>                                     |
| Objects with effective Chern character at a base point                       | closed by <a href="#">Theorem 18.5</a>                                                                                                         | objects indecomposable in $D^b$ , <a href="#">Remark 18.8</a>                              |

*continued on the next page*

Table 6, continued.

| Route                                                                                           | Status here                                                                                                                                                                                                               | Not covered                                                                                                   |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| The weakened criterion for a secant object of positive rank                                     | reduced to one identity on the factor, <a href="#">Theorem 18.29</a> ; the identity fails for Markman's candidate, <a href="#">Theorem 18.34</a> , and at every transversal double point, <a href="#">Corollary 18.36</a> | objects whose support has no such point, <a href="#">Remark 18.37</a>                                         |
| A secant object of rank one with smooth support                                                 | invariants forced and the discriminant bounded by 9, <a href="#">Theorem 18.40</a> , <a href="#">Corollary 18.41</a>                                                                                                      | existence for $d \in \{1, 3, 5, 7\}$ , and one vanishing in $H^1(S, N_{S/X})$ , <a href="#">Remark 18.42</a>  |
| The local part of the weakened criterion                                                        | it is a condition on depth: no obstruction at a Cohen–Macaulay point, <a href="#">Theorem 18.43</a> , <a href="#">Corollary 18.45</a>                                                                                     | Cohen–Macaulay supports, where the criterion becomes one global vanishing                                     |
| A support that is a complete intersection                                                       | closed for every $b$ and every $d$ , <a href="#">Theorem 18.49</a>                                                                                                                                                        | supports of higher Hilbert–Burch rank, <a href="#">Theorem 18.50</a>                                          |
| A support with a Hilbert–Burch resolution by line bundles or by simple semi-homogeneous bundles | closed for every rank, twist and $d$ , <a href="#">Theorem 18.52</a> , <a href="#">Theorem 18.55</a>                                                                                                                      | resolutions over bundles that are not projectively flat                                                       |
| The secant route, granted every demand it makes                                                 | reaches $\delta = \delta_0$ , and every $\delta$ by <a href="#">Proposition 20.13</a> , <a href="#">Theorem 20.40(iv)</a>                                                                                                 | the CM fields of degree at least four, which give the quadratic ones by <a href="#">Proposition 20.13(ii)</a> |
| Propagation from a base point                                                                   | closes every imaginary quadratic family at once, <a href="#">Theorem 16.9</a> , <a href="#">Theorem 16.28</a>                                                                                                             | (P2) not proved, <a href="#">Corollary 20.41</a>                                                              |
| The bounded form (P1), read on the transported cycle                                            | closed: the transport spreads each component by $c^{2n}/m_i$ and the data is unbounded whenever a component has finite stabiliser, <a href="#">Lemma 16.14</a> , <a href="#">Theorem 16.15</a>                            | nothing; the other reading is closed too                                                                      |

continued on the next page

Table 6, *continued*.

| Route                                                     | Status here                                                                                                                                                                                                                                                     | Not covered                                                                                                                        |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| The bounded form (P1), read on the minimal representative | closed: equivalent to the conjecture for the family, <a href="#">Theorem 16.17</a>                                                                                                                                                                              | nothing; it is a restatement, so it is not a route                                                                                 |
| The semiregularity form (P2), for a given representative  | forces the evaluation map to vanish on the $4n^2$ -dimensional annihilator, and excludes every representative that is a sheaf on a smooth germ of codimension two of its support, <a href="#">Theorem 16.30</a> , <a href="#">Corollary 16.31</a>               | representatives that are genuine complexes along every such component; members with an abelian subvariety tangent to an eigenspace |
| The same form, tested by a number                         | $\dim \operatorname{Ext}^2(E, E) = 2n(2n - 1)$ at one base point would suffice, <a href="#">Theorem 16.36</a> , and no representative attains it, <a href="#">Theorem 17.5</a>                                                                                  | nothing                                                                                                                            |
| The same form, for a representative in codimension $n$    | closed at every member of every family: the support must have a component of codimension less than $n$ , and at a general member every component of codimension at least two has multiplicity zero, <a href="#">Lemma 16.37</a> , <a href="#">Theorem 16.38</a> | representatives glued along components of lower codimension with alternating length zero                                           |
| The Hochschild action on such a representative            | the annihilator is the mixed part of a splitting of $HH^1$ and the action factors through a ring of dimension $2^{2n+1} - 1$ in one of dimension $2^{4n}$ , <a href="#">Theorem 16.33</a> , <a href="#">Corollary 16.34</a>                                     | whether a complex with that action exists                                                                                          |

*continued on the next page*

Table 6, continued.

| Route                                                       | Status here                                                                                                                                                                                                                                                                            | Not covered                                                                                                                                      |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| The shape of such a representative                          | an indecomposable summand carries the class; at $n = 2$ , and at $n = 3$ granting the compatibility of <a href="#">Corollary 16.34(iii)</a> , its endomorphism algebra has dimension at least three, so no simple object serves, <a href="#">Theorem 16.40</a>                         | nothing: no object of any shape meets the pure form, <a href="#">Theorem 17.5</a>                                                                |
| The pure form of (P2), Chern character exactly a Weil class | false for every imaginary quadratic field and every $n \geq 2$ : a semiregular such complex would deform to very general non-algebraic Weil tori, whose coherent sheaves have no Chern character below the top degree, <a href="#">Proposition 17.4</a> , <a href="#">Theorem 17.5</a> | the corrected form (P2), with powers of the polarisation allowed in the Chern character, <a href="#">Remark 17.6</a>                             |
| The reduction to Weil classes, and to abelian varieties     | false as a statement about Hodge rings in dimension eight, <a href="#">Proposition 20.44</a> ; impossible Hodge-theoretically, <a href="#">Proposition 20.45</a>                                                                                                                       | the algebraicity of the classes beyond the Weil lines, which is (F2), and (F3), which is the whole conjecture, <a href="#">Proposition 20.35</a> |
| The two classes on the self-product of a Mumford fourfold   | a real multiplication by a cubic field on the primitive part of $H^2$ , algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is, <a href="#">Proposition 21.2</a> , <a href="#">Theorem 21.4</a>                                                           | that Kuga–Satake class, <a href="#">Remark 21.5</a>                                                                                              |

continued on the next page

Table 6, *continued*.

| Route                                                           | Status here                                                                                                                                                                                                                                                                                        | Not covered                                                                                  |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Bypasses for the same two classes                               | degeneration and larger families closed off by rigidity; semiregularity reduced to $\dim \text{Ext}^2 = 119$ at one CM point; specialisation reduced to a degree bound, <a href="#">Theorem 21.21</a> , <a href="#">Theorem 21.27</a> , <a href="#">Corollary 21.28</a>                            | an object with $\dim \text{Ext}^2 = 119$ , or the degree bound, <a href="#">Remark 21.29</a> |
| Objects built from line bundles, for the same two classes       | closed at every point of the curve: the Chern characters stay invariant under the Lefschetz group, <a href="#">Theorem 21.33</a>                                                                                                                                                                   | steps that break that symmetry, <a href="#">Remark 21.34</a>                                 |
| Twistor deformation of the square, for the same two classes     | closed: the component of the Hodge locus among all complex tori is the Mumford curve, which contains no twistor line, <a href="#">Proposition 21.24</a>                                                                                                                                            | the Kuga–Satake class, which moves in a seven-dimensional family                             |
| The five routes left open, for the same two classes             | one statement: three of them are equivalent to the target and two imply it, and every minimal set of open statements yielding the target has one element, <a href="#">Corollary 21.40</a> , item (XLIX)                                                                                            | the target itself, which is only the first case of (F2), <a href="#">Remark 21.41</a>        |
| Pull-back from a hyperkähler manifold, for the same two classes | the square is holomorphic symplectic and the classes are the twisted dual forms of the part of $H^2$ its symplectic class generates; a symplectic rational map to some $S_\mu^{[n]}$ , $n \geq 5$ , suffices, and no K3 surface or hyperkähler eightfold serves, <a href="#">Proposition 21.25</a> | such a map, <a href="#">Remark 21.26</a>                                                     |

*continued on the next page*

Table 6, continued.

| Route                                           | Status here                                                                                                                                                                                                                                                                                                                                                                     | Not covered                                                                                                                                                                  |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reduction modulo $p$ , for the same two classes | represented by cycles at every closed point of every reduction; the target is equivalent to a bound on their Hilbert data on a Zariski dense set, <a href="#">Proposition 21.37</a> , <a href="#">Theorem 21.38</a>                                                                                                                                                             | the bound, <a href="#">Remark 21.39</a>                                                                                                                                      |
| The Weil classes of every CM field              | the same propagation problem, with a CM base point in every family, <a href="#">Theorem 19.4</a> , <a href="#">Theorem 19.6</a>                                                                                                                                                                                                                                                 | (P2) not proved for any field                                                                                                                                                |
| What is left, as a whole                        | five minimal sets of open statements: $\{F3\}$ and $\{L, M\}$ , each equivalent to the conjecture, and three through the conjecture modulo abelian varieties, $\{L, F3'\}$ , $\{V, F3'\}$ and this route, $\{P2_{\text{split}}, F2, F3'\}$ , <a href="#">Theorem 20.40</a> , <a href="#">Proposition 20.35</a> , <a href="#">Corollary 20.41</a> , <a href="#">Remark 20.43</a> | the Lefschetz standard conjecture, (M), (V), (F2), (F3'), all open                                                                                                           |
| The conjecture modulo abelian varieties, (F3')  | holds on the class $\mathcal{A}$ of varieties whose cohomology is reached from abelian varieties by algebraic correspondences (curves, abelian varieties, products, surjective images), in degrees 0, 2, $2n - 2$ , $2n$ , and in dimension at most three, <a href="#">Proposition 20.37</a>                                                                                    | degree four on fourfolds outside $\mathcal{A}$ and beyond, for instance squares of K3 surfaces without an algebraic Kuga–Satake correspondence, <a href="#">Remark 20.38</a> |

## 21. HODGE CLASSES BEYOND THE WEIL LINES

**21.1. The first classes beyond the Weil lines.** Statement (F2) is the Hodge conjecture for the Hodge classes on abelian varieties that no Weil line reaches, and [Proposition 20.44](#) exhibits the first of them: two classes in  $H^4(X \times X, \mathbb{Q})$  for an abelian fourfold  $X$  of Mumford type. None of the constructions of the earlier sections acts on them. In this subsection we identify them as a real multiplication on the primitive part of  $H^2(X)$ , and we prove that they are algebraic as soon as the Kuga–Satake class of one K3 surface of

Picard number thirteen attached to  $X$  is algebraic. We do not prove that the Kuga–Satake class is algebraic, and we know of no proof. What changes is the kind of question: two classes on an eightfold are traded for one correspondence of a well-studied type, between a surface and its Kuga–Satake variety. The field enters through the Clifford relations, which force the Kuga–Satake map to see all of it.

**Setup 21.1** (Mumford type). Let  $X$  be as in [Proposition 20.44](#), with  $V = H^1(X, \mathbb{Q})$ , polarisation  $\psi$  and polarisation class  $\theta \in H^2(X, \mathbb{Q})$ , and let  $G = \text{Hg}(X)$ . Over  $\bar{\mathbb{Q}}$  the Hodge cocharacter is nontrivial on exactly one of the three factors of  $G$ : on  $V_i$  it has weights  $\pm a_i/2$  with  $a_i \geq 0$ , and  $V$  has only the two weights  $\pm \frac{1}{2}$ , so exactly one  $a_i$  is nonzero. Because  $\text{End}^0(X) = \mathbb{Q}$  the group  $G$  is therefore  $\mathbb{Q}$ -simple: the Lie algebra of a  $\mathbb{Q}$ -factor not containing that geometric factor would consist of Hodge classes in  $\text{End}(V)$ , that is of endomorphisms of  $X$ . So  $G$  is the image in  $\text{GL}(V)$  of  $\text{Res}_{F/\mathbb{Q}} D^1$ , where  $F$  is a totally real cubic field,  $D$  is a quaternion algebra over  $F$  split at one real place  $\sigma_3$  and definite at the other two,  $\sigma_1$  and  $\sigma_2$ , with  $\text{Cor}_{F/\mathbb{Q}} D \cong M_8(\mathbb{Q})$ , and  $D^1$  is the group of elements of reduced norm one [[Mum69](#), [Gal00](#)]; the kernel is the central subgroup of triples of signs with product one. Over  $\bar{\mathbb{Q}}$ ,  $V = V_1 \otimes V_2 \otimes V_3$ , where  $V_i$  is the standard representation of  $D \otimes_{F, \sigma_i} \mathbb{Q} = M_2(\bar{\mathbb{Q}})$ ; the Galois group permutes the three factors as it permutes the three embeddings, and  $H^{1,0}(X) = V_1 \otimes V_2 \otimes V_3^{1,0}$ . Put

$$T = \text{Lie } G = D^0, \quad q_\lambda(x, y) = \text{Tr}_{F/\mathbb{Q}}(\lambda \text{trd}(xy)) \quad (\lambda \in F^\times),$$

where  $D^0$  is the space of quaternions of reduced trace zero, regarded as a  $\mathbb{Q}$ -vector space of dimension nine, and  $\text{trd}$  is the reduced trace. Over  $\bar{\mathbb{Q}}$ ,  $T = T_1 \oplus T_2 \oplus T_3$  with  $T_i = \mathfrak{sl}(V_i)$ , and  $a \in F$  acts on  $T_i$  by  $\sigma_i(a)$ . Write  $m_a$  for multiplication by  $a$ . The space  $T \subset \text{End}(V)$  is a sub-Hodge structure of weight zero.

**Proposition 21.2** (The two classes are a real multiplication). *In [Setup 21.1](#):*

- (i)  $H^2(X, \mathbb{Q}) = \mathbb{Q}\theta \oplus U$ , where  $U$  is the primitive part, of dimension 27, and over  $\bar{\mathbb{Q}}$

$$U = U_{12} \oplus U_{13} \oplus U_{23}, \quad U_{ij} = S^2 V_i \otimes S^2 V_j \otimes \wedge^2 V_k, \quad \{i, j, k\} = \{1, 2, 3\},$$

each of dimension nine. The summand  $U_{12}$  is of type  $(1, 1)$ , and  $U_{13}$  and  $U_{23}$  have Hodge numbers  $(3, 3, 3)$ .

- (ii) The algebra of Hodge endomorphisms of  $H^2(X, \mathbb{Q})$  is  $\mathbb{Q} \times F$ , acting by scalars on  $\mathbb{Q}\theta$ , with  $a \in F$  acting on  $U_{ij}$  by  $\sigma_k(a)$ . Through Poincaré duality and the Lefschetz isomorphism  $\theta^2: H^2 \rightarrow H^6$  this algebra is the space of Hodge classes in the Künneth component  $H^2(X) \otimes H^2(X)$  of  $H^4(X \times X, \mathbb{Q})$ ; the subalgebra  $\mathbb{Q} \times \mathbb{Q}$  is spanned by the products of divisor classes that lie there, and the two classes of [Proposition 20.44\(ii\)](#) may be taken to be the elements of  $F$  of trace zero.
- (iii)  $T(-1)$  is a Hodge structure of K3 type with Hodge numbers  $(1, 7, 1)$ , and its Hodge endomorphisms are the  $m_a$ ,  $a \in F$ . The  $G$ -invariant symmetric forms on  $T$  are the  $q_\lambda$ , and  $q_\lambda$  polarises  $T(-1)$  if and only if  $\lambda$  is totally positive.

*Proof.* (i) is the decomposition of  $\wedge^2(V_1 \otimes V_2 \otimes V_3)$  in the proof of [Proposition 20.44](#), where  $U_{ij}$  appears as the summand of highest weight 2 on the  $i$ -th and  $j$ -th factors. The Hodge cocharacter acts trivially on  $V_1$  and  $V_2$  and splits  $V_3 = V_3^{1,0} \oplus V_3^{0,1}$ , so  $\wedge^2 V_3$  has type  $(1, 1)$  and  $S^2 V_3$  has one dimension of each of the types  $(2, 0)$ ,  $(1, 1)$ ,  $(0, 2)$ .

(ii) A Hodge endomorphism commutes with the Mumford–Tate group  $G \cdot \mathbb{G}_m$ , and  $\mathbb{G}_m$  acts on  $H^2$  by scalars, so the Hodge endomorphisms are the  $G$ -endomorphisms. Over  $\bar{\mathbb{Q}}$  the four summands  $\mathbb{Q}\theta$ ,  $U_{12}$ ,  $U_{13}$ ,  $U_{23}$  are irreducible and pairwise non-isomorphic, so the  $G$ -endomorphisms are the combinations of the four projections. The Galois group

fixes the projection onto  $\mathbb{Q}\theta$  and permutes the other three as it permutes the indices  $k$ , that is as it permutes the embeddings of  $F$ . An endomorphism acting on  $U_{ij}$  by  $\sigma_k(a)$  is therefore Galois invariant, hence rational, and these exhaust the rational points, which are  $\mathbb{Q} \times F$ . Poincaré duality identifies  $\text{End } H^2(X, \mathbb{Q})$  with  $H^6(X) \otimes H^2(X)$ , and  $\theta^2 \otimes 1$  carries  $H^2 \otimes H^2$  onto it; both are given by algebraic classes, and on an abelian variety so is the inverse of the Lefschetz isomorphism [Kle68, §2]. The products of divisor classes in  $H^2 \otimes H^2$  are  $\psi_{11}\psi_{22}$  and  $\psi_{12}^2$  in the notation of the proof of Proposition 20.44. They are invariant under the whole of  $\text{Sp}(V, \psi)$ , whose endomorphisms of  $H^2 = \mathbb{Q}\theta \oplus U$  are spanned by the identity and the projection onto  $\mathbb{Q}\theta$ , and they are linearly independent by item (XXXV); so they span  $\mathbb{Q} \times \mathbb{Q}$ , and any complement of  $\mathbb{Q} \times \mathbb{Q}$  in  $\mathbb{Q} \times F$  may serve as the two remaining classes.

(iii) The adjoint action of the Hodge cocharacter is trivial on  $T_1$  and  $T_2$  and has weights  $-1, 0, 1$  on  $T_3$ , so  $T$  has Hodge numbers  $(1, 7, 1)$  in the types  $(-1, 1)$ ,  $(0, 0)$ ,  $(1, -1)$ , and  $T(-1)$  is of K3 type. Its Hodge endomorphisms are again the  $G$ -endomorphisms, and the argument of (ii), applied to the three pairwise non-isomorphic summands  $T_i$ , gives  $F$  acting by multiplication. On each  $T_i$  the invariant symmetric forms are the multiples of  $\text{tr}(xy)$ , and  $\text{tr}(x_i y_i) = \sigma_i(\text{trd}(xy))$  for  $x, y \in D^0$ , so the Galois-invariant ones are the  $q_\lambda$ . At  $\sigma_1$  and  $\sigma_2$  the algebra  $D \otimes \mathbb{R}$  is Hamilton's quaternions,  $T_i$  is of type  $(1, 1)$ , and  $\text{trd}(x^2) = -2\text{nrd}(x)$  is negative on nonzero pure quaternions. At  $\sigma_3$  the form  $\text{tr}(xy)$  on  $\mathfrak{sl}_2(\mathbb{R})$  is negative on the line of type  $(1, 1)$ , the Lie algebra of the rotations fixed by the Hodge cocharacter, and positive on its orthogonal complement, the real points of the types  $(2, 0)$  and  $(0, 2)$ . The Hodge–Riemann relations for a structure of K3 type ask for exactly these signs after multiplication by  $\sigma_i(\lambda)$ , so  $q_\lambda$  polarises  $T(-1)$  precisely when every  $\sigma_i(\lambda)$  is positive. Item (XLI) of Appendix D computes the four projections from the Casimir operators of the three factors on an explicit model and checks (i) and the Hodge numbers. ■

The two classes are thus the endomorphisms of  $U$  given by the elements of  $F$  of trace zero. A Hodge structure of K3 type with the same field sits beside them, and the next lemma connects the two.

**Lemma 21.3** (The product map). *For  $x, y \in T$  let  $\mu(x \otimes y) \in H^2(X, \mathbb{Q})$  be the class that corresponds under  $\psi$  to the alternating form  $(v, w) \mapsto \psi(xv, yw) + \psi(yv, xw)$  on  $V$ . Then  $\mu$  is a  $G$ -equivariant map  $T \otimes T \rightarrow H^2(X, \mathbb{Q})$ , and:*

- (i) *for  $i \neq j$  it restricts to an isomorphism  $T_i \otimes T_j \rightarrow U_{ij}$ , and it carries  $T_i \otimes T_i$  onto  $\mathbb{Q}\theta$ ;*
- (ii) *for  $e, \lambda \in F^\times$  put  $\mu_e = \mu \circ (m_e \otimes m_e)$  and let  $\mu_e^\dagger$  be its adjoint for the form  $q_\lambda \otimes q_\lambda$  on  $T \otimes T$  and the Lefschetz pairing  $B(\alpha, \beta) = \int_X \alpha \beta \theta^2$  on  $H^2(X, \mathbb{Q})$ . Then  $\mu_e \mu_e^\dagger$  preserves  $\mathbb{Q}\theta$  and acts on  $U$  as the element*

$$\nu N_{F/\mathbb{Q}}(g) g^{-1} \in F, \quad g = e^2/\lambda,$$

*where  $\nu$  is a nonzero rational number independent of  $e$  and  $\lambda$ .*

*Proof.* Equivariance is clear, since  $\mu$  is built from  $\psi$  and the action of  $\text{End}(V)$  on  $V$ . An element  $x \in T_i$  acts on  $V$  through the  $i$ -th factor and lies in the symplectic Lie algebra of  $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$ , so  $\psi(xv, yw) = -\psi(v, xyw)$  and  $\mu(x \otimes y)$  is the form  $-\psi(v, (xy + yx)w)$ . If  $x \in T_i$  and  $y \in T_j$  with  $i \neq j$ , then  $xy = yx$  acts as  $x \otimes y$  on  $V_i \otimes V_j$ , and the form is  $-2$  times the product of  $\epsilon_i(\cdot, x \cdot)$ ,  $\epsilon_j(\cdot, y \cdot)$  and  $\epsilon_k$ ; the first two are symmetric because  $x$  and  $y$  lie in  $\mathfrak{sp}(\epsilon) = \mathfrak{sl}_2$ , and  $x \mapsto \epsilon(\cdot, x \cdot)$  is an isomorphism  $\mathfrak{sl}_2 \rightarrow S^2 V^\vee$ , so  $\mu$  maps

$T_i \otimes T_j$  isomorphically onto  $U_{ij}$ . If  $i = j$ , then  $xy + yx = \text{tr}(xy)$  on  $V_i$ , because  $x$  and  $y$  are trace-free  $2 \times 2$  matrices, and  $\mu(x \otimes y) = -\text{tr}(xy)\theta$ .

For (ii),  $m_e$  is self-adjoint for  $q_\lambda$ , and the adjoint of a map out of  $(T \otimes T, q_\lambda \otimes q_\lambda)$  is  $(m_\lambda \otimes m_\lambda)^{-1}$  times its adjoint for  $q_1 \otimes q_1$ , so  $\mu_e \mu_e^\dagger = \mu(m_g \otimes m_g) \mu^{\dagger 1}$  with  $\mu^{\dagger 1}$  the adjoint for  $q_1 \otimes q_1$ . This map is  $G$ -equivariant, so it preserves  $\mathbb{Q}\theta$  and each  $U_{ij}$ , and  $\mu^{\dagger 1}$  carries  $U_{ij}$  into  $(T_i \otimes T_j) \oplus (T_j \otimes T_i)$ , on which  $m_g \otimes m_g$  is multiplication by  $\sigma_i(g)\sigma_j(g) = N(g)/\sigma_k(g)$ . Hence  $\mu_e \mu_e^\dagger$  is  $N(g)\sigma_k(g)^{-1}$  times  $\mu \mu^{\dagger 1}$  on  $U_{ij}$ . Item (XLI) computes  $\mu \mu^{\dagger 1}$  on the split model and finds one and the same nonzero rational number  $\nu$  on the three summands  $U_{ij}$ ; it also checks the formula for  $\mu_e \mu_e^\dagger$  directly for several values of the conjugates of  $e$  and  $\lambda$ . By Proposition 21.2(ii) an endomorphism of  $U$  acting on  $U_{ij}$  by  $\nu N(g)/\sigma_k(g)$  is the element  $\nu N(g)g^{-1}$  of  $F$ . ■

By the Kuga–Satake class of a projective K3 surface  $S$  we mean the Hodge class on  $S \times \text{KS}(S) \times \text{KS}(S)$  that induces the embedding of Proposition 11.7 for  $T(S)_\mathbb{Q}$ ; its algebraicity is the Kuga–Satake Hodge conjecture for  $S$  of Remark 11.8.

**Theorem 21.4** (One K3 surface for the two classes). *In Setup 21.1, let  $\lambda \in F$  be totally positive.*

- (i) *There is a projective K3 surface  $S_\lambda$  of Picard number thirteen and an isomorphism of Hodge structures  $T(S_\lambda)_\mathbb{Q} \cong T(-1)$  carrying the intersection form to  $q_\lambda$ . Its Kuga–Satake variety is isogenous to  $X^{32}$ .*
- (ii) *If the Kuga–Satake class of  $S_\lambda$  is algebraic, then the real multiplication by  $F$  on  $T(S_\lambda)$  is induced by algebraic cycles on  $S_\lambda \times S_\lambda$ , and every Hodge class in  $H^4(X \times X, \mathbb{Q})$  is algebraic; in particular the two classes of Proposition 20.44(ii) are algebraic.*

*Proof.* (i) The quadratic space  $(T, q_\lambda)$  has dimension nine and signature  $(2, 7)$  by Proposition 21.2(iii), and  $H^2$  of a K3 surface with its intersection form has dimension 22 and signature  $(3, 19)$ . Over  $\mathbb{Q}_p$  a nondegenerate quadratic space embeds isometrically in every nondegenerate space of dimension at least three larger, and over  $\mathbb{R}$  the signatures allow it, so by the Hasse–Minkowski theorem for representations of forms by forms [OMe00, §§63, 66] there is an isometric embedding of  $(T, q_\lambda)$  into  $\Lambda_\mathbb{Q}$ , where  $\Lambda$  is the K3 lattice. The line  $\ell = T(-1)^{2,0} \subset T_\mathbb{C} \subset \Lambda_\mathbb{C}$  satisfies  $(\ell, \ell) = 0$  and  $(\sigma, \bar{\sigma}) > 0$  for  $0 \neq \sigma \in \ell$ , so by the surjectivity of the period map there is a K3 surface  $S_\lambda$  with a marking  $H^2(S_\lambda, \mathbb{Z}) \cong \Lambda$  that carries  $H^{2,0}(S_\lambda)$  to  $\ell$  [Huy16, Chs. 6–7]. A rational class is of type  $(1, 1)$  exactly when it is orthogonal to  $\ell$ , and its component in  $T$  is then a rational class of type  $(1, 1)$  in  $T(-1)$ , that is a  $G$ -invariant vector of  $T = \text{Lie } G$ ; there is none, since  $G$  is semisimple. So  $\text{NS}(S_\lambda)_\mathbb{Q} = T^\perp$ , of dimension 13 and signature  $(1, 12)$ , and  $T(S_\lambda)_\mathbb{Q} = T$  with the form  $q_\lambda$ . A class of positive square in  $\text{NS}(S_\lambda)$  makes  $S_\lambda$  projective [Huy16, Ch. 8].

For the Kuga–Satake variety, note first that over  $\bar{\mathbb{Q}}$  the space  $T$  is the orthogonal sum of the three ternary spaces  $T_i$ , so  $C(T)$  is the graded tensor product of the  $C(T_i)$ . Scaling a ternary form does not change its even Clifford algebra, so  $D^1$  is the spin group of  $(D^0, \lambda \text{trd})$  for every  $\lambda$ , and  $\text{Res}_{F/\mathbb{Q}} D^1$  maps to  $\text{Spin}(T, q_\lambda)$  by multiplication in the Clifford algebra, with the same kernel as its map to  $\text{GL}(V)$ , so that  $G$  is a subgroup of  $\text{Spin}(T, q_\lambda)$ ; as a representation of the  $i$ -th factor,  $C(T_i)$  is four copies of  $V_i$ . Hence  $C^+(T)$  is  $V^{\oplus 32}$  as a representation of  $G$  over  $\bar{\mathbb{Q}}$ , and over  $\mathbb{Q}$  as well, because  $V$  is absolutely irreducible. The Hodge structures on  $C^+(T) = H^1(\text{KS}(S_\lambda), \mathbb{Q})$  and on  $V^{\oplus 32} = H^1(X^{32}, \mathbb{Q})$  are both given by lifts of weight one of the same homomorphism  $\mathbb{S} \rightarrow \text{MT}(T)_\mathbb{R}$  through  $G \cdot \mathbb{G}_m$ , and two

such lifts coincide.<sup>62</sup> So  $H^1(\mathrm{KS}(S_\lambda), \mathbb{Q}) \cong H^1(X^{32}, \mathbb{Q})$  as Hodge structures, and  $\mathrm{KS}(S_\lambda)$  is isogenous to  $X^{32}$ . For the form Galluzzi uses this is her theorem [Gal00], and van Geemen proves the corresponding splitting for every K3 surface with real multiplication [vG08].

(ii) Write  $S = S_\lambda$  and  $A = \mathrm{KS}(S)$ , and let  $Z_S$  be an algebraic cycle on  $S \times A \times A$  inducing the Kuga–Satake embedding  $T(S) \rightarrow \mathrm{End} H^1(A)$ . Composing  $Z_S$  with the graph of an isogeny  $A \rightarrow X^{32}$  and with the graphs of the inclusions and projections of the factors of  $X^{32}$  gives algebraic cycles on  $S \times X \times X$  inducing the  $32^2$  block components of the embedding, which are morphisms of Hodge structures  $T(S) \rightarrow \mathrm{End}(V)$ , up to a Tate twist; both lie in the tensor category generated by  $H^1(X)$ , so the blocks are equivariant for  $\mathrm{MT}(X) = G \cdot \mathbb{G}_m$ . Over  $\bar{\mathbb{Q}}$ ,  $\mathrm{End}(V) = S^2 V \oplus \wedge^2 V$ , and  $T_i \cong S^2 V_i$  occurs once in  $S^2 V$ , as  $S^2 V_i \otimes \wedge^2 V_j \otimes \wedge^2 V_k$ , and not at all in  $\wedge^2 V$ . So each block takes values in  $T$ , and through the identification of (i) it is a  $G$ -endomorphism of  $T$ , that is  $m_{e_{ab}}$  for some  $e_{ab} \in F$  by Proposition 21.2(iii). Over  $\bar{\mathbb{Q}}$  the embedding is therefore  $v \mapsto v \otimes N_i$  on  $T_i$ , acting on  $H^1(A) = V \otimes \mathbb{Q}^{32}$ , with  $N_i = (\sigma_i(e_{ab}))_{a,b}$ . The embedding  $\Psi$  of Proposition 11.7 satisfies the Clifford relation  $\Psi(v)\Psi(w) + \Psi(w)\Psi(v) = 2q_\lambda(v_0, v_0)q_\lambda(v, w)$ , because  $\Psi(v)\Psi(w)x = vwxv_0^2$ . Take nonzero  $v \in T_i$  and  $w \in T_j$  with  $i \neq j$ . As endomorphisms of  $V$  they commute and  $vw \neq 0$ , so the left side is  $vw \otimes (N_i N_j + N_j N_i)$ , while the right side is zero because  $T_i$  and  $T_j$  are orthogonal; hence  $N_i$  and  $N_j$  anticommute. For  $v \in T_i$  with  $q_\lambda(v, v) \neq 0$  the relation reads  $v^2 \otimes N_i^2 = q_\lambda(v_0, v_0)q_\lambda(v, v)$ , and  $v^2$  is a nonzero scalar on  $V_i$ , so  $N_i^2$  is a nonzero scalar. Three pairwise anticommuting invertible matrices are linearly independent: multiplying a relation  $\sum_i c_i N_i = 0$  by  $N_j$  on either side and adding gives  $2c_j N_j^2 = 0$ . If the  $e_{ab}$  spanned a subspace of  $F$  of dimension at most two, the  $N_i$  would lie in the span of at most two rational matrices; so the  $e_{ab}$  span  $F$ , and two of them,  $e$  and  $e'$ , are not rational multiples of each other. Then  $(e/e')^2$  is irrational, since a cubic field contains no irrational element with rational square, so  $e^2/\lambda$  and  $e'^2/\lambda$  are not both rational. Fix a block  $\kappa = m_e$  with  $e^2/\lambda \notin \mathbb{Q}$ , induced by a cycle  $\Gamma$ . Item (XLI) builds the Clifford algebra of the split model and checks that  $\Psi$  has this shape, with three anticommuting  $N_i$ .

The real multiplication first. The trace form  $(a, b) \mapsto \mathrm{tr}(ab)$  on  $\mathrm{End}(V)$  is induced by an algebraic class: for correspondences  $a, b$  on  $X$  the Lefschetz trace formula gives  $\mathrm{tr}(a_* b_* | H^1) = -\deg((a \circ b \circ \pi_1) \cdot \Delta_X)$ , where  $\pi_1$  is the Künneth projector onto  $H^1$ , algebraic on an abelian variety [Kle68, §2]. So the adjoint  $\kappa^\dagger$  of  $\kappa$ , for this form and the intersection form on  $S$ , is induced by a cycle built from the transpose of  $\Gamma$ ,  $\pi_1$  and  $\Delta_X$ , and  $\kappa^\dagger \kappa$  is an algebraic endomorphism of  $T(S)$ . An element  $x$  of  $T_i$  acts on the eight-dimensional space  $V$  through a two-dimensional factor, with multiplicity four, so  $\mathrm{tr}_V(xy) = 4 \mathrm{Tr}_{F/\mathbb{Q}} \mathrm{trd}(xy)$ , and

$$q_\lambda(\kappa^\dagger \kappa x, y) = \mathrm{tr}_V(ex \cdot ey) = 4 \mathrm{Tr}_{F/\mathbb{Q}}(e^2 \mathrm{trd}(xy)) = q_\lambda(4e^2 \lambda^{-1} x, y),$$

that is  $\kappa^\dagger \kappa = m_{4e^2/\lambda}$ . By the choice of  $e$  this element is irrational, so it generates the cubic field  $F$  and every  $m_a$  is a polynomial in  $\kappa^\dagger \kappa$ . Composing with the projector of  $H^2(S)$  onto  $T(S)$ , which is algebraic because it is the diagonal minus  $[\mathrm{pt} \times S]$ ,  $[S \times \mathrm{pt}]$  and  $\sum_r d_r \times d_r^\vee$  over dual bases of  $\mathrm{NS}(S)_\mathbb{Q}$ , gives algebraic cycles on  $S \times S$  inducing the real multiplication.

The classes on  $X \times X$  next. The map  $\mu_\kappa = \mu \circ (\kappa \otimes \kappa)$  from  $T(S) \otimes T(S)$  to  $H^2(X, \mathbb{Q})$  is induced by an algebraic cycle on  $S \times S \times X$ , obtained from  $\Gamma \times \Gamma$ , the divisor  $m^* \theta - p_1^* \theta - p_2^* \theta$  on  $X \times X$ , whose class is  $\psi$  in  $H^1 \otimes H^1$ , and the diagonal of  $X$  by pull-back, push-forward and intersection, because  $\mu$  uses nothing but  $\psi$  and the composition of endomorphisms

<sup>62</sup>They differ by a homomorphism from the anisotropic torus  $\mathbb{S}/\mathbb{G}_m$  to the centre of  $G \cdot \mathbb{G}_m$ , whose identity component is  $\mathbb{G}_m$ , and such a homomorphism is trivial.

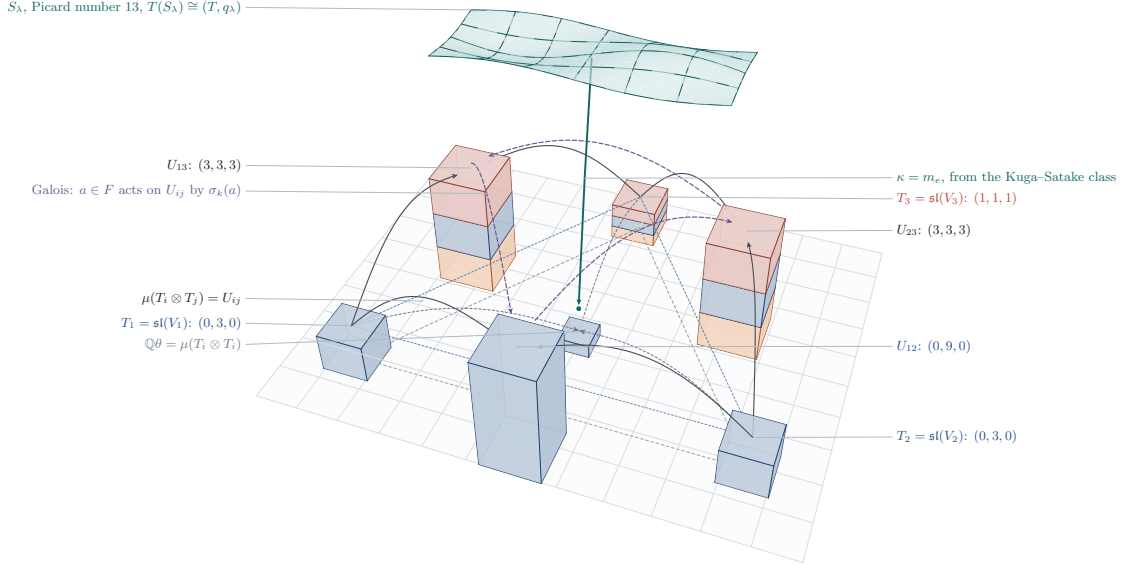


FIGURE 32. [Proposition 21.2](#) and [Lemma 21.3](#), drawn over the triangle of the three factors of the Hodge group; every height is a dimension. At the corners stand the summands  $T_i = \mathfrak{sl}(V_i)$  of  $T = \text{Lie } G$ , banded by Hodge type after the Tate twist  $T(-1)$ : blue for  $(1, 1)$ , clay for  $(2, 0)$  and amber for  $(0, 2)$ , so that only  $T_3$  carries the K3 type. On the edges stand the three summands  $U_{ij}$  of the primitive part of  $H^2(X)$ , of dimension nine,  $U_{12}$  of pure type  $(1, 1)$  and the other two in three bands of three; in the middle is the line of  $\theta$ . The solid arcs are the product map  $\mu$ , which carries  $T_i \otimes T_j$  onto  $U_{ij}$ , and the dashed ones carry  $T_i \otimes T_i$  to  $\mathbb{Q}\theta$ . The violet arcs are the Galois group permuting the  $U_{ij}$  as it permutes the embeddings of  $F$ , which is why the two exceptional classes may be taken to be the trace-zero part of a cubic field. Above, the K3 surface  $S_\lambda$  of [Theorem 21.4](#), whose Kuga–Satake class, if algebraic, comes down to  $T$  as the map  $\kappa = m_e$ .

of  $V$ . The form  $q_\lambda \otimes q_\lambda$  is the intersection form of  $S \times S$  on  $T(S) \otimes T(S)$  and  $B$  is the intersection form of  $X$  twisted by  $\theta^2$ , so the adjoint  $\mu_\kappa^\dagger$  is induced by cup product with  $\theta^2$  followed by the transposed cycle, and  $E = \mu_\kappa \mu_\kappa^\dagger$  is an algebraic endomorphism of  $H^2(X, \mathbb{Q})$ . Since  $\kappa = m_e$ , [Lemma 21.3\(ii\)](#) says that  $E$  preserves  $\mathbb{Q}\theta$  and acts on  $U$  as  $\nu N(g)g^{-1}$  with  $g = e^2/\lambda$ , and this element of  $F$  is not rational, since  $g$  is not. The projection  $\pi_\theta$  onto  $\mathbb{Q}\theta$  is algebraic, being given by the class  $\theta \times \theta$  suitably normalised, so the algebraic Hodge endomorphisms of  $H^2(X, \mathbb{Q})$  contain 1,  $\pi_\theta$  and  $E(1 - \pi_\theta)$ , which generate  $\mathbb{Q} \times F$ , that is all of them by [Proposition 21.2\(ii\)](#). By the same part every Hodge class in  $H^2(X) \otimes H^2(X)$  is then algebraic. The Hodge classes of  $H^4(X \times X, \mathbb{Q})$  in the other Künneth components number one in each of  $H^a \otimes H^{4-a}$ ,  $a \neq 2$ , by the count in the proof of [Proposition 20.44](#), and are products of divisor classes. ■

*Remark 21.5* (What is known about the Kuga–Satake class). The Kuga–Satake class of  $S_\lambda$  is not known to be algebraic, and [Theorem 21.4](#) does not make it so. The known cases of the Kuga–Satake Hodge conjecture lie elsewhere. They include the K3 surfaces of Picard number at least seventeen, which Morrison relates to abelian surfaces [[Mor84](#)], and

the double planes branched along six lines, of Picard number at least sixteen, settled by Paranjape’s cycle; see [vG00]. Floccari proves the conjecture for the K3 surfaces attached to projective sixfolds of generalised Kummer type [Flo24], whose transcendental lattices embed, after scaling, in the rational second cohomology of such a sixfold, of rank seven and signature  $(3, 4)$ ; their transcendental rank is therefore at most six, and the surfaces  $S_\lambda$  have transcendental rank nine. Schlickewei made the real multiplication algebraic on those double planes that have it, necessarily by a field of degree at most two, from Paranjape’s cycle and van Geemen’s description of their Kuga–Satake varieties [Sch10, vG08], and Varesco showed that when the Kuga–Satake Hodge conjecture holds for two hyperkähler varieties, every Hodge similarity between their transcendental lattices is algebraic up to the Lefschetz isomorphism [Var23], which reaches a quadratic field through its square root. A cubic field contains no irrational element with rational square, so none of its elements acts as a similarity; the element  $\kappa^\dagger \kappa = m_{4e^2/\lambda}$  in the proof takes the place that the similarities occupy there, and the Clifford relations are what make it irrational. Buskin’s theorem that rational Hodge isometries between K3 surfaces are algebraic [Bus19] does not give the real multiplication either. A Hodge isometry of  $T(S_\lambda)$  is a Hodge endomorphism, hence some  $m_a$  by Proposition 21.2(iii), and it is an isometry only when  $a^2 = 1$ . For  $b \in F^\times$  the theorem does give algebraic cycles between  $S_\lambda$  and  $S_{\lambda b^{-2}}$  that realise  $m_b$ , but every composite of them that returns to  $S_\lambda$  is again a Hodge isometry of  $T(S_\lambda)$ , so it multiplies by an element of square one.

The surfaces  $S_\lambda$  include those of Zhu, who computes the integral structure that the fourfold induces on  $T$  and studies the K3 surfaces of Picard number thirteen that carry it [Zhu18]; its form is the corestriction of the reduced norm of  $D^0$ , which up to sign is  $\frac{1}{2}q_1$ . Theorem 21.4 applies to them as it stands, because the irrationality of  $e^2/\lambda$  comes from the anticommuting blocks and not from  $\lambda$ . The closest explicit families are those of van Geemen and Schütt, with real multiplication by the cyclic cubic fields  $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$  and  $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$  and Picard numbers four and ten, on which the real multiplication is induced by explicit algebraic cycles [vGS25, Theorem 1.2 and Section 4.8]; their Picard numbers are not thirteen, and the Kuga–Satake class, which Theorem 21.4 needs and algebraic real multiplication does not supply, is not treated there. The first case of (F2) is therefore implied by one instance of the Kuga–Satake Hodge conjecture, for one K3 surface of Picard number thirteen; by Proposition 21.12 below that instance is equivalent to the first case of (F2) at  $X$  together with a link, and it is not a reduction of it.

Theorem 21.4 trades the two exceptional classes for one Kuga–Satake class. The next results say what the trade buys and what it costs. If the exceptional classes are algebraic, so is every Hodge class on every power of  $X$  (Theorem 21.7); if the Kuga–Satake class is algebraic, so is every Hodge class on every product  $S_\lambda^a \times X^b$  (Corollary 21.11). The Kuga–Satake class is algebraic exactly when the exceptional classes are and, in addition, some algebraic cycle links  $S_\lambda$  to a power of  $X$  (Proposition 21.12), and no such link can pass through an abelian variety of dimension at most five (Proposition 21.13). None of this is a new case of the Hodge conjecture: the theorem and the corollary are implications whose hypotheses are open, and the first case of (F2) remains open (Remark 21.17). The computations are those of item (LV) of Appendix D, made by the scripts in `code/attack/mumford_target`.

**Proposition 21.6** (Where  $T$  occurs). *In Setup 21.1, write  $\mathrm{Hom}_{\mathrm{Hdg}}$  for morphisms of rational Hodge structures.*

- (i)  *$T$  is isomorphic to no subquotient of  $H^k(X, \mathbb{Q})(j)$ , for any  $k$  and  $j$ .*

- (ii) For  $n = 2, 4, \dots, 14$  the  $F$ -vector space  $\mathrm{Hom}_{\mathrm{Hdg}}(T, H^n(X \times X, \mathbb{Q})(n/2))$  has dimension 1, 6, 16, 24, 16, 6, 1, and for  $n \notin \{2, 4, \dots, 14\}$  the Hodge structure  $T$  is isomorphic to no subquotient of any  $H^n(X \times X, \mathbb{Q})(j)$ . The copy of  $T$  in  $H^2(X \times X, \mathbb{Q})(1)$  lies in  $H^1(X) \otimes H^1(X)$ .
- (iii)  $\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{End} V)$  is a one-dimensional  $F$ -vector space, spanned by the inclusion. For  $\lambda \in F$  totally positive, after an isogeny  $\mathrm{KS}(S_\lambda) \rightarrow X^{32}$  the Kuga–Satake embedding of  $T(S_\lambda)$  is a  $32 \times 32$  matrix of maps  $m_{e_{ab}}$  with  $e_{ab} \in F$ , and  $\mathrm{KS}(S_\lambda)$  is isogenous to  $B^4$ , where  $B$  is isogenous to  $X^8$  and  $\mathrm{End}^0(B) = M_8(\mathbb{Q})$ .

*Proof.* After the twists all the Hodge structures involved have weight zero and lie in the tensor category generated by  $V$ , whose Mumford–Tate group is  $G \cdot \mathbb{G}_m$  with  $\mathbb{G}_m$  acting trivially in weight zero; so their morphisms and subquotients are those of representations of  $G$ . Over  $\mathbb{Q}$ ,  $T = T_1 \oplus T_2 \oplus T_3$  with  $T_i \cong S^2 V_i$ , and the Galois group permutes the three summands transitively. Hence  $T$  is irreducible, a representation of  $G$  defined over  $\mathbb{Q}$  contains  $T_1$ ,  $T_2$  and  $T_3$  with one and the same multiplicity  $m$ , and its space of  $G$ -maps from  $T$ , of  $\mathbb{Q}$ -dimension  $3m$ , is a vector space over  $\mathrm{End}_G(T) = F$  (Proposition 21.2(iii)) of dimension  $m$ . Item (XXXV) decomposes every  $\wedge^k V$ , and item (LV) checks twice, by peeling highest weights and by Weyl’s alternating sum over the Weyl group of  $\mathrm{SL}_2^3$ , that no summand  $S^2 V_i$  occurs; that is (i). For (ii),  $H^n(X \times X) = \bigoplus_{a+b=n} \wedge^a V \otimes \wedge^b V$ , and item (LV) computes by the same two methods the multiplicity of  $S^2 V_1$  in it: 0, 0, 1, 0, 6, 0, 16, 0, 24, 0, 16, 0, 6, 0, 1, 0, 0 for  $n = 0, \dots, 16$ . In  $H^2(X \times X) = \wedge^2 V \oplus (V \otimes V) \oplus \wedge^2 V$  the copy lies in  $V \otimes V$  by (i). For (iii),  $\mathrm{End} V = S^2 V \oplus \wedge^2 V$  contains each  $T_i$  once, and the proof of Theorem 21.4(ii) shows that the blocks of the Kuga–Satake embedding are maps  $m_{e_{ab}}$ . Finally  $\mathrm{KS}(S_\lambda)$  is isogenous to  $X^{32} = (X^8)^4$  by Theorem 21.4(i), and  $\mathrm{End}^0(X^8) = M_8(\mathrm{End}^0 X) = M_8(\mathbb{Q})$ . This is the splitting  $\mathrm{KS} \sim B^{2^{d-1}}$ ,  $d = [F : \mathbb{Q}]$ , which van Geemen proves for K3 surfaces with real multiplication [vG08]; there  $H^1(B, \mathbb{Q})$  is the corestriction to  $\mathbb{Q}$  of the even Clifford algebra of  $T$  over  $F$ , here  $D$ , and over  $\bar{\mathbb{Q}}$  one has  $\mathrm{Cor}_{F/\mathbb{Q}}(D) = \bigotimes_k D \otimes_{F, \sigma_k} \bar{\mathbb{Q}} = \bigotimes_k \mathrm{End}(V_k) = \mathrm{End}(V)$ , which is  $V^{\oplus 8}$  under left multiplication by  $G$ . ■

**Theorem 21.7** (Every power of a Mumford fourfold). *In Setup 21.1, suppose that the two exceptional classes of Proposition 20.44(ii) are algebraic on  $X \times X$ . Then every Hodge class on  $X^n$  is algebraic, for every  $n \geq 1$ .*

*Proof.* Over  $\bar{\mathbb{Q}}$  keep  $V = V_1 \otimes V_2 \otimes V_3$  and  $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$ , as in the proof of Lemma 21.3. For  $m \geq 1$  regard  $V^{\otimes m} = H^1(X)^{\otimes m}$  as the Künneth component of  $H^m(X^m, \mathbb{Q})$  of multidegree  $(1, \dots, 1)$ , whose projector is algebraic [Kle68, §2]. Call an endomorphism of  $V^{\otimes m}$  algebraic if it is the action on  $V^{\otimes m}$  of an algebraic cycle on  $X^m \times X^m$  that preserves  $V^{\otimes m}$ . Composing with the Künneth projector, the algebraic endomorphisms form a  $\mathbb{Q}$ -algebra, and they carry algebraic classes in  $V^{\otimes m}$  to algebraic classes.

*Step 1: reduction to multilinear invariants.* Let  $n \geq 1$  and  $p \geq 0$ . The map  $V^{\otimes 2p} \otimes (\mathbb{Q}^n)^{\otimes 2p} \rightarrow \wedge^{2p}(V \otimes \mathbb{Q}^n) = H^{2p}(X^n, \mathbb{Q})$  sending  $(v_1 \otimes \dots \otimes v_{2p}) \otimes (a_1 \otimes \dots \otimes a_{2p})$  to  $(v_1 \otimes a_1) \wedge \dots \wedge (v_{2p} \otimes a_{2p})$  is surjective and  $G$ -equivariant for the trivial action on  $\mathbb{Q}^n$ , so it is surjective on  $G$ -invariants,  $G$  being reductive. For  $a_1, \dots, a_{2p} \in \mathbb{Z}^n$  the image of  $t \otimes (a_1 \otimes \dots \otimes a_{2p})$  is  $f^*t$ , where  $f: X^n \rightarrow X^{2p}$  is the homomorphism with rows  $a_1, \dots, a_{2p}$ , and such tensors span  $(\mathbb{Q}^n)^{\otimes 2p}$ . So  $\mathrm{Hdg}^p(X^n)$  is spanned by the classes  $f^*t$  with  $f$  a homomorphism and  $t \in (V^{\otimes 2p})^G$ , and it suffices to show that every element of  $(V^{\otimes 2p})^G$  is algebraic.

*Step 2: degree four.* The space  $(V^{\otimes 4})^G$  has dimension eight, the product of the dimensions of the three spaces of  $\mathrm{SL}_2$ -invariants in  $V_k^{\otimes 4}$ . Let  $\mathcal{P}_0 \subset (V^{\otimes 4})^G$  be the span of the multilinear

components of the classes  $f^*\xi$ , for  $\xi \in \text{Hdg}^2(X \times X)$  and  $f: X^4 \rightarrow X^2$  a homomorphism, and of the products of two divisor classes of  $X^4$ . By [Proposition 20.44\(ii\)](#)  $\text{Hdg}^2(X \times X)$  is spanned by the six products of divisor classes and the two exceptional classes, so under the hypothesis every element of  $\mathcal{P}_0$  is algebraic. For  $t \in V^{\otimes 4}$ , read as a correspondence from the first two factors of  $X^4$  to the last two, let  $\hat{t}$  be the endomorphism  $v \mapsto t_*((\theta^3 \times \theta^3) \cup v)$  of  $V \otimes V$ . The pairing  $(v, w) \mapsto \int_X v \cup w \cup \theta^3$  on  $V$  is nondegenerate by hard Lefschetz and  $G$ -invariant, so  $t \mapsto \hat{t}$  is an isomorphism  $(V^{\otimes 4})^G \rightarrow \text{End}_G(V \otimes V)$ , and  $\hat{t}$  is algebraic when  $t$  is. Item (LV) computes exactly, on the split model over  $\bar{\mathbb{Q}}$ , that  $\mathcal{P}_0$  has dimension 7 and that the  $\hat{t}$  together with the composites  $\hat{u}\hat{t}$ , for  $t, u \in \mathcal{P}_0$ , span  $\text{End}_G(V \otimes V)$ . Since composition is bilinear and  $\mathcal{P}_0 \otimes \bar{\mathbb{Q}}$  is spanned by the same pull-backs and products of a basis of Hodge classes, the  $\mathbb{Q}$ -algebra of algebraic  $G$ -endomorphisms of  $V \otimes V$  has dimension 8: every  $G$ -endomorphism of  $V \otimes V$  is algebraic.

*Step 3: the partial swaps.* Over  $\bar{\mathbb{Q}}$ ,  $\text{End}_G(V \otimes V) = \bigotimes_k \text{End}_{\text{SL}_2}(V_k \otimes V_k)$ , and  $\text{End}_{\text{SL}_2}(V_k \otimes V_k)$  is spanned by 1 and the exchange  $s_k$  of the two copies of  $V_k$ ; item (LV) checks that  $s_k = \text{Cas}_k + \frac{1}{2}$ , with  $\text{Cas}_k$  the Casimir operator of  $\mathfrak{sl}(V_k)$  for the trace form, so that the  $s_k$  come from the three summands  $T_k$  of  $T$ . The Galois group permutes the  $s_k$  as it permutes the embeddings  $\sigma_k$ , so for  $a \in F$  the endomorphism  $r_a = \sum_k \sigma_k(a)s_k$  is rational, hence algebraic by Step 2. For  $m \geq 2$  and  $1 \leq i < m$  let  $r_a^{(i)}$  and  $s_k^{(i)}$  act as  $r_a$  and  $s_k$  on the factors  $i$  and  $i+1$  of  $V^{\otimes m}$  and as the identity on the others;  $r_a^{(i)}$  is algebraic, being induced by the product of a cycle inducing  $r_a$  with the diagonals of the other factors. By Schur–Weyl duality  $\text{End}_{\text{SL}_2}(V_k^{\otimes m})$  is the image of  $\bar{\mathbb{Q}}[S_m]$ , generated by the adjacent transpositions, and the commutant of  $\text{SL}_2^3$  in  $\bigotimes_k V_k^{\otimes m}$  is the tensor product of the three commutants; so  $\text{End}_G(V^{\otimes m}) \otimes \bar{\mathbb{Q}}$  is generated by the  $s_k^{(i)}$ . For a basis  $a_1, a_2, a_3$  of  $F$  the matrix  $(\sigma_k(a_j))$  is invertible, the square of its determinant being the discriminant of  $F$ , so each  $s_k^{(i)}$  is a  $\bar{\mathbb{Q}}$ -linear combination of the  $r_{a_j}^{(i)}$ . The  $\mathbb{Q}$ -algebra generated by the  $r_a^{(i)}$  therefore lies in  $\text{End}_G(V^{\otimes m})$  and spans it over  $\bar{\mathbb{Q}}$ , so it is  $\text{End}_G(V^{\otimes m})$ : every  $G$ -endomorphism of  $V^{\otimes m}$  is algebraic. Item (LV) checks the dimensions, 8 for  $m = 2$  and  $125 = 5^3$  for  $m = 3$ .

*Step 4: all multilinear invariants.* By the first fundamental theorem for  $\text{SL}_2$  the invariants in  $V_k^{\otimes 2m}$  are spanned by the images of  $\epsilon_k^{\otimes m}$  under the permutations of the  $2m$  factors. So  $(V^{\otimes 2m})^G \otimes \bar{\mathbb{Q}}$  is spanned by the images of  $\psi^{\otimes m} = \bigotimes_k \epsilon_k^{\otimes m}$  under the operators that permute the factors of the three  $V_k^{\otimes 2m}$  separately, which lie in  $\text{End}_G(V^{\otimes 2m}) \otimes \bar{\mathbb{Q}}$ , and the map  $\text{End}_G(V^{\otimes 2m}) \rightarrow (V^{\otimes 2m})^G$ ,  $e \mapsto e(\psi^{\otimes m})$ , is surjective. The class  $\psi$  in  $V \otimes V$  is that of the divisor  $m^*\theta - p_1^*\theta - p_2^*\theta$  on  $X \times X$  (proof of [Theorem 21.4\(ii\)](#)), up to a nonzero factor, and  $\psi^{\otimes m}$  is, up to sign, the product of its pull-backs along the projections of  $X^{2m}$  to the pairs of factors  $(2j-1, 2j)$ ; so it is algebraic, and by Step 3 so is every element of  $(V^{\otimes 2m})^G$ . Step 1 concludes.  $\blacksquare$

*Remark 21.8 (Why a composite is needed).* Step 2 cannot be carried out with pull-backs and products alone. The skew Cauchy decomposition  $\Lambda^4(V \otimes \mathbb{Q}^4) = \bigoplus_\lambda S_\lambda V \otimes S_{\lambda'} \mathbb{Q}^4$ , over the partitions  $\lambda$  of 4 with conjugates  $\lambda'$ , is functorial in  $\mathbb{Q}^4$ , and its summand  $S^4 V \otimes \Lambda^4 \mathbb{Q}^4$  contains exactly one line of Hodge classes of  $X^4$ . It is spanned by a class  $\text{Det} \in (V^{\otimes 4})^G$  whose restriction to the diagonal, on the split model  $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$ , is a nonzero multiple of Cayley’s hyperdeterminant (item (LV)), and the permutations of the factors of  $X^4$  act on it by the sign. Pull-backs along homomorphisms  $X^4 \rightarrow X^2$  have no component in that summand, because  $\Lambda^4 \bar{\mathbb{Q}}^2 = 0$ , and products of divisor classes have none, because they are invariant under  $\text{Sp}(V, \psi)$  and  $S^4 V$  has no  $\text{Sp}(V, \psi)$ -invariant. So the space  $\mathcal{P}_0$  of Step 2, of dimension 7, is the space of Hodge classes of  $X^4$  in the multilinear parts of

the other summands, and  $\text{Det}$  is reached only through a composite of correspondences, which involves a push-forward; item (LV) finds one already of the form  $\hat{t}\hat{t}$ . This is where [Theorem 21.7](#) differs from [Proposition 20.25](#), in which the first fundamental theorem for  $\text{SL}(W)$  lets pull-backs and products suffice. We have not found [Theorem 21.7](#) in the literature and make no claim of priority; it is an implication between open statements, not a new case of the Hodge conjecture.

At the CM points the powers are unconditional.

**Proposition 21.9** (Powers at a CM point). *Let  $W \rightarrow C$  be a family as in [Corollary 20.23](#) and  $c \in C$  a CM point. Then the Hodge conjecture holds for  $X_c^n$  for every  $n \geq 1$ .*

*Proof.* By the proof of [Theorem 20.30](#),  $H^1(X_c, \mathbb{C})$  has a basis of vectors  $x_w$  of the eight pairwise distinct weights  $w \in \{\pm 1\}^3$  of the Hodge group, and  $\text{End}^0(X_c) \otimes \mathbb{C}$  is the algebra of diagonal matrices in this basis. So  $\text{Hdg}^*(X_c^n) \otimes \mathbb{C}$  is spanned by the products of vectors  $x_w \otimes u$ ,  $u \in \mathbb{C}^n$ , in  $\bigwedge^*(H^1(X_c, \mathbb{C}) \otimes \mathbb{C}^n)$  in which the weights  $w$ , counted with multiplicity, sum to zero. Put  $w_1 = (1, 1, 1)$ ,  $w_2 = (1, -1, -1)$ ,  $w_3 = (-1, 1, -1)$  and  $w_4 = (-1, -1, 1)$ ; the weights are the  $\pm w_i$ ,  $w_1 + w_2 + w_3 + w_4 = 0$ , and any three of the  $w_i$  are linearly independent, so the integer relations among them are the multiples of  $(1, 1, 1, 1)$ . Let  $\alpha_i$  and  $\beta_i$  be the multiplicities of  $w_i$  and  $-w_i$  in a nonempty multiset of weights with sum zero. Then  $\alpha_i - \beta_i$  is independent of  $i$ ; so either some  $\alpha_i$  and  $\beta_i$  are both positive, or all  $\alpha_i$  are positive, or all  $\beta_i$  are. In each case the multiset contains one of the sets  $\{w_i, -w_i\}$ ,  $\{w_1, w_2, w_3, w_4\}$ ,  $\{-w_1, -w_2, -w_3, -w_4\}$ , each of sum zero, and by induction every product above is, up to sign, a product of monomials  $\bigwedge_{w \in P} (x_w \otimes u_w)$  over such sets  $P$ , whose weights are pairwise distinct; item (LV) confirms that these six sets are the irreducible multisets of sum zero with multiplicities at most five. As the diagonal entries of an element of  $\text{End}^0(X_c) \otimes \mathbb{C}$  can be prescribed independently, there is  $f \in \text{Hom}(X_c^n, X_c) \otimes \mathbb{C}$  with  $f^*x_w = x_w \otimes u_w$  for  $w \in P$ , and then  $\bigwedge_{w \in P} (x_w \otimes u_w) = f^*(\bigwedge_{w \in P} x_w)$ , with  $\bigwedge_{w \in P} x_w$  a Hodge class of  $X_c$  tensored with  $\mathbb{C}$ . As in the proof of [Theorem 20.30](#), Zariski density lets  $f$  and the class be taken rational, so the ring  $\text{Hdg}^*(X_c^n)$  is generated by pull-backs of Hodge classes of the fourfold  $X_c$  along homomorphisms, and these are algebraic [[Mar25a](#)]. ■

For  $n = 2$  this is [Theorem 20.30](#) again, without the computation of item (XLIII). The statement is also contained in the theorem of [[Li26](#)] on the powers of CM fourfolds.

**Corollary 21.10** (Powers and fibre powers of a Mumford family). *Let  $W \rightarrow C$  be as in [Corollary 20.23](#) and, for  $n \geq 1$ , let  $W^{(n)} = W \times_C \cdots \times_C W$ , with  $n$  factors. The conditions of [Corollary 20.31](#) are equivalent to each of the following.*

- (e) *The Hodge conjecture holds for  $X_t^n$  for every  $t \in C$  and every  $n \geq 1$ .*
- (f)  *$B(W^{(n)})$  holds for  $L_{W^{(n)}}$  for every  $n \geq 1$ .*

*Proof.* (b) implies (e): by [Corollary 20.31](#), (b) gives (c), so the exceptional classes of  $X_t \times X_t$  are algebraic for every  $t$ . At a point  $t$  that is not a CM point the Hodge group of  $X_t$  is  $G$ , as shown in the proof of [Corollary 20.31](#), and [Theorem 21.7](#) applies; at a CM point [Proposition 21.9](#) does. (e) implies (f):  $W^{(n)}$  is an abelian scheme over  $C$  with projective total space, and, as in the proof of [Corollary 20.26](#), its monodromy invariant classes are Hodge classes on every fibre; by (e) they are algebraic on every fibre, hence restrictions of algebraic classes on  $W^{(n)}$  by [Lemma 20.28](#), and [Theorem 20.22](#) gives  $B(W^{(n)})$ . (f) implies (a): take  $n = 2$ . ■

**Corollary 21.11** (Products with the K3 surface). *In Setup 21.1, let  $\lambda \in F$  be totally positive and  $S = S_\lambda$  as in Theorem 21.4. If the Kuga–Satake class of  $S$  is algebraic, then the Hodge conjecture holds for  $S^a \times X^b$  for all  $a, b \geq 0$ .*

*Proof.* By Theorem 21.4(ii) the exceptional classes are algebraic, so by Theorem 21.7 the Hodge conjecture holds for every power of  $X$ . In the proof of Theorem 21.4(ii) a block  $\kappa = m_e$ ,  $e \neq 0$ , of the Kuga–Satake embedding, a map from  $T(S)$  to  $\text{End } V$ , which is a Künneth component of  $H^*(X \times X, \mathbb{Q})$  up to a twist, is induced by a cycle on  $S \times X \times X$ ; its adjoint  $\kappa^\dagger$  is algebraic,  $\kappa^\dagger \kappa = m_{4e^2/\lambda}$ , and every  $m_a$ ,  $a \in F$ , is induced by a cycle on  $S \times S$ . So  $L = m_{\lambda/(4e^2)} \kappa^\dagger$  is algebraic and  $L\kappa$  is the identity of  $T(S)$ . The decomposition  $H^*(S, \mathbb{Q}) = \mathbb{Q} \oplus \text{NS}(S)_\mathbb{Q} \oplus T(S) \oplus H^4(S, \mathbb{Q})$  has algebraic projectors, as in that proof, so  $H^*(S^a \times X^b, \mathbb{Q})$  is the sum of the summands  $N \otimes T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q})$ ,  $N$  a tensor product of  $a - r$  of the three other pieces, each cut out by an algebraic projector. Each such  $N$  is spanned by products of algebraic classes and is of Tate type, so a Hodge class in the summand is a sum of products  $p^* \eta \cup q^* \xi$  with  $\eta$  algebraic on  $S^{a-r}$  and  $\xi$  a Hodge class in  $T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q}) \subset H^*(S^r \times X^b, \mathbb{Q})$ . For such  $\xi$  the class  $(\kappa^{\otimes r} \otimes 1)\xi$  is a Hodge class on  $X^{2r+b}$ , hence algebraic, and  $\xi = (L^{\otimes r} \otimes 1)(\kappa^{\otimes r} \otimes 1)\xi$  is algebraic. ■

Corollary 21.11 strengthens Theorem 21.4(ii), which gives  $X \times X$  only. Varesco proves an implication of the same kind, from the algebraicity of the Kuga–Satake correspondence to the Hodge conjecture for all powers, for families of K3 surfaces of generic Picard number sixteen [Var22]; we have seen that paper only in summary. The converse question, what the Kuga–Satake class asks for beyond the target, has a precise answer.

**Proposition 21.12** (What the Kuga–Satake route asks). *In the situation of Corollary 21.11, the Kuga–Satake class of  $S$  is algebraic if and only if*

- (i) *the two exceptional classes of  $X \times X$  are algebraic, and*
- (ii) *(a link) for some  $n \geq 1$  there is an algebraic cycle  $Z$  on  $S \times X^n$  whose action  $Z_*$  is nonzero on  $T(S)$ .*

*Proof.* If the Kuga–Satake class is algebraic, (i) is Theorem 21.4(ii), and a nonzero block of the embedding, induced by a cycle on  $S \times X \times X$ , is a link with  $n = 2$ . Conversely assume (i) and (ii). By Theorem 21.7 the Hodge conjecture holds for every power of  $X$ . Since  $T(S) \cong T(-1)$  is irreducible (proof of Proposition 21.6),  $Z_*$  maps  $T(S)$  isomorphically onto a sub-Hodge structure of some  $H^{2k}(X^n, \mathbb{Q})(k-1)$ , which is a direct summand, polarisable Hodge structures being semisimple. Projecting onto it and composing with an isomorphism onto the copy of  $T(-1)$  in  $H^2(X \times X, \mathbb{Q})$  (Proposition 21.6(ii)) gives a morphism of Hodge structures  $H^{2k}(X^n, \mathbb{Q})(k-1) \rightarrow H^2(X \times X, \mathbb{Q})$ , that is a Hodge class on  $X^{n+2}$ , which is algebraic. Its composite with  $Z_*$  is an algebraic nonzero morphism  $T(S) \rightarrow T(-1) \subset H^2(X \times X, \mathbb{Q})$ , which is some  $m_e$ ,  $e \neq 0$ , by Proposition 21.2(iii). The endomorphisms of  $H^2(X \times X, \mathbb{Q})$  that act by  $m_a$  on this copy of  $T(-1)$  and by zero on a Hodge complement are Hodge classes on  $X^4$ , hence algebraic, so every  $m_a: T(S) \rightarrow H^1(X) \otimes H^1(X)$ ,  $a \in F$ , is algebraic, and so is its composite with cup product with  $\theta^3$  on one factor, which carries  $H^1(X) \otimes H^1(X)$  onto the Künneth component of  $H^8(X \times X, \mathbb{Q})$  that represents  $\text{End } V$ . By Proposition 21.6(iii), after an isogeny  $\text{KS}(S) \rightarrow X^{32}$  the Kuga–Satake embedding is a matrix of such maps, and the Kuga–Satake class is recovered from its blocks through the graphs of the isogeny and of the inclusions and projections of the factors of  $X^{32}$ ; so it is algebraic. ■

So the Kuga–Satake route is at least as strong as the first case of (F2) and is not a reduction of it: at the point  $X$  it asks for (i), which is the first case of (F2) there, and in

addition for a link. Whether (i) implies (ii) is not known; the Hodge conjecture implies both. The route helps only if the Kuga–Satake class of  $S_\lambda$  can be made algebraic by arguments proper to K3 surfaces. A link has to be a genuinely new correspondence:  $T$  occurs in no  $H^k(X)$  (Proposition 21.6(i)), and the next proposition shows that it occurs in the cohomology of no abelian variety of dimension below six.

**Proposition 21.13** (No small abelian carrier of  $T$ ). *In Setup 21.1, let  $Y$  be an abelian variety such that  $T(-1)$  is isomorphic to a subquotient of  $H^k(Y, \mathbb{Q})(j)$  for some  $k$  and  $j$ . Then  $\dim Y \geq 6$ . Consequently, if  $\dim Y \leq 5$  and  $S = S_\lambda$  is as in Theorem 21.4, every Hodge class on  $S \times Y$ , algebraic or not, acts by zero on  $T(S)$ , and no link as in Proposition 21.12(ii) is a composite  $Z_2 \circ Z_1$  of algebraic cycles on  $S \times Y$  and on  $Y \times X^n$ .*

*Proof.* Write  $W = T(-1)$ . As  $W$  is a subquotient of a tensor construction on  $H^1(Y)$ ,  $\mathrm{MT}(Y)$  maps onto  $\mathrm{MT}(W)$ , whose derived group is the adjoint group  $G^{\mathrm{ad}}$  of  $G$ , acting on  $W$  through the adjoint representation. So the derived group of  $\mathrm{Hg}(Y)$ , an almost direct product of  $\mathbb{Q}$ -simple groups, maps onto  $G^{\mathrm{ad}}$ . The image of each factor is a normal subgroup of the  $\mathbb{Q}$ -simple group  $G^{\mathrm{ad}}$ , the images commute, and  $G^{\mathrm{ad}}$  is not commutative, so exactly one factor  $M$  maps onto  $G^{\mathrm{ad}}$ , with finite kernel. Over  $\bar{\mathbb{Q}}$ ,  $M$  is isogenous to  $\mathrm{SL}_2^3$ , with factors indexed by the embeddings  $\sigma_1, \sigma_2, \sigma_3$  and permuted transitively by the Galois group, and the irreducible constituents of  $H^1(Y, \bar{\mathbb{Q}})$  under  $M$  are products  $R_1 \otimes R_2 \otimes R_3$  of irreducible representations of the three factors.

Let  $\mu_Y$  be the Hodge cocharacter of  $Y$ , with the two weights 0 and 1 on  $H^1(Y)$ , and write  $\mu_Y = \mu' \mu''$  with fractional cocharacters  $\mu'$  of  $M$  and  $\mu''$  commuting with  $M$ . The image of  $\mu''$  in  $\mathrm{MT}(W)$  commutes with  $G^{\mathrm{ad}}$ , so it is central, and the image of  $\mu'$  in  $G^{\mathrm{ad}}$  is the component there of the Hodge cocharacter of  $W$ , which by Proposition 21.2(iii) is trivial on the factors at  $\sigma_1$  and  $\sigma_2$  and has weights  $-1, 0, 1$  on  $T_3$ . So  $\mu'$  is trivial on the first two factors of  $M$  and has weights  $\pm \frac{1}{2}$  on the standard representation of the third. On  $R_1 \otimes R_2 \otimes R_3$ , tensored with an eigenvector of  $\mu''$  in its multiplicity space, the weights of  $\mu_Y$  therefore fill an interval of length  $\dim R_3 - 1$ , so  $\dim R_3 \leq 2$ . The constituents form a Galois-stable multiset and the Galois group carries the third factor to each of the others, so every constituent is  $V_I = \bigotimes_{i \in I} V_i$  for a subset  $I \subseteq \{1, 2, 3\}$ , with a multiplicity  $m_I$  that depends only on  $|I|$ , and  $m_I \neq 0$  for some nonempty  $I$ , since  $M$  acts nontrivially.

If  $m_I \neq 0$  for some  $I$  with  $|I| = 2$ , then  $\dim H^1(Y) \geq 3 \cdot 4 = 12$ . If  $m_{\{1\}} \neq 0$ , let  $U = V_1 \otimes N$ ,  $\dim N = m_{\{1\}}$ , be the  $V_1$ -isotypic summand of  $H^1(Y, \mathbb{C})$ . The embedding  $\sigma_1$  is real, so complex conjugation preserves the first factor of  $M$  and  $U$  is defined over  $\mathbb{R}$ . The connected group  $\mathrm{MT}(Y)$  acts on  $M$  preserving each factor, and  $\mathrm{SL}_2$  has a single irreducible representation of dimension two, so  $U$  is stable under  $\mathrm{MT}(Y)(\mathbb{R})$  and is a real sub-Hodge structure of  $H^1(Y, \mathbb{R})$ , and  $U^{1,0}$  and  $U^{0,1}$  are complex conjugate and both nonzero. Since  $\mu'$  acts trivially on  $U$ ,  $\mu_Y$  acts on  $V_1 \otimes N$  through  $N$ , so  $N$  carries both weights,  $m_{\{1\}} \geq 2$ , and  $\dim H^1(Y) \geq 3 \cdot 2 \cdot 2 = 12$ . Otherwise  $H^1(Y, \bar{\mathbb{Q}})$  is  $V^{\oplus m}$ ,  $m \geq 1$ , plus a trivial representation of  $M$ . If  $m = 1$ , every  $H^k(Y)$  is, as a representation of  $M$ , a sum of copies of the  $\bigwedge^a V$ , none of which contains an  $S^2 V_i$  (Proposition 21.6(i)), while  $W$  contains all three; so  $m \geq 2$  and  $\dim H^1(Y) \geq 16$ . In every case  $\dim Y = \frac{1}{2} \dim H^1(Y) \geq 6$ .

For the rest, a Hodge class on  $S \times Y$  acting nontrivially on the irreducible  $T(S) \cong W$  carries it isomorphically onto a sub-Hodge structure of some  $H^k(Y, \mathbb{Q})(j)$ , and if  $Z_2 \circ Z_1$  acts nontrivially on  $T(S)$ , so does  $Z_1$ . ■

Among the powers of  $X$  the smallest carrier of  $T$  is  $X \times X$ , of dimension eight, where  $T(-1)$  sits in  $H^2$  (Proposition 21.6(ii)). Whatever is known about abelian varieties of

dimension at most five, in particular Markman's theorem in dimension four [Mar25a], no link passes through one.

*Remark 21.14* (The bound six as a statement about Hodge structures). The bound of [Proposition 21.13](#) is, as far as we know, the best that Hodge theory gives. Let  $K \subset D$  be a CM quadratic extension of  $F$ ,  $\chi_i$  the character of the  $i$ -th factor of the torus  $\text{Res}_{F/\mathbb{Q}} K^1$  over  $\mathbb{Q}$ , and  $H = \bigoplus_i (V_i \chi_i \oplus V_i \chi_i^{-1})$ , of dimension twelve, with types  $(1, 0)$  on  $V_i \chi_i$  and  $(0, 1)$  on  $V_i \chi_i^{-1}$  for  $i = 1, 2$ , and with the Hodge cocharacter of  $V_3$  on the two summands with  $i = 3$ : the signature  $(2, 0)$ ,  $(2, 0)$ ,  $(1, 1)$ . Item (LV) computes that the part of  $\bigwedge^2 H$  on which the torus acts trivially is  $T_1 \oplus T_2 \oplus T_3$  plus three trivial summands, and that this copy of  $T$  has Hodge numbers  $(1, 7, 1)$ , those of  $T(S_\lambda)$ . We recall, without having re-read the sources, that these are the Hodge structures on  $H^1(Z, \mathbb{Q}) \cong D$  of the abelian sixfolds  $Z$  with multiplication by  $K$  over the Shimura curves of PEL type attached to  $K$  and  $D$ , which go back to Shimura, and that at a suitable member the copy of  $T$  is isomorphic to  $T(S_\lambda)$ , the Hodge structure on it depending only on the image of the Hodge cocharacter in  $G^{\text{ad}}$ . Nothing in the paper uses this. It would not give a link: that would still need an algebraic cycle on  $S_\lambda \times Z$  acting nontrivially on  $T(S_\lambda)$ .

*Remark 21.15* (The K3 surfaces at the CM points). Apply the construction of [Theorem 21.4\(i\)](#) at a CM point  $c$  of a family as in [Corollary 20.23](#), with the Hodge structure that  $X_c$  induces on  $T$ , and call the surface  $S_c$ . By the proof of [Theorem 20.30](#) the Hodge group of  $X_c$  is  $\text{Res}_{F/\mathbb{Q}} L^1$  for a quadratic extension  $L$  of  $F$  contained in  $D$ ;  $L$  is a CM field, since the torus is compact at  $\sigma_1$  and  $\sigma_2$ , where  $D$  is definite, and at  $\sigma_3$ , where it contains the nontrivial image of the circle of the Hodge structure. It acts on  $T = D^0$  by conjugation, with invariants  $D^0 \cap L$ , the three-dimensional space  $L^0$  of elements of  $L$  of reduced trace zero. So  $\text{NS}(S_c)_{\mathbb{Q}} = T^\perp \oplus L^0$  has rank sixteen. Write  $D = L \oplus Lj$  with  $jx = \bar{x}j$  for  $x \in L$ . Then  $Lj$  is the  $q_\lambda$ -orthogonal complement of  $L^0$  in  $D^0$ , because  $\text{trd}(yj) = 0$  for  $y \in L$ , so  $T(S_c)_{\mathbb{Q}} = Lj$  has rank six;  $t \in L^1$  acts on it by multiplication by  $t^2$ , so left multiplication by  $L$  consists of Hodge endomorphisms and  $T(S_c)$  has complex multiplication by the sextic CM field  $L$ . At the CM points the Kuga–Satake question therefore lies among K3 surfaces of Picard number sixteen, among which the known cases recalled in [Remark 21.5](#) are particular families; we have not checked whether some  $S_c$  belongs to one of them. It would not help: at  $c$  the exceptional classes are already algebraic ([Theorem 20.30](#)), and what is missing is the propagation along  $C$ .

The last reduction replaces the eightfold  $X \times X$  by a fourfold.

**Proposition 21.16** (A surface in  $X$ ). *In [Setup 21.1](#), let  $i: \Sigma \rightarrow X$  be a smooth surface that is the intersection of two ample divisors, and  $L$  the cup product with  $\theta$ . Then  $i_* i^* = cL^2$  on  $H^2(X, \mathbb{Q})$  for a rational number  $c > 0$ . For  $a \in F$  let  $\rho_a$  be the Hodge endomorphism of  $H^2(X, \mathbb{Q})$  that is zero on  $\mathbb{Q}\theta$  and acts on  $U$  as a ([Proposition 21.2\(ii\)](#)), and put  $\rho'_a = c^{-1} i^* \rho_a L^{-2} i_*$ , a Hodge endomorphism of  $H^2(\Sigma, \mathbb{Q})$ , where  $L^{-2}$  inverts  $L^2: H^2(X) \rightarrow H^6(X)$ . Then  $\rho_a$  is induced by an algebraic cycle on  $X \times X$  if and only if  $\rho'_a$  is induced by an algebraic cycle on  $\Sigma \times \Sigma$ . So the two exceptional classes of  $X \times X$  are algebraic if and only if the Hodge classes in  $H^2(\Sigma) \otimes H^2(\Sigma) \subset H^4(\Sigma \times \Sigma, \mathbb{Q})$  that correspond to the  $\rho'_a$ ,  $a \in F$ , are algebraic on the fourfold  $\Sigma \times \Sigma$ .*

*Proof.* By [Proposition 20.44\(i\)](#) the two divisors have classes  $d_1\theta$  and  $d_2\theta$  with  $d_1, d_2 > 0$ , so  $i_* i^* \alpha = \alpha \cup [\Sigma] = d_1 d_2 \theta^2 \alpha$  and  $c = d_1 d_2$ . By hard Lefschetz  $L^2: H^2(X) \rightarrow H^6(X)$  is an isomorphism, so  $i^*$  is injective on  $H^2(X)$ , and  $L^{-2}$  is induced by an algebraic cycle,  $X$  being an abelian variety [[Kle68](#), §2]. For  $\alpha \in H^2(X)$ ,  $\rho'_a(i^* \alpha) = c^{-1} i^* \rho_a L^{-2}(cL^2 \alpha) = i^* \rho_a \alpha$ . If

$\beta \in H^2(\Sigma)$  is orthogonal to  $i^*H^2(X)$ , then  $\int_X i_*\beta \cup \gamma = \int_\Sigma \beta \cup i^*\gamma = 0$  for every  $\gamma \in H^2(X)$ , so  $i_*\beta = 0$  and  $\rho'_a\beta = 0$ . Hence  $\rho'_a$  is  $i^*\rho_a(i^*)^{-1}$  on  $i^*H^2(X)$  and zero on its orthogonal complement, and  $\rho_a = c^{-1}L^{-2}i_*\rho'_ai^*$ . Both formulas are composites of correspondences with the graph of  $i$ , its transpose and  $L^{-2}$ , so  $\rho_a$  is algebraic if and only if  $\rho'_a$  is. The exceptional classes are algebraic if and only if every  $\rho_a$  is, by [Proposition 21.2\(ii\)](#), since 1 and the projection onto  $\mathbb{Q}\theta$  are algebraic. The last statement uses Poincaré duality on  $\Sigma$  and the algebraicity of the Künneth projectors of a surface [[Mur90](#)]. ■

This is the usual reduction to a complete intersection surface, and it does not make the problem easier. By adjunction  $K_\Sigma$  is the restriction of the sum of the two ample divisors, so  $\Sigma$  is a surface of general type, and the  $\rho'_a$  are Hodge classes of degree four on its self-product, for which the Hodge conjecture is not known.

*Remark 21.17* (What remains open). Nothing proved from [Proposition 21.6](#) to [Proposition 21.16](#) is a new case of the Hodge conjecture. [Theorem 21.7](#), [Corollary 21.11](#) and one direction of [Proposition 21.12](#) are implications whose hypotheses are open; [Proposition 21.9](#) is a consequence of Markman's theorem, and of [[Li26](#)]; the rest are statements about Hodge structures and correspondences. The first case of (F2) is open. By [Corollary 20.31](#), [Corollary 21.10](#) and [Proposition 21.16](#) it is equivalent to each of the following: the algebraicity of the exceptional classes of  $X_t \times X_t$  for uncountably many  $t \in C$ ; the Hodge conjecture for every power of every fibre  $X_t$ ;  $B(W \times_C W)$ ;  $B$  for every fibre power of  $W$ ; the Hodge conjecture for  $W \times_C W$ , for the curves of [Corollary 20.31\(d\)](#); and, for uncountably many  $t$ , the algebraicity of the classes  $\rho'_a$  on  $\Sigma_t \times \Sigma_t$  for one complete intersection surface  $\Sigma_t \subset X_t$ . It is implied by the Kuga–Satake Hodge conjecture for the surfaces  $S_\lambda$  attached to uncountably many fibres ([Corollary 21.40\(ii\)](#)), which by [Proposition 21.12](#) is equivalent to it together with a link, and that is open too: none of the cases of the Kuga–Satake Hodge conjecture known to us ([Remark 21.5](#)) covers a K3 surface of transcendental rank nine with real multiplication by a cubic field, and we do not claim that this survey is complete.

Two limits explain why the constructions at hand do not reach the target. First, the Kuga–Satake class has a nonzero component in  $T(S) \otimes H^*(A)$ ,  $A$  a power of  $X$ , whereas a product  $p^*\alpha \cup q^*\beta$  of pull-backs of algebraic classes of  $S$  and of  $A$  has none, the algebraic classes of  $S$  lying in  $H^0 \oplus \text{NS}(S)_\mathbb{Q} \oplus H^4$ ; so no product of pull-backs gives it, whatever is known on  $A$ , and a new correspondence between  $S$  and an abelian variety is needed. Second,  $\text{Sp}(V, \psi)$  acts on  $H^*(X^m, \mathbb{Q}) = \bigwedge^*(V \otimes \mathbb{Q}^m)$  through  $V$ , compatibly with products, with pull-back and push-forward along homomorphisms and with integration, hence with the composition of correspondences. The divisor classes of  $X^m$ , the Hodge classes of  $X$  and the Weil classes of every abelian variety isogenous to a power of  $X$  are invariant under it: for a CM field  $E \subset M_m(\mathbb{Q}) = \text{End}^0(X^m)$ , each component over  $\mathbb{C}$  of the top exterior power over  $E$  of  $V \otimes \mathbb{Q}^m$  is the top exterior power of  $V \otimes U$  for a complex vector space  $U$  on which  $V$  plays no part, that is  $(\det V)^{\otimes \dim U} \otimes (\det U)^{\otimes 8}$ , and  $\text{SL}(V)$  acts trivially on it. So every class obtained from these by the operations above is  $\text{Sp}(V, \psi)$ -invariant, while the exceptional classes are not, the  $\text{Sp}(V, \psi)$ -invariants of  $H^4(X \times X, \mathbb{Q})$  being the six products of divisor classes (proof of [Proposition 20.44](#)). This argument says nothing about Hodge classes on products  $X^m \times Z$  with  $Z$  not isogenous to a power of  $X$ , for instance with  $T(-1)$  in  $H^2(Z)$  as in [Remark 21.14](#); the Hodge classes of  $X \times X \times Z$  that pair the two copies of  $T(-1)$  escape it, and their algebraicity is not known either.

The missing statement can be taken in either of two forms, and neither is proved here. One is (V) along  $C$  for the exceptional classes, from a CM point, where they are algebraic

(Theorem 20.30); this is the form of Remark 20.34, and Section 21.2 studies it. The other is (V) for the Kuga–Satake class along the seven-dimensional family of K3 surfaces whose Néron–Severi group contains  $\mathrm{NS}(S_\lambda)$ , on which it stays a Hodge class, from a member at which it is algebraic; it implies the Kuga–Satake Hodge conjecture for  $S_\lambda$  and with it everything above. The members of Picard number at least seventeen, to which [Mor84] applies, are the natural starting points, and we have not checked that the class is algebraic there.

### 21.2. The Mumford target: rigidity, a numerical criterion, and the bypasses.

The first case of (F2) is now a propagation problem along one compact Shimura curve: the exceptional classes are algebraic at every CM point (Theorem 20.30), and on every member exactly when  $B(W \times_C W)$  holds (Corollary 20.31). This subsection asks whether the propagation can be obtained without passing through the Lefschetz standard conjecture, by one of the mechanisms that have closed other cases of the Hodge conjecture: a degeneration, a deformation into a larger family on which the class stays of Hodge type, a specialisation to special points, or semiregularity. It proves that the first two cannot work, that the third works exactly when the representing cycles have bounded degree, and it reduces the fourth to one dimension count at one CM point. It does not prove the classes algebraic at a very general point.

**Setup 21.18** (The classes  $\omega_a$ ). Keep Setup 21.1 and let  $W \rightarrow C$  be a family as in Corollary 20.23, with fibres  $X_t$ . Over  $\mathbb{Q}$  let  $\pi_0, \pi_{12}, \pi_{13}, \pi_{23} \in H^2(X) \otimes H^2(X) \subset H^4(X \times X)$  be the classes that correspond, under the isomorphism  $H^2 \otimes H^2 \cong \mathrm{End} H^2$  given by the form  $\wedge^2 \psi$  on  $H^2(X) = \wedge^2 V^\vee$ , to the projections onto  $\mathbb{Q}\theta$ ,  $U_{12}$ ,  $U_{13}$  and  $U_{23}$  of Proposition 21.2(i), and for  $a = (a_0, a_1, a_2, a_3)$  put

$$\omega_a = a_0 \pi_0 + a_1 \pi_{12} + a_2 \pi_{13} + a_3 \pi_{23}, \quad L_0(a) = a_0 + a_1 + a_2 + a_3.$$

The four tensors are invariant under  $G$  and under the exchange  $\tau$  of the two factors of  $X \times X$ , because  $\wedge^2 \psi$  is symmetric; so each  $\omega_a$  is a flat section over  $C$  of Hodge type at every point. The argument of Proposition 21.2(ii) shows that the rational ones are those with  $a_0 \in \mathbb{Q}$  and  $(a_1, a_2, a_3) = (\sigma_3(\alpha), \sigma_2(\alpha), \sigma_1(\alpha))$  for some  $\alpha \in F$ ; we call such a class *exceptional* when  $\alpha \notin \mathbb{Q}$ . The products of divisor classes in  $H^2 \otimes H^2$  are the rational  $\omega_a$  with  $\alpha \in \mathbb{Q}$ , and  $\pi_0$  is a multiple of  $\mathrm{pr}_1^* \theta \cup \mathrm{pr}_2^* \theta$ ; so adding a rational multiple of  $\pi_0$  to an exceptional class changes  $L_0$  and not the question of algebraicity. For  $c \in C$  write  $Y = X_c \times X_c$ , let  $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\wedge^2 T_Y)$  act on  $H^*(Y, \mathbb{C})$  by contraction as in Setup 18.27, of dimension  $28 + 64 + 28 = 120$ , and let  $\kappa \in H^1(T_Y)$  be the Kodaira–Spencer class at  $c$  of the family  $W \times_C W \rightarrow C$ . Put

$$e(\omega) = \dim HT^2(Y) - \dim \mathrm{Ann}_{HT^2(Y)}(\omega) = 120 - \dim \mathrm{Ann}_{HT^2(Y)}(\omega).$$

**Lemma 21.19** (Linear relations among conjugates). *Let  $F$  be a cubic number field with embeddings  $\sigma_1, \sigma_2, \sigma_3$ , let  $\alpha \in F \setminus \mathbb{Q}$ , and let  $c_0, c_1, c_2, c_3, a_0 \in \mathbb{Q}$ . The three conjugates  $\sigma_i(\alpha)$  are pairwise distinct and nonzero, and if  $c_0 a_0 + \sum_i c_i \sigma_i(\alpha) = 0$  then  $c_1 = c_2 = c_3$ .*

*Proof.* The minimal polynomial of  $\alpha$  has degree three, since  $[F : \mathbb{Q}]$  is prime, and is separable, so the conjugates are distinct; they are nonzero because  $\alpha \neq 0$ . Let  $\Gamma$  be the Galois group of a normal closure; it permutes the  $\sigma_i$  transitively by  $\gamma \circ \sigma_i = \sigma_{\gamma(i)}$ . Put  $\beta = \alpha - \mathrm{Tr}_{F/\mathbb{Q}}(\alpha)/3$ , so that  $\beta \neq 0$  has trace zero, and  $r = -c_0 a_0 - (c_1 + c_2 + c_3) \mathrm{Tr}_{F/\mathbb{Q}}(\alpha)/3 \in \mathbb{Q}$ ; the hypothesis reads  $\sum_i c_i \sigma_i(\beta) = r$ . Applying every  $\gamma \in \Gamma$  and averaging, each  $\sigma_j$  occurs  $|\Gamma|/3$  times in the place of each  $\sigma_i$ , so  $r = \frac{1}{3}(\sum_i c_i) \sum_j \sigma_j(\beta) = 0$ . The relations  $c \in \mathbb{Q}^3$  with  $\sum_i c_i \sigma_i(\beta) = 0$  form a  $\Gamma$ -stable subspace  $N$  containing  $(1, 1, 1)$ . The plane  $\{\sum c_i = 0\}$  is an irreducible

representation of  $\Gamma$  over  $\mathbb{Q}$ , since  $\Gamma$  contains the 3-cycles, which have no rational eigenvector in that plane; so if  $N$  were larger than the line of  $(1, 1, 1)$  it would contain the plane and be all of  $\mathbb{Q}^3$ , and  $\sigma_1(\beta) = 0$ . Hence  $c \in \mathbb{Q}(1, 1, 1)$ .<sup>63</sup> ■

**Lemma 21.20** (One exceptional class gives the others). *Let  $t \in C$ . If one rational exceptional class  $\omega$  is algebraic on  $X_t \times X_t$ , then every Hodge class in  $H^2(X_t) \otimes H^2(X_t)$  is algebraic; in particular so are the two exceptional classes of Proposition 20.44.*

*Proof.* At a CM point every Hodge class of  $X_t \times X_t$  is algebraic (Theorem 20.30); so let  $t$  be another point, where the Hodge group is  $G$ , and write  $X = X_t$ . For  $x \in H^2(X) \otimes H^2(X)$  let  $E_x$  be the endomorphism  $y \mapsto x_*(\theta^2 \cup y)$  of  $H^2(X, \mathbb{Q})$ , where  $x_*$  is the action of  $x$  as a correspondence. By Poincaré duality and the hard Lefschetz isomorphism  $\theta^2: H^2 \rightarrow H^6$ , the map  $x \mapsto E_x$  is a  $G$ -equivariant isomorphism onto  $\text{End } H^2(X, \mathbb{Q})$ , so it carries the Hodge classes onto the Hodge endomorphisms, which are  $\mathbb{Q} \times F$  by Proposition 21.2(ii). It carries the products of divisor classes, which are invariant under  $\text{Sp}(V, \psi)$ , onto the  $\text{Sp}(V, \psi)$ -equivariant endomorphisms, which are  $\mathbb{Q} \times \mathbb{Q}$ , so it carries  $\omega$  to an element  $(r, \beta)$  with  $\beta \notin \mathbb{Q}$ . The algebraic Hodge classes form a subspace  $A$  containing the products of divisor classes, and  $A$  is closed under composition: if  $x_1, x_2$  are algebraic, then the composite correspondence  $x_1 \circ \Delta_*(\theta^2) \circ x_2$  is algebraic, and so is its Künneth component in  $H^2 \otimes H^2$ , since the Künneth projectors of  $X \times X$  are polynomials in the graphs of  $[2] \times 1$  and  $1 \times [2]$ , which act on  $H^i(X) \otimes H^j(X)$  by  $2^i$  and  $2^j$  (as in Step 1 of the proof of Theorem 20.22); that component  $x$  has  $E_x = E_{x_1} E_{x_2}$ . So the image of  $A$  is a subalgebra of  $\mathbb{Q} \times F$  containing  $\mathbb{Q} \times \mathbb{Q}$  and  $(r, \beta)$ , hence  $\mathbb{Q} \times \mathbb{Q}[\beta] = \mathbb{Q} \times F$ , because  $\beta$  generates the cubic field  $F$ . ■

**Theorem 21.21** (Rigidity of the exceptional classes). *Let  $c \in C$ ,  $Y = X_c \times X_c$ , and let  $\omega = \omega_a$  be a rational exceptional class.*

- (i)  $\text{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$  and  $\text{Ann}_{H^{0,2}(Y)}(\omega) = 0$ .
- (ii)  $\text{Ann}_{H^0(\wedge^2 T_Y)}(\omega) = 0$  if and only if  $L_0(a) = a_0 + \text{Tr}_{F/\mathbb{Q}}(\alpha) \neq 0$ . So  $\text{Ann}_{HT^2(Y)}(\omega) = \mathbb{C}\kappa$  and  $e(\omega) = 119$  whenever  $L_0(a) \neq 0$ , which can always be arranged by adding a rational multiple of  $\pi_0$ .
- (iii) Let  $\mathfrak{S}$  be the Siegel space of weight one Hodge structures on  $H^1(Y, \mathbb{Q})$  polarised by  $\psi \oplus \psi$ , of dimension 36. The germ at  $Y$  of the locus in  $\mathfrak{S}$  where  $\omega$  stays of Hodge type is the germ of the curve  $\{X_t \times X_t\}$  through  $Y$ .

*Proof.* Over  $\mathbb{C}$  the Hodge decomposition at  $c$  is  $H^{1,0}(X_c) = V_1 \otimes V_2 \otimes V_3^{1,0}$ , as at every point of  $C$ , since the Hodge cocharacter lies in the third factor. A maximal torus  $S$  of  $\text{SL}(V_1) \times \text{SL}(V_2)$  therefore preserves the Hodge decomposition of  $H^1(Y, \mathbb{C}) = V \oplus V$  and acts on  $HT^2(Y)$  and on  $H^*(Y, \mathbb{C})$ , and so does  $\tau$ . Contraction is equivariant for both, and  $\omega$  is fixed by both, so  $x \mapsto x \lrcorner \omega$  preserves the decomposition of  $HT^2(Y)$  by type ( $H^{0,2}$ ,  $H^1(T)$  or  $H^0(\wedge^2 T)$ ), which land in the distinct summands  $H^{2,4}$ ,  $H^{1,3}$  and  $H^{0,2}$  of  $H^*(Y)$ , by weight of  $S$ , and by parity under  $\tau$ . The annihilator is the sum of the kernels on the 46 nonzero blocks. Item (XLIV) of Appendix D writes each block as a matrix whose entries are linear in  $a$  and computes the Gram determinant of its columns, a polynomial in  $a$  that, for real  $a$ , vanishes exactly where the columns become dependent. Every Gram determinant is a positive constant times a product of the following factors, and the one

<sup>63</sup>For a cyclic cubic field  $\Gamma$  is the group of 3-cycles; otherwise it is the full symmetric group. The argument uses only that  $\Gamma$  contains the 3-cycles, which holds for every transitive subgroup of the symmetric group on three letters.

block with a kernel at a generic point is the  $\tau$ -even block of weight zero in  $H^1(T)$ , whose kernel is spanned, for every  $a$ , by a vector independent of  $a$ , which is dropped before the determinant is taken.

- (a) Eleven quadratic forms  $P_1, P_2, Q_1, \dots, Q_4, R, S_1, \dots, S_4$ , each a positive combination of squares of rational linear forms whose common zeros have two of  $a_1, a_2, a_3$  equal or one of them zero, for instance  $2P_1 = \sum_{i < j} (a_i - a_j)^2$ ,  $Q_4 = (3a_1 - a_0 - a_2 - a_3)^2 + 16(a_2 - a_3)^2$  and  $S_2 = (a_0 + a_2)^2 + 3(a_1 + a_3)^2 + 8a_2^2$ .
- (b) Six positive definite forms  $Z_1, \dots, Z_6$ , each a positive combination of the squares of four independent linear forms.
- (c) The linear forms  $a_2, a_3, a_0 - 3a_1 + a_2 - 3a_3, a_0 - 3a_1 - 3a_2 + a_3, a_0 + 9a_1 - 3a_2 - 3a_3$ , whose coefficients on  $a_1, a_2, a_3$  are not all equal.
- (d) The linear form  $L_0$ , which divides the Gram determinants of the nine  $\tau$ -odd blocks of  $H^0(\wedge^2 T)$  and no other.

The sums of squares are certified in exact arithmetic by item (XLIV) and by the Lean theorem `mumford_rigidity_sos`. For a rational exceptional class the coefficients are real, and by Lemma 21.19 the factors of (a), (b) and (c) do not vanish: those of (a) would force two conjugates of  $\alpha$  to coincide or one to vanish, those of (b) vanish only at  $a = 0$ , and those of (c) would give a rational relation among the conjugates with unequal coefficients. So every block is injective except the one named, and  $L_0$  decides the bivector blocks: if  $L_0(a) \neq 0$  they are injective, and if  $L_0(a) = 0$  the four blocks of dimension one, whose Gram determinants are  $L_0^2/16$ , lie in the annihilator.

The vector of the exceptional block is  $\kappa$ . The family of Hodge structures along  $C$  is the orbit of the Hodge structure at  $c$  under the real points of the third factor of  $G$ , so the Kodaira–Spencer class  $\kappa$  is the component of type  $(-1, 1)$  of an element of  $\mathfrak{sl}(V_3)$ , the lowering operator acting on both copies of  $V$ . It kills  $\omega$  because  $\omega$  is invariant under  $G$ , it is fixed by  $\tau$  and has weight zero for  $S$ , and item (XLIV) checks that it spans the kernel. This proves (i) and (ii).

For (iii), a point of  $\mathfrak{S}$  is a Hodge filtration  $F^1$  on  $H^1(Y, \mathbb{C})$ , and  $\omega$  is of Hodge type at it when  $\omega \in F^2 H^4$ , a holomorphic condition. For weight one every tangent direction of  $\mathfrak{S}$  at  $Y$  is an element  $\xi$  of  $\mathfrak{sp}(\psi \oplus \psi)^{-1,1} \subset H^1(T_Y)$ , and it moves  $F^2 H^4$  by the derivation  $\xi$  of  $\wedge^* H^1$ , which maps  $H^{2,2}$  into  $H^{1,3}$ ; so the Zariski tangent space of the Hodge locus is  $\{\xi : \xi \lrcorner \omega = 0\}$ , which is  $\mathbb{C}\kappa$  by (i).<sup>64</sup> The Hodge locus is an analytic germ containing the smooth curve germ  $\{X_t \times X_t\}$ , whose tangent line is  $\mathbb{C}\kappa$ , and its Zariski tangent space is that line; so the two germs are equal. ■

*Remark 21.22* (What the divisor classes do instead). The same computation at classes with  $\alpha \in \mathbb{Q}$  shows what rigidity excludes. At  $a = (3, 5, 5, 5)$ , a combination of products of divisor classes, the tangent space of the Hodge locus in  $\mathfrak{S}$  has dimension 10, and it contains the tangent space of the diagonal Siegel space  $\{X' \times X'\}$  of dimension 10, because that class is invariant under the whole of  $\mathrm{Sp}(V, \psi)$ ; at  $a = (1, 1, 1, 1)$  it has dimension 20. So the products of divisor classes move with  $X' \times X'$  for every  $X'$ , the exceptional classes move with  $X_t \times X_t$  for  $t \in C$  only, and every deformation of  $Y$  off the curve, however small, makes them cease to be of Hodge type. Figure 33 draws the three loci.

<sup>64</sup>This is Griffiths transversality in its simplest case: for a variation of weight one there is no horizontality condition, so the tangent space of the Hodge locus of a class of type  $(p, p)$  is the kernel of the contraction into  $H^{p-1, p+1}$ , see [Voi03, Chapter 5]. For a class of higher weight on a general variety only the horizontal directions enter.

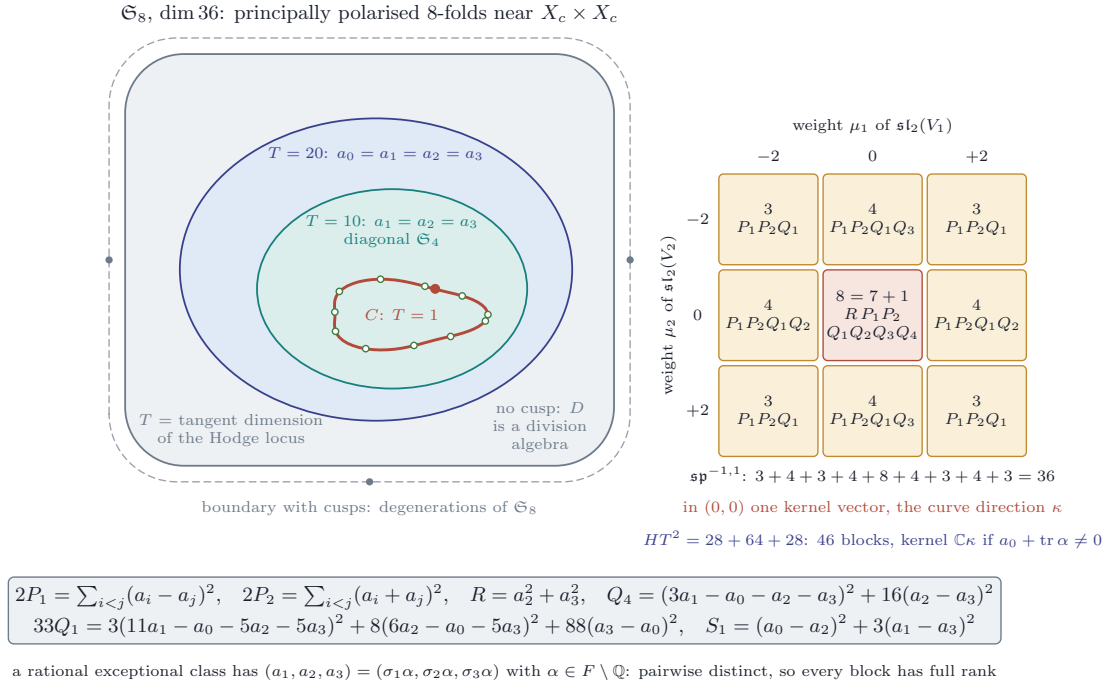


FIGURE 33. **Theorem 21.21.** Left: the Siegel space  $\mathfrak{S}$  of dimension 36 near  $Y = X_c \times X_c$  and three Hodge loci through it, labelled by the tangent dimension  $T$  at  $Y$ : the class with equal coefficients ( $T = 20$ ), the products of divisor classes with  $a_1 = a_2 = a_3$  ( $T = 10$ , the diagonal Siegel space), and a rational exceptional class ( $T = 1$ ), whose locus is the compact Mumford curve with its CM points (hollow). The dashed rim is the boundary of  $\mathfrak{S}$ , which the curve does not reach. Right: the nine weight blocks of  $\mathfrak{sp}^{-1,1}$  under the torus of the first two factors, with their dimensions and the forms whose product is the Gram determinant of each; only the block of weight  $(0, 0)$  has a kernel, the direction  $\kappa$  of the curve. The same holds in the 46 blocks of  $HT^2(Y)$  off the hyperplane  $L_0 = 0$ . Below: the sums of squares that make the determinants nonzero at the conjugates of an irrational element of  $F$ .

**Corollary 21.23** (No degeneration and no larger family). *Let  $\omega$  be a rational exceptional class on  $X_c \times X_c$ . Every connected complex analytic family of polarised abelian eightfolds through  $X_c \times X_c$  on which  $\omega$  stays of Hodge type lies, near each point of the curve, inside the curve  $\{X_t \times X_t : t \in C\}$ , which is compact. In particular no such family degenerates, and on no member of it other than the CM points of the curve is  $\omega$  a combination of products of divisor classes or of Weil classes of CM fields acting on the member or on its quotients.*

*Proof.* By **Theorem 21.21**(iii) the Hodge locus of  $\omega$  coincides with the curve in a neighbourhood of every point of the curve, so a connected family through  $Y$  on which  $\omega$  stays of Hodge type maps into the curve near each of its points over the curve; by connectedness and the identity theorem its image is the curve. The curve is compact because  $D$  is a

division algebra, so the arithmetic group of the Shimura curve is cocompact [Mum69].<sup>65</sup> For the last sentence: a member of the curve other than a CM point has Hodge group  $G$  (proof of Corollary 20.31), and on it  $\omega$  lies outside the span of the products of divisor classes and of the Weil classes of quotients, by Proposition 20.44(ii) and (iii). At a CM point the class is algebraic already (Theorem 20.30), and the countably many CM points do not carry algebraicity to the other members (Lemma 20.28). ■

The same computation controls deformations that are not algebraic at all, and so decides whether the twistor lines along which Markman moves sheaves for Weil classes can be used here.

**Proposition 21.24** (The Hodge locus among all complex tori, and twistor lines). *Let  $\omega$  be a rational exceptional class,  $Y = X_c \times X_c$  with  $c \in C$ , and  $\Lambda = H_1(Y, \mathbb{Z})$ . Let  $\mathcal{T}$  be the space of complex structures on the real torus  $\Lambda_{\mathbb{R}}/\Lambda$ , a complex manifold of dimension 64 with tangent space  $H^1(T_Y)$  at  $Y$ , and  $\mathcal{T}_{\omega} \subset \mathcal{T}$  the closed analytic subset on which  $\omega$  is of type  $(2, 2)$ .*

- (i) *The connected component of  $\mathcal{T}_{\omega}$  that contains  $Y$  is the universal cover  $\mathfrak{H}$  of the Mumford curve, the set of squares of members of the family, biholomorphic to the upper half plane. Every connected complex analytic family of complex tori, algebraic or not, through a member of the Mumford family on which  $\omega$  stays of Hodge type lies in the family.*
- (ii) *No twistor line through a member of the Mumford family, and no compact connected complex curve in  $\mathcal{T}$  through it, keeps  $\omega$  of Hodge type.*
- (iii)  *$\mathcal{T}_{\omega}$  contains twistor lines nevertheless. The algebra  $D \otimes_{F, \sigma_1} \mathbb{R}$  is Hamilton's quaternions and acts on  $V_{\mathbb{R}}$  through the first factor of  $G(\mathbb{R})$ ; for a unit quaternion  $j$  with  $j^2 = -1$ , the complex structure  $j \oplus j$  on  $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$  makes  $\omega$  of type  $(2, 2)$ , and these structures form a twistor line. On each factor  $\psi(x, jy)$  has signature  $(4, 4)$ , and a member whose Mumford–Tate group is  $G$  is not an abelian variety. These twistor lines lie in components of  $\mathcal{T}_{\omega}$  other than  $\mathfrak{H}$ .*

*Proof.* (i) The complex structures on  $\Lambda_{\mathbb{R}}$  for which all the tensors invariant under  $G$  are of Hodge type are those  $J$  with  $J^2 = -1$  in the real points of  $G \cdot \mathbb{G}_m$ , since a reductive group containing the homotheties is the stabiliser of its invariant tensors. Let  $\mathfrak{H}$  be the connected component of this closed set that contains  $Y$ ; it is closed in  $\mathcal{T}$  and consists of the  $J = 1 \otimes 1 \otimes j_3$  with  $j_3 \in \mathrm{SL}_2(\mathbb{R})$ ,  $j_3^2 = -1$ , in the half plane that contains the structure of  $Y$ , that is of the squares of the members of the Mumford family. At each of its points the first order condition for  $\omega$  to stay of type  $(2, 2)$  along  $\xi \in H^1(T_Y)$  is  $\xi \lrcorner \omega = 0$ , as in the proof of Theorem 21.21(iii), where no polarisation was used, and by Theorem 21.21(i) the solutions form the line  $\mathbb{C}\kappa$  tangent to  $\mathfrak{H}$ . So near each of its points  $\mathcal{T}_{\omega}$  is the smooth curve germ of  $\mathfrak{H}$ , and  $\mathfrak{H}$  is open in  $\mathcal{T}_{\omega}$ . Being also closed and connected, it is a connected component. A connected analytic family on which  $\omega$  stays of Hodge type maps into  $\mathcal{T}_{\omega}$ , hence into  $\mathfrak{H}$  once one of its points does.

(ii) A twistor line is a nonconstant holomorphic image of  $\mathbb{P}^1$  in  $\mathcal{T}$ . If  $\omega$  stayed of Hodge type along one through a point of  $\mathfrak{H}$ , or along any compact connected curve through it,

<sup>65</sup>By Setup 21.1 the algebra  $D$  is definite at  $\sigma_1$  and  $\sigma_2$ , so  $D \otimes_{F, \sigma_1} \mathbb{R}$  is Hamilton's quaternions and  $D$  is not  $M_2(F)$ : it is a division algebra, split at the one real place  $\sigma_3$ . The norm one group of an order in it, embedded in  $\mathrm{SL}_2(\mathbb{R})$  through  $\sigma_3$ , is a Fuchsian group of finite covolume with no parabolic element: a parabolic  $\gamma \neq \pm 1$  would make  $\gamma \mp 1$  a nonzero nilpotent element of  $D$ . A Fuchsian group of finite covolume without parabolic elements has compact quotient.

the curve would lie in  $\mathfrak{H}$  by (i), and a holomorphic map from a compact connected curve to the upper half plane is constant.

(iii) At  $\sigma_1$  the quaternion algebra  $D$  is definite (Setup 21.1), so the first factor of  $G(\mathbb{R})$  is the group of unit quaternions, and  $V_{\mathbb{R}}$  restricted to it is a module over  $D \otimes_{F, \sigma_1} \mathbb{R} \cong \mathbb{H}$ . The unit quaternions  $j$  with  $j^2 = -1$  form a two-sphere, and the complex structures given by left multiplication by them on a quaternionic vector space form a twistor line. Each  $j$  lies in  $G(\mathbb{R})$ , and so does the circle  $\cos \vartheta + j \sin \vartheta$ , which fixes  $\omega$ ; so  $\omega$  is of type  $(2, 2)$ . Item (XLVII) of Appendix D realises  $V_{\mathbb{R}}$  inside  $V_1 \otimes V_2 \otimes V_3$  as the fixed space of the real structure that is quaternionic on the first two factors and real on the third, and finds that  $\psi(x, Jy)$  is definite for the structures of the Mumford family and of signature  $(4, 4)$  for  $J = j \otimes 1 \otimes 1$ . At a member whose Mumford–Tate group is  $G$ , the Hodge classes of degree two on  $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$  are the forms  $\psi \otimes s$  with  $s$  a symmetric  $2 \times 2$  matrix, by Proposition 20.44(ii), and the signature of  $\psi(\cdot, J\cdot) \otimes s$  is  $(4(p+q), 4(p+q))$  when  $s$  has signature  $(p, q)$ , so none is definite, none is a polarisation, and the member is not an abelian variety. The members of the Mumford family are not on these lines, so by (i) the lines lie in other components. ■

Markman’s proof for Weil classes on abelian fourfolds moves a sheaf along twistor lines on which its Chern character stays of Hodge type. By Proposition 21.24 no such line passes through the square of a member of the Mumford family: the twistor lines on which  $\omega$  is of Hodge type exist, but they carry non-algebraic tori whose Hodge cocharacter lies in a definite factor of  $D$ , in components of the Hodge locus that the Mumford family never meets. The Kuga–Satake class of Theorem 21.4 behaves differently. The Kuga–Satake construction applies to every Hodge structure of K3 type on  $(T, q_{\lambda})$ , so that class stays of Hodge type over the whole period domain of  $(T, q_{\lambda})$ , of dimension seven, and not only along the curve; it is the one class attached to the target that deformation methods of this kind are not excluded from.

The same space  $T$  also makes the square a holomorphic symplectic variety, and this decides what the mechanism of van Geemen and Rapagnetta would have to supply here. They prove the Hodge conjecture for the general abelian fourfold of Weil type with trivial discriminant by pulling back the second Chern class of a hyperkähler sixfold along a map from the fourfold, which gives an algebraic class of codimension two that is not a product of divisor classes [vGR26]. For a projective hyperkähler manifold  $M$  write  $\sigma_M$  for its symplectic form,  $q_M$  for its Beauville–Bogomolov form and  $T(M)$  for the orthogonal complement of  $\text{NS}(M)_{\mathbb{Q}}$  in  $H^2(M, \mathbb{Q})$  for  $q_M$ .

**Proposition 21.25** (The square as a holomorphic symplectic variety). *Let  $c \in C$  be a point at which the Mumford–Tate group of  $X = X_c$  is  $G \cdot \mathbb{G}_m$ , let  $Y = X \times X$ , and let  $\Psi \in \bigwedge^2 V$  be the  $G$ -invariant bivector dual to  $\psi$ . For  $x \in T$  put  $\iota(x) = (x \otimes 1)\Psi$  in  $V \otimes V = H^1(X, \mathbb{Q}) \otimes H^1(X, \mathbb{Q})$ , a Künneth component of  $H^2(Y, \mathbb{Q})$ , and let  $\iota_2: S^2 T \rightarrow H^4(Y, \mathbb{Q})$  be the map  $xy \mapsto \iota(x) \cup \iota(y)$ . Let  $C_i \in S^2 T_i$  be the Casimir element of the trace form of  $\mathfrak{sl}(V_i)$ , and for  $\nu \in F$  let  $c_{\nu} = \sum_i \sigma_i(\nu) C_i$ , a rational element of  $S^2 T$ ; for  $\nu \neq 0$  it is the element dual to the form  $q_{1/\nu}$ .*

- (i)  $\iota$  is an embedding of Hodge structures  $T(-1) \rightarrow H^2(Y, \mathbb{Q})$ , its image is the only irreducible sub-Hodge structure of  $H^2(Y, \mathbb{Q})$  with  $h^{2,0} = 1$ , and the holomorphic 2-form  $\sigma_0$  spanning its part of type  $(2, 0)$  is symplectic.
- (ii)  $\iota_2(c_{\nu}) = \omega_a$  with  $a_0 = 3 \text{Tr}_{F/\mathbb{Q}}(\nu)$  and  $\alpha = 4\nu - \text{Tr}_{F/\mathbb{Q}}(\nu)$ , in the notation of Setup 21.18. So  $\iota_2(c_{\nu})$  is exceptional if and only if  $\nu \notin \mathbb{Q}$ , and every rational exceptional class is some  $\iota_2(c_{\nu})$  plus a rational multiple of  $\pi_0$ .

- (iii) Let  $M$  be a projective hyperkähler manifold and  $f: Y \dashrightarrow M$  a rational map with  $f^*\sigma_M \neq 0$ . Then  $f^*$  maps  $T(M)$  isomorphically onto  $\iota(T)$  and is compatible with products of classes in  $T(M)$ ;  $f$  is generically finite onto a subvariety of dimension eight on which  $\sigma_M$  is generically nondegenerate;  $b_2(M) \geq 10$ ; and  $\dim M \geq 10$ .
- (iv) Suppose in (iii) that  $M = S^{[n]}$  is the Hilbert scheme of  $n$  points on a projective K3 surface  $S$ , and let  $q_\lambda$  be the form on  $T$  that corresponds to the intersection form of  $S$  on  $T(S) = T(M)$  under  $\iota^{-1}f^*$  (Proposition 21.2(iii)). If  $\lambda \notin \mathbb{Q}$ , or if the real multiplication of  $T(S)$  is induced by algebraic cycles on  $S \times S$ , then the Hodge conjecture holds for  $Y$ .

*Proof.* (i) The map  $\iota$  is  $G$ -equivariant because  $\Psi$  is invariant, and injective. Its image lies in  $S^2V$ : for  $x$  in the symplectic Lie algebra of  $\psi$  one has  $(x \otimes 1)\Psi = -(1 \otimes x)\Psi$ , and the exchange of the two factors carries  $(x \otimes 1)\Psi$  to  $-(1 \otimes x)\Psi$  because  $\Psi$  is alternating. An element  $g$  of  $G \cdot \mathbb{G}_m$  multiplies  $\Psi$  by the multiplier  $\chi(g)$  of  $\psi$ , so  $(g \otimes g)\iota(x) = \chi(g)\iota(gxg^{-1})$ : the map  $\iota$  is equivariant for the adjoint action on  $T$  twisted by  $\chi$ , which is the action defining  $T(-1)$ . Both Hodge structures come from the Hodge cocharacter through  $G \cdot \mathbb{G}_m$ , so  $\iota$  is a morphism of Hodge structures from  $T(-1)$ . The sub-Hodge structures of  $H^2(Y, \mathbb{Q})$  are its  $G$ -stable subspaces, since  $\mathbb{G}_m$  acts by scalars. Over  $\bar{\mathbb{Q}}$ , item (XLVIII) of Appendix D splits  $H^2(Y) = \Lambda^2(V \oplus V)$  by the three Casimir operators into the isotypic pieces of highest weights  $(0, 0, 0)$ ,  $(2, 2, 0)$ ,  $(2, 0, 2)$ ,  $(0, 2, 2)$ ,  $(2, 2, 2)$ ,  $(2, 0, 0)$ ,  $(0, 2, 0)$ ,  $(0, 0, 2)$ , of dimensions 3, 27, 27, 27, 27, 3, 3, 3, whose parts of type  $(2, 0)$  have dimensions 0, 0, 9, 9, 9, 0, 0, 1; the last three pieces are  $\iota(T_1)$ ,  $\iota(T_2)$ ,  $\iota(T_3)$ . So an irreducible summand of highest weight  $(2, 0, 2)$  or  $(0, 2, 2)$  contributes 3 to  $h^{2,0}$ , one of weight  $(2, 2, 2)$  contributes 9, one of weight  $(0, 0, 2)$  contributes 1, and the others contribute nothing. A rational sub-Hodge structure  $W$  is stable under the Galois group, which permutes the highest weights as it permutes the factors, transitively. So if  $W \otimes \bar{\mathbb{Q}}$  has a summand of weight  $(2, 2, 0)$ ,  $(2, 0, 2)$  or  $(0, 2, 2)$ , it has summands of all three and  $h^{2,0}(W) \geq 6$ ; if it has one of weight  $(2, 2, 2)$ , then  $h^{2,0}(W) \geq 9$ . If  $h^{2,0}(W) = 1$ , therefore,  $W \otimes \bar{\mathbb{Q}}$  contains the piece  $\iota(T_3)$  of weight  $(0, 0, 2)$ , which has multiplicity one, and with it  $\iota(T_1)$  and  $\iota(T_2)$ ; so  $W \supseteq \iota(T)$ , and  $W = \iota(T)$  if  $W$  is irreducible. The part of type  $(2, 0)$  of  $\iota(T)$  is spanned by  $\sigma_0 = \iota(E_3)$ , where  $E_3 \in \mathfrak{sl}(V_3) \otimes \mathbb{C}$  maps  $V_3^{0,1}$  onto  $V_3^{1,0}$ , and item (XLVIII) finds that  $\sigma_0$  pairs  $H^{1,0}$  of the first factor with  $H^{1,0}$  of the second through an invertible matrix. So  $\sigma_0$  has rank eight: it is a holomorphic symplectic form on  $Y$ .

(ii) Item (XLVIII) computes  $\iota_2(C_1) = 3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23}$  and the two cyclic permutations of this identity, in the split model; both sides are built from  $\psi$  and the actions of the three factors, so the identities hold over  $\bar{\mathbb{Q}}$  in Setup 21.1. Summing with the coefficients  $\sigma_i(\nu)$  gives  $\iota_2(c_\nu) = 3 \operatorname{Tr}(\nu)\pi_0 + \sum \sigma_k(4\nu - \operatorname{Tr}(\nu))\pi_{ij}$ , the sum over  $\{i, j, k\} = \{1, 2, 3\}$  with  $i < j$ , which is  $\omega_a$  with the stated  $a_0$  and  $\alpha$ . Since  $\operatorname{Tr}(\nu) \in \mathbb{Q}$ , the element  $\alpha$  is rational exactly when  $\nu$  is; and  $\operatorname{Tr}(\alpha) = \operatorname{Tr}(\nu)$ , so  $\nu = (\alpha + \operatorname{Tr}(\alpha))/4$  recovers  $\nu$  from any  $\alpha \in F$ . Two rational classes with the same  $\alpha$  differ by a multiple of  $\pi_0$ . The Casimir element  $C_i$  is dual to the trace form on  $T_i$ , so  $c_\nu$  is dual to  $\sum_i \sigma_i(\nu)^{-1} \operatorname{tr}(x_i y_i) = q_{1/\nu}(x, y)$ .

(iii) Resolve the indeterminacy of  $f$  by a sequence of blow-ups along smooth centres,  $\pi: \tilde{Y} \rightarrow Y$  and  $\tilde{f}: \tilde{Y} \rightarrow M$ , and put  $f^* = \pi_* \tilde{f}^*$ , a morphism of Hodge structures. The kernel of  $\pi_*$  on  $H^2(\tilde{Y}, \mathbb{Q})$  is spanned by the classes of the exceptional divisors, so  $\alpha \mapsto \tilde{f}^* \alpha - \pi^* f^* \alpha$  is a morphism of Hodge structures from  $H^2(M, \mathbb{Q})$  to a space of Hodge classes. The Hodge structure  $T(M)$  is irreducible. A sub-Hodge structure  $W \subseteq T(M)$  with  $W^{2,0} = 0$  consists of rational classes of type  $(1, 1)$ , which lie in  $\operatorname{NS}(M)_{\mathbb{Q}} \cap T(M) = 0$ ; if  $W^{2,0} \neq 0$ , then  $\sigma_M \in W_{\mathbb{C}}$ , the orthogonal complement of  $W$  in  $H^2(M, \mathbb{Q})$  is a sub-Hodge structure orthogonal to  $\sigma_M$ .

and  $\bar{\sigma}_M$ , hence of type  $(1, 1)$  and contained in  $\text{NS}(M)_{\mathbb{Q}}$ , and  $W \supseteq T(M)$ . It is not spanned by Hodge classes, so that morphism vanishes on it, and  $\tilde{f}^*\alpha = \pi^*f^*\alpha$  for  $\alpha \in T(M)$ . Since  $\tilde{f}^*$  is a ring homomorphism and  $\pi_*\pi^*$  is the identity,  $f^*(\alpha\beta) = f^*\alpha \cup f^*\beta$  for  $\alpha, \beta \in T(M)$ . The kernel of  $f^*$  on  $T(M)$  is a sub-Hodge structure not containing  $\sigma_M$ , hence zero, and the image is irreducible with  $h^{2,0} = 1$ , hence  $\iota(T)$  by (i). So  $f^*\sigma_M$  is a nonzero multiple of  $\sigma_0$  and has rank eight at every point of  $Y$  where  $f$  is defined; the rank of a pulled-back form is at most the dimension of the image, so  $f$  is generically finite onto its image  $Z$ , of dimension eight, and  $\sigma_M$  is generically nondegenerate on  $Z$ . As  $M$  is projective,  $b_2(M) = \dim T(M) + \dim \text{NS}(M)_{\mathbb{Q}} \geq 9 + 1$ ; this excludes the generalised Kummer varieties, whose second Betti number is seven [Bea83]. Finally  $\dim M \geq \dim Z = 8$ , and suppose  $\dim M = 8$ . Then  $f$  is dominant, and for  $\alpha \in T(M)_{\mathbb{C}}$  the relation of Fujiki and Beauville [Bea83, Fuj87] gives

$$\int_Y (f^*\alpha)^8 = \int_{\tilde{Y}} \tilde{f}^*(\alpha^8) = \deg(f) \int_M \alpha^8 = \deg(f) c_M q_M(\alpha)^4, \quad c_M > 0.$$

Write  $f^*\alpha = \iota(x)$  with  $x = x_1 + x_2 + x_3 \in T_{\mathbb{C}}$  and  $N_i = -\det x_i$ . The form  $x \mapsto q_M((f^*)^{-1}\iota(x))$  is invariant under  $G$ , so it is  $q_{\lambda}(x, x) = \sum_i \sigma_i(\lambda) \text{tr}(x_i^2) = 2 \sum_i \sigma_i(\lambda) N_i$  for some  $\lambda \in F^{\times}$  by Proposition 21.2(iii). On the other side, item (XLVIII) finds  $\iota(x)^8 = 8! \det(x)$  times a volume form, and  $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$  with  $\Delta = N_1^2 + N_2^2 + N_3^2 - 2N_1N_2 - 2N_1N_3 - 2N_2N_3$ ; this is also the product of the eight numbers  $\pm r_1 \pm r_2 \pm r_3$ , where  $\pm r_i$  are the eigenvalues of  $x_i$  and  $r_i^2 = N_i$ . The  $N_i$  are algebraically independent functions on  $T_{\mathbb{C}}$ , so  $\Delta^2$  would be a constant multiple of the fourth power of the linear form  $\sum_i \sigma_i(\lambda) N_i$ . But  $\Delta$  is a nondegenerate ternary quadratic form, hence irreducible, and a contradiction results. So  $\dim M \geq 10$ .

(iv) By [Bea83] the map  $H^2(S, \mathbb{Q}) \rightarrow H^2(S^{[n]}, \mathbb{Q})$  induced by the universal subscheme  $Z_n \subset S \times S^{[n]}$  is injective, carries the intersection form to the Beauville–Bogomolov form, and has  $H^2(S^{[n]}, \mathbb{Q}) = H^2(S, \mathbb{Q}) \oplus \mathbb{Q}\delta$  with  $\delta$  algebraic, so  $T(M) = T(S)$ . Let  $\Delta_T$  be the Künneth component in  $T(S) \otimes T(S)$  of the class of the diagonal of  $S$ ; it is algebraic, being the class of the diagonal minus  $[\text{pt} \times S]$ ,  $[S \times \text{pt}]$  and  $\sum_r d_r \times d_r^{\vee}$  over dual bases of  $\text{NS}(S)_{\mathbb{Q}}$ . Its image under the correspondence  $Z_n \times Z_n$ , pulled back along the diagonal of  $M$ , is the class  $\sum_a e_a \cup e_a^{\vee}$  over dual bases of  $T(S)$  for the intersection form, which is therefore algebraic on  $M$ , and by (iii) its image under  $f^*$  is  $\iota_2$  of the element of  $S^2T$  dual to  $q_{\lambda}$ , that is  $\iota_2(c_{1/\lambda})$ , an algebraic class on  $Y$ . If  $\lambda \notin \mathbb{Q}$  it is exceptional by (ii). If instead the real multiplication is algebraic, apply it to the first factor of  $\Delta_T$ : for  $b \in F$  the class  $\sum_a (m_b e_a) \cup e_a^{\vee}$  is algebraic, and its image is  $\iota_2(c_{b/\lambda})$ , which is exceptional for  $b \notin \mathbb{Q}\lambda$ . In both cases Lemma 21.20 makes every Hodge class in  $H^2(X) \otimes H^2(X)$  algebraic; the Hodge classes in the other Künneth components of  $H^4(Y, \mathbb{Q})$  are products of divisor classes, as at the end of the proof of Theorem 21.4; and the Hodge classes of  $Y$  form a ring generated in degrees two and four (proof of Corollary 20.31). ■

*Remark 21.26* (What the mechanism of van Geemen and Rapagnetta would need). By Proposition 21.25(ii) the exceptional classes are the classes  $\iota_2(c_{\nu})$  with  $\nu \notin \mathbb{Q}$ , the images of the elements dual to the forms  $q_{\lambda}$ ,  $\lambda = 1/\nu$ , on the part  $\iota(T)$  of  $H^2(Y)$  generated by the symplectic class; the form with  $\lambda$  rational gives a product of divisor classes. They are the analogues for  $(Y, \sigma_0)$  of the dual Beauville–Bogomolov class, whose pull-back is what [vGR26] uses. Part (iii) says what a transfer of that argument needs: a projective hyperkähler manifold of dimension at least ten with transcendental part  $T(-1)$ , such as  $S_{\mu}^{[n]}$  with  $n \geq 5$  for a surface  $S_{\mu}$  of Theorem 21.4, and a rational map from  $Y$  onto a symplectic subvariety of dimension eight. For  $M = S_{\mu}^{[n]}$  the form of (iv) is  $q_{\mu/e^2}$  for the  $e \in F^{\times}$  with

$\iota^{-1}f^* = m_e$  on  $T(S_\mu) = T$ , so  $\lambda \notin \mathbb{Q}$  for every such map as soon as  $\mu \notin \mathbb{Q}^\times F^{\times 2}$ . No such map is known. It would give an algebraic cycle on  $S_\mu \times X \times X$  inducing an isomorphism of  $T(S_\mu)$  onto  $\iota(T) \subset H^1(X) \otimes H^1(X)$ , a correspondence of the kind that the Kuga–Satake class supplies in the proof of [Theorem 21.4](#); so this is the Kuga–Satake route in geometric form, not an independent one. Maps to K3 surfaces and to hyperkähler eightfolds cannot serve, by (iii).

**Theorem 21.27** (The Mumford criterion is a dimension). *Let  $c \in C$ ,  $Y = X_c \times X_c$ ,  $\omega = \omega_a$  a rational exceptional class with  $L_0(a) \neq 0$ , and  $E$  a perfect complex on  $Y$  with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ , and no other component.*

- (i)  $\dim \text{Ext}^2(E, E) \geq 119$ .
- (ii) *If  $\dim \text{Ext}^2(E, E) = 119$ , then  $\text{ev}_E: HT^2(Y) \rightarrow \text{Ext}^2(E, E)$  is surjective with kernel  $\mathbb{C}\kappa$ , and  $\sigma_E$  is injective. Conversely, if  $\sigma_E$  is injective, then  $\text{ev}_E$  has rank 119 and kernel  $\mathbb{C}\kappa$ , and  $E$  extends to first order along the curve.*
- (iii) *If some such  $E$  at one point  $c$  has  $\dim \text{Ext}^2(E, E) = 119$ , then  $\omega_t$  is algebraic for every  $t \in C$ , the Hodge conjecture holds for  $X_t \times X_t$  for every  $t$ , and  $B(W \times_C W)$  holds.*
- (iv) *At every CM point  $c$  some perfect complex  $E$  on  $Y$  has  $\text{ch}(E) = N\omega$  for some  $N \neq 0$  and no other component.*

*Proof.* (i) and (ii). By the square  $\sigma_E \circ \text{ev}_E = (\cdot) \lrcorner \text{ch}(E)$  of [Setup 18.27](#),  $\ker \text{ev}_E \subseteq \text{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$  by [Theorem 21.21\(ii\)](#), so  $\text{ev}_E$  has rank at least 119, which is (i). If  $\dim \text{Ext}^2(E, E) = 119$ , then  $\text{ev}_E$  is onto, its kernel has dimension one and lies in  $\mathbb{C}\kappa$ , so equals it; and  $\sigma_E(\text{ev}_E(x)) = N x \lrcorner \omega$  vanishes only for  $x \in \mathbb{C}\kappa = \ker \text{ev}_E$ , so  $\sigma_E$  is injective on  $\text{im } \text{ev}_E = \text{Ext}^2(E, E)$ . Conversely, if  $\sigma_E$  is injective, then  $\sigma_E(\text{ev}_E(\kappa)) = N\kappa \lrcorner \omega = 0$  forces  $\text{ev}_E(\kappa) = 0$ , so the kernel is  $\mathbb{C}\kappa$  and the rank is 119; and  $\text{ev}_E$  restricted to  $H^1(T_Y)$  is the obstruction map [\[Tod09\]](#), which vanishes at  $\kappa$ .

(iii) By (ii)  $\sigma_E$  is injective. The Chern character of  $E$  is  $N\omega$ , a flat section of Hodge type over the whole of  $C$ , so [Theorem 16.27](#) applies to every infinitesimal deformation of  $Y$  inside  $W \times_C W \rightarrow C$ . The proof of [Theorem 16.28](#) now applies word for word, with the stack of perfect complexes on the fibres of  $W \times_C W \rightarrow C$  [\[Lie06\]](#) in place of that of  $\mathcal{A} \rightarrow \mathcal{S}$ , or the analytic Kuranishi germ if  $\text{Ext}^{<0}(E, E) \neq 0$ : it is smooth over  $C$  at  $[E]$ , its image contains a nonempty open subset  $U \subseteq C$ , and for  $t \in U$  a perfect complex  $E_t$  on  $X_t \times X_t$  in the same connected component has  $\text{ch}(E_t) = N\omega_t$ , so  $\omega_t$  is algebraic, and by [Lemma 21.20](#) so are both exceptional classes.  $U$  is uncountable, and [Corollary 20.31](#) gives the rest.

(iv) By [Theorem 20.30](#) there is  $\xi \in \text{CH}^2(Y)_\mathbb{Q}$  with class  $\omega$ , and the argument of [Proposition 16.26](#) gives  $E$ : the Chern character is an isomorphism  $K_0(Y)_\mathbb{Q} \rightarrow \text{CH}^*(Y)_\mathbb{Q}$  [\[Ful98, Ex. 15.2.16\(b\) and Cor. 18.3.2\]](#), so  $\xi = \text{ch}(u)$  for a unique  $u$ , and  $Nu$  is the class of a perfect complex for a suitable  $N$ . ■

Part (iii) is the Mumford analogue of [Theorem 16.36\(iii\)](#): one object, at one point, and one number, which is 119 here and  $2n(2n-1)$  for a Weil family. For a Weil family that number is attained by no object ([Theorem 17.5](#)), because an object attaining it would have to deform to non-algebraic tori; the classes here are rigid, so that argument does not apply ([Remark 17.6](#)). The objects of (iv) exist, and the only open question is whether one of them, or another representative of the same class in  $K_0$ , attains the bound of (i). We have not found one, and we do not claim that one exists. The rigidity of [Theorem 21.21](#) is not an obstacle to it: semiregularity moves an object along the Hodge locus of its Chern

character, and the curve is exactly that locus, so a semiregular object would propagate along the curve and nowhere else, which is all that is needed.

**Corollary 21.28** (Bounded data at CM points). *Fix a relatively ample line bundle on  $W \times_C W \rightarrow C$ , and let  $\omega$  be a rational exceptional class. The following are equivalent.*

- (a)  $\omega_t$  is algebraic for every  $t \in C$ ; equivalently, by [Lemma 21.20](#) and [Corollary 20.31](#), the Hodge conjecture holds for  $X_t \times X_t$  for every  $t \in C$ .
- (b) There are a finite set  $\Phi$  of pairs  $(\underline{P}, \underline{q})$ , each a tuple of Hilbert polynomials with a tuple of rational numbers of the same length, and infinitely many points  $c \in C$  at each of which  $\omega_c = \sum_i q_i [Z_i]$  for subschemes  $Z_i \subset X_c \times X_c$  with Hilbert polynomials  $P_i$ , for some  $(\underline{P}, \underline{q}) \in \Phi$ .

*In particular the classes are algebraic on every member as soon as they are represented at infinitely many CM points by cycles of bounded degree.*

*Proof.* For a pair  $(\underline{P}, \underline{q})$  let  $\Sigma_{\underline{P}, \underline{q}} \subseteq C$  be the set of points at which  $\omega$  is represented with these data. The proof of [Theorem 16.4](#) applies verbatim to  $(W \times_C W, C, \omega)$ , as in [Lemma 20.28](#), and shows that each  $\Sigma_{\underline{P}, \underline{q}}$  is Zariski closed. If (b) holds, one pair of  $\Phi$  serves at infinitely many points, so its  $\Sigma$  is an infinite closed subset of the irreducible curve  $C$ , hence  $C$ , and (a) holds. If (a) holds,  $C$  is the union of the countably many  $\Sigma_{\underline{P}, \underline{q}}$ , and  $C$  is uncountable, so one of them is infinite, hence equal to  $C$ ; then (b) holds with  $\Phi$  that one pair and every point of  $C$ . ■

[Theorem 20.30](#) gives a representation at every CM point, through Markman’s theorem for abelian fourfolds [[Mar25a](#)] and the ring of [Theorem 20.30](#), with no control of degree, and isogenies between CM members, which move CM points along their Hecke orbits, do not bound it either. [Corollary 21.28](#) says that such a bound is not a convenience but the whole remaining content: with it the target falls, and without it the CM points, being countable, say nothing about a very general member ([Lemma 20.28](#) needs uncountably many points).

*Remark 21.29* (The bypass mechanisms, one by one). [Figure 34](#) collects what is known about each route to the Mumford target that avoids the Lefschetz standard conjecture.

- (1) *Degeneration.* Blocked: the base is compact and the Hodge locus is the base ([Corollary 21.23](#)), so no family on which  $\omega$  stays of Hodge type approaches the boundary, and there is no limit mixed Hodge structure to specialise to.
- (2) *Deformation into a larger Hodge locus,* on which  $\omega$  would become a product of divisor classes or a Weil class and be algebraic by an earlier theorem, then be carried back. Blocked by [Theorem 21.21](#)(iii), already to first order.
- (3) *Hochschild deformations:* twisting  $D^b(Y)$  by a  $B$ -field or a Poisson bivector, as in the gerby and noncommutative deformations used for Weil classes. Blocked: an object with  $\text{ch}(E) = N\omega$  extends along no direction of  $HH^2(Y)$  except that of the curve ([Corollary 21.36](#)(i)).
- (4) *Constructions from line bundles:* cones, tensor products, duals, pull-backs and push-forwards along homomorphisms, and Fourier–Mukai functors whose kernels are built in the same way. Blocked at every point of the curve: everything so built has a Chern character invariant under the Lefschetz group, and the exceptional classes are not ([Theorem 21.33](#)).
- (5) *Twistor and hyperholomorphic deformation,* the mechanism of Markman’s proof for Weil classes: move a sheaf along twistor lines on which its Chern character stays

of Hodge type. Blocked on the square: no twistor line through a member of the Mumford family keeps  $\omega$  of Hodge type, and those that do carry non-algebraic tori, in other components of the Hodge locus (Proposition 21.24).

- (6) *Specialisation*. The class is algebraic at every CM point (Theorem 20.30), on every power of  $X_c$  [Li26, Theorem 1.3]; this suffices exactly when the degrees are bounded (Corollary 21.28), and no bound is known.
- (7) *Reduction modulo  $p$* . At every closed point of every reduction of the curve the class is a  $\mathbb{Q}_\ell$ -combination of algebraic cycles (Proposition 21.37), by Li's theorem [Li26, Theorem 1.1]; this suffices, with no lifting step, exactly when the Hilbert data of the cycles are bounded on a Zariski dense set of closed points of the arithmetic model (Theorem 21.38), and no bound is known.
- (8) *Semiregularity*. Reduced to one dimension count at one CM point (Theorem 21.27); an object attaining it has large odd self-extensions (Theorem 21.31), is not built from line bundles (Proposition 21.32) and splits only in a constrained way (Corollary 21.36); no such object is known.
- (9) *Kuga–Satake*. Reduced to the Kuga–Satake class of one K3 surface of Picard number thirteen (Theorem 21.4), which the known cases do not reach (Remark 21.5). Its geometric form is the mechanism of van Geemen and Rapagnetta [vGR26], pull-back of the dual Beauville–Bogomolov class of a hyperkähler manifold: the exceptional classes are exactly such twisted dual forms of the part of  $H^2$  of the square generated by its symplectic class, and a rational map onto a symplectic subvariety of some  $S_\mu^{[n]}$ ,  $n \geq 5$ , would suffice; K3 surfaces and hyperkähler eightfolds cannot serve (Proposition 21.25, Remark 21.26).
- (10) *Motivated classes*. Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], so  $\omega_t$  is motivated for every  $t$ ; a motivated class is algebraic once the Lefschetz involutions of the auxiliary varieties are (Proposition 20.17), and for the auxiliary varieties of André's proof that is not known. Along the family the auxiliary variety that suffices is  $W \times_C W$  (Theorem 20.29), and its Lefschetz standard conjecture is equivalent to the target (Corollary 20.31): the route moves the problem and does not shorten it.

Five of the ten are closed off by theorems; two work at special points only, the CM points and the closed points of the reductions modulo  $p$ , and each of them suffices exactly when a degree bound holds; two are reduced to a single object, one Kuga–Satake class or one perfect complex with  $\dim \text{Ext}^2 = 119$  at one CM point; and one is equivalent to the target. None of them is carried out here, and the target remains open. The five that are not closed off are one statement and two stronger inputs, not five obligations: any one of them would suffice (Corollary 21.40, Remark 21.41).

**21.3. What an object meeting the Mumford criterion must look like, and the target modulo  $p$ .** Theorem 21.27 reduces the semiregularity route to one object at one CM point. This subsection derives what any object with the right Chern character must satisfy, whether or not it is semiregular: no sequence of standard operations produces it from line bundles, it extends along no first order deformation of  $D^b(Y)$  other than the direction of the curve, a semiregular one splits only into summands with no  $\text{Ext}^2$  between them, and its self-extensions are large in every degree, with an odd part that the Euler characteristic forces beyond what Hochschild cohomology supplies. It then turns to the reduction modulo  $p$ , after the recent proof of the Tate conjecture for powers of abelian

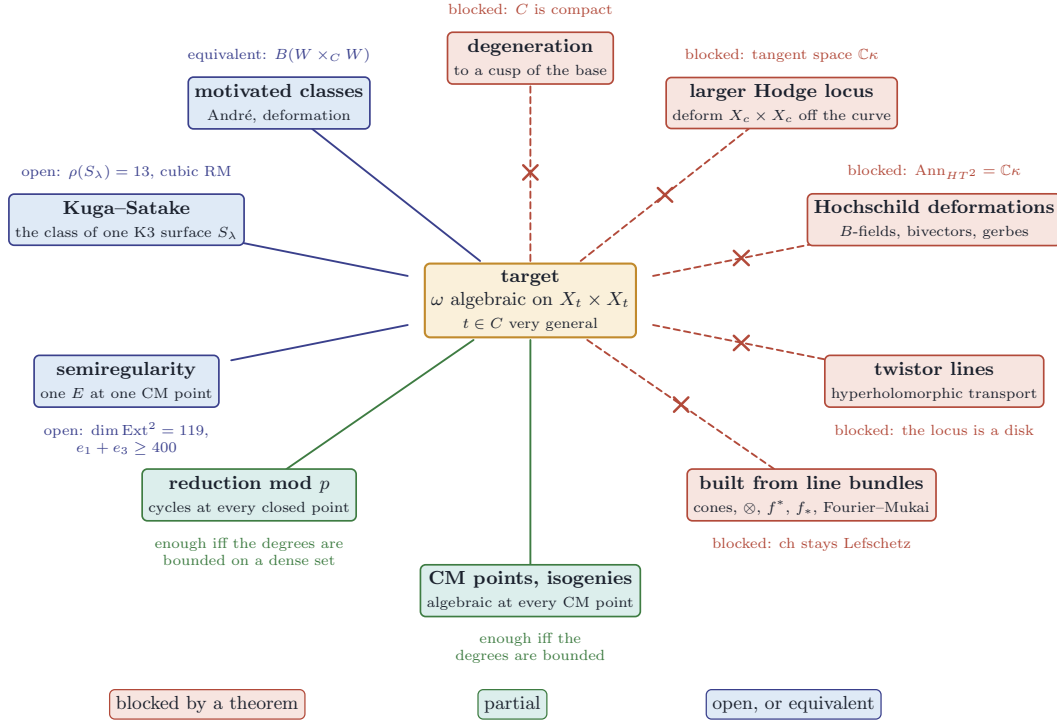


FIGURE 34. **Remark 21.29:** the ten routes to the Mumford target that avoid the Lefschetz standard conjecture. Clay, with a crossed dashed spoke: a route closed off by a theorem of [Section 21.2](#) or [Section 21.3](#). Grass: a route that works at special points only, at the CM points or at the closed points of the reductions modulo  $p$ , and suffices exactly when a degree bound holds ([Corollary 21.28](#), [Theorem 21.38](#)). Indigo: a route reduced to one open statement, of the same kind as the target or equivalent to it.

fourfolds over finite fields: the classes are represented by cycles at every closed point of every reduction of the curve, and the target is equivalent to a bound on those cycles on a Zariski dense set of closed points. None of this constructs the object or closes the target.

**Lemma 21.30** (The contraction ranks are symmetric). *Let  $A$  be an abelian variety of dimension  $g$  and  $c \in H^*(A, \mathbb{C})$  a class each of whose components has Hodge type  $(p, p)$  for some  $p$ . For  $k \geq 0$  put  $r_k(c) = \dim(HT^k(A) \lrcorner c)$ , with the action of [Setup 16.32](#). Then  $r_k(c) = r_{g-k}(c)$  for  $0 \leq k \leq g$ , and  $r_k(c) = 0$  for  $k > g$ .*

*Proof.* Write  $H^{1,0} = H^{1,0}(A)$  and  $A' = H^{0,1}(A)$ , both of dimension  $g$ , so that  $H^*(A, \mathbb{C}) = \Lambda^* H^{1,0} \otimes \Lambda^* A'$  and  $HT^*(A) = \Lambda^*(A' \oplus T)$  with  $T = (H^{1,0})^\vee$ . Fix a generator of  $\Lambda^g A'$  and let  $\star: \Lambda^q A' \rightarrow \Lambda^{g-q} A'^\vee$  send  $\eta$  to  $\zeta \mapsto \eta \wedge \zeta$ . Under  $\Phi = \text{id} \otimes \star$ , wedging with  $z \in A'$  becomes, up to a sign fixed on each bigraded piece, contraction with  $z$  on the second factor, while  $T$  still contracts the first. So  $\Phi$  identifies the action of  $HT^*(A)$  with the action of  $\Lambda^* U^\vee$  by interior products on  $\Lambda^* U$ , where  $U = H^{1,0} \oplus A'^\vee$  has dimension  $2g$ , up to signs that do not change ranks. A class of type  $(p, p)$  goes to  $\Lambda^p H^{1,0} \otimes \Lambda^{g-p} A'^\vee \subset \Lambda^g U$ , so  $\eta = \Phi(c)$  lies in the middle degree whatever the  $p$  of its components. Hence  $r_k(c)$  is the rank of  $\Lambda^k U^\vee \rightarrow \Lambda^{g-k} U$ ,  $x \mapsto \iota_x \eta$ , which is zero for  $k > g$ . Its transpose for the natural

pairings is  $\bigwedge^{g-k} U^\vee \rightarrow \bigwedge^k U$ ,  $y \mapsto \pm \iota_y \eta$ , because  $\langle \iota_x \eta, y \rangle = \langle \eta, x \wedge y \rangle = \pm \langle \iota_y \eta, x \rangle$ . A map and its transpose have the same rank. ■

**Theorem 21.31** (The self-extensions of an object with an exceptional Chern character). *Let  $c \in C$  and  $Y = X_c \times X_c$ , and write  $e_k = \dim \operatorname{Ext}^k(E, E)$  for a perfect complex  $E$  on  $Y$ .*

- (i) *For every  $E$  and every  $k$ ,  $e_k \geq r_k(\operatorname{ch} E)$  and  $e_k = e_{8-k}$ .*
- (ii) *For a rational exceptional class  $\omega = \omega_a$  with  $L_0(a) \neq 0$ :  $r_0 = r_8 = 1$ ,  $r_1 = r_7 = 16$  and  $r_2 = r_6 = 119$ .*
- (iii) *There is a set of rational exceptional classes, Zariski dense in the space of rational Hodge classes in  $H^2 \otimes H^2$ , on which*

$$(r_0, \dots, r_8)(\omega) \geq (1, 16, 119, 328, 560, 328, 119, 16, 1)$$

*term by term. When  $F$  is the cubic subfield of  $\mathbb{Q}(\zeta_7)$  it contains the class with  $a_0 = 2$  and  $(a_1, a_2, a_3)$  the conjugates of  $\zeta_7 + \zeta_7^{-1}$ .*

- (iv) *Let  $\omega$  be in the set of (iii) and  $\operatorname{ch}(E) = N\omega$  with  $N \neq 0$ . Then*

$$e_1 + e_3 = e_0 + e_2 + \frac{1}{2}e_4 + \sum_{j \geq 1} (-1)^j e_{-j}.$$

*If  $\operatorname{Ext}^{<0}(E, E) = 0$ , for instance if  $E$  is a sheaf, then  $\sum_{k \text{ odd}} e_k = \sum_{k \text{ even}} e_k \geq 800$ ,  $e_1 + e_3 \geq 400$ , and  $\sum_k e_k \geq 1600$ , while the bounds of (iii) add up to 1488: the odd degrees carry at least 112 dimensions beyond the Hochschild bound  $2(16 + 328) = 688$ .*

*Proof.* (i) The square  $\sigma_E \circ \operatorname{ev}_E = (\cdot) \lrcorner \operatorname{ch}(E)$  of [Setup 16.32](#) holds in every degree, so the kernel of  $\operatorname{ev}_E$  on  $HT^k(Y)$  lies in the annihilator of  $\operatorname{ch}(E)$  and its image has dimension at least  $r_k(\operatorname{ch} E)$ . Serre duality with the trivial canonical bundle gives  $e_k = e_{8-k}$ .

(ii)  $r_0 = 1$  because  $\omega \neq 0$ , and  $r_2 = 119$  is [Theorem 21.21\(ii\)](#). If  $w \in HT^1(Y)$  annihilates  $\omega$ , so does  $w' \wedge w$  for every  $w' \in HT^1(Y)$ , since the annihilator is a left ideal; for  $w \neq 0$  these products span a space of dimension 15, which cannot lie in  $\operatorname{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$ . So  $r_1 = 16$ , and [Lemma 21.30](#) gives  $r_8, r_7, r_6$ .

(iii) For each  $k$  the set of  $a \in \mathbb{C}^4$  at which the contraction on  $HT^k(Y)$  into  $\omega_a$  has rank at least the displayed number is Zariski open, being the complement of the common zeros of minors, and item (XLV) of [Appendix D](#) shows that it contains  $a = (3, 5, -7, 11)$ , by an exact computation over  $\mathbb{Q}$  block by block. The rational Hodge classes in  $H^2 \otimes H^2$  form a  $\mathbb{Q}$ -vector space of dimension four whose complex span is  $\mathbb{C}^4$  ([Setup 21.18](#)), hence Zariski dense, and so they meet the intersection of these open sets in a Zariski dense set; the classes with  $\alpha \in \mathbb{Q}$  form a proper linear subspace and are removed. Item (XLV) also computes the ranks modulo a prime at which the minimal polynomial of  $\zeta_7 + \zeta_7^{-1}$  splits, for  $a_0 = 2$  and the three conjugates in each of the six orders, and finds the displayed numbers; a rank modulo a prime of good reduction is a lower bound for the rank in characteristic zero.

(iv) Since  $\operatorname{ch}(E)$  has only a component in  $H^4$ ,  $\operatorname{ch}(E^\vee) \operatorname{ch}(E)$  lies in  $H^8(Y)$  and  $\chi(E, E) = \int_Y \operatorname{ch}(E^\vee) \operatorname{ch}(E) = 0$ , the Todd class being trivial. By (i) and Serre duality, which also pairs  $\operatorname{Ext}^{-j}$  with  $\operatorname{Ext}^{8+j}$ ,

$$0 = \chi(E, E) = 2(e_0 - e_1 + e_2 - e_3) + e_4 + 2 \sum_{j \geq 1} (-1)^j e_{-j},$$

which is the identity. With no negative extensions the odd and even parts are equal, and the even part is at least  $2 \cdot 1 + 2 \cdot 119 + 560 = 800$  by (i) and (iii), so the odd part is at least 800 as well, and  $e_1 + e_3 \geq 400$  by Serre duality. The Hochschild bound on the odd part is  $2(16 + 328) = 688$ . ■

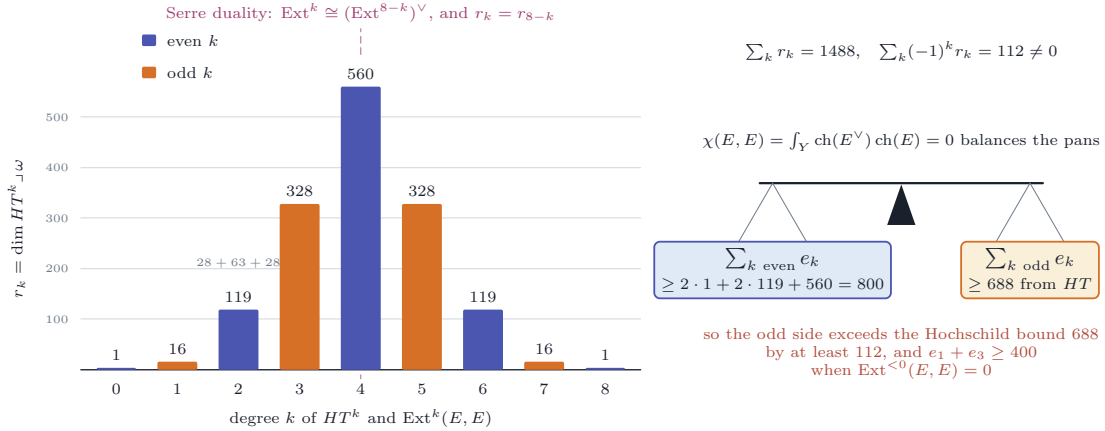


FIGURE 35. **Theorem 21.31.** Left: the lower bounds  $r_k$  for  $\dim \text{Ext}^k(E, E)$  when  $\text{ch}(E) = N\omega$ , one bar per degree, symmetric about  $k = 4$  as **Lemma 21.30** predicts; the bar at  $k = 2$  is the 119 of **Theorem 21.21**, made of 28 classes in  $H^{0,2}$ , 63 in  $H^1(T)$  and 28 bivectors. On the hyperplane  $L_0 = 0$  the bars at 2, 4 and 6 drop to 110, 548 and 110 at a sample point. Right:  $\chi(E, E) = 0$  balances the even and the odd degrees, so the odd side, which Hochschild cohomology fills only to 688, carries at least 800.

**Proposition 21.32** (Divisor classes do not reach the exceptional classes). *Let  $t \in C$  and let  $\omega$  be a rational exceptional class. Then  $\omega_t$  is not in the  $\mathbb{Q}$ -span of products of divisor classes of  $X_t \times X_t$ : that span meets the rational Hodge classes of  $H^2 \otimes H^2$  exactly in the classes with  $\alpha \in \mathbb{Q}$ . Consequently no perfect complex whose Chern character lies in the subring generated by divisor classes, in particular no bounded complex of direct sums of line bundles, has  $\text{ch}(E) = N\omega$  with  $N \neq 0$ . At a CM point  $c$  the fourfold  $X_c$  has eight Hodge classes in  $H^4$ , of which the products of divisor classes span six.*

*Proof.* At a point with Hodge group  $G$ , the divisor classes of  $X_t \times X_t$  are the three classes of **Proposition 20.44(ii)**, all invariant under  $\text{Sp}(V, \psi)$ , and so are their products; the  $\text{Sp}(V, \psi)$ -invariant part of  $H^2 \otimes H^2$  is spanned by  $\pi_0$  and  $\pi_{12} + \pi_{13} + \pi_{23}$ , the classes with  $a_1 = a_2 = a_3$ . At a CM point the Hodge group is a maximal torus of  $G$  (proof of **Theorem 20.30**), so the Hodge classes are the monomials of weight zero, and item (XLV) computes over  $\mathbb{Q}$ : 16 divisor classes, 132 Hodge classes in degree four, products of divisor classes spanning 100 of them, and a span meeting that of the four  $\pi$  in exactly the classes with  $a_1 = a_2 = a_3$ . For a rational class these coordinates are the conjugates of  $\alpha$ , which by **Lemma 21.19** are equal only if  $\alpha \in \mathbb{Q}$ . The Chern character of a complex in the subring generated by divisor classes has its component in  $H^4$  in the span of products of two divisor classes, which excludes  $N\omega$ . The last sentence is the same computation for  $X_c$ . ■

At a CM point the Hodge classes of  $X_c$  in  $H^4$  that are not products of divisor classes are among the classes that **Theorem 20.30** pulls back along homomorphisms  $X_c \times X_c \rightarrow X_c$ ; they are algebraic by Markman's theorem [Mar25a]. Li proves, independently and more generally, that every Hodge class on every power of every complex CM abelian fourfold is algebraic [Li26, Theorem 1.3], which gives **Theorem 20.30** for every power  $X_c^n$  and not only the square. **Proposition 21.32** says that an object with  $\text{ch}(E) = N\omega$  cannot have its Chern

character in the subring generated by divisor classes, so it cannot be assembled from line bundles alone. The next theorem shows that no sequence of the standard operations of the derived category, applied to line bundles on powers of  $X_t$ , gets any further.

**Theorem 21.33** (What line bundles generate). *Let  $t \in C$ ,  $X = X_t$ , and let  $L(X) \subseteq \mathrm{Sp}(V, \psi)$  be the Lefschetz group of  $X$ , the centraliser of  $\mathrm{End}^0(X)$  in  $\mathrm{Sp}(V, \psi)$ . Let  $\mathcal{A}$  be the class of abelian varieties isogenous to a power of  $X$ ; through an isogeny  $X^n \rightarrow B$  the group  $L(X)$  acts diagonally on  $H^*(B, \mathbb{Q})$ , and the action does not depend on the isogeny. Let  $\mathcal{L}$  be the smallest class of objects of the categories  $D^b(B)$ ,  $B \in \mathcal{A}$ , that contains the line bundles and is closed under shifts, cones, derived tensor products, derived duals, translations, the functors  $Lf^*$  and  $Rf_*$  for homomorphisms  $f$  between members of  $\mathcal{A}$ , and the Fourier–Mukai functors  $\Phi_K: D^b(B) \rightarrow D^b(B')$  with kernel  $K \in \mathcal{L}$  on  $B \times B'$ .*

- (i) *Every object of  $\mathcal{L}$  has an  $L(X)$ -invariant Chern character.*
- (ii)  *$L(X) = \mathrm{Sp}(V, \psi)$  if  $t$  is not a CM point, and  $L(X)$  is a maximal torus of  $\mathrm{Sp}(V, \psi)$  if it is. In both cases the  $L(X)$ -invariant classes on  $X \times X$  form the subring generated by divisor classes, of dimensions 1, 3, 6, 10, 15, 10, 6, 3, 1, respectively 1, 16, 100, 304, 454, 304, 100, 16, 1, in the degrees 0, 2,  $\dots$ , 16.*
- (iii) *No rational exceptional class  $\omega_t$  is  $L(X)$ -invariant. Hence for no object  $E \in \mathcal{L}$  on  $X \times X$  and no  $N \neq 0$  is  $\mathrm{ch}(E) - N\omega_t$  invariant under  $L(X)$ ; in particular  $\mathrm{ch}(E) \neq N\omega_t$ .*
- (iv) *The  $\mathrm{Sp}(V, \psi)$ -module generated by  $\pi_{12} - \pi_{13}$  is irreducible, of highest weight  $2\varpi_2$  and dimension 308, and it meets the span of the four  $\pi$  in the plane of the classes with  $a_0 = 0$  and  $a_1 + a_2 + a_3 = 0$ .*

*Proof.* (i) Two isogenies  $X^n \rightarrow B$  differ by an element of  $\mathrm{End}^0(X^n) = M_n(\mathrm{End}^0(X))$ , which commutes with  $L(X)$  acting diagonally, so the action is well defined. For the same reason a homomorphism  $f$  between members of  $\mathcal{A}$  is  $L(X)$ -equivariant on  $H^1$ , so  $f^*$  is equivariant on  $H^*$ , and so is  $f_*$ , which is  $f^*$  conjugated by Poincaré duality, since  $L(X) \subseteq \mathrm{SL}(V)$  fixes the orientation classes. Translations act trivially on cohomology. The Chern character is additive on distinguished triangles and multiplicative on derived tensor products,  $\mathrm{ch}(E^\vee) = \sum_i (-1)^i \mathrm{ch}_i(E)$ ,  $\mathrm{ch}(Lf^*E) = f^* \mathrm{ch}(E)$ , and  $\mathrm{ch}(Rf_*E) = f_* \mathrm{ch}(E)$  by the Grothendieck–Riemann–Roch theorem [Ful98, Theorem 15.2], the Todd class of an abelian variety being 1; and  $\Phi_K = Rp_{2*}(p_1^*(\cdot) \otimes^L K)$ . So the objects with  $L(X)$ -invariant Chern character form a class with all the closure properties, and it remains to see that it contains the line bundles, that is, that  $c_1(\mathcal{O}(D))$  is  $L(X)$ -invariant. Pulled back along an isogeny, which is injective on rational cohomology,  $c_1$  becomes a Hodge class of degree two on some  $X^n$ , an element of the invariants of the Hodge group in  $\wedge^2(V^\vee \otimes \mathbb{Q}^n) = \wedge^2 V^\vee \otimes S^2 \mathbb{Q}^n \oplus S^2 V^\vee \otimes \wedge^2 \mathbb{Q}^n$ . If  $t$  is not a CM point the Hodge group is  $G$ , whose invariants in  $\wedge^2 V$  are the multiples of  $\psi$  and in  $S^2 V$  are zero, since every summand of  $S^2(V_1 \otimes V_2 \otimes V_3)$  contains a factor  $S^2 V_i$ ; so the Hodge classes of degree two are  $\mathrm{Sp}(V, \psi)$ -invariant. If  $t$  is a CM point, a monomial of degree two in a weight basis has weight zero for the Hodge torus exactly when its two weights are opposite, and then it has weight zero for  $L(X)$ , by the description in (ii).

(ii) If  $t$  is not a CM point,  $\mathrm{End}^0(X) = \mathbb{Q}$  and  $L(X) = \mathrm{Sp}(V, \psi)$ . At a CM point  $\mathrm{End}^0(X) \otimes \mathbb{C}$  is the algebra of diagonal matrices in a weight basis (proof of Theorem 20.30), and  $\psi$  pairs each weight with its opposite only, so  $L(X)_\mathbb{C}$  is the torus of diagonal matrices  $\mathrm{diag}(\lambda_w)$  with  $\lambda_w \lambda_{-w} = 1$ , of rank four. For  $t$  not CM, item (XLVI) of Appendix D bounds the  $\mathrm{Sp}(V, \psi)$ -invariants of  $\wedge^*(V \oplus V)$  from above by the stated dimensions and shows that products of the three divisor classes span spaces of those dimensions. At a CM point the

invariants are spanned by the monomials of weight zero for  $L(X)$ ; a set of weights  $\pm e_j$  with sum zero is a union of opposite pairs, so each such monomial is, up to sign, a product of monomials of degree two and weight zero, which are divisor classes, and item (XLVI) counts them: their numbers are the coefficients of  $(1 + 4y + y^2)^4$ .

(iii) If  $t$  is not a CM point, the  $\mathrm{Sp}(V, \psi)$ -invariant combinations of the  $\pi$  are those with  $a_1 = a_2 = a_3$  (Proposition 21.32), and the coefficients of  $\omega_t$  are the distinct conjugates of  $\alpha$  (Lemma 21.19). At a CM point use the map  $x \mapsto E_x$  of the proof of Lemma 21.20, which is  $\mathrm{Sp}(V, \psi)$ -equivariant: it carries  $\omega_t$  to an element  $(r, \beta)$  of  $\mathbb{Q} \times F$  with  $\beta \notin \mathbb{Q}$ , acting on  $U_{ij}$  by the conjugate  $\sigma_k(\beta)$  (Proposition 21.2(ii)), and  $\omega_t$  is  $L(X)$ -invariant only if  $E_{\omega_t}$  preserves the weight spaces of  $L(X)$  in  $\Lambda^2 V$ . Take weight vectors  $v = x \otimes y \otimes z$  and  $w = x \otimes y' \otimes z'$  of weights  $(1, 1, 1)$  and  $(1, -1, -1)$  for the Hodge torus. They are not opposite, so  $v \wedge w$  spans a weight space of  $L(X)$ , every weight of  $L(X)$  on  $V$  having multiplicity one. But  $v \wedge w = (x \cdot x) \otimes (y \cdot y' \otimes z \wedge z' + y \wedge y' \otimes z \cdot z')$  has nonzero components in  $U_{12}$  and in  $U_{13}$ , on which  $E_{\omega_t}$  acts by  $\sigma_3(\beta) \neq \sigma_2(\beta)$ . So the line is not preserved. Item (XLVI) confirms the conclusion in exact arithmetic: the components of  $\pi_{12}, \pi_{13}, \pi_{23}$  of nonzero weight for  $L(X)$  span a space of dimension two, with the relation  $(1, 1, 1)$  only. The rest of (iii) follows from (i).

(iv) Item (XLVI) generates the module by applying the 36 elements  $x \mapsto \psi(u, x)v + \psi(v, x)u$  of  $\mathfrak{sp}(V, \psi)$  until the span stops growing: it has dimension 308 and highest weight  $(2, 2, 0, 0)$  for the torus of diagonal matrices in  $\mathrm{Sp}(V, \psi)$ , and 308 is, by Weyl's dimension formula, the dimension of the irreducible module of highest weight  $2\varpi_2$ ; and adding the four  $\pi$  raises its dimension by two. ■

*Remark 21.34* (Where a construction has to leave  $\mathcal{L}$ ). Theorem 21.33 extends Proposition 21.32 from sums of line bundles to everything the standard operations make out of them, at every point of the curve, including the CM points, where the exceptional classes are algebraic: the cycles that represent them there are not Chern characters of objects of  $\mathcal{L}$ , and neither are the two Hodge classes of  $X_c$  in degree four that are not products of divisor classes. At a point that is not CM every Chern character of an object of  $\mathcal{L}$  lies in the isotypic component of the trivial representation of  $\mathrm{Sp}(V, \psi)$ , while by (iv) an object with  $\mathrm{ch}(E) = N\omega$  must have a nonzero component in the one of highest weight  $2\varpi_2$ . Every construction of such an object therefore contains a step that does not commute with the Lefschetz group on cohomology: passing to a direct summand, a subobject or a Jordan–Hölder factor, choosing a point of a moduli space, or deforming from a member with more endomorphisms. The last is what semiregularity does along the curve (Theorem 21.27).

**Lemma 21.35** (The summands of a semiregular object). *Let  $Z$  be a smooth projective variety and  $E_1, E_2$  perfect complexes on  $Z$ . On  $\mathrm{Ext}^2(E_1 \oplus E_2, E_1 \oplus E_2) = \bigoplus_{i,j} \mathrm{Ext}^2(E_j, E_i)$  the semiregularity map is  $\sigma_{E_1 \oplus E_2}(\zeta) = \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$ . So  $\sigma_{E_1 \oplus E_2}$  is injective if and only if  $\sigma_{E_1}$  and  $\sigma_{E_2}$  are, and  $\mathrm{Ext}^2(E_1, E_2) = \mathrm{Ext}^2(E_2, E_1) = 0$ .*

*Proof.* The Atiyah class is natural, so  $\mathrm{At}(E_1 \oplus E_2)$  is the diagonal sum of  $\mathrm{At}(E_1)$  and  $\mathrm{At}(E_2)$ . In  $\sigma(\zeta) = \sum_q \frac{1}{q!} \mathrm{tr}(\zeta \circ \mathrm{At}^q)$  of Theorem 16.27 the composite  $\zeta \circ \mathrm{At}^q$  has diagonal blocks  $\zeta_{ii} \circ \mathrm{At}(E_i)^q$ , and the trace of a block matrix of morphisms is the sum of the traces of its diagonal blocks. ■

**Corollary 21.36** (No Hochschild bypass, and no free splitting). *Let  $c \in C$ ,  $Y = X_c \times X_c$ ,  $\omega = \omega_a$  a rational exceptional class with  $L_0(a) \neq 0$ , and  $E$  a perfect complex on  $Y$  with  $\mathrm{ch}(E) = N\omega$ ,  $N \neq 0$ .*

- (i)  $E$  extends to the first order deformation of  $D^b(Y)$  given by  $x \in HH^2(Y) = HT^2(Y)$  only if  $x \in \mathbb{C}\kappa$ . No  $B$ -field, no Poisson bivector and no commutative deformation off the curve carries it.
- (ii) If  $\sigma_E$  is injective and  $E = E_1 \oplus E_2$  with both summands nonzero, then  $\text{Ext}^2(E_1, E_2) = \text{Ext}^2(E_2, E_1) = 0$ , each  $\sigma_{E_i}$  is injective, each  $E_i$  extends to first order along the curve, and  $\kappa \lrcorner \text{ch}(E_i) = 0$ . If moreover  $\dim \text{Ext}^2(E, E) = 119$ , then  $\dim \text{Ext}^2(E_i, E_i) = 120 - \dim \text{Ann}_{HT^2}(\text{ch } E_i)$ , and

$$\text{Ann}_{HT^2}(\text{ch } E_1) + \text{Ann}_{HT^2}(\text{ch } E_2) = HT^2(Y), \quad \text{Ann}_{HT^2}(\text{ch } E_1) \cap \text{Ann}_{HT^2}(\text{ch } E_2) = \mathbb{C}\kappa.$$

*Proof.* (i) The kernel of  $\text{ev}_E$  on  $HH^2(Y)$  is the space of deformations of  $D^b(Y)$  along which  $E$  deforms to first order [Tod09], [Mar25a, Section 8.3]. If  $\text{ev}_E(x) = 0$  then  $N x \lrcorner \omega = \sigma_E(\text{ev}_E(x)) = 0$ , so  $x \in \mathbb{C}\kappa$  by Theorem 21.21(ii). The three kinds of deformation named are the three summands of  $HT^2(Y)$ .

(ii) The first two assertions are Lemma 21.35. By Theorem 21.27(ii),  $\text{ev}_E(\kappa) = 0$ , and  $\text{ev}_E$  is the diagonal sum of  $\text{ev}_{E_1}$  and  $\text{ev}_{E_2}$ , so each vanishes at  $\kappa$ ; then  $\kappa \lrcorner \text{ch}(E_i) = \sigma_{E_i}(\text{ev}_{E_i}(\kappa)) = 0$ . Put  $e(x) = 120 - \dim \text{Ann}_{HT^2}(x)$ , the rank of contraction into  $x$ . It is subadditive, and by Theorem 21.31(i)  $\dim \text{Ext}^2(E_i, E_i) \geq e(\text{ch } E_i)$ . The cross terms vanish, so

$$119 = \dim \text{Ext}^2(E, E) = \sum_i \dim \text{Ext}^2(E_i, E_i) \geq e(\text{ch } E_1) + e(\text{ch } E_2) \geq e(N\omega) = 119,$$

and every inequality is an equality. Hence  $\dim \text{Ann}(\text{ch } E_1) + \dim \text{Ann}(\text{ch } E_2) = 121$ . The intersection of the two annihilators lies in  $\text{Ann}(N\omega) = \mathbb{C}\kappa$  and contains  $\kappa$ , so it is  $\mathbb{C}\kappa$ , and the sum has dimension 120. ■

Part (ii) is the Mumford form of Theorem 20.33: a semiregular object attaining the bound either does not split, or splits as two pieces whose annihilators fill  $HT^2(Y)$  and meet only in the direction of the curve, the analogue of the two branches  $P$  and  $Q$  of Remark 16.35.

**Proposition 21.37** (Tate classes at every closed point of every reduction). *There are a finitely generated subring  $R \subset \mathbb{C}$ , a model  $W_R \rightarrow C_R$  of  $W \rightarrow C$  over  $R$ , with  $C_R$  smooth and projective over  $R$  with geometrically connected fibres and  $W_R$  an abelian scheme, projective over  $C_R$ , and a finite surjective morphism  $C'_R \rightarrow C_R$  from a second curve of the same kind, with the following properties. Put  $Y' = (W_R \times_{C_R} W_R) \times_{C_R} C'_R$ , with projection  $\pi': Y' \rightarrow C'_R$ , and let  $\omega$  be a rational exceptional class. For every prime  $\ell$  there is a global section  $\omega_\ell$  of the lisse sheaf  $R^4\pi'_*\mathbb{Q}_\ell(2)$  on  $C'_R[1/\ell]$  whose restriction to  $C'_R \otimes_R \mathbb{C}$  is the pull-back of  $\omega \otimes 1$ ; write  $\omega_\ell(\bar{x})$  for its value at a geometric point over a point  $x$ .*

- (i) *At a closed point  $x$  of  $C'_R[1/\ell]$  the residue field  $\kappa(x)$  is finite,  $Y'_x$  is the square of an abelian fourfold over it, and  $\omega_\ell(\bar{x})$  is fixed by  $\text{Gal}(\bar{\kappa}/\kappa(x))$ .*
- (ii) *At such a point  $\omega_\ell(\bar{x})$  is a  $\mathbb{Q}_\ell$ -linear combination of classes of algebraic cycles of codimension two on  $Y'_x$ , defined over  $\kappa(x)$ .*
- (iii) *Let  $s$  be a closed point of  $\text{Spec } R[1/\ell]$  and  $\eta_s$  the generic point of the reduction  $C'_s$ . Then  $\omega_\ell(\bar{\eta}_s)$  is fixed by the Galois group of the function field  $\kappa(\eta_s)$ , and it is not a  $\mathbb{Q}_\ell$ -linear combination of products of divisor classes of  $Y'_{\eta_s}$ .*

*Every closed point  $x$  of a reduction  $C_s$  of  $C$  is the image of a closed point  $x'$  of  $C'_s$ , and  $Y'_{x'} = Y_x \otimes_{\kappa(x)} \kappa(x')$ ; so (i) and (ii) hold at every closed point of every reduction of  $C$ , after a finite extension of its residue field.*

*Proof. The models.* The family  $W \rightarrow C$  is defined over a subfield  $k_1 \subset \mathbb{C}$  finitely generated over  $\mathbb{Q}$ , say as  $W_1 \rightarrow C_1$ . Let  $K_1$  be the function field of  $C_1$  and fix an embedding over  $k_1$  of an algebraic closure  $\bar{K}_1$  into  $\mathbb{C}$ ; it sends the geometric generic point of  $C_1$  to a point  $t \in C$ , and  $Y_t = X_t \times X_t$  is the base change of  $Y_{\bar{K}_1}$ . The class  $\omega_t$  is a Hodge class on an abelian variety, hence absolutely Hodge, so it comes from an absolutely Hodge class of  $Y_{\bar{K}_1}$ , and  $\text{Gal}(\bar{K}_1/K_1)$  acts on these through a finite quotient [Del82, Theorem 2.11 and Proposition 2.9]. Let  $K' \subset \bar{K}_1$  be a finite extension of  $K_1$  on which the action is trivial,  $k'$  the algebraic closure of  $k_1$  in  $K'$ , and  $C'$  the smooth projective curve over  $k'$  with function field  $K'$ ; it is geometrically connected over  $k'$  and maps finitely onto  $C_1 \otimes_{k_1} k'$ . Spreading out  $W_1 \rightarrow C_1$ ,  $C'$  and this map over a finitely generated subring of  $k'$ , and inverting one element of it, gives a normal  $R$  and models with the stated properties.

*The section.* By smooth and proper base change  $R^4\pi'_*\mathbb{Q}_\ell(2)$  is lisse on  $C'_R[1/\ell]$ . Its stalk at the geometric generic point  $\text{Spec } \bar{K}_1$  is  $H^4(Y_{\bar{K}_1}, \mathbb{Q}_\ell(2)) = H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell$ , and  $\omega_t \otimes 1$  is the  $\ell$ -adic component of an absolutely Hodge class, so it is fixed by  $\text{Gal}(\bar{K}_1/K')$ . The scheme  $C'_R[1/\ell]$  is normal and integral with function field  $K'$ , so  $\text{Gal}(\bar{K}_1/K')$  maps onto its fundamental group [Stacks, Tag 0BQI], and the vector extends to a global section  $\omega_\ell$ . On the connected curve  $C'_R \otimes_R \mathbb{C}$  it agrees with the pull-back of  $\omega \otimes 1$ , the two being global sections of the same local system that agree at a point over  $t$ .

(i) and (ii). A scheme of finite type over  $\mathbb{Z}$  has finite residue fields at its closed points. A global section of a lisse sheaf is fixed by the fundamental group of every point, which is (i); and (ii) is Li's theorem [Li26, Theorem 1.1], with  $n = r = 2$ , for the abelian fourfold that is the fibre of  $W_R$  at the image of  $x$ , base changed to  $\kappa(x)$ .

(iii) The first assertion holds for the same reason as (i). For the second, let  $\bar{\eta}$  be a geometric generic point of  $C'_R$  and  $\bar{s}$  a geometric point over  $s$ . The morphism  $C'_R \rightarrow \text{Spec } R$  is flat and proper with geometrically connected and geometrically reduced fibres, so the specialisation map  $\pi_1(C'_{\bar{\eta}}) \rightarrow \pi_1(C'_{\bar{s}})$  is surjective [Stacks, Tag 0C0P], and it is compatible with the actions of the two groups on the first cohomology of the fibres of  $W_R$ , identified by cospecialisation. The image of  $\pi_1(C'_{\bar{\eta}})$  contains that of  $\pi_1(C'_C)$ , the closure of the image of the topological fundamental group of a finite cover of  $C$ , whose Zariski closure contains  $G_{\mathbb{Q}_\ell}$  because the connected monodromy group of  $W \rightarrow C$  is  $G$  (proof of Corollary 20.23). So the Zariski closure of the monodromy of  $W_R$  over  $C'_s$  contains  $G_{\mathbb{Q}_\ell}$ . A divisor class of  $Y'_{\bar{\eta}_s}$  is defined over a finite extension of  $\kappa(\eta_s)$ , so it is fixed by an open subgroup of the Galois group, whose image in  $\pi_1(C'_s)$  is open [Stacks, Tag 0BQI] and meets  $\pi_1(C'_s)$  in an open subgroup; the Zariski closure of the image of an open subgroup has the same identity component, so the class is  $G_{\mathbb{Q}_\ell}$ -invariant. The  $G$ -invariants of  $\wedge^2(V \oplus V)$  are the three divisor classes of Proposition 20.44(ii), which are  $\text{Sp}(V, \psi)$ -invariant, so every product of divisor classes of  $Y'_{\bar{\eta}_s}$  is  $\text{Sp}(V, \psi)$ -invariant. The class  $\omega_\ell(\bar{\eta}_s)$  corresponds to  $\omega \otimes 1$ , whose coefficients  $a_1, a_2, a_3$  are the distinct conjugates of  $\alpha$  (Lemma 21.19), and the  $\text{Sp}(V, \psi)$ -invariant classes among the  $\omega_a$  are those with  $a_1 = a_2 = a_3$  (Proposition 21.32). The last sentence of the statement holds because  $C'_R \rightarrow C_R$  is finite and surjective. ■

At the generic point of each reduction the class is therefore a Tate class over a global function field that no product of divisor classes explains, the exact analogue of the situation over  $\mathbb{C}$ , while at every closed point it is explained by cycles. The next theorem says that a bound on those cycles is all that is missing, with no passage through  $p$ -adic Hodge theory.

**Theorem 21.38** (The Mumford target is a degree bound in positive characteristic). *Keep Proposition 21.37, fix a prime  $\ell$  and a relatively ample line bundle on  $Y' \rightarrow C'_R$ , and for a finite set  $\Phi$  of tuples of Hilbert polynomials say that  $\omega$  is  $\Phi$ -represented at a point  $x$  of*

$C'_R[1/\ell]$  if  $\omega_\ell(\bar{x}) = \sum_i q_i \text{cl}(Z_i)$  for some  $q_i \in \mathbb{Q}_\ell$  and closed subschemes  $Z_i \subset Y'_x$  of pure codimension two whose Hilbert polynomials form a tuple in  $\Phi$ . The following are equivalent.

- (a)  $\omega_t$  is algebraic for every  $t \in C$ .
- (b) For some finite  $\Phi$ , the closed points of  $C'_R[1/\ell]$  at which  $\omega$  is  $\Phi$ -represented are Zariski dense in  $C'_R$ .
- (c) For some finite  $\Phi$  and some Zariski dense set  $S$  of closed points of  $\text{Spec } R[1/\ell]$ , the class  $\omega$  is  $\Phi$ -represented at infinitely many closed points of  $C'_s$  for every  $s \in S$ .

For a single closed point  $s$  of  $\text{Spec } R[1/\ell]$ , the class  $\omega$  is  $\Phi$ -represented at infinitely many closed points of  $C'_s$ , for some finite  $\Phi$ , if and only if  $\omega_\ell(\bar{\eta}_s)$  is a  $\mathbb{Q}_\ell$ -linear combination of classes of algebraic cycles on  $Y'_{\bar{\eta}_s}$ .

*Proof.* One standard property of the  $\ell$ -adic cycle class is used beyond those over a field: if  $\mathcal{O}$  is a discrete valuation ring in which  $\ell$  is invertible,  $\mathcal{Y} \rightarrow \text{Spec } \mathcal{O}$  is smooth and projective, and  $\mathcal{Z} \subset \mathcal{Y}$  is a closed subscheme flat over  $\mathcal{O}$ , then the specialisation isomorphism of  $\ell$ -adic cohomology carries the cycle class of the geometric generic fibre of  $\mathcal{Z}$  to that of its geometric special fibre [SGA 4 $\frac{1}{2}$ , Cycle]. The specialisation isomorphism also carries the values of a global section of a lisse sheaf to each other, by its construction.

*Step 1: the represented loci are closed.* For a tuple  $\underline{P} = (P_1, \dots, P_m)$  let  $\mathcal{H}_{\underline{P}}$  be the fibre product over  $C'_R[1/\ell]$  of the relative Hilbert schemes  $\text{Hilb}^{P_i}(Y'/C'_R[1/\ell])$ . It is projective over  $C'_R[1/\ell]$  [Gro61], so it has finitely many connected components. On a component  $T$ , with universal subschemes  $\mathcal{Z}_i$ , put  $A(h) = \{q \in \mathbb{Q}_\ell^m : \sum_i q_i \text{cl}(\mathcal{Z}_{i,\bar{h}}) = \omega_\ell(\bar{h})\}$  for  $h \in T$ . If  $h'$  is a specialisation of  $h$ , there is a discrete valuation ring mapping to  $T$  with generic point to  $h$  and closed point to  $h'$  [Stacks, Tag 054F], and the property above, applied to the pull-backs of  $Y'$  and of the  $\mathcal{Z}_i$ , gives  $A(h) = A(h')$ . Any two points of the connected noetherian scheme  $T$  are joined by a chain of specialisations and generisations, through the generic points of its irreducible components, so  $A$  is constant on  $T$ , say  $A_T$ . Let  $\Sigma_{\underline{P}}$  be the union of the images in  $C'_R[1/\ell]$  of the components  $T$  with  $A_T \neq \emptyset$ . It is closed, as a finite union of images of proper morphisms, and it is the set of points at which  $\omega$  is  $\{\underline{P}\}$ -represented: at a point  $x$  of the image of such a  $T$ , a closed point of the fibre of  $T$  over  $x$  has residue field finite over  $\kappa(x)$  and gives subschemes of  $Y'_x$ . Put  $\Sigma_\Phi = \bigcup_{\underline{P} \in \Phi} \Sigma_{\underline{P}}$ .

*Step 2: (b) implies (a).*  $\Sigma_\Phi$  is closed and contains a dense set, so it is  $C'_R[1/\ell]$  and contains the generic point  $\xi$  of  $C'_R$ , whose residue field is  $K'$ . So  $\omega_t \otimes 1 = \omega_\ell(\xi) = \sum_i q_i \text{cl}(Z_i)$  with  $q \in \mathbb{Q}_\ell^m$  and subschemes  $Z_i$  of  $Y_{\bar{K}_1}$ , and so of  $Y_t$ . The cycle class maps are compatible with the comparison isomorphism  $H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell \cong H^4(Y_t, \mathbb{Q}_\ell)$  [Del82, Section 1 and Example 2.1(a)], so the linear system  $\omega_t = \sum_i q_i [Z_i]$ , which has rational coefficients, is solvable over  $\mathbb{Q}_\ell$ , hence over  $\mathbb{Q}$ , and  $\omega_t$  is algebraic. The  $Z_i$  are defined over a finite extension of  $K'$ , so they spread out to subschemes, flat over a nonempty open subset [Stacks, Tag 052A], of the pull-back of  $W \times_C W$  to a finite cover of  $C$ , and by the local constancy of the cycle class used in the proof of Theorem 16.4,  $\omega$  is algebraic at every point of a nonempty Zariski open subset of  $C$ , which is uncountable. Corollary 20.31 gives (a).

*Step 3: (c) implies (b).* Let  $D$  be the set of closed points at which  $\omega$  is  $\Phi$ -represented and  $\bar{D}$  its closure. For  $s \in S$  the set  $D \cap C'_s$  is infinite, so its closure is the irreducible curve  $C'_s$ , and  $C'_s \subseteq \bar{D}$ . The points  $u$  of  $\text{Spec } R[1/\ell]$  over which the fibre of  $\bar{D}$  has dimension at least one form a closed set, by the upper semicontinuity of the dimension of the fibres of a proper morphism [Stacks, Tag 0D4I]; it contains the dense set  $S$ , so it contains the generic point. The generic fibre of  $\bar{D}$  is then a closed subset of dimension one of the irreducible curve  $C'_R \otimes_R k'$ , so it contains  $\xi$ , and  $\bar{D} = C'_R[1/\ell]$ .

*Step 4: (a) implies (c).* By (a) and Step 1, applied over  $\bar{K}_1$  with the Betti class in place of the  $\ell$ -adic one,  $\omega_t$  is the class of a rational combination of subschemes of  $Y_{\bar{K}_1}$ , defined over a finite extension  $L$  of  $K'$ . Enlarge  $R$  to a finitely generated subring of the algebraic closure of  $k'$  in  $L$ , let  $C''_R \rightarrow C'_R$  be the normalisation in  $L$ , and spread the subschemes out over  $C''_R$ ; by generic flatness [Stacks, Tag 052A] they are flat over a dense open subset  $U$  of  $C''_R[1/\ell]$ , with constant Hilbert polynomials. By Step 1 applied on  $U$ ,  $\omega$  is represented by their fibres at every point of  $U$ . The image of  $U$  in  $C'_R[1/\ell]$  contains a dense open subset  $U'$ , every closed point of  $U'$  is the image of a closed point of  $U$ , and  $U'$  meets  $C'_s$  in a nonempty open, hence infinite, subset for every  $s$  in a dense open subset of  $\text{Spec } R[1/\ell]$ .

*The single reduction.* If  $\omega$  is  $\Phi$ -represented at infinitely many closed points of  $C'_s$ , then  $\Sigma_\Phi \cap C'_s$  is closed and infinite, so it is  $C'_s$  and contains  $\eta_s$ . Conversely, cycles at  $\bar{\eta}_s$  are defined over a finite extension of  $\kappa(\eta_s)$ , and spreading them out as in Step 4 over a finite cover of  $C'_s$  represents  $\omega$ , with fixed Hilbert polynomials, at every closed point of a nonempty open subset of  $C'_s$ . ■

*Remark 21.39* (What is left in positive characteristic). By Proposition 21.37(ii) the class is represented at every closed point of  $C'_R[1/\ell]$ , so Theorem 21.38 turns the Mumford target into a statement of uniformity alone: the Hilbert data of some representation must stay in one finite set on a Zariski dense set of closed points, for instance at infinitely many closed points of infinitely many reductions. The passage to characteristic zero is then done by the specialisation of cycle classes along  $C'_R$ , and neither the  $p$ -adic variational Hodge conjecture nor a lifting of cycles is needed. Li's theorem gives no bound, and where its proof carries cycles into a prescribed isogeny class it does so by an isogeny whose degree it does not control [Li26, proof of Theorem 1.2, Step 3]. Within one reduction the bound is equivalent to the class being algebraic at the generic point of  $C'_s$ , a Tate class over a global function field that is not a combination of products of divisor classes (Proposition 21.37(iii)); this is the codimension two case of the variational Tate conjecture along  $C'_s$ , which Morrow proves for divisors in its crystalline form [Mor19]. From one reduction alone the way back to characteristic zero would be the  $p$ -adic variational Hodge conjecture for  $W \times_C W$  over the Witt vectors of  $\kappa(s)$ , with rational rather than  $\ell$ -adic coefficients and with the crystalline class of the cycle matching the de Rham class of  $\omega$ ; Bloch, Esnault and Kerz lift such a class in  $K_0 \otimes \mathbb{Q}$  of the special fibre to the formal completion for  $p > 15$ , the dimension being nine, and leave the algebraisation open [BEK14, Theorem 1.3]. Neither the uniform bound nor this conjecture is proved.

The five routes of Remark 21.29 that are not closed off are *specialisation*, *reduction modulo  $p$* , *semiregularity*, *Kuga–Satake* and *motivated classes*. They are not five obligations.

**Corollary 21.40** (The open routes are one statement). *Let  $W \rightarrow C$  be as in Corollary 20.23, let  $\omega$  be a rational exceptional class, and let the Mumford target be the first case of (F2): the Hodge conjecture for  $X_t \times X_t$  for every  $t \in C$ .*

- (i) *Each of the following is equivalent to the Mumford target:  $\omega_t$  is algebraic for uncountably many  $t \in C$ ;  $B(W \times_C W)$  holds for  $L_{W \times_C W}$ ;  $\omega$  is represented with Hilbert data in one finite set at infinitely many points of  $C$ , for instance at infinitely many CM points;  $\omega$  is represented with Hilbert data in one finite set on a Zariski dense set of closed points of the arithmetic model of Theorem 21.38.*
- (ii) *Each of the following implies the Mumford target: a perfect complex  $E$  at one point of  $C$  with  $\text{ch}(E) = N\omega$ ,  $N \neq 0$ , no other component, and  $\dim \text{Ext}^2(E, E) = 119$ ; the algebraicity, for uncountably many  $t$ , of the Kuga–Satake class of the surface  $S_\lambda$  of*

*Theorem 21.4* attached to  $X_t$ ; a rational map as in *Proposition 21.25(iv)*, under its hypotheses, from  $X_t \times X_t$  for uncountably many  $t$ .

- (iii) *There is a countable set  $\Sigma_0 \subset C$  containing the CM points such that the Mumford target holds as soon as the Hodge conjecture holds for  $X_c \times X_c$  at one point  $c \notin \Sigma_0$ .*

*Proof.* (i) The first two are [Corollary 20.31](#), with [Lemma 21.20](#) for one class in place of two, the third is [Corollary 21.28](#) and the fourth is [Theorem 21.38](#). (ii) The first is [Theorem 21.27\(iii\)](#). The second and the third give the Hodge conjecture for  $X_t \times X_t$  at each of the uncountably many points, by [Theorem 21.4\(ii\)](#) and [Proposition 21.25\(iv\)](#), and then (i) applies. (iii) In the proof of [Corollary 21.28](#) the set of points at which  $\omega$  is algebraic is the union of the countably many closed sets  $\Sigma_{P,q}$ , each finite or equal to  $C$ . Let  $\Sigma_0$  be the union of the finite ones and of the CM points. If  $\omega_c$  is algebraic at a point  $c \notin \Sigma_0$ , then  $c$  lies in some  $\Sigma_{P,q}$  that is not finite, which is  $C$ , and (i) applies. ■

*Remark 21.41* (Which routes are needed). Of the five routes, specialisation, reduction modulo  $p$  and motivated classes are the Mumford target itself in three other forms, by [Corollary 21.40\(i\)](#) and, for the motivated classes, [Corollary 20.31](#); semiregularity and Kuga–Satake are stronger inputs that imply it. So one route suffices and the other four can be dropped, and item (XLIX) of [Appendix D](#) computes this from the rules of the paper: every minimal set of open statements from which the target follows has one element, and the statements equivalent to it over what is proved are exactly the four forms of (i) and the target. Nothing in the paper proves any of them, and each of the three forms is the whole of the problem: a bound that no known construction gives, or a case of (L) that is equivalent to the target. The two stronger inputs ask for an object and a cycle that nothing in the paper constructs.

Nor is the target, once proved, a step that finishes the conjecture. It is a special case of the conjecture, and so it is needed in the sense that every proof of the conjecture proves it. But it is only the first case of (F2), and a proof of it gives the conjecture for every power of every fibre ([Corollary 21.10](#)) but leaves (F1), the other cases of (F2), and (F3) open. The last of these is not a remainder of the conjecture but the conjecture itself ([Proposition 20.35](#)); its weaker form (F3') is open as well ([Remark 20.38](#)). Of the five minimal sufficient sets of [Theorem 20.40\(ii\)](#), only the route through this paper,  $\{P2_{\text{split}}, F2, F3'\}$ , has the target as an input, as the first case of (F2). On the other four it is not an input: (F3) is the conjecture, and (L) implies (V) ([Proposition 20.18](#)), which carries the algebraicity at the CM points of [Theorem 20.30](#) to every member. Item (XLIX) checks both: that (F3), being equivalent to the conjecture, yields the target, and that (L) and (V) do.

## 22. CLOSURE

A paper on a problem that remains open has to say where it stops, and saying so is not a matter of tone. It is a theorem, and this section states it. Every clause below is proved in the body, none of them depends on a hypothesis that is not itself proved here or quoted with a reference, and the last clause is checked by machine.<sup>66</sup> Taken together they fix the

<sup>66</sup>The quoted results are theorems and are used as such: the semiregularity theorem of Buchweitz and Flenner [[BF03](#)], in the form for perfect complexes on a deforming variety with every obstruction annihilated that Pridham proves [[Pri24](#)], the theorem of Cattani, Deligne and Kaplan on the locus of Hodge classes [[CDK95](#)], the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [[MZ99](#)], Markman's theorems on abelian varieties of Weil type [[Mar25a](#), [Mar25b](#)], and, in the last clause, the theorems of Deligne, André and Milne recorded in [Section 20.6](#) [[Del71](#), [And96b](#), [Mil20](#)] and Li's theorem on the powers of abelian fourfolds over finite fields [[Li26](#)], and, for the refutation in clause (iv), the

boundary of the subject as this paper leaves it, and the fixing is what we mean by closure: not that the conjecture is settled, which it is not, but that the distance from here to it is now measured rather than estimated.

**Theorem 22.1** (Closure). *Let  $K = \mathbb{Q}(\sqrt{-d})$  be imaginary quadratic with  $d > 0$  squarefree, let  $n \geq 2$ , and let  $\delta \in \mathbb{Q}_{>0}^\times / N_{K/\mathbb{Q}}(K^\times)$ . The following hold unconditionally.*

- (i) One class. *For an abelian  $2n$ -fold  $A$  of  $(K, -1, n)$ -Weil type with  $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$  one has  $\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = 1$  for  $k \neq n$  and 3 for  $k = n$ , and the Hodge conjecture for  $A$  in every degree is the algebraicity of the single class  $\omega_1$  of (11), written out with  $2^{2n-1}$  integer coefficients (Theorem 4.14, Corollary 4.15).*
- (ii) One class, one connected domain, and no loss. *For the family of all members of the triple  $(K, n, \delta)$  the conjecture is equivalent to the algebraicity of  $\omega_s$  at every point  $s$  of the period domain  $\mathcal{D}_{n,n}$ , which is connected of dimension  $n^2$ ; the implication runs in both directions, so nothing is given away by passing to this form (Theorem 15.14, Corollary 15.35).*
- (iii) A base point in every family. *Every triple  $(K, n, \delta)$  contains a member whose Weil classes are algebraic, with an explicit representative (36) as a rational polynomial in two divisor classes, the member being one with quaternionic multiplication extending  $K$  (Theorem 15.25, Theorem 15.31). The hypothesis of nonemptiness in (ii) is therefore never the obstacle, for any field, any dimension or any discriminant.*
- (iv) One finite sufficient condition, false in its first form, and one criterion that turns out to be a restatement. *If at  $s_0$  the semiregularity map of a perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation is injective, then  $\Sigma = \mathcal{D}_{n,n}$  and the Hodge conjecture holds in every degree for every member of the triple with full Hodge group (Theorem 16.28, Remark 17.6). When the Chern character is exactly a multiple of the Weil class, no complex at any member of any family has injective semiregularity map (Theorem 17.5): it would deform to very general non-algebraic Weil tori, on which every coherent sheaf has vanishing Chern character in degrees 1 to  $2n - 1$  (Proposition 17.4). So the numerical form of that criterion,  $\dim \mathrm{Ext}^2(E, E) = 2n(2n - 1)$ , which would suffice (Theorem 16.36) and is the least value possible (Corollary 16.34(i)), is attained by no complex, and the shape it would force (Theorem 16.40) is the shape of nothing. In the corrected form, with powers of the polarisation added, the number becomes  $(4 + \rho)n^2 - 2n$  for  $n \geq 3$ , with  $\rho \leq 3$  the rank of a Hankel matrix of the coefficients;<sup>67</sup> an object attaining it satisfies the criterion, and the shapes  $Ae^{t\theta} + B\theta^{2n}$  with  $t\theta$  the class of a line bundle, of Hankel rank at most two, are excluded as the pure form was (Theorem 17.9). For every other shape and  $n \geq 3$  the Hodge locus of the character is the polarised family, so the pure-form obstruction does not extend (Proposition 17.11); at  $n = 2$  the same holds except at one ratio of the coefficients, where the locus is five-dimensional and reaches non-algebraic tori but the argument of Theorem 17.5 does not finish (Remark 17.12). The other criterion, that  $\omega$  be represented along the*

theorem of Bando and Siu on stable reflexive sheaves [BS94], Kuranishi's theorem in the form of Goldman and Millson [Kur65, GM88], Grivaux's Chern classes of coherent sheaves [Gri10], and Voisin's theorem on Weil tori [Voi02b]. Unconditionally in Theorem 22.1 means that no open statement is assumed anywhere in it; in particular none of the statements named in its last clause is.

<sup>67</sup>Writing the Chern character as  $N\omega_1 + \sum_k c_k \theta^k$ ,  $\rho$  is the rank of the  $(2n - 1) \times 3$  Hankel matrix  $[\mu_{i+j}]$ ,  $0 \leq j \leq 2n - 2$ ,  $0 \leq i \leq 2$ , of the moments  $\mu_m = m! c_m$  (Setup 17.8); so  $\rho \leq 2$  exactly when the moments satisfy a linear recurrence of order at most two.

orbit by cycles of bounded Hilbert data, implies the same conclusion ([Theorem 16.9](#)) and is implied by it ([Theorem 16.17](#)), so it is a reformulation of the conjecture for the family and not a route to it; read instead on the transported cycle, the only representative one can name, it is false whenever a component of the base cycle has finite stabiliser ([Theorem 16.15](#)).

- (v) The routes that are closed, and in what sense. Each of the following routes to the Weil class on a general member is closed, in the precise sense stated by the theorem cited and delimited in [Section 20](#): a construction is obstructed if it is
  - (a) one whose input has Chern character in the subring generated by divisor classes, transported by the Poincaré kernel ([Theorem 1.1](#));
  - (b) one whose output is a  $K^\times$ -eigenclass for the norm character ([Theorem 1.3](#));
  - (c) one carried by an object whose Chern character lies on the Weil line, whether through semiregularity or through the weakened criterion, in every dimension and by  $n^2$  dimensions ([Theorem 9.7](#), [Corollary 9.18](#), [Theorem 9.19](#));
  - (d) one carried by an object with a line bundle summand off the polarisation line, hence by any sum of line bundles ([Theorem 9.20](#), [Corollary 18.22](#));
  - (e) one carried by an object whose Chern character is effective at a base point ([Theorem 18.5](#));
  - (f) induction through hyperplane sections, in either direction ([Theorem 1.4](#), [Theorem 1.5](#));
  - (g) Kuga–Satake from a weight two sub-structure of  $H^2(A, \mathbb{Q})$ , which is available only at  $n = 1$  ([Proposition 11.11](#), [Corollary 11.12](#));
  - (h) a divisorial argument at a general member of a family of nontrivial discriminant ([Proposition 15.18](#));
  - (i) a secant object whose support fails to be Cohen–Macaulay at a point where two components meet only there or where there is an embedded point, which removes the candidate of [\[Mar25b, Section 12\]](#) ([Theorem 18.34](#), [Corollary 18.36](#), [Corollary 18.45](#));
  - (j) a secant object of rank one whose support is a complete intersection of two divisors, for every  $b$  and every  $d$  ([Theorem 18.49](#));
  - (k) a secant object quasi-isomorphic to a two-term complex of sums of powers of the polarisation, or of simple semi-homogeneous bundles, in particular one whose support has a Hilbert–Burch resolution by such bundles, in every dimension  $n \geq 3$  and for every rank, twist and discriminant ([Theorem 18.52](#), [Theorem 18.55](#), [Corollary 18.57](#));
  - (l) the semiregularity form of the propagation statement, for any representative of the Weil  $K$ -theory class that is a locally free sheaf on a smooth germ of codimension at least two of its support, at every member with no abelian subvariety tangent to a  $K$ -eigenspace; for every representative, full rank forces the evaluation map to vanish on the  $4n^2$ -dimensional annihilator of the Weil class, which is the mixed part of a splitting  $HH^1(A) = P \oplus Q$  into the annihilators of the two conjugate pieces of that class, so that the whole Hochschild action factors through a ring of dimension  $2^{2n+1} - 1$  inside one of dimension  $2^{4n}$  ([Theorem 16.30](#), [Corollary 16.31](#), [Theorem 16.33](#), [Corollary 16.34](#));
  - (m) the same form of the propagation statement for any representative whose support has codimension at least  $n$ , at every member of every family whatever its endomorphisms, hence for every complex carried by a cycle that represents

the Weil class; and, at a member with no abelian subvariety tangent to a  $K$ -eigenspace, for any representative with a component of codimension at least two along which its cohomology has nonzero alternating length: two independent jet classes on a complex of finite length multiply to a class that the semiregularity map of a torus detects through the Euler characteristic, and the Weil line misses every class pulled back from a quotient by an abelian subvariety tangent to an eigenspace ([Lemma 16.37](#), [Theorem 16.38](#));

- (n) the bounded form of the propagation statement read on the transported cycle, for every base cycle with a component of finite stabiliser, hence for every cycle this paper writes down: the transport multiplies the degree of a component by  $c^{2n}/|\ker \varphi \cap G(Z)|$ , and the denominator is bounded while the numerator is not ([Lemma 16.14](#), [Theorem 16.15](#)); and the same form read on a representative chosen afresh at each point, which is equivalent to the conjecture for the family and so carries no information ([Theorem 16.17](#)).
- (vi) Every CM field, not only the quadratic ones. For every CM field  $F$  of degree  $2m$ , every  $n \geq 2$  and every discriminant, the family of abelian varieties of  $(F, n, \delta)$ -Weil type has a connected period domain of dimension  $mn^2$  carrying a flat space of Weil classes of rank  $2m$ ; it contains a member, isogenous to a product of CM abelian  $m$ -folds, at which those classes are polynomials with rational coefficients in divisor classes, hence algebraic ([Theorem 19.4](#)); the algebraic locus is stable under the rational points of  $\text{Res}_{F_0/\mathbb{Q}} \text{SU}(V, H)$ , whose orbits are dense, and is a countable union of Zariski closed sets; and each of the two criteria of (iv) closes the whole family ([Theorem 19.6](#)). So the Weil classes of the higher CM fields are not a separate problem: they are the same propagation statement, with the base case supplied ([Proposition 19.3](#), [Proposition 19.8](#)). The families are linked as well: for each CM field the split families in infinitely many dimensions give all of its families, and the fields of degree at least four give the imaginary quadratic ones ([Proposition 20.13](#)).
- (vii) What is left, and that it is exactly that. The Hodge conjecture follows from the results of this paper together with three further statements, namely the propagation statement of (iv), read for the split families of the CM fields of degree at least four, which by [Proposition 20.13](#) carry the Weil classes of every CM field with them, the algebraicity of the Hodge classes on abelian varieties that divisor classes and Weil classes do not generate, and the conjecture for the varieties that are not abelian. The last of these is the conjecture itself, since it covers  $A \times \mathbb{P}^1$  for every abelian variety  $A$  and so contains the abelian case. Its weaker form, the conjecture modulo the images of Hodge classes of abelian varieties under algebraic correspondences, holds on every variety whose cohomology is reached from abelian varieties in that way, on every variety of dimension  $m$  in degrees  $0, 2, 2m - 2$  and  $2m$ , and on every variety of dimension at most three, and is open beyond. Over the rule set of the paper and the literature there are exactly five minimal sets: the conjecture for the varieties that are not abelian alone; the Lefschetz standard conjecture for every variety with the motivatedness of every Hodge class; and the weaker form together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes, or with the propagation statement and the

classes beyond the Weil lines, the last being the route of this paper.<sup>68</sup> The first two are each equivalent to the conjecture, every statement in the five except the propagation statement is a consequence of it, and the secant construction of [Section 14](#) lies outside all five ([Proposition 20.35](#), [Definition 20.36](#), [Proposition 20.37](#), [Theorem 20.40](#), [Corollary 20.41](#), [Proposition 20.17](#), [Proposition 20.18](#), [Theorem 20.19](#)). Nor are the last two of the three statements reductions: the self-product of a Mumford fourfold carries two Hodge classes outside the subring of divisor and Weil classes, and the  $H^3$  of a very general quintic threefold is not of abelian type ([Proposition 20.44](#), [Proposition 20.45](#)). Those two classes are the real multiplication by a totally real cubic field on the primitive part of  $H^2$ , and they are algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen attached to the fourfold is algebraic ([Proposition 21.2](#), [Theorem 21.4](#)). They are algebraic at every CM point, and on every member of a Mumford family exactly when the Lefschetz standard conjecture holds for the ninefold  $W \times_C W$ ; one Weil family is likewise equivalent to the Lefschetz standard conjecture for one total space ([Theorem 20.30](#), [Corollary 20.31](#), [Corollary 20.32](#)). The Mumford classes are rigid, their Hodge locus being the compact base curve, and they are algebraic on every member as soon as one perfect complex at one CM point has  $\text{Ext}^2$  of dimension 119, or as soon as they are represented at infinitely many CM points by cycles of bounded degree ([Theorem 21.21](#), [Theorem 21.27](#), [Corollary 21.28](#)); any complex with their Chern character is not built from line bundles, and, for a Zariski dense set of those classes, it has self-extensions of total dimension at least 1600 when it has no negative ones and extends along no Hochschild direction but the curve ([Theorem 21.31](#), [Proposition 21.32](#), [Corollary 21.36](#)). No construction from line bundles by cones, tensor products, pull-backs, push-forwards and Fourier–Mukai functors reaches them at any point of the curve, and no twistor line through a member of the family keeps them of Hodge type; they are the twisted dual forms of the part of  $H^2$  of the square generated by its symplectic class, and outside the CM points no rational map from the square to a K3 surface or to a hyperkähler eightfold pulls back a symplectic form to a nonzero form. At every closed point of every reduction of the curve modulo a prime they are represented by algebraic cycles, so that they are algebraic on every member exactly when those cycles can be chosen with bounded Hilbert data on a Zariski dense set of closed points ([Theorem 21.33](#), [Proposition 21.24](#), [Proposition 21.25](#), [Proposition 21.37](#), [Theorem 21.38](#)). The routes to them that are not closed off are that one statement in three forms and two stronger inputs, any one of which would suffice ([Corollary 21.40](#)).

*Proof.* Each clause is the theorem or theorems cited with it, and no clause is used in the proof of another. For (i), [Theorem 4.14](#) counts the Hodge classes and [Corollary 4.15](#) reduces every degree to  $\omega_1$ , the coefficient count being the rank of  $\bigwedge^{2n} H^1(A, \mathbb{Z})$  in the basis of [Theorem 4.9](#). For (ii), [Theorem 15.14\(a\)](#) and (b) are the two implications and [Proposition B.8](#) gives the connectedness and the dimension. For (iii), [Theorem 15.25\(v\)](#) supplies the cycle at a quaternionic point and [Theorem 15.31](#) places such a point in every family, by the discriminant computation of [Theorem 15.25\(iv\)](#) together with [Lemma 15.30](#). For (iv), the sufficient condition is [Theorem 16.28](#) with [Remark 17.6](#), the refutation of its

<sup>68</sup>These are the statements (L), (M) and (V) of [Definition 20.15](#):  $B(X)$  of [Definition 10.5](#) for every smooth projective  $X$ ; every Hodge class motivated in the sense of André [[And96b](#), §2.1]; and, for every smooth projective morphism  $f$  to a smooth connected base, a flat section of  $R^{2p}f_*\mathbb{Q}$  that is algebraic at one point is algebraic at every point.

first form is [Theorem 17.5](#), which rests on [Lemma 17.2](#), [Lemma 17.3](#) and [Proposition 17.4](#) and on item (LI), the lower bound is [Corollary 16.34\(i\)](#), and the sufficiency of the minimum is [Theorem 16.36\(iii\)](#), with the shape of the object in [Theorem 16.40](#); the number of the corrected form, the sufficiency of attaining it and the excluded shapes are [Theorem 17.9\(i\)](#), (iv) and (v), and the Hodge locus of the other shapes is [Proposition 17.11\(iii\)](#) and (iv) with [Remark 17.12](#), all with item (LII); the bounded criterion implies the conclusion by [Theorem 16.9](#), is implied by it by [Theorem 16.17](#), and fails on the transported cycle by [Theorem 16.15](#). For (v), each item is the statement of the theorem cited, under its own hypotheses, which [Section 20](#) delimits item by item. For (vi), [Proposition 19.3](#) gives the family and the reduction to one class, [Theorem 19.4](#) the base point and the representative, [Theorem 19.6](#) the density and the two criteria, and [Proposition 20.13](#), with item (LIV), the links between the families. For (vii), [Proposition 20.35](#) shows that the third statement is the conjecture, [Proposition 20.37](#) gives what is known of its weaker form, [Theorem 20.40\(ii\)](#) and (iii) give the five minimal sets and the necessity of each of their elements, [Theorem 20.40\(vi\)](#) the consequences of the conjecture, with [Proposition 20.17](#), [Proposition 20.18](#) and [Theorem 20.19](#) for the rules of the literature, and [Theorem 20.40\(iv\)](#) places the secant construction outside them; the computation is item (XXXIII) of [Appendix D](#) and [Section 24](#) of the Lean certificate. The remaining sentences of (vii), on the classes beyond the Weil lines, are in order [Proposition 20.44](#) and [Proposition 20.45](#), with item (XXXV); [Proposition 21.2](#) and [Theorem 21.4\(ii\)](#), with item (XLI); [Theorem 20.30](#), [Corollary 20.31](#) and [Corollary 20.32](#), with item (XLIII); [Theorem 21.21](#), [Theorem 21.27](#), [Corollary 21.28](#), [Theorem 21.31](#), [Proposition 21.32](#) and [Corollary 21.36](#), with items (XLIV) and (XLV) and the Lean theorems `mumford_rigidity_sos` and `mumford_ext_profile`; [Theorem 21.33](#), with item (XLVI) and the Lean theorem `lefschetz_counts`, [Proposition 21.24](#), with item (XLVII), [Proposition 21.25](#), with item (XLVIII), and [Proposition 21.37](#) and [Theorem 21.38](#), which rest on [\[Li26, Theorem 1.1\]](#); and [Corollary 21.40](#), with item (XLIX). ■

*Remark 22.2* (What this closes). Four things are settled by [Theorem 22.1](#), and further work along the lines of this paper will not change them.

The problem has a smallest form and it has been reached. By (i) and (ii) the Hodge conjecture for a whole family of abelian varieties of Weil type is the algebraicity of one explicitly written class at every point of one connected domain, and the reduction loses nothing, since the converse holds as well. No smaller statement is equivalent to it, because the equivalence is already an equivalence.

The base case is not the difficulty. By (iii) every family has a base point, for every field, every dimension and every discriminant, and the cycle is a polynomial in two divisor classes rather than an existence statement. Nor is the discriminant: by [Proposition 20.13](#) the split families in all dimensions give every discriminant, and the fields of degree at least four give the imaginary quadratic ones, so neither the base points of nontrivial discriminant nor those of the quadratic fields carry weight in any minimal derivation. What separates a base point from the family is propagation.

Propagation has one criterion, and in its first form it cannot be met. The bounded criterion is equivalent to its own conclusion, so it restates the problem rather than reducing it. The semiregularity criterion would be met by one perfect complex at one base point with  $\text{Ext}^2$  of dimension  $2n(2n - 1)$ , but no complex whose Chern character is exactly a Weil class is semiregular at all, because semiregularity would carry it to non-algebraic Weil tori, where the Weil class is Hodge and the Chern class of nothing. The criterion survives only with powers of the polarisation added to the Chern character, and in that form it is again one number: [Theorem 17.9](#) computes the rank that  $\dim \text{Ext}^2(E, E)$  must reach,

$(4+\rho)n^2-2n$  for  $n \geq 3$  with  $\rho$  the rank of a Hankel matrix of the coefficients, and shows that an object reaching it satisfies the criterion. No finite direct sum of shifted push-forwards of line bundles along isogenies, among them the simple semi-homogeneous bundles, meets the criterion ([Proposition 17.14](#)); and injectivity is more than propagation needs, since a flatness condition on the deformations of the object suffices ([Proposition 17.15](#)). What none of this says is whether an object meeting either condition exists.

The ways that do not work are known, and they are known for a reason rather than by exhaustion. Each item of (v) fails for an identifiable structural cause: a divisor-generated Chern character cannot leave the divisor subring, a norm eigenclass sits in the wrong summand of the character grading, a Chern character on the Weil line makes the annihilator too large by exactly the dimension of the family, a support that is not Cohen–Macaulay carries a local  $\mathcal{E}xt^2$  that the compensating scalar cannot meet, a two-term complex of powers of the polarisation has summand inclusions that  $H^2(\mathcal{O})$  does not kill, and a component of nonzero multiplicity makes two transverse translation classes multiply to a class that the Euler characteristic detects. These are not obstacles that a more careful version of the same construction would avoid.

What is not settled is the conjecture, and (vii) says precisely how far from it this is. Along this route there are three statements, and the third as first stated, the conjecture for the varieties that are not abelian, is the conjecture itself ([Proposition 20.35](#)). An earlier version of this paper counted three statements and no fewer; that count rested on a rule set that omitted the elementary implication of [Proposition 20.35](#), and it was wrong. With the third statement weakened to the conjecture modulo abelian varieties the route is minimal again, and three statements remain: the propagation statement in the corrected form of (iv), read for the split families of the CM fields of degree at least four, which by [Proposition 20.13](#) carry the Weil classes of every CM field with them; the conjecture for the Hodge classes on abelian varieties that no Weil line reaches, which is not a reduction; and the conjecture modulo abelian varieties, which holds wherever the cohomology is reached from abelian varieties by algebraic correspondences and is open beyond ([Proposition 20.37](#), [Remark 20.38](#)). The shorter routes of the literature replace the first two by the Lefschetz standard conjecture or the variational statement, and the Lefschetz standard conjecture is the statement at their centre ([Remark 20.43](#)). The smallest case of the second statement is implied by the algebraicity of a single correspondence, the Kuga–Satake class of one K3 surface of Picard number thirteen, which asks for that case and a link besides ([Proposition 21.12](#)), and a standard open case of the third, the square of a K3 surface, meets the same kind of correspondence, the Kuga–Satake correspondence of that surface; that is the place where a new source of algebraic cycles would have to act first. We do not claim any of the three, nor the Lefschetz standard conjecture beyond the total spaces of the abelian schemes over curves of [Corollary 20.23](#) and [Corollary 20.26](#), the motivatedness of Hodge classes or the variational statement, and we have said so at each place where a reader might reasonably expect otherwise. The value of (vii) is that the boundary is a computation over a finite rule set rather than a judgement, so that it can be checked and not merely believed.

#### APPENDIX A. THE EXTERIOR ALGEBRA MODEL IN COORDINATES

We record the computation behind [Theorem 6.5](#) in coordinates, both because it is the only place in the paper where a sign has to be pinned down and because the model is the one the verification script implements.

Let  $g \geq 1$  and let  $W$  be the  $\mathbb{Q}$ -vector space with basis  $e_1, \dots, e_{2g}, f_1, \dots, f_{2g}$ . We identify  $V = \langle e_1, \dots, e_{2g} \rangle$  with  $H^1(A, \mathbb{Q})$  and  $V^\vee = \langle f_1, \dots, f_{2g} \rangle$  with  $H^1(A^\vee, \mathbb{Q})$ , so that  $\Lambda^* W = H^*(A \times A^\vee, \mathbb{Q})$ .

In  $\Lambda^* W$  a basis is indexed by subsets  $S$  of  $\{1, \dots, 4g\}$ , with the convention that the generator numbered  $i$  is  $e_i$  for  $i \leq 2g$  and  $f_{i-2g}$  for  $i > 2g$ , and that the basis element  $w_S$  is the wedge of the generators of  $S$  in increasing order. The product is  $w_S w_T = \varepsilon(S, T) w_{S \cup T}$  when  $S \cap T = \emptyset$  and zero otherwise, where

$$\varepsilon(S, T) = (-1)^{\#\{(s,t) \in S \times T : s > t\}}. \quad (68)$$

The three distinguished elements are

$$\ell = \sum_{i=1}^{2g} e_i \wedge f_i, \quad \theta = \sum_{j=1}^g e_j \wedge e_{g+j}, \quad \theta' = \sum_{j=1}^g f_j \wedge f_{g+j},$$

and  $p_*$  is the linear map  $\Lambda^* W \rightarrow \Lambda^* V$  that sends  $w_S$  to  $\varepsilon(S \cap [1, 2g], S \setminus [1, 2g]) w_{S \cap [1, 2g]}$  when  $S \supseteq \{2g+1, \dots, 4g\}$  and to zero otherwise.

**Lemma A.1.** *Let  $0 \leq k \leq g$ , let  $T \subseteq \{1, \dots, g\}$  with  $|T| = k$ , let  $U = \{1, \dots, g\} \setminus T$  and  $S = U \cup (g + U)$ . In the product  $\theta'^k \cdot e^\ell$  the unique term contributing  $w_S$  after  $p_*$  is*

$$k! \left( \prod_{j \in T} f_j f_{g+j} \right) \cdot \left( \prod_{i \in S} e_i f_i \right),$$

and its image under  $p_*$  is  $k! \sigma(g, k) \prod_{j \in U} e_j \wedge e_{g+j}$  with  $\sigma(g, k) = (-1)^{g(g+1)/2+k}$ , independent of  $T$ .

*Proof.* Uniqueness of the term and the identification of  $S$  are established in the proof of [Theorem 6.5](#). The sign is the product of the two shuffle signs (68) relating the displayed word to the target word  $(\prod_{j \in U} e_j e_{g+j})(f_1 \cdots f_{2g})$ . Relabelling the symplectic pairs by a permutation  $\pi$  of  $\{1, \dots, g\}$  carries the word for  $T$  to the word for  $\pi(T)$  and multiplies both the displayed and the target word by the sign of the induced permutation of generators, so the quotient is unchanged; hence  $\sigma$  depends only on  $(g, k)$ . Evaluating at  $T = \{1, \dots, k\}$  gives the stated value, and the evaluation is carried out for all  $g \leq 12$  and all  $k$  in [Appendix D](#), item (II).  $\blacksquare$

*Remark A.2.* The computation is small enough to do by hand at  $g = 2$ . There  $\ell = e_1 f_1 + e_2 f_2 + e_3 f_3 + e_4 f_4$ ,  $\theta = e_1 e_3 + e_2 e_4$  and  $\theta' = f_1 f_3 + f_2 f_4$ , and one finds  $\Phi_{\mathcal{P}}(1) = -\frac{1}{2}\theta^2$ ,  $\Phi_{\mathcal{P}}(\theta') = \theta$  and  $\Phi_{\mathcal{P}}(\theta'^2) = -2$ , in agreement with (19) for  $g = 2$ , where  $(-1)^{g(g+1)/2} = (-1)^3 = -1$ .

## APPENDIX B. THE LOCUS OF HODGE CLASSES

Every deformation argument about Hodge classes rests on one theorem: the locus in a family where a given class stays a Hodge class is an algebraic subvariety, not merely an analytic one. This is the theorem of Cattani, Deligne and Kaplan [\[CDK95\]](#). The present appendix states it, proves the part of it that is elementary, isolates the two analytic inputs that the rest needs, computes the example that matters here, and then says precisely why the theorem is not a route to the Hodge conjecture. The last point is the one that bears on [Section 14](#), and it is the reason the appendix is here rather than in a reference.

### B.1. Variations of Hodge structure.

**Definition B.1** (Variation of Hodge structure). Let  $S$  be a smooth connected complex quasi-projective variety. A *polarised variation of  $\mathbb{Q}$ -Hodge structure* of weight  $w$  on  $S$  consists of

- (a) a local system  $\mathbb{V}$  of finite dimensional  $\mathbb{Q}$ -vector spaces on  $S$ , with associated holomorphic bundle  $\mathcal{V} = \mathbb{V} \otimes_{\mathbb{Q}} \mathcal{O}_S$  carrying the flat connection  $\nabla$  whose flat sections are  $\mathbb{V}$ ;
- (b) a decreasing filtration  $\mathcal{V} = F^0 \supseteq F^1 \supseteq \dots$  by holomorphic subbundles, inducing on each fibre  $\mathbb{V}_s$  a  $\mathbb{Q}$ -Hodge structure of weight  $w$ ;
- (c) a flat nondegenerate  $(-1)^w$ -symmetric pairing  $Q: \mathbb{V} \otimes \mathbb{V} \rightarrow \mathbb{Q}_S$  polarising each fibre;

subject to Griffiths transversality

$$\nabla F^p \subseteq F^{p-1} \otimes \Omega_S^1. \quad (69)$$

The example to keep in mind is the geometric one: for  $f: \mathcal{X} \rightarrow S$  smooth projective,  $\mathbb{V} = R^w f_* \mathbb{Q}$  with its Gauss–Manin connection is such a variation, and (69) is Griffiths’ theorem [Voi02, Ch. 10]. Our other example is the Weil family of Proposition B.8 below, which is not defined by a family of varieties in the first instance but by a period domain, and which carries a variation all the same.

*Notation B.2.* Write  $\mathcal{V}^{\text{an}}$  for the total space of  $\mathcal{V}$ , a complex manifold fibred over  $S$  with fibre  $\mathbb{V}_s \otimes \mathbb{C}$ , and  $\mathbb{V}^{\text{tot}} \subset \mathcal{V}^{\text{an}}$  for the subset of pairs  $(s, \alpha)$  with  $\alpha \in \mathbb{V}_s$ , a covering space of  $S$  with countable fibres.

### B.2. The locus of Hodge classes.

**Definition B.3.** Let  $\mathbb{V}$  be a polarised variation of weight  $2p$  on  $S$ . Its *locus of Hodge classes* is

$$\text{Hdg}^p(\mathbb{V}) = \{(s, \alpha) \in \mathbb{V}^{\text{tot}} : \alpha \in F^p \mathbb{V}_s\}.$$

For  $N \in \mathbb{Q}$  write  $\text{Hdg}_N^p$  for the part with  $Q(\alpha, \alpha) = N$ .

The condition  $\alpha \in F^p$  for a rational  $\alpha$  of weight  $2p$  is exactly the condition that  $\alpha$  be of type  $(p, p)$ , by conjugation symmetry. The following two lemmas are elementary and give everything about  $\text{Hdg}^p$  except algebraicity.

**Lemma B.4** (Analyticity).  *$\text{Hdg}^p(\mathbb{V})$  is a closed analytic subset of  $\mathbb{V}^{\text{tot}}$ , and the projection  $\pi: \text{Hdg}^p(\mathbb{V}) \rightarrow S$  is locally biholomorphic onto its image on each connected component. In particular each component is a closed analytic subvariety of an open subset of  $S$ , and there are countably many components.*

*Proof.* Work on a simply connected open  $U \subseteq S$  over which  $\mathbb{V}$  is trivialised,  $\mathbb{V}|_U \cong V_0 \times U$  for a fixed  $\mathbb{Q}$ -vector space  $V_0$ . Then  $\mathbb{V}^{\text{tot}}|_U$  is the disjoint union over  $\alpha \in V_0$  of copies of  $U$ . Fix  $\alpha$ . The quotient  $\mathcal{V}/F^p$  is a holomorphic vector bundle on  $U$  and the image  $\bar{\alpha}(s)$  of the flat section  $\alpha$  in  $(\mathcal{V}/F^p)_s$  is a holomorphic section of it. The locus in question is the zero locus of  $\bar{\alpha}$ , a closed analytic subset of  $U$ . Taking the union over the countable set  $V_0$  gives the first statement, and shows that on the sheet indexed by  $\alpha$  the projection  $\pi$  is the inclusion of that zero locus into  $U$ , hence injective. Countability of the set of components follows because  $V_0$  is countable and each zero locus has countably many components. ■

**Lemma B.5** (Finiteness at bounded norm). *Fix  $s \in S$  and  $N \in \mathbb{Q}$ . The set of  $\alpha \in \mathbb{V}_s$  with  $\alpha$  of type  $(p, p)$ ,  $Q(\alpha, \alpha) = N$ , and  $\alpha$  in a fixed lattice  $\mathbb{V}_{s, \mathbb{Z}}$ , is finite. The number of such  $\alpha$  is locally constant on each component of  $\text{Hdg}_N^p$ .*

*Proof.* Decompose  $\alpha$  into its primitive parts for the Lefschetz decomposition attached to the polarisation. On the primitive part of type  $(p, p)$  in weight  $2p$  the Hodge–Riemann bilinear relations read  $(-1)^p Q(\alpha, \bar{\alpha}) > 0$  for  $\alpha \neq 0$ , so the form  $(-1)^p Q$  is positive definite on the real span of the primitive real  $(p, p)$ -classes; the same holds on each Lefschetz summand with an alternating sign, and the resulting form  $\tilde{Q}$  obtained by flipping signs summand by summand is positive definite on the whole space of real  $(p, p)$ -classes. The conditions  $Q(\alpha, \alpha) = N$  and  $\alpha$  of type  $(p, p)$  therefore confine  $\alpha$  to a compact set for  $\tilde{Q}$ , and a compact set meets a lattice in finitely many points. The count is locally constant because the lattice and  $Q$  are flat and the Hodge filtration varies continuously, so no Hodge class can appear or disappear without its  $Q$ -norm changing. ■

**Theorem B.6** (Cattani, Deligne and Kaplan [CDK95]).<sup>69</sup> *Let  $S$  be a smooth complex quasi-projective variety and  $\mathbb{V}$  a polarised variation of  $\mathbb{Z}$ -Hodge structure of weight  $2p$  on  $S$ . Then each connected component of  $\mathrm{Hdg}^p(\mathbb{V})$  is a closed algebraic subvariety of  $\mathbb{V}^{\mathrm{tot}}$ , and its image in  $S$  is a closed algebraic subvariety of  $S$ .*

We do not reproduce the proof. What we do record is exactly where the difficulty lies and which two theorems remove it, because the shape of the argument is what a reader needs in order to judge whether a deformation argument in this area is legitimate.

By Lemma B.4 a component  $Z$  of  $\mathrm{Hdg}^p$  is already a closed analytic subvariety of an open subset of  $S$ , and by Lemma B.5 it carries a locally constant  $Q$ -norm and is one sheet of a finite cover of its image. Choose a smooth projective compactification  $\bar{S} \supseteq S$  with  $\bar{S} \setminus S$  a normal crossing divisor. By Chow’s theorem it is enough to show that  $Z$  extends to an analytic subvariety of  $\bar{S}$ , that is, that  $Z$  does not oscillate wildly as one approaches the boundary. That is an asymptotic question about the Hodge filtration, and the two theorems that answer it are:

- (A) *The nilpotent orbit theorem* of Schmid [Sch73]: near a boundary point the period map is approximated, to exponentially small error in the natural coordinates, by the orbit of a nilpotent Lie algebra element built from the local monodromy logarithms.
- (B) *The  $\mathrm{SL}_2$ -orbit theorem* in several variables, of Cattani, Kaplan and Schmid [CKS86]: that nilpotent orbit is in turn approximated by an orbit of a fixed real reductive group, with uniform estimates in the several-variable setting.

Together these give a two-sided estimate for the Hodge norm of a flat section near the boundary, of the shape  $\|\alpha\|_s \sim \prod_j (\log |t_j|^{-1})^{m_j}$  with integers  $m_j$  determined by the weight filtration. A flat section that is Hodge on  $Z$  has  $\tilde{Q}$ -norm bounded by Lemma B.5, and the estimate converts that bound into a statement that the possible boundary behaviours form a finite list. Algebraicity then follows from the analytic extension across the boundary together with Chow’s theorem in  $\bar{S}$ .

*Remark B.7.* Both (A) and (B) are theorems about degenerating Hodge structures and neither mentions algebraic cycles. This is worth saying plainly: the algebraicity in Theorem B.6 is algebraicity of a locus of periods, obtained from analysis, and it carries no information about algebraic cycles whatever. The point is developed in Appendix B.4.

**B.3. The Weil family.** The relevant example for this paper is the family of complex structures in Setup 4.1.

<sup>69</sup>The paper is short and the proof is not. It draws on the whole asymptotic theory of degenerating Hodge structures built in the two decades before it, and the two inputs isolated below are the load-bearing ones. Deligne had earlier shown that Hodge classes on abelian varieties are absolute by a different route; the theorem here is what carries the relevant consequences beyond abelian varieties.

**Proposition B.8** (The Weil period domain). *Fix  $K$ ,  $n$ , the  $K$ -space  $V = K^{2n}$ , and the hermitian form  $H$  of signature  $(n, n)$  from Lemma 4.5. The set of Hodge structures of weight one on  $V$  that are polarised by  $Q$ , commute with the  $K$ -action, and are of  $(K, -1, n)$ -Weil type is in bijection with*

$$\mathcal{D}_{n,n} = \{P \subset V_+ : \dim_{\mathbb{C}} P = n, H|_P \text{ negative definite}\},$$

*an open subset of the Grassmannian  $\mathrm{Gr}(n, 2n)$  and a bounded symmetric domain isomorphic to  $\mathrm{U}(n, n)/(\mathrm{U}(n) \times \mathrm{U}(n))$ . In particular*

$$\dim_{\mathbb{C}} \mathcal{D}_{n,n} = n^2,$$

*against  $\dim \mathcal{A}_{2n} = n(2n + 1)$  for the moduli of principally polarised abelian  $2n$ -folds.*

*Proof.* A Hodge structure of weight one on  $V$  is a subspace  $V^{1,0} \subset V_{\mathbb{C}}$  with  $V_{\mathbb{C}} = V^{1,0} \oplus \overline{V^{1,0}}$ . Commuting with the  $K$ -action means that  $V^{1,0}$  is stable under  $S$ , hence splits as  $V^{1,0} = P \oplus P'$  with  $P = V^{1,0} \cap V_+$  and  $P' = V^{1,0} \cap V_-$ . Conjugation interchanges  $V_+$  and  $V_-$ , so  $\overline{V^{1,0}} = \overline{P} \oplus \overline{P'}$  with  $\overline{P} \subset V_-$ , and the direct sum condition becomes  $V_- = P' \oplus \overline{P}$ , which determines  $P'$  up to the choice of a complement; the Weil condition  $\dim P = \dim P' = n$  of Lemma 3.5 together with the polarisation condition pins it down as  $P' = \overline{P}^{\perp_H}$ . So the structure is determined by  $P$ .

For the positivity, extend  $H$  to the  $\mathbb{C}$ -bilinear pairing between  $V_+$  and  $V_-$ ; the computation (9) in the proof of Theorem 4.4 says that the second Riemann relation holds exactly when  $iQ$  is positive on  $V^{1,0}$ , which on the  $V_+$  part is the condition that  $H$  be negative definite on  $P$ , the sign being fixed by the convention in Lemma 4.5. The set of  $n$ -dimensional negative definite subspaces of a hermitian space of signature  $(n, n)$  is the standard realisation of the symmetric domain of  $\mathrm{U}(n, n)$ : the unitary group acts transitively on it and the stabiliser of a point is  $\mathrm{U}(n) \times \mathrm{U}(n)$ , the unitary groups of  $P$  and of  $P^{\perp}$ . It is open in  $\mathrm{Gr}(n, 2n)$  because negative definiteness is an open condition, and  $\dim \mathrm{Gr}(n, 2n) = n(2n - n) = n^2$ .

The comparison dimension is  $\dim \mathcal{A}_g = g(g + 1)/2$  with  $g = 2n$ , which is  $n(2n + 1)$ . ■

**Corollary B.9** (The Weil locus is algebraic). *Inside  $\mathcal{A}_{2n}$  the image of  $\mathcal{D}_{n,n}$ , that is the locus of principally polarised abelian  $2n$ -folds admitting an action of an order of  $K$  of  $(K, -1, n)$ -Weil type, is a closed algebraic subvariety of dimension  $n^2$ .*

*Proof.* The condition of admitting such an action is the condition that a certain class in  $\mathrm{End}(H^1)$ , namely the graph of  $\iota(\sqrt{-d})$ , be a Hodge class, together with the balanced signature, which is locally constant. So the locus is a component of a locus of Hodge classes for the variation  $\mathrm{End}(R^1 f_* \mathbb{Q})$  over  $\mathcal{A}_{2n}$  with its level structure, and Theorem B.6 applies. The dimension is Proposition B.8. ■

#### B.4. What the theorem does not give.

*Remark B.10* (Algebraic locus, not algebraic class). Theorem B.6 says that the set of parameters where  $\alpha$  is Hodge is cut out by polynomial equations. It says nothing about whether  $\alpha$  is the class of a cycle at any one of those parameters. The two statements are independent in both directions: the Hodge locus of a class that happens to be algebraic everywhere is still only known to be algebraic by Theorem B.6, and conversely knowing the locus is algebraic supplies no cycle. It is easy to slide from one to the other in words, and Corollary B.9 is exactly the kind of statement that invites the slide: the Weil locus is an algebraic subvariety, it is even a Shimura subvariety, and none of that produces the class  $\omega_1$  of (11).

*Remark B.11* (What would be needed: the variational conjecture). The property that would turn [Theorem B.6](#) into a tool is the variational Hodge conjecture of Grothendieck: if  $\mathcal{X} \rightarrow S$  is smooth projective over a connected base,  $\alpha$  is a flat section of  $R^{2p}f_*\mathbb{Q}$  that is Hodge at every point, and  $\alpha_{s_0}$  is algebraic at one point  $s_0$ , then  $\alpha_s$  is algebraic at every point. This is open. What is known is the infinitesimal statement under a semiregularity hypothesis: if the cycle at  $s_0$  is semiregular in the sense of Bloch [[Blo72](#)], then it deforms to first order, and hence formally, along the locus where its class stays Hodge. That is the theorem discussed in [Section 9](#), and [Corollary 9.3](#) records that its numerical hypothesis is not what obstructs the present problem.

*Remark B.12* (Where this leaves the construction of [Section 14](#)). The construction of [Section 14](#) produces, for every  $n$ , every  $d$  and every nondegenerate  $\beta$ , an abelian variety of Weil type together with a sheaf whose second Chern class is computed in closed form. Its output is algebraic on the split locus. The missing ingredient recorded in [Remark 14.17](#) is a statement along the algebraic locus of [Corollary B.9](#): one needs the Hodge type of the deformed object to be controlled as the parameter moves off the split locus. [Theorem B.6](#) guarantees that the locus one is deforming along is an algebraic variety, which is what makes the question well posed; it does not answer it. Stating this precisely is the purpose of [Question 13.7](#).

## APPENDIX C. ENDOMORPHISM ALGEBRAS AND WHERE WEIL TYPE SITS

[Definition 3.1](#) asks for three things: an even dimension, an imaginary quadratic field acting with balanced eigenvalue multiplicities, and compatibility with the polarisation. This appendix places those three conditions inside Albert’s classification of endomorphism algebras. Two statements come out of it that are used in the body. The first is that the compatibility condition is automatic once the field is stable under the Rosati involution, so Weil type really has only two conditions ([Lemma C.4](#)). The second is that the balanced condition is not a normalisation but the exact condition under which the Weil line carries a Hodge class ([Proposition C.7](#)), which is what makes the whole subject about balanced signatures and nothing else.

### C.1. Positive involutions. <sup>70</sup>

**Definition C.1.** Let  $D$  be a finite dimensional semisimple  $\mathbb{Q}$ -algebra. An involution  $\dagger$  of  $D$ , that is a  $\mathbb{Q}$ -linear anti-automorphism with  $\dagger^2 = \text{id}$ , is *positive* if

$$\text{Tr}_{D/\mathbb{Q}}(xx^\dagger) > 0 \quad \text{for every } x \in D, x \neq 0,$$

where  $\text{Tr}_{D/\mathbb{Q}}$  is the trace of left multiplication on  $D$  as a  $\mathbb{Q}$ -vector space.

For an abelian variety  $A$  with a polarisation  $\eta$ , the algebra  $D = \text{End}^0(A) = \text{End}(A) \otimes \mathbb{Q}$  is semisimple and the Rosati involution  $\dagger$  attached to  $\eta$  is positive [[BL04](#), Ch. 5]. Every statement below is a statement about the pair  $(D, \dagger)$ .

**Lemma C.2** (Totally real centre). *Let  $(D, \dagger)$  be as above with  $D$  a division algebra, let  $F$  be the centre of  $D$  and  $F_0 = \{z \in F : z^\dagger = z\}$ . Then  $F_0$  is a totally real number field.*

<sup>70</sup>The numbering of the four types is Albert’s and has stuck, although it corresponds to no natural order: types I to III have a totally real centre and type IV a CM one, so the natural division is two against two rather than three against one. Shimura’s normalisation of the same classification, in terms of the associated symmetric domains, is the one that makes the moduli side transparent.

*Proof.* For  $z \in F_0$  the positivity condition reads  $\mathrm{Tr}_{D/\mathbb{Q}}(z^2) > 0$ , and  $\mathrm{Tr}_{D/\mathbb{Q}}$  restricted to  $F_0$  is  $[D : F_0] \cdot \mathrm{Tr}_{F_0/\mathbb{Q}}$ , so  $\mathrm{Tr}_{F_0/\mathbb{Q}}(z^2) > 0$  for every nonzero  $z \in F_0$ . Suppose  $F_0$  had a complex place  $\sigma$ . The embedding of  $F_0$  into the product of its archimedean completions has dense image, so there is  $z \in F_0$  with  $\sigma(z)$  within  $\frac{1}{10}$  of  $i$  and  $|\rho(z)| < \varepsilon$  for every other archimedean place  $\rho$ . Then  $\mathrm{Tr}_{F_0/\mathbb{Q}}(z^2) = \sum_{\rho} \rho(z)^2$ , where a complex place contributes twice the real part of  $\rho(z)^2$ ; the term from  $\sigma$  is at most  $2 \cdot (-1 + \frac{1}{2}) < -1$  and the remaining terms are bounded by  $2[F_0 : \mathbb{Q}]\varepsilon^2$ , so the total is negative for  $\varepsilon$  small. This contradicts positivity, so every place of  $F_0$  is real. ■

**Theorem C.3** (Albert [Alb34, Alb35a, Alb35b]). *Let  $(D, \dagger)$  be a division algebra over  $\mathbb{Q}$  with a positive involution,  $F$  its centre,  $F_0$  the subfield fixed by  $\dagger$ ,  $e_0 = [F_0 : \mathbb{Q}]$  and  $[D : F] = d^2$ . Then  $F_0$  is totally real,  $[F : F_0] \in \{1, 2\}$ , and exactly one of the following holds.*

- Type I.  $D = F = F_0$  is a totally real field and  $\dagger$  is the identity.
- Type II.  $F = F_0$  is totally real and  $D$  is a quaternion algebra over  $F$  with  $D \otimes_{F, \rho} \mathbb{R} \cong M_2(\mathbb{R})$  at every real place  $\rho$ ;  $\dagger$  is of orthogonal type.
- Type III.  $F = F_0$  is totally real and  $D$  is a quaternion algebra over  $F$  with  $D \otimes_{F, \rho} \mathbb{R} \cong \mathbb{H}$  at every real place  $\rho$ ;  $\dagger$  is the canonical involution.
- Type IV.  $F$  is a CM field with maximal totally real subfield  $F_0$ , and  $\dagger$  induces complex conjugation on  $F$ .

If moreover  $D = \mathrm{End}^0(A)$  for an abelian variety  $A$  of dimension  $g$  with  $\dagger$  the Rosati involution of a polarisation, then

$$\text{I: } e_0 \mid g, \quad \text{II: } 2e_0 \mid g, \quad \text{III: } 2e_0 \mid g, \quad \text{IV: } e_0 d^2 \mid g.$$

We use Theorem C.3 as a quoted classification; Lemma C.2 proves the part of it that we appeal to directly, and the four types with the divisibility conditions are in [BL04, Ch. 9].

## C.2. The polarisation condition is automatic.

**Lemma C.4** (Rosati on an imaginary quadratic subfield). *Let  $A$  be an abelian variety with a polarisation  $\eta$  and Rosati involution  $\dagger$ , and let  $\iota: K \hookrightarrow \mathrm{End}^0(A)$  be an embedding of an imaginary quadratic field such that  $\iota(K)$  is stable under  $\dagger$ . Then*

$$\iota(\tau)^{\dagger} = \iota(\bar{\tau}) \quad \text{for every } \tau \in K.$$

That is, condition (iii) of Definition 3.1 holds automatically, and Weil type is the conjunction of (i), (ii) and the  $\dagger$ -stability of  $\iota(K)$ .

*Proof.* By stability,  $\dagger$  restricts to a  $\mathbb{Q}$ -algebra involution of the field  $\iota(K) \cong K$ . A field automorphism of  $K$  of order dividing two is either the identity or complex conjugation, since  $\mathrm{Gal}(K/\mathbb{Q})$  has order two and an anti-automorphism of a commutative ring is an automorphism. Suppose it were the identity, and put  $x = \iota(\sqrt{-d})$ . Then  $x x^{\dagger} = x^2 = \iota(-d) = -d \cdot 1$ , so

$$\mathrm{Tr}_{D/\mathbb{Q}}(x x^{\dagger}) = -d \cdot \dim_{\mathbb{Q}} D < 0,$$

contradicting positivity of  $\dagger$ . Hence  $\dagger$  restricts to complex conjugation. ■

**Remark C.5.** Stability is a genuine hypothesis and not a triviality: an arbitrary subfield of  $\mathrm{End}^0(A)$  need not be  $\dagger$ -stable. What Lemma C.4 removes is the second half of the condition, the identification of  $\dagger|_K$  with the specific automorphism. In particular, in the model of Setup 4.1 one does not have to choose the polarisation to match the field; any  $\dagger$ -stable embedding does the same work.

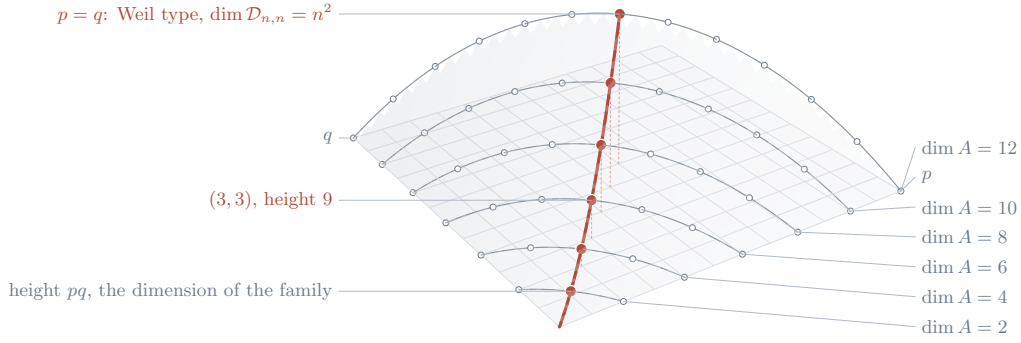


FIGURE 36. The signatures  $(p, q)$  of an imaginary quadratic action, with  $p + q = \dim A$ , drawn on the surface of height  $pq$ , which is the dimension of the family of abelian varieties carrying that action, the unitary Shimura variety of signature  $(p, q)$ . Each curve on the surface collects the signatures available in one dimension, and on it the diagonal point is the highest. The divisibility conditions of [Theorem C.3](#) put no constraint on which of them occur when  $\text{End}^0(A) = K$ , by [Proposition C.9\(i\)](#). By [Proposition C.7](#) the Weil line consists of Hodge classes exactly at the filled points on the ridge  $p = q$ , and at every open point it is a rational sub-Hodge structure of type  $\{(p, 2n - p), (2n - p, p)\}$  carrying no Hodge class at all. The diagonal is the locus that [Corollary B.9](#) shows to be an algebraic subvariety of the moduli.

### C.3. The signature, and why balance is forced.

**Definition C.6** (Signature). Let  $\iota: K \hookrightarrow \text{End}^0(A)$  be as above,  $g = \dim A$ . The operator  $\iota(\sqrt{-d})$  acts on the tangent space  $T_0 A \cong \mathbb{C}^g$  with the two eigenvalues  $\pm i\sqrt{d}$ ; write  $p$  and  $q = g - p$  for their multiplicities. The pair  $(p, q)$  is the *signature* of  $(A, \iota)$ .

The signature is an isogeny invariant, since an isogeny induces an isomorphism on tangent spaces commuting with the action, and it is locally constant in families, being the dimension of a kernel of a fixed operator on a bundle. By duality  $\iota(\tau)^*$  acts on  $H^{1,0}(A)$  with the same multiplicities, so  $\dim V_+^{1,0} = p$  and  $\dim V_-^{1,0} = q$  in the notation of [Section 3](#).

**Proposition C.7** (The balanced signature is forced). *Let  $A$  be an abelian variety of dimension  $2n$  with a  $\dagger$ -stable embedding  $\iota: K \hookrightarrow \text{End}^0(A)$  of signature  $(p, q)$ ,  $p + q = 2n$ , and let*

$$W(A) = \left( \bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_- \right) \cap H^{2n}(A, \mathbb{Q})$$

as in [Definition 3.6](#). Then  $W(A)$  is a rational sub-Hodge structure of  $H^{2n}(A, \mathbb{Q})$  of dimension two and of Hodge type  $\{(p, 2n - p), (2n - p, p)\}$ , and the following are equivalent:

- (i)  $p = q = n$ , that is  $(A, \iota, \eta)$  is of  $(K, -1, n)$ -Weil type;
- (ii)  $W(A)$  contains a nonzero Hodge class;
- (iii)  $W(A)$  consists of Hodge classes.

*Proof.* The space  $\bigwedge^{2n} V_+$  is a line, and by (3) it equals  $\bigwedge^p V_+^{1,0} \otimes \bigwedge^{2n-p} V_+^{0,1}$ , which is of pure Hodge bidegree  $(p, 2n - p)$ ; the conjugate line  $\bigwedge^{2n} V_-$  is of bidegree  $(2n - p, p)$ . Their sum is stable under conjugation and defined over  $\mathbb{Q}$  by the argument of [Proposition 3.7\(i\)](#), which never used the balanced condition, so  $W(A)$  is a two-dimensional rational sub-Hodge structure of the stated type.

A Hodge class is an element of bidegree  $(n, n)$ . A nonzero element of  $W(A) \otimes \mathbb{C}$  has a nonzero component in one of the two lines, so it is of bidegree  $(n, n)$  only if  $(p, 2n-p) = (n, n)$  or  $(2n-p, p) = (n, n)$ , both of which say  $p = n$ . This gives (ii) implies (i). If  $p = n$  then both lines are of bidegree  $(n, n)$  and every element of  $W(A)$  is a Hodge class, which is (i) implies (iii); and (iii) implies (ii) because  $\dim W(A) = 2 > 0$ . ■

*Remark C.8.* [Proposition C.7](#) is the reason the literature speaks of abelian varieties of Weil type rather than of imaginary quadratic multiplication in general. For an unbalanced signature the space  $W(A)$  still exists, is still a canonical two-dimensional rational sub-Hodge structure, and is still not in the divisor subring, but it contains no Hodge class and therefore predicts nothing. The exceptional Hodge classes appear exactly at balance, and the locus of balance is the algebraic subvariety of [Corollary B.9](#). The statement is verified over all signatures  $p$  and all  $n \leq 3$  in `explicit_weil.py`, check (W4).

#### C.4. Which algebras can carry Weil classes.

**Proposition C.9** (Weil type in the classification). *Let  $A$  be an abelian variety of dimension  $2n$  admitting a  $\dagger$ -stable embedding of an imaginary quadratic field  $K$  with balanced signature, and suppose  $\text{End}^0(A)$  is a division algebra  $D$  with centre  $F$  as in [Theorem C.3](#). Then:*

- (i) *if  $D = K$  then  $D$  is of type IV with  $e_0 = 1$  and  $d = 1$ , and the divisibility condition of [Theorem C.3](#) is vacuous, so every even dimension occurs;*
- (ii)  *$D$  can also be of type III with  $F_0 = \mathbb{Q}$ , since a totally definite quaternion algebra over  $\mathbb{Q}$  contains imaginary quadratic subfields on which the canonical involution is conjugation;*
- (iii)  *$D$  cannot be of type I unless  $D \subsetneq \text{End}^0(A)$ , because a totally real field contains no imaginary quadratic subfield.*

*Proof.* (i) If  $D = K$  then  $F = K$  is a CM field with  $F_0 = \mathbb{Q}$ , so  $e_0 = 1$ ; and  $[D : F] = 1$  gives  $d = 1$ . The involution is conjugation by [Lemma C.4](#), which is type IV. The condition  $e_0 d^2 \mid g$  reads  $1 \mid 2n$ .

(ii) Let  $D$  be a quaternion algebra over  $\mathbb{Q}$  ramified at the infinite place, with standard generators  $u, v$  satisfying  $u^2 = a < 0$ ,  $v^2 = b < 0$ ,  $uv = -vu$ , and canonical involution  $x \mapsto \bar{x}$  fixing  $\mathbb{Q}$  and negating  $u, v, uv$ . Then  $\mathbb{Q}(u) \cong \mathbb{Q}(\sqrt{a})$  is imaginary quadratic, stable under the involution, and the involution restricts to conjugation on it. Whether such a  $D$  actually occurs as  $\text{End}^0(A)$  together with a balanced signature is a separate question about the moduli, but nothing in the classification excludes it.

(iii) Immediate, a totally real field has no element of negative square. ■

*Remark C.10* (The generic case is the one that matters). Case (ii) of [Proposition C.9](#) is a special position: there  $\text{End}^0(A)$  is larger than  $K$ , the Hodge group is correspondingly smaller, and  $\text{Hdg}^n(A)$  can be larger than the three dimensions of [Theorem 4.14](#). It also tends to be the case where the Weil classes are accessible; [Remark 20.5](#) records that at such points the Weil line can meet the divisor subring, which is the split situation. The hard case, and the one all of [Sections 5, 7 and 10](#) address, is case (i) with  $\text{Hg}(A) = \text{SU}(V, H)$ .

*Remark C.11* (Why dimension four is where the subject starts). For  $n = 1$  the Weil line lies in  $H^2(A, \mathbb{Q})$  and consists of divisor classes, algebraic by the Lefschetz theorem on  $(1, 1)$  classes, so it predicts nothing new; this is the same boundary as in [Proposition 11.11](#). Moonen and Zarhin determined the Hodge classes on abelian varieties of dimension at most five [\[MZ99\]](#), and on simple abelian fourfolds in [\[MZ95\]](#): through dimension three the Hodge ring is generated by divisor classes, and the first Hodge classes not of that form occur in dimension four, on abelian fourfolds of Weil type. So  $n = 2$  is simultaneously the

first case of [Definition 3.1](#) with anything in it, the first case where [Theorem 4.14](#) gives three rather than one, and the case that was the first open one of the Hodge conjecture for abelian varieties until Markman settled it [[Mar25a](#), Corollary 1.6.1]; the first open case is now  $n = 3$  at nontrivial discriminant, by [Proposition 20.12\(ii\)](#).

## APPENDIX D. VERIFICATION

Every numerical assertion in this paper is checked by exact arithmetic.<sup>71</sup> The scripts are described in [Appendix D](#). They use the standard library of Python, together with `sympy` for exact symbolic linear algebra in several of them, `numpy` for integer arrays in a few and `python-flint` for exact rational matrices in item (LV); they work over  $\mathbb{Q}$  and its finite extensions with the `fractions` module or exact symbolic arithmetic, and they perform no floating-point operation; items (XXIX), (XXXI) and (XL) are Ext computations over a polynomial ring, carried out in Macaulay2. We describe what is checked.

- (I) *The Fourier–Mukai transform on powers of the polarisation.* The exterior algebra  $\wedge^*(V \oplus V^\vee)$  of [Notation 6.2](#) is constructed explicitly for  $g \leq 4$ , with basis elements indexed by bit masks and structure constants computed from the shuffle sign. The class  $\ell$  of (18), the forms  $\theta$  and  $\theta'$  of (16) and (17), and the exponential  $e^\ell$  are formed, and  $\Phi_{\mathcal{P}}(\theta'^k)$  is computed by extracting the coefficient of the top form in the dual variables. The output is checked to be an exact rational multiple of  $\theta^{g-k}$  with no other component, and the multiple is compared with (19). All values agree.
- (II) *The sign.* The uniformity of the sign  $\sigma(g, k)$  over the index sets  $T$ , which is the step in the proof of [Theorem 6.5](#) that the reader may wish to see confirmed, is checked as a permutation computation for every  $g \leq 12$  and every  $0 \leq k \leq g$ , together with the closed form (22). All values agree.
- (III) *The character separation.* The grading (15) is formed for  $n \leq 4$  and the positions of the Weil line and of  $\eta^n$  are computed, confirming [Lemma 5.2](#), [Lemma 5.3](#) and the disjointness of [Theorem 1.3](#).
- (IV) *The numerical comparison.* The values of  $2n^2 + 2n$ ,  $2^{2n}$  and  $3^{2n}$  are tabulated for  $2 \leq n \leq 8$ , confirming [Lemma 8.2](#); the script also records, as a counterexample to the inference that [Remark 8.5](#) retracts, the types (1, 3) and (1, 5) on an abelian surface with their Euler characteristics 3 and 5.
- (V) *The semiregularity target.* The identity  $\sum_q \binom{2n}{q} \binom{2n}{q+2} = \binom{4n}{2n-2}$  of [Proposition 9.2](#) is verified for  $2 \leq n \leq 7$ , with the growth rates of [Corollary 9.3](#).
- (VI) *The coordinate model.* The script `explicit_weil.py` builds the model of [Setup 4.1](#) over  $\mathbb{Q}(\sqrt{-d})$  for  $(n, d)$  in a table covering  $n \leq 3$  and  $d \in \{1, 2, 3, 5, 7\}$ , and checks eight things for each: that the  $u_j^s$  of (7) are eigenvectors of  $S$ ; that  $\omega_+ + \omega_-$  and  $(\omega_+ - \omega_-)/\delta$  are rational and agree with the closed form (11); that  $\iota(\tau)^*$  multiplies  $\omega_\pm$  by  $\tau^{2n}$  and  $\bar{\tau}^{2n}$  at five values of  $\tau$ ; that  $\omega_\pm$  is of Hodge type  $(n, n)$  for the balanced structure and of type  $(p, 2n - p)$  for every other one, which is [Proposition C.7](#); that  $\eta$  is rational, of type  $(1, 1)$  and a norm eigenclass; that the two Riemann relations hold with the sign vector of [Setup 4.1](#); that  $\eta^{2n} \neq 0$ ; and that the supports of  $\omega_1, \omega_2$  and  $\eta^n$  are disjoint, which is [Proposition 4.11](#).

<sup>71</sup>Exact arithmetic is not a scruple here. Several of the quantities are differences of large binomial coefficients, and at  $n = 7$  the semiregularity target exceeds  $10^{11}$ ; in double precision the comparison in item (V) would be meaningless. Everything uses Python’s `fractions` module or integers, and nothing converts to a float.

- (VII) *The Hodge class count.* The script `hodge_invariants.py` computes, for  $n \leq 4$  and every  $(r, s)$ , the dimension of the space of  $\mathfrak{sl}_{2n}$ -invariants in  $\bigwedge^r V_+ \otimes \bigwedge^s V_+^\vee$ , as the kernel of the raising operators on the weight-zero subspace, in exact rational arithmetic. It confirms  $\dim \mathrm{Hdg}^n = 3$  and  $\dim \mathrm{Hdg}^k = 1$  for  $k \neq n$ , which is [Theorem 4.14](#), and the codimension count of [Proposition B.8](#).
- (VIII) *The Clifford algebra.* The script `kugasatake.py` builds  $C(T, q)$  for forms of signature  $(3, b-3)$  with  $b \leq 8$ , checks the Clifford relations and  $v^2 = q(v)$  for a general  $v$ , checks  $J^2 = -1$  and that left multiplication by  $J$  squares to  $-\mathrm{id}$  on  $C^+$ , which is [Lemma 11.4](#), checks  $C^- C^+ C^- \subseteq C^+$ , which is what makes  $\Psi$  of [Proposition 11.7](#) land where it should, and tabulates the Hodge numbers of [Proposition 11.10](#) together with the solution  $n = 1$  of [Proposition 11.11](#).
- (IX) *The family and the secant plane.* The scripts `weiltype_family.py` and `secant_plane.py` verify [Theorem 14.2](#) symbolically for  $n \leq 5$  and several  $d$ , and derive the closed forms of [Theorem 14.5](#) and [Theorem 14.7](#).
- (X) *The split member.* The script `split_locus.py` builds  $A = X \times \widehat{X}$  with the action (28) in coordinates and does two independent things. First it solves, by exact rational Gaussian elimination, for an expression of each of  $\omega_1, \omega_2$  as a rational combination of the degree  $2n$  monomials in  $\beta, \widehat{\beta}, \ell$ , and prints the solution; it succeeds for  $n \leq 3$  and  $d \in \{1, 2, 3, 7\}$ . Second it verifies [Theorem 15.2](#) directly: that  $\gamma \mp \sqrt{-d}\ell$  is an eigenclass for  $\tau^2$  respectively  $\bar{\tau}^2$ , and that (33) holds on the nose, for  $n \leq 5$  and several  $d$ . Third it checks the formulas of [Example 15.3](#) as they are printed, at  $n = 2$  for  $d \leq 3$  and at  $n = 3$  for  $d \in \{1, 2, 3, 7\}$ ; this check was added after an earlier printing of  $\omega_1$  at  $n = 3$  was found to lack the factor  $d$ , and it fails on that printing for  $d \neq 1$ .
- (XI) *The reach of the base case.* The script `split_geometry.py` settles the four assertions of [Proposition 15.19](#). It verifies  $M^2 = -d$  and  $M^\top Q + QM = 0$  entry by entry for the block form  $Q = \beta \oplus (\widehat{\beta}/d)$  on  $H_1(A, \mathbb{Q})$ , for  $n \leq 4$ , and checks that the action induced on  $H^2(A, \mathbb{Q})$  multiplies  $d\beta + \widehat{\beta}$  by  $d$ , multiplies  $\gamma$  and  $\ell$  by  $-d$ , and has  $\beta + d\widehat{\beta}$  as an eigenclass only when  $d = 1$ , for six pairs  $(n, d)$ ; it computes  $\det H = (-d)^n (d_1 \cdots d_n)^2$  for ten choices of  $(n, d, \beta)$  including non-principal  $\beta$ , and certifies in each case that  $(-1)^n \det H$  is a norm from  $K$ , together with the two facts the proof uses, that  $\det B$  is a square and that  $d$  is a norm; it checks the codimension formula  $n^2 - n(n+1)/2 = n(n-1)/2$  for  $n \leq 10$ ; and it computes both the dimension of the space of  $\mathrm{Sp}$ -invariants and the rank of the set of monomials  $\beta^i \widehat{\beta}^j \ell^k$  in degree  $2n$ , finding rank three in degree two for  $n \leq 4$  and agreement in the middle degree for  $n \leq 3$ .
- (XII) *Quaternionic multiplication.* The script `quaternionic.py` builds  $B = (-d, b)$  and the module  $B^n$  in coordinates and verifies [Theorem 15.25](#) for nine choices of  $(n, d, b, a)$  with  $n \leq 3$ : the relations  $i^2 = -d$ ,  $j^2 = b$ ,  $ij = -ji$ ; that  $E$  of (37) is alternating and nondegenerate with  $i$  anti-self-adjoint and  $j$  self-adjoint; that the hermitian form on the  $K$ -basis is  $\mathrm{diag}(2a_k d, -2a_k b d)$  with signature  $(n, n)$ ; the determinant formula and hence the class  $b^n$ ; and that  $g(x, y) = E(jx, y)$  is alternating,  $K$ -bilinear and nondegenerate, which is the hypothesis of [Theorem 15.16](#). It then checks [Theorem 15.31](#) numerically: for each of several fields, every nontrivial class in the tested range is realised in every dimension  $n \leq 8$  with signature  $(n, n)$ , by a single factor when  $n$  is odd and by the product of [Lemma 15.30](#) when  $n$  is even. Last it records the two dimensions behind [Remark 16.25](#): the quaternionic locus has

dimension  $n(n+1)/2$  inside a period domain of dimension  $n^2$ , so its codimension is  $n(n-1)/2$ , which is checked for  $n \leq 10$  and is 1 exactly at  $n = 2$ .

- (XIII) *The semiregularity map of a sum of line bundles.* The script `semiregularity.py` builds the split member with  $X = E_i^n$ , so that the Hodge decomposition is defined over  $\mathbb{Q}(i)$  and every step is exact, and verifies: that the six degree four monomials in  $\beta, \hat{\beta}, \ell$  are independent for  $n \leq 4$ , which is [Lemma 18.3](#); that  $\theta^+\theta^-$  is never a multiple of  $\eta^2$ , for six pairs  $(n, d)$ , which is [Lemma 18.4](#); that the coefficient of  $\theta^+\theta^-$  in  $\text{ch}_2$  is  $\sum_i m_i N(u_i)$ , which is the identity behind [Theorem 18.5](#); that no multiset with positive multiplicities satisfies the resulting equations with some  $u_i$  nonzero, for five pairs  $(n, d)$ ; that the map  $\Phi$  of [Lemma 18.1\(iii\)](#) is injective at  $n = 2$  for  $s \leq 3$  and of rank 22 rather than 24 at  $s = 4$ ; that an exhaustive search over the classes  $a\beta + b\hat{\beta} + c\ell$  with  $|a|, |b|, |c| \leq 3$  and  $d \in \{1, 2\}$  produces no signed tuple of at most three line bundles with a flat Chern character and a nonzero Weil component; and that every signed tuple of at most four in the same box with a flat Chern character and a nonzero Weil component has exactly four terms and a map  $\Phi$  with nonzero kernel, of dimension 2, 3, 4 or 6 at  $d = 1$  and 3 or 6 at  $d = 2$ . It then turns to  $n = 3$ : it finds the Prouhet–Tarry–Escott pair of [Lemma 18.11](#) on the norm circle  $N = 4$  of  $\mathbb{Q}(\sqrt{-3})$  by exhaustive search, verifies that the six line bundles of [Proposition 18.14](#) have  $\text{ch}_k = 0$  for  $k \neq 3$  and  $\text{ch}_3 = 12\omega_1$ , and verifies that every difference of the six classes is nondegenerate of index three. The companion `semireg_fast.py` repeats the rank computations modulo a prime congruent to 1 mod 4, which reaches  $n = 3$ : it reproduces the  $n = 2$  ranks exactly, finds  $\Phi$  injective at  $n = 3$  for a general choice of up to ten classes, and finds rank 75 of 90 for the explicit object.
- (XIV) *The annihilator and the rank-zero obstruction.* The script `weil_annihilator.py` computes, modulo the same prime, the dimension of the annihilator of each rational generator of the Weil line in  $H^{0,2}$  at  $n = 2, 3, 4$  and finds  $n^2$  in every case, which is [Lemma 9.6](#); splits the semiregularity map of the explicit object of [Section 14](#) by parity and finds an odd kernel of dimension  $9 = n^2$  equal to the annihilator, an even kernel of dimension 6, and a total of 15, against 0 for a generic triple, which is [Corollary 9.8](#); and runs the semiregularity rank over every configuration  $s_i\gamma + c_i\ell$  with  $\sum s_i = \sum c_i = 0$  in a box of side 7 whose virtual Chern character lies on the Weil line, finding eighteen configurations and rank 75 of 90 for all of them, which is [Remark 9.9](#).
- (XV) *The Weil class under products.* The script `weil_product.py` verifies, in exact arithmetic over  $\mathbb{Q}(\sqrt{-d})$ , the two identities (35) for every pair  $n_1, n_2 \leq 3$ ; their  $K$ -valued form  $\omega(A_1)\omega(A_2) = 2\omega(A_1 \times A_2)$  over the same range; the  $k$ -factor identity with constant  $2^{k-1}$  for six partitions with  $k = 3$  and  $k = 4$ ; that the Weil class lies in  $H^{2n}$ , so that at  $n = 1$  it is a divisor class, which is [Lemma 15.9](#); and the dimension comparison of [Remark 15.11](#), that the product locus has dimension  $n$  inside a family of dimension  $n^2$  and is smaller than the quaternionic locus for every  $n \geq 2$ .
- (XVI) *The annihilator as the tangent space.* The script `weil_tangent.py` builds a polarised Hodge structure of  $(K, -1, n)$ -Weil type from scratch over the Gaussian rationals, with no reduction modulo a prime and no floating point anywhere, and checks five things at  $n = 2, 3, 4$ . That the construction is what it claims to be: the polarisation has exactly two nonzero blocks on  $V_+ \times V_-$ , each a perfect pairing of  $n$ -dimensional spaces, and the two rational generators of the Weil line are real of type  $(n, n)$ . That the annihilator of each generator in  $H^{0,2}$  has dimension exactly  $n^2$ , which

is [Lemma 9.6](#) again, now in a model built independently of the one used in item (XIV). That the polarisation-preserving first-order deformations have dimension  $n(2n+1) = \dim \mathcal{A}_{2n}$  and those commuting with  $K$  have dimension  $n^2 = \dim \mathcal{D}_{n,n}$ . That a polarised deformation contracts the whole Weil line to zero exactly on that  $n^2$ -dimensional subspace, and that one generator already cuts it out, which is [Proposition 9.14](#). And that the map of [Theorem 9.17](#) is injective, independent of the basis of  $V_+^{1,0}$  used to write it, and has image exactly the annihilator. The five statements are then rerun for eight distinct choices of the  $n$ -plane  $V_+^{1,0}$  at  $n = 2$  and at  $n = 3$ , and hold in all sixteen.

- (XVII) *The quaternionic locus as a Lagrangian Grassmannian.* The script `lagrangian_locus.py` verifies, over  $\mathbb{Q}$  in exact arithmetic and for  $2 \leq n \leq 5$ , the three statements of [Proposition 15.28](#). That the form  $\omega_\psi(x, y) = E(x, \psi y)$  attached to a Rosati-symmetric  $\psi$  is alternating on  $V_+$ , which is the identity  $E(\psi x, y) = E(x, \psi y)$  read twice. That the graph of a symmetric form is isotropic and the graph of a form that is not symmetric is not, so the chart of  $\mathrm{LG}(n, 2n)$  by symmetric forms is exactly right and not merely contained in the locus. That the chart has dimension  $n(n+1)/2$ , computed as the rank of the symmetric basis rather than quoted, and that the codimension in  $\mathcal{D}_{n,n}$  is therefore  $n(n-1)/2$ ; at  $n = 4$  these are 10 and 6. And that a random  $n$ -plane, completed to a basis of  $V_+$ , is Lagrangian for the form that is standard in that basis, which is the construction behind [Remark 15.29](#): over the reals the loci sweep the whole family, and only the rationality of  $\psi$  prevents them from covering it.
- (XVIII) *The size forced on an object carrying the class.* The script `object_size.py` verifies, in the exterior algebra over  $\mathbb{Q}$  for  $1 \leq n \leq 4$ , that  $\alpha_\pm^2 = 0$  and hence  $\omega_1^2 = 2\alpha_+\alpha_-$ , that  $\int \omega_1^2 \neq 0$ , and that  $\chi(E, E) = (-1)^n N^2 \int \omega^2$  for every rank  $r$  and every multiplicity  $N$  in the ranges tested, the rank cancelling out of the identity. That is [Proposition 9.11](#), and with it the lower bound  $\sum_i \dim \mathrm{Ext}^i(E, E) \geq N^2 |\int \omega^2|$ . The script also records that the constant is not dimension-free, since it depends on the normalisation of  $\omega$  against  $\eta$ , so that no growth rate in  $n$  should be read off it; what the bound needs, and all that it needs, is that the Weil class is not isotropic.
- (XIX) *Infinitesimal rigidity of divisor classes.* The script `rigidity.py` builds the four blocks of  $H^{1,1}$  and the tangent space of  $\mathcal{D}_{n,n}$  over  $\mathbb{Q}(i)$  and verifies, for  $n = 2, 3, 4$ , the three parts of [Proposition 9.15](#). That the real span of the classes killed by contraction has dimension exactly one and is the polarisation, checked against the full real space of  $(1, 1)$  classes, of dimension  $4n^2$ , which is 16 at  $n = 2$  and 64 at  $n = 4$ . That the kernel attached to a norm-character class with block  $a$  has dimension  $n(n - \mathrm{rank} a)$  for every rank; at  $n = 4$  the four values are 12, 8, 4, 0. And that a class drawn at random has kernel zero, so it rigidifies the family to a point. The dimension  $n^2$  of the tangent space is recovered in the same computation rather than assumed.
- (XX) *The invariants of the required object.* The script `integrality.py` computes, for  $1 \leq n \leq 8$ , every squarefree  $d \leq 11$  and every rank and twist in a range, the Chern classes of the Chern character  $au + bv$  by Newton's identities, and finds every one of them integral; it also confirms the closed form  $\Delta = a^2d + b^2$  of [Theorem 14.7](#) and its positivity. That is [Proposition 15.26](#): nothing numerical forbids the object the construction needs, in any dimension tested.

- (XXI) *A sheaf with the required Chern character exists.* The script `secant_exists.py` carries out, exactly over  $\mathbb{Q}$  and for  $1 \leq n \leq 10$ , the computation behind [Theorem 14.13](#). For every squarefree  $d \leq 11$  and every rank and twist in a range it solves the Vandermonde system that expresses the rational point  $au_t + bv_t$  of the secant plane as a rational combination of the Chern characters  $e^{jt} = \text{ch}(\mathcal{O}_X(jt))$ , clears denominators to obtain the least positive multiple  $M$  and the integer witness (31), and then verifies that witness by recomputing every moment  $\sum_j n_j j^k$  and comparing it with  $Mc_k$ , rather than trusting the solve. It checks that the rank  $\sum_j n_j$  of the witness equals  $Ma$ , which is the positive rank hypothesis of [Lemma 14.12](#), and it records the least  $M$  in each case: 6 at  $n = 4$  with  $d = 3$  and  $a = b = 1$ , at most 2016 over the whole range at  $n = 8$ , and never 1 once  $n \geq 7$ , which is [Remark 14.14](#). Finally it expands  $\theta^k/k!$  in the exterior algebra of a symplectic lattice and finds every coefficient integral, which is why the coordinates  $c_k$  are integers and hence why the problem is a lattice problem at all.
- (XXII) *Which split objects can be semiregular.* The script `pte_search.py` settles [Theorem 18.20](#) in exact integer arithmetic. It enumerates the norms represented by  $\mathcal{O}_K$  below a bound, forms for every set of  $n + 1$  of them the forced level sums (51) and the least integral scale, and tests them against the supply  $r_K(N_j)$  of elements of that norm; at  $n + 2$  norms the solution space is two dimensional and its integer points in the same box are swept coordinate by coordinate. For  $4 \leq n \leq 8$  and  $d \leq 19$  nothing survives, with norms to 110 at  $n = 4$  and to 30 at  $n = 8$ . At  $n = 3$  and norms to 200 exactly eleven sets survive, all at  $d = 1$  or  $d = 2$ , of the three shapes recorded in [Theorem 18.20\(iii\)](#). The script then runs the whole system (44) on those sets: for each it enumerates every assignment of signs to the elements of  $\mathcal{O}_K$  at those norms with the prescribed level sums, splitting the four levels into a tabulated half and a streamed half so that the enumeration is complete, and finds no solution for eight of them. The companion script `pte_remaining.py` settles the other three,  $d = 1$  at  $(13, 52, 117, 130)$ ,  $(17, 68, 153, 170)$  and  $(25, 50, 100, 125)$ , whose largest level has sixteen elements: it streams three levels against the fourth through a linear hash of the ten off-diagonal coordinates, re-checks every match exactly, and finds no solution. It also reruns two of the eight as a control. The largest search covers  $4.29 \times 10^{14}$  sign patterns.
- (XXIII) *The quaternionic Weil cycle.* The script `quaternionic_divisibility.py` checks [Proposition 16.19](#) on the lattice  $\mathcal{O}_b^2$ . It confirms the ratios  $-b/3$  and  $4b^2/3$  and the value  $1/12$  for  $d \in \{1, 3\}$ , two weight vectors and fifteen values of  $b$  up to 101; the decomposition  $E_b = E_u + bE_v$ ,  $g_b = bg_1$ ,  $\lambda_b = b\lambda_1$  with  $g_1$  and  $\lambda_1$  integral; the equalities  $Z_b = b^2 Z_1$  and  $Z'_b = b^2 Z'_1$  in  $\wedge^4 \Lambda^*$ ; the growth  $\int \eta_b^4 = b^2 \int \eta_1^4$ ; and the effect of replacing  $j$  by  $\alpha j$ . It then computes the saturation of the Weil plane in  $\wedge^4 \Lambda^*$  and the sublattice spanned by products of two integral divisor classes, for  $d \in \{1, 2, 3, 7\}$ , three weight vectors and seven values of  $b$ , and finds Gram matrix  $\text{diag}(8d, 8d^2)$  and index  $2(a_1 a_2)^2$  throughout.
- (XXIV) *The divisor route on a fixed lattice.* The script `divisor_route.py` checks [Proposition 16.24](#) on  $\Lambda = \mathcal{O}^2$  with  $j^2 = 1$ , for  $d \in \{1, 3\}$ . It computes the lattice of  $K$ -bilinear classes, of rank 12, checks  $t(x) = 0$  on it and the identity  $-\int xy\eta^2/\int \eta^4 = \text{tr}(\phi_x \phi_y)/24$  on all 78 pairs of basis vectors, and finds signature  $(8, 4)$  for  $I$ . It verifies that  $\phi_x^2$ ,  $\phi_y^2$  and  $\phi_x \phi_y + \phi_y \phi_x$  are scalar for the pencil, and rechecks  $\phi_{x+ky}^2 = \beta(k)$  and  $I(x+ky) = \beta(k) \int \eta^4/3$  directly at eight values of  $k$ . For  $0 \leq k \leq 10$  it saturates  $\mathbb{Q}\eta + \mathbb{Q}x_k + \mathbb{Q}x_k(i \cdot, \cdot)$  in  $H^2(\Lambda, \mathbb{Z})$ , checks that  $I$  is positive

definite on the result, and computes the minimum  $\mu_k$  of  $I$  off  $\mathbb{Q}\eta$  by an exhaustive search in a box derived from the inverse Gram matrix. For the bound valid at every  $k$  it forms the  $2 \times 2$  minors of the matrix with rows  $x_k$  and  $x_k(i, \cdot)$  as polynomials in  $k$ , computes resultants of pairs of them, and obtains  $N_1 = 2$  and  $N_3 = 4$ ; it checks directly that the gcd of the minors divides  $N_d$  for  $|k| \leq 200$ . Finally, for  $0 \leq k \leq 4$  it computes the centraliser of  $i$  and  $\phi_{x_k}$  in  $\mathfrak{sp}(V, E)$ , of dimension 10, and its invariants in  $\wedge^2 V^*$ , of dimension 3; and at the one value of  $k$  in  $[-6, 6]$  for each  $d$  at which  $\beta(k)$  is a rational square it constructs a rational complex structure  $J_0$  on the pencil member, commuting with  $i$  and  $\phi_{x_k}$  and compatible with  $E$ , and checks that  $[\mathfrak{c}, J_0]$  has dimension 6 and that  $[\mathfrak{c}, J_0]$  together with its brackets spans all of  $\mathfrak{c}$ , which is the Lie algebra step in the proof of [Proposition 16.24\(iv\)](#).

- (XXV) *The obstruction map of a split object.* The script `split_obstruction.py` works in the model of item (XIII), modulo the same prime, and checks [Proposition 18.16](#). For  $(n, d) = (2, 1), (2, 3), (3, 3)$  it computes the space  $T$  of first-order deformations killing  $\eta$  and both generators of the Weil line and finds dimension  $n^2$ ; computes the space  $T_{\text{spl}}$  of deformations keeping  $\beta, \hat{\beta}, \ell$  of type  $(1, 1)$  and finds dimension  $n(n+1)/2$  and  $T_{\text{spl}} \subseteq T$ ; as a control, replaces  $\eta$  by  $\beta + d\hat{\beta}$  and finds that  $T$  collapses to dimension  $n(n+1)/2$  for  $d > 1$ , which is the mismatch of conventions described in the footnote to [Setup 14.1](#); and for three classes spanning  $\text{NS}(A_0)_{\mathbb{Q}}$  together with  $\eta$  computes the obstruction map  $v \mapsto (v \lrcorner \lambda_i)_i$  on  $T$ , finding kernel exactly  $T_{\text{spl}}$  and rank  $n(n-1)/2$ ; it also finds that each of the seven single classes  $\gamma, \ell, \gamma + \ell, \gamma + 2\ell, \beta, \hat{\beta}, \beta + \ell$  already has  $\{v \in T : v \lrcorner \theta = 0\} = T_{\text{spl}}$ , and that each of 150 random rational classes off  $\mathbb{Q}\eta$  is contracted nontrivially by some vector of  $T$  while  $\eta$  and  $3\eta$  are contracted to zero by all of  $T$ , which is the mechanism of [Theorem 9.20](#). For the six line bundles of [Proposition 18.14](#) at  $n = 3$  it finds kernel  $T_{\text{spl}}$  of dimension 6 and rank 3; checks that  $\sigma_E$  kills the image; computes the annihilator of  $\omega_1$  in  $H^{0,2}$ , of dimension 9, and checks that  $\sigma_E$  kills  $z \cdot \text{id}_E$  for  $z$  in it; finds that the sum of that nine-dimensional space and the three-dimensional image has dimension 12; recomputes the kernel of  $\sigma_E$  as 15; and checks that the components of every obstruction vector on  $L_i$  and on  $L_i^{-1}[3]$  are negatives of each other.
- (XXVI) *The evaluation map.* The script `evaluation_map.py` rewrites the model of item (XIII) in its Hodge basis and checks [Proposition 18.18](#) for the explicit object at  $n = 3$ . It builds a basis of  $HT^2$ , of dimension  $15+36+15 = 66$ , implements the three actions of (48) as multiplication, a derivation and a double interior product, and verifies the identity  $\sigma_E(\text{ev}_E(x)) = x \lrcorner \text{ch}(E)$  on every basis vector and on random elements; confirms that  $\text{ch}(E)$  is concentrated in degree six and proportional to  $\omega_1$ ; computes the rank of  $\text{ev}_E$ , which is 45, against  $\dim \text{Ext}^2(E, E) = 90$ ; computes the three kernels of contraction with  $\text{ch}(E)$ , of dimensions 9, 18 and 9; finds that their images under  $\text{ev}_E$  have dimensions 9, 6 and 9, that the first and third coincide, that the first and second meet in zero, and that the sum of all three has dimension 15, equal to the dimension of  $\ker \sigma_E$ , which it recomputes; and repeats the last comparison for every one of the eighteen configurations of item (XIV), finding  $\dim \ker \sigma_E = 15$  and  $\dim(\ker \sigma_E \cap \text{im ev}_E) = 15$  in every case.
- (XXVII) *Markman's candidate in dimension eight.* The script `markman_candidate.py` works in  $\mathbb{Q}[\Theta]/(\Theta^5)$  with  $\int \Theta^4 = 24$  and checks [Proposition 18.25](#) for every odd squarefree  $d \leq 101$ : the Euler characteristics 14,  $-36$  and  $24$  from the Koszul complexes; the count  $12N(N-5) = (d+9)(3d-3)$  of isolated points and its comparison with the  $(d+9)(2d-1)$  normalised;  $\chi(\mathcal{O}_{\tilde{Z}}) = (d+9)(27-d)$  from the Mayer–Vietoris

complex; the equality  $\text{ch}(\mathcal{F}) = u_t + 3v_t$  degree by degree for the twist  $3\Theta$ , and the failure by  $-4\Theta^2$  in degree two for the twist  $\Theta$ ;  $N(3 + \sqrt{-d}) = 2N$  and the discriminant  $(d+9)\Theta^2$ ; and the ranks  $8d(d-9)$  and  $8d(d+9)$  of the two transforms, with the threefold control  $-8q$  that reproduces Markman's rank at  $n = 3$ .

- (XXVIII) *The classes preserving a secant Chern character.* The script `secant_kernel.py` works in the Hodge algebra of an abelian  $n$ -fold with  $t = \sum_a p_a \wedge q_a$ , modulo a large prime, and checks [Theorem 18.29](#)(i), (ii) and [Lemma 18.28](#)(i) for  $(n, d, a, b)$  equal to  $(3, 3, 1, 1)$ ,  $(3, 3, 1, 3)$ ,  $(4, 3, 1, 3)$ ,  $(4, 5, 1, 3)$ ,  $(4, 7, 2, 1)$  and  $(5, 3, 1, 3)$ , the third being Markman's candidate. It builds  $\text{ch}(F) = au_t + bv_t$ , implements the actions of  $HT^1$  and  $HT^2$  as multiplication, interior product, derivation and double interior product, and finds: that the kernel of  $y \mapsto y \lrcorner \text{ch}(F)$  on  $HT^1$ , of dimension  $2n$ , is zero in every case; that the kernel on  $HT^2$  has dimension exactly  $n^2$ , contains the  $n(n+1)/2$  polarised deformations and the  $n(n-1)/2$  classes  $(\frac{d}{2}\pi \lrcorner t^2, 0, \pi)$ , and is spanned by them; and, as controls, that the bare classes  $(0, 0, \pi)$  and the classes  $(-\frac{d}{2}\pi \lrcorner t^2, 0, \pi)$  with the compensation of the wrong sign all act nontrivially on  $\text{ch}(F)$ . It then verifies the identity  $x \lrcorner (we^B) = ((e^B x) \lrcorner w)e^B$  of [\(53\)](#) on random even classes  $w$ , random  $x \in HT^2$  and  $B = kt$  for  $n = 3, 4$ , and the count  $n(n+1)/2 + n(n-1)/2 = n^2$  for  $n < 40$ .
- (XXIX) *The local products at a double point.* The Macaulay2 script `m2/local_products.m2` recomputes [Lemma 18.33](#) over  $\mathbb{Q}[x_1, \dots, x_4]$ . For each of the two local models, the ideal  $I = (x_1, x_2) \cap (x_3, x_4)$  and the complex  $K = [R \rightarrow R/(x_1, x_2) \oplus R/(x_3, x_4)]$ , resolved by the cone of the lifted map, it builds the jet chain maps  $-\partial_a(d)$  on the free resolution, checks that they anticommute with the differential and are not null homotopic, and for every pair  $(a, b)$  decides whether the composite is null homotopic. It finds  $\text{Ext}^2 = k^4$  for  $I$ , with the four mixed products nonzero and spanning it and the same-plane products zero;  $\text{Ext}^2 = k^2$  for  $K$ , with the two same-plane products nonzero and spanning it and the mixed products zero; anticommutation  $a_a a_b + a_b a_a = 0$  in both models; and, for  $I$ , the same vanishing pattern from the jet modules and `yonedaProduct`. This item needs Macaulay2 (version 1.22 with the package `Complexes`) and is not part of `verify_all.py`; its transcript `m2/local_products.txt` is included with the code.
- (XXX) *A rank one secant object with smooth support.* The script `smooth_support.py` expands  $\text{ch}(\mathcal{O}_S) = 1 - \text{ch}(\mathcal{F})e^{-b\Theta}$  from scratch in  $\mathbb{Q}[\Theta]/(\Theta^5)$ , for every  $b \leq 6$  and every squarefree  $d < 60$ , and reads off the class  $N\Theta^2$  with  $N = N_{K/\mathbb{Q}}(b + \sqrt{-d})/2$ , the pushforward  $\iota_* K_S = (4bN/3)\Theta^3$  and the Euler characteristic  $\chi(\mathcal{O}_S) = 4N(2b^2 - N)$ , checking each against the closed form rather than quoting it. It then verifies that Noether's formula and the self-intersection formula together give  $K_S^2 = 12N(4b^2 - N)$  and  $e(S) = 12N(4b^2 - 3N)$ , that the Hodge index bound  $9N \geq 4b^2$  is implied by  $d > 0$  and so never binds, that  $e(S) \geq 0$  is  $d \leq \frac{5}{3}b^2$  and that  $3e(S) - K_S^2 = 24(b^4 - d^2) \geq 0$  is  $d \leq b^2$ , and that equality in the Hodge index would need  $9N = 4b^2$ , which no admissible pair reaches. At  $b = 3$  it prints the six surviving discriminants, the four of them with  $N$  an integer, and [Table 3](#). That is [Theorem 18.40](#) and [Corollary 18.41](#).
- (XXXI) *Where the local obstruction lives.* The Macaulay2 script `m2/lci_products.m2` computes, for ten codimension two germs in  $\mathbb{C}^4$ , the number of minimal generators, the projective dimension of  $R/I$ , the ranks of a minimal free resolution, and  $\mathcal{E}xt^1$ ,  $\mathcal{E}xt^2$  and  $\mathcal{E}xt^3$  of the ideal against itself; for the germs with  $\mathcal{E}xt^2 \neq 0$  it also decides, by null homotopy of the composite of the two jet chain maps, which products of

translation classes are nonzero. The seven Cohen–Macaulay germs, namely the smooth one, two and three planes meeting along a line, the double plane, the quadric cone, the fat plane and the cone over the twisted cubic, all have  $\mathcal{E}xt^2 = 0$ , the last two despite needing three generators in codimension two. The three that are not Cohen–Macaulay, namely two planes meeting at a point, three planes meeting pairwise at a point and a plane with an embedded point, have  $\mathcal{E}xt^2 \neq 0$  and nonzero products. That is [Theorem 18.43](#) and [Remark 18.44](#). The script runs separately from `verify_all.py` because it needs Macaulay2; its transcript is distributed with it.

- (XXXII) *Cohen–Macaulay supports with a split resolution.* The script `hilbert_burch.py` reads the coefficients  $(1, b, -d, -db, d^2)$  off  $u_\Theta + bv_\Theta$  rather than quoting them, confirms the three identities  $c_{k+2} + dc_k = 0$  that make the weighted measures of [Theorem 18.49](#) share mass, mean and variance, and checks that the quadratic of that theorem has discriminant  $-\frac{2}{9}b^2 - 2d < 0$  for every  $b \leq 11$  and every squarefree  $d < 200$ , so that no complete intersection can occur. It then enumerates (65) exactly, by meet in the middle on the four moments, over every multiset of twists in the boxes of [Table 4](#), imposing the two conditions of [Lemma 18.48](#). Nothing survives at  $r = 2$  or  $r = 3$ , for  $b = 3$  and every squarefree  $d < 24$  and also for every  $b \leq 6$  over seven fields; at  $r = 4, 5, 6$  only  $d = 15$  and  $d = 23$  survive, and the first candidate at each rank is printed. The script also records that the moment system alone is not enough: at  $r = 4$ ,  $d = 23$  it has the solution  $p = (-8, -5, 0, 1, 2)$ ,  $q = (-7, -6, -4, 4)$ , which only the injectivity of  $\varphi$  removes.
- (XXXIII) *The closure graph.* The script `closure_graph.py` carries the rule set of [Setup 20.39](#) as data: forty-two statements, each marked proved here, quoted, open or derived, and twenty-two rules, each carrying the label of the theorem that proves it. It checks that every statement named in a rule is declared and every declared statement is used, that the rule set is acyclic, and that every one of the seventeen theorem labels it names, for its rules and for the consequences of **HC** it records, is a label of this paper, against the list of all 584 labels generated from the sources and distributed with the script as `paper_labels.txt`. It then computes the forward closure by iterating the one step map to a fixed point, and verifies that no statement marked proved rests at any depth on an open one, that **HC** is not in the closure of the proved and quoted statements, and, by exhaustive search over all  $2^{16}$  subsets of the sixteen open statements, that there are exactly the five minimal sufficient sets of [Theorem 20.40\(ii\)](#),  $\{F3\}$ ,  $\{L, M\}$ ,  $\{L, F3'\}$ ,  $\{V, F3'\}$  and  $\{P2_{\text{split}}, F2, F3'\}$ , with  $P2_{\text{split}}$  the propagation statement for the split families of the CM fields of degree at least four, that  $(F3)$  is the one open statement that suffices alone, that  $(F3')$  alone gives neither the conjecture for abelian varieties nor **HC**, that the sets  $\{P2_{\text{split}}, F2, F3\}$ ,  $\{L, F3\}$  and  $\{V, F3\}$  suffice and are not minimal, that the route through this paper is the only minimal set with three elements and the only one whose derivation uses a statement proved here, that each element of each set is necessary, that every set contains  $(F3)$ ,  $(F3')$  or  $(M)$ , that every element of every set other than  $P2_{\text{split}}$  is recorded as a consequence of **HC**, and that  $\{F3\}$  and  $\{L, M\}$  are each equivalent to it. It confirms that  $(V)$  alone gives the conjecture for abelian varieties and not **HC**, and that  $(L)$  gives  $(V)$ . It confirms that the six open demands of the secant route lie outside every minimal set, that granting the two of them that matter reaches  $\mathbf{W}(K, n, \delta_0)$  and, through the descent rule of [Proposition 20.13](#),  $\mathbf{W}(K, n, \delta)$  for every  $\delta$ , but not the Weil classes of the fields of degree at least

four; that the trivial discriminant closes the imaginary quadratic case and the split families of the fields of degree at least four close it as well; that the propagation statement for the split imaginary quadratic families reaches all of that case, and for the split families of the higher fields the Weil classes of every CM field, so that those are not open; that no minimal set contains a propagation statement for a family of nontrivial discriminant or for an imaginary quadratic field; and that the bounded criterion is absent from the rule set, being equivalent to its own conclusion. It then removes the rule of [Proposition 20.35](#) and the statement (F3') with its rule and repeats the search, which returns the four sets of the earlier version of this paper,  $\{L, M\}$ ,  $\{L, F3\}$ ,  $\{V, F3\}$  and  $\{P2_{\text{split}}, F2, F3\}$ ; so the change in the answer is those two additions and nothing else. Finally it checks that the numbering under which the same graph is carried into Section 24 of the Lean certificate maps these rules onto those. The full derivation of **HC** from the proved and quoted statements and each minimal sufficient set is printed, one rule per line. That is [Theorem 20.40](#), [Proposition 20.35](#), [Proposition 20.37](#) and [Corollary 20.41](#).

- (XXXIV) *No split resolution runs the criterion.* The script `split_resolution.py` records the data behind [Theorem 18.52](#): that the scalar  $(e^2 + d)/2$  by which the compensated Poisson class acts on  $\mathcal{O}(e\Theta)$  is nonzero for every twist and every  $d > 0$  in a range; the vanishing pattern  $H^2(\mathcal{O}(m\Theta)) = 0$  for  $m \neq 0$  on an  $n$ -fold with  $3 \leq n \leq 6$ , from the index theorem; that the three split resolutions of [Table 4](#) have no twist in common between  $E_0$  and  $E_1$  and satisfy the moment identities (65), so that the theorem removes them as they stand; and, as a control, that no Koszul resolution of a complete intersection, twisted by any  $b \in [-6, 12]$ , is secant for  $d < 30$ , which is [Theorem 18.49](#) in that range.
- (XXXV) *The two statements that are not reductions, and what the rank forces.* The script `exceptional_classes.py` has three parts. For an abelian fourfold  $X$  of Mumford type it decomposes every  $\wedge^k V$ ,  $V = V_1 \otimes V_2 \otimes V_3$ , under  $\text{SL}_2^3$  by peeling highest weights, finds one invariant in  $\wedge^2 V$  and one in  $\wedge^4 V$ , counts eight invariants in  $\wedge^4(V \oplus V)$  distributed 1, 1, 4, 1, 1 over the Künneth summands and three in  $\wedge^2(V \oplus V)$ , multiplies the three divisor classes  $\psi_{11}, \psi_{12}, \psi_{22}$  in the exterior algebra of  $V \oplus V$  and finds their six products independent, and counts the  $\text{Sp}_8$ -invariants of  $\wedge^4(V \oplus V)$  by the Weyl character formula over the 384 signed permutations, finding six. That is [Proposition 20.44](#). For the quintic threefold it computes the Hodge numbers  $(1, 101, 101, 1)$  from the Jacobian ring of the Fermat quintic, checks in a four-dimensional model that the map from the  $(0, 3)$  part to the  $(3, 0)$  part is symplectic of adjoint weight 3, and lists the adjoint weights,  $\{-1, 0, 1\}$  in weight one against  $\{-3, \dots, 3\}$  here. That is [Proposition 20.45](#). For  $n = 2$  and  $n = 3$  it builds  $H^*(A, \mathbb{C})$  with the eigenspace decomposition, the Weil class  $\alpha_+ + \alpha_-$ , and the action of all of  $HT^2$  on it, finds the annihilator of dimension  $4n^2$  spanned by the mixed block, the  $K$ -linear deformations and the mixed bivectors with the remaining classes acting injectively, computes the ideal generated by the mixed block in the exterior algebra on  $2n$  generators, of codimension  $2^{n+1} - 1$  with quotient of dimension  $2\binom{n}{k}$  in degree  $k \geq 1$ , and checks over  $\mathbb{F}_3$  in dimensions two and three that two subspaces spanning the space with all mixed wedge products zero cannot both be nonzero. That is [Theorem 16.30](#).
- (XXXVI) *The Weil classes of a CM field of degree four and six.* The script `cm_fields.py` builds the cyclotomic fields  $\mathbb{Q}(\zeta_5)$ ,  $\mathbb{Q}(\zeta_7)$ ,  $\mathbb{Q}(\zeta_8)$  and  $\mathbb{Q}(\zeta_9)$  exactly, as  $\mathbb{Q}[x]/(p)$  with elements stored as rational coordinate vectors, and carries out the construction

of [Theorem 19.4](#) for six pairs  $(F, n)$ , with  $m = 2$  and  $m = 3$  and  $n = 1, 2, 3$ . For each it checks that the CM type and its conjugate partition the embeddings, that  $\zeta - \zeta^{-1}$  is totally imaginary, that the trace-dual bases satisfy  $\sum_k \sigma(\omega_k) \sigma'(\omega_k^*) = \delta_{\sigma\sigma'}$  over every pair of embeddings, that  $V^{1,0}$  is half-dimensional and isotropic for  $E = \text{Tr}(\xi H)$  so that the point lies in the period domain, that every eigenspace meets  $V^{1,0}$  in dimension  $n$  so that the Weil classes are of type  $(n, n)$ , that  $\delta_i(f) = \sum_{\sigma} \sigma(f) u_{i,\sigma} \wedge u_{i',\sigma}$  and is therefore of type  $(1, 1)$ , that every  $\alpha_{\sigma}$  is nonzero, and that the balanced  $n$ -fold product of the  $\delta_i$  is  $w(f) = \sum_{\sigma} \sigma(f) \alpha_{\sigma}$ , rational, spanning  $W_F$ . For  $F = \mathbb{Q}(\zeta_5)$  at  $n = 2$  it also computes, in the exterior algebra of the sixteen-dimensional space, the space of rational  $(1, 1)$  classes, of dimension 32, and records that the balanced sum is necessary: the plain product  $\delta_1(1)\delta_2(1)$  is not a Weil class. Finally, for  $F = \mathbb{Q}(\zeta_8) \supset K = \mathbb{Q}(\sqrt{-1})$  it checks that each  $K$ -Weil class is the product of the two  $F$ -Weil classes above it and that  $W_K$  lies in the span of the products of pairs from  $W_F$ , which is [Proposition 19.8](#).

(XXXVII) *What the transport does along the orbit.* The script `transport_growth.py` works on the lattice  $\Lambda = \mathbb{Z}^{2G}$  with the standard alternating form. For a sample of rational symplectic elements drawn from the unipotent radical of the stabiliser of an isotropic line, with denominators up to 29, it verifies the three scaling identities of [Lemma 16.13](#), namely  $\det \varphi = c^{2G}$ ,  $\varphi^* E = c^2 E$  and  $\bigwedge^k \varphi = c^k \bigwedge^k g$ , the last on the exterior algebra for  $G = 2, 3, 4$ , and records that the denominators are unbounded. It then takes two sublattices of rank  $2n = 4$  in an abelian fourfold, one preserved by the sample elements and one not, computes the multiplicity  $m = [\varphi^{-1}(\Lambda) \cap W : \Lambda_W]$  by Smith normal form and the degrees as Pfaffians, and checks  $\deg \varphi(B) = c^{2n} \deg(B)/m$  in every case. The two cases separate: for the preserved subtorus  $m = c^{2n}$  exactly and the image degree is constant, while for the other  $m = c^{2n-1}$  and the image degree grows linearly in  $c$ . Finally it tabulates the bound  $c^{2n} \deg(Z)/|G(Z)|$  of [Theorem 16.15\(ii\)](#) for stabilisers of order 1, 2 and 4, which is strictly increasing and unbounded. That is [Lemma 16.13](#), [Lemma 16.14](#) and [Theorem 16.15](#).

(XXXVIII) *The annihilator of a Weil class in Hochschild cohomology.* The script `hochschild_annihilator.py` builds  $HH^*(A) = \bigwedge^*(H^{0,1} \oplus H^0(T_A))$ , of dimension  $2^{4n}$ , as an exterior algebra on  $4n$  operators indexed by bit masks, acting on  $H^*(A, \mathbb{C})$  by wedging and contracting, and computes the annihilator of  $\omega = a\alpha_+ + b\alpha_-$  in every degree by exact rational elimination. It confirms that  $P = V_+^{0,1} \oplus T_-$  is exactly the annihilator of  $\alpha_+$  in degree one and  $Q = V_-^{0,1} \oplus T_+$  that of  $\alpha_-$ , that  $\text{Ann}_{HH^1}(\omega) = 0$ , that  $\text{Ann}_{HH^2}(\omega) = P \wedge Q$  of dimension  $4n^2$ , that  $\dim \text{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$  in every degree  $k \geq 1$  (and 0 in degree 0), that the ideal generated by  $P \wedge Q$  consists of the monomials meeting both halves and has codimension  $2^{2n+1} - 1$ , that contraction with  $\alpha_-$  is injective on  $\bigwedge^* P$  and with  $\alpha_+$  on  $\bigwedge^* Q$  with the two images meeting in exactly one line in degree  $2n$ , and that  $\dim HH^2 - \dim \text{Ann}_{HH^2} = 2n(2n - 1)$ . The computation is run in every degree for  $n = 1$ , where degree two is the exceptional degree  $2n$  and the annihilator is one larger, and for  $n = 2$  with two choices of  $(a, b)$ , and in degrees up to three for  $n = 3$ . That is [Theorem 16.33](#) and [Corollary 16.34](#).

(XXXIX) *The criterion as a number, and the support of an object meeting it.* The script `p2_support.py` checks the finite linear algebra of [Theorem 16.36](#), [Lemma 16.37](#) and [Theorem 16.38](#) in five parts. (A) In the exterior algebra of  $H^*(A, \mathbb{C})$  with  $HH^1$  acting by wedging and contracting, the map  $\xi \mapsto \xi \lrcorner \omega$  on  $HH^2$  has rank exactly

$2n(2n-1)$  for  $n = 2$  and  $n = 3$ , that is 12 and 30, and it is injective on  $\bigwedge^2 P \oplus \bigwedge^2 Q$ ; and  $\binom{4n}{2} - 4n^2 = 2\binom{2n}{2} = 2n(2n-1)$  for every  $n \leq 60$ . (B) On a torus of dimension  $c$ , for  $2 \leq c \leq 6$ , the class of a point contracted with  $a \wedge b$  is nonzero for all coordinate pairs and for twenty-five random rational pairs of independent vectors, and zero for twenty-five dependent pairs. (C) For  $n = 2, 3, 4$  and random subspaces  $W$  of  $T_+ \oplus T_-$  of codimension at least two, together with the boundary cases where  $W$  contains  $T_+$  or  $T_-$ , the images of  $T_+$  and  $T_-$  in the quotient have all mixed wedges zero exactly when  $W$  contains  $T_+$  or  $T_-$ ; this is the dichotomy used in the proof of [Theorem 16.38\(i\)](#). (D) In an explicit rational model of  $\mathbb{R}^{4n}$  with complex structure  $J$  and an action  $M$  of  $\mathbb{Q}(i)$  of signature  $(n, n)$ , scrambled by a random rational change of basis, the forms vanishing on  $\ker(J - M)$ , the real form of  $T_+$ , are exactly  $V_-^{1,0} \oplus V_+^{0,1}$ , computed over the Gaussian rationals for  $n = 1, 2$ . (E) For  $2 \leq n \leq 6$  neither  $\alpha_+$  nor  $\alpha_-$  is a monomial of  $\bigwedge^*(V_-^{1,0} \oplus V_+^{0,1})$  or of  $\bigwedge^*(V_+^{1,0} \oplus V_-^{0,1})$ , so the Weil line meets the sum of the two subrings in zero.

(XL) *Two jet classes on a complex of finite length.* The Macaulay2 script `m2/finite_length_products.m2` builds, for a bounded complex  $M$  of free modules over  $\mathbb{Q}[x_1, x_2]$  or  $\mathbb{Q}[x_1, x_2, x_3]$  with finite length cohomology, the jet chain maps  $-\sum_i a_i \partial_i(d)$  and decides whether the composite of two of them is null homotopic. It first confirms that each input has finite length, and then tests every pair of coordinate directions and a random pair of independent rational directions on eighteen modules: in two variables the residue field,  $R/\mathfrak{m}^2$ ,  $R/\mathfrak{m}^3$ , four monomial ideals, the complete intersection  $(x_1^2 + x_2^2, x_1 x_2)$ , the ideal  $(x_1^3 - x_2^3, x_1^2 x_2, x_1 x_2^2)$ , the non-cyclic modules  $\mathfrak{m}/\mathfrak{m}^3$  and the cokernel of  $\begin{pmatrix} x_1 & x_2 & 0 \\ 0 & x_1 & x_2 \end{pmatrix}$ , and  $k \oplus R/\mathfrak{m}^2$ ; in three variables the residue field,  $R/\mathfrak{m}^2$ , two monomial complete intersections, a monomial ideal of colength five and the Gorenstein algebra  $R/(y_1^2 - y_2^2, y_2^2 - y_3^2, y_1 y_2, y_2 y_3, y_1 y_3)$ ; and three complexes with several cohomology modules. In every case the Euler characteristic is nonzero and every product of two independent jet classes is nonzero, which is [Lemma 16.37](#). It then builds the cone  $C$  of  $J(\partial_2)J(\partial_1): K \rightarrow K[2]$  on the Koszul complex  $K$  of the residue field in two variables, checks that  $C$  is a complex with cohomology of lengths 1 and 1 and Euler characteristic zero, and that  $J(\partial_2)J(\partial_1)$  is null homotopic on  $C$ , while on  $k \oplus k[1]$ , also of Euler characteristic zero, it is not. That is the sharpness of the lemma. The transcript `m2/finite_length_products.txt` is distributed with the script.

(XLI) *The two classes on a Mumford fourfold, and the K3 surface that would carry them.* The script `mumford_rm.py` works on the split model  $V = V_1 \otimes V_2 \otimes V_3$  of [Setup 21.1](#), with  $\mathfrak{sl}_2^3$  acting through its nine generators and  $\psi = \epsilon \otimes \epsilon \otimes \epsilon$ , in five parts. (A) The commutant of  $\mathfrak{sl}_2^3$  in  $\text{End } \bigwedge^2 V$ , computed as the null space of a linear system in 784 unknowns, has dimension four, and the four projections built from the three Casimir operators are orthogonal idempotents of ranks 1, 9, 9, 9 that sum to the identity, commute with the action and span the commutant. (B) The Hodge numbers are (6, 16, 6) on  $H^2$ , (0, 9, 0) on  $U_{12}$ , (3, 3, 3) on  $U_{13}$  and  $U_{23}$ , and (1, 7, 1) on  $T(-1)$ . (C) The product map  $\mu$  of [Lemma 21.3](#) carries  $T_i \otimes T_j$  onto  $U_{ij}$  with rank nine for  $i \neq j$  and  $T_i \otimes T_i$  onto the line of  $\theta$ , and its image is all of  $H^2$ . (D) The Lefschetz pairing  $\int \alpha \beta \theta^2$  is nondegenerate on  $H^2$ ;  $\mu \mu^\dagger$  acts on the three  $U_{ij}$  by one and the same nonzero rational number  $\nu$ , which is  $-1$  in the normalisation of the script; for three choices of the conjugates of  $e$  and  $\lambda$ ,  $\mu_e \mu_e^\dagger$  acts on  $U_{ij}$  by  $\nu \sigma_i(g) \sigma_j(g)$  with  $g = e^2/\lambda$  and preserves the line of  $\theta$ ; the algebra it generates on  $U$  is spanned by

the three projections when the conjugates of  $g$  are distinct and is the scalars when they coincide; and  $\kappa^\dagger \kappa$ , for the trace form on  $\text{End}(V)$ , is multiplication by  $4e^2/\lambda$ . (E) In the Clifford algebra of  $T$  with the form  $\sum_i l_i \text{tr}(x_i y_i)$ ,  $(l_1, l_2, l_3) = (1, 2, 5)$ , built on bit masks of an orthogonal basis, the spin lifts of the three copies of  $\mathfrak{sl}_2$  act on  $C^+(T)$ , of dimension 256; the vectors of highest weight  $(1, 1, 1)$  span 32 dimensions and lowering them gives a basis, so  $C^+(T) \cong V^{\oplus 32}$ . In that basis the map  $\Psi(v)x = vxv_0$  of [Proposition 11.7](#) is  $v \otimes N_i$  for  $v \in T_i$ , the three matrices  $N_i$  anticommute pairwise, square to the scalars 4, 8, 20, proportional to the  $l_i$ , and are linearly independent. That is [Proposition 21.2](#), [Lemma 21.3](#) and the shape of the Kuga–Satake map used in the proof of [Theorem 21.4](#).

- (XLII) *The Lefschetz operator of an abelian scheme over a curve.* The script `lefschetz_family.py` works in exact exterior algebra over  $\mathbb{Q}$  and assumes no sign convention. (A) On ten abelian varieties with  $g \leq 4$  and polarisation types from  $(1)$  to  $(1, 2, 4)$  and  $(1, 1, 2, 2)$ , the contraction by the dual bivector satisfies  $[L, \Lambda] = H$ , and the Pontryagin product, computed from  $x \star y = \mu_*(p_1^* x \cup p_2^* y)$  with  $\mu_*$  defined against  $\mu^*$  by the projection formula, gives  $D^{-1}(x \star \gamma) = \Lambda x$  in every degree, with  $\ell \star \gamma = gD$ ; without the factor  $D^{-1}$  the identity fails at type  $(1, 2)$ . (B) On four product families  $E \times A$  over an elliptic curve, where every class is invariant, the operator  $D^{-1}T + \Lambda_C$  of the proof of [Theorem 20.22](#), with  $\Lambda_C$  built from bases of invariant classes and the inverse of their pairing matrix, satisfies  $[L_W, \Lambda] = H$ , and the cross relations  $[L_{\text{rel}}, \Lambda_C] = 0$  and  $[L_C, T] = 0$  hold. (C) The invariants of  $\mathfrak{sl}_2^3$  in  $\Lambda^q(V_1 \otimes V_2 \otimes V_3)$ , computed as the joint kernel of the six generators, have dimensions 1, 0, 1, 0, 1, 0, 1, 0, 1. That is [Proposition 20.21](#), the algebra of [Theorem 20.22](#) and the count in [Corollary 20.23](#).
- (XLIII) *The two remaining targets, reduced.* The script `targets_reduction.py` works in the split model  $H^*(X \times X) = \Lambda^*(V \oplus V)$ , sixteen generators, with exact rational arithmetic where a space is determined and ranks modulo the prime 1000003 of integer vectors where a span is to be shown full, a rank modulo a prime being a lower bound for the rank over  $\mathbb{Q}$ . (A) The invariants of  $\mathfrak{sl}_2^3$  in  $\Lambda^k(V \oplus V)$  have dimensions 1, 0, 3, 0, 8, 0, 16, 0, 28, 0, 16, 0, 8, 0, 3, 0, 1; in degree four the products of divisor classes give six, so two are exceptional; and the ring of invariants is generated in degrees two and four. (B) At a CM point the Hodge classes of  $X_c$  number 1, 0, 4, 0, 8, 0, 4, 0, 1 and those of  $X_c \times X_c$  number 1, 0, 16, 0, 132, 0, 432, 0, 648,  $\dots$ ; the divisor classes of  $X_c \times X_c$  span only 100 of the 132 in degree four, while the ring generated by them and by the pull-backs of the Hodge classes of  $X_c$  along  $(\varphi, \psi): X_c \times X_c \rightarrow X_c$  spans every Hodge class in every degree. (C) On an abelian  $2n$ -fold of Weil type,  $n = 1, 2, 3$ , the classes of type  $(p, p)$  annihilated by  $Q \wedge HH^1$ , by  $P \wedge HH^1$  and by  $HH^1 \wedge HH^1$  are  $\mathbb{C}\alpha_-$ ,  $\mathbb{C}\alpha_+$  and 0. That is [Theorem 20.30](#), the ring structure used in [Corollary 20.31](#), and the annihilators in the proof of [Theorem 20.33](#).
- (XLIV) *The exceptional classes of a Mumford square are rigid.* The script `mumford_rigidity.py` works in the model of [Setup 21.18](#) at a point of the Mumford curve, with  $V^{1,0}$  the vectors of third weight  $+1$ , and in exact rational arithmetic with the coefficients  $a_0, \dots, a_3$  as symbols. (A) The Casimir operators split  $\Lambda^2 V$  as  $1 + 9 + 9 + 9$  and the four tensors  $\pi$  are invariant. (B)  $\mathfrak{sp}^{-1,1}$  has dimension 36 and splits into nine weight blocks of dimensions 3, 4, 3, 4, 8, 4, 3, 4, 3; the generic kernel of contraction into  $\omega_a$  is one vector, independent of  $a$ ; every Gram determinant is a positive constant times a product of  $P_1, P_2, Q_1, \dots, Q_4, R$ ; each is a sum of squares whose zeros force two of  $a_1, a_2, a_3$  to coincide; the tangent dimensions at

$(3, 5, -7, 11)$ ,  $(3, 5, 5, 5)$  and  $(1, 1, 1, 1)$  are 1, 10 and 20. (C)  $HT^2 = 28 + 64 + 28$  splits into 46 blocks by type, weight and exchange parity,  $\omega_a$  being exchange invariant; every Gram determinant is a positive constant times a product of the forms of (B),  $S_1, \dots, S_4$ ,  $Z_1, \dots, Z_6$ ,  $a_2$ ,  $a_3$  and four further linear forms; the only block with a kernel at a generic point is the even block of weight zero in  $H^1(T)$ , and its kernel is the lowering derivation of the third factor on both copies;  $L_0$  divides only the nine odd bivector blocks. (D) Certificates: each  $S$  is a sum of squares whose zeros force a coincidence or a zero among  $a_1, a_2, a_3$ , each  $Z$  is a positive combination of four independent squares, and the linear factors other than  $L_0$  have unequal coefficients on  $a_1, a_2, a_3$ . (E) As a cross-check, the annihilator in  $HT^2$  has dimension one at forty random admissible points, 36 at  $(1, 1, 1, 1)$ , 7 at  $a_2 = 0$ , 4 on the hyperplane  $a_0 - 3a_1 - 3a_2 + a_3 = 0$ , and 10 at three points of  $L_0 = 0$ . That is [Theorem 21.21](#), [Remark 21.22](#) and the annihilator used in [Theorem 21.27](#).

(XLV) *What an object with an exceptional Chern character must look like.* The script `mumford_object.py` works in the same model. (A) At  $a = (3, 5, -7, 11)$  the ranks of contraction into  $\omega_a$  on  $HT^k$ , computed exactly over  $\mathbb{Q}$  block by block, are 1, 16, 119, 328, 560, 328, 119, 16, 1 for  $k = 0, \dots, 8$  and 0 above. (B) The same ranks hold modulo the prime 1000133, at which the minimal polynomial of  $\zeta_7 + \zeta_7^{-1}$  splits, at  $a_0 = 2$  and the three conjugates in each of the six orders; at a point of  $L_0 = 0$  they are 1, 16, 110, 328, 548, 328, 110, 16, 1. (C) For a random class with components of types  $(0, 0)$  to  $(3, 3)$  the ranks are palindromic and vanish above degree eight. (D) The alternating sum of the ranks is 112 and  $2 \cdot 1 + 2 \cdot 119 + 560 = 800$ . (E) At a CM point there are 16 divisor classes and 132 Hodge classes in degree four; products of divisor classes span 100, and meet the span of the four  $\pi$  exactly where  $a_1 = a_2 = a_3$ . (F) At a CM point  $X_c$  has 8 Hodge classes in degree four, of which products of divisor classes span 6. That is [Lemma 21.30](#), [Theorem 21.31](#) and [Proposition 21.32](#).

(XLVI) *What line bundles generate on a Mumford square.* The script `lefschetz_closure.py` works in the same model, with  $\mathfrak{sp}(V, \psi)$  spanned by the 36 maps  $x \mapsto \psi(u, x)v + \psi(v, x)u$  and the torus of diagonal matrices in  $\mathrm{Sp}(V, \psi)$ , of rank four. (A) The 36 maps and the Lie algebra of  $G$  lie in  $\mathfrak{sp}(V, \psi)$ . (B) The  $\mathrm{Sp}(V, \psi)$ -invariants of  $\bigwedge^k(V \oplus V)$  have dimension at most 1, 3, 6, 10, 15, 10, 6, 3, 1 for  $k = 0, 2, \dots, 16$ , as a kernel modulo 1000003, and products of the three invariant divisor classes span spaces of exactly these dimensions. (C) The  $G$ -invariants have dimension 1, 3, 8, 16, 28, 16, 8, 3, 1, by weight multiplicities and as a kernel, so the Hodge classes that are not Lefschetz classes number 0, 0, 2, 6, 13, 6, 2, 0, 0, twenty-nine in all. (D) At a CM point the monomials of weight zero for the diagonal torus number 1, 16, 100, 304, 454, 304, 100, 16, 1, the coefficients of  $(1 + 4y + y^2)^4$ ; each is a product of the 16 divisor classes, and those products span exactly them; the Hodge classes number 1, 16, 132, 432, 648, 432, 132, 16, 1. (E)  $\pi_0$  and  $\pi_{12} + \pi_{13} + \pi_{23}$  are  $\mathrm{Sp}(V, \psi)$ -invariant and  $\pi_{12}$  is not; the components of  $\pi_{12}, \pi_{13}, \pi_{23}$  of nonzero weight for the diagonal torus have rank two over  $\mathbb{Q}$ , with the relation  $(1, 1, 1)$  only. (F) The modules generated by  $\pi_{12} - \pi_{13}$  and by  $\pi_{13} - \pi_{23}$  have dimension 308 and highest weight  $(2, 2, 0, 0)$ , and Weyl's formula gives 1, 27, 42, 308 for the highest weights 0,  $\varpi_2, \varpi_4, 2\varpi_2$ , with sum 378. (G) The module meets the span of the four  $\pi$  in a plane. That is [Theorem 21.33](#).

(XLVII) *The Hodge locus of an exceptional class among all complex tori.* The script `twistor_locus.py` realises  $V_{\mathbb{R}}$  as the fixed space in  $V_1 \otimes V_2 \otimes V_3$  of the real structure that is quaternionic on the first two factors and real on the third, in exact Gaussian

integer arithmetic. (A) It is a real structure,  $V_{\mathbb{R}}$  has dimension 8, and  $\psi$  is real and alternating on it. (B) The complex structures  $1 \otimes 1 \otimes j_3$  of the Mumford family and  $j \otimes 1 \otimes 1$ ,  $1 \otimes j \otimes 1$  of the compact factors preserve  $V_{\mathbb{R}}$ , and  $\psi(x, Jy)$  has signature  $(0, 8)$  for the first and  $(4, 4)$  for the other two. The unit quaternion  $k \otimes 1 \otimes 1$  with  $kj = -jk$  preserves  $V_{\mathbb{R}}$  and  $\psi$  and carries  $\psi(x, Jy)$ , for  $J = j \otimes 1 \otimes 1$ , to its negative, so every form  $\psi \otimes s$  on  $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$  with  $s \neq 0$  is congruent to its negative and indefinite. (C) At  $j \otimes 1 \otimes 1$  the tensors  $\pi$  have type  $(2, 2)$  and the annihilator of  $\omega_a$  in  $H^1(T)$ , of dimension 64, is spanned by the lowering derivation of the first factor; for the Mumford family it is spanned by that of the third, as in item (XLIV). (D) The exchange of the first and third factors carries  $\pi_{12}$  to  $\pi_{23}$  and fixes  $\pi_0$  and  $\pi_{13}$ . That is [Proposition 21.24](#).

(XLVIII) *The square as a holomorphic symplectic variety.* The script `hk_pullback.py` works in the same split model, with  $\iota(x) = (x \otimes 1)\Psi$  for  $x$  in  $T = \bigoplus_i \mathfrak{sl}(V_i)$ . (A)  $\iota(x)$  is symmetric and  $\iota$  is equivariant, for all pairs of the nine generators. (B) The three Casimir operators split  $\wedge^2(V \oplus V)$ , of dimension 120, into isotypic pieces of highest weights  $(0, 0, 0)$ ,  $(2, 2, 0)$ ,  $(2, 0, 2)$ ,  $(0, 2, 2)$ ,  $(2, 2, 2)$ ,  $(2, 0, 0)$ ,  $(0, 2, 0)$ ,  $(0, 0, 2)$  and dimensions 3, 27, 27, 27, 27, 3, 3, 3, with parts of type  $(2, 0)$  of dimensions 0, 0, 9, 9, 9, 0, 0, 1; the last three pieces are spanned by  $\iota(T_1)$ ,  $\iota(T_2)$ ,  $\iota(T_3)$ . (C) The form  $\sigma_0 = \iota(E_3)$  is of type  $(2, 0)$  and pairs the vectors of type  $(1, 0)$  of the two copies through an invertible  $4 \times 4$  block. (D)  $\iota_2(C_1) = 3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23}$  and cyclically, exactly, and hence the formula for  $\iota_2(\sum s_i C_i)$  with symbolic  $s_i$ . (E)  $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$  as a polynomial identity in the nine coordinates. (F)  $\iota(x)^8 = 8! \det(x)$  times the volume form at three random integral points. (G)  $\Delta$  is a nondegenerate ternary form, of Gram determinant  $-4$ , and irreducible. That is [Proposition 21.25](#).

(XLIX) *Which of the open routes to the Mumford target are needed.* The script `mumford_routes.py` carries the implications into and out of the Mumford target as data, in the manner of item (XXXIII): sixteen statements, one of them proved (the Hodge conjecture at the CM points), and twenty-two Horn rules, each labelled by the theorem that proves it; unlike that rule set this one contains equivalences. (A) Its eleven labels are labels of the paper and its statements are all used. (B) The target is not in the closure of what is proved. (C) The statements equivalent to the target over what is proved are exactly the target, algebraicity at uncountably many points,  $B(W \times_C W)$ , bounded data at infinitely many points, and bounded data modulo  $p$ . (D) The minimal sets of open statements that yield the target, found by exhaustive search, number twelve and each has one element. (E) The target yields neither the complex with  $\dim \text{Ext}^2 = 119$  nor the two Kuga–Satake statements. (F) (L) and (V) each yield the target. (G) The Hodge conjecture and (F2) yield the target, and the target yields none of the Hodge conjecture, (F1), (F2), (F3), (L), (V), (M). (H) (F3) is equivalent to the Hodge conjecture by [Proposition 20.35](#), so it is the twelfth one-element route to the target, and the minimal sets of open statements that yield the Hodge conjecture in this rule set are the conjecture itself, (F3), and (L) with (M), none of which contains the target. That is [Corollary 21.40](#) and [Remark 21.41](#).

(L) *What an object meeting the numerical criterion must look like.* The script `criterion_shape.py` checks the finite ingredients of [Theorem 16.40](#). (A) In the coordinate model of [Section 4](#), for  $n = 2$  with  $d = 1, 2, 3, 5, 7$  and for  $n = 3$  with  $d = 1, 2, 3$ , the rational Weil classes  $\omega_1$ ,  $\omega_2$  satisfy  $\eta \wedge \omega_1 = \eta \wedge \omega_2 = 0$ , and in the orientation in which  $\eta^{2n}$  is positive the intersection form on the rational Weil plane

is positive definite at  $n = 2$  and negative definite at  $n = 3$ , as the Hodge–Riemann relations require; its Gram matrix in the basis  $\omega_1, \omega_2$  is  $\text{diag}(8d^2, 8d)$  at  $n = 2$  and  $\text{diag}(-32d^3, -32d^2)$  at  $n = 3$ . (B) In the model of item (XXXVIII), for  $n = 2$  and  $n = 3$  and in every degree, the classes of type  $(p, p)$  killed by all  $4n^2$  monomials of  $P \wedge Q$  are exactly  $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$ . (C) The generators of  $\wedge^{2n} P$  and  $\wedge^{2n} Q$  contract  $\omega$  to nonzero multiples of the generator of  $H^{0,2n}$ ; contraction with  $\omega$  is injective on  $\wedge^{2n-2} P \oplus \wedge^{2n-2} Q$ ; and the pairing  $\wedge^2 P \times \wedge^{2n-2} P \rightarrow \wedge^{2n} P$  is perfect. (D) The alternating sum of the lower bounds  $1, 2\binom{2n}{1}, \dots, 1$  is  $-2$  for every  $n \leq 60$ ; at  $n = 2$  and  $n = 3$ , exhaustive search over the admissible profiles shows  $2e_0 \geq \chi + 4$ , hence  $e_0 \geq 3$ , lists the smallest profiles, and gives  $e_1 \leq e_0 + 9$  at  $n = 3$ ; at  $n = 4$  and  $n = 5$  a profile with  $e_0 = 1$  satisfies the same constraints, so the bound on End is special to  $n \leq 3$ ; and the same identity with  $\chi = 0$  on the Mumford square returns the 800 odd self-extensions of item (XLV).

- (LI) *The Hodge classes of a very general Weil torus.* The script `weil_tori.py` checks the finite inputs of Lemma 17.2, Lemma 17.3 and Theorem 17.5, in the coordinate model of Section 4, for  $n = 2$  with  $d = 1, 3$  and for  $n = 3$  with  $d = 2$ . The members of the family  $S$  of Setup 17.1 used are  $T_{X,Y}$  with  $X$  and  $Y$  spanned by explicit  $K$ -rational combinations of the eigenvectors, three of them for  $n = 2$  and five for  $n = 3$ . (A) The rational Weil classes  $\omega_1, \omega_2$  are of type  $(n, n)$  at every member used. (B) The polarisation  $\eta$  is of type  $(1, 1)$  at the balanced member and at none of the members used, which therefore lie off the polarised family. (C) The rational classes of degree  $2k$ ,  $1 \leq k \leq n$ , that are of type  $(k, k)$  at all the members used form a space of dimension 0 for  $k < n$  and 2 for  $k = n$ , the Weil plane. A class of degree  $2k$  is of type  $(k, k)$  exactly when its product with every product of  $2n - k + 1$  holomorphic 1-forms vanishes; the ranks are computed modulo a prime  $q < 2^{31}$  in which  $-d$  is a square, with  $\sqrt{-d}$  sent to both square roots so that the conjugate conditions are imposed as well, and since reduction can only lower a rank the dimensions found are upper bounds, attained by the Weil classes. Degrees above  $2n$  are not checked; the proof of the lemma covers them. (D) For an element  $\kappa$  of  $V_+ \otimes V_-$  with random coefficients,  $\alpha_+ \wedge \kappa^n = \alpha_- \wedge \kappa^n = 0$ , and  $\alpha_+ \wedge (\kappa + \beta)^n = \alpha_+ \wedge \beta^n$  becomes nonzero once a component  $\beta \in \wedge^2 V_-$  with  $\beta^n \neq 0$  is added (while  $\alpha_- \wedge (\kappa + \beta)^n$  stays zero, every term having a factor from  $V_-$ ; symmetrically for  $\beta' \in \wedge^2 V_+$  and  $\alpha_-$ ), so the vanishing used for the block diagonal Kähler class  $\kappa_0$  is a property of  $V_+ \otimes V_-$  and not of the coordinates.
- (LII) *The corrected criterion as a number.* The script `p2prime.py` works in the eigenbasis model of item (XXXVIII), with  $\theta = i \sum_j s_j e_j \wedge f_j$  and  $\gamma = \sum_k c_k \theta^k + u\alpha_+ + \bar{u}\alpha_-$ , builds the image  $\xi \lrcorner \gamma$  of every element of a basis of  $HT^2$ , and computes ranks exactly over  $\mathbb{Q}(i)$  by blocks of torus weight, which is legitimate because contraction is equivariant for the torus of Theorem 17.9 and elements whose weights differ by other than a multiple of  $(1, \dots, 1)$  have images in different weight spaces; nothing else about the structure is used. (A) For  $n = 3, 4$ , ten shapes (pure,  $c_0$  only,  $c_{2n}$  only,  $e^\theta$ ,  $c_1$  only,  $c_0$  and  $c_1$ ,  $c_0$  and  $c_{2n}$ , two exponentials,  $c_n$  only, and random rational coefficients) and two values of  $u$ :  $\dim \text{Ann}_{HT^2}(\gamma) = n^2(4 - \rho)$  and  $r = (4 + \rho)n^2 - 2n$ , with  $\rho$  computed separately as the rank of the Hankel matrix. (B) For  $n = 2, 3, 4$  and a general shape, the annihilator is spanned by the  $n^2$  vectors  $v_{ab}^1 + v_{ab}^2$ , lies in  $H^1(T_A)$ , and these vectors kill  $\gamma$ . (C)  $\rho = 1$  on  $\mathbb{C}e^{t\theta}$  and  $\mathbb{C}\theta^{2n}$ , and multiplying  $\gamma$  by  $e^{t\theta}$  changes neither  $\rho$  nor  $r$ . (D) At  $n = 2$ ,  $\dim \text{Ann} = 4(4 - \rho) + \dim \ker((HK)^2 - |u|^2)$  on six values of  $|u|$  for  $\theta^2 + \omega$  and on four random shapes, and  $r = 24, 22$  and

- 23 at  $|u|^2 = 1, 4, 16$ . (E) The first-order Hodge locus  $\text{Ann}_{H^1(T_A)}(\gamma)$  has dimension  $n^2$  or, when  $c_1 = \cdots = c_{2n-1} = 0$ ,  $2n^2$ , for  $n = 3, 4$  and all ten shapes; at  $n = 2$  it is 5 exactly at  $c_1 = c_3 = 0$ ,  $|u| = 4|c_2|$ . (F)  $\int \text{ch}^\vee \text{ch} = (2n)!Q(c) + 2|u|^2$  in the normalisation  $\int \theta^{2n} = (2n)!$  for  $n = 2, 3$ , and in the rational model of `explicit_weil.py` at  $n = 2$ ,  $\int \omega_1^2 = 8d^2$  and  $\int \eta^4 = 24$  for  $d = 1, 2, 3, 5, 7$ . (G) At  $n = 2$  and the exceptional ratio  $|u| = 4|c_2|$ , the stabiliser of  $\omega + c_2\theta^2$  in  $\mathfrak{gl}(H^1)$  has dimension 21, Hodge pieces 5, 11, 5, commutant on  $H^1$  the scalars, a real form whose trace form has signature (12, 9), and invariants of dimensions 1, 0, 0, 0, 1, 0, 0, 0, 1 in  $\bigwedge^k H^1$ ; at an ordinary ratio the stabiliser has dimension 15 and invariants 1, 0, 1, 0, 3, 0, 1, 0, 1; at the exceptional ratio  $\int \gamma \kappa_h^2 = \frac{1}{6}(\sigma_2(h) + 4 \text{Re det } h_{01,23}) \int \theta^4$  at eight random  $h$ , and the sign of  $\int \gamma \kappa^2$  is that of  $c_2$  at 150 random Kähler classes, while at the ratio  $R = 3/16$  both signs occur. That is [Proposition 17.11](#) and [Remark 17.12](#). With `-extreme` the closed form of (A) is checked exactly for  $n = 5, \dots, 9$  on five shapes, and with `-nmax=10` also at  $n = 10$ , where  $HT^2$  has dimension 780 and  $\gamma$  has  $2^{20} + 2$  terms; that run takes 142 minutes and its transcript is in `code/extreme`. That is [Theorem 17.9](#).
- (LIII) *The Hochschild profile of a corrected character.* The script `p2prime_profile.py` computes, in the same model and by the same torus blocks, the rank  $\rho_k$  of contraction into  $\gamma$  on every  $HT^k$ , exactly over  $\mathbb{Q}(i)$ . (A) For  $n = 2, 3$ , in every degree  $0 \leq k \leq 4n$ , on eight shapes and two values of  $u$ , the ranks equal  $2\binom{2n}{k} + M_k \min(k+1, 2n+1-k, r') - [k=n]d$  for  $1 \leq k \leq 2n-1$ , with  $\rho_0 = \rho_{2n} = 1$  and  $\rho_k = 0$  above  $2n$ ; the general profiles are (1, 8, 24, 8, 1) and (1, 12, 57, 112, 57, 12, 1), and the pure case gives  $2\binom{2n}{k}$ . (B)  $\rho_k = \rho_{2n-k}$  on every shape. (C) For  $c_n\theta^n$  alone the middle rank drops by  $d$  exactly at  $|u| = \binom{n}{a}n!|c_n|$  and in no other degree. (D) On  $\bigwedge^{2n} \mathbb{Z}^{4n}$ ,  $n \leq 3$ , the form  $\int x \wedge y$  pairs each monomial only with its complement, so  $\int x^2$  is even; at  $n = 2$  the point  $|u| = 4|c_2|$  of  $\theta^2 + \omega$  has  $\rho_2 = 23$ . With `-extreme` (A) is run for  $n = 4$  in every degree, where  $HT^8$  has dimension 12870, and for  $n = 5$  up to degree three. That is [Proposition 17.10](#).
- (LIV) *Descent and scalar extension.* The script `descent.py` models the cohomology of an abelian variety of  $(F, n)$ -Weil type,  $F$  a CM field of degree  $2m$ , by alternating  $\mathbb{Q}$ -multilinear forms on  $H_1 = F^N$ , its Weil classes by the forms  $\text{Tr}_{F/\mathbb{Q}}(a \det_F)$  and its polarisation by  $\text{Tr}_{F/\mathbb{Q}}(\xi H)$ . (A) For  $F = \mathbb{Q}(i)$ ,  $\mathbb{Q}(\sqrt{-2})$ ,  $\mathbb{Q}(\sqrt{-3})$ ,  $\mathbb{Q}(\zeta_5)$  and  $\mathbb{Q}(\zeta_8)$ , and  $n = 1, 2$  (and  $n = 3$  over  $\mathbb{Q}(i)$ ), the correspondence  $P(x) = pr_{B*}(x \cdot pr_Y^*(\eta_Y^{2m-2}y'))$  maps the  $2m$ -dimensional space  $W_F(B \times Y)$  onto  $W_F(B)$ , and it kills  $W_F(B \times Y)$  when the Weil class  $y'$  of  $Y$  is replaced by  $\eta_Y$ . (B) For  $H_Y = \text{diag}(1, -t)$  the form  $H_B \oplus H_Y$  has signature  $(n+1, n+1)$  and normalised discriminant  $t$  times that of  $H_B$ . (C) For  $\mathbb{Q}(i) \subset \mathbb{Q}(\zeta_8)$ ,  $\mathbb{Q}(\sqrt{-7}) \subset \mathbb{Q}(\zeta_7)$  and  $\mathbb{Q}(\sqrt{-3}) \subset \mathbb{Q}(\zeta_9)$ , and  $n = 1, 2$ , the pull-back  $j^*W_F(B_F)$  is all of  $W_K(B)$ . With `-extreme`, (A) is run up to  $n = 5$  over  $\mathbb{Q}(i)$ , up to  $n = 4$  over  $\mathbb{Q}(\sqrt{-2})$  and  $\mathbb{Q}(\sqrt{-3})$ , up to  $n = 3$  over  $\mathbb{Q}(\sqrt{-5})$ ,  $\mathbb{Q}(\zeta_5)$  and  $\mathbb{Q}(\zeta_8)$ , and up to  $n = 2$  over the sextic field  $\mathbb{Q}(\zeta_7)$ , and (C) up to  $n = 3$ . That is [Proposition 20.13](#).
- (LV) *Quartic fields, the Mumford fourfold, and the objects of the criterion.* The directory `code/attack` holds four sets of scripts, each with the scripts of an independent re-implementation in `verify/` and the unedited transcripts of their runs in `transcripts/`; its `README.md` states each claim with its status (proved, computed, conditional or sketched). The item `attack_checks.py` runs a fast subset, in about two and a half minutes, and turns the printed numbers into 48 checks; the longer

runs are not repeated, and their transcripts are distributed. Where a space of invariants is bounded above by a rank modulo a prime, which can only lower a rank, the bound is matched by explicit invariants. (A) `quartic_cm`, where the scripts write  $T$  for  $R_F$ : for  $F = \mathbb{Q}(\zeta_5)$  (cyclic, with no imaginary quadratic subfield),  $\mathbb{Q}(\zeta_8)$ ,  $\mathbb{Q}(\zeta_{12})$  (biquadratic) and  $\mathbb{Q}(\sqrt{-(3 + \sqrt{2})})$  (not Galois), and six hermitian forms in all,  $\mathfrak{su}(V, H)$  has dimension 30 and kills  $W_F$ , and its invariants have dimension 2 in degree two and 7 in degree four, with  $W_F$  meeting the products of divisor classes in 0;  $\phi$  is semilinear with  $\phi^2 = (\det H)^{-1}$ , it commutes with  $\mathfrak{su}(V, H)$  on  $R_F$  and the commutant has dimension 8, and over  $\mathbb{Q}(\zeta_5)$  with a hermitian form that is not diagonal  $\phi g = \overline{\det g} g \phi$  for an element  $g$  of  $U(V, H)$  with  $\det g \notin \mathbb{Q}$ ; (66) holds,  $\Delta^* C_q = 6c\Omega$ , and the pull-backs of  $W_F$  along five maps span a space of dimension 20 containing  $C_q$ ; on the split locus for three fields  $\det H$  is a square in  $F_0$  and the polarisation of  $X$  gives a Hodge class in  $R_F$ ; on the scalar-extension locus for three biquadratic fields and four forms the pull-backs  $g_F^* W_K(C)$  span a space of dimension 10 containing  $W_F(B_F)$ , and the invariants of  $\mathfrak{su}(V_C, H_C)$  in degree two meet  $R_F$  in 0;  $\dim HT^2 = 120$ , the annihilator of a Weil class has dimensions 0, 16, 24, the eight-dimensional tangent space kills  $\gamma$ , and  $r(\gamma) = 112$  for nine general  $p$ , in two independent implementations, and at six elements of the Hodge ring; the part of the contraction that no  $p$  reaches has rank 68; and on the 25 monomials and all 300 pairs of them, with seven ratios,  $r$  takes the values 80, 84, 88, 104, 108, 112, and 92, 96 on pairs, with no value below 80 in about 1800 further exact trials. That is [Proposition 19.9](#), [Proposition 19.10\(iii\)](#), [Proposition 19.12](#), [Proposition 19.13](#) and [Theorem 19.16](#). (B) `mumford_target`: by peeling highest weights and, independently, by Weyl's alternating sum, no  $\bigwedge^k V$  contains a summand  $S^2 V_i$ , the multiplicity of  $S^2 V_1$  in  $H^n(X \times X)$  is 1, 6, 16, 24, 16, 6, 1 for  $n = 2, 4, \dots, 14$  and 0 otherwise, and the copy in  $H^2$  lies in  $H^1 \otimes H^1$  with Hodge numbers (1, 7, 1), as it does in the twelve-dimensional unitary model of [Remark 21.14](#); on the split model, in the exterior algebras of  $V \otimes \mathbb{Q}^m$ , the degree-four Hodge classes of  $X \times X$  have dimensions 1, 1, 4, 1, 1 in the Künneth components,  $(V^{\otimes 4})^G$  has dimension 8, the pull-backs along 45 integer homomorphisms  $X^4 \rightarrow X^2$  and the products of pulled-back divisor classes span exactly 7 dimensions and miss  $\text{Det}$ , six times Cayley's hyperdeterminant on the diagonal, and composites of two correspondences fill all 8;  $s_k = \text{Cas}_k + \frac{1}{2}$ , the images of  $\mathbb{Q}[S_2]$ ,  $\mathbb{Q}[S_3]$ ,  $\mathbb{Q}[S_4]$  have dimensions 2, 5, 14, so  $\text{End}_G(V \otimes V)$  and  $\text{End}_G(V^{\otimes 3})$  have dimensions 8 and 125; and the irreducible zero-sum multisets of the weights  $\{\pm 1\}^3$  are the four pairs  $\{w, -w\}$  and two sets of four. That is [Proposition 21.6](#), [Theorem 21.7](#), [Remark 21.8](#) and [Proposition 21.9](#); [Corollary 21.11](#), [Proposition 21.12](#), [Proposition 21.13](#) apart from its computation, and [Proposition 21.16](#) are proved in the text and have no script. (C) `explicit_objects`: the Hochschild profile of a secant class for  $n = 2, \dots, 5$  and the Euler characteristic for  $n = 2, \dots, 6$ , at values of  $d$  the first run did not use; the certificates of [Theorem 17.19](#), with profile (1, 8, 18, 8, 1) for seven values of  $d$  and (1, 12, 48, 74, ...) for five; the flatness of  $\text{ch}(E)e^{\ell/2}$ , the recursion  $\mu_{m+2} = -\mu_m/(4d)$  and the Weil part over the function field  $\mathbb{Q}(d)$  for  $n \leq 5$ , the  $B$ -field in the eight conventions for  $\Phi$ , and the dimensions 5 and 9 of the Hodge loci;  $r(\text{ch } G) = 18, 48, 88$  at  $n = 2, 3, 4$ ; the parity numbers of [Remark 17.21](#); and the correction to [Example 15.3](#) at  $n = 3$ . (D) `natural_objects`: over  $\mathbb{Q}(i)$ , exactly,  $\dim T = 16$ ,  $T$  kills  $\eta$  and the Weil plane, and  $T \lrcorner \text{ch } F$  has rank 6 for the natural objects off  $C_\eta$  and 0 on it; modulo a prime,

$r(\gamma) = 56, 72, 88, 104$  for  $\rho = 0, \dots, 3$ ; the Hodge ring of dimension 55, the fit of  $\Phi_4$  and the geometry of  $C_\eta$ ,  $l_W$  and  $L_\pm$  in [Setup 17.22](#); the relation of [Example 17.24](#) in both sign conventions, its minimality among the 12870 eight-sets, the configurations of [Theorem 17.23](#) (eight points on one conic give a pure relation, and the splittings  $7, 4 + 4, 5 + 3, 6 + 2, 6 + 1 + 1, 7 + 1, 3 + 3 + 2$  give none), the lattice spanned by the classes of the subvarieties, of index 2612736000, and the intersection numbers of the sum of [Example 17.24](#); and the agreement with the exhaustive search of item (XIII) at  $n = 2$ , where the Weil tuples all use four points off  $C_\eta$  and the pure ones are those on a conic through  $l_W$ . [Lemma 17.16](#), [Corollary 17.17](#), [Proposition 17.18](#) and [Proposition 17.20](#) are proved in the text and checked here in examples.

Items (I), (II), (III), (VI), (VII), (VIII) and (X) to (LV) are the ones on which theorems depend. Items (IV) and (V) support statements that no obstruction exists, and a failure there would strengthen rather than weaken the paper, so they are included for completeness rather than for safety. Running `verify_all.py` executes the fifty Python items and prints 951 checks passed, 0 failed; items (XXIX), (XXXI) and (XL) need Macaulay2 and are run separately, and their transcripts are distributed with the scripts.

## APPENDIX E. MACHINE VERIFICATION IN LEAN 4

The theorems of this paper are statements of algebraic geometry and are not formalised. What is formalised is the finite arithmetic on which many of them turn, which the kernel can settle by computation; the rest of their computational content is checked in exact arithmetic by the scripts of [Appendix D](#). The file `HodgeObstruction.lean` accompanies the paper. It compiles against a bare Lean 4 toolchain, has no `sorry`, and depends on no library, `Mathlib` included; the report at its foot states, for each of its eighty theorems, that it depends on no axiom whatever, or on propositional extensionality alone, which is what `decide` uses when it reduces an equality of Booleans, or when a bound is proved from the lemmas on natural numbers in the core library.<sup>72</sup> The package also builds with `lake` build under the `lakefile.toml` distributed with it, with the same report; each run takes about two minutes. The results of [Section 9.1](#), the annihilator dimension and the rank-zero obstruction, are not among the eighty: the lemma is proved by hand, and its dimension count is verified in exact arithmetic modulo a prime by item (XIV) of [Appendix D](#).

### E.1. What is checked.

Table 7: The eighty machine-checked theorems.

| theorem                              | statement                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------|
| <code>qnorm_mul_check</code>         | the norm on $\mathbb{Z}[\sqrt{-d}]$ is multiplicative on the data used                |
| <code>qmul_conj</code>               | $\tau\bar{\tau}$ equals the norm, as an element of $\mathbb{Z}[\sqrt{-d}]$            |
| <code>characters_differ</code>       | $\tau^{2n} \neq N(\tau)^n$ at $\tau = 2 + \sqrt{-d}$ , for nine fields and $n \leq 8$ |
| <code>characters_differ'</code>      | the same at five further witnesses                                                    |
| <code>bad_witness</code>             | the witness $\tau = 1 + \sqrt{-d}$ fails, at $(d, n) = (1, 4)$ and $(3, 3)$           |
| <code>bad_witness_exceptions</code>  | and fails exactly when $d = 1, 4 \mid n$ or $d = 3, 3 \mid n$                         |
| <code>ratio_not_root_of_unity</code> | $\tau^n \neq \bar{\tau}^n$ at $\tau = 2 + \sqrt{-d}$                                  |

*continued on the next page*

<sup>72</sup>Sixty-nine of the eighty report does not depend on any axioms; the remaining eleven report `[propxt]`. No theorem reaches for anything else, and none mentions `sorryAx`. The workflow in `.github/workflows/lean.yml` fails the build if either changes.

Table 7, continued.

| theorem                   | statement                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| sign_uniform_and_closed   | the sign of <a href="#">Theorem 6.5</a> is uniform over index sets and equals $(-1)^{g(g+1)/2+k}$ , for $g \leq 5$    |
| subsets_count             | the subset enumeration used there has $\binom{g}{k}$ members                                                          |
| semiregularity_target     | $\sum_q \binom{2n}{q} \binom{2n}{q+2} = \binom{4n}{2n-2}$ for $n \leq 7$                                              |
| secant_count_             | $2n^2 + 2n < 2^{2n} < 3^{2n}$ for $2 \leq n \leq 8$                                                                   |
| below_thresholds          |                                                                                                                       |
| thresholds_not_necessary  | $3 < 2^2$ and $5 < 3^2$ , the two retractions of <a href="#">Section 8</a>                                            |
| grading_positions_differ  | $(2n, 0)$ , $(0, 2n)$ and $(n, n)$ are distinct cells for $n \geq 1$                                                  |
| weil_delta_parity         | $\sqrt{-d}^k$ is rational exactly when $k$ is even, for $k \leq 20$                                                   |
| weil_delta_closed_even    | $\sqrt{-d}^{2m} = (-d)^m$                                                                                             |
| weil_delta_closed_odd     | $\sqrt{-d}^{2m+1} = (-d)^m \sqrt{-d}$                                                                                 |
| weil_term_counts          | each of $\omega_1, \omega_2$ has $2^{2n-1}$ monomials                                                                 |
| supports_disjoint         | no monomial of the Weil line occurs in a power of $\eta$                                                              |
| hodge_dim_count           | $\dim \text{Hdg}^k = 3$ for $k = n$ and 1 otherwise, for $n \leq 6$                                                   |
| weil_line_hodge_type      | the Weil line is of type $(n, n)$ exactly at balanced signature                                                       |
| clifford_dimensions       | $\dim C = 2^b$ , $\dim C^+ = 2^{b-1}$ , $\dim \text{KS} = 2^{b-2}$                                                    |
| k3_type_only_at_one       | $h^{2,0} = 1$ on the pieces of $H^2$ forces $n = 1$                                                                   |
| moduli_codimension        | the Weil family has codimension $n(n+1)$ in $\mathcal{A}_{2n}$                                                        |
| discriminant_is_a_norm    | $d^n p^2$ is a norm from $K$ , with an explicit witness                                                               |
| split_codimension         | the split locus has codimension $n(n-1)/2$ , zero only at $n = 1$                                                     |
| quat_class_odd            | for $n$ odd the quaternionic class is $b$ times a square                                                              |
| quat_class_even           | for $n$ even it is a square                                                                                           |
| quat_product_class        | $d^{n-1}$ is a norm for $n$ even, with a witness                                                                      |
| quat_locus_codimension    | the quaternionic locus has dimension $n(n+1)/2$ ,<br>codimension $n(n-1)/2$                                           |
| quat_locus_divisor_at_two | at $n = 2$ that locus is a divisor: the three numbers are 4, 3, 1                                                     |
| irrational_witness_exists | some $m \leq 7$ makes $(m + \sqrt{-d})^{2n}$ irrational, for $n \leq 8$                                               |
| weil_line_spanned         | the determinant of $w, \tau w$ is $b(w_1^2 + dw_2^2)$ , hence nonzero                                                 |
| norm_form_positive        | $a^2 + db^2 > 0$ for every nonzero $(a, b)$ , for ten fields                                                          |
| norm_sum_forces_zero      | a positive combination of norms vanishes only when every term does                                                    |
| semiregularity_           | at $n = 2$ the source of $\sigma$ overtakes its target at $s = 5$                                                     |
| source_target             |                                                                                                                       |
| tensor_rank_gap           | $\binom{2n}{2} \geq 6 > 1$ for $n \geq 2$ , the separation of <a href="#">Lemma 18.4</a>                              |
| secant_witness_at_n_four  | the line bundle witness of <a href="#">Theorem 14.13</a> reproduces<br>$6(u_t + v_t)$ in every degree                 |
| secant_witness_rank       | that witness has rank $6 = Ma$ , which is positive, so<br><a href="#">Lemma 14.12</a> applies                         |
| vandermonde_              | the Vandermonde product on the nodes $0, \dots, n$ is nonzero                                                         |
| nodes_distinct            | for $n \leq 8$                                                                                                        |
| level_sums_forced         | the level sums of <a href="#">Theorem 18.20</a> satisfy $\sum_j M_j N_j^r = 0$ ,<br>$r = 1, 2, 3$ , at six level sets |
| level_sums_totals         | their totals are 1, 3 and $-2$ , so none of those objects has<br>rank zero                                            |
| level_matrix_nonsingular  | with at most $n$ distinct norms the matrix $(N_j^r)$ is nonsingular                                                   |

continued on the next page

Table 7, continued.

| theorem                   | statement                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------|
| gauss_norm_counts         | the number of Gaussian integers of norm 1, 2, 3, 4, 5, 9, 45                                           |
| norm_supply_              | at the nodes 1, 2, 3, 4 the forced $ M_3  = 4$ while $\mathbb{Z}[i]$ has no                            |
| blocks_small_case         | element of norm 3                                                                                      |
| pencil_index_d1           | at $d = 1$ two minors of the pencil differ by the constant 2 for                                       |
|                           | $ k  \leq 200$ , so the index of $\mathbb{Z}[i]x_k$ divides $N_1 = 2$                                  |
| pencil_index_d3           | at $d = 3$ a constant minor $-4$ and a second minor have gcd                                           |
|                           | dividing 4 for $ k  \leq 200$ , so $N_3 = 4$                                                           |
| pencil_beta_positive      | $4\beta_1(k) = 2 + 2k + k^2$ and $9\beta_3(k) = 1 + 2k^2$ are positive for                             |
|                           | $ k  \leq 200$ , with discriminants $-4$ and $-8$                                                      |
| pencil_square_members     | $\beta_1(-1) = 1/4$ and $\beta_3(-2) = 1$ , the members of the pencil                                  |
|                           | with a rational point                                                                                  |
| split_obstruction_rank    | $n^2 - n(n+1)/2 = n(n-1)/2$ and $n(n-1) = 2 \cdot n(n-1)/2$                                            |
|                           | for $n \leq 30$ : the rank of the obstruction map of                                                   |
|                           | <a href="#">Proposition 18.16(ii)</a> and the even side of <a href="#">Corollary 9.8</a>               |
| split_obstruction_count   | the bookkeeping of <a href="#">Proposition 18.16(iv)</a> at $n = 3$ : kernel                           |
|                           | $15 = 9 + 6$ , image 3, direct sum 12, complement 3                                                    |
| evaluation_not_surjective | $s \binom{2n}{2} > 2 \binom{2n}{2} + 4n^2$ for $5 \leq s \leq 40$ and $2 \leq n \leq 30$ , so the      |
|                           | evaluation map of <a href="#">Proposition 18.18(iii)</a> is not surjective; at                         |
|                           | $n = 3$ , $s = 6$ the dimensions are 66 and 90                                                         |
| markman_candidate_        | for odd $d \leq 201$ and $N = (d+9)/2$ : the point count                                               |
| identities                | $12N(N-5) = (d+9)(3d-3)$ , the Euler characteristic                                                    |
|                           | $(d+9)(27-d)$ , the two identities that place $\text{ch}(\mathcal{F})$ at $(1, 3)$ ,                   |
|                           | $N(3 + \sqrt{-d}) = 2N$ , and the nonvanishing of the ranks,                                           |
|                           | <a href="#">Proposition 18.25</a>                                                                      |
| secant_kernel_arithmetic  | the matrix $\begin{pmatrix} a & b \\ b & -ad \end{pmatrix}$ has determinant $-(a^2d + b^2) \neq 0$ for |
|                           | $(a, b) \neq (0, 0)$ , $a, b < 13$ , $d \leq 20$ ; the coefficients $C_k = k! c_k$ of                  |
|                           | $au_t + bv_t$ satisfy $C_{k+1} = -dC_{k-1}$ ; and                                                      |
|                           | $n(n+1)/2 + n(n-1)/2 = n^2$ for $n < 60$ , <a href="#">Theorem 18.29</a>                               |
| candidate_unnormalised_   | for odd $d \leq 201$ :                                                                                 |
| points                    | $(d+9)(3d-3) - (d+9)(2d-1) = (d+9)(d-2) \geq 12$ , so an                                               |
|                           | unnormalised double point exists; normalising all of them                                              |
|                           | gives $\chi = 50N$ , and $50N \neq 72N - 4N^2$ for every integer                                       |
|                           | $N > 0$ , <a href="#">Theorem 18.34</a> , <a href="#">Remark 18.35</a>                                 |
| mumford_rigidity_sos      | a positive multiple of each of the seventeen forms                                                     |
|                           | $P_1, P_2, Q_1, \dots, Q_4, R, S_1, \dots, S_4, Z_1, \dots, Z_6$ is a sum of                           |
|                           | squares of integral linear forms with positive coefficients,                                           |
|                           | checked on $\{0, 1, 2\}^4$ , which suffices for polynomials of degree                                  |
|                           | at most two in each variable, <a href="#">Theorem 21.21</a>                                            |
| smooth_invariants         | Noether and the self-intersection formula for a smooth                                                 |
|                           | support, $K^2 + e = 12\chi$ and $K^2 - e = 6(b^2 + d)^2$                                               |
| smooth_bmy_defect         | the Bogomolov-Miyaoka-Yau defect of a smooth support is                                                |
|                           | $24(b^4 - d^2)$                                                                                        |
| smooth_bmy_is_d_le_bsq    | so the inequality is exactly $d \leq b^2$ , over a box in $b$ and $d$                                  |
| smooth_index_vacuous      | the Hodge index bound $9(b^2 + d) \geq 8b^2$ holds for every $d \geq 1$                                |
| smooth_four_discriminants | the squarefree odd $d \leq 9$ are exactly 1, 3, 5, 7, which is                                         |
|                           | <a href="#">Corollary 18.41</a>                                                                        |
| smooth_table              | the invariants of <a href="#">Table 3</a> at $b = 3$                                                   |

continued on the next page

Table 7, continued.

| theorem                     | statement                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| burch_weight_identities     | the weight $x^2 + d$ annihilates three moments, $c_{k+2} + dc_k = 0$ , which is <a href="#">Theorem 18.49</a>                                                                                                                                                                                                                                                                                     |
| burch_rank_one_discriminant | nine times the discriminant of the rank one quadratic is $-2b^2 - 18d$ , always negative                                                                                                                                                                                                                                                                                                          |
| burch_rank_four_witness     | the rank four candidate at $d = 23$ solves the moment system and is removed only by injectivity                                                                                                                                                                                                                                                                                                   |
| closure_omits_conjecture    | the Hodge conjecture is not in the forward closure of what this paper proves and quotes                                                                                                                                                                                                                                                                                                           |
| frontier_suffices           | it is in the closure once (F3) alone is adjoined, which is <a href="#">Proposition 20.35</a> ; once $\{L, M\}$ , $\{L, F3'\}$ , $\{V, F3'\}$ or $\{P2_{\text{split}}, F2, F3'\}$ is, with $P2_{\text{split}}$ the propagation statement for the split families of the CM fields of degree at least four; and, not minimally, once $\{P2_{\text{split}}, F2, F3\}$ , $\{L, F3\}$ or $\{V, F3\}$ is |
| frontier_minimal            | in each of the five minimal sets of <a href="#">Theorem 20.40(ii)</a> every element is necessary: dropping any one leaves the conjecture underivable                                                                                                                                                                                                                                              |
| frontier_smallest           | of the sixteen open statements exactly one, (F3), suffices alone, and of the one hundred and twenty pairs exactly those containing (F3) and the pairs $\{L, M\}$ , $\{L, F3'\}$ , $\{V, F3'\}$ do                                                                                                                                                                                                 |
| variational_gives_abelian   | (V) alone puts the conjecture for abelian varieties in the closure, and not <b>HC</b> , <a href="#">Theorem 20.19</a>                                                                                                                                                                                                                                                                             |
| lefschetz_gives_abelian     | (L) gives (V), hence the conjecture for abelian varieties, and not <b>HC</b> , <a href="#">Proposition 20.18</a>                                                                                                                                                                                                                                                                                  |
| secant_route_descends       | granting both open demands of the secant route reaches $\delta = \delta_0$ and, through <a href="#">Proposition 20.13</a> , every imaginary quadratic family, and neither the CM fields of higher degree nor the conjecture                                                                                                                                                                       |
| propagation_closes          | (P2) for the split imaginary quadratic families puts the whole imaginary quadratic case in the closure, and (P2) for the split families of the fields of degree at least four puts the Weil classes of every CM field there                                                                                                                                                                       |
| p2_minimal_count            | $\binom{4n}{2} - 4n^2 = 2\binom{2n}{2} = 2n(2n - 1)$ for every $n \leq 60$ , the dimension at which the first form of the criterion would be met, <a href="#">Theorem 16.36</a> , and which no complex attains, <a href="#">Theorem 17.5</a>                                                                                                                                                      |
| mumford_invariants          | the invariants of $\mathfrak{sl}_2^3$ in $\bigwedge^q(V_1 \otimes V_2 \otimes V_3)$ have dimensions 1, 0, 1, 0, 1, 0, 1, 0, 1, counted from weight multiplicities, <a href="#">Corollary 20.23</a>                                                                                                                                                                                                |
| two_branch_annihilator      | for $n = 2, 3$ the monomials of type $(p, p)$ killed by $Q \wedge HH^1$ , $P \wedge HH^1$ and $HH^1 \wedge HH^1$ are $\alpha_-$ , $\alpha_+$ and none, <a href="#">Theorem 20.33</a>                                                                                                                                                                                                              |
| weil_monomials_separated    | neither Weil monomial lies in $\bigwedge^*(V_-^{1,0} \oplus V_+^{0,1})$ or $\bigwedge^*(V_+^{1,0} \oplus V_-^{0,1})$ , for $n \leq 12$ , which is the last step of <a href="#">Theorem 16.38(iii)</a>                                                                                                                                                                                             |

continued on the next page

Table 7, continued.

| theorem                                 | statement                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>mumford_ext_profile</code>        | the bounds 1, 16, 119, 328, 560, 328, 119, 16, 1 are palindromic, with alternating sum 112, sum 1488 and $2 + 238 + 560 = 800$ , <a href="#">Theorem 21.31</a>                                                                                                                                                                                                            |
| <code>lefschetz_counts</code>           | the $\mathrm{Sp}_8$ -invariants of $\bigwedge^*(V \oplus V)$ number 1, 3, 6, 10, 15, 10, 6, 3, 1 by the decomposition of each $\bigwedge^i V$ , the Hodge classes that are not Lefschetz 0, 0, 2, 6, 13, 6, 2, 0, 0, the Lefschetz classes at a CM point are counted by $(1 + 4y + y^2)^4$ , and Weyl's formula gives 308 for $2\varpi_2$ , <a href="#">Theorem 21.33</a> |
| <code>criterion_endomorphisms_n2</code> | for all natural numbers, $2e_0 + 12 = \chi + 2e_1$ with $e_1 \geq 8$ and $\chi \geq 1$ gives $e_0 \geq 3$ : the Euler characteristic of an object meeting the criterion at $n = 2$ , <a href="#">Theorem 16.40(iv)</a>                                                                                                                                                    |
| <code>criterion_endomorphisms_n3</code> | for all natural numbers, $2e_0 + 60 = \chi + 2e_1 + e_3$ with $e_1 \geq 12$ , $e_3 \geq 40$ and $\chi \geq 1$ gives $e_0 \geq 3$ : the same at $n = 3$ , <a href="#">Theorem 16.40(iv)</a>                                                                                                                                                                                |

An element of  $\mathbb{Z}[\sqrt{-d}]$  is a pair of integers, multiplication and conjugation are the evident operations, and every statement in the table but the last two is a closed Boolean expression evaluated by the kernel.<sup>73</sup> Nothing is quoted; the kernel recomputes each assertion from the definitions.

Two of the entries deserve a word about what they do and do not settle. `hodge_dim_count` verifies the bookkeeping of [Theorem 4.14](#), namely that summing the contributions  $(n, n)$ ,  $(2n, 0)$  and  $(0, 2n)$  over the degrees gives three in the middle and one elsewhere; the representation theory that identifies those contributions is in the proof of [Theorem 4.14](#) and is not formalised. Likewise `k3_type_only_at_one` verifies that the equation  $h^{2,0} = 1$  has  $n = 1$  as its only solution once the three possible values 0,  $n^2$  and  $n(n - 1)$  are known; the identification of those values is [Proposition 11.11](#). In both cases the division of labour is the same as everywhere else in this section: the kernel settles the computation, and the mathematics around it is written out and must be read.

**E.2. Two statements that look true and are not.** Two theorems of the file record statements that a careful reader would accept and that are false, and both bear on [Section 5](#).

The first concerns the witness  $\tau = 1 + \sqrt{-d}$  for [Lemma 5.4](#). It is the obvious choice, and it does not work at every field: `bad_witness` shows that at  $d = 1$  the ratio  $\tau/\bar{\tau}$  is  $\sqrt{-1}$  and at  $d = 3$  it is a primitive cube root of unity, so at those fields the two characters agree at that particular  $\tau$  for suitable  $n$ . The lemma is true, but not by that route, which is why the proof given in [Section 5](#) uses only the finiteness of  $\mu(K)$  and avoids witnesses altogether.

The second concerns the exceptional set of that witness at  $d = 3$ , which is  $3 \mid n$  and not  $6 \mid n$ , as a hasty reading of the order of  $\mu(K)$  would suggest; `bad_witness_exceptions` records the correct condition. Neither affects [Theorem 1.3](#). Both are the kind of slip that

<sup>73</sup>For those the tactic is `decide`, which asks the kernel to evaluate a closed Boolean expression by reduction. That is the strongest check available and the least flexible: it works only over finite ranges, which is why each of those theorems carries an explicit bound, and it is also why none of them can be the general statement the paper proves. The last two, `criterion_endomorphisms_n2` and `criterion_endomorphisms_n3`, are stated for all natural numbers and proved from the lemmas on natural numbers in the core library, with `decide` only for a final numerical inequality.

a careful reading does not catch and a kernel check does, which is the reason for keeping them in the file.

**E.3. What is not formalised.** Everything else, and we are explicit about it. The cohomology of an abelian variety as an exterior algebra, the Hodge decomposition, the Fourier–Mukai transform as a functor, Grothendieck–Riemann–Roch, the Lefschetz hyperplane theorem, hard Lefschetz, the invariant theory of  $SL_{2n}$ , the Clifford algebra as a functor, and the Lefschetz standard conjecture are all used as mathematics and none is formalised here. A formalisation of those would require a library of complex algebraic geometry that does not presently exist. What the Lean file establishes is that where this paper computes, it computes correctly; where it reasons, the reader must still read.

#### DECLARATIONS

**Funding.** None.

**Competing interests.** None.

**Data availability.** All code and data supporting the results of this paper are openly available. The verification package comprises the Python scripts of [Appendix D](#), the Lean 4 certificate of [Appendix E](#), the Macaulay2 computations of local Ext algebras with their transcripts, and the generators and TikZ sources of every figure. It is held in the public GitHub repository <https://github.com/creelie/H8>, where it was developed on the branches `claude/hodge-conjecture-obstructions-2z011a` and `claude/elegant-gates-nc3yph`. Release v1.0.0 of that repository,

<https://github.com/creelie/H8/releases/tag/v1.0.0>,

is archived on Zenodo under DOI [10.5281/zenodo.22950276](https://doi.org/10.5281/zenodo.22950276),

<https://doi.org/10.5281/zenodo.22950276>;

it accompanies an earlier version of this paper, and the items added since are on the second branch named above, to be archived with a later release under its own version DOI. Running `python3 code/verify_all.py` prints 951 checks passed, 0 failed. Running `lean HodgeObstruction.lean` checks eighty theorems, each of which uses no axiom or propositional extensionality alone. The Python computations and the Lean formalisation were carried out with the assistance of Claude, an AI system developed by Anthropic. Every result they certify is reproduced from the repository by the scripts and by the Lean kernel, and the mathematical statements and their proofs are the responsibility of the authors. No other data were generated or analysed.

#### REFERENCES

- [Alb34] A. A. Albert, *On the construction of Riemann matrices. I*, Ann. of Math. (2) **35** (1934), no. 1, 1–28. [doi:10.2307/1968116](https://doi.org/10.2307/1968116).
- [Alb35a] A. A. Albert, *On the construction of Riemann matrices. II*, Ann. of Math. (2) **36** (1935), no. 2, 376–394. [doi:10.2307/1968578](https://doi.org/10.2307/1968578).
- [Alb35b] A. A. Albert, *Involutorial simple algebras and real Riemann matrices*, Ann. of Math. (2) **36** (1935), no. 4, 886–964. [doi:10.2307/1968595](https://doi.org/10.2307/1968595).
- [And92] Y. André, *Mumford–Tate groups of mixed Hodge structures and the theorem of the fixed part*, Compositio Math. **82** (1992), no. 1, 1–24. [numdam.org/item/CM\\_1992\\_\\_82\\_1\\_1\\_0](https://numdam.org/item/CM_1992__82_1_1_0).
- [And96a] Y. André, *On the Shafarevich and Tate conjectures for hyperkähler varieties*, Math. Ann. **305** (1996), no. 2, 205–248. [doi:10.1007/BF01444219](https://doi.org/10.1007/BF01444219).
- [And96b] Y. André, *Pour une théorie inconditionnelle des motifs*, Inst. Hautes Études Sci. Publ. Math. **83** (1996), 5–49. [doi:10.1007/BF02698643](https://doi.org/10.1007/BF02698643).

- [BB66] W. L. Baily, Jr. and A. Borel, *Compactification of arithmetic quotients of bounded symmetric domains*, Ann. of Math. (2) **84** (1966), no. 3, 442–528. doi:10.2307/1970457.
- [Bea83] A. Beauville, *Variétés Kähleriennes dont la première classe de Chern est nulle*, J. Differential Geom. **18** (1983), no. 4, 755–782. doi:10.4310/jdg/1214438181.
- [BF03] R.-O. Buchweitz and H. Flenner, *A semiregularity map for modules and applications to deformations*, Compositio Math. **137** (2003), no. 2, 135–210. doi:10.1023/A:1023999012081.
- [BF08] R.-O. Buchweitz and H. Flenner, *The global decomposition theorem for Hochschild (co-)homology of singular spaces via the Atiyah–Chern character*, Adv. Math. **217** (2008), no. 1, 243–281. doi:10.1016/j.aim.2007.06.013.
- [BLM23] R. Bandiera, E. Lepri and M. Manetti,  *$L_\infty$  liftings of semiregularity maps via Chern–Simons classes*, Adv. Math. **435** (2023), 109358. doi:10.1016/j.aim.2023.109358. arXiv:2111.12985.
- [BL04] C. Birkenhake and H. Lange, *Complex Abelian Varieties*, 2nd ed., Grundlehren der mathematischen Wissenschaften **302**, Springer, Berlin, 2004. doi:10.1007/978-3-662-06307-1.
- [BEK14] S. Bloch, H. Esnault and M. Kerz,  *$p$ -adic deformation of algebraic cycle classes*, Invent. Math. **195** (2014), 673–722. doi:10.1007/s00222-013-0461-4.
- [BS94] S. Bando and Y.-T. Siu, *Stable sheaves and Einstein–Hermitian metrics*, in: Geometry and Analysis on Complex Manifolds, World Sci. Publ., River Edge, NJ, 1994, pp. 39–50. doi:10.1142/9789814350112\_0002.
- [Blo72] S. Bloch, *Semi-regularity and de Rham cohomology*, Invent. Math. **17** (1972), 51–66. doi:10.1007/BF01390023.
- [Bus19] N. Buskin, *Every rational Hodge isometry between two K3 surfaces is algebraic*, J. Reine Angew. Math. **755** (2019), 127–150. doi:10.1515/crelle-2017-0027.
- [Cal05] A. Căldăraru, *The Mukai pairing, II: the Hochschild–Kostant–Rosenberg isomorphism*, Adv. Math. **194** (2005), no. 1, 34–66. doi:10.1016/j.aim.2004.05.012.
- [CDK95] E. Cattani, P. Deligne and A. Kaplan, *On the locus of Hodge classes*, J. Amer. Math. Soc. **8** (1995), no. 2, 483–506. doi:10.1090/S0894-0347-1995-1273413-2.
- [CKS86] E. Cattani, A. Kaplan and W. Schmid, *Degeneration of Hodge structures*, Ann. of Math. (2) **123** (1986), no. 3, 457–535. doi:10.2307/1971333.
- [CRV12] D. Calaque, C. A. Rossi and M. Van den Bergh, *Căldăraru’s conjecture and Tsygan’s formality*, Ann. of Math. (2) **176** (2012), no. 2, 865–923. doi:10.4007/annals.2012.176.2.4.
- [Del71] P. Deligne, *Théorie de Hodge. II*, Inst. Hautes Études Sci. Publ. Math. **40** (1971), 5–57. doi:10.1007/BF02684692.
- [Del72] P. Deligne, *La conjecture de Weil pour les surfaces K3*, Invent. Math. **15** (1972), 206–226. doi:10.1007/BF01404126.
- [Del74] P. Deligne, *La conjecture de Weil. I*, Inst. Hautes Études Sci. Publ. Math. **43** (1974), 273–307. doi:10.1007/BF02684373.
- [Del82] P. Deligne, *Hodge cycles on abelian varieties* (notes by J. S. Milne), in: Hodge Cycles, Motives, and Shimura Varieties, Lecture Notes in Math. **900**, Springer, Berlin, 1982, pp. 9–100. doi:10.1007/978-3-540-38955-2\_3.
- [FM98] B. Fantechi and M. Manetti, *Obstruction calculus for functors of Artin rings, I*, J. Algebra **202** (1998), no. 2, 541–576. doi:10.1006/jabr.1997.7239.
- [FF26] S. Floccari and L. Fu, *The Hodge conjecture for Weil fourfolds with discriminant 1 via singular OG6-varieties*, J. Math. Pures Appl. (9) **210** (2026), paper no. 103876. doi:10.1016/j.matpur.2026.103876.
- [Flo24] S. Floccari, *Sixfolds of generalized Kummer type and K3 surfaces*, Compos. Math. **160** (2024), no. 2, 388–410. doi:10.1112/S0010437X23007625.
- [Fuj87] A. Fujiki, *On the de Rham cohomology group of a compact Kähler symplectic manifold*, in: Algebraic Geometry, Sendai, 1985, Adv. Stud. Pure Math. **10** (1987), 105–165. doi:10.2969/aspm/01010105.
- [Ful98] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **2**, Springer, New York, 1998. doi:10.1007/978-1-4612-1700-8.
- [Gal00] F. Galluzzi, *Abelian fourfold of Mumford-type and Kuga–Satake varieties*, Indag. Math. (N.S.) **11** (2000), no. 4, 547–560. doi:10.1016/S0019-3577(00)80024-5.
- [GM88] W. M. Goldman and J. J. Millson, *The deformation theory of representations of fundamental groups of compact Kähler manifolds*, Inst. Hautes Études Sci. Publ. Math. **67** (1988), 43–96. doi:10.1007/BF02699127.
- [GP98] M. Gross and S. Popescu, *Equations of  $(1, d)$ -polarized abelian surfaces*, Math. Ann. **310** (1998), no. 2, 333–377. doi:10.1007/s002080050151.

- [Gri10] J. Grivaux, *Chern classes in Deligne cohomology for coherent analytic sheaves*, Math. Ann. **347** (2010), no. 2, 249–284. doi:10.1007/s00208-009-0430-9.
- [Gro61] A. Grothendieck, *Techniques de construction et théorèmes d’existence en géométrie algébrique IV: les schémas de Hilbert*, Séminaire Bourbaki, Vol. 6 (1960/61), Exp. No. 221, 249–276; reprinted by Soc. Math. France, Paris, 1995.
- [HM73] G. Horrocks and D. Mumford, *A rank 2 vector bundle on  $\mathbf{P}^4$  with 15,000 symmetries*, Topology **12** (1973), 63–81. doi:10.1016/0040-9383(73)90022-0.
- [Hua21] S. Huang, *A note on a question of Markman*, J. Pure Appl. Algebra **225** (2021), no. 9, 106673. doi:10.1016/j.jpaa.2021.106673.
- [Huy16] D. Huybrechts, *Lectures on K3 Surfaces*, Cambridge Studies in Advanced Mathematics **158**, Cambridge University Press, Cambridge, 2016. doi:10.1017/CB09781316594193.
- [Kle68] S. L. Kleiman, *Algebraic cycles and the Weil conjectures*, in: Dix exposés sur la cohomologie des schémas, North-Holland, Amsterdam, 1968, pp. 359–386.
- [Kle94] S. L. Kleiman, *The standard conjectures*, in: Motives (Seattle, WA, 1991), Proc. Sympos. Pure Math. **55**, Part 1, Amer. Math. Soc., Providence, RI, 1994, pp. 3–20. doi:10.1090/pspum/055.1/1265519.
- [KS67] M. Kuga and I. Satake, *Abelian varieties attached to polarized K3-surfaces*, Math. Ann. **169** (1967), 239–242. doi:10.1007/BF01399540.
- [Kur65] M. Kuranishi, *New proof for the existence of locally complete families of complex structures*, in: Proc. Conf. Complex Analysis (Minneapolis, 1964), Springer, Berlin, 1965, pp. 142–154. doi:10.1007/978-3-642-48016-4\_13.
- [Li26] N. Li, *The Tate conjecture for powers of abelian fourfolds and the Hodge conjecture for powers of CM fourfolds*, preprint, 2026. arXiv:2609.27916.
- [Lie06] M. Lieblich, *Moduli of complexes on a proper morphism*, J. Algebraic Geom. **15** (2006), no. 1, 175–206. doi:10.1090/S1056-3911-05-00418-2.
- [Mar25a] E. Markman, *Cycles on abelian 2n-folds of Weil type from secant sheaves on abelian n-folds*, preprint, arXiv:2502.03415 (2025). doi:10.48550/arXiv.2502.03415.
- [Mar25b] E. Markman, *Secant sheaves and Weil classes on abelian varieties*, preprint, arXiv:2509.23403 (2025). doi:10.48550/arXiv.2509.23403.
- [Mar25c] E. Markman, *Secant sheaves on abelian n-folds with real multiplication and Weil classes on abelian 2n-folds with complex multiplication*, preprint, 2025. arXiv:2509.23079.
- [Mat86] H. Matsumura, *Commutative Ring Theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge, 1986. doi:10.1017/CB09781139171762.
- [Mil20] J. S. Milne, *Hodge classes on abelian varieties*, preprint, 2020. arXiv:2010.08857.
- [Mil22] J. S. Milne, *The Tate and standard conjectures for certain abelian varieties*, preprint, arXiv:2112.12815, version 2 (2022). doi:10.48550/arXiv.2112.12815.
- [Mor84] D. R. Morrison, *On K3 surfaces with large Picard number*, Invent. Math. **75** (1984), no. 1, 105–121. doi:10.1007/BF01403093.
- [Mor19] M. Morrow, *A variational Tate conjecture in crystalline cohomology*, J. Eur. Math. Soc. **21** (2019), no. 11, 3467–3511. doi:10.4171/JEMS/907.
- [Muk78] S. Mukai, *Semi-homogeneous vector bundles on an abelian variety*, J. Math. Kyoto Univ. **18** (1978), no. 2, 239–272. doi:10.1215/kjm/1250522574.
- [Muk81] S. Mukai, *Duality between  $D(X)$  and  $D(\hat{X})$  with its application to Picard sheaves*, Nagoya Math. J. **81** (1981), 153–175. doi:10.1017/S002776300001922X.
- [Mum69] D. Mumford, *A note of Shimura’s paper “Discontinuous groups and abelian varieties”*, Math. Ann. **181** (1969), 345–351. doi:10.1007/BF01350672.
- [Mur90] J. P. Murre, *On the motive of an algebraic surface*, J. Reine Angew. Math. **409** (1990), 190–204. doi:10.1515/crll.1990.409.190.
- [MZ95] B. Moonen and Yu. G. Zarhin, *Hodge classes and Tate classes on simple abelian fourfolds*, Duke Math. J. **77** (1995), no. 3, 553–581. doi:10.1215/S0012-7094-95-07717-5.
- [MZ99] B. Moonen and Yu. G. Zarhin, *Hodge classes on abelian varieties of low dimension*, Math. Ann. **315** (1999), no. 4, 711–733. doi:10.1007/s002080050333.
- [OMe00] O. T. O’Meara, *Introduction to Quadratic Forms*, Classics in Mathematics, Springer, Berlin, 2000 (reprint of the 1973 edition). doi:10.1007/978-3-642-62031-7.
- [Per26] A. Perry, *The semiregularity theorem for equivariant noncommutative varieties*, preprint, arXiv:2604.00511 (2026). doi:10.48550/arXiv.2604.00511.
- [PR94] V. P. Platonov and A. S. Rapinchuk, *Algebraic Groups and Number Theory*, Pure and Applied Mathematics **139**, Academic Press, Boston, MA, 1994.

- [Pri24] J. P. Pridham, *Semiregularity as a consequence of Goodwillie’s theorem*, Forum Math. Sigma **12** (2024), e126. arXiv:1208.3111. doi:10.1017/fms.2024.132.
- [Sat66] I. Satake, *Clifford algebras and families of abelian varieties*, Nagoya Math. J. **27** (1966), 435–446. doi:10.1017/S0027763000026295.
- [Sch10] U. Schlickewei, *The Hodge conjecture for self-products of certain K3 surfaces*, J. Algebra **324** (2010), no. 3, 507–529. doi:10.1016/j.jalgebra.2010.04.010.
- [Sch73] W. Schmid, *Variation of Hodge structure: the singularities of the period mapping*, Invent. Math. **22** (1973), 211–319. doi:10.1007/BF01389674.
- [Sch85] W. Scharlau, *Quadratic and Hermitian Forms*, Grundlehren der mathematischen Wissenschaften **270**, Springer, Berlin, 1985. doi:10.1007/978-3-642-69971-9.
- [Ser00] J.-P. Serre, *Local Algebra*, Springer Monographs in Mathematics, Springer, Berlin, 2000. doi:10.1007/978-3-662-04203-8.
- [SGA 4 $\frac{1}{2}$ ] P. Deligne, *Cohomologie étale* (SGA 4 $\frac{1}{2}$ ), with the exposé [Cycle], *La classe de cohomologie associée à un cycle*, by A. Grothendieck, rédigé par P. Deligne, pp. 129–153, Lecture Notes in Math. **569**, Springer, Berlin, 1977. doi:10.1007/BFb0091516.
- [Stacks] The Stacks project authors, *The Stacks project*, <https://stacks.math.columbia.edu>, 2026.
- [Tan03] S. G. Tankeev, *On the standard conjecture for complex abelian schemes over smooth projective curves*, Izv. Math. **67** (2003), no. 3, 597–635. doi:10.1070/IM2003v067n03ABEH000439.
- [Tod09] Y. Toda, *Deformations and Fourier–Mukai transforms*, J. Differential Geom. **81** (2009), no. 1, 197–224. doi:10.4310/jdg/1228400631.
- [Tot26] B. Totaro, *On the Hodge conjecture*, guest post on the weblog of T. Tao, 11 September 2026. [terrytao.wordpress.com/2026/09/11/on-the-hodge-conjecture](https://terrytao.wordpress.com/2026/09/11/on-the-hodge-conjecture).
- [Var22] M. Varesco, *The Hodge conjecture for powers of K3 surfaces of Picard number 16*, Michigan Math. J., advance publication (2024), 1–32; arXiv:2203.09778. doi:10.1307/mmj/20236349.
- [Var23] M. Varesco, *Hodge similarities, algebraic classes, and Kuga–Satake varieties*, Math. Z. **305** (2023), paper no. 69. doi:10.1007/s00209-023-03390-8.
- [vG00] B. van Geemen, *Kuga–Satake varieties and the Hodge conjecture*, in: The Arithmetic and Geometry of Algebraic Cycles (Banff, 1998), NATO Sci. Ser. C Math. Phys. Sci. **548**, Kluwer, Dordrecht, 2000, pp. 51–82. doi:10.1007/978-94-011-4098-0\_3.
- [vG08] B. van Geemen, *Real multiplication on K3 surfaces and Kuga Satake varieties*, Michigan Math. J. **56** (2008), no. 2, 375–399. doi:10.1307/mmj/1224783519.
- [vGR26] B. van Geemen and A. Rapagnetta, *Hyperkähler sixfolds, abelian fourfolds of Weil type and a Hodge class*, preprint, 2026. arXiv:2607.18341.
- [vGS25] B. van Geemen and M. Schütt, *On families of K3 surfaces with real multiplication*, Forum Math. Sigma **13** (2025), Paper No. e2. doi:10.1017/fms.2024.146.
- [Voi02] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge Studies in Advanced Mathematics **76**, Cambridge University Press, Cambridge, 2002. doi:10.1017/CB09780511615344.
- [Voi02b] C. Voisin, *A counterexample to the Hodge conjecture extended to Kähler varieties*, Int. Math. Res. Not. **2002**, no. 20, 1057–1075. doi:10.1155/S1073792802111135. arXiv:math/0112247.
- [Voi03] C. Voisin, *Hodge Theory and Complex Algebraic Geometry II*, Cambridge Studies in Advanced Mathematics **77**, Cambridge University Press, Cambridge, 2003. doi:10.1017/CB09780511615177.
- [Voi26] C. Voisin, *The status of the Hodge conjecture*, guest post on the weblog of T. Tao, 12 September 2026. [terrytao.wordpress.com/2026/09/12/the-status-of-the-hodge-conjecture](https://terrytao.wordpress.com/2026/09/12/the-status-of-the-hodge-conjecture).
- [Wei77] A. Weil, *Abelian varieties and the Hodge ring*, in: Œuvres Scientifiques, Collected Papers, Vol. III, Springer, New York, 1979, pp. 421–429.
- [Zar83] Yu. G. Zarhin, *Hodge groups of K3 surfaces*, J. Reine Angew. Math. **341** (1983), 193–220. doi:10.1515/crll.1983.341.193.
- [Zhu18] Y. Zhu, *K3 surfaces associated to abelian fourfolds of Mumford’s type*, preprint, arXiv:1812.06882 (2018). doi:10.48550/arXiv.1812.06882.
- [Zuc79] S. Zucker, *Hodge theory with degenerating coefficients:  $L_2$  cohomology in the Poincaré metric*, Ann. of Math. (2) **109** (1979), no. 3, 415–476. doi:10.2307/1971221.

FORMERLY, ELECTRO-GRAVITATIONAL SPACE PROPULSION LABORATORY (EGSPL), BHUBANESWAR,  
ODISHA 751030, INDIA

*Email address:* [d.bhattacharjee@erl-forschung.de](mailto:d.bhattacharjee@erl-forschung.de)

*Email address:* [itsdeep@live.com](mailto:itsdeep@live.com)

*URL:* ORCID [0000-0003-0466-750X](https://orcid.org/0000-0003-0466-750X)

DEPARTMENT OF MATHEMATICS, BIOINFORMATICS AND COMPUTER APPLICATIONS, MAULANA AZAD  
NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL, BHOPAL 462003, INDIA

*Email address:* [priyabrata@manit.ac.in](mailto:priyabrata@manit.ac.in)

*URL:* ORCID [0000-0001-6472-6239](https://orcid.org/0000-0001-6472-6239)

FORMERLY, NATIONAL TAIWAN UNIVERSITY, TAIPEI 106319, TAIWAN

*Email address:* [rai.bhattacharya.in@gmail.com](mailto:rai.bhattacharya.in@gmail.com)

*URL:* ORCID [0009-0002-2254-3914](https://orcid.org/0009-0002-2254-3914)